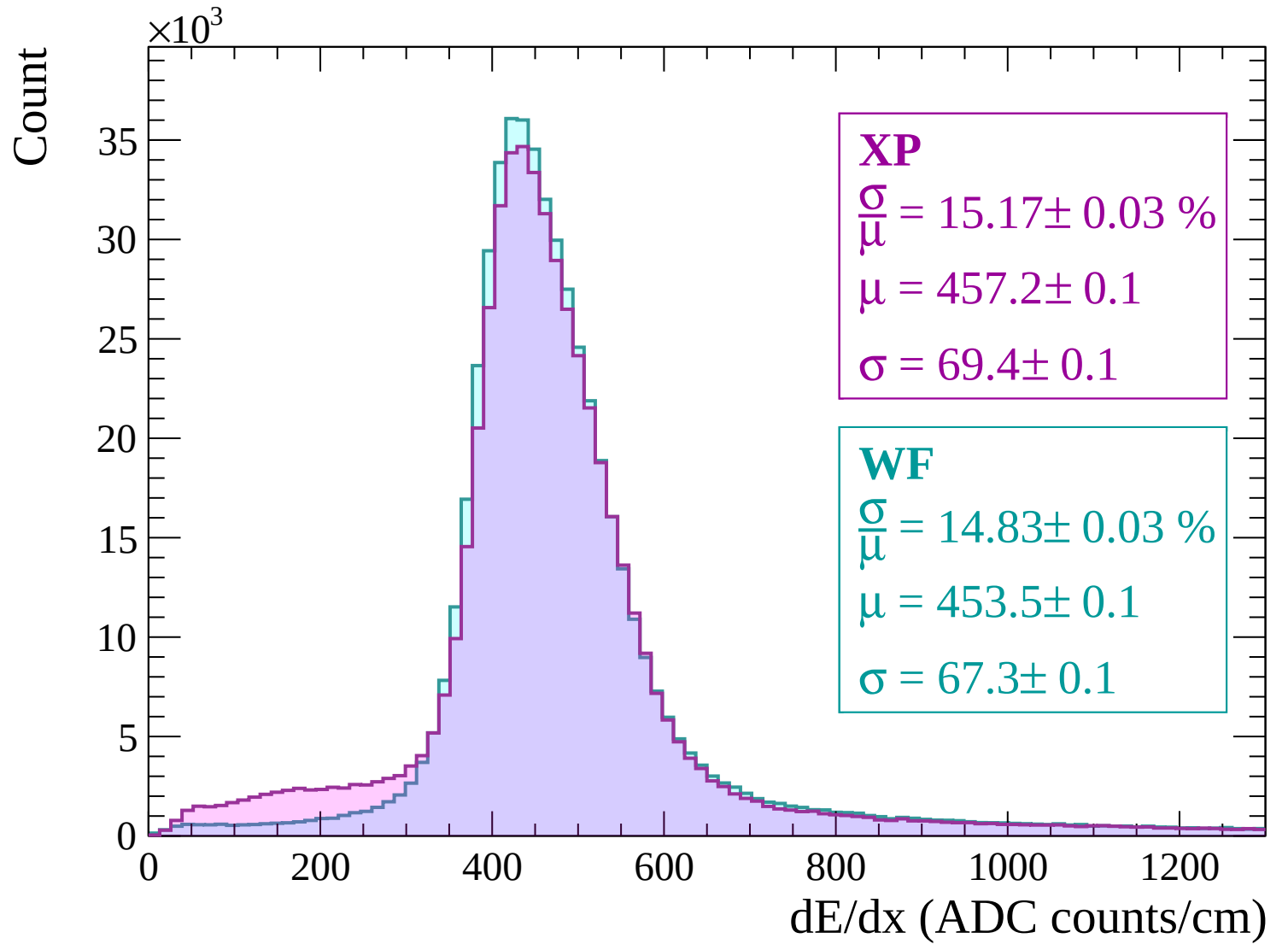
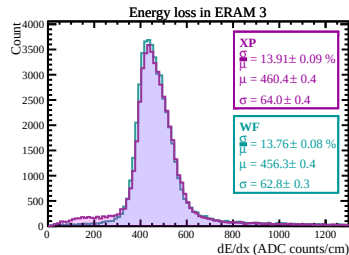
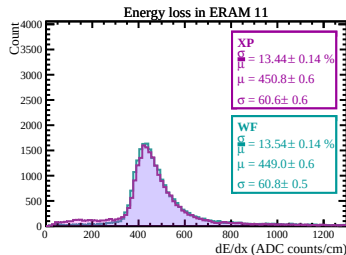
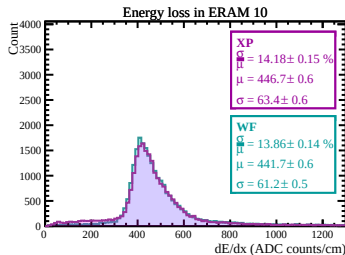
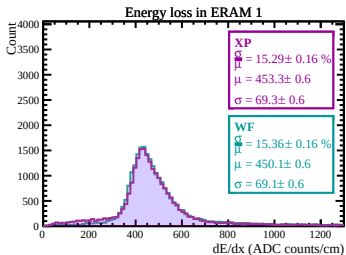
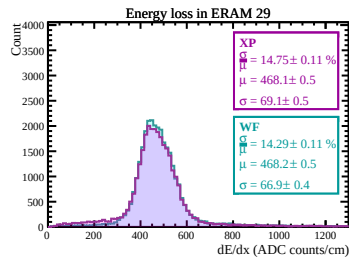
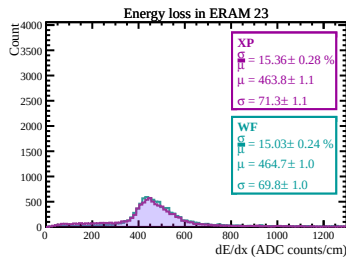
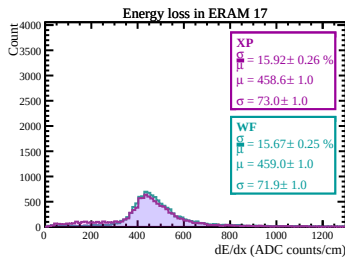
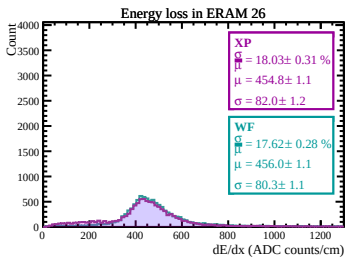
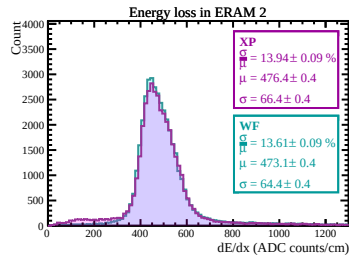
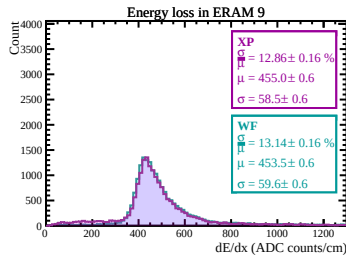
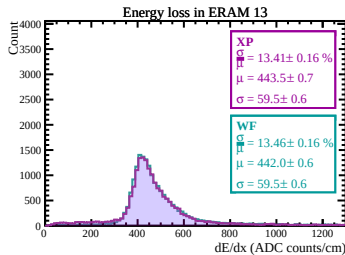
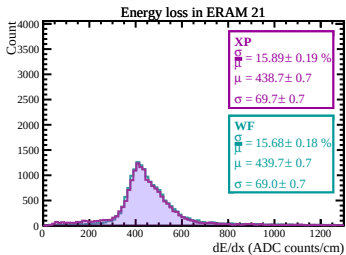
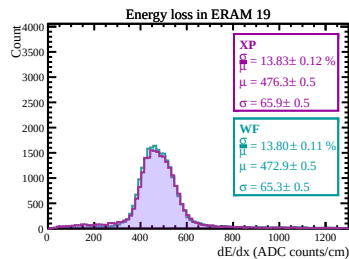
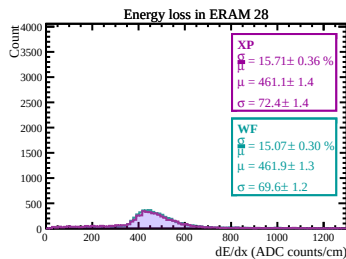
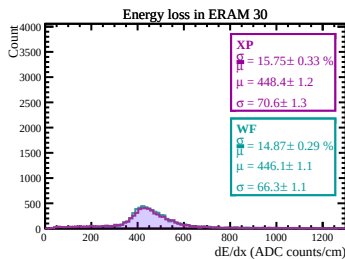
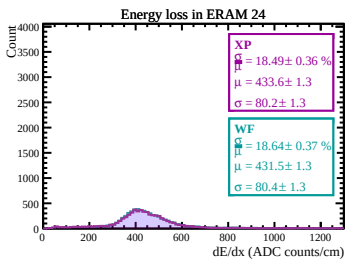
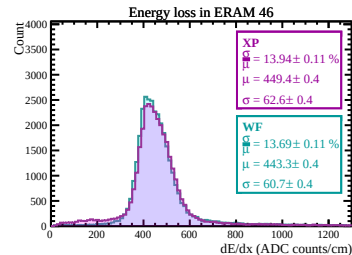
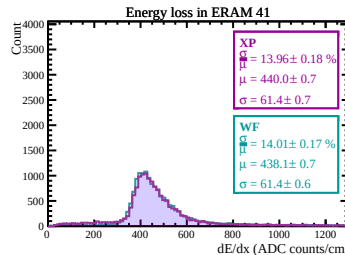
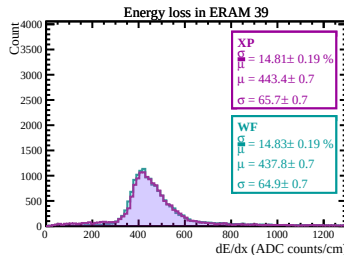
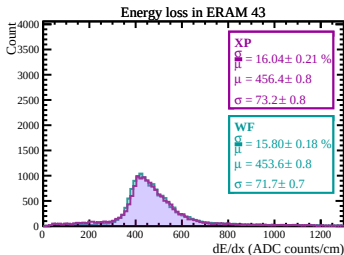
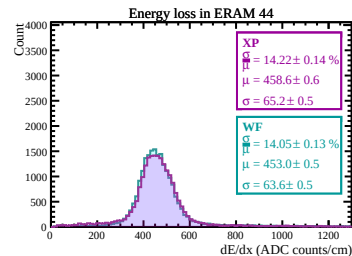
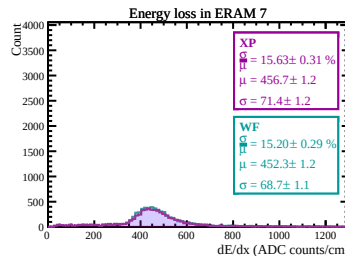
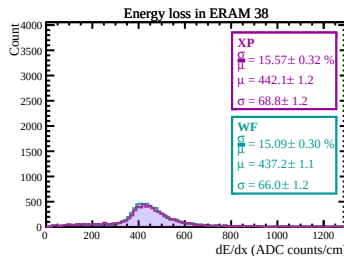
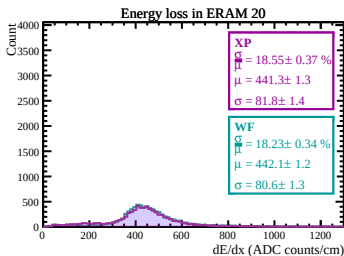
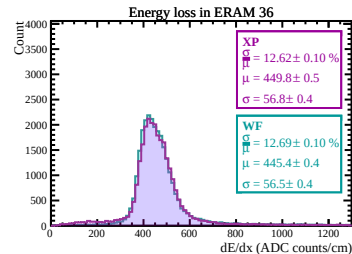
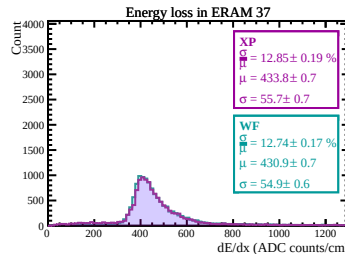
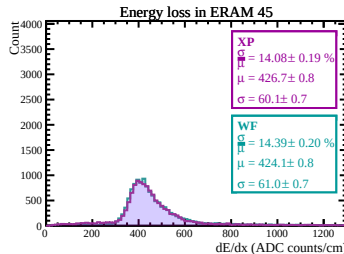
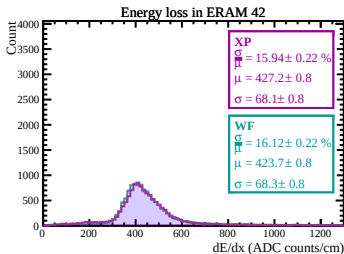
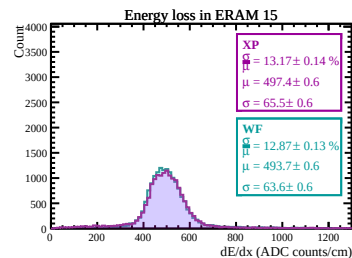
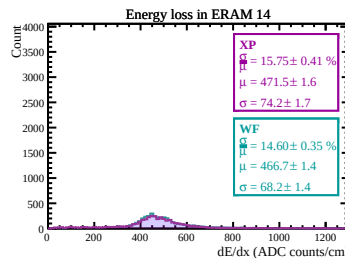
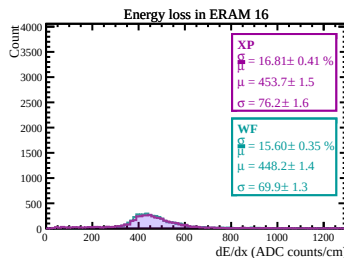
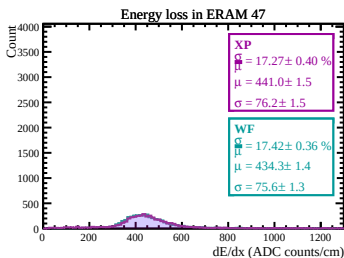
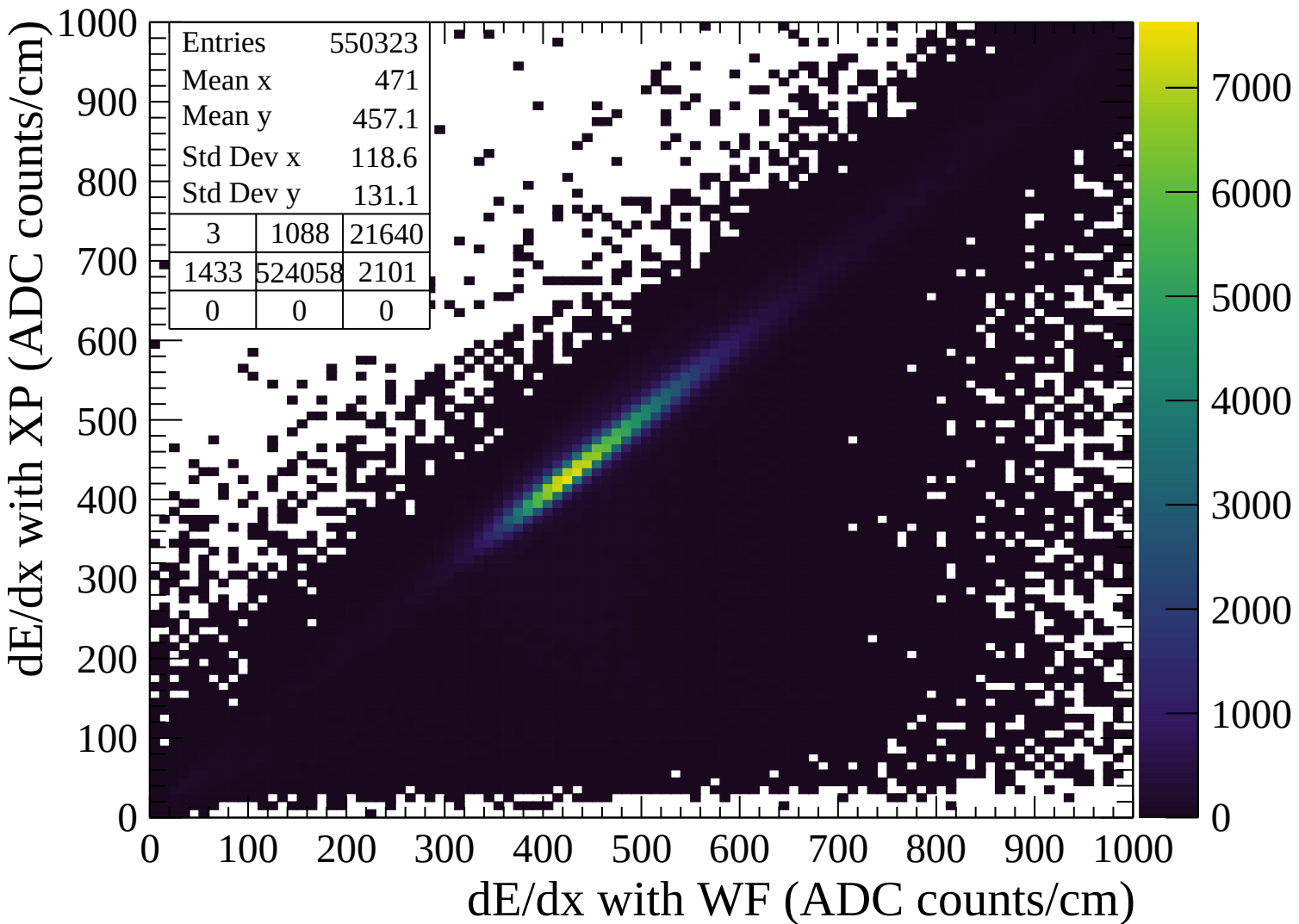
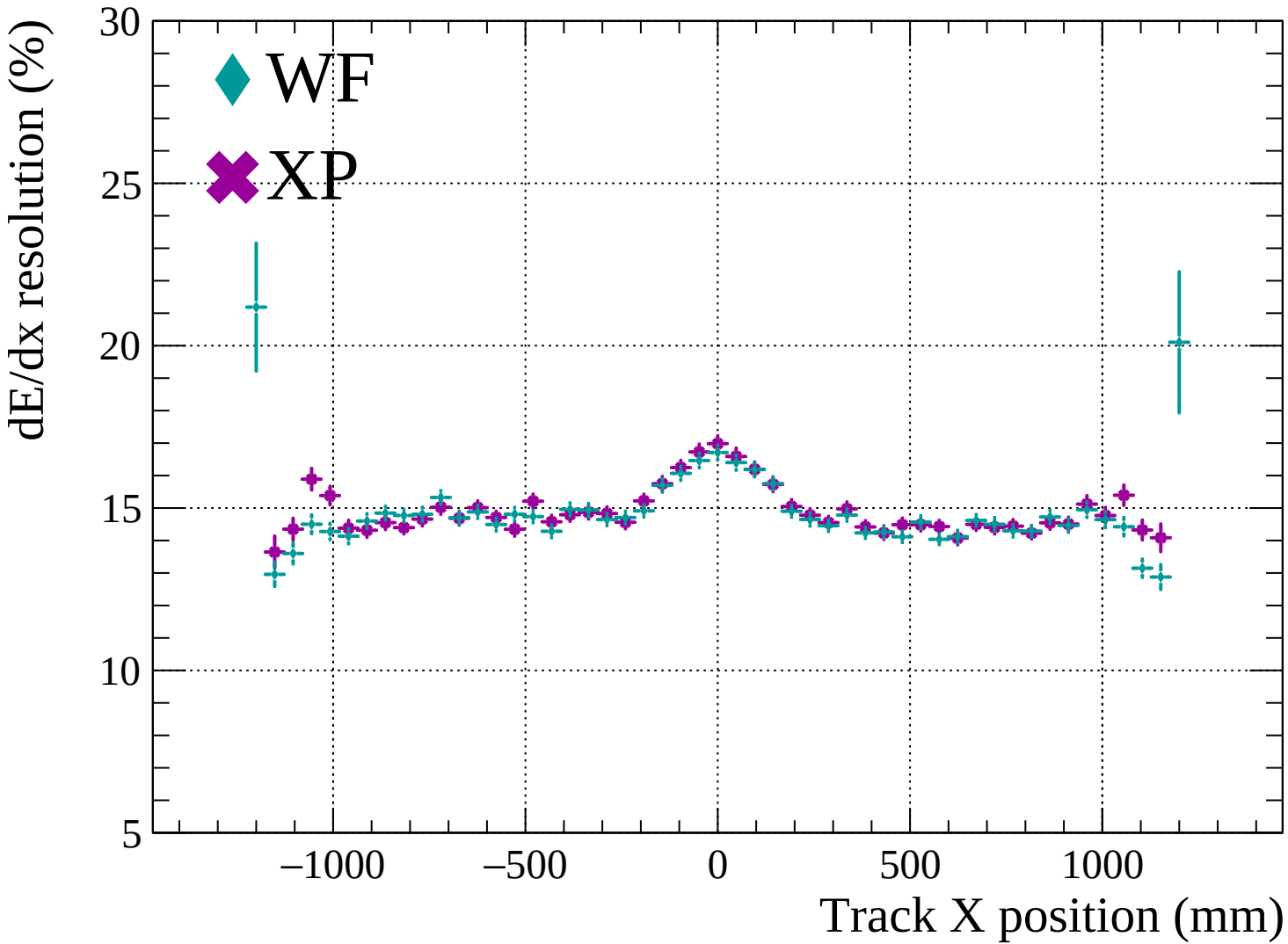


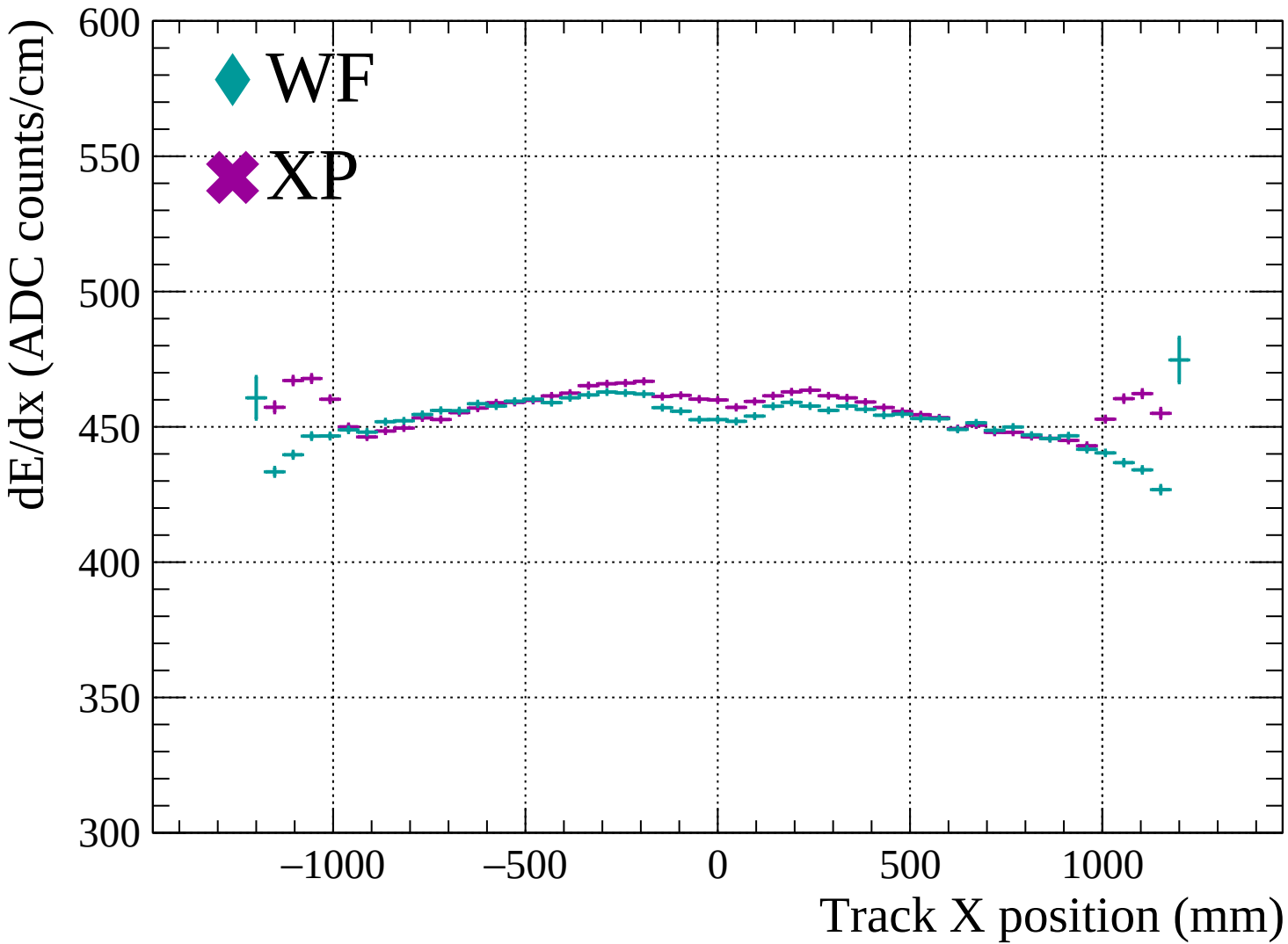


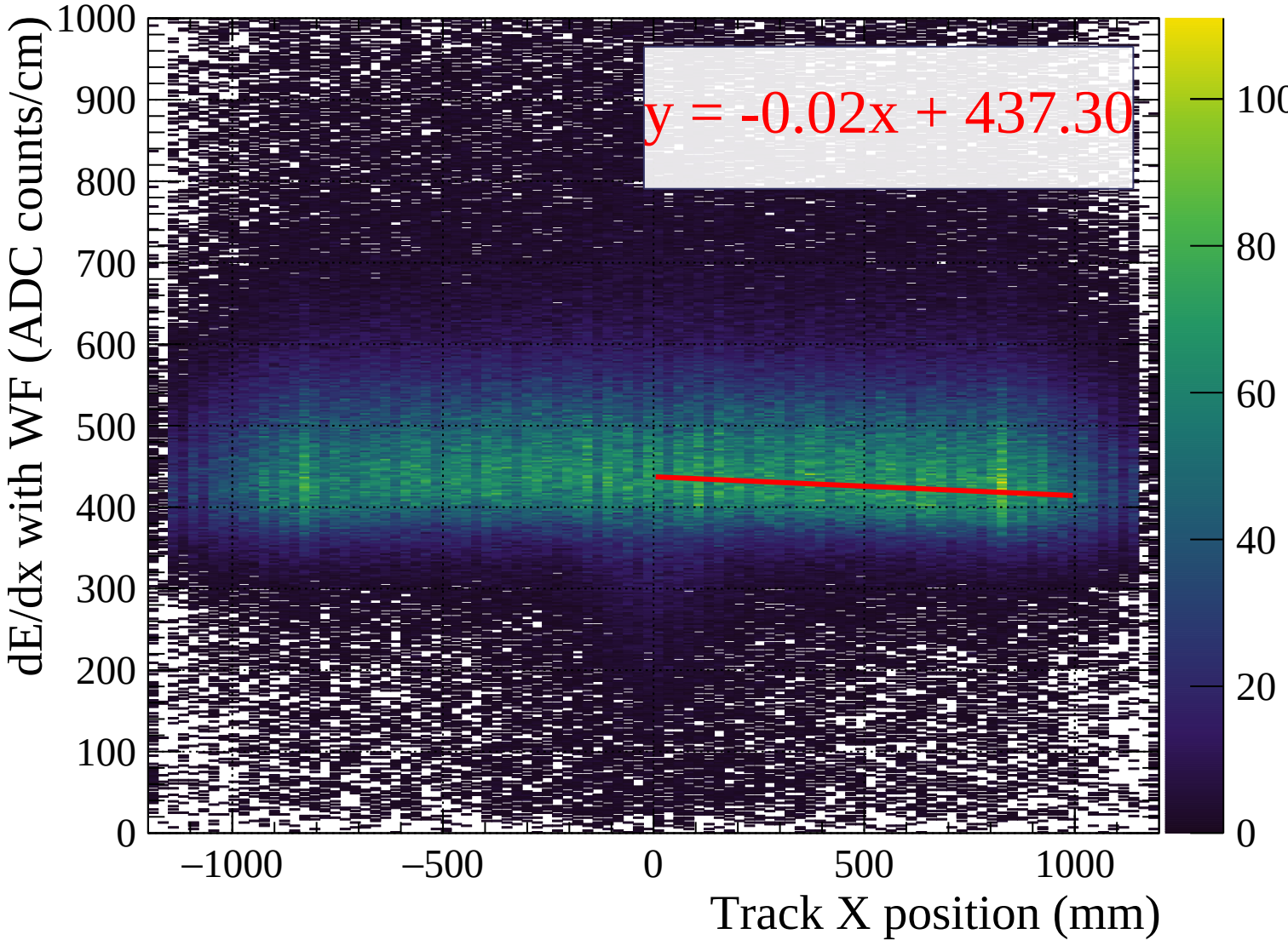


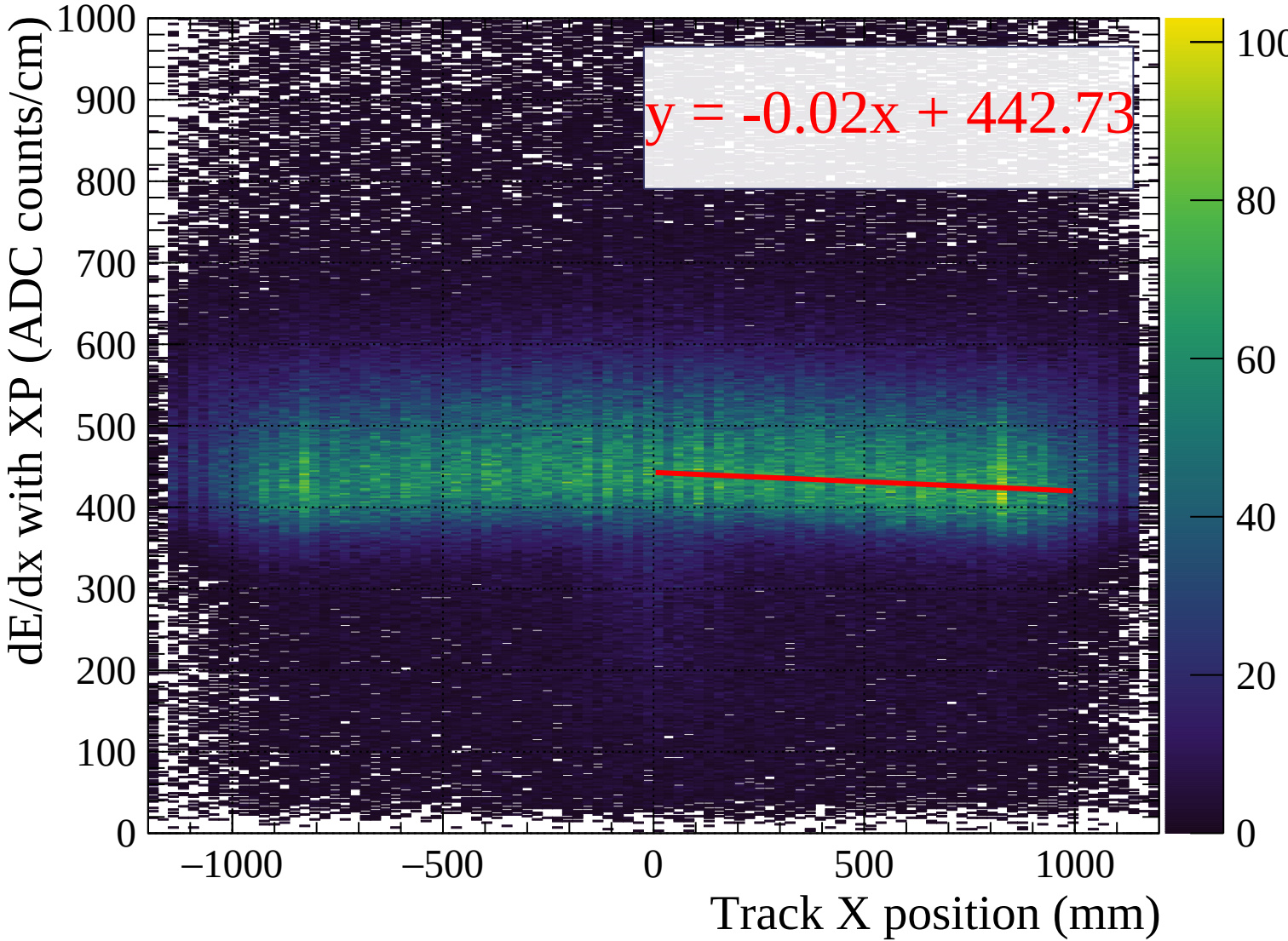


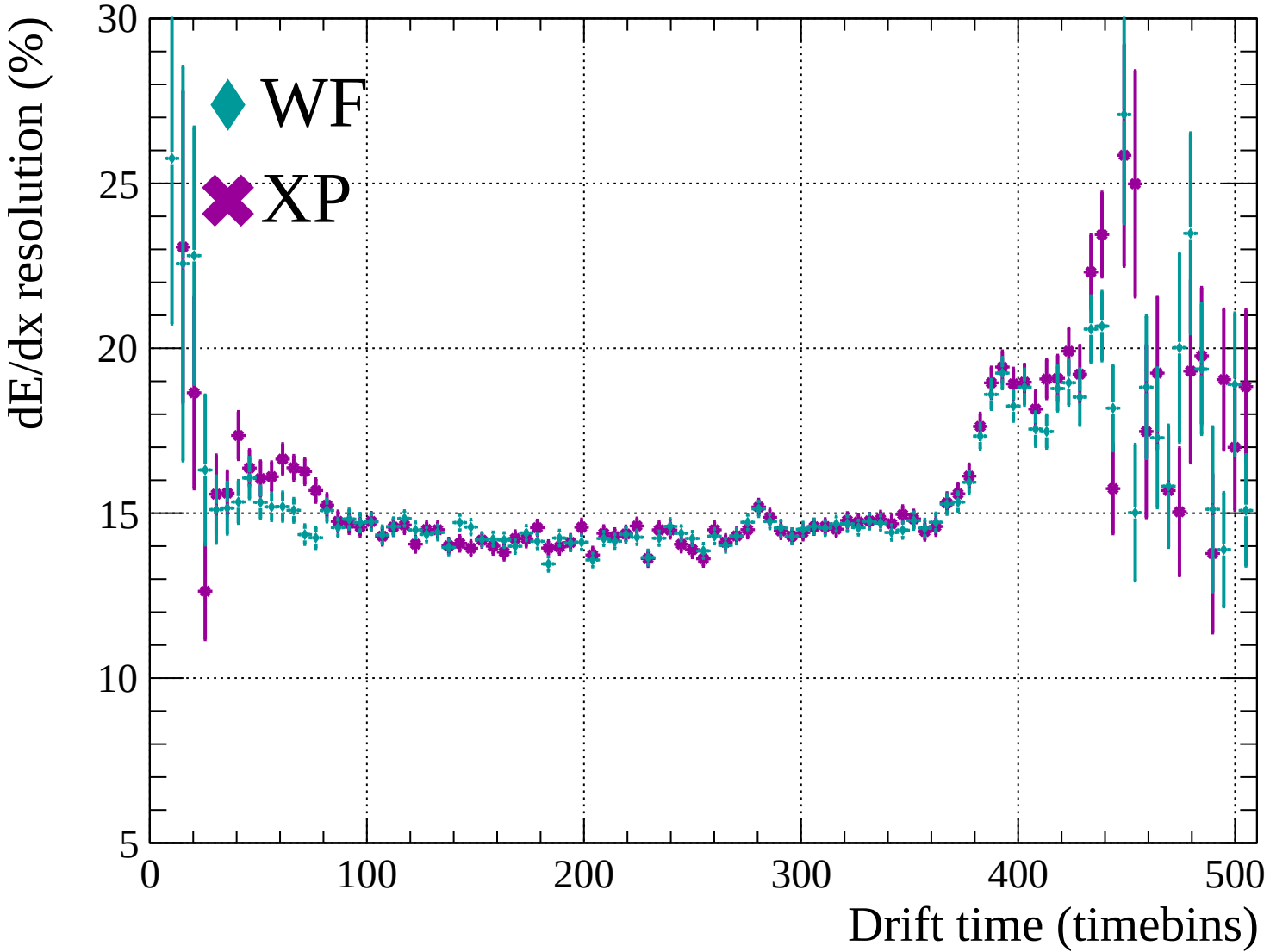


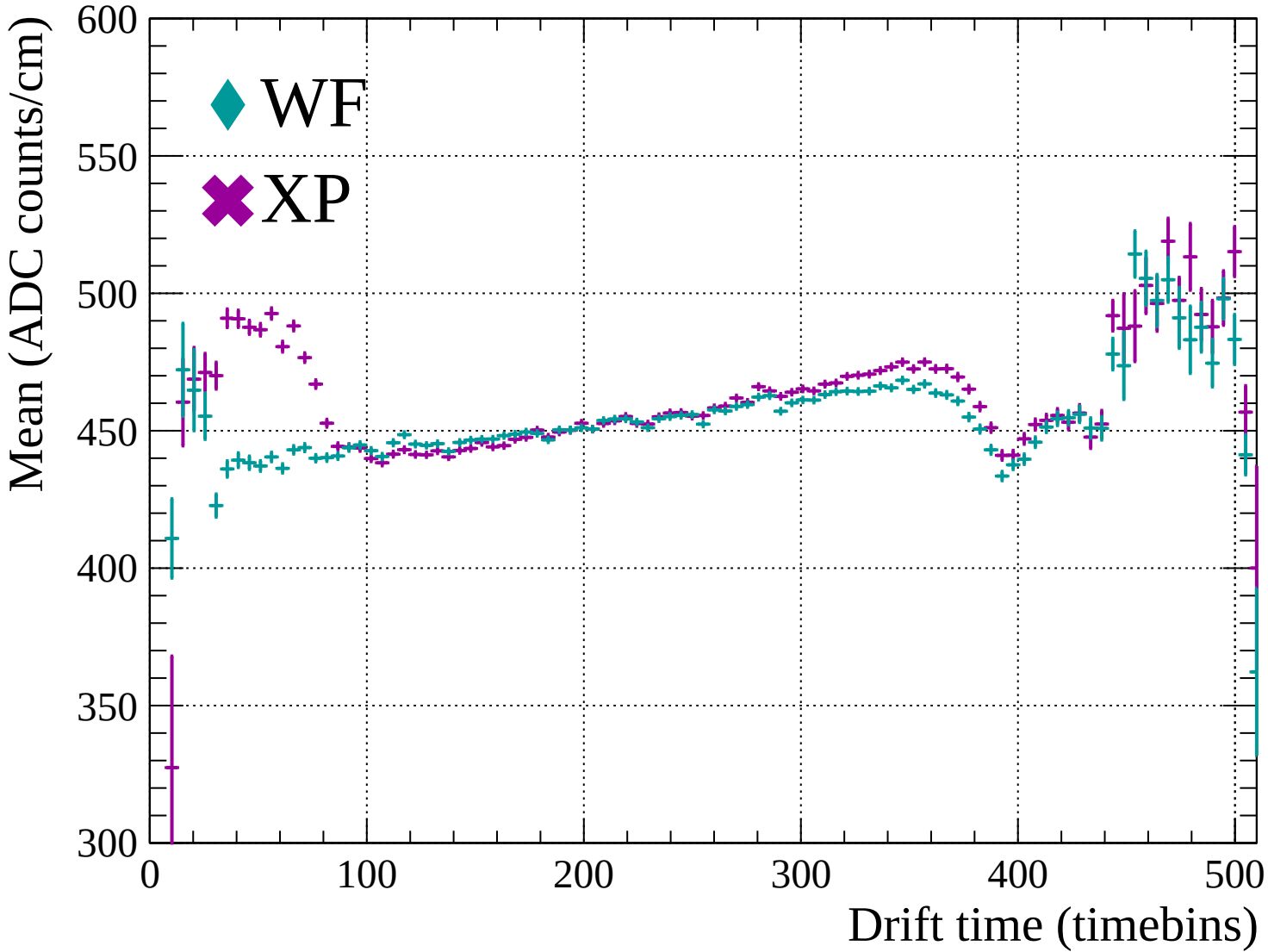


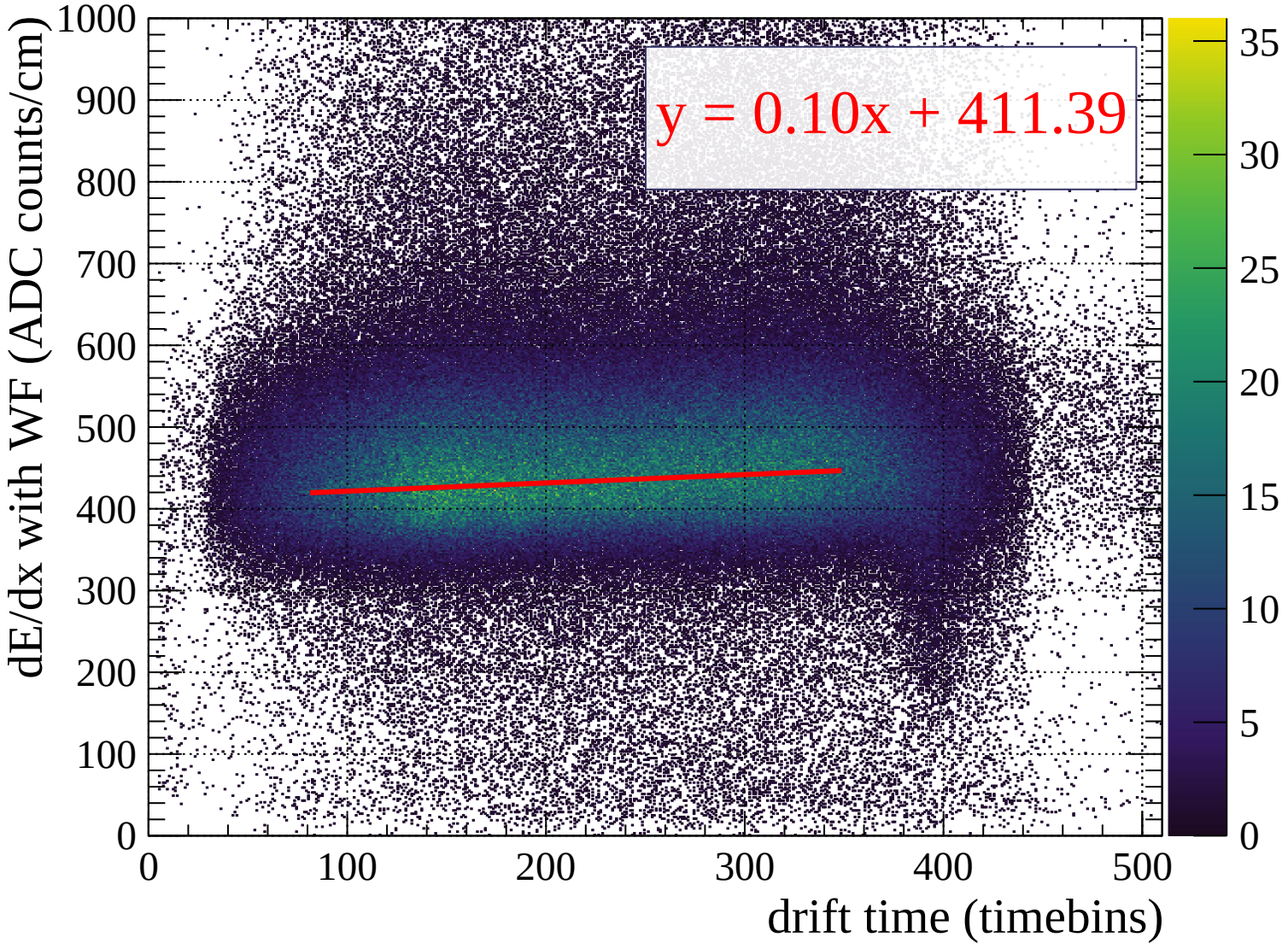


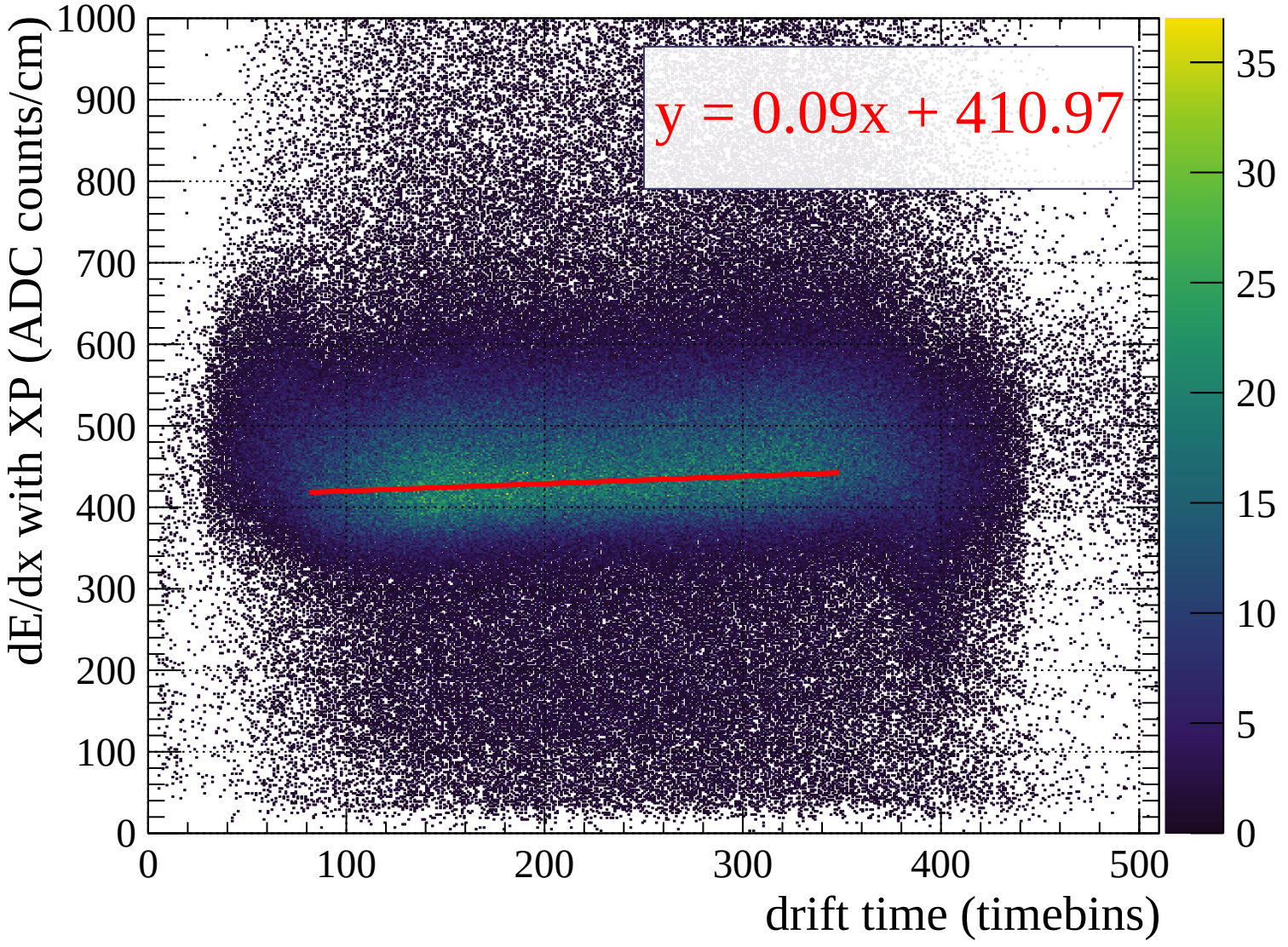


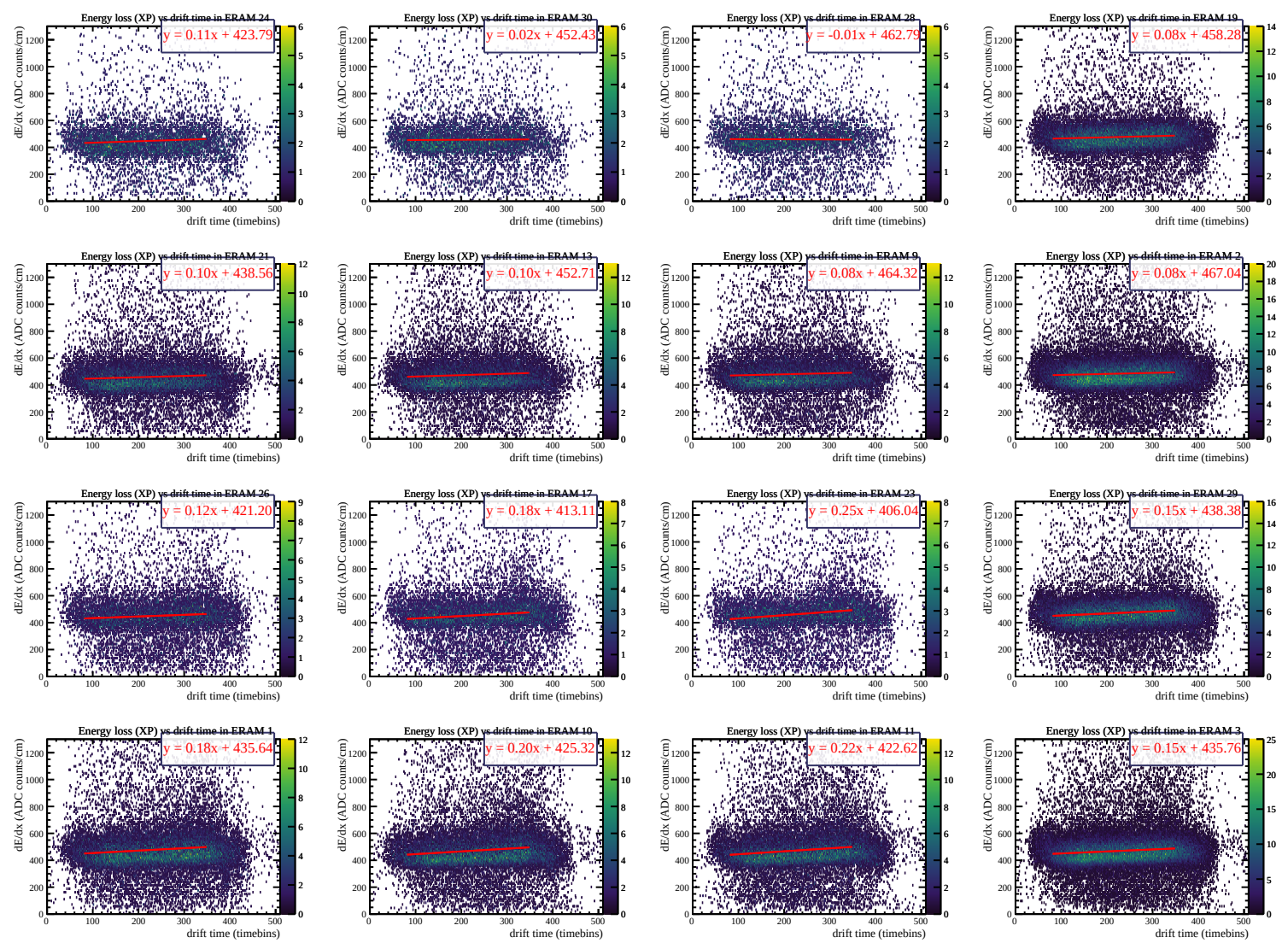


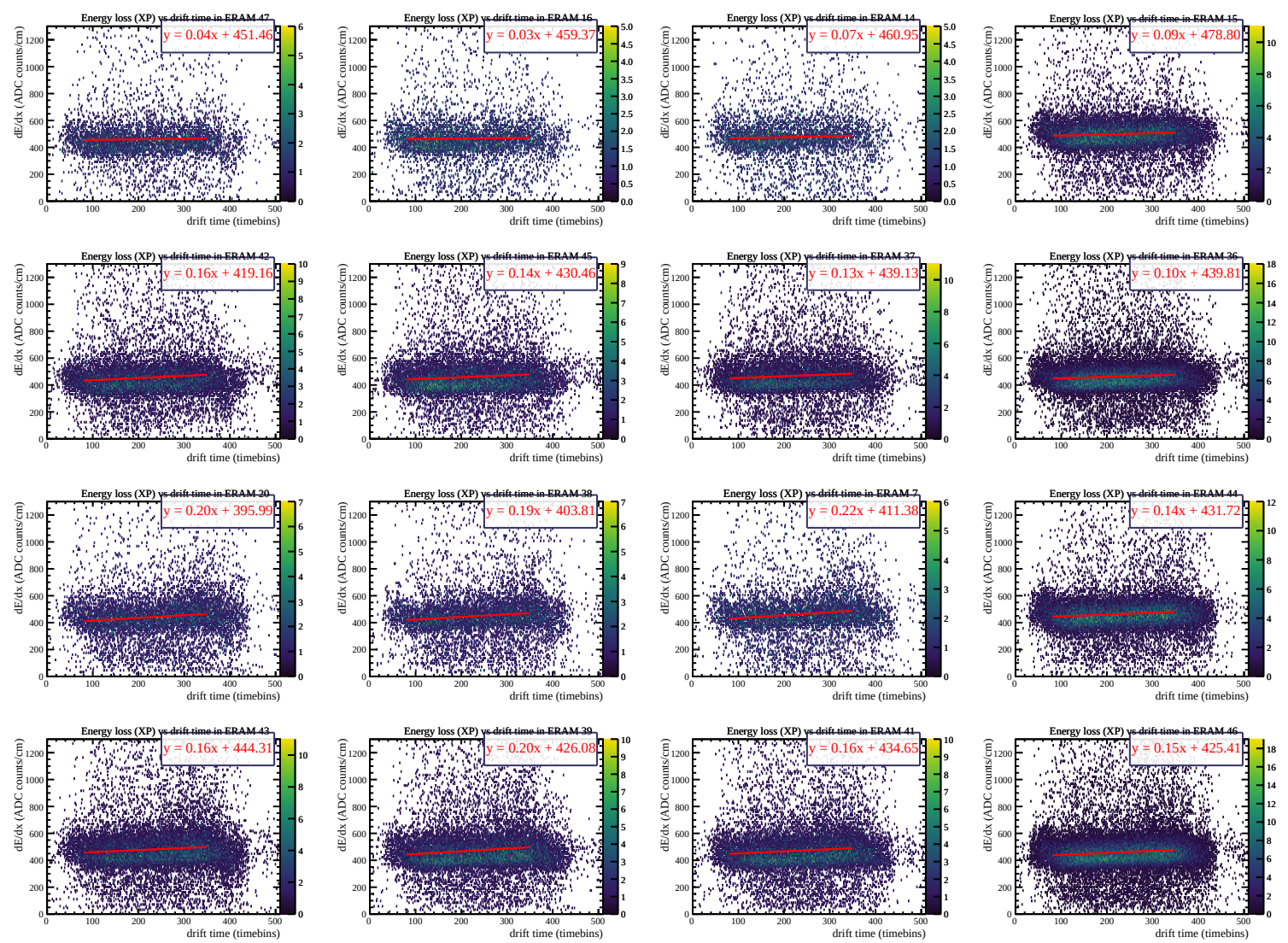


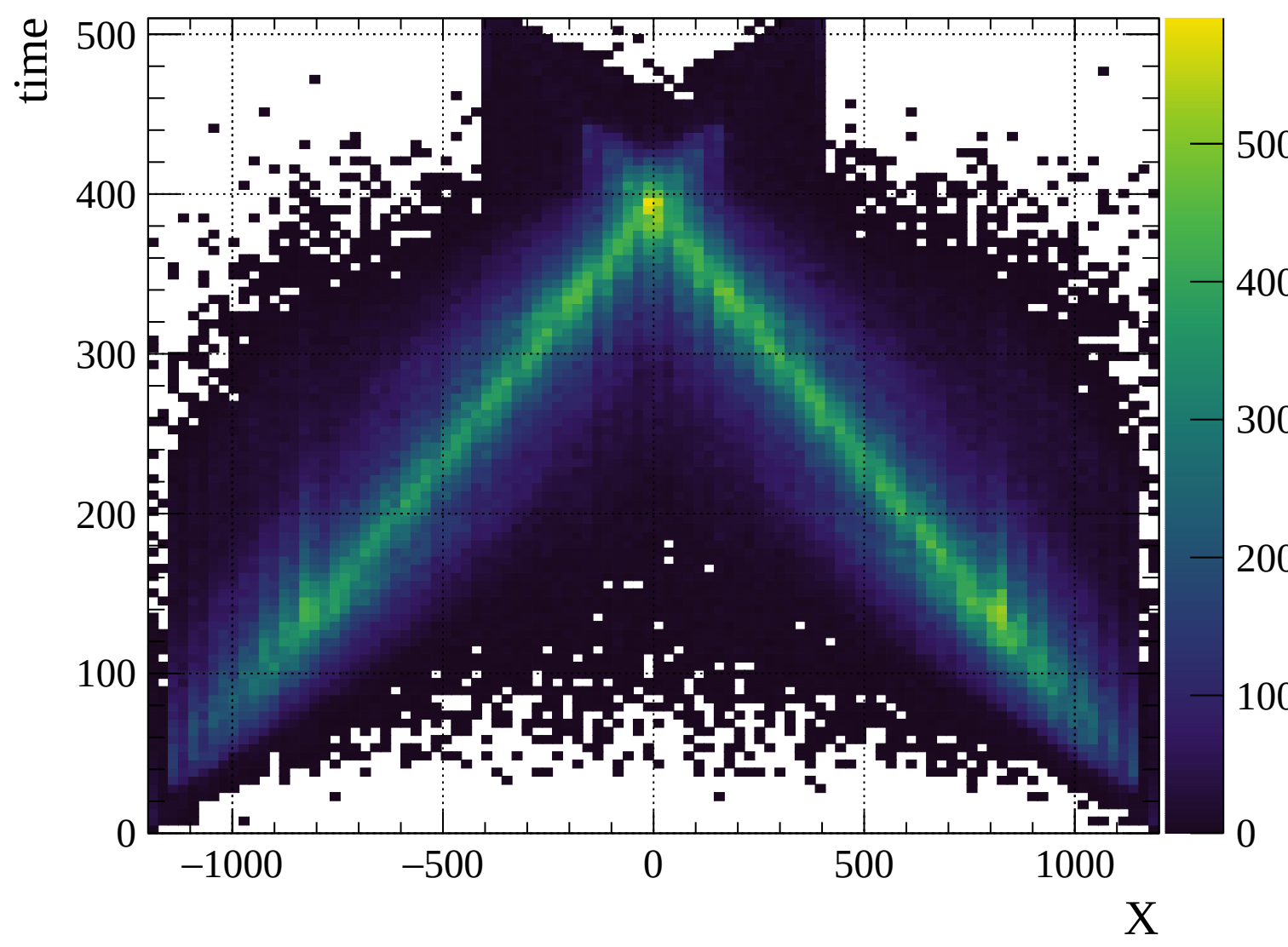


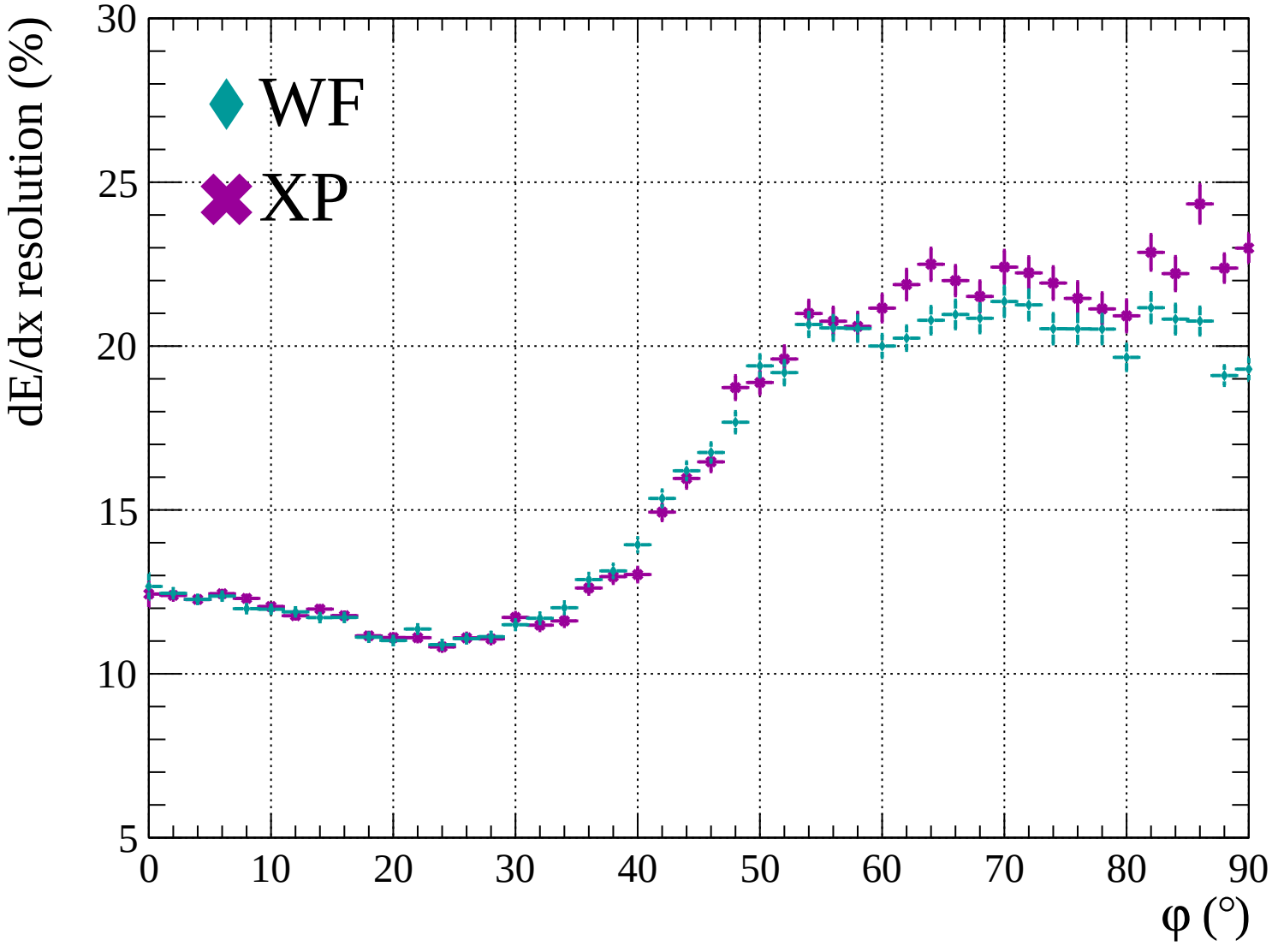


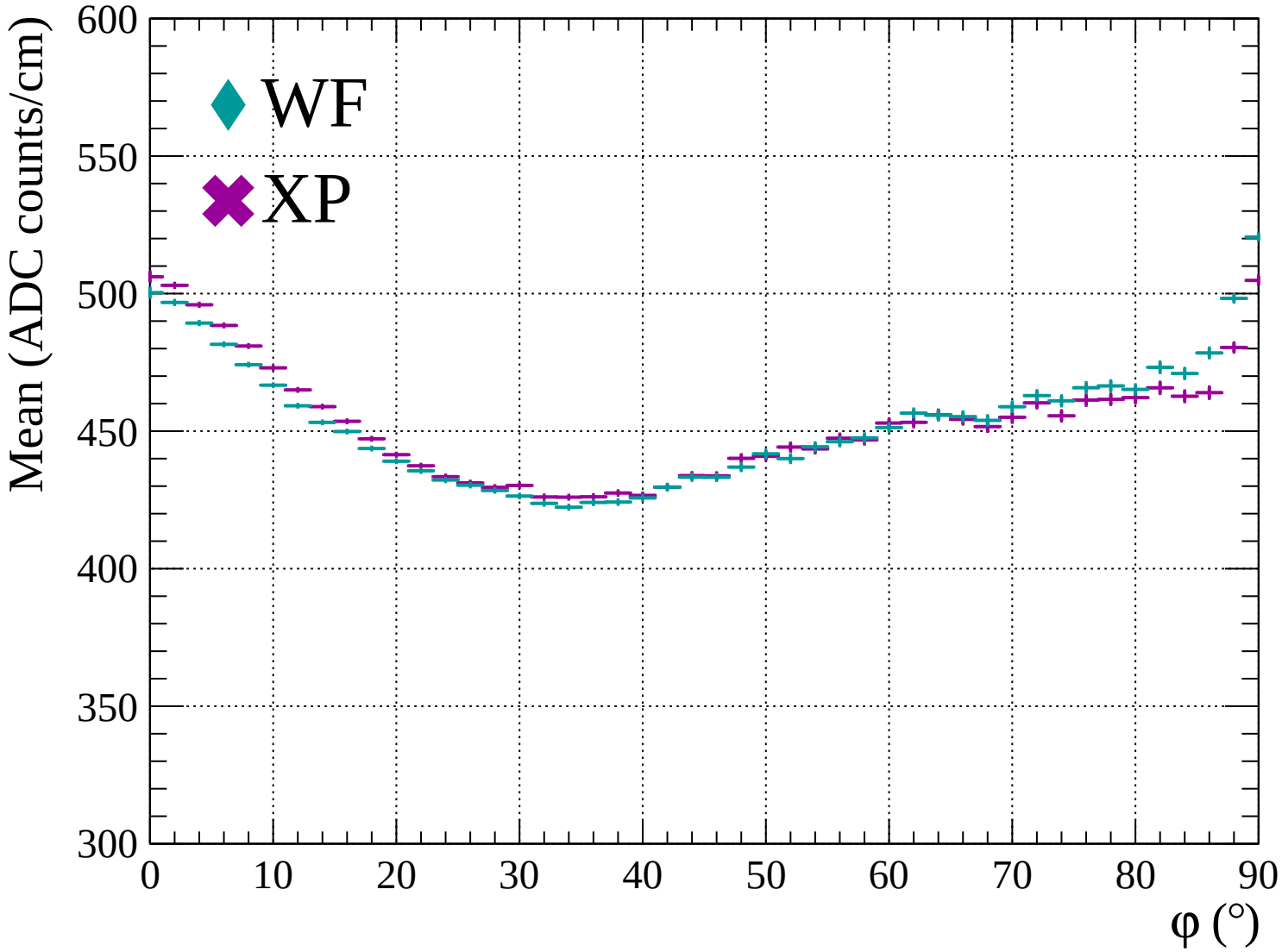


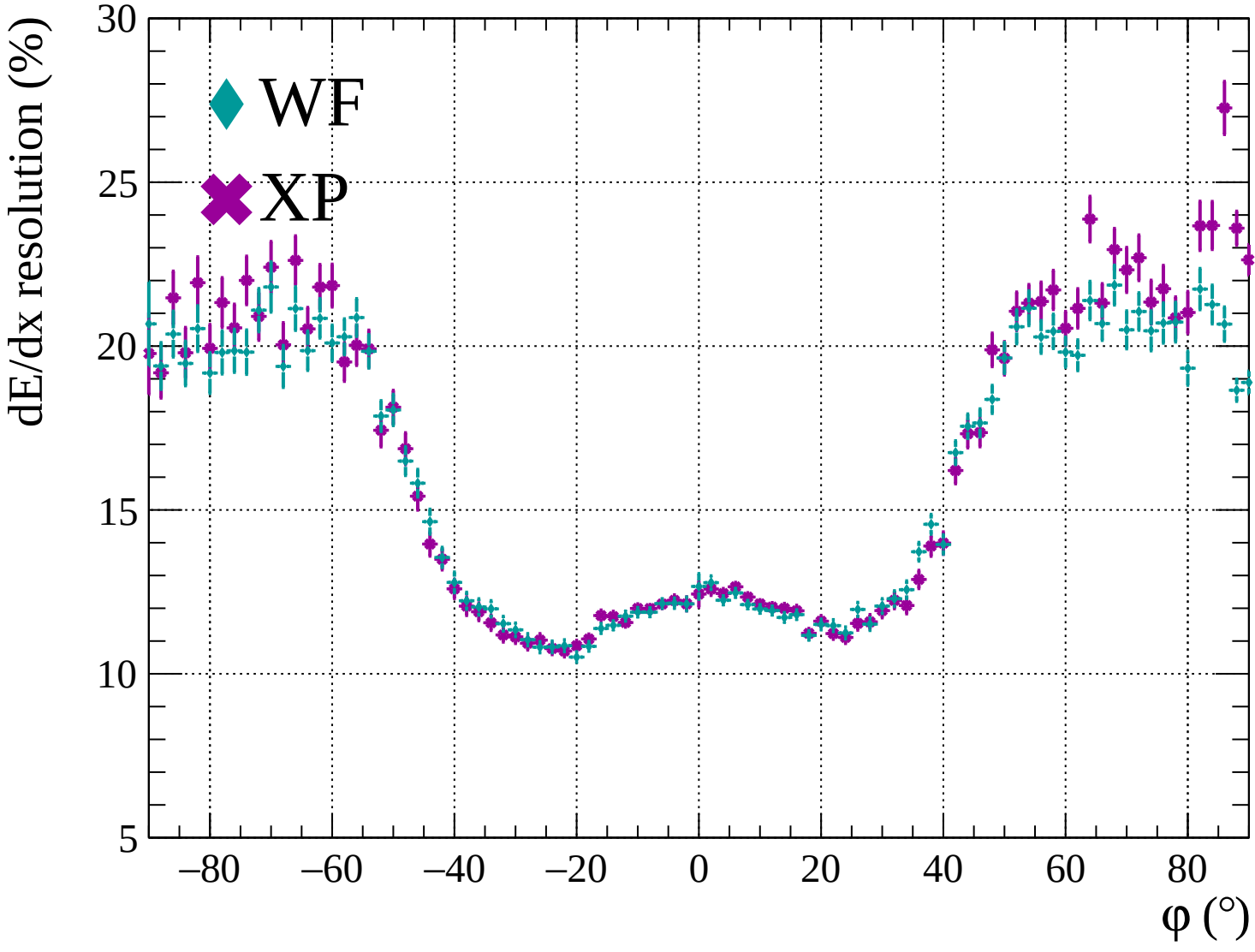


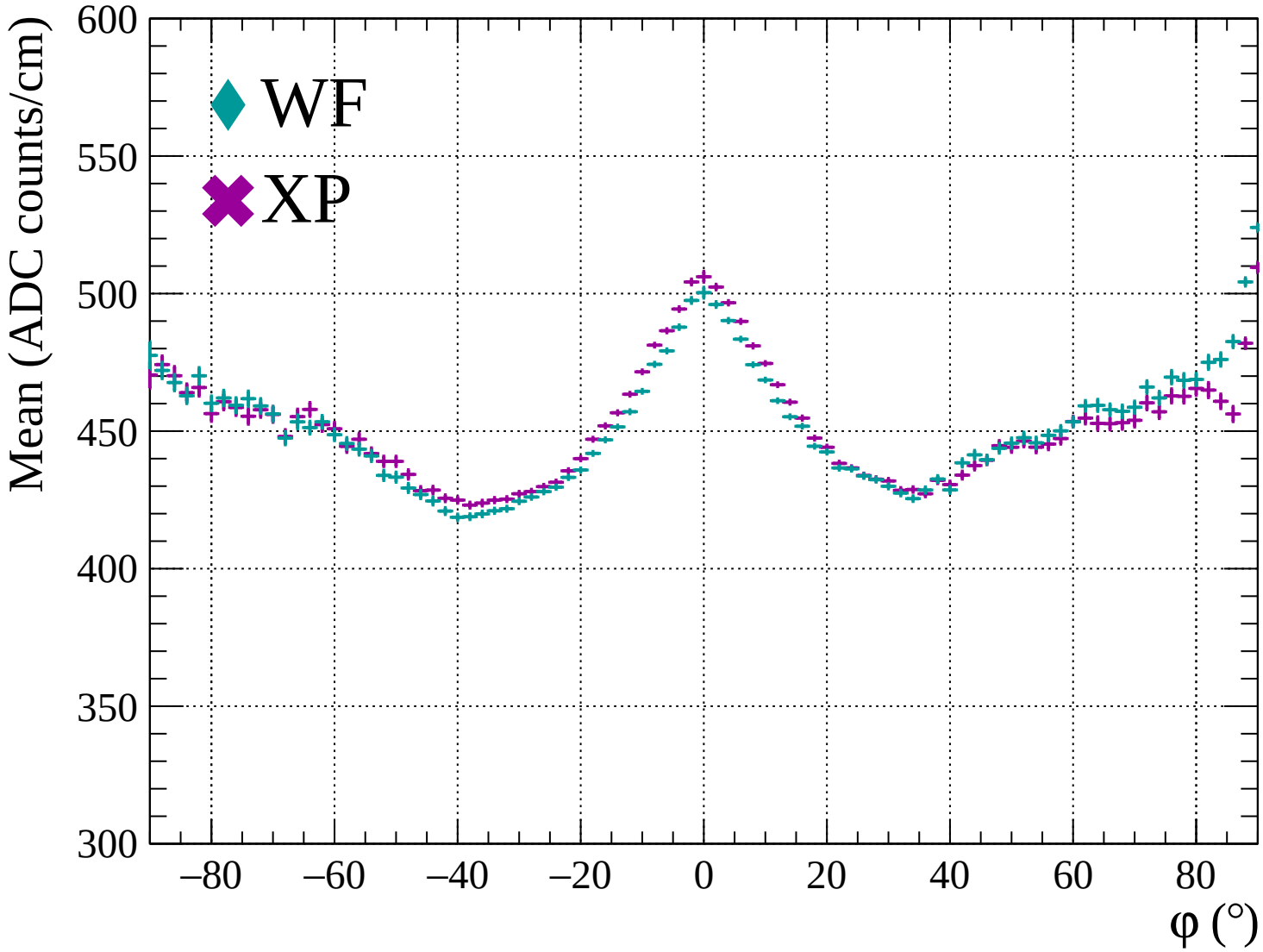


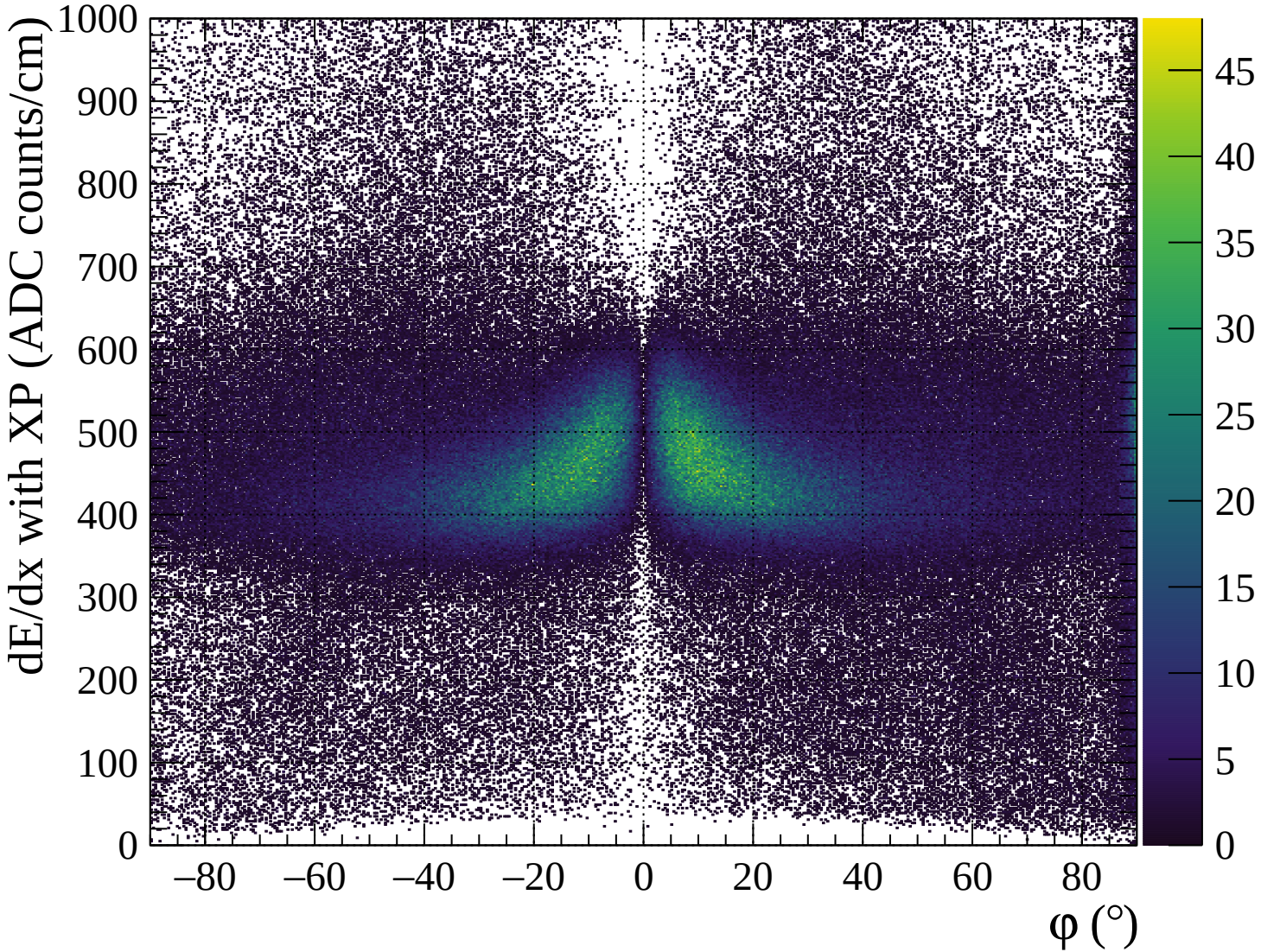


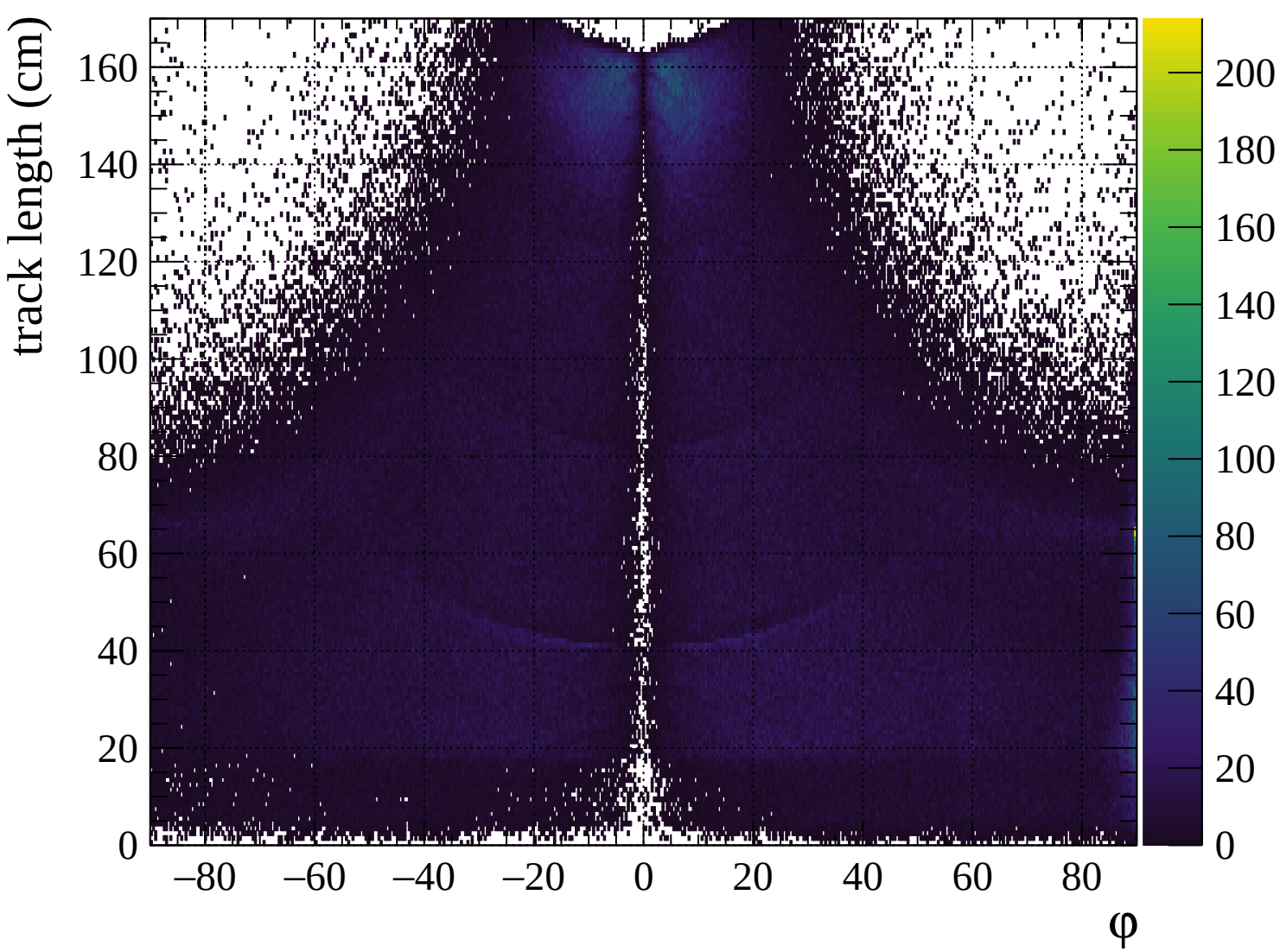


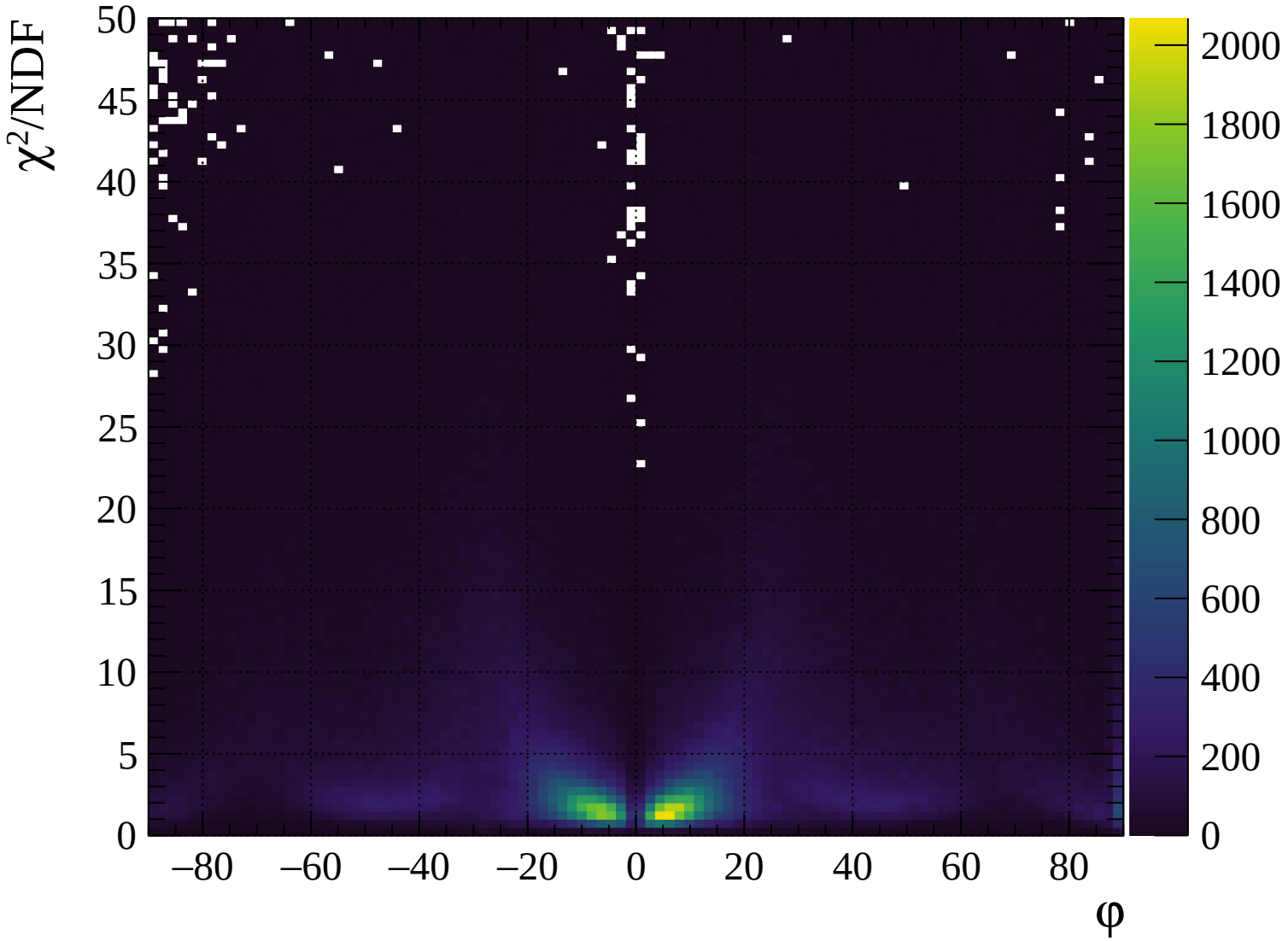


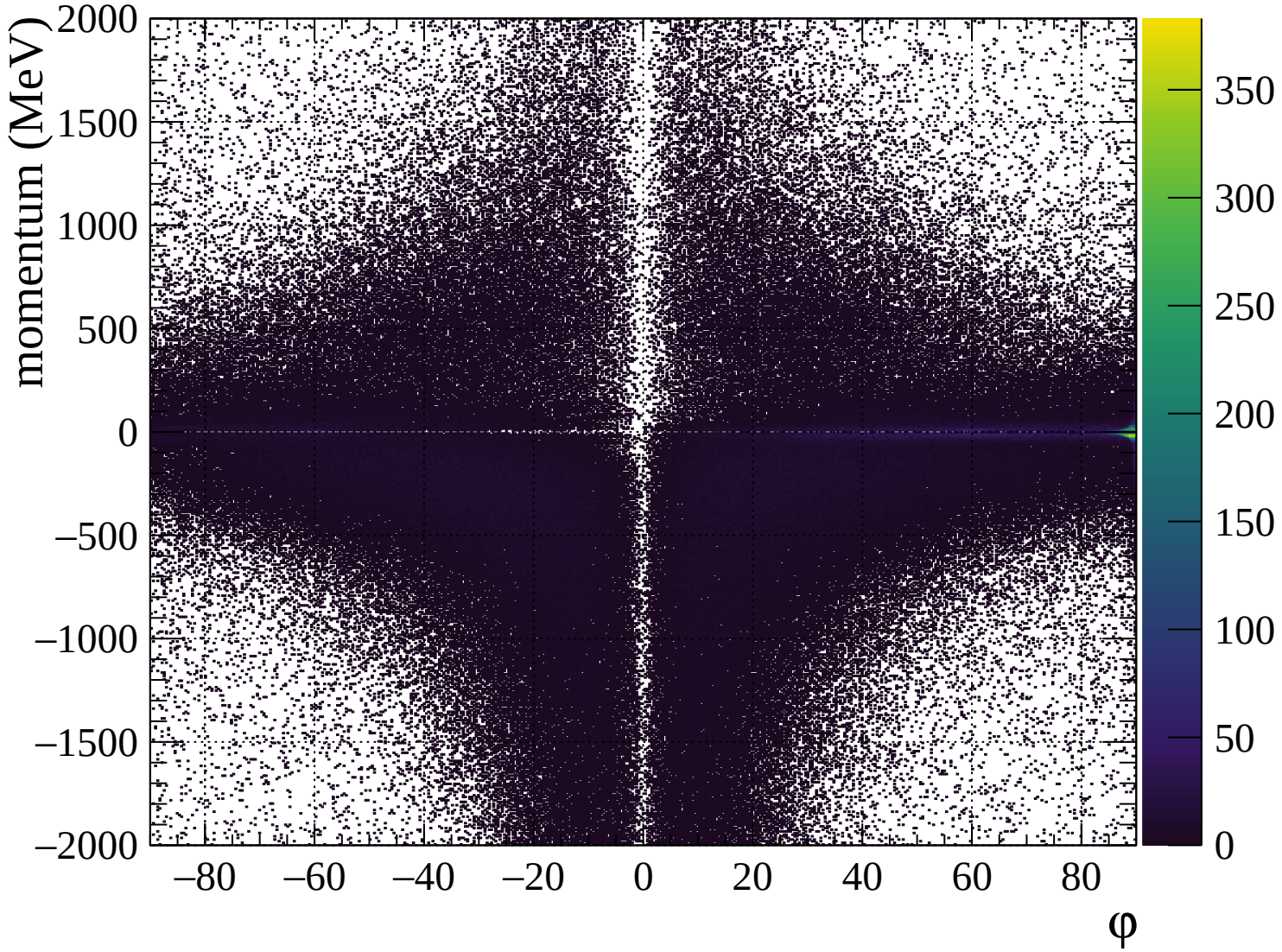


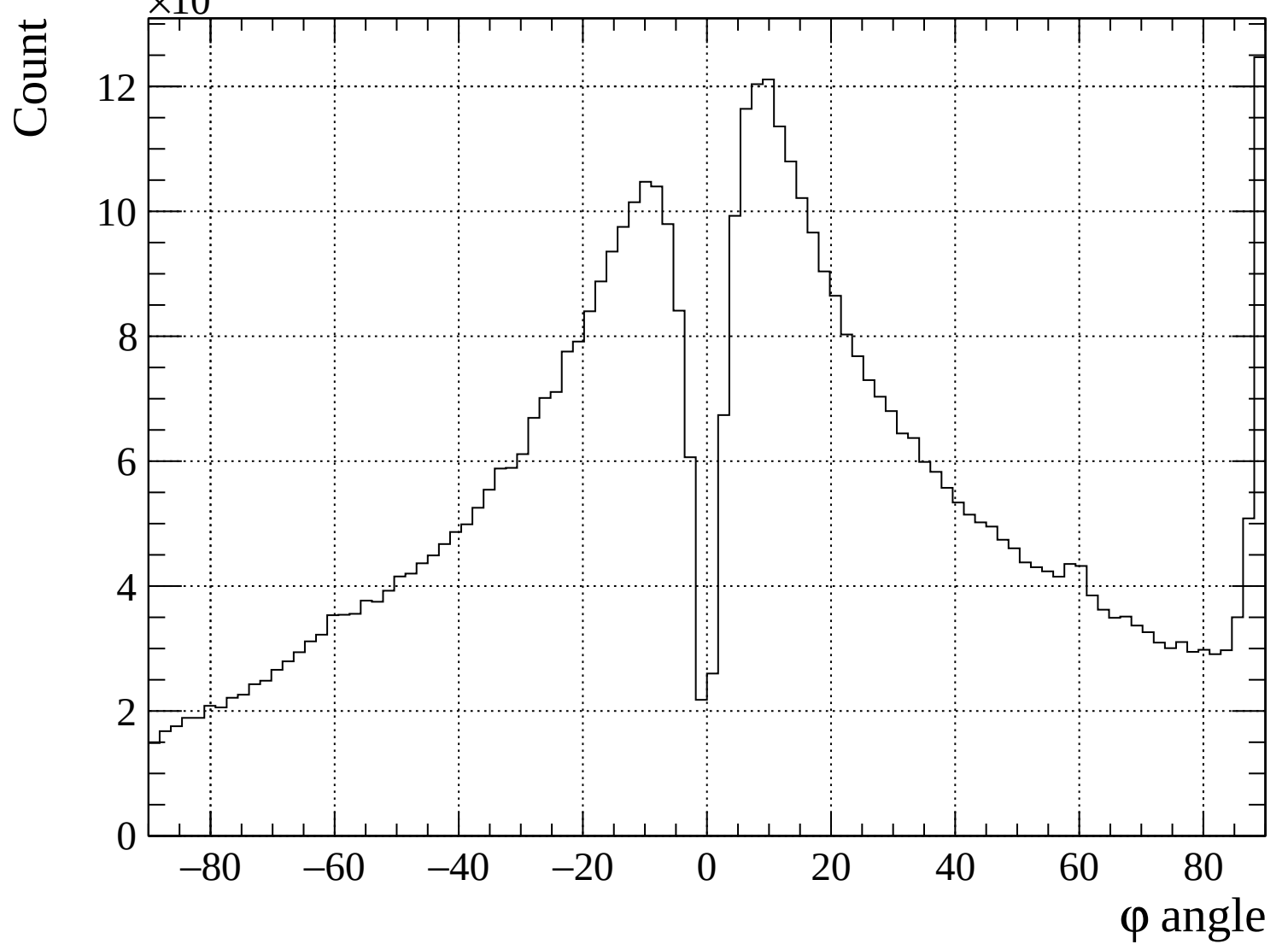


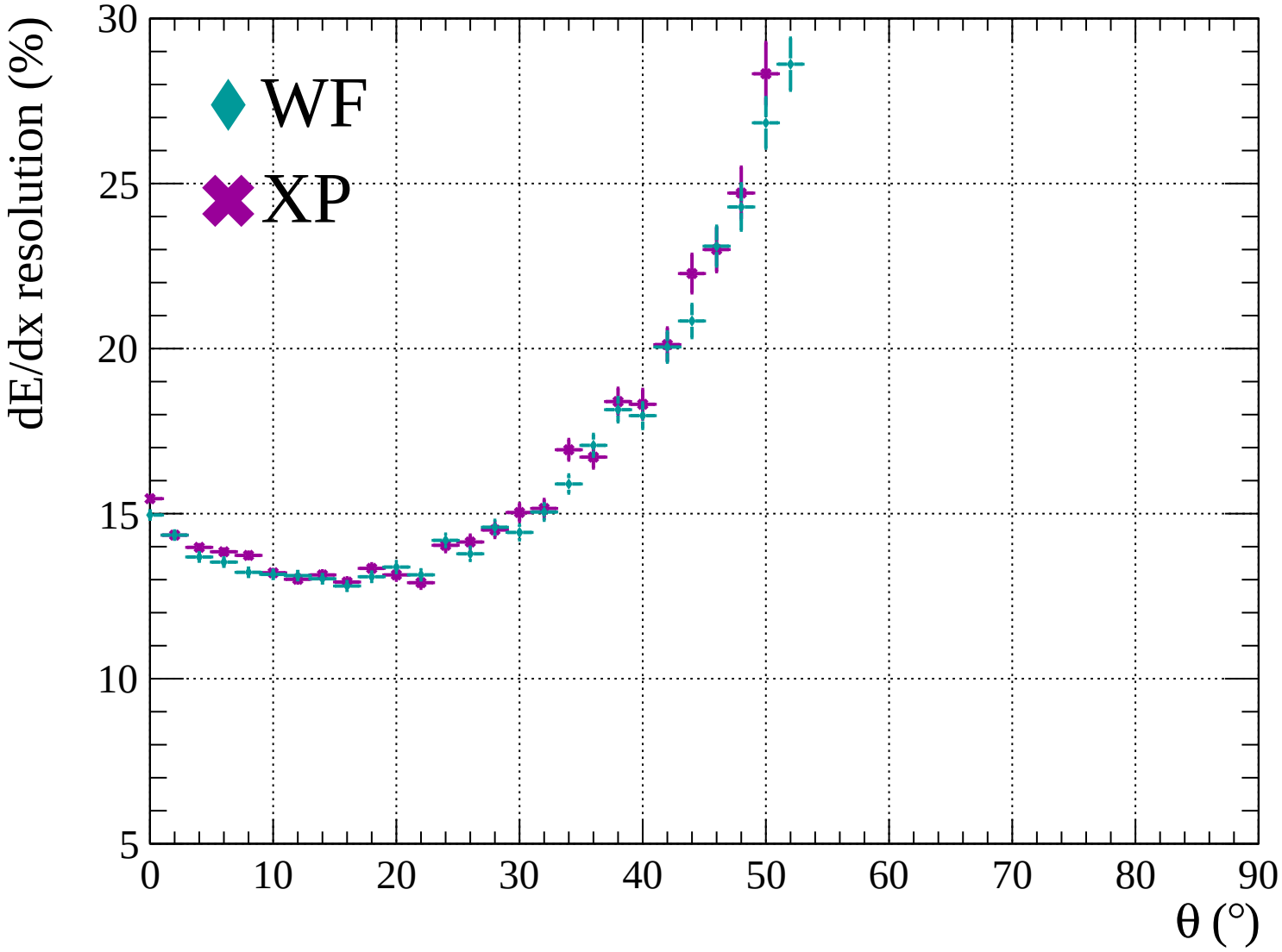


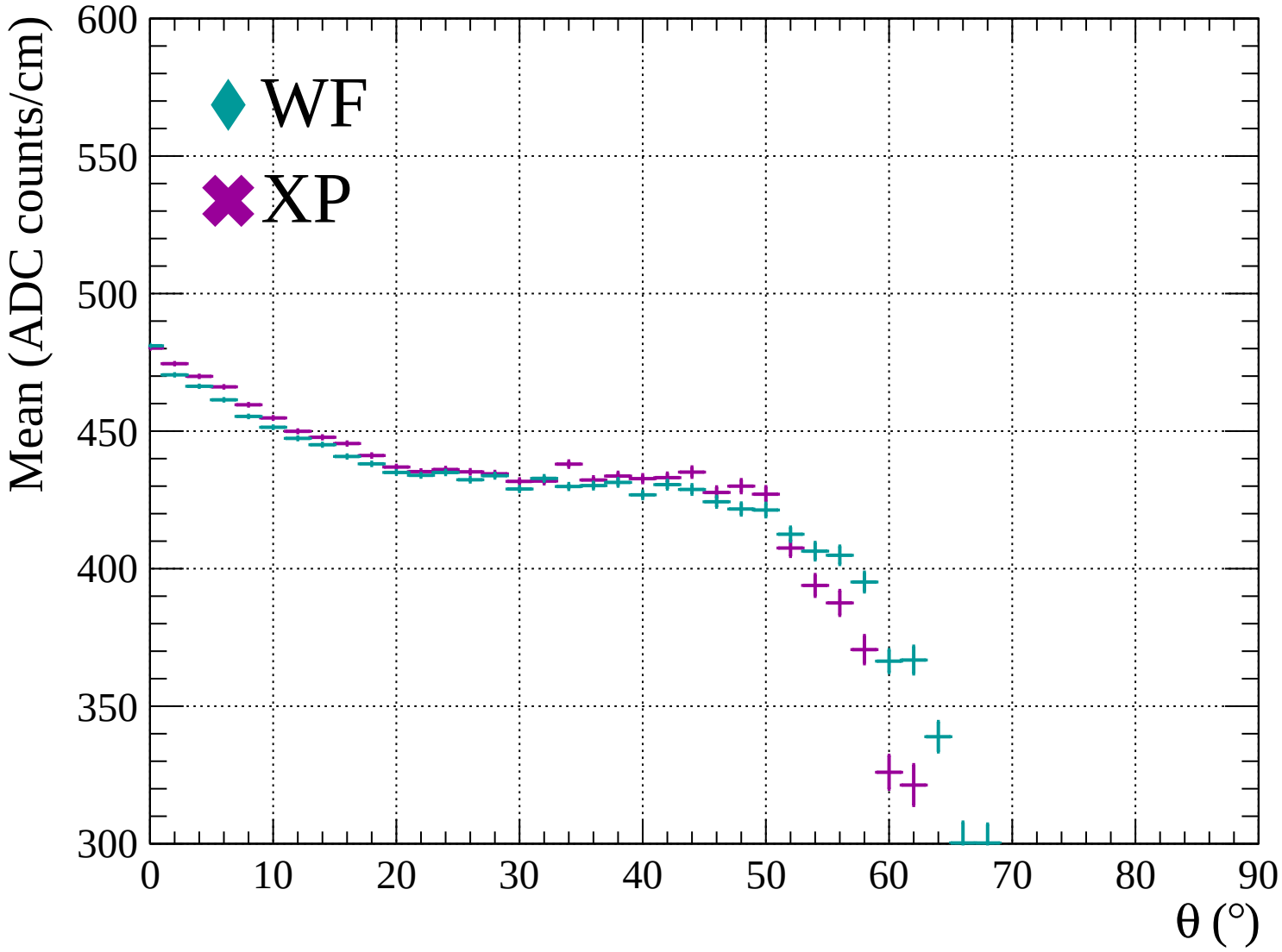


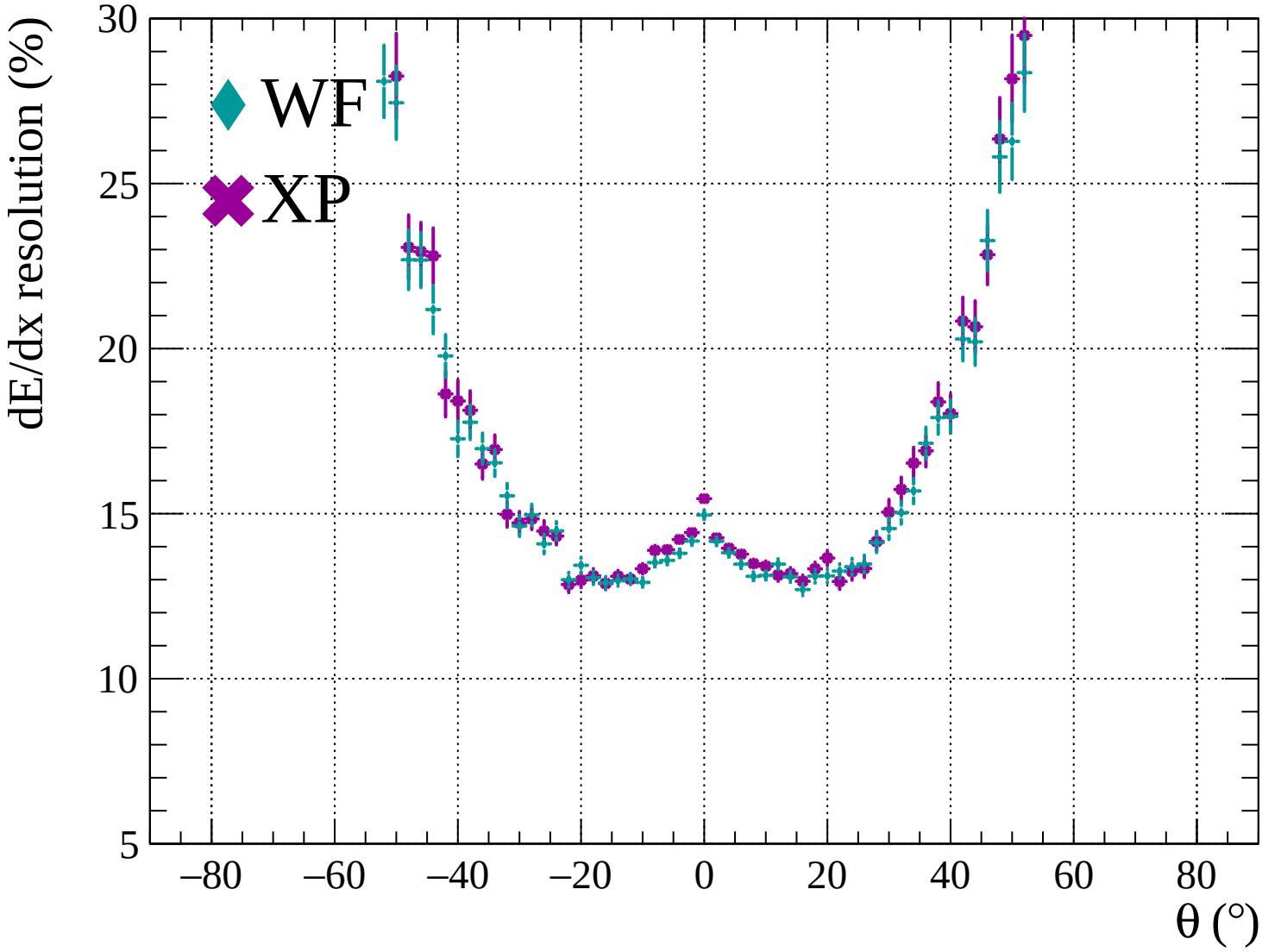


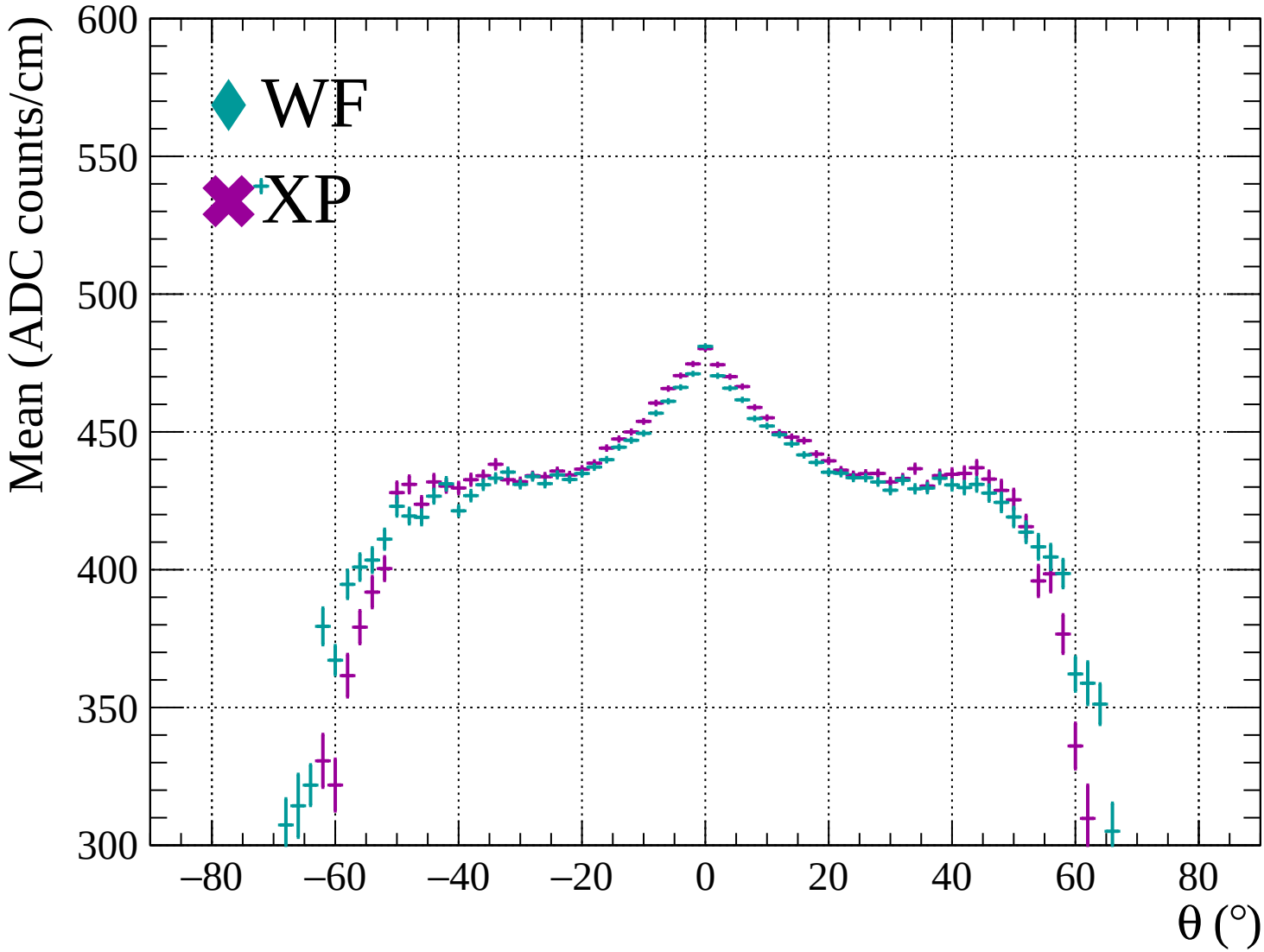


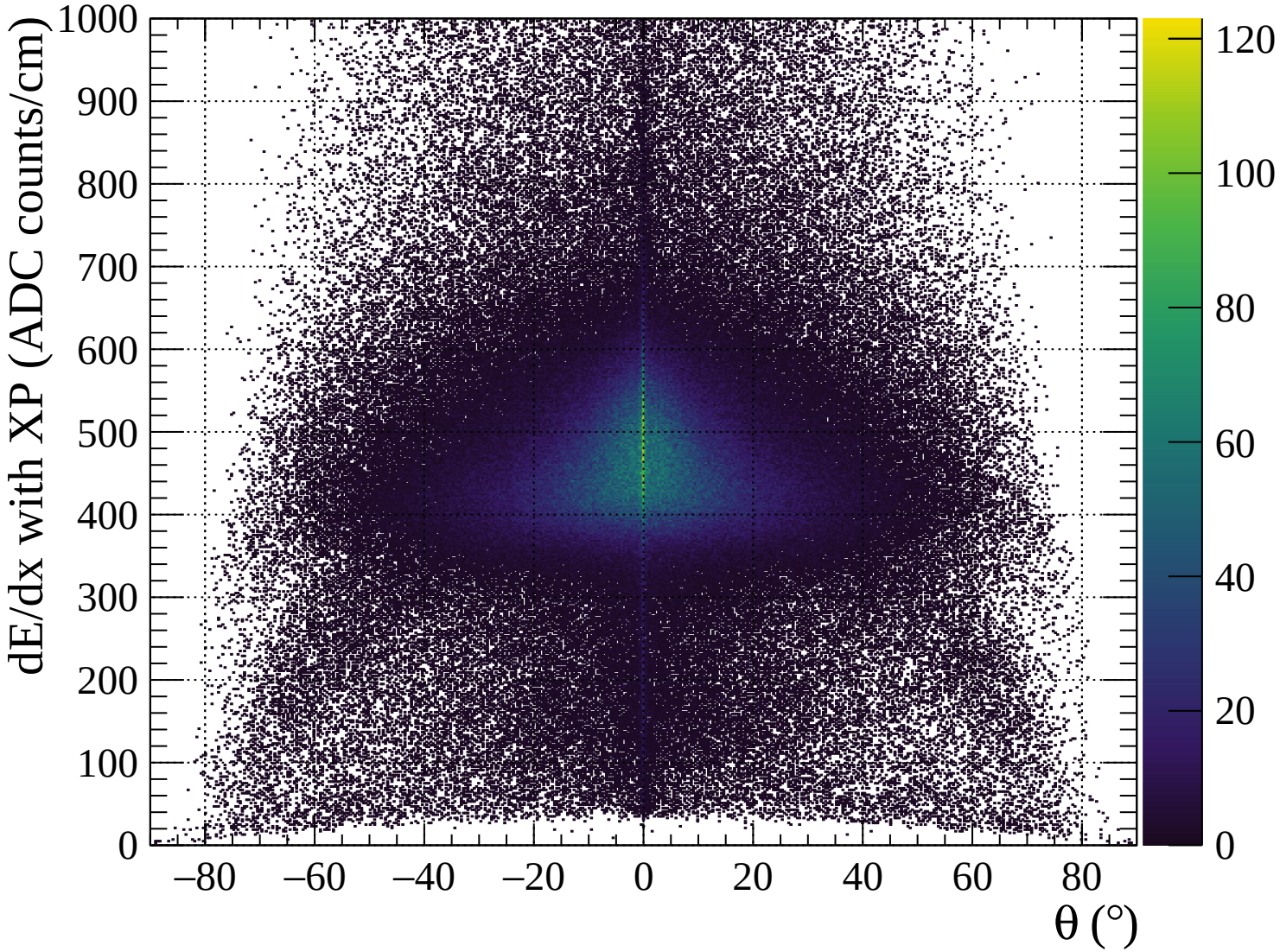


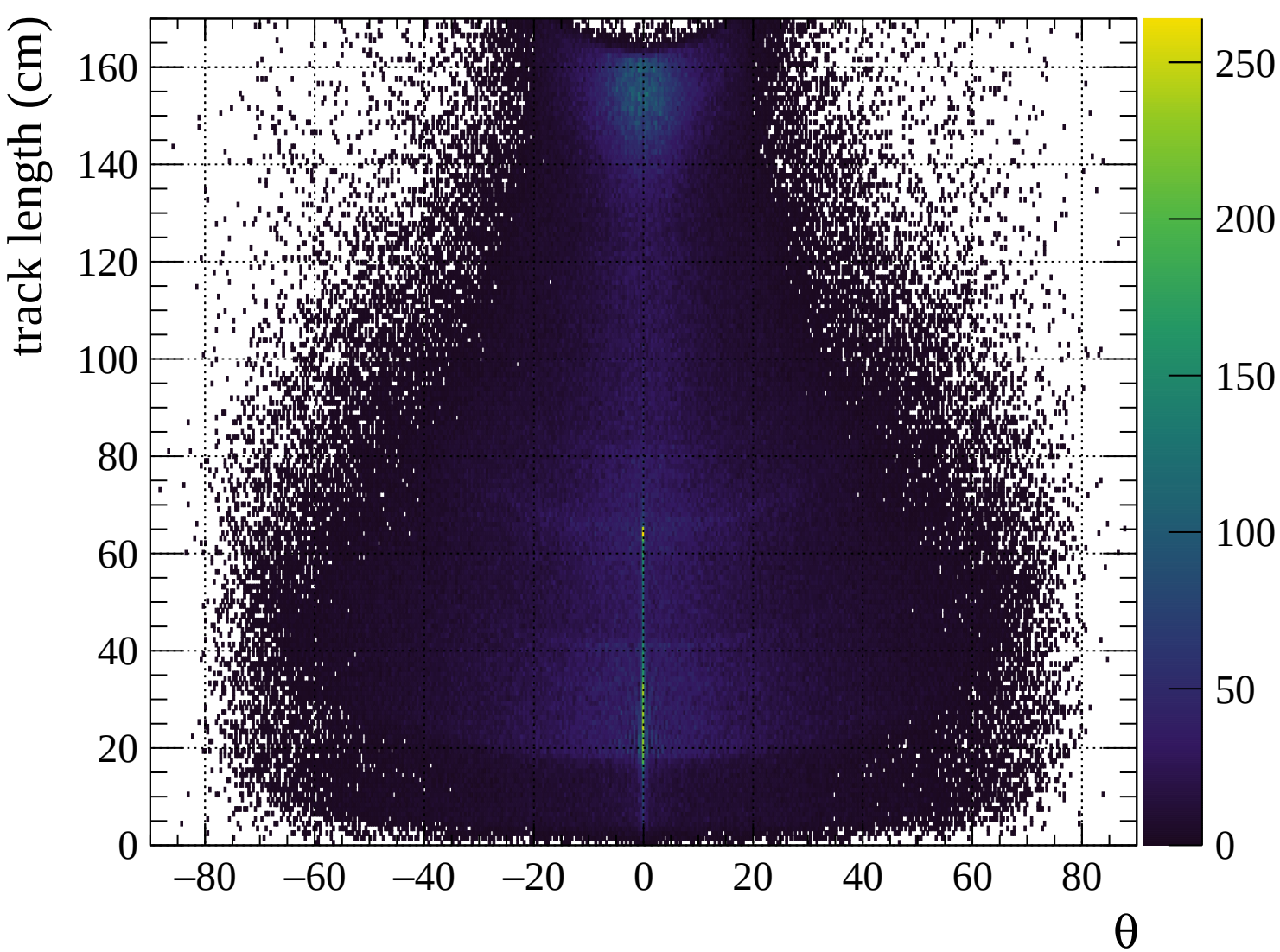


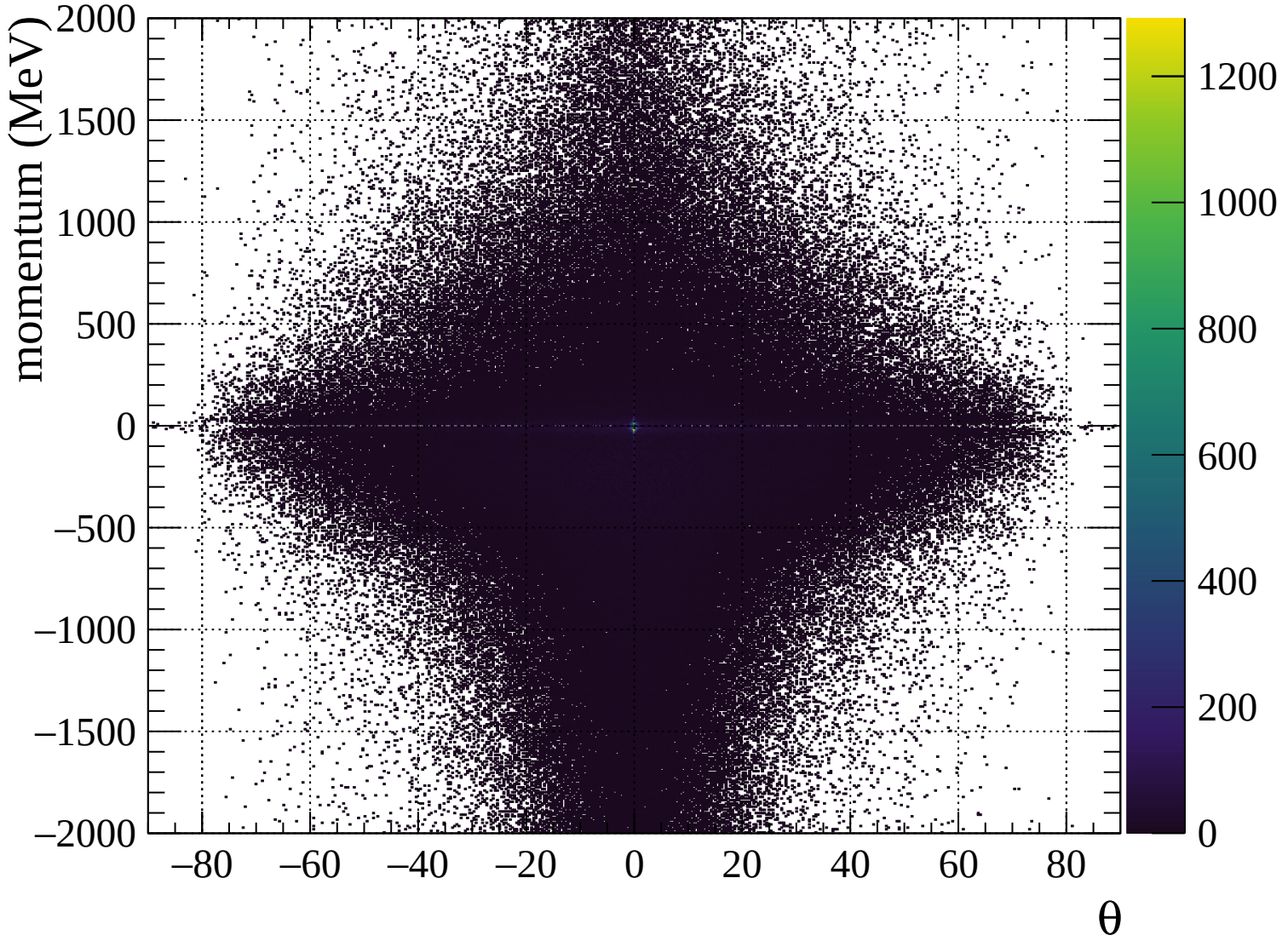


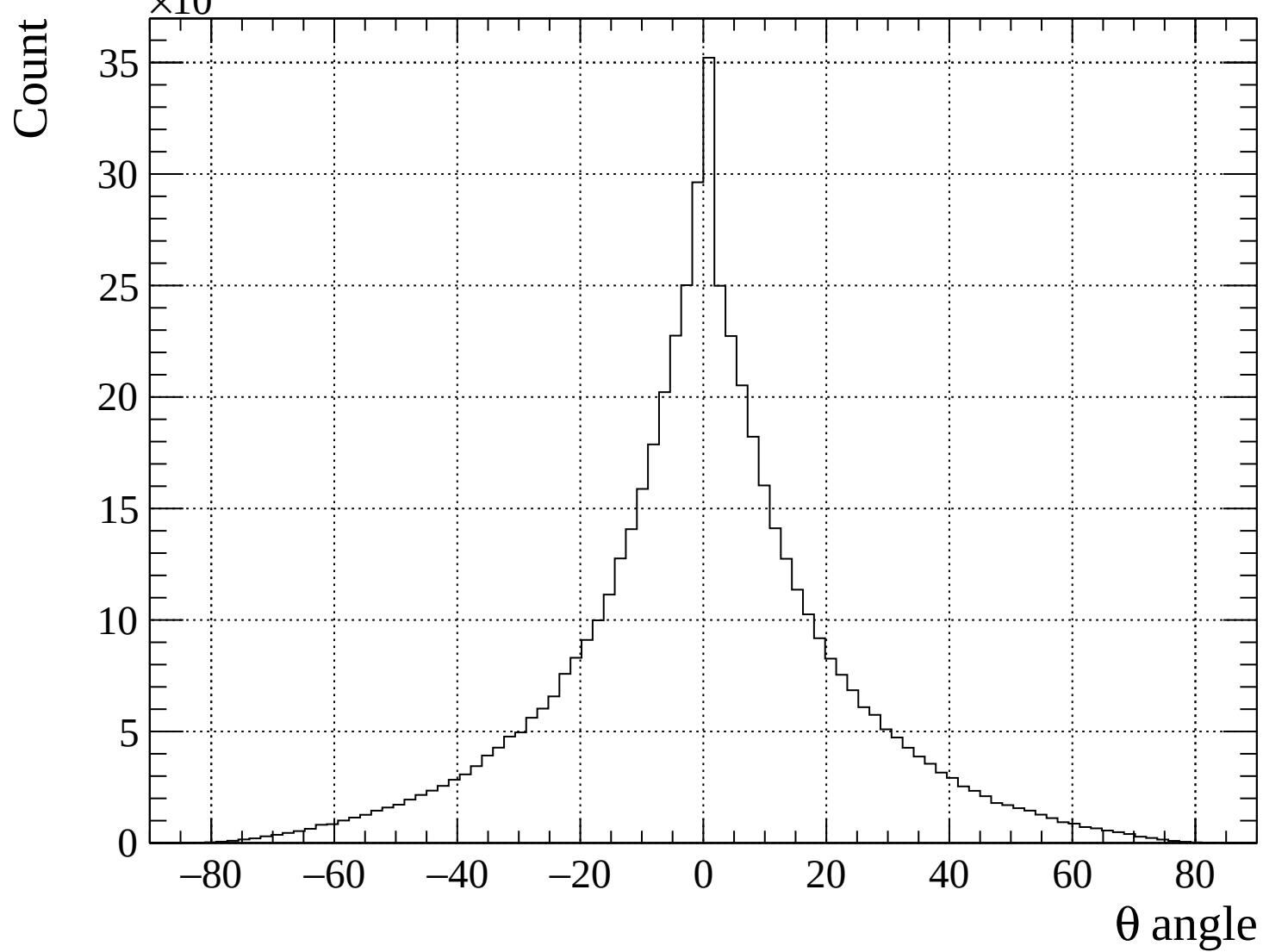


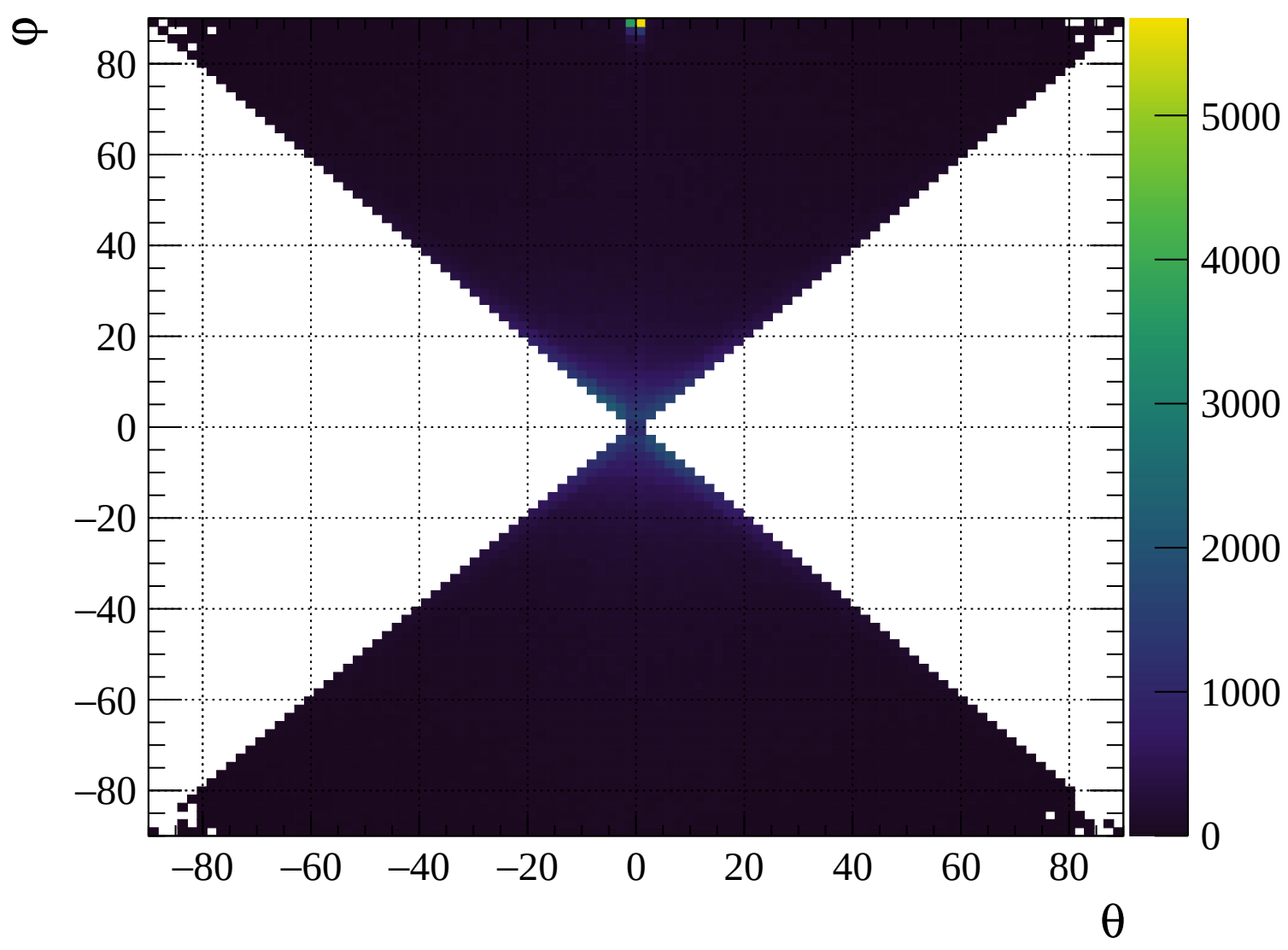


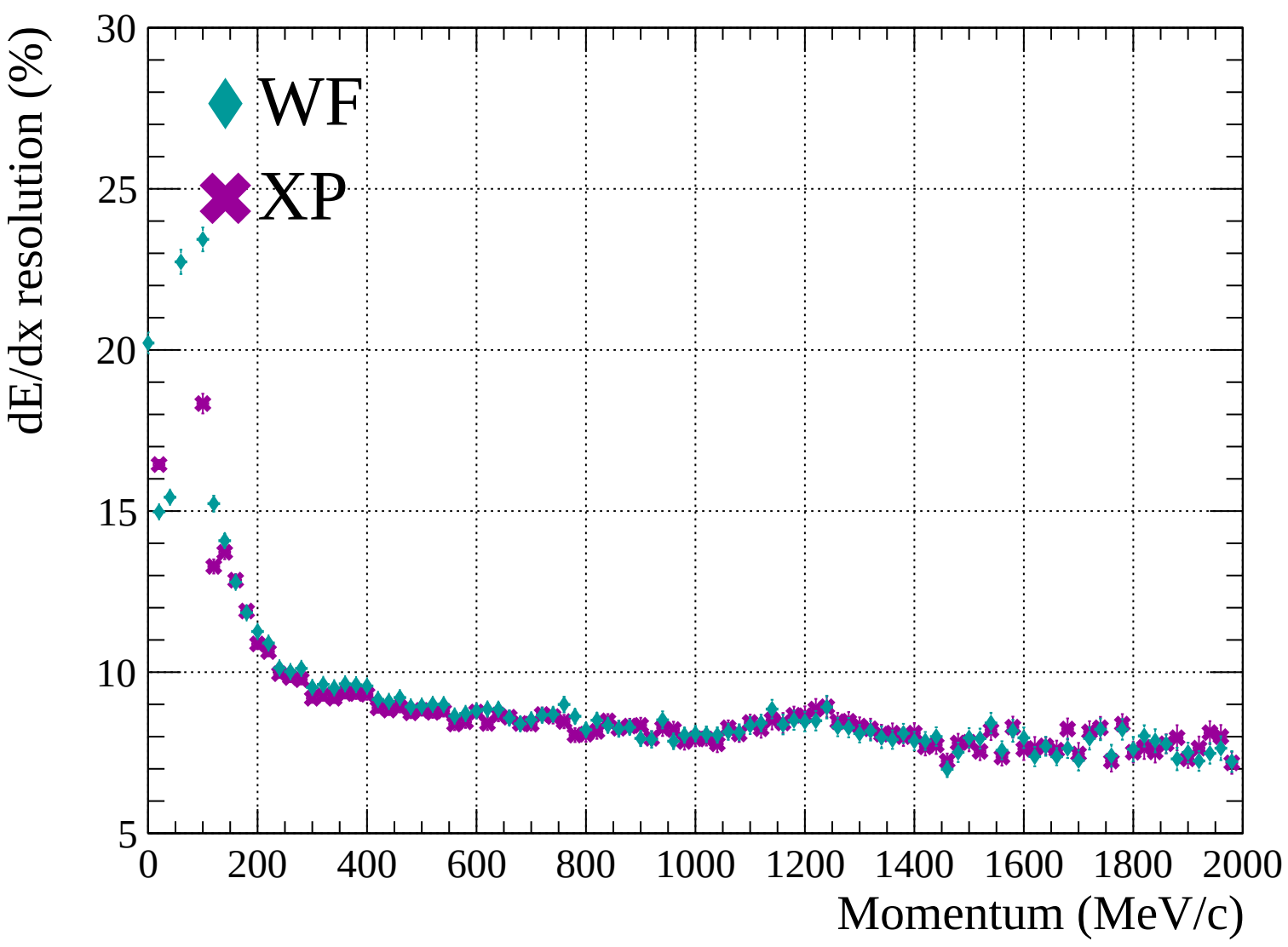


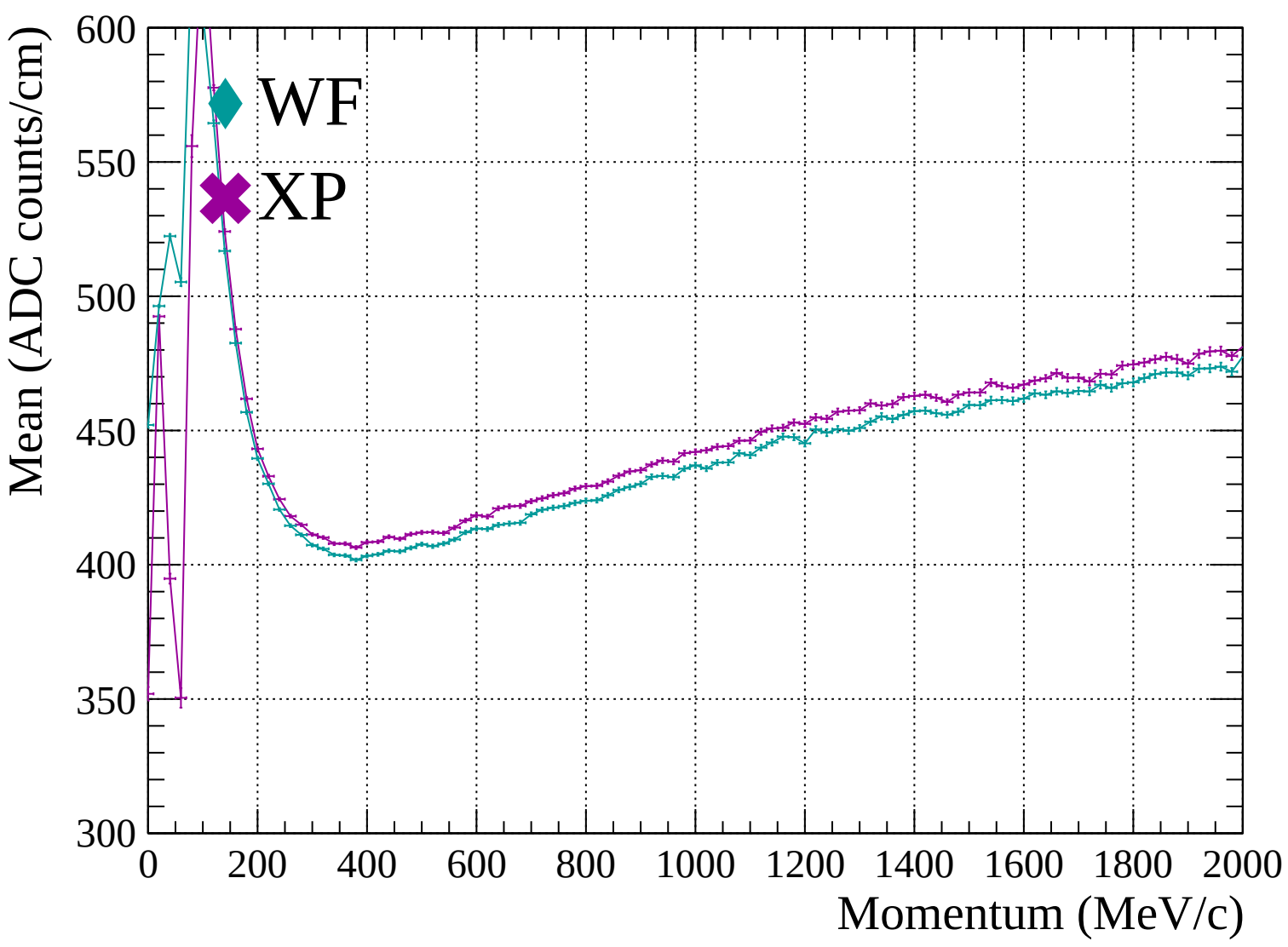


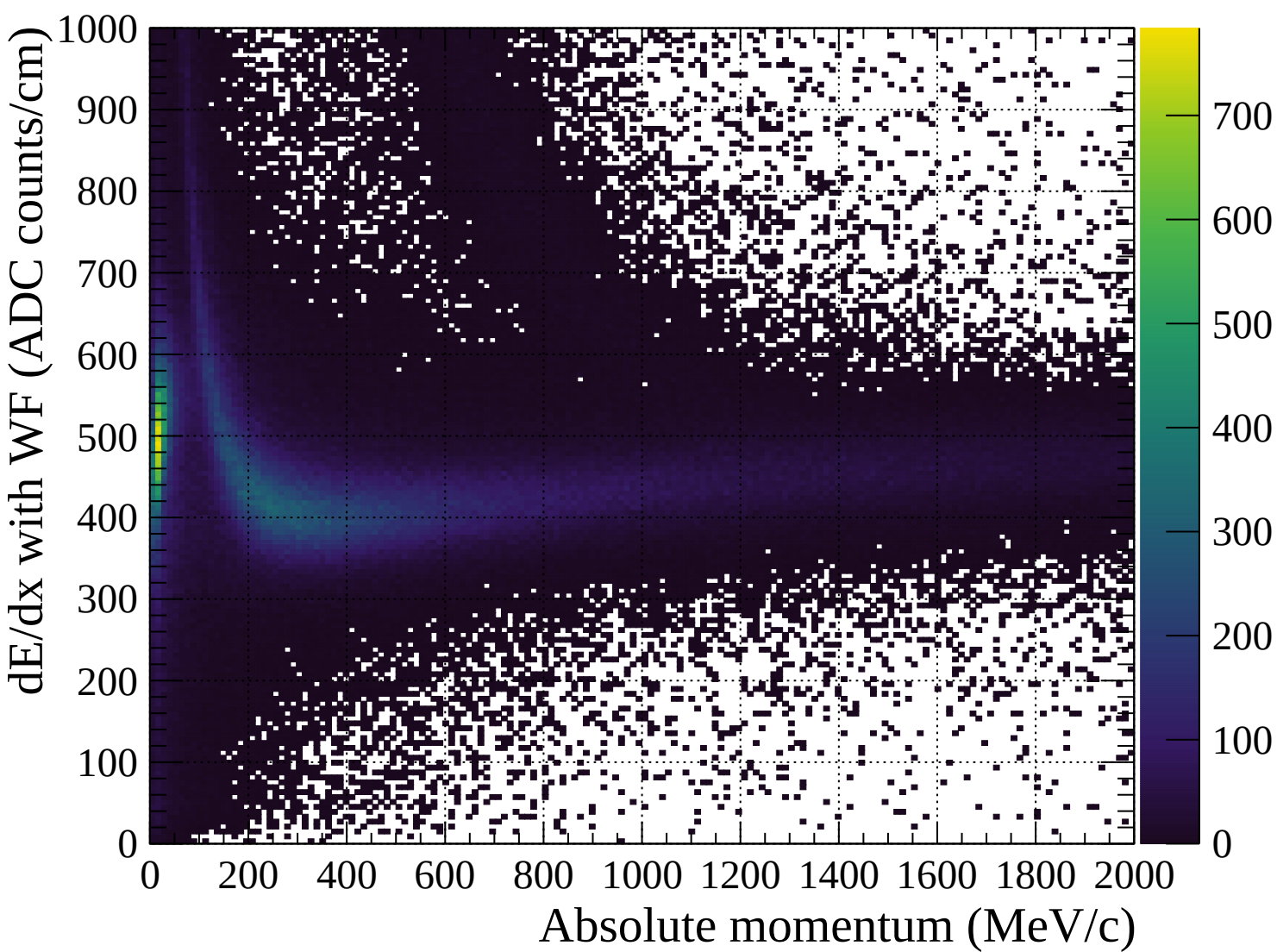


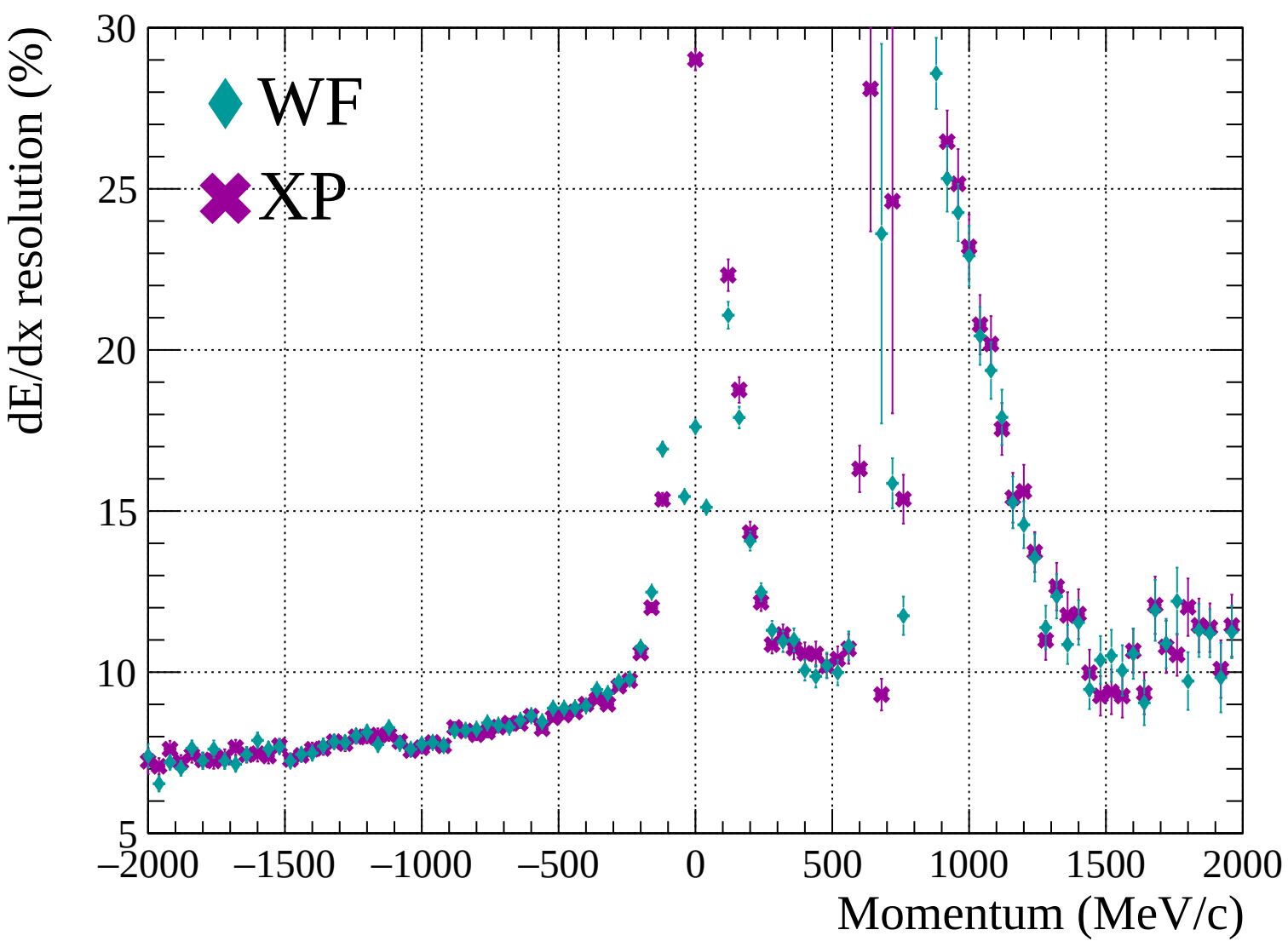


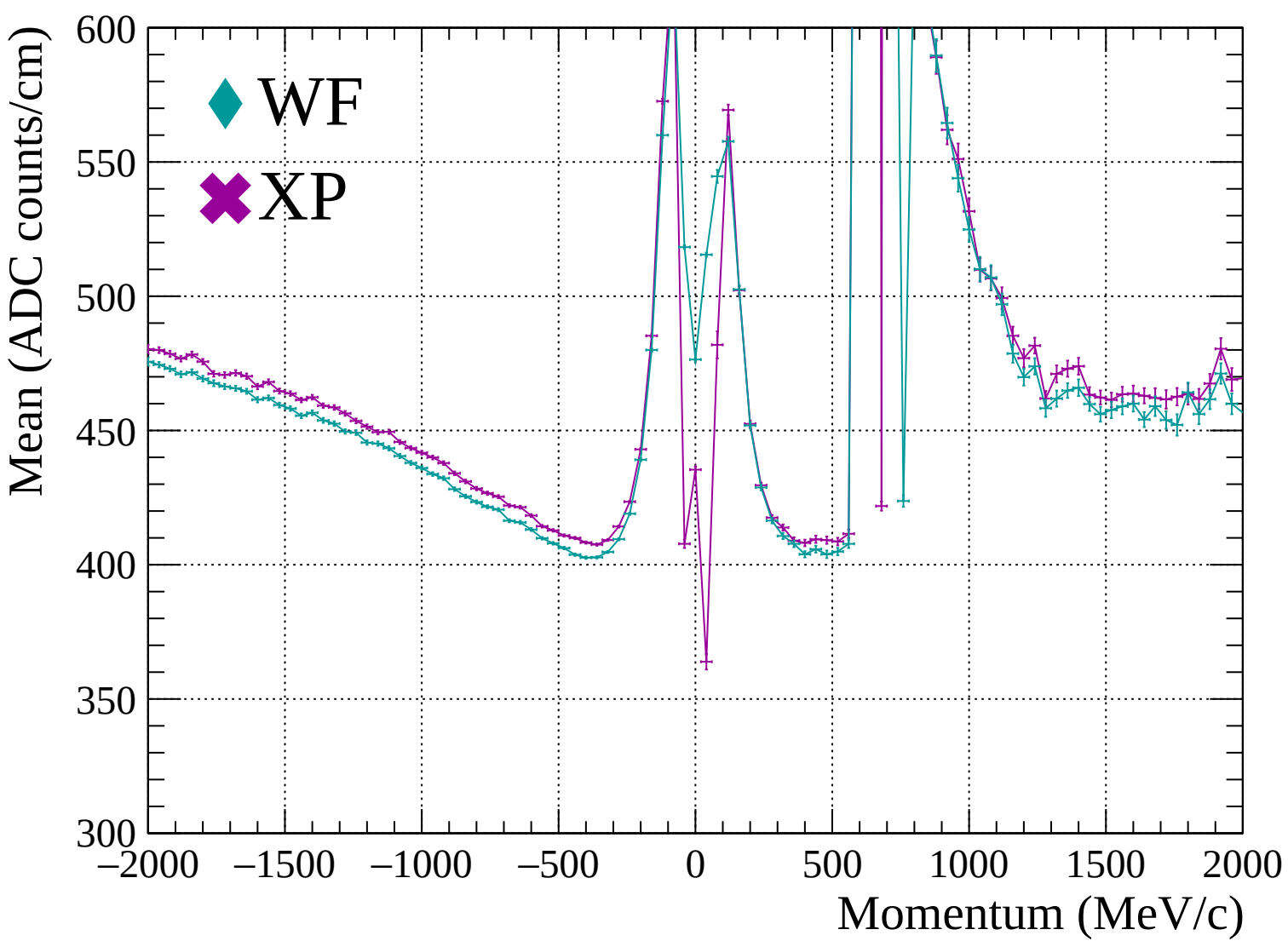


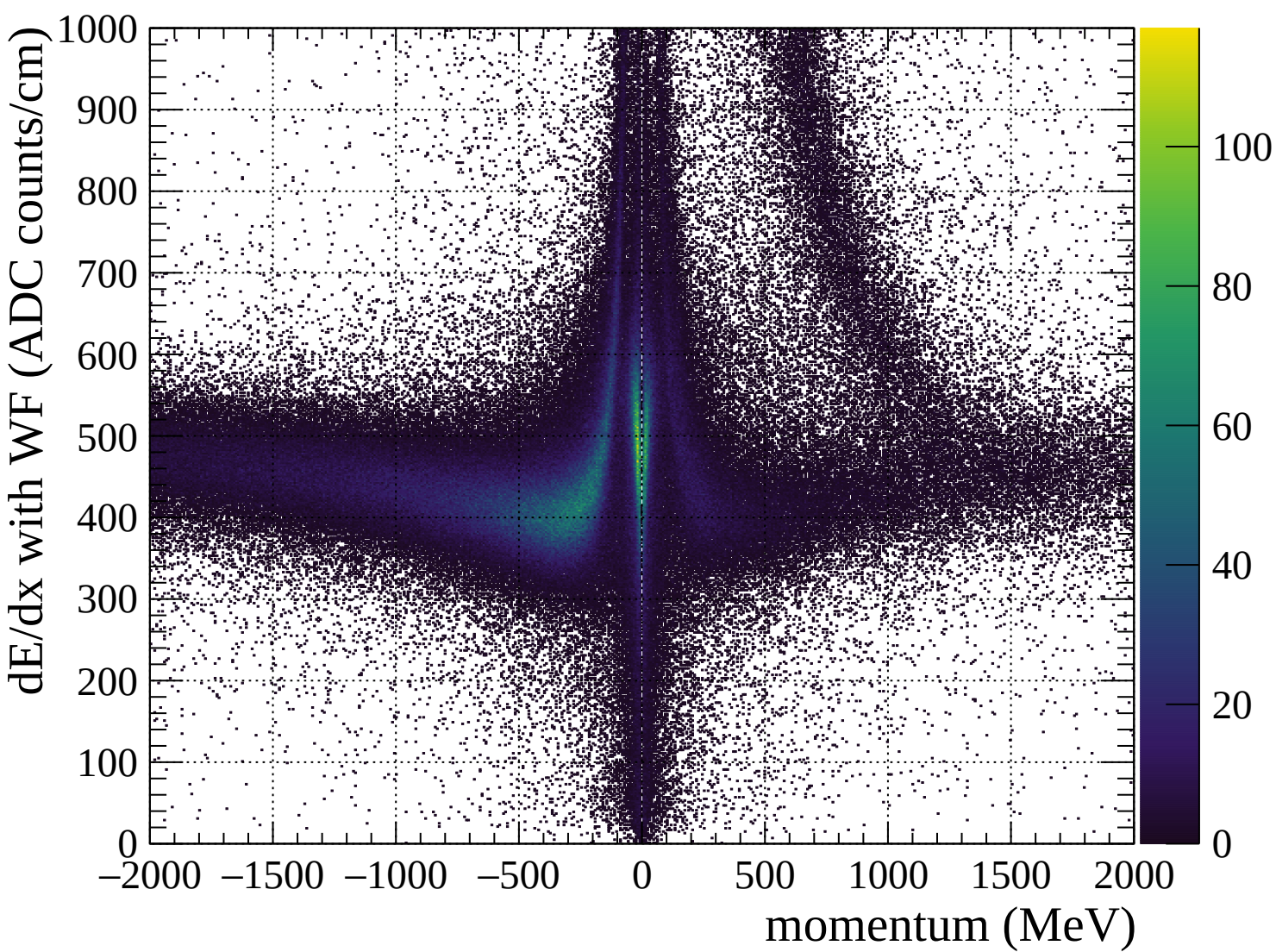


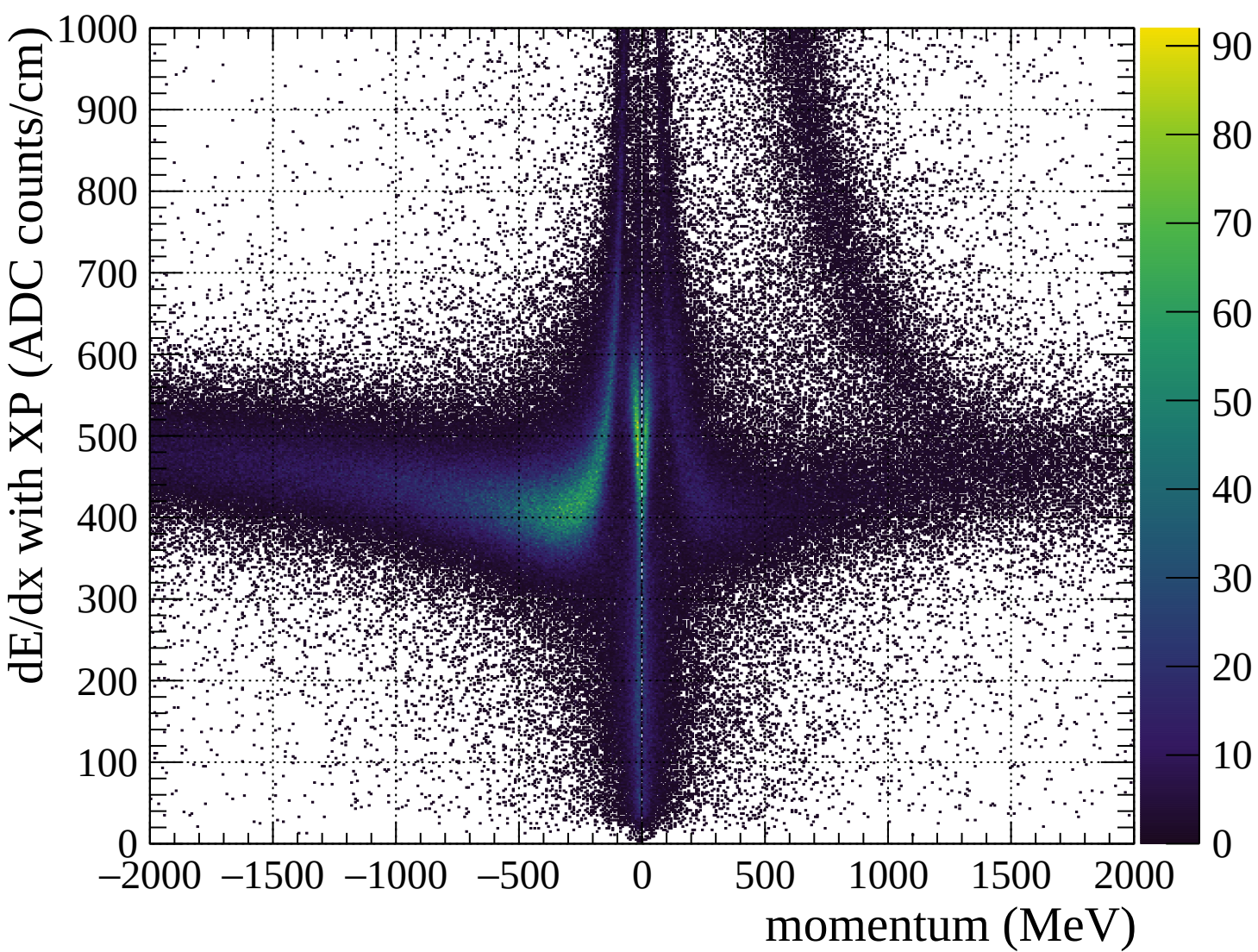


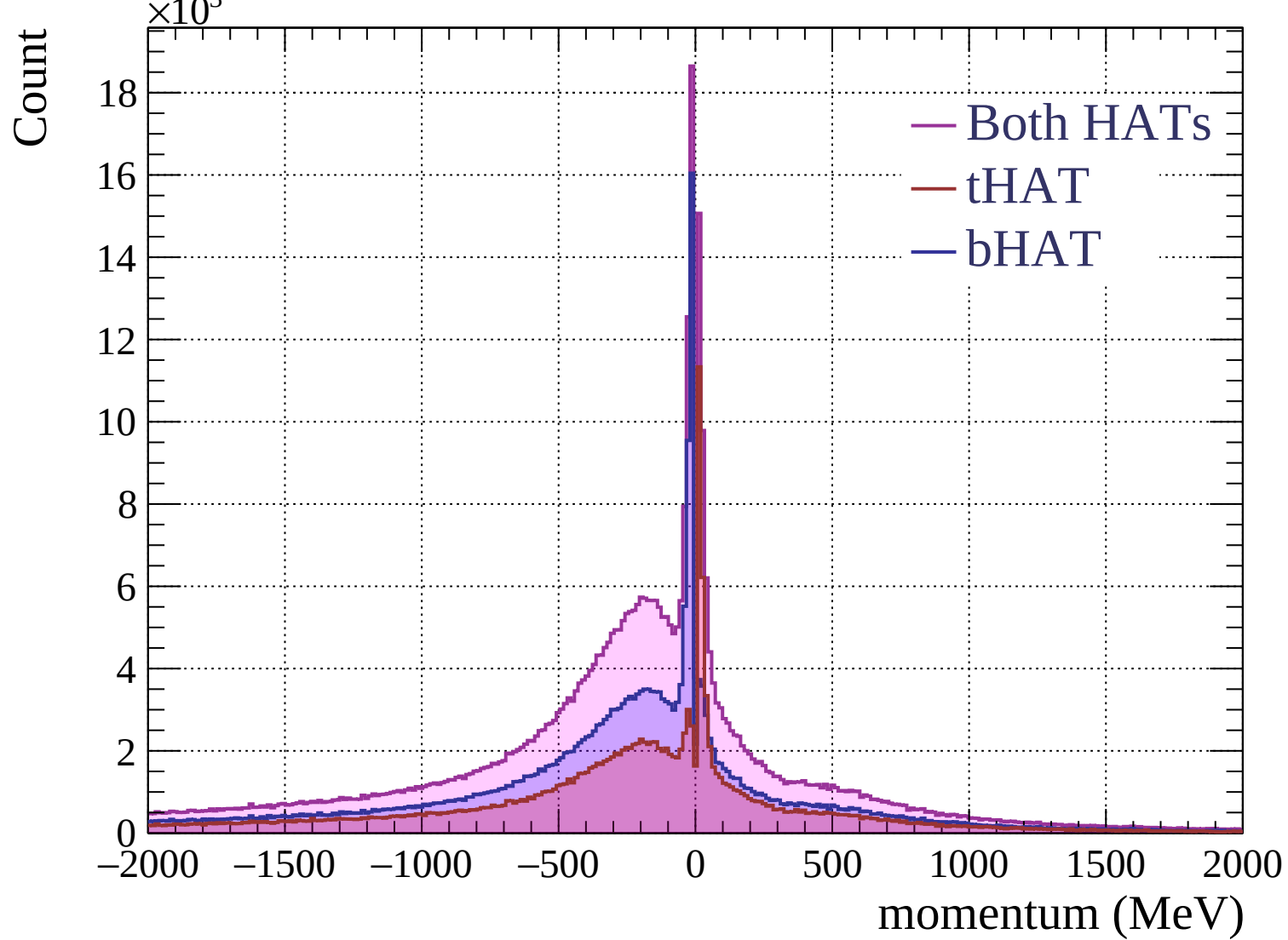


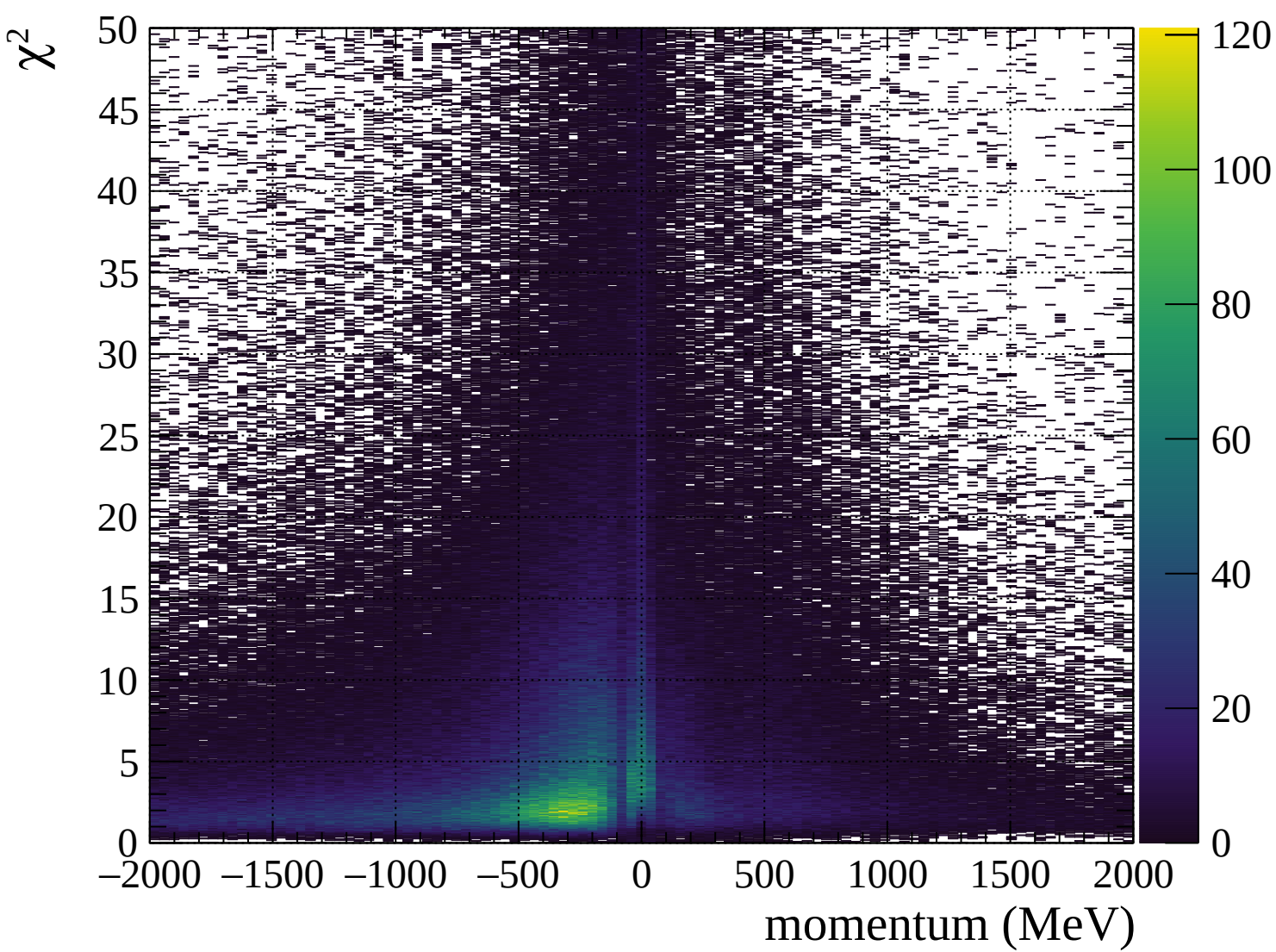




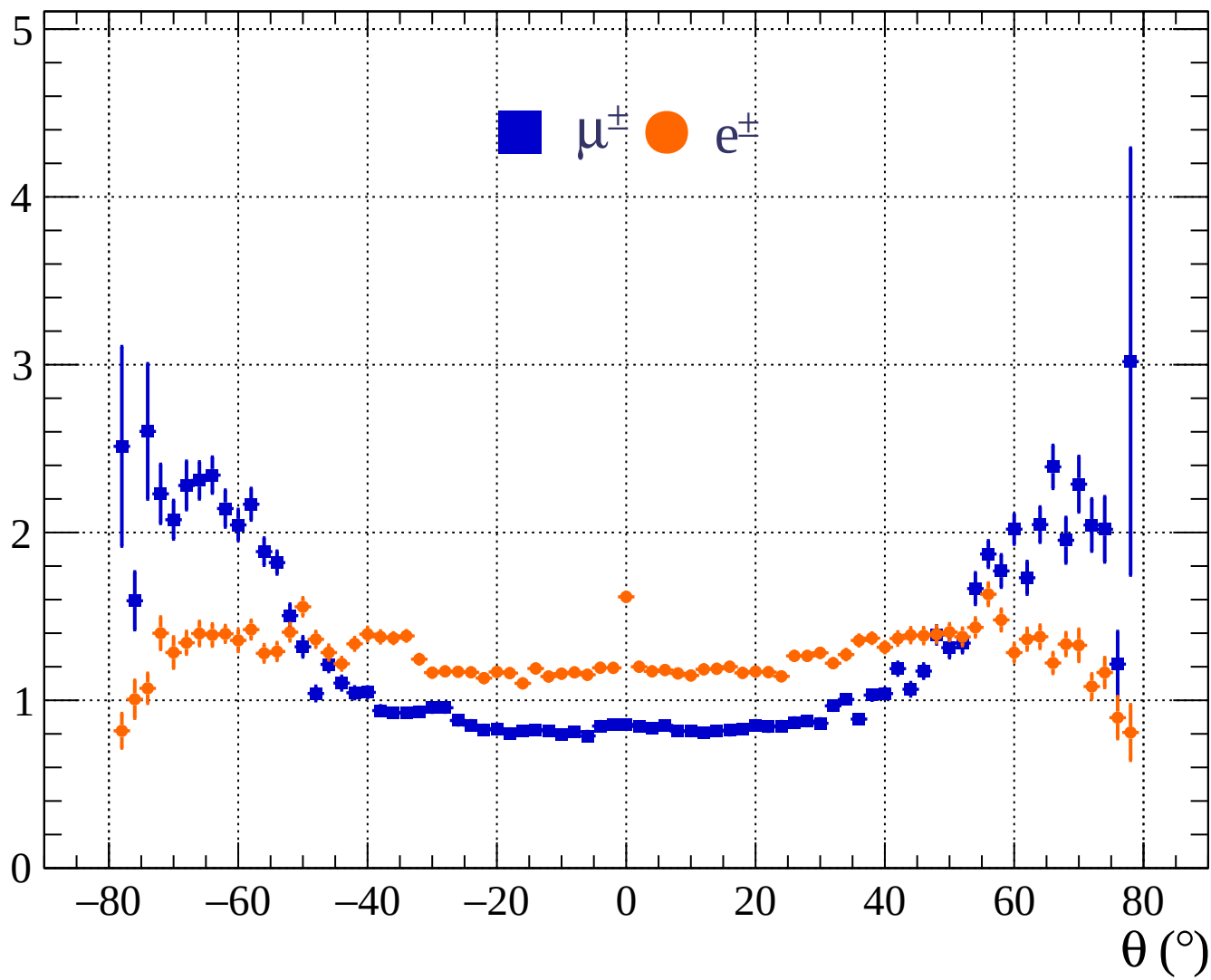




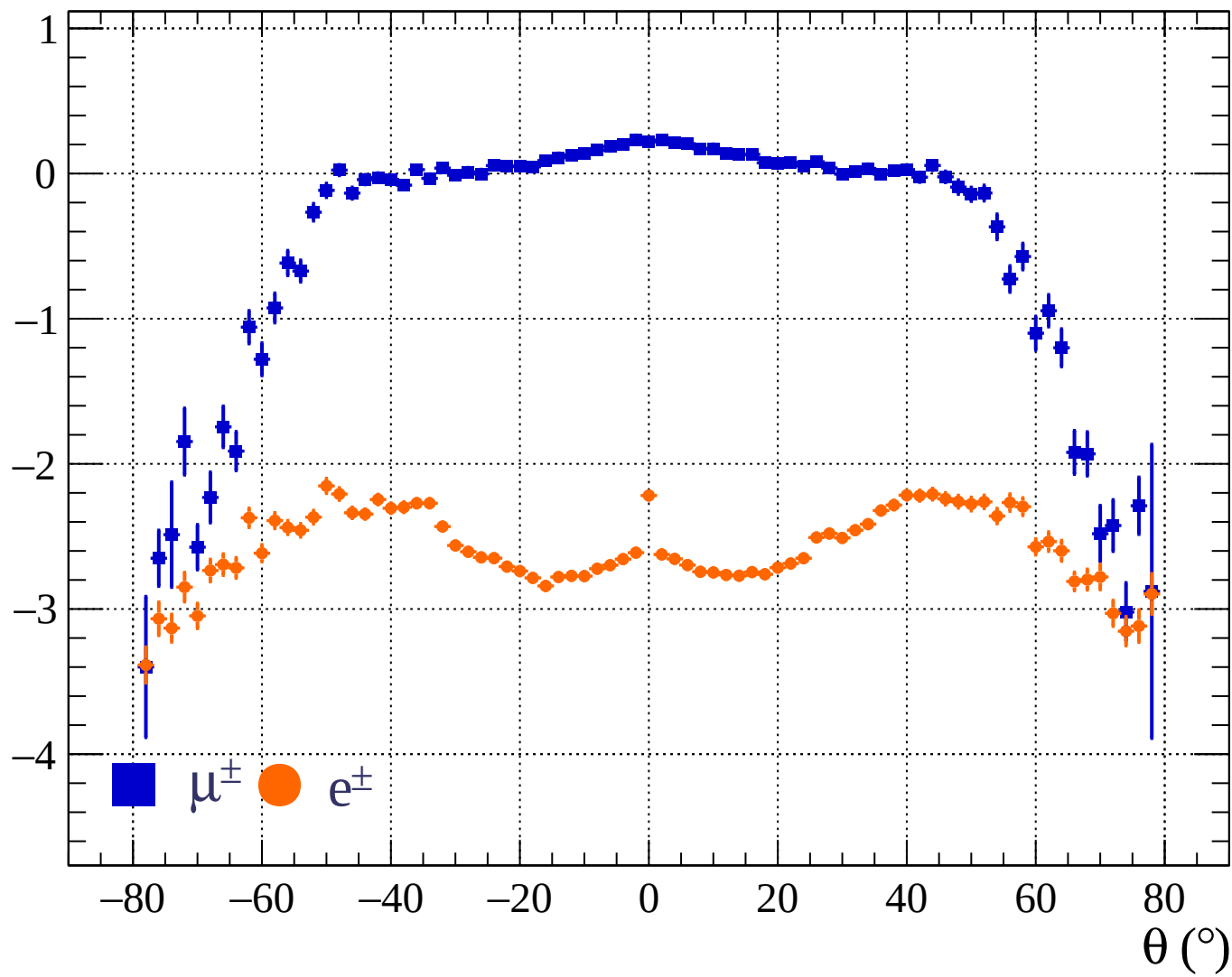




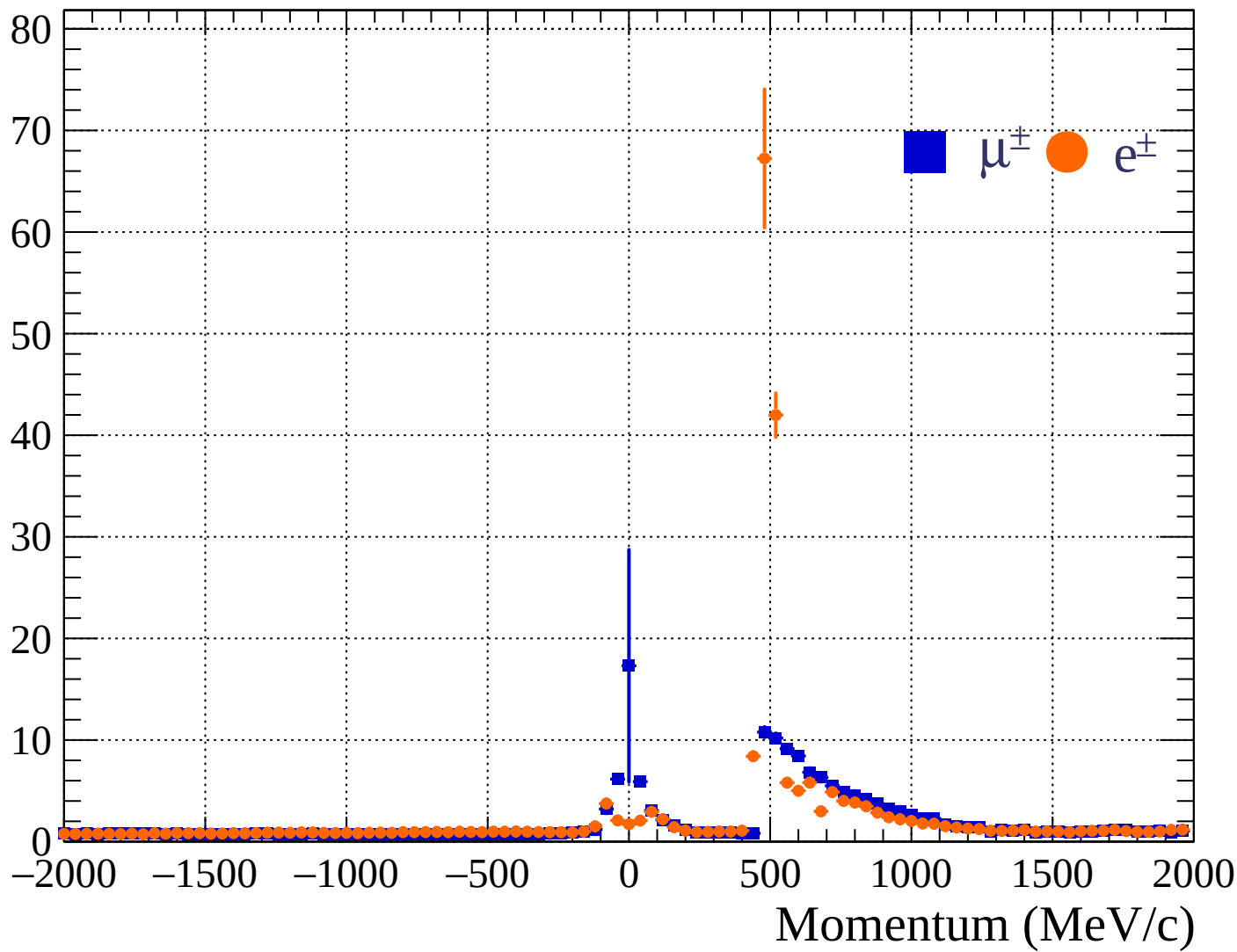
Pulls standard deviation



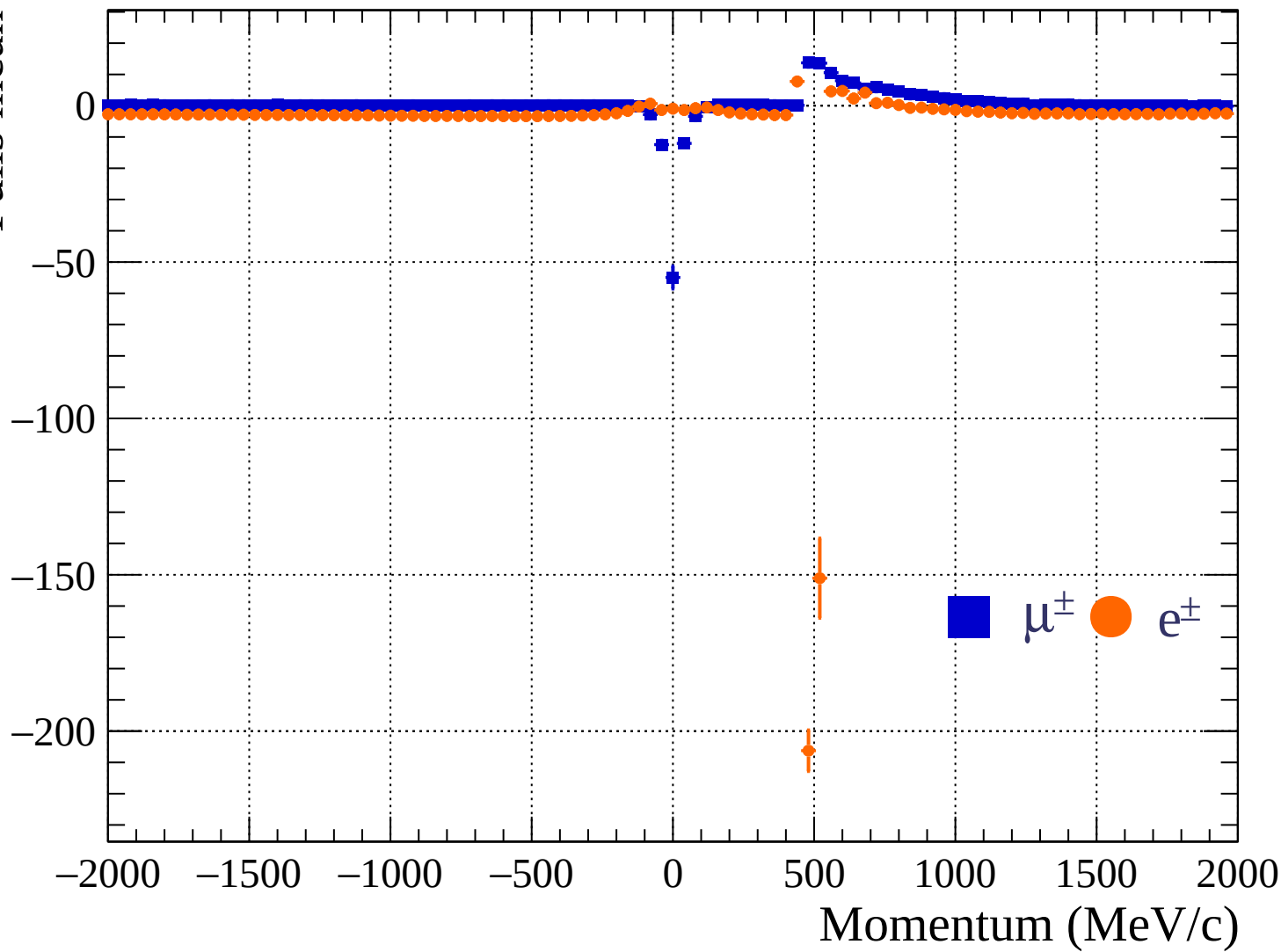
Pulls mean

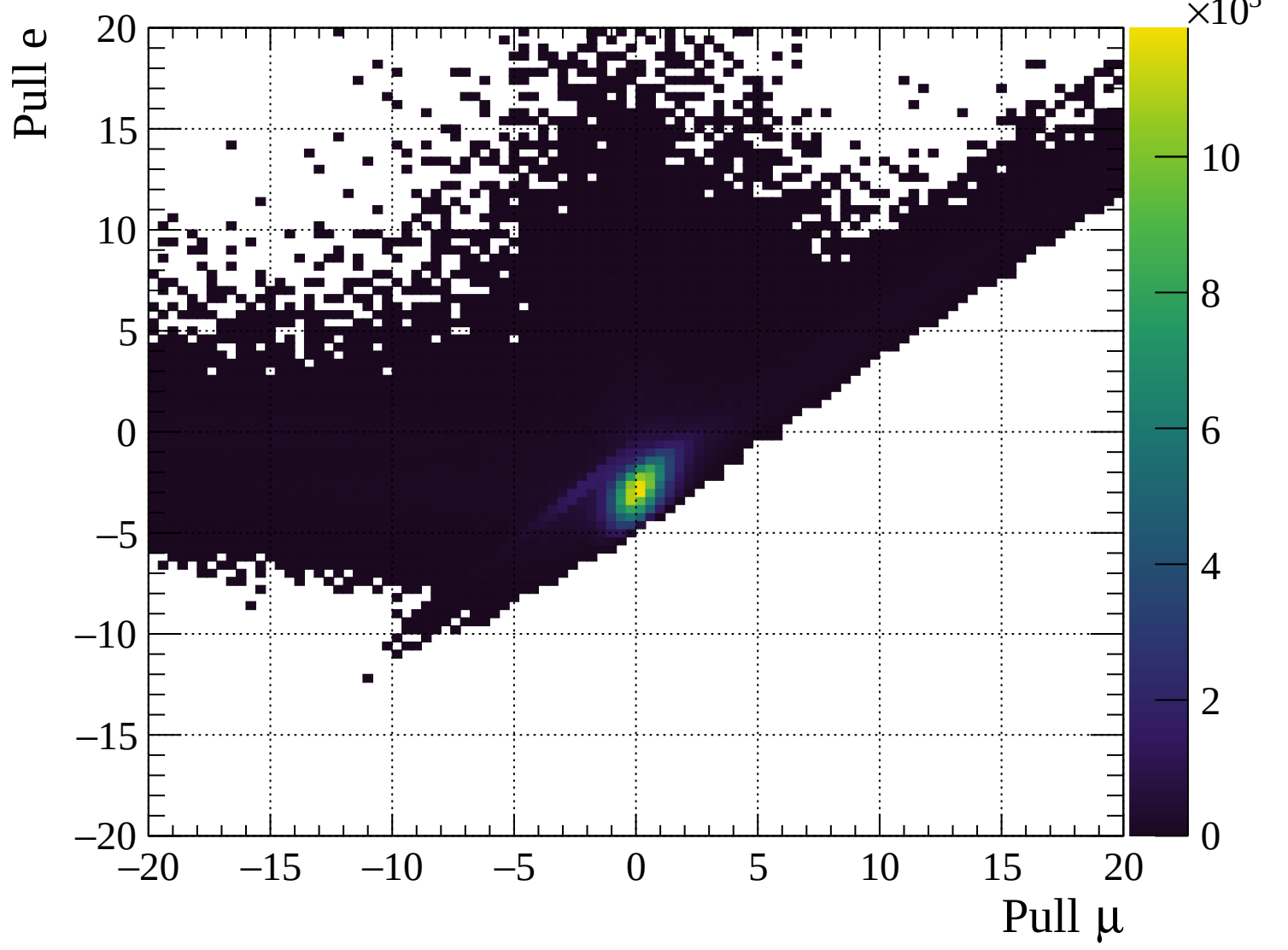


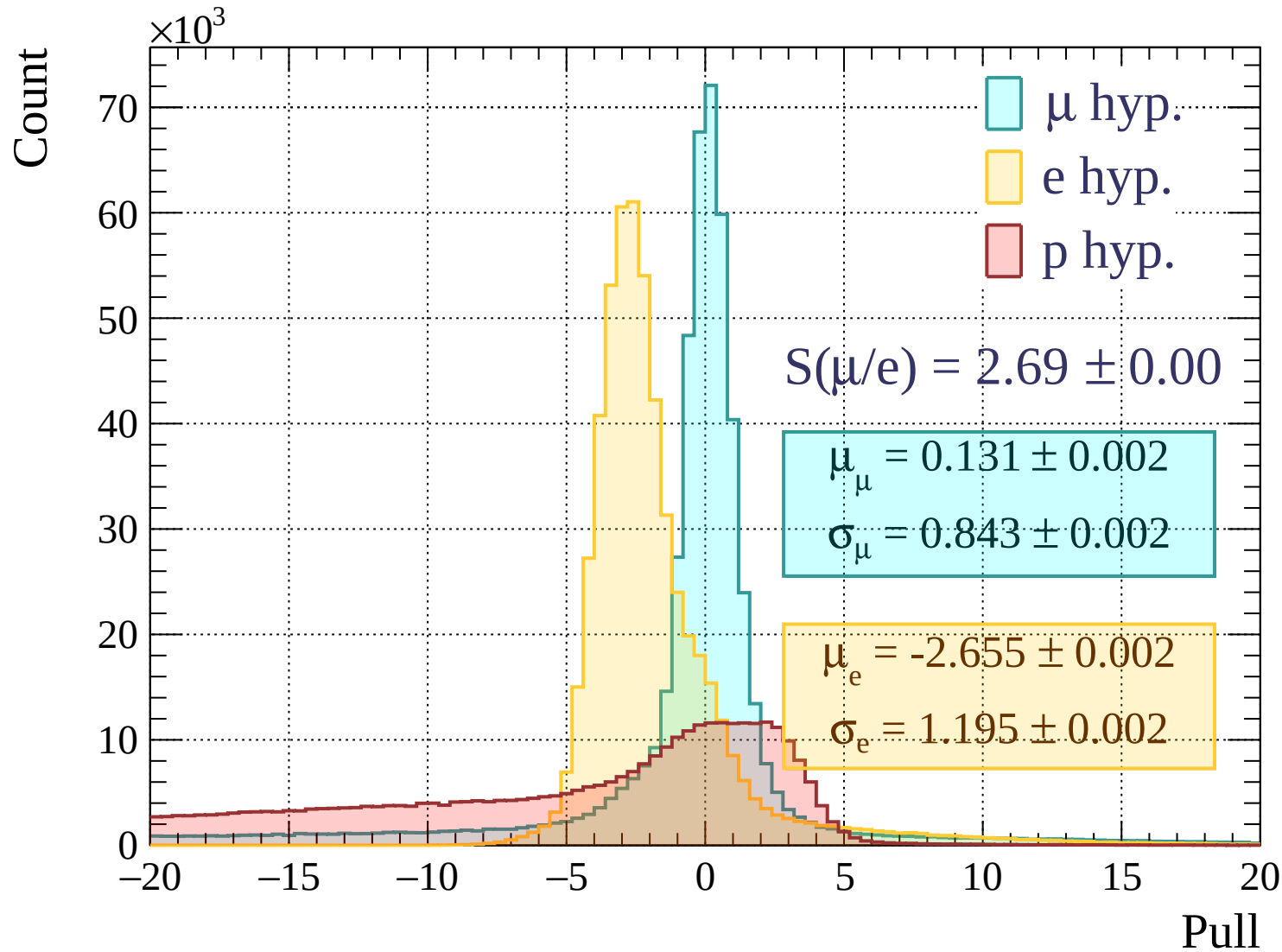
Pulls standard deviation

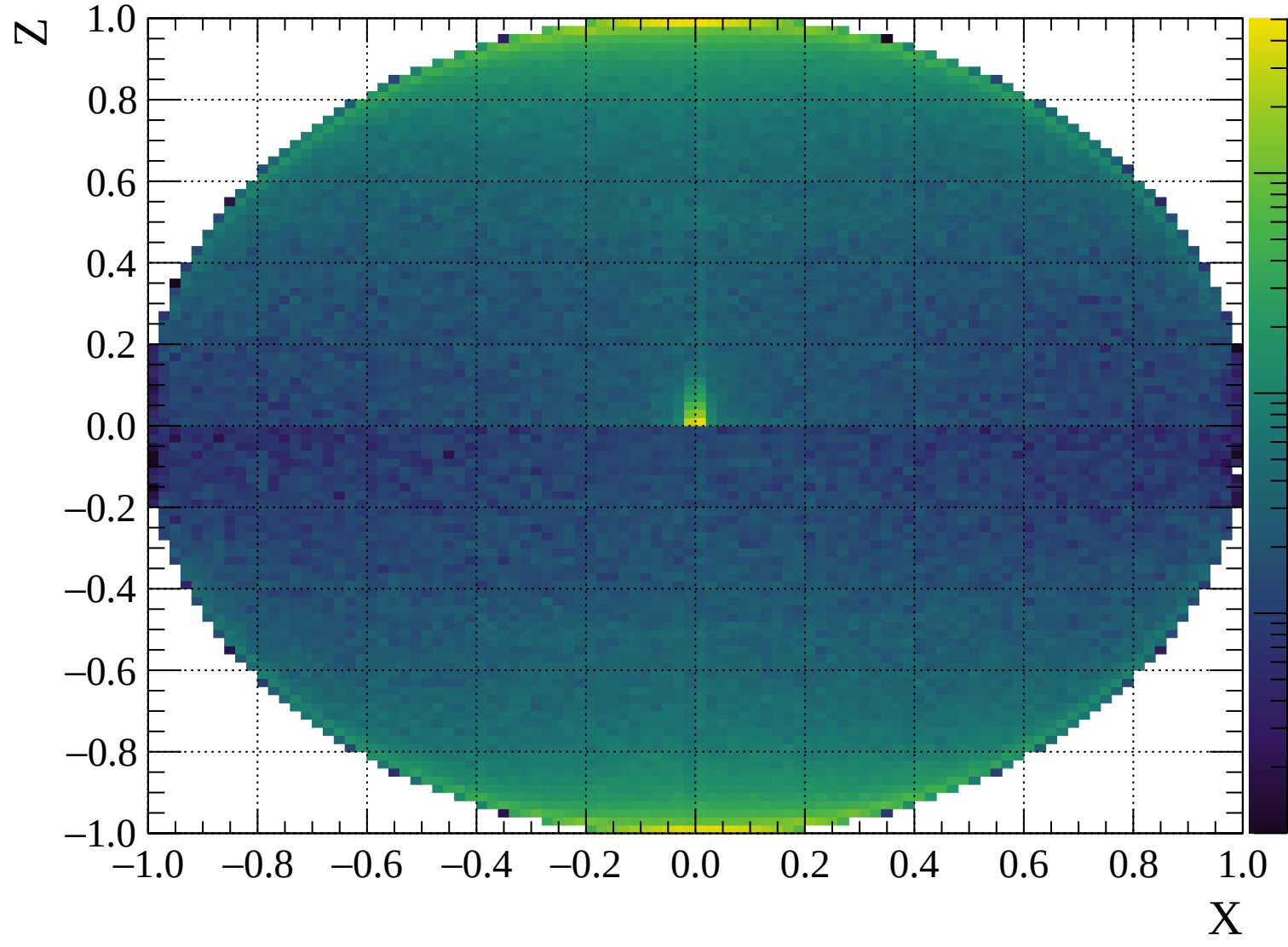


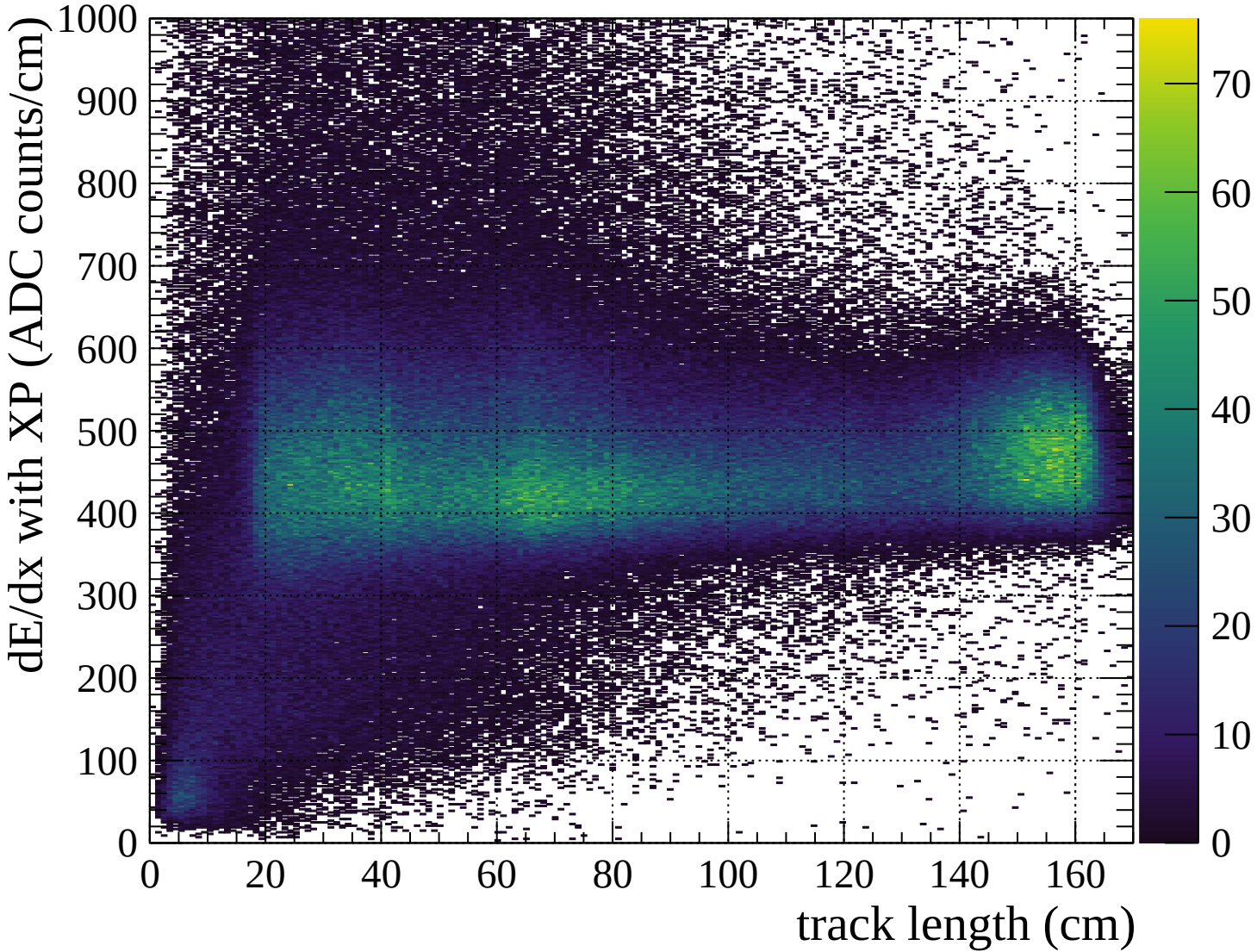
Pulls mean

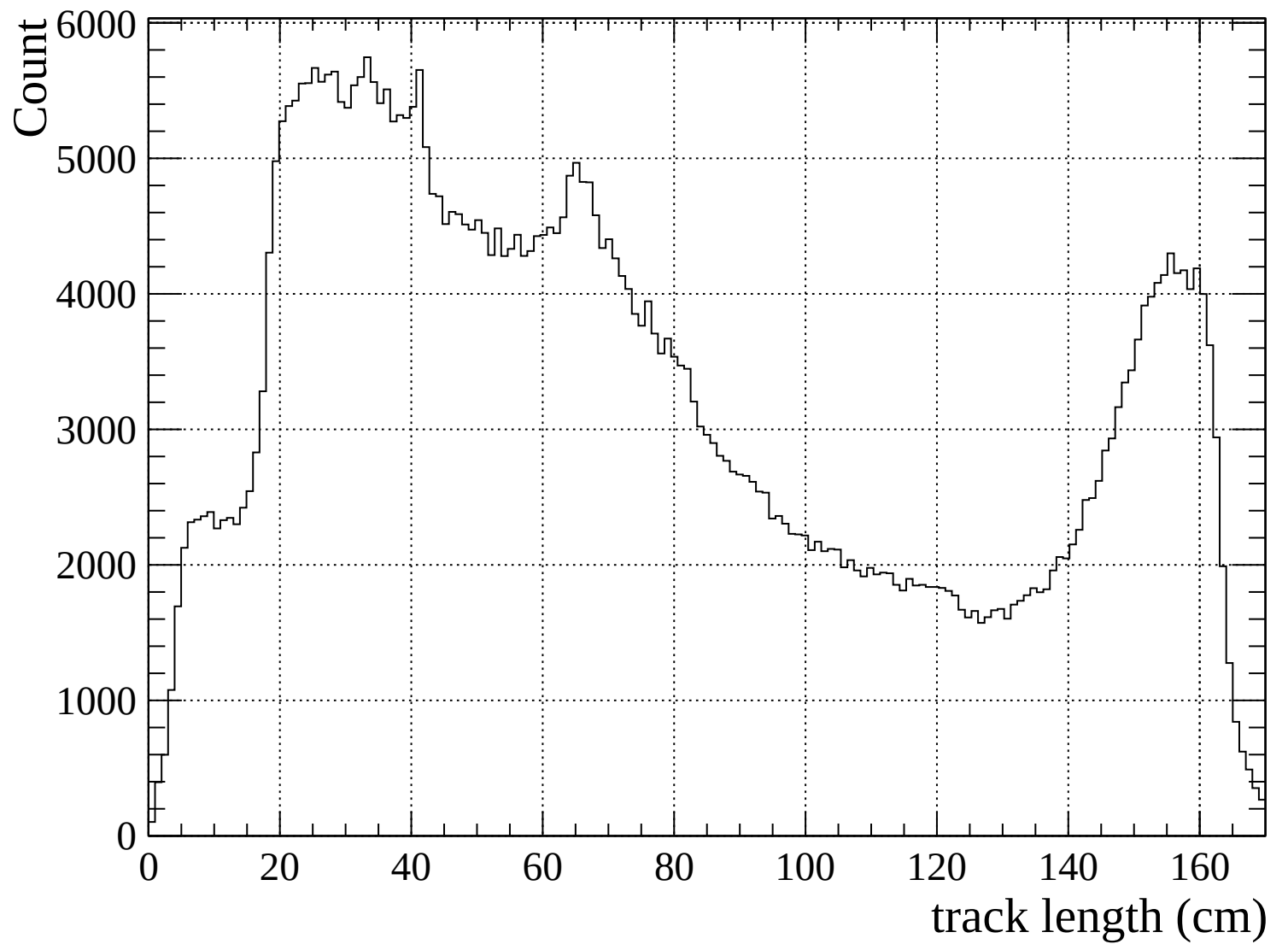


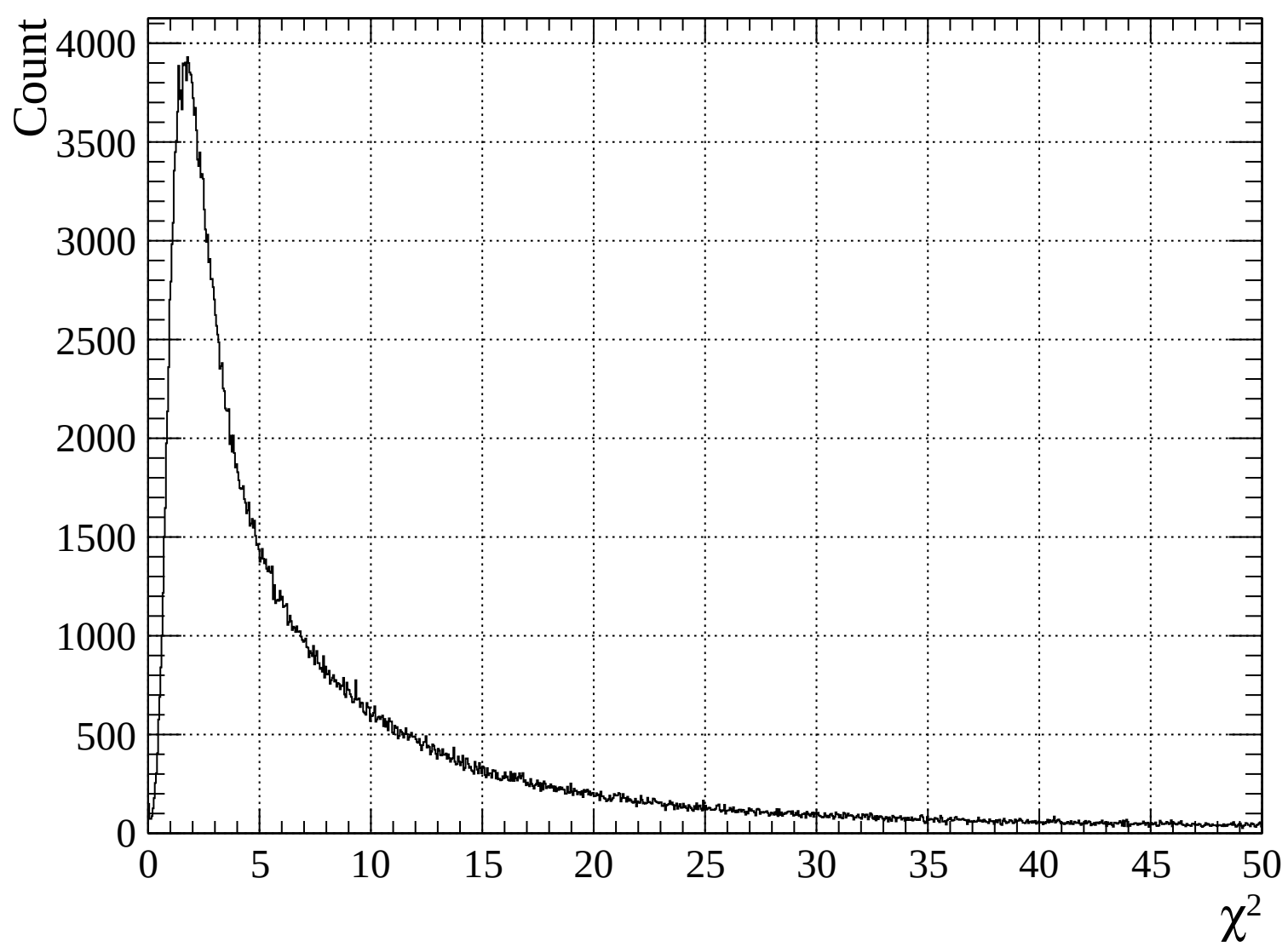


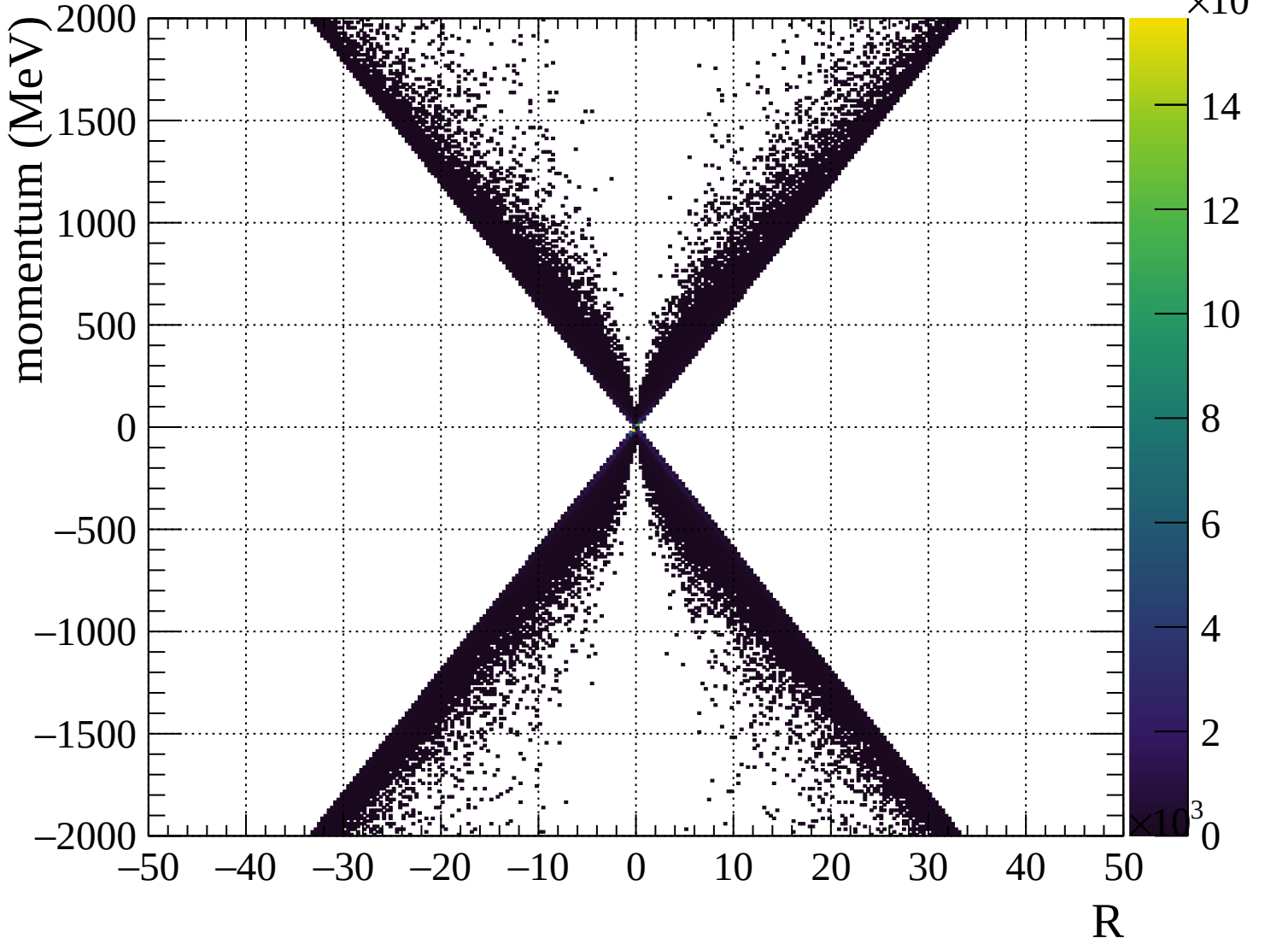


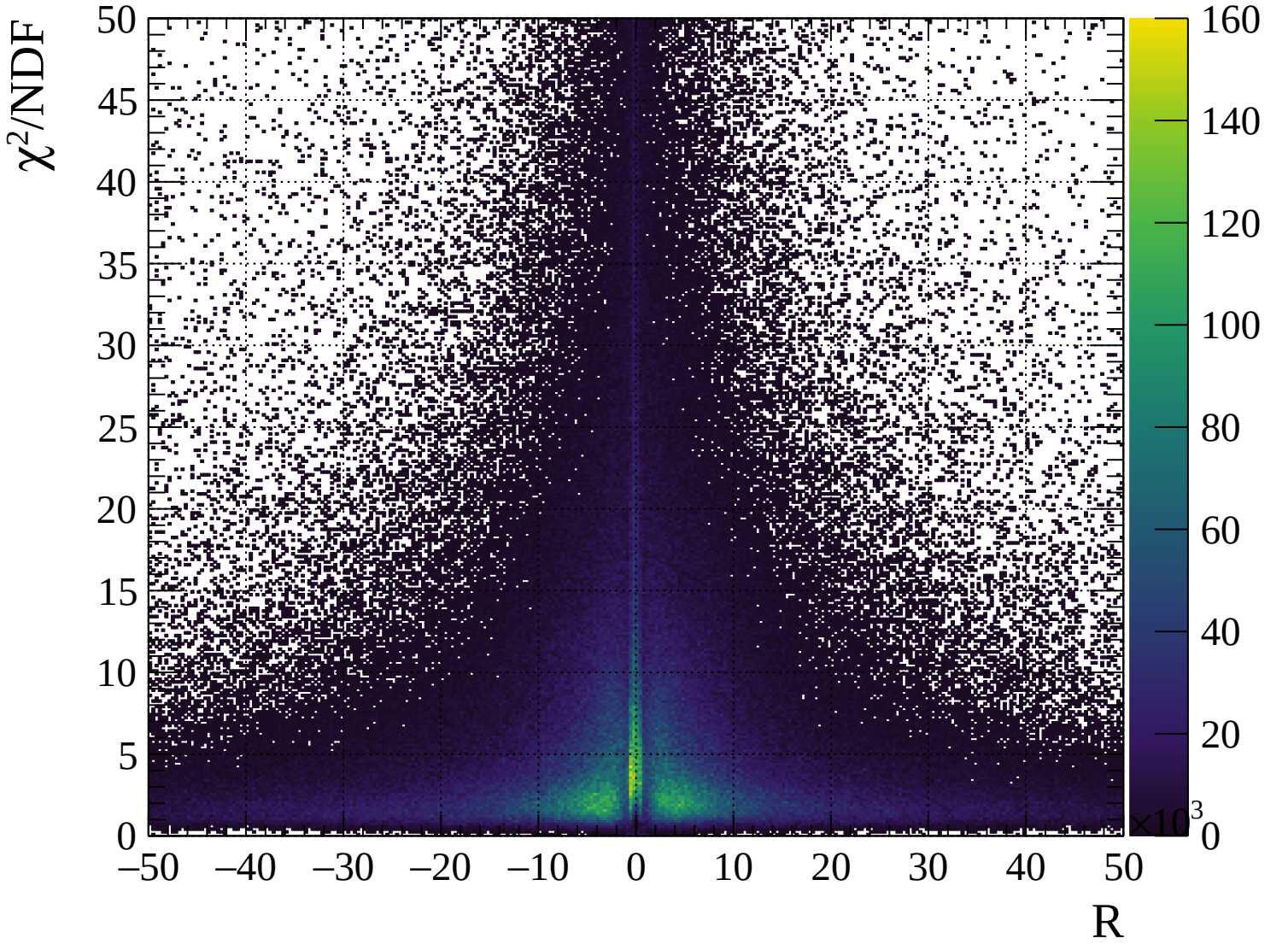






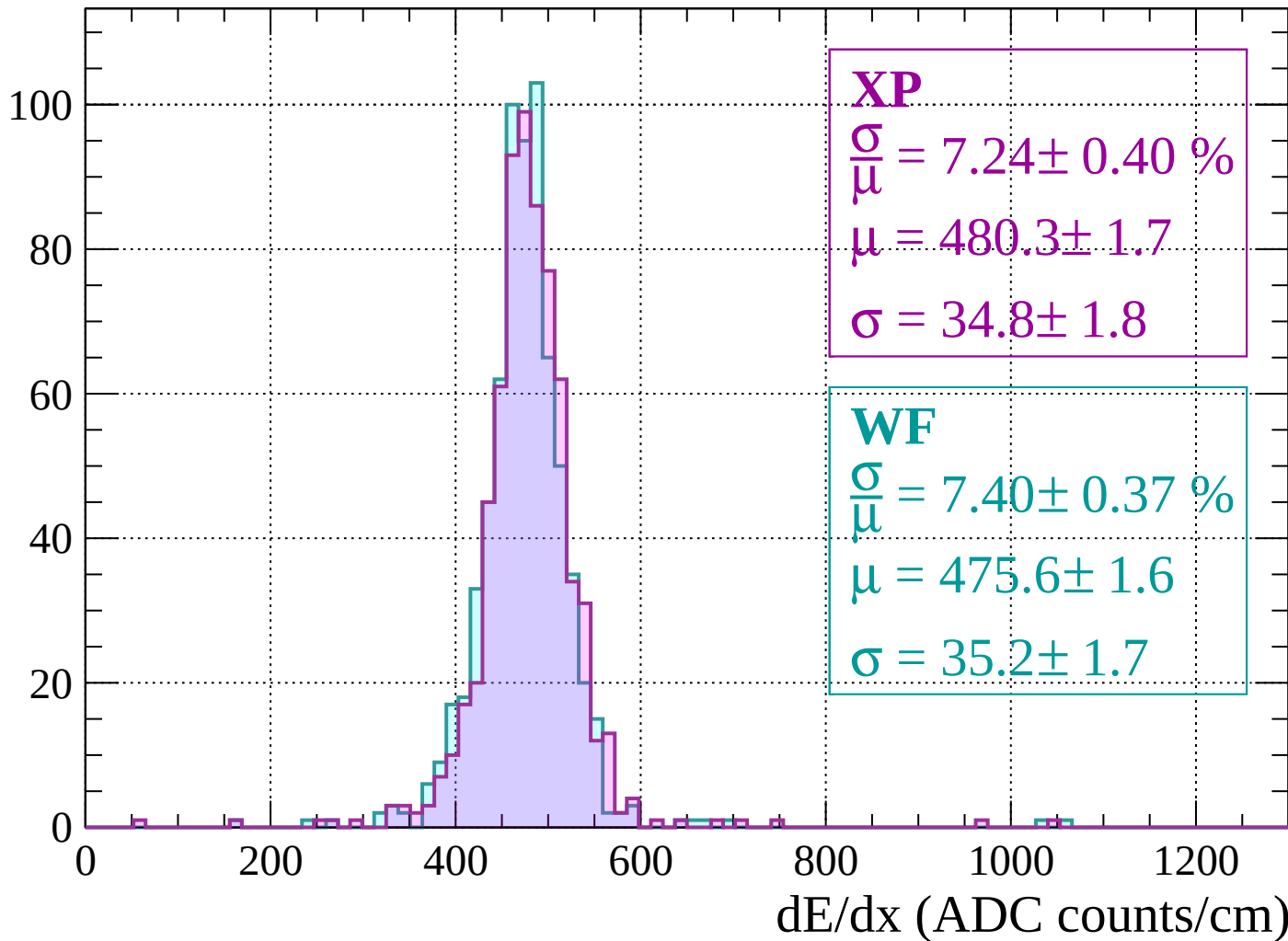




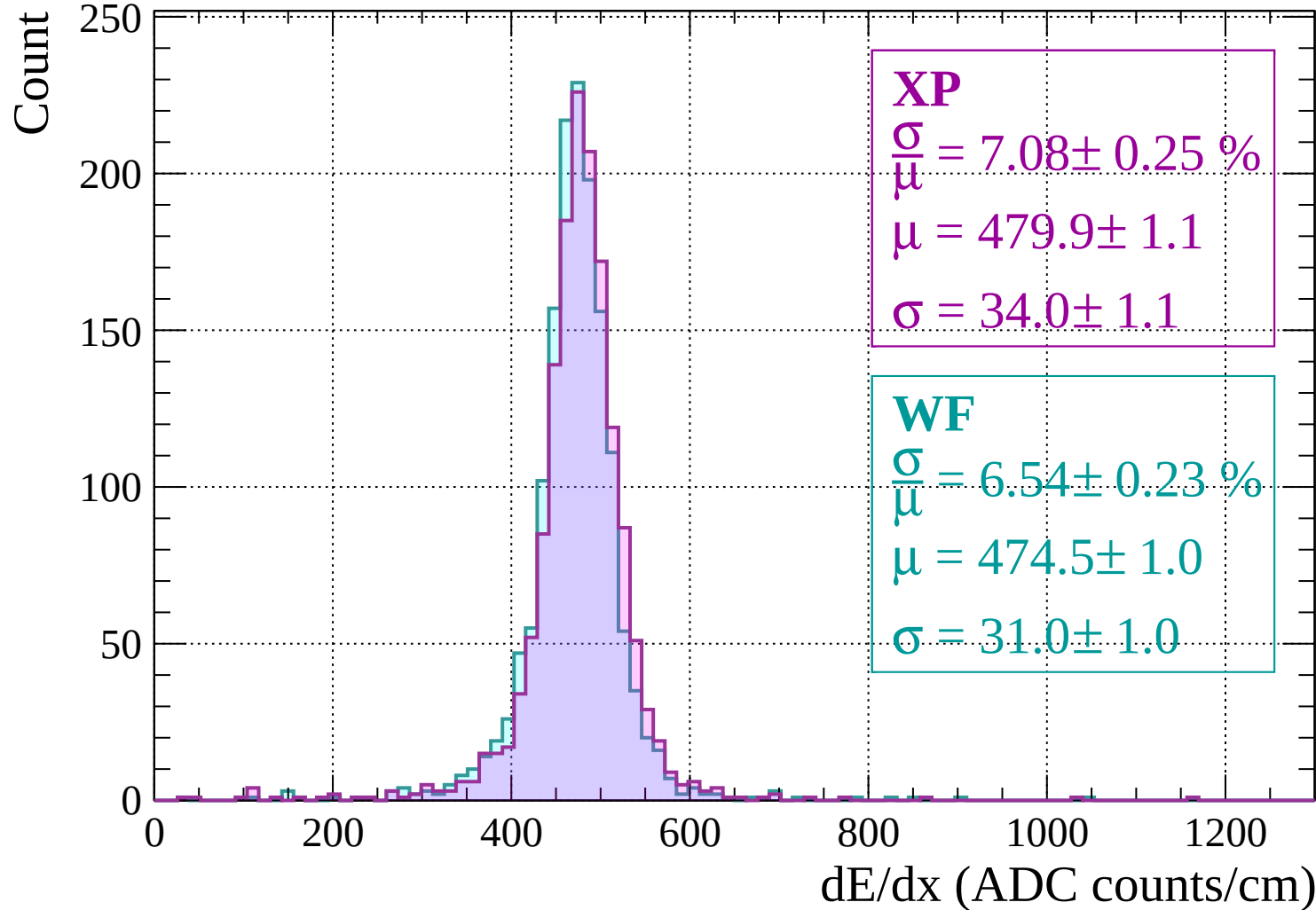


Energy loss | $-2000 < p < -1960$

Count

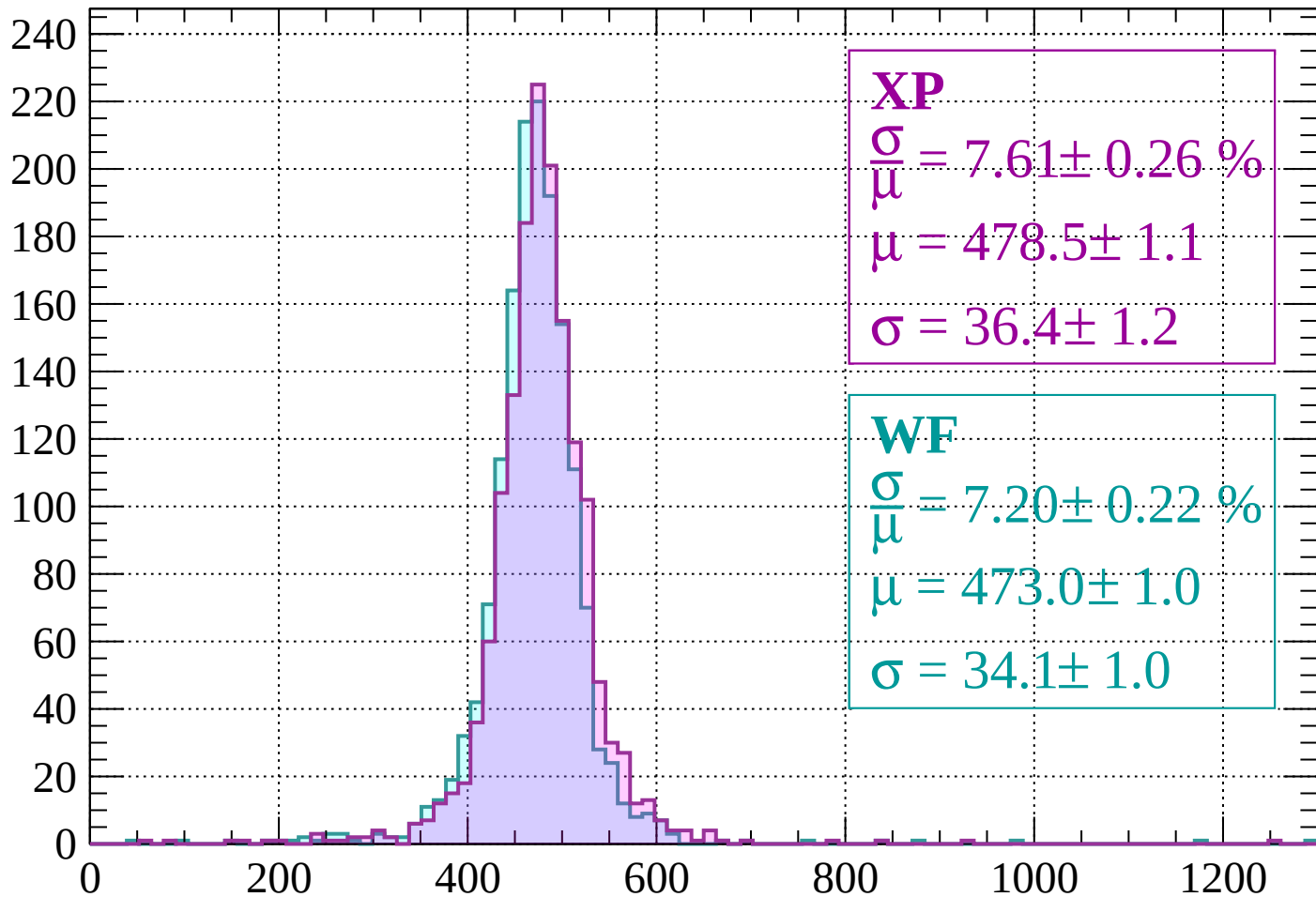


Energy loss | -1960 < p < -1920



Energy loss | -1920 < p < -1880

Count



XP

$$\frac{\sigma}{\mu} = 7.61 \pm 0.26 \%$$

$$\mu = 478.5 \pm 1.1$$

$$\sigma = 36.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.20 \pm 0.22 \%$$

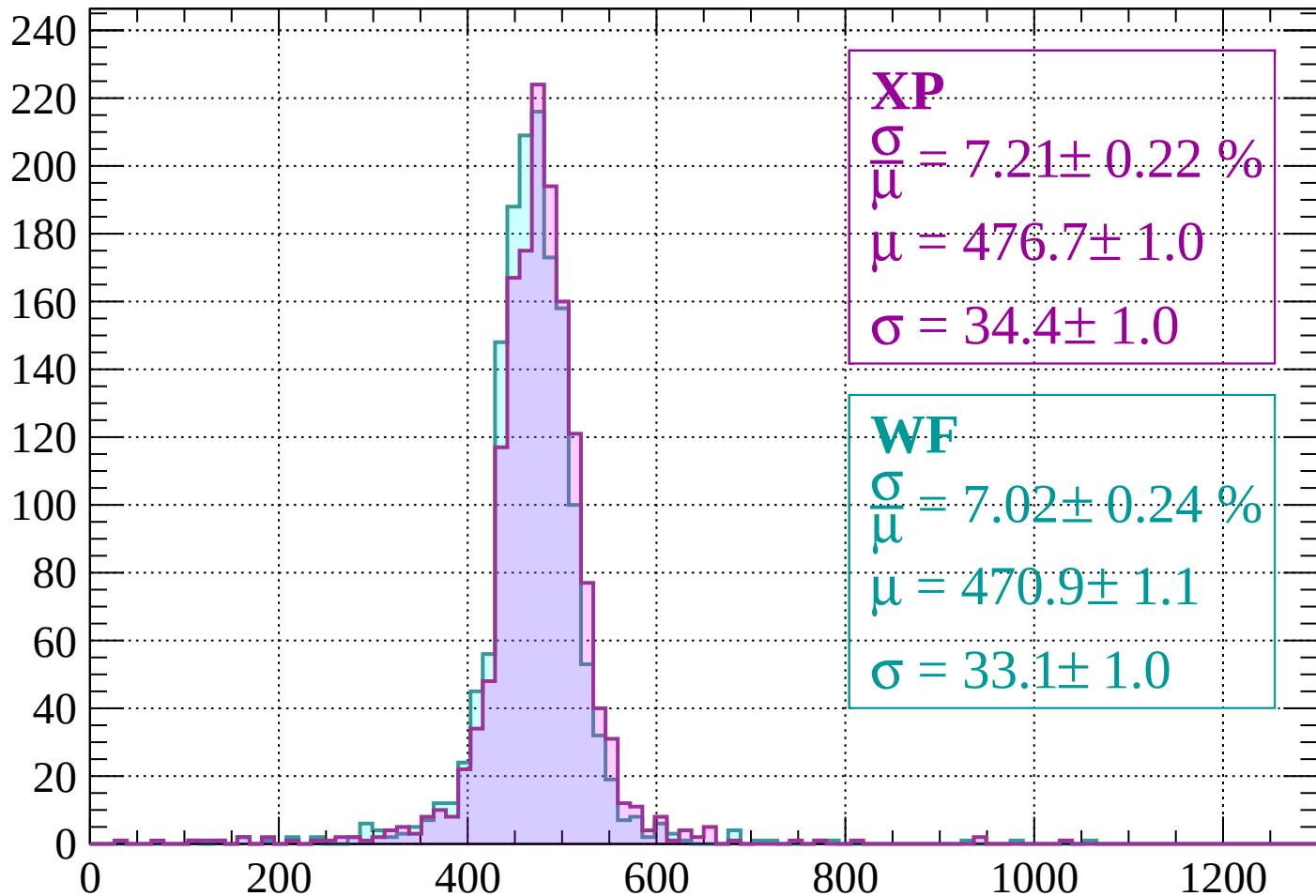
$$\mu = 473.0 \pm 1.0$$

$$\sigma = 34.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.22 \%$$

$$\mu = 476.7 \pm 1.0$$

$$\sigma = 34.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.02 \pm 0.24 \%$$

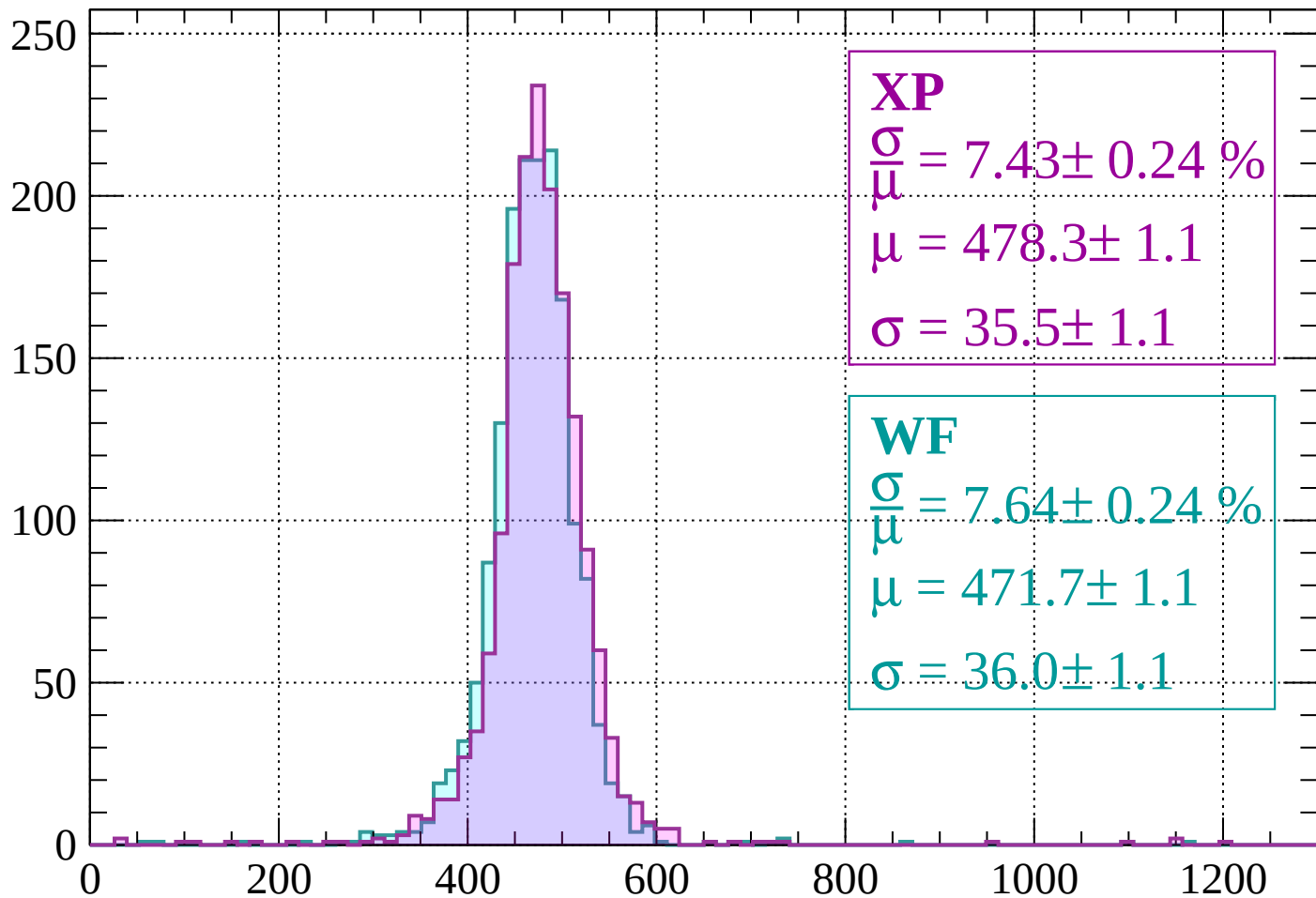
$$\mu = 470.9 \pm 1.1$$

$$\sigma = 33.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 7.43 \pm 0.24 \%$$

$$\mu = 478.3 \pm 1.1$$

$$\sigma = 35.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.64 \pm 0.24 \%$$

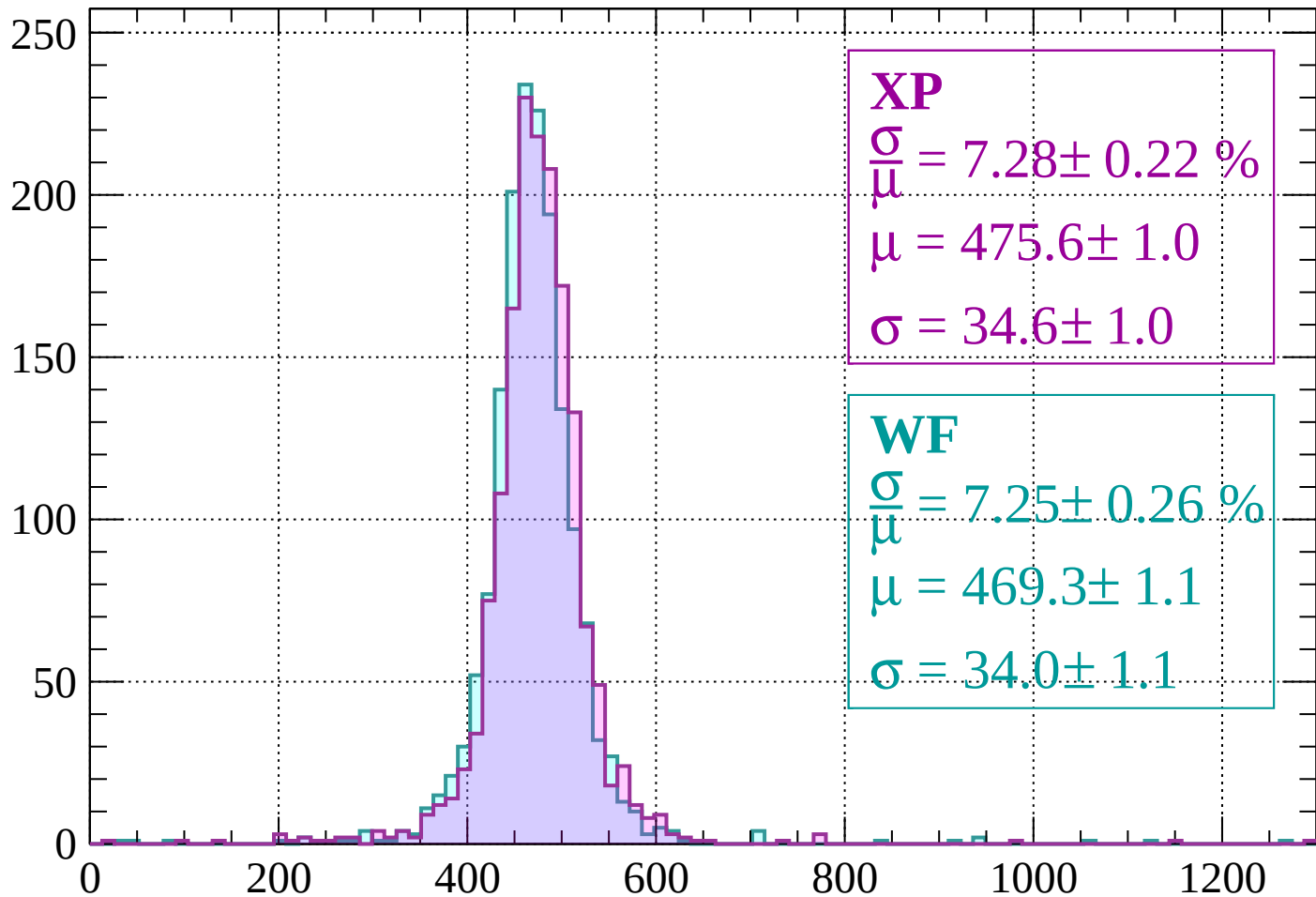
$$\mu = 471.7 \pm 1.1$$

$$\sigma = 36.0 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | -1800 < p < -1760

Count



XP

$$\frac{\sigma}{\mu} = 7.28 \pm 0.22 \%$$

$$\mu = 475.6 \pm 1.0$$

$$\sigma = 34.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.25 \pm 0.26 \%$$

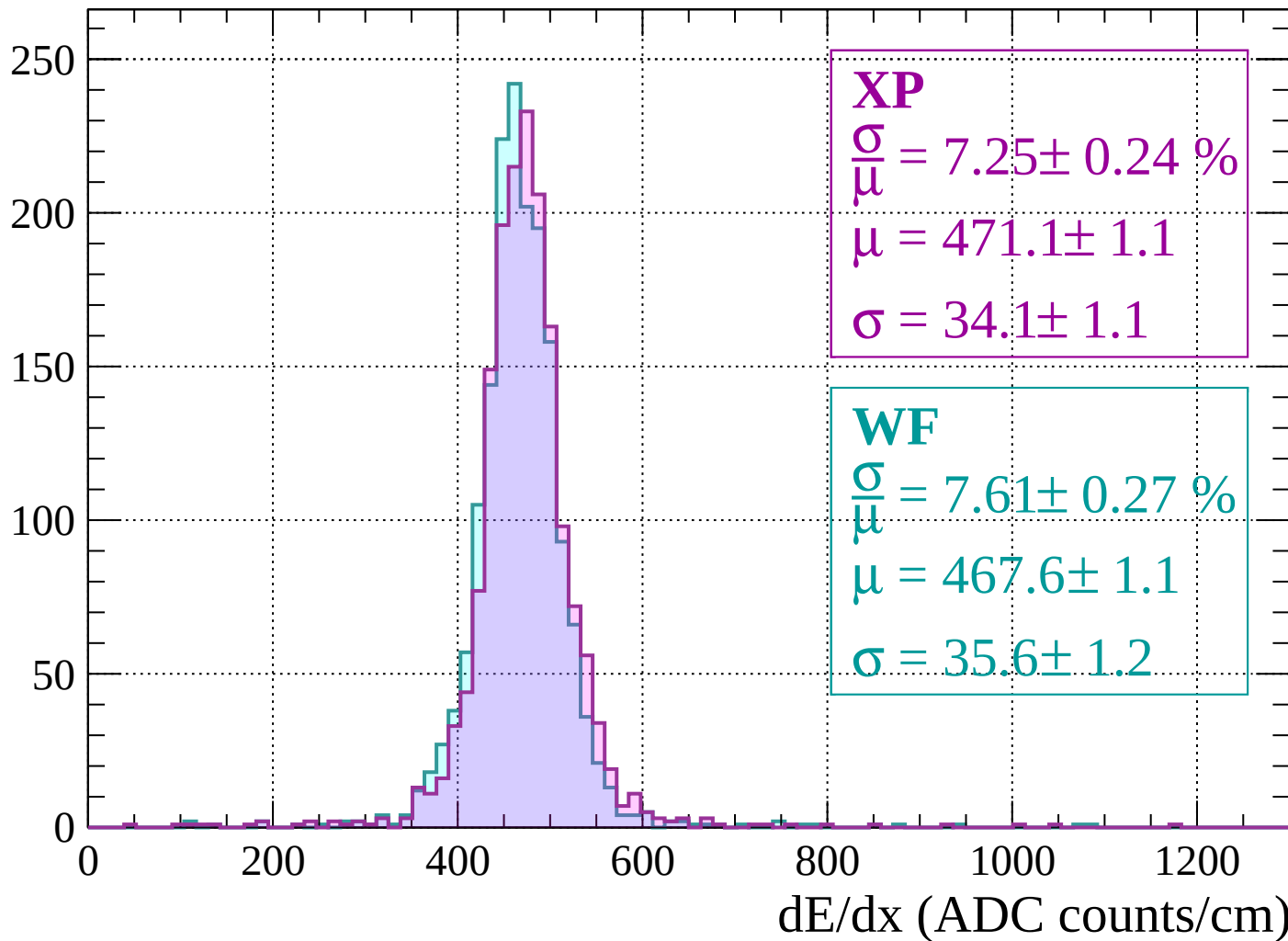
$$\mu = 469.3 \pm 1.1$$

$$\sigma = 34.0 \pm 1.1$$

dE/dx (ADC counts/cm)

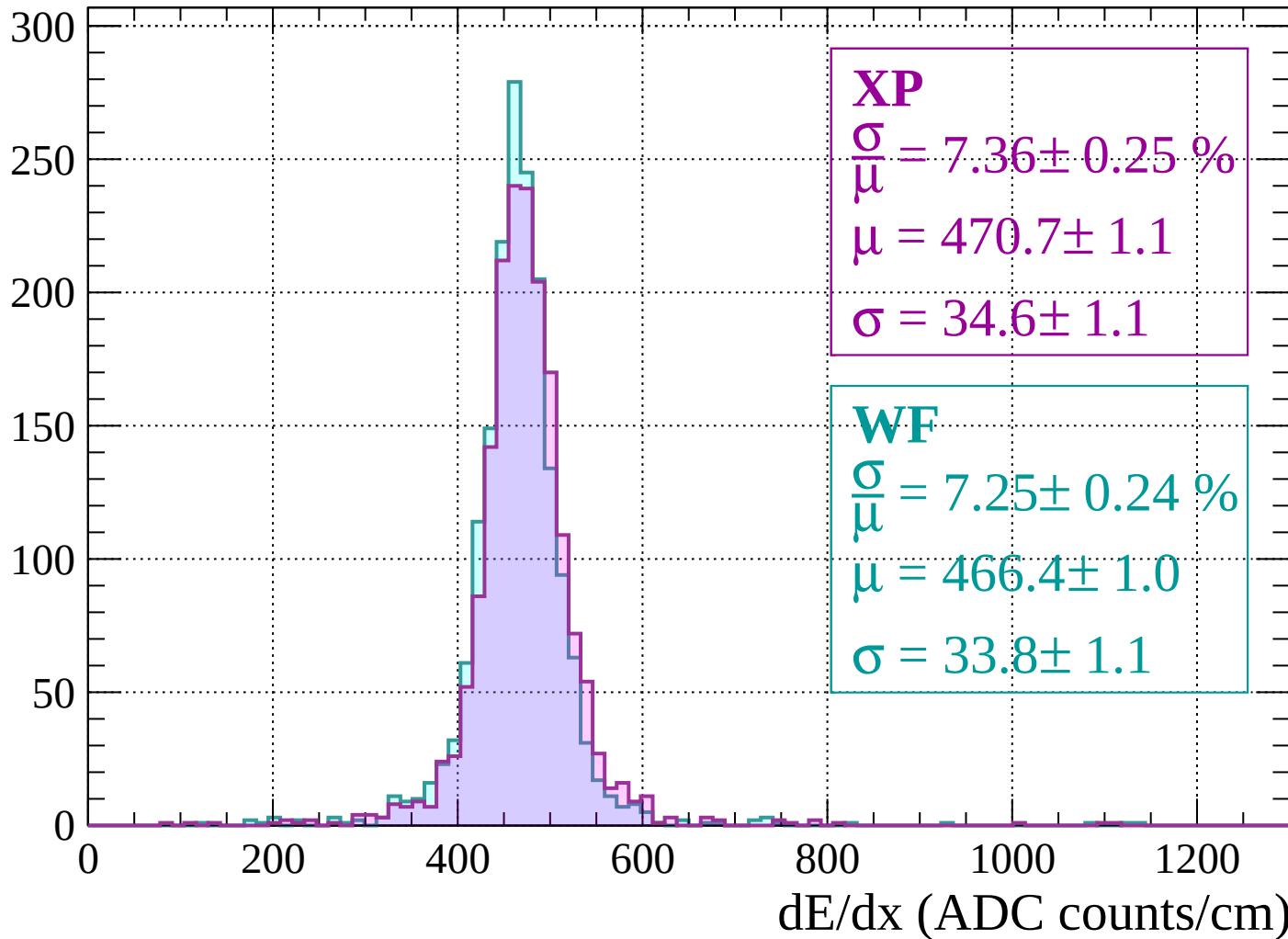
Energy loss | $-1760 < p < -1720$

Count



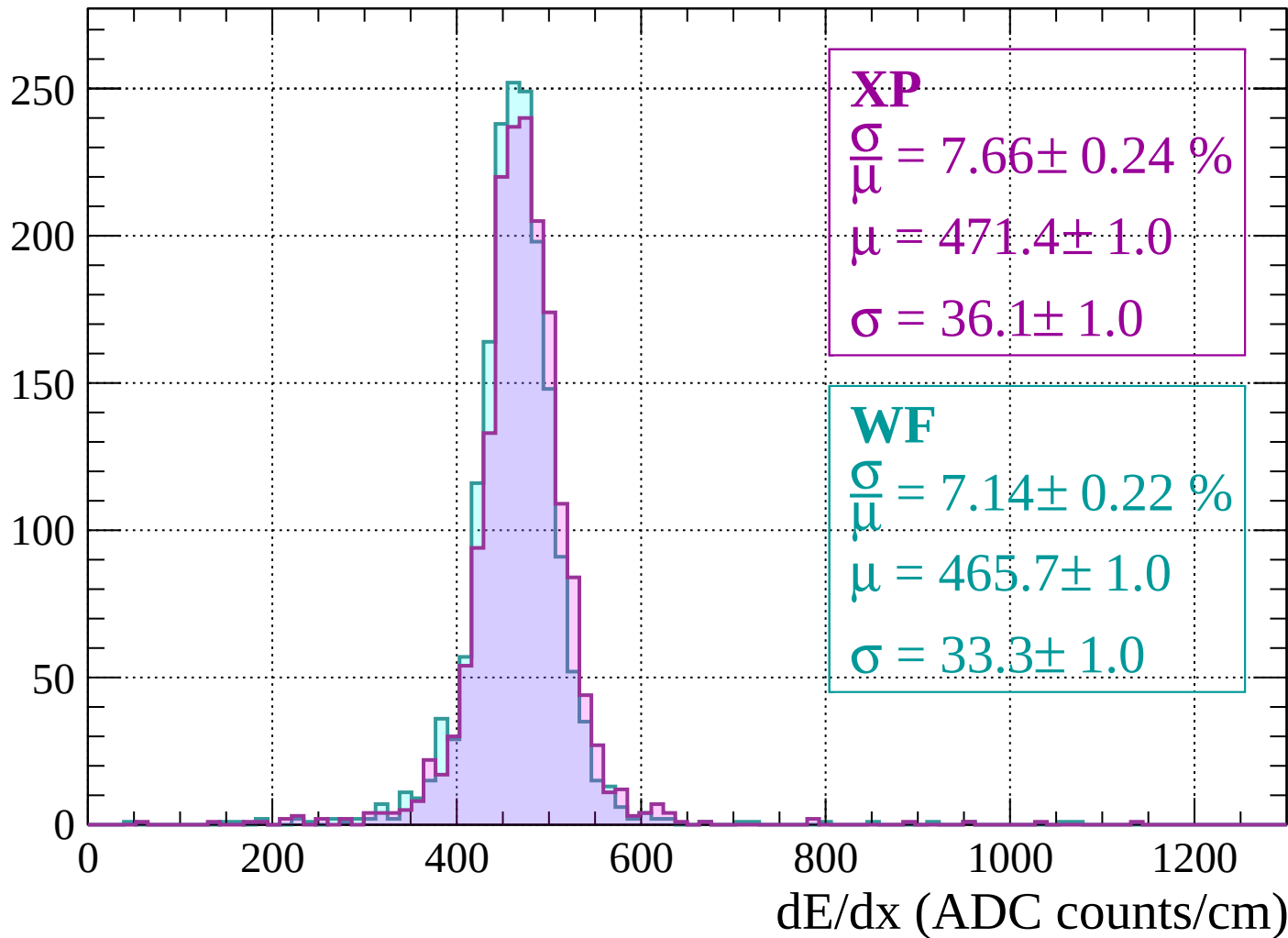
Energy loss | $-1720 < p < -1680$

Count



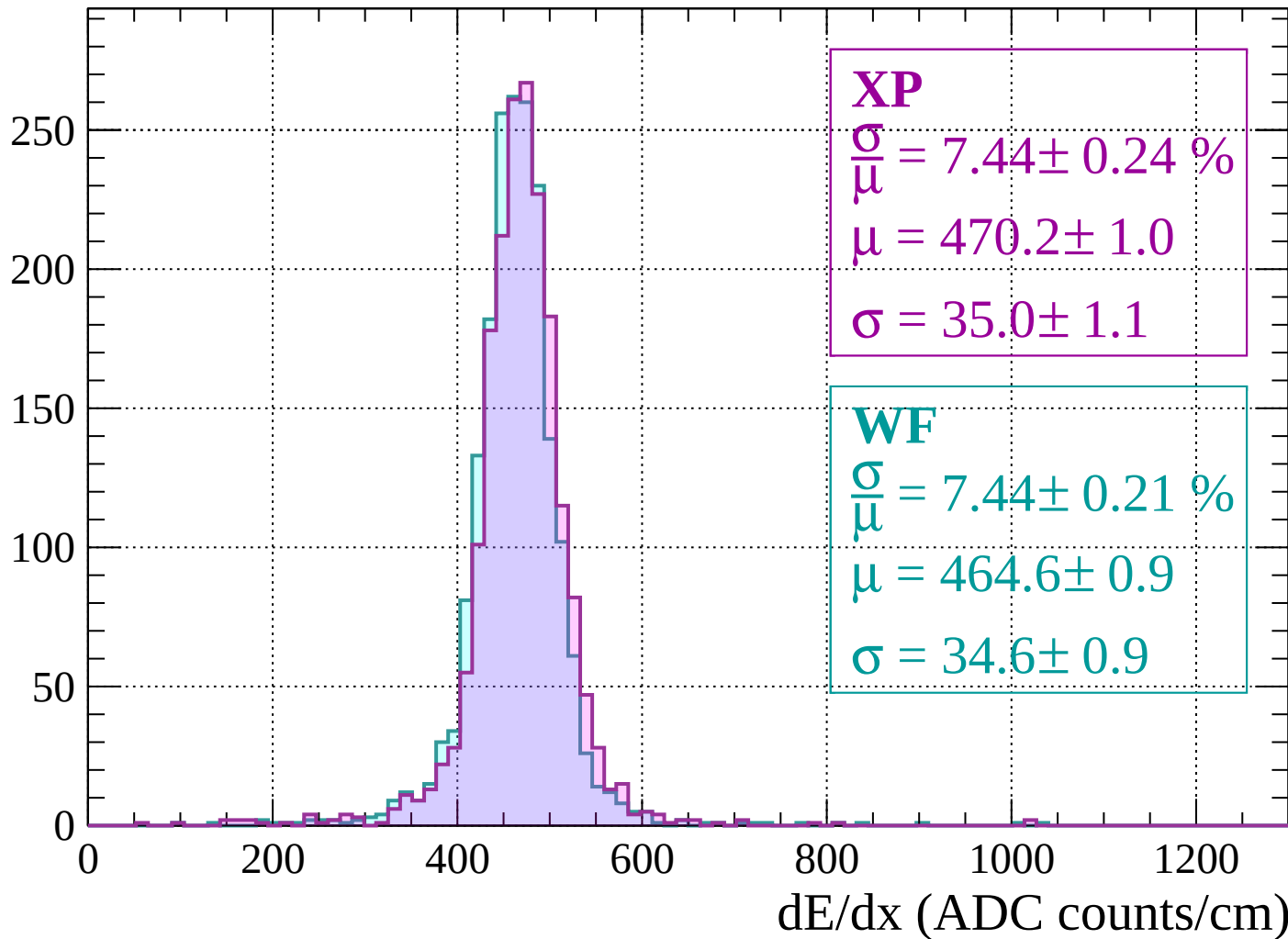
Energy loss | $-1680 < p < -1640$

Count



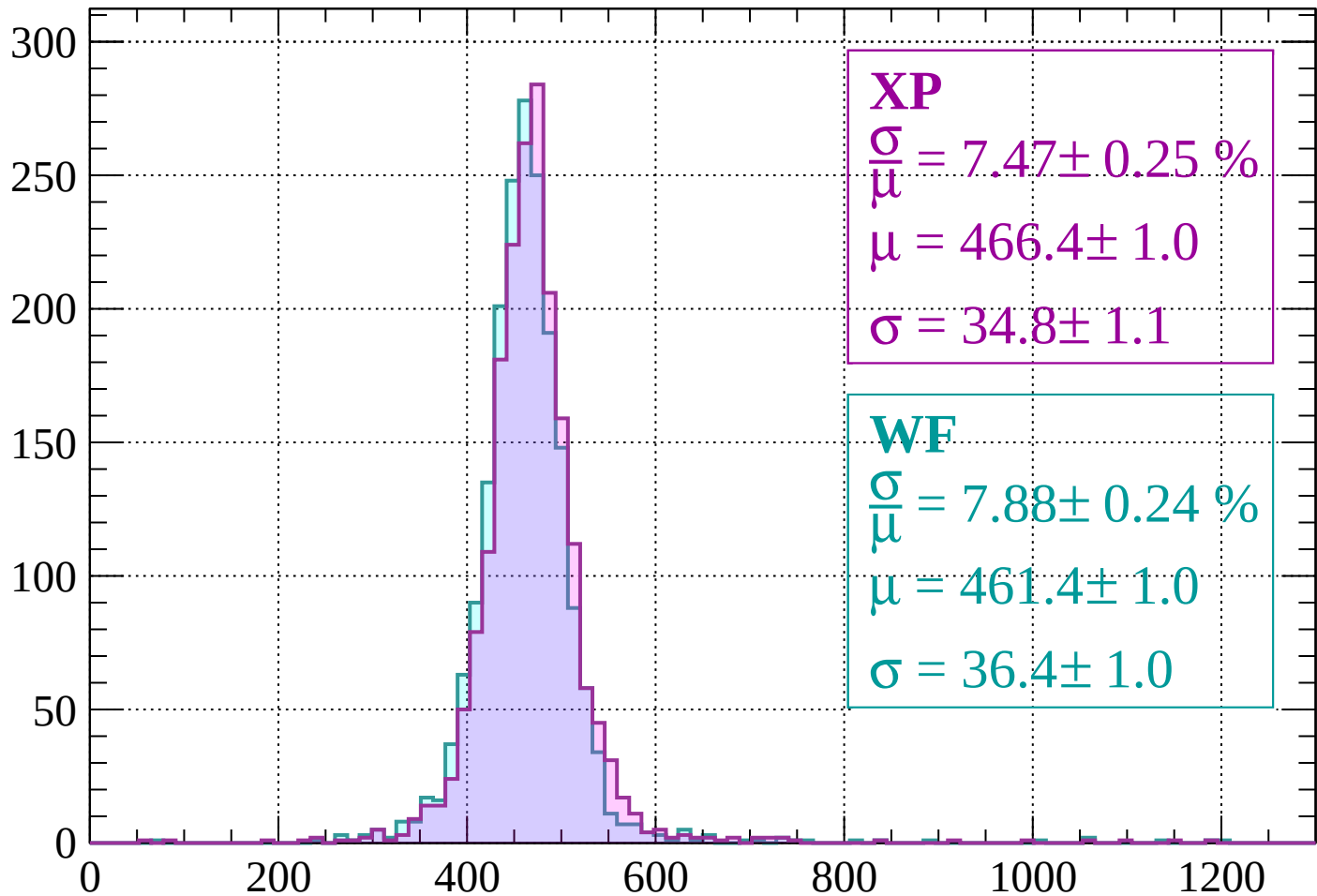
Energy loss | $-1640 < p < -1600$

Count



Energy loss | $-1600 < p < -1560$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.25 \%$$

$$\mu = 466.4 \pm 1.0$$

$$\sigma = 34.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.24 \%$$

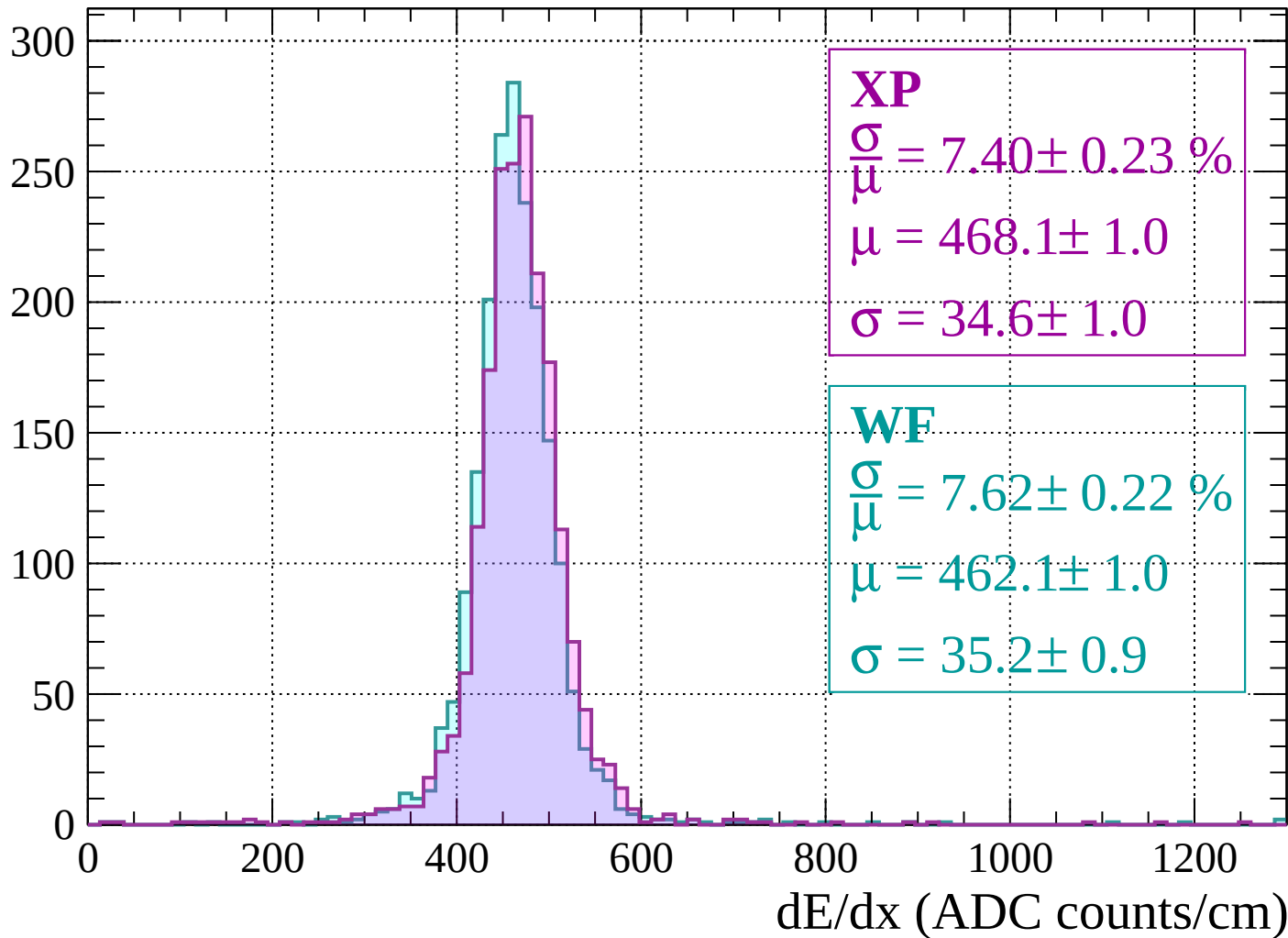
$$\mu = 461.4 \pm 1.0$$

$$\sigma = 36.4 \pm 1.0$$

dE/dx (ADC counts/cm)

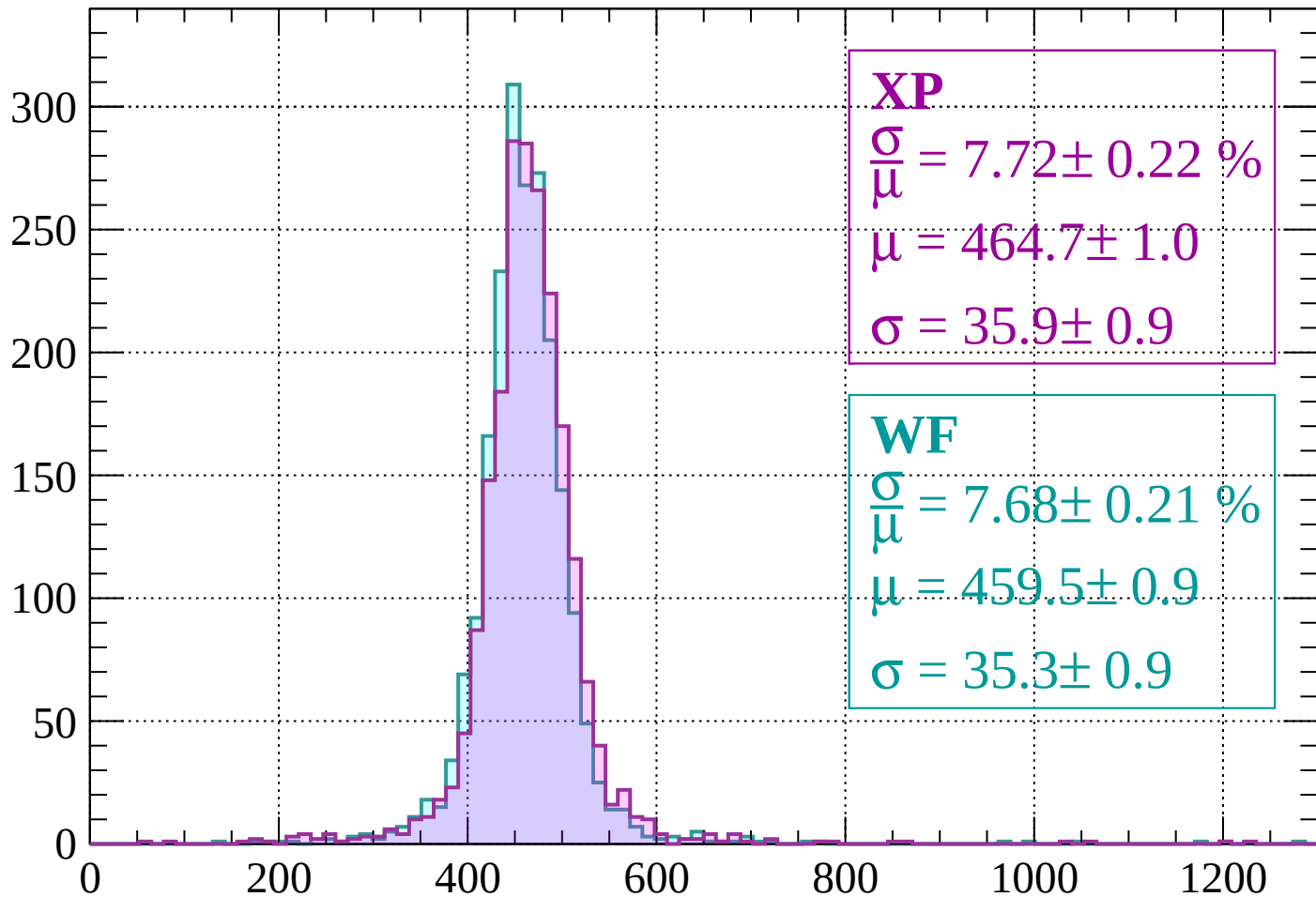
Energy loss | $-1560 < p < -1520$

Count



Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 7.72 \pm 0.22 \%$$

$$\mu = 464.7 \pm 1.0$$

$$\sigma = 35.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.68 \pm 0.21 \%$$

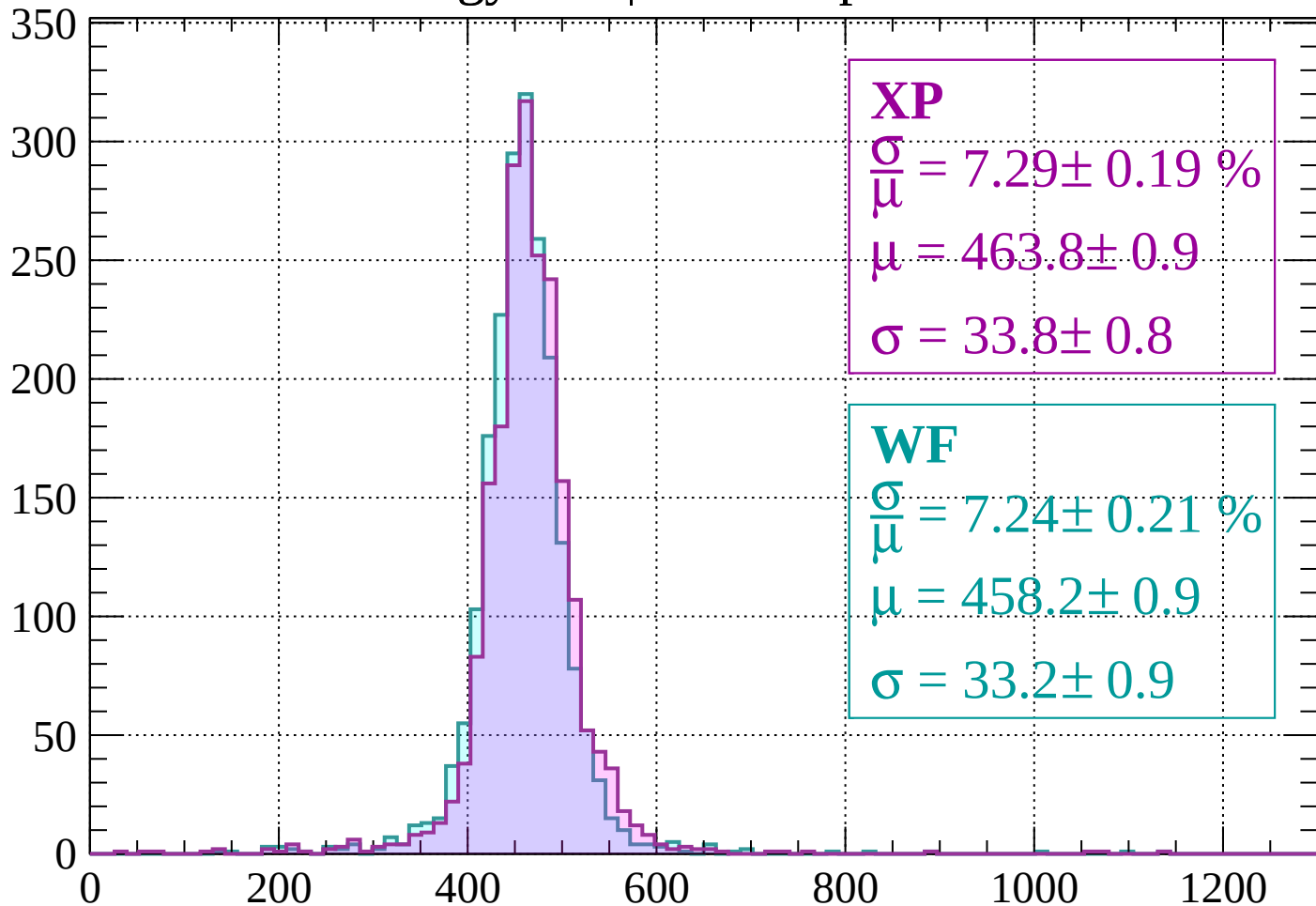
$$\mu = 459.5 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.29 \pm 0.19 \%$$

$$\mu = 463.8 \pm 0.9$$

$$\sigma = 33.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.24 \pm 0.21 \%$$

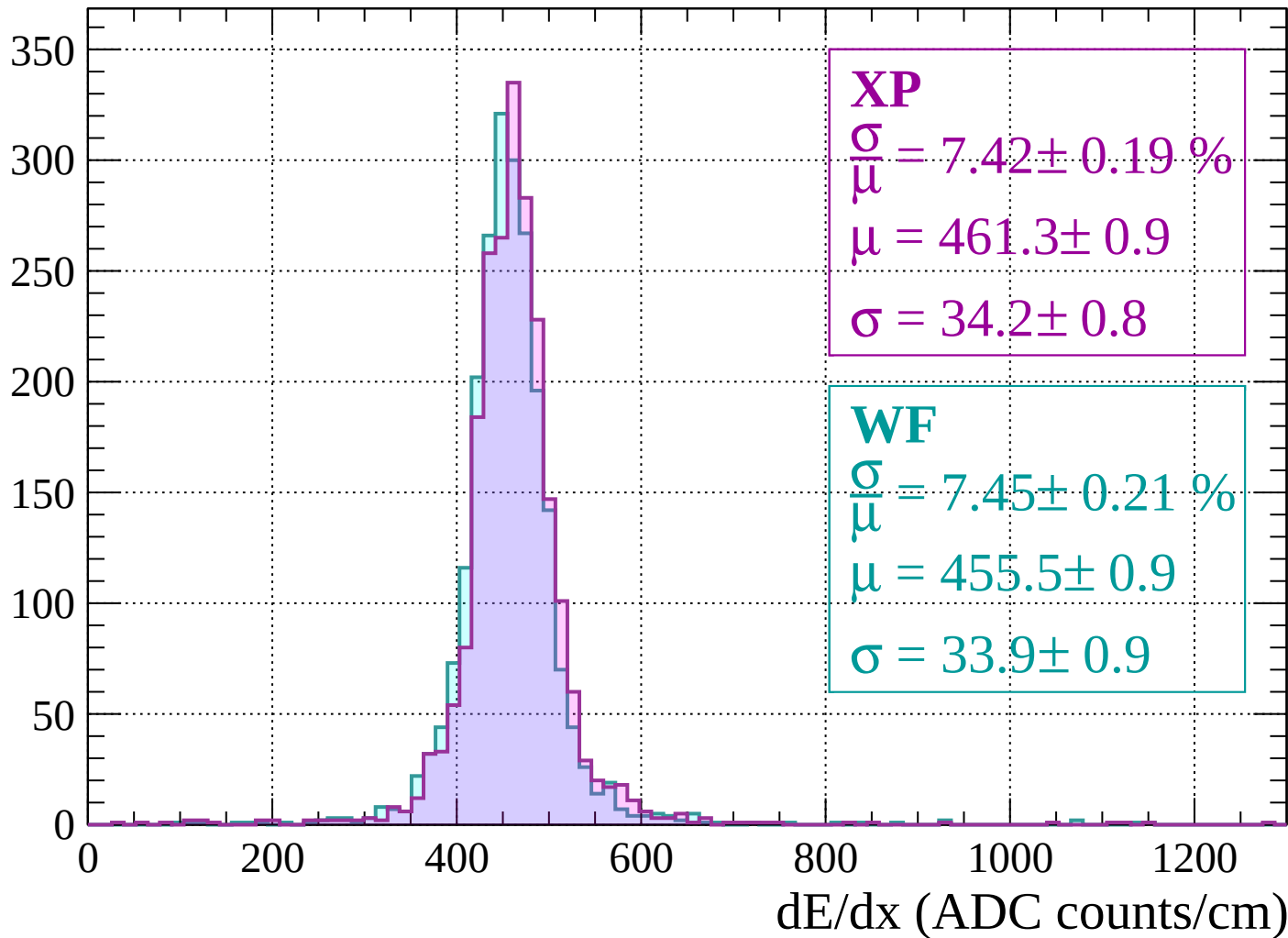
$$\mu = 458.2 \pm 0.9$$

$$\sigma = 33.2 \pm 0.9$$

dE/dx (ADC counts/cm)

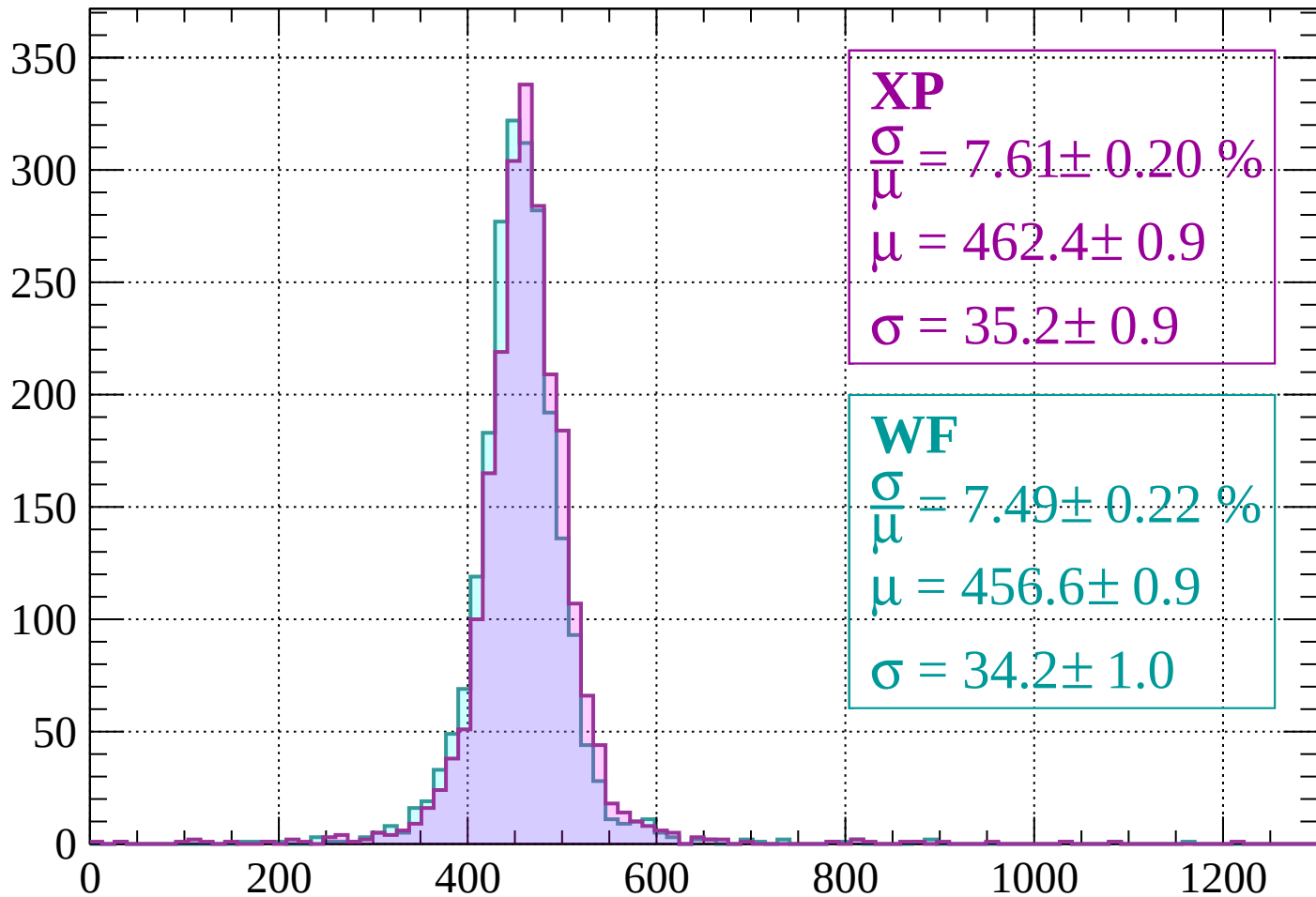
Energy loss | $-1440 < p < -1400$

Count



Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 7.61 \pm 0.20 \%$$

$$\mu = 462.4 \pm 0.9$$

$$\sigma = 35.2 \pm 0.9$$

WF

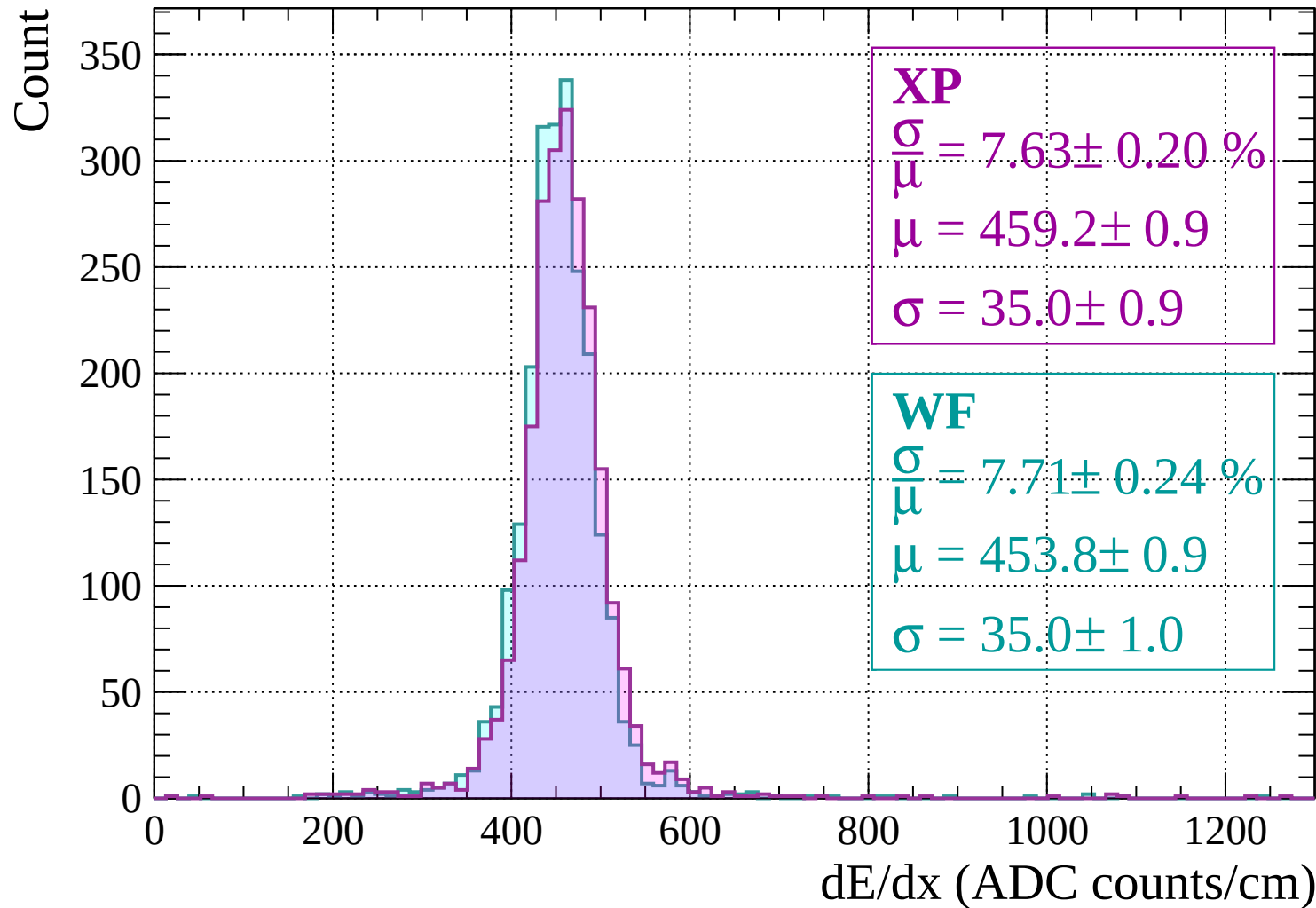
$$\frac{\sigma}{\mu} = 7.49 \pm 0.22 \%$$

$$\mu = 456.6 \pm 0.9$$

$$\sigma = 34.2 \pm 1.0$$

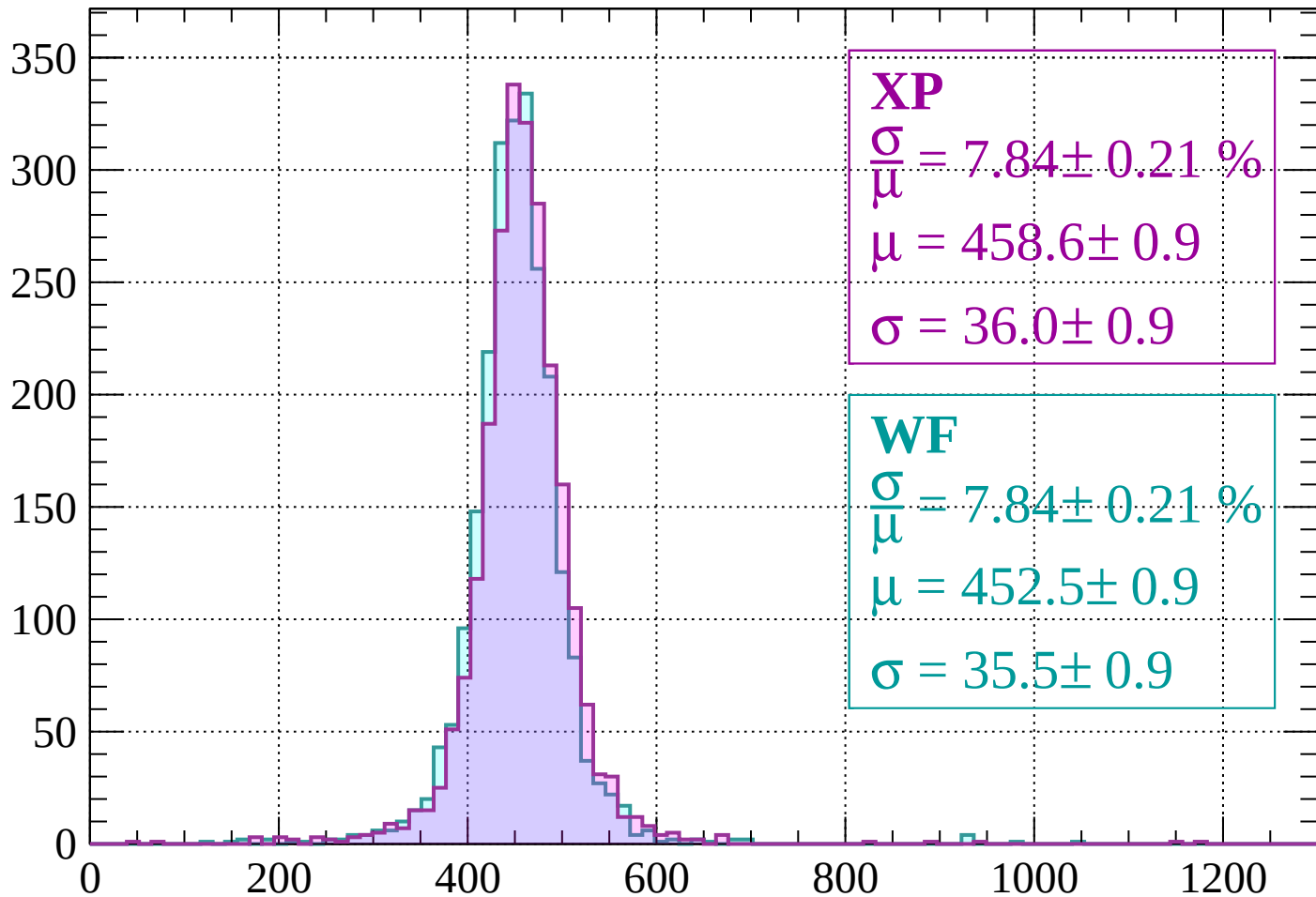
dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.21 \%$$

$$\mu = 458.6 \pm 0.9$$

$$\sigma = 36.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.84 \pm 0.21 \%$$

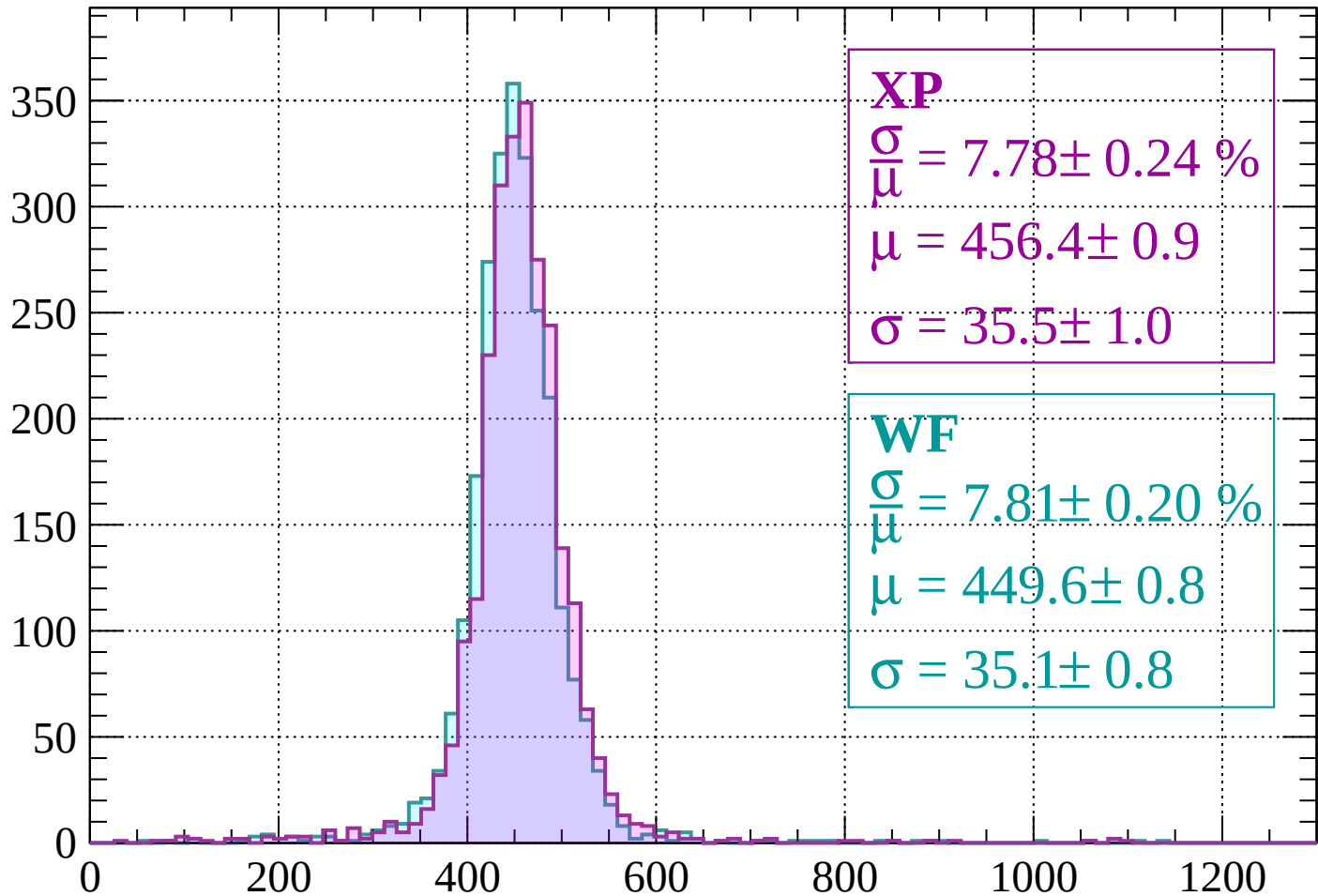
$$\mu = 452.5 \pm 0.9$$

$$\sigma = 35.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.24 \%$$

$$\mu = 456.4 \pm 0.9$$

$$\sigma = 35.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.20 \%$$

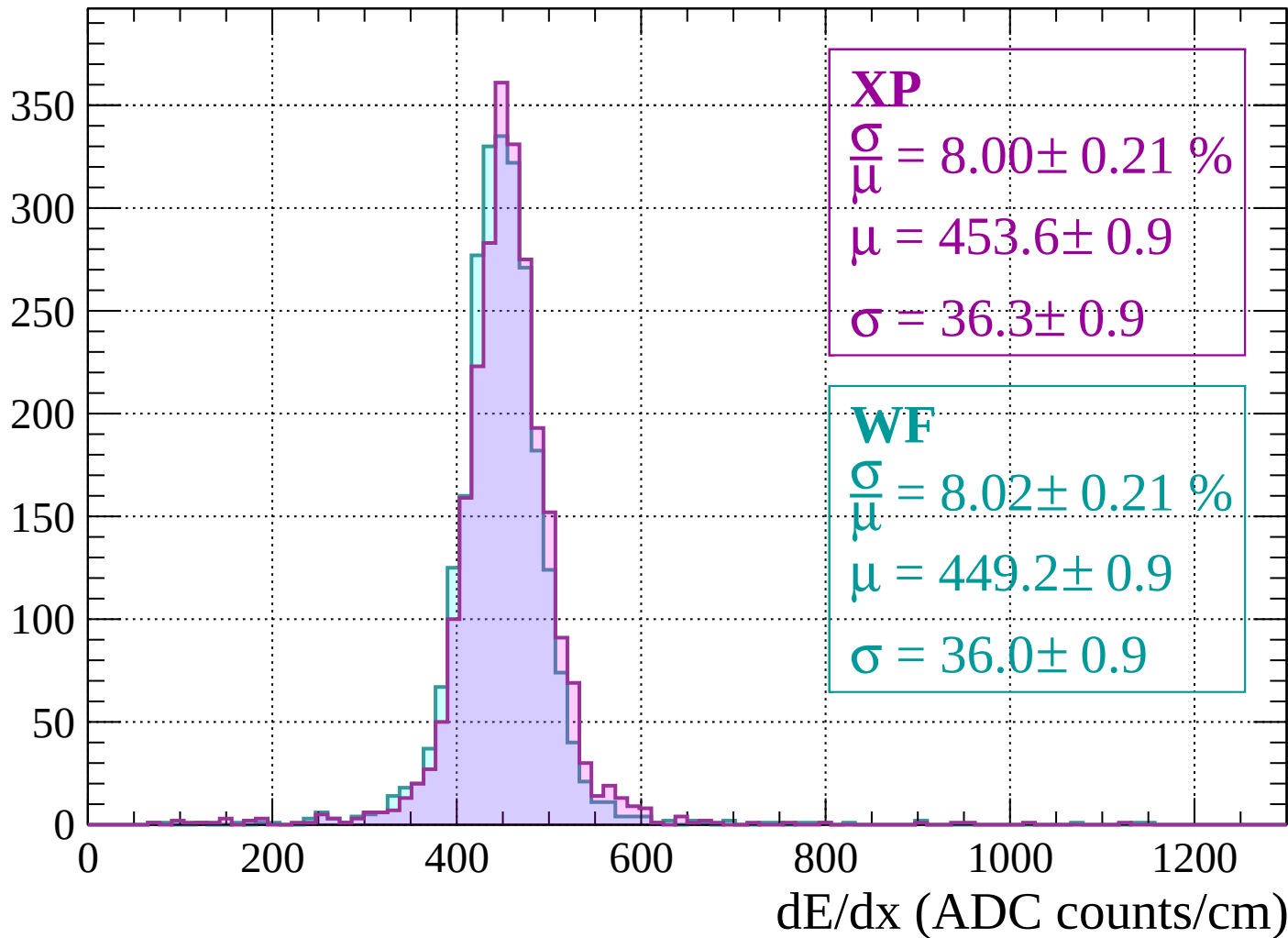
$$\mu = 449.6 \pm 0.8$$

$$\sigma = 35.1 \pm 0.8$$

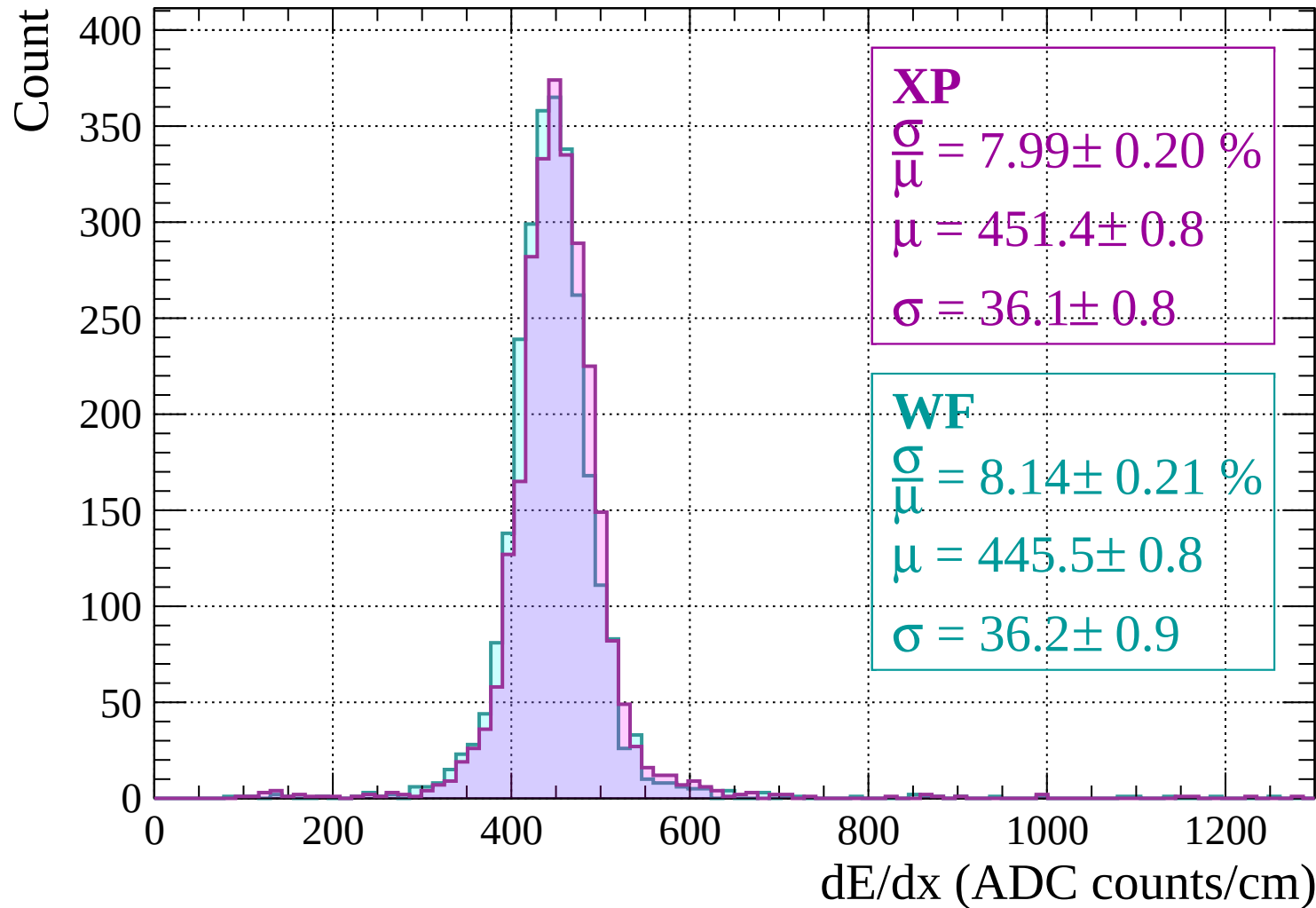
dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

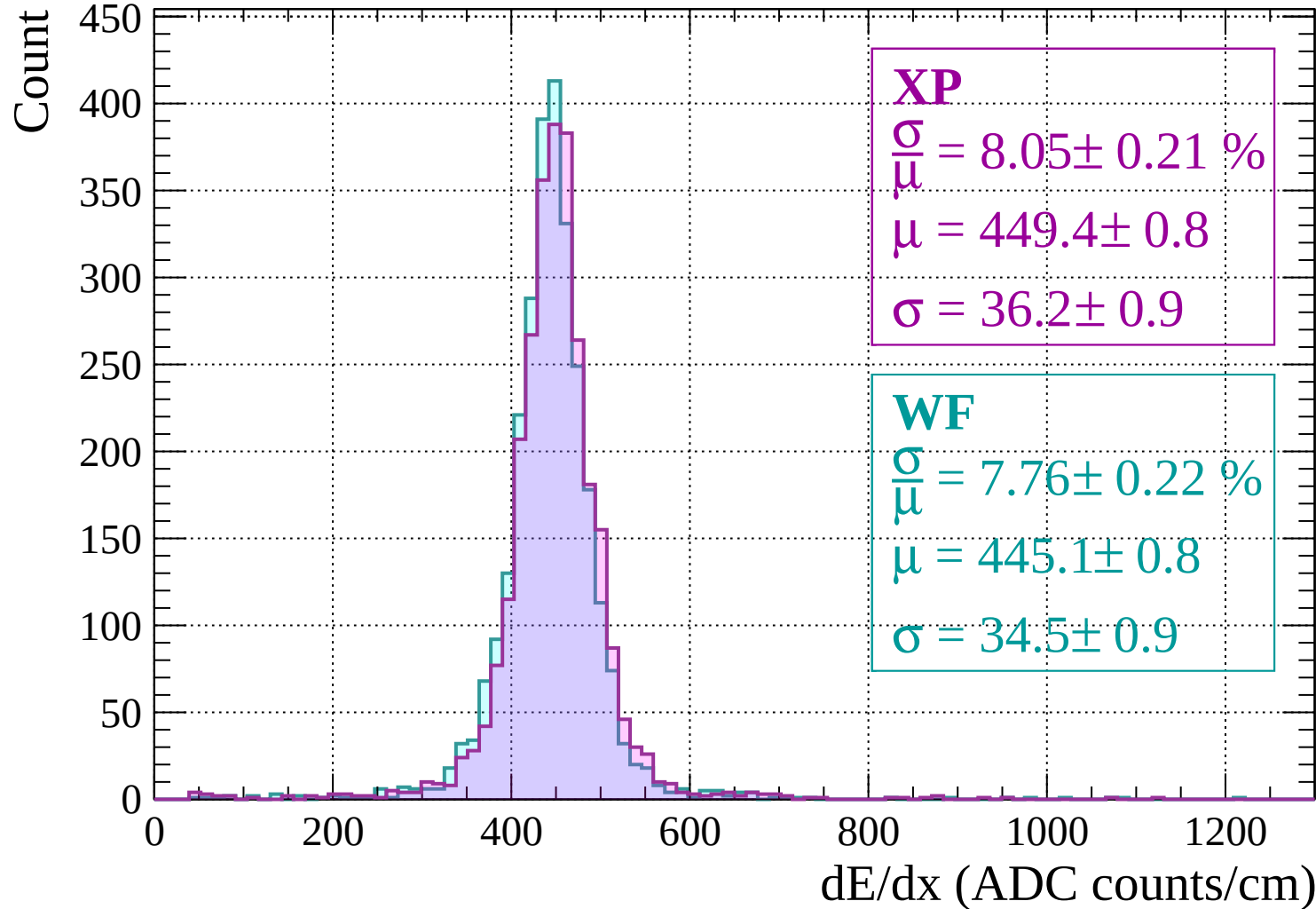
Count



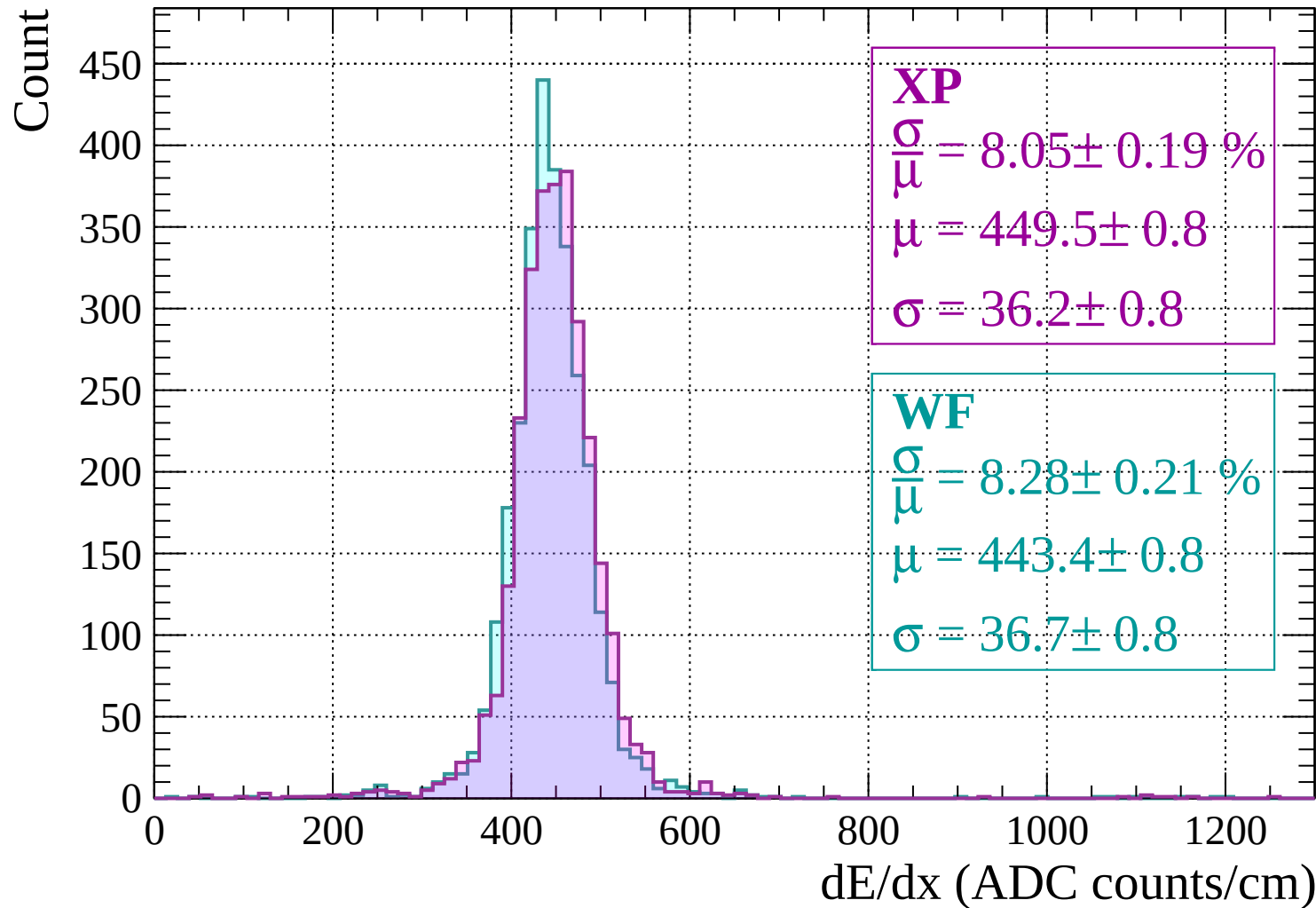
Energy loss | -1200 < p < -1160



Energy loss | -1160 < p < -1120

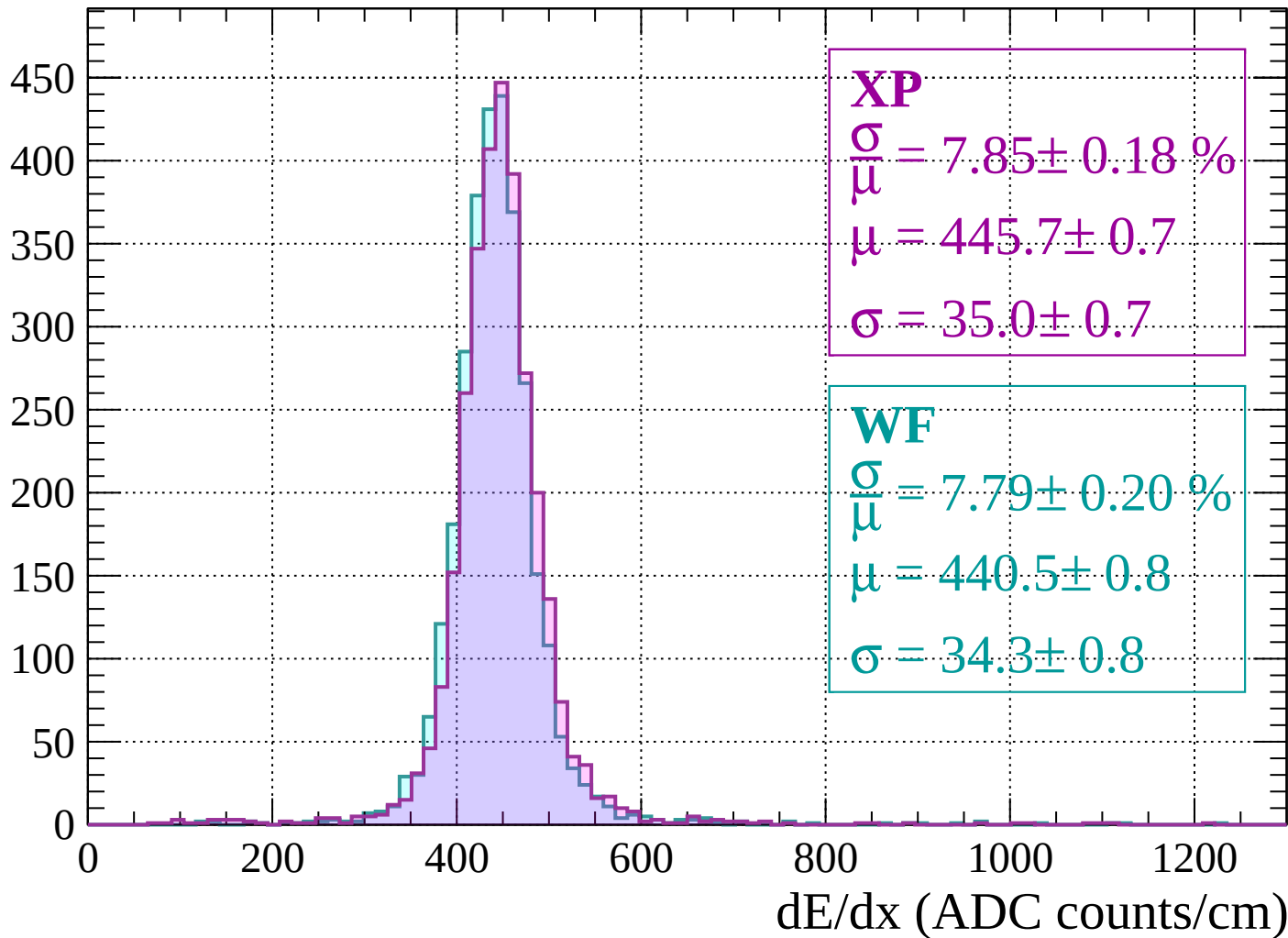


Energy loss | -1120 < p < -1080



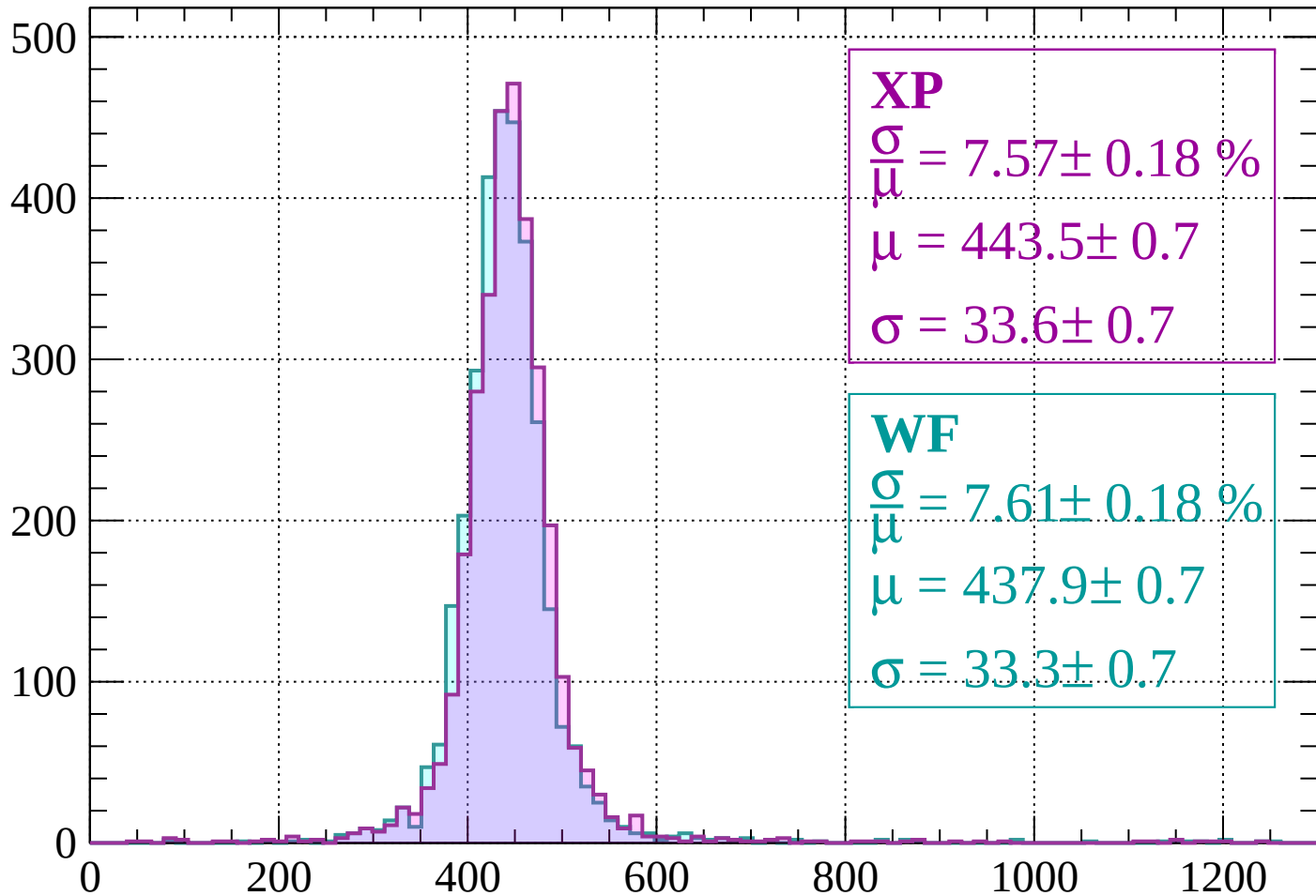
Energy loss | $-1080 < p < -1040$

Count



Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.18 \%$$

$$\mu = 443.5 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.61 \pm 0.18 \%$$

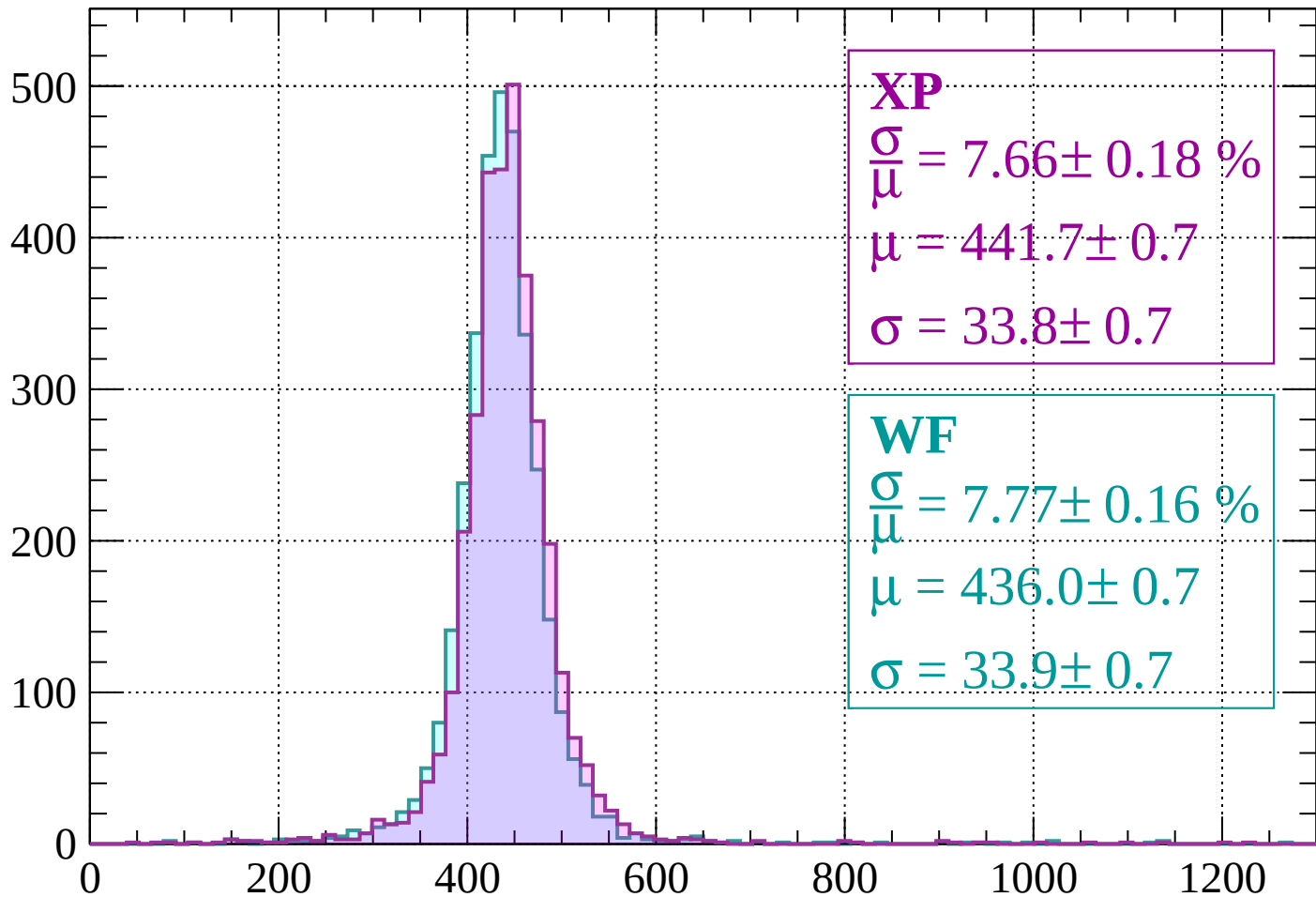
$$\mu = 437.9 \pm 0.7$$

$$\sigma = 33.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 7.66 \pm 0.18 \%$$

$$\mu = 441.7 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

WF

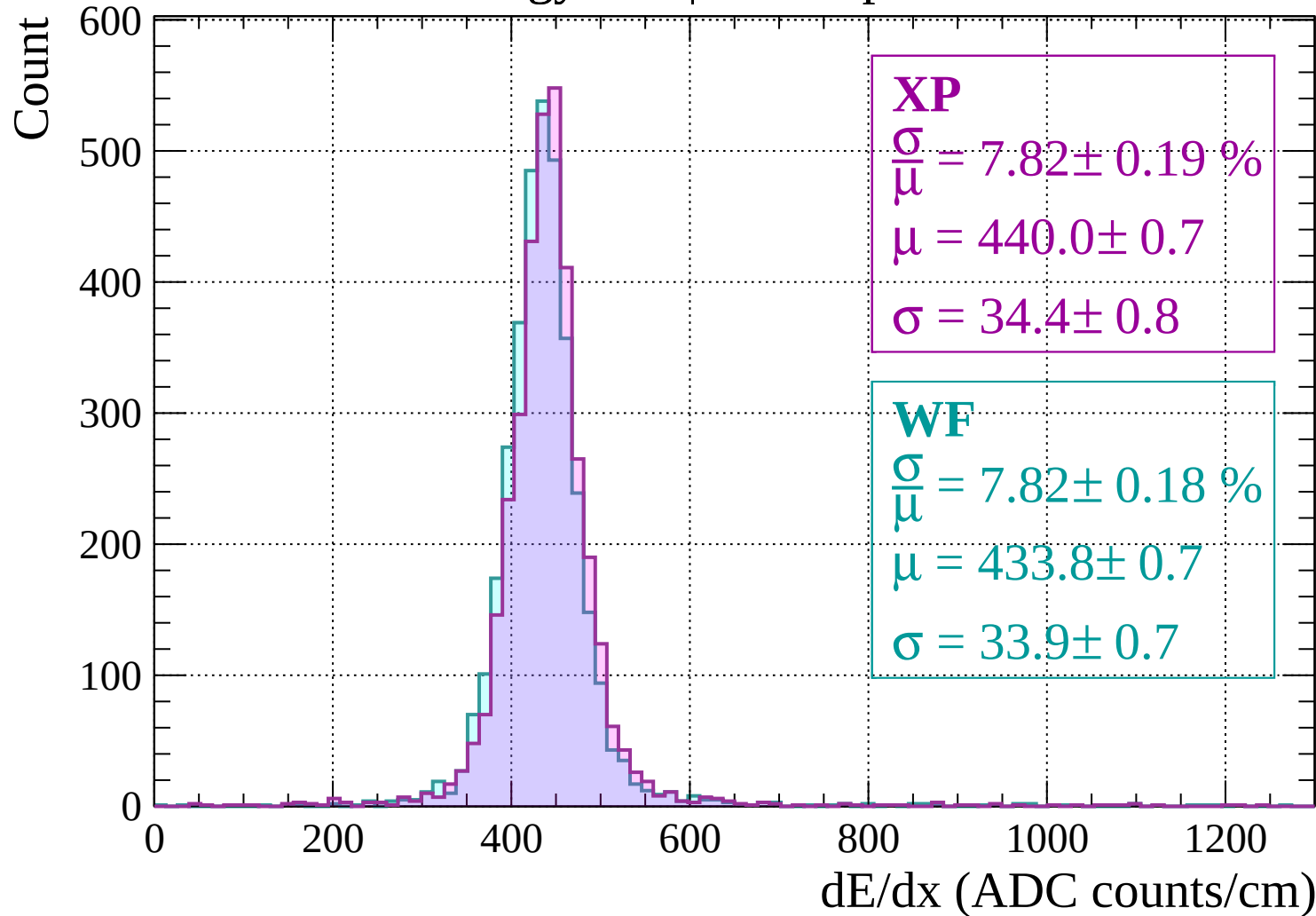
$$\frac{\sigma}{\mu} = 7.77 \pm 0.16 \%$$

$$\mu = 436.0 \pm 0.7$$

$$\sigma = 33.9 \pm 0.7$$

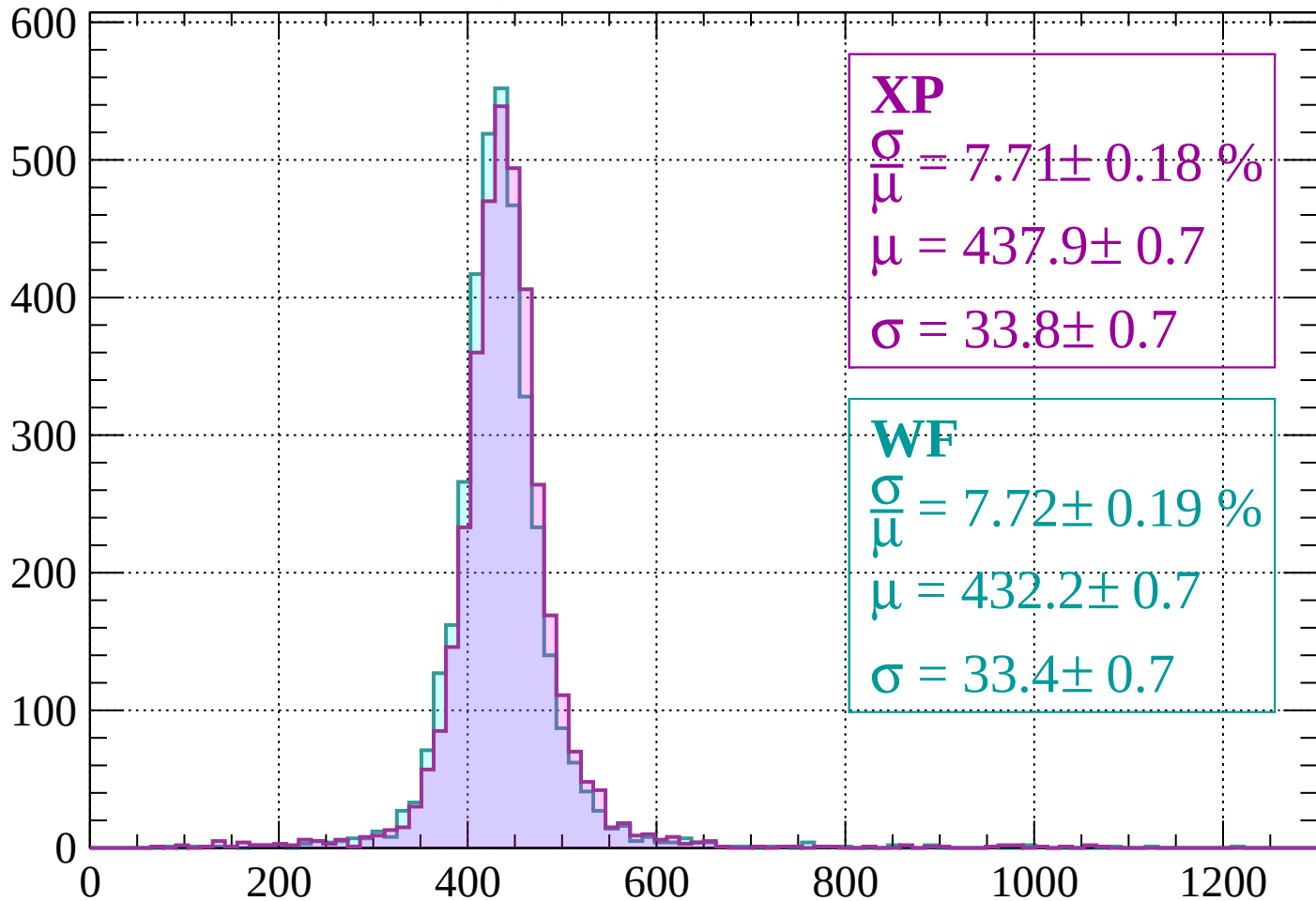
dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$



Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 7.71 \pm 0.18 \%$$

$$\mu = 437.9 \pm 0.7$$

$$\sigma = 33.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.72 \pm 0.19 \%$$

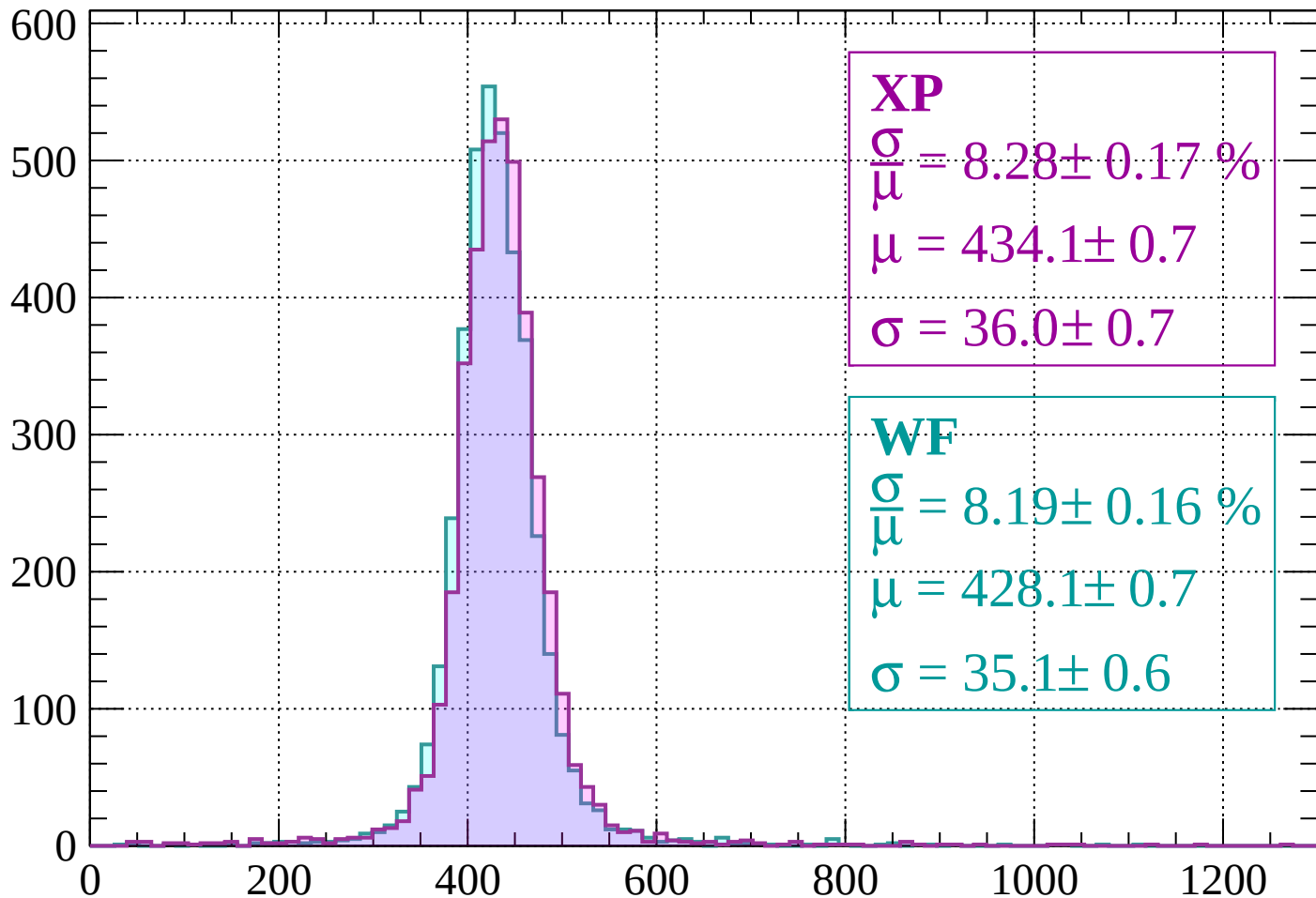
$$\mu = 432.2 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.17 \%$$

$$\mu = 434.1 \pm 0.7$$

$$\sigma = 36.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.19 \pm 0.16 \%$$

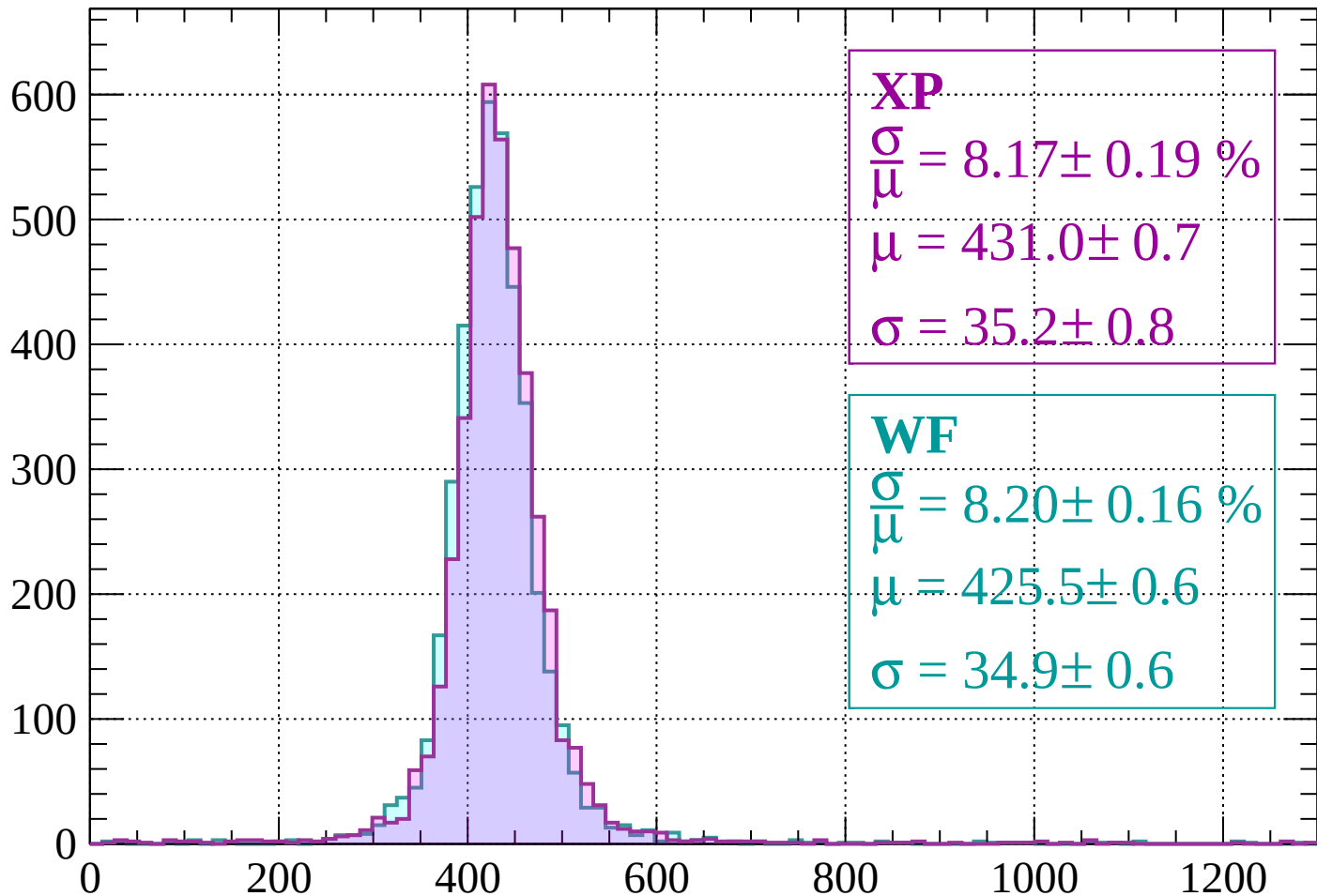
$$\mu = 428.1 \pm 0.7$$

$$\sigma = 35.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 8.17 \pm 0.19 \%$$

$$\mu = 431.0 \pm 0.7$$

$$\sigma = 35.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.16 \%$$

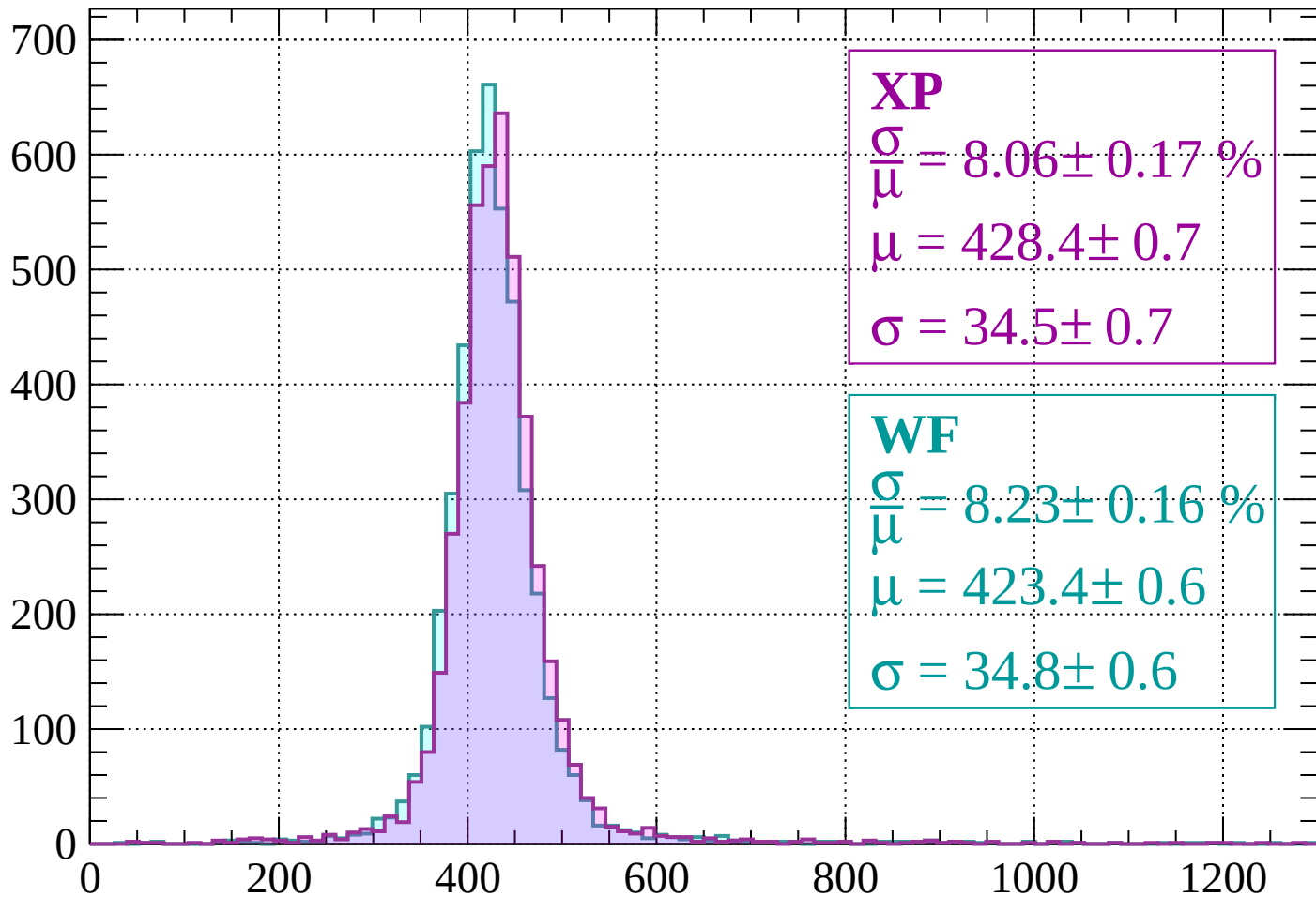
$$\mu = 425.5 \pm 0.6$$

$$\sigma = 34.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.06 \pm 0.17 \%$$

$$\mu = 428.4 \pm 0.7$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.23 \pm 0.16 \%$$

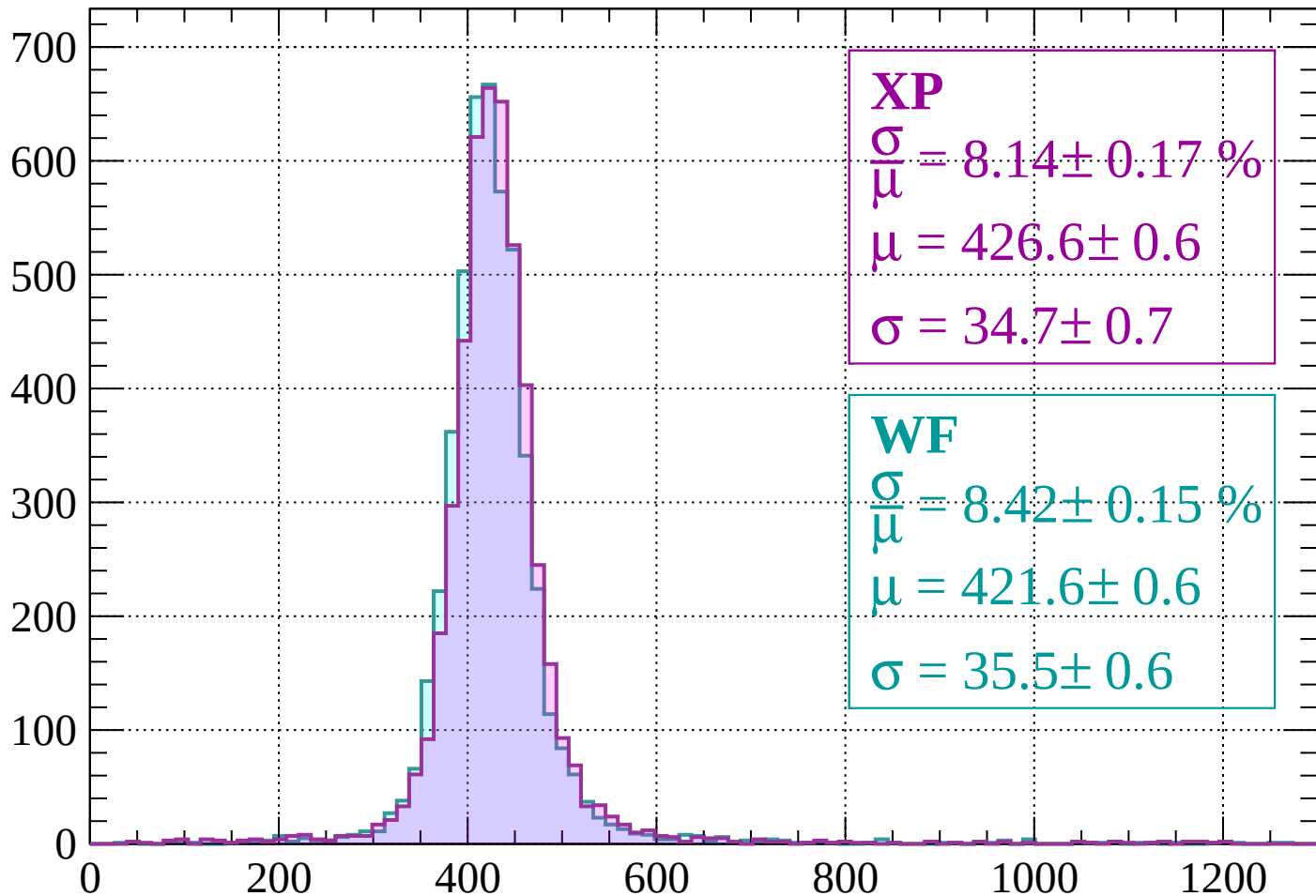
$$\mu = 423.4 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.17 \%$$

$$\mu = 426.6 \pm 0.6$$

$$\sigma = 34.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.42 \pm 0.15 \%$$

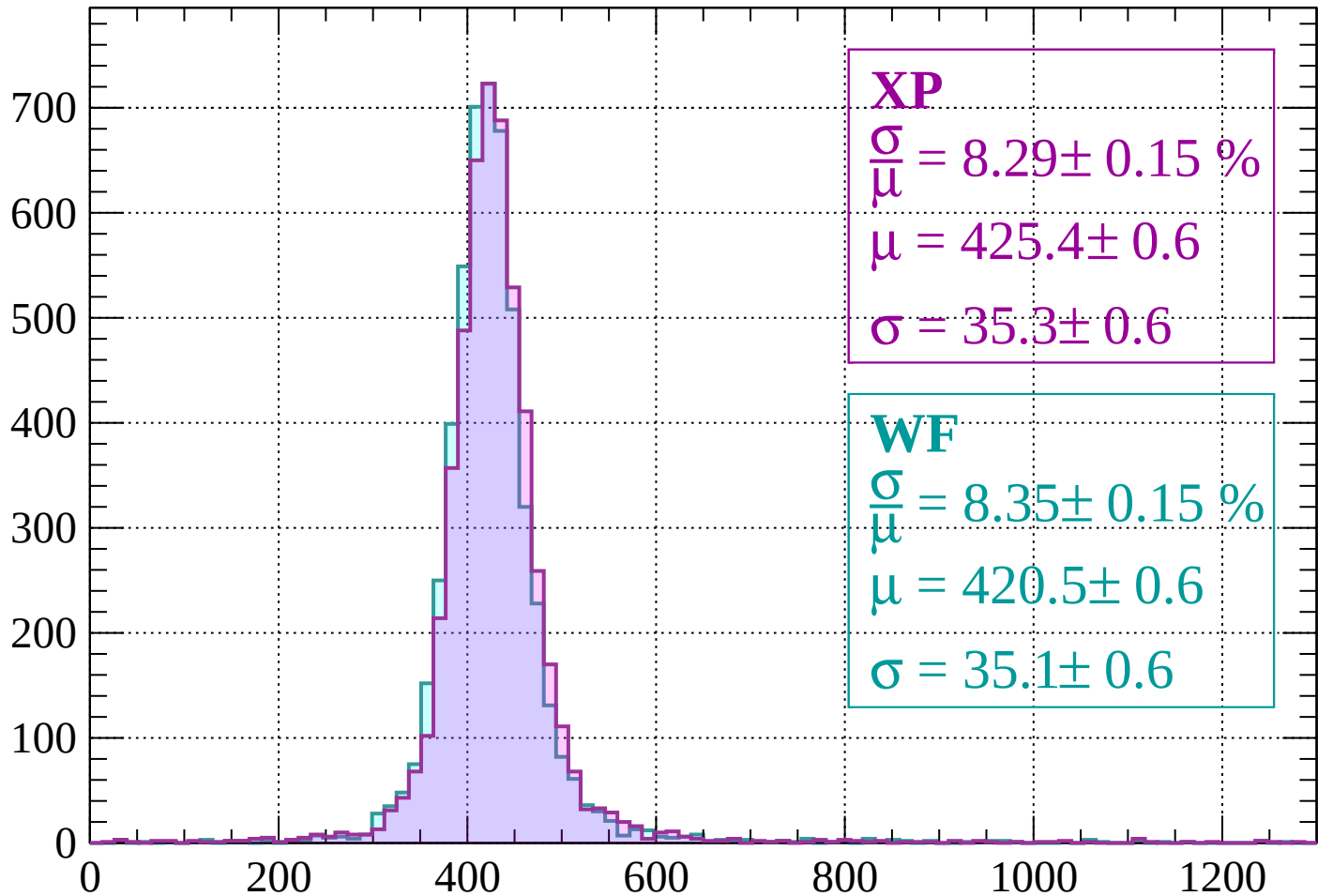
$$\mu = 421.6 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.15 \%$$

$$\mu = 425.4 \pm 0.6$$

$$\sigma = 35.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.35 \pm 0.15 \%$$

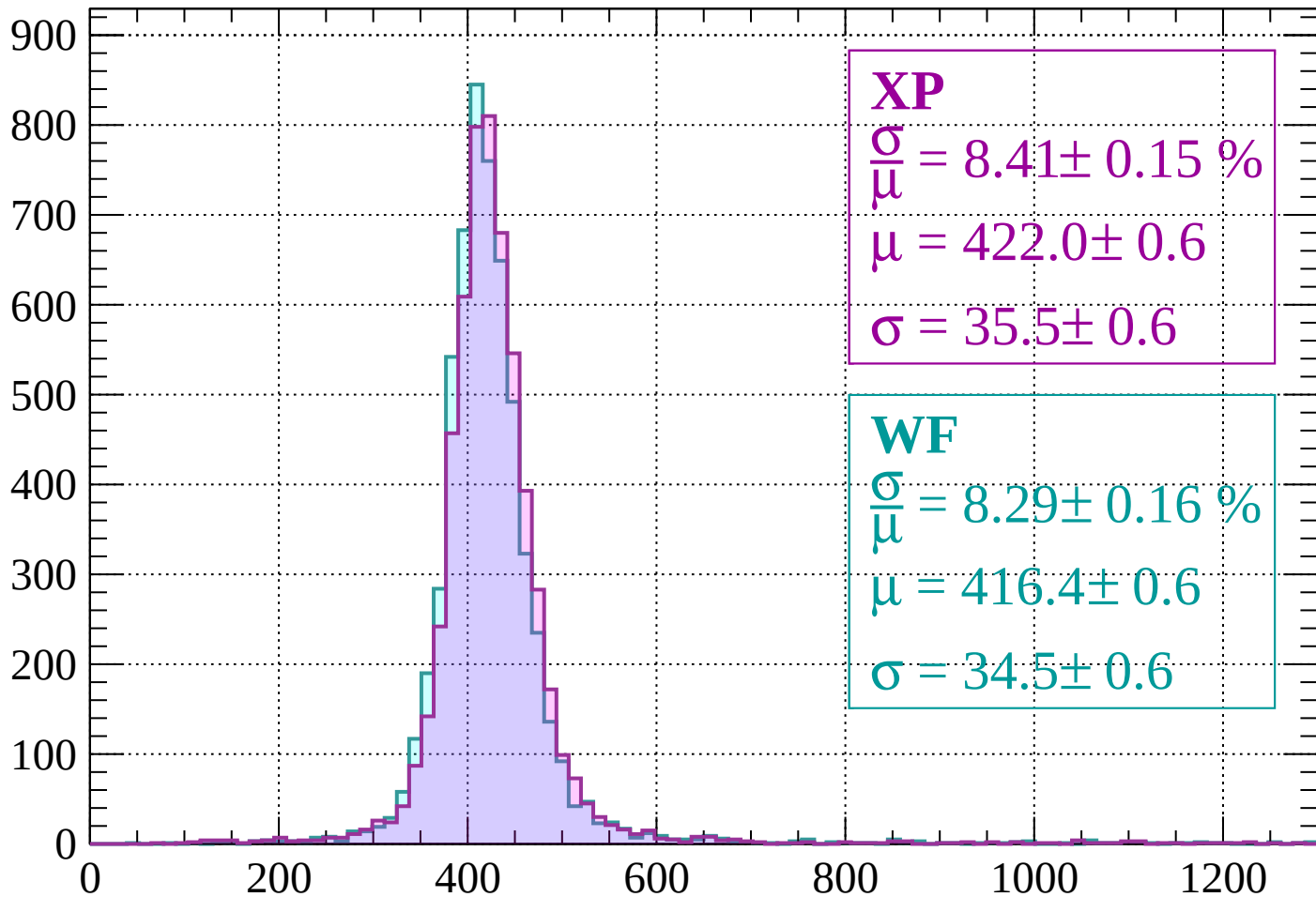
$$\mu = 420.5 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.15 \%$$

$$\mu = 422.0 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.29 \pm 0.16 \%$$

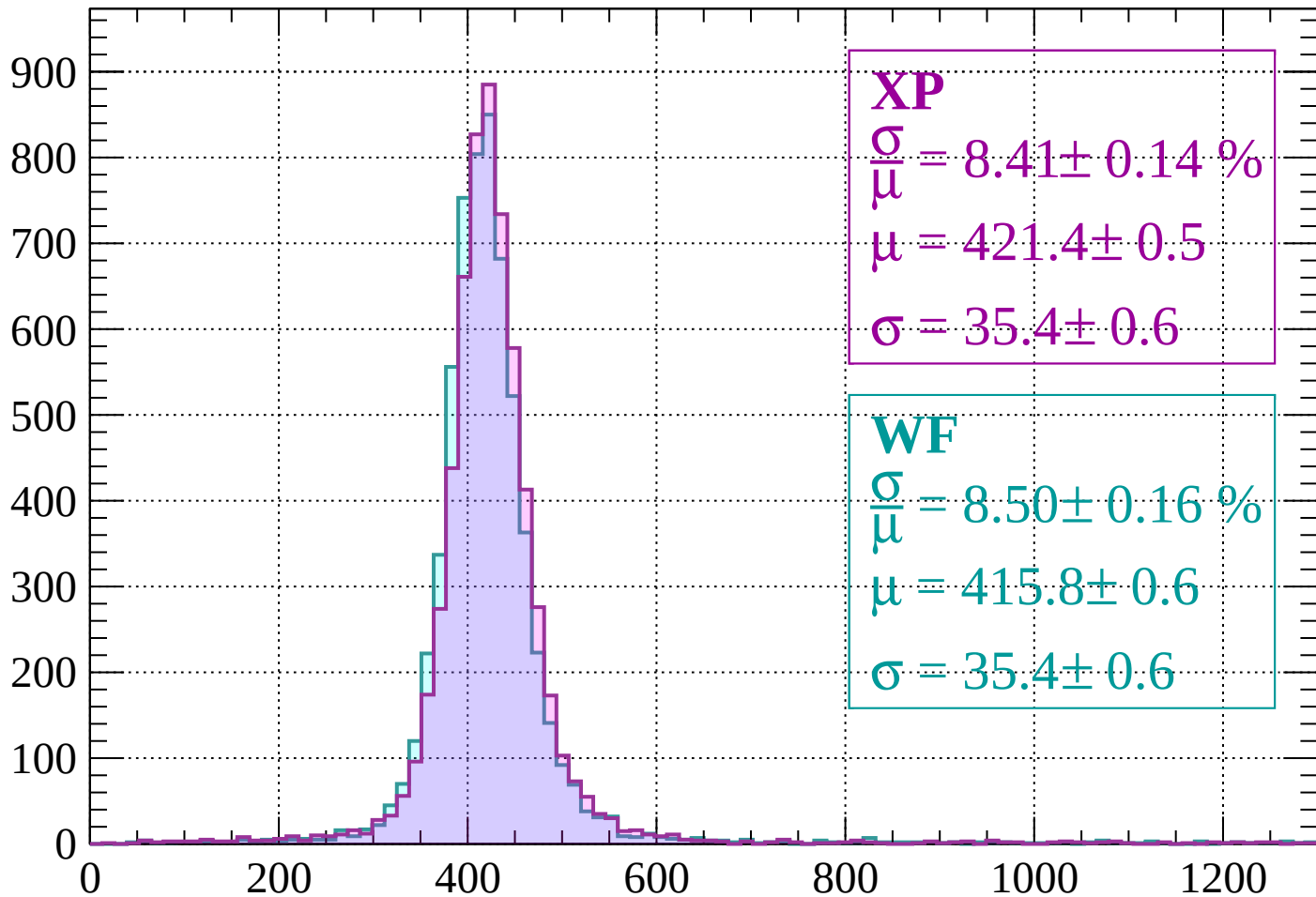
$$\mu = 416.4 \pm 0.6$$

$$\sigma = 34.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 8.41 \pm 0.14 \%$$

$$\mu = 421.4 \pm 0.5$$

$$\sigma = 35.4 \pm 0.6$$

WF

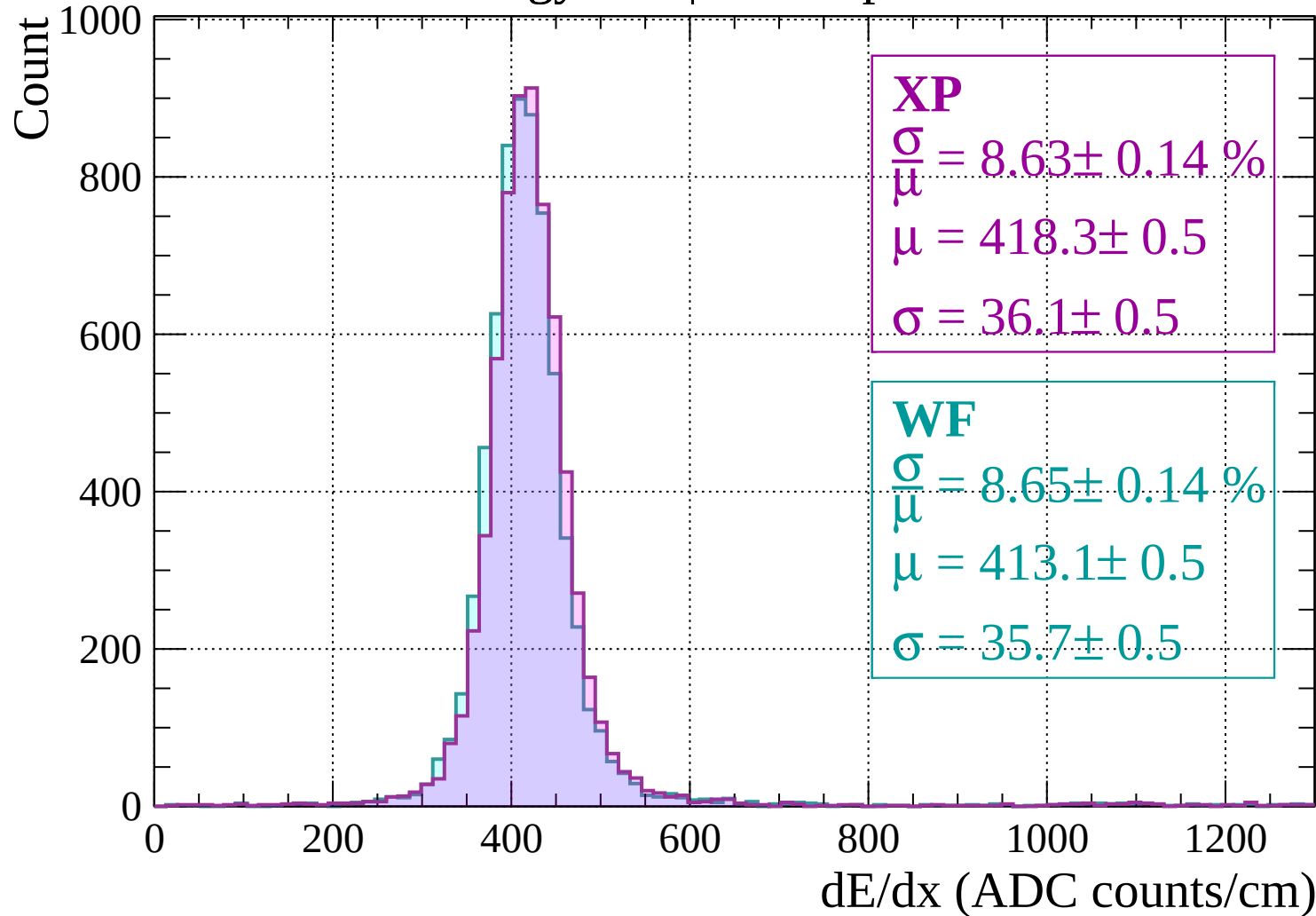
$$\frac{\sigma}{\mu} = 8.50 \pm 0.16 \%$$

$$\mu = 415.8 \pm 0.6$$

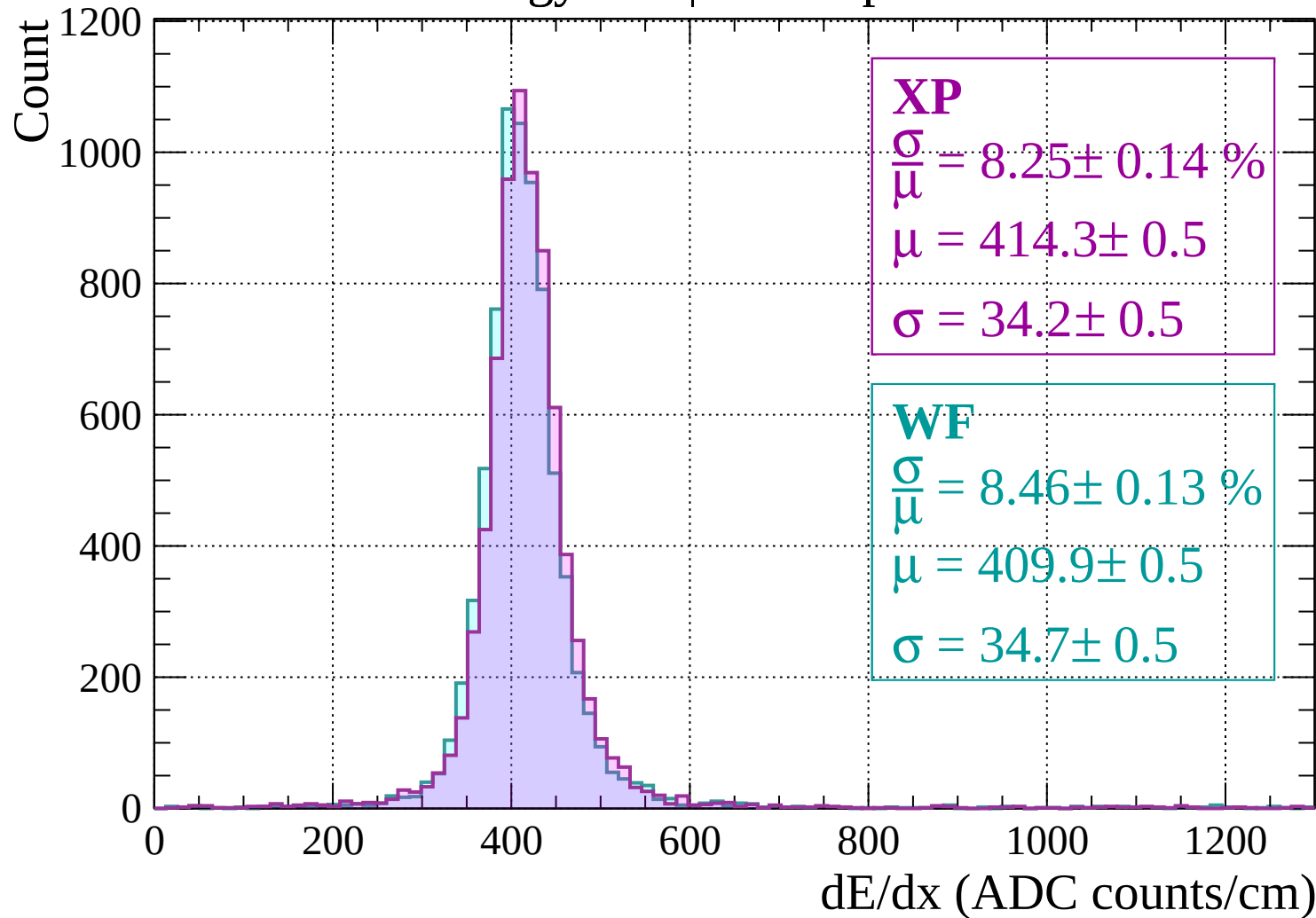
$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

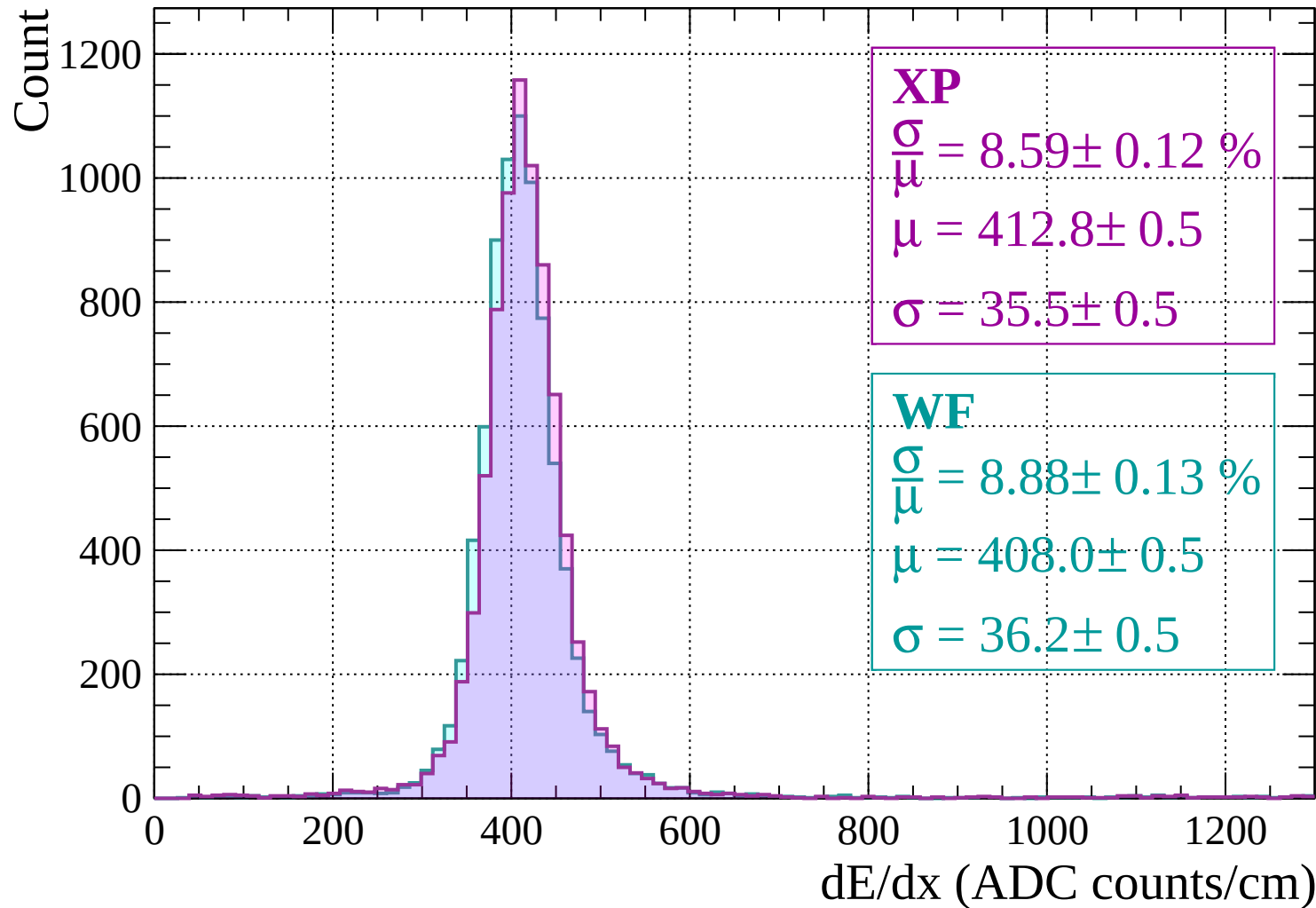
Energy loss | $-600 < p < -560$



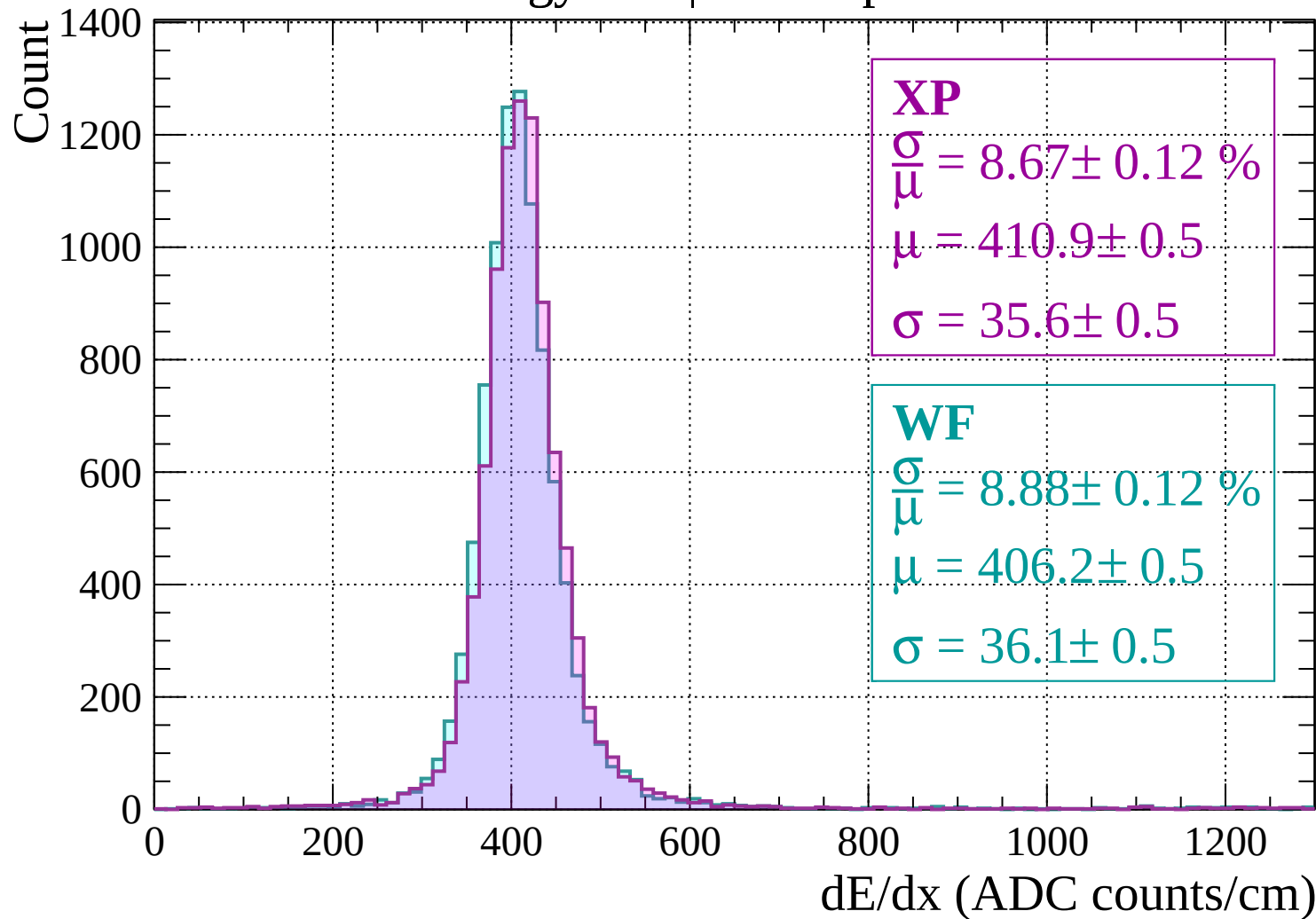
Energy loss | $-560 < p < -520$



Energy loss | $-520 < p < -480$

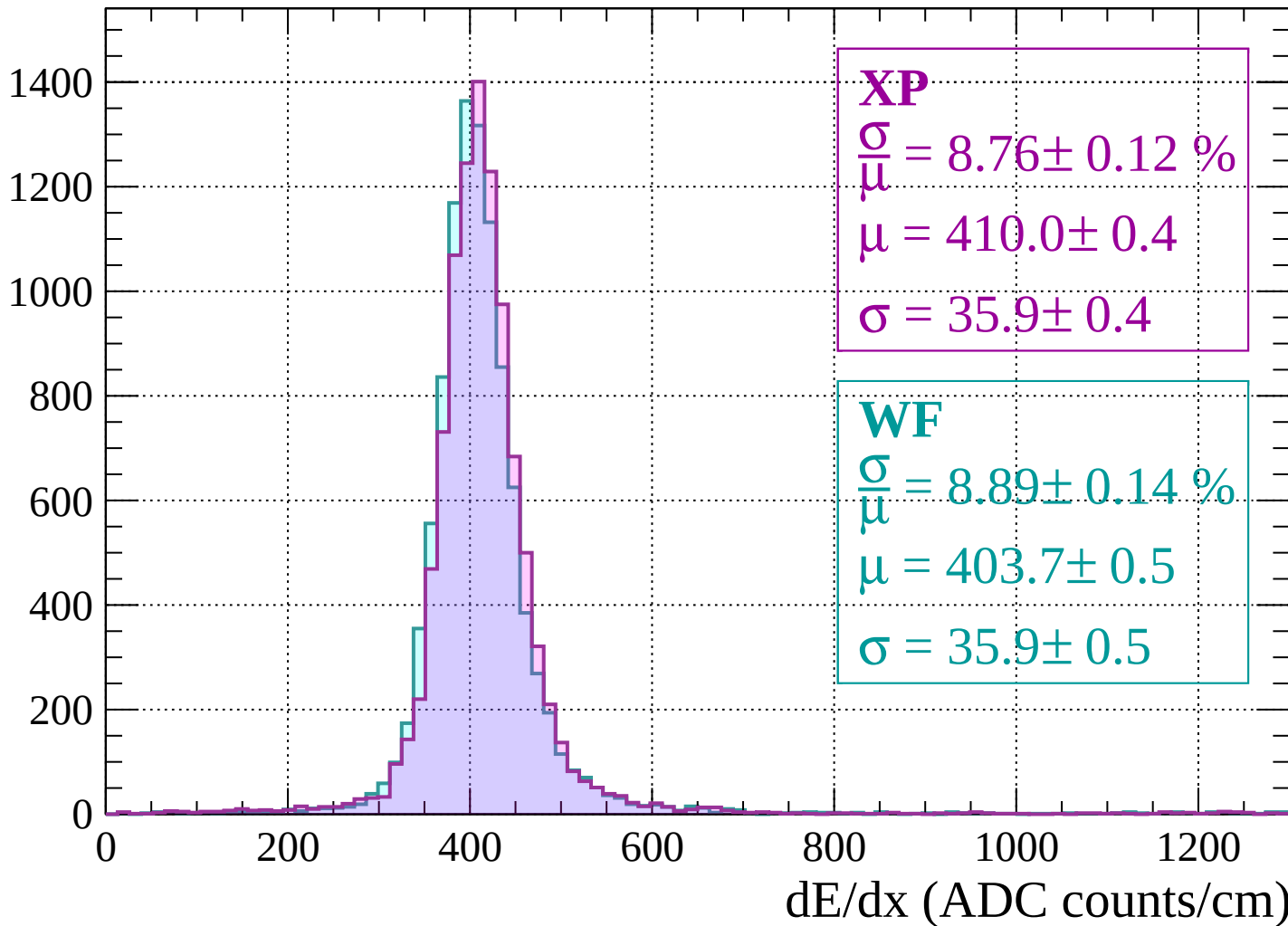


Energy loss | $-480 < p < -440$

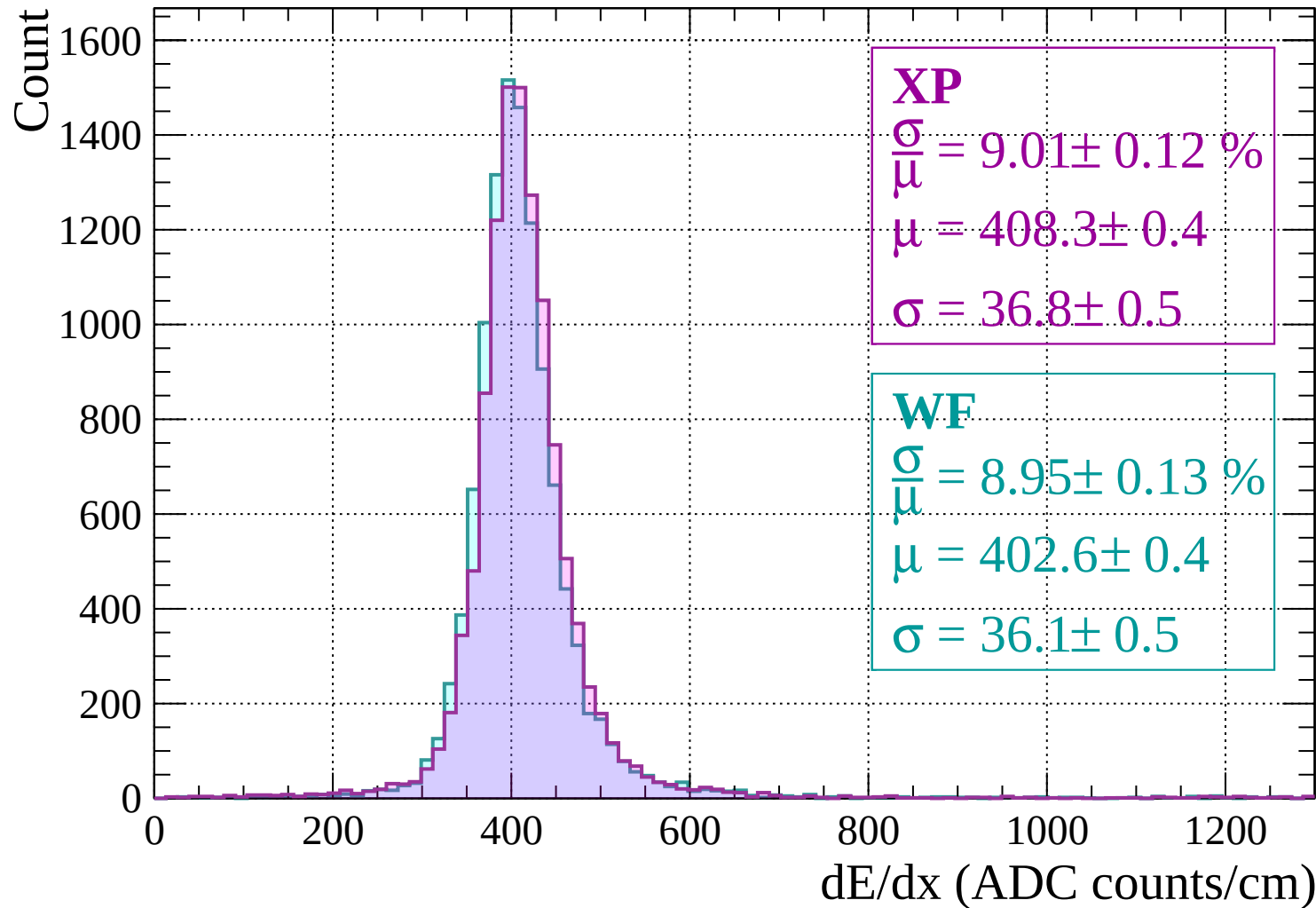


Energy loss | $-440 < p < -400$

Count



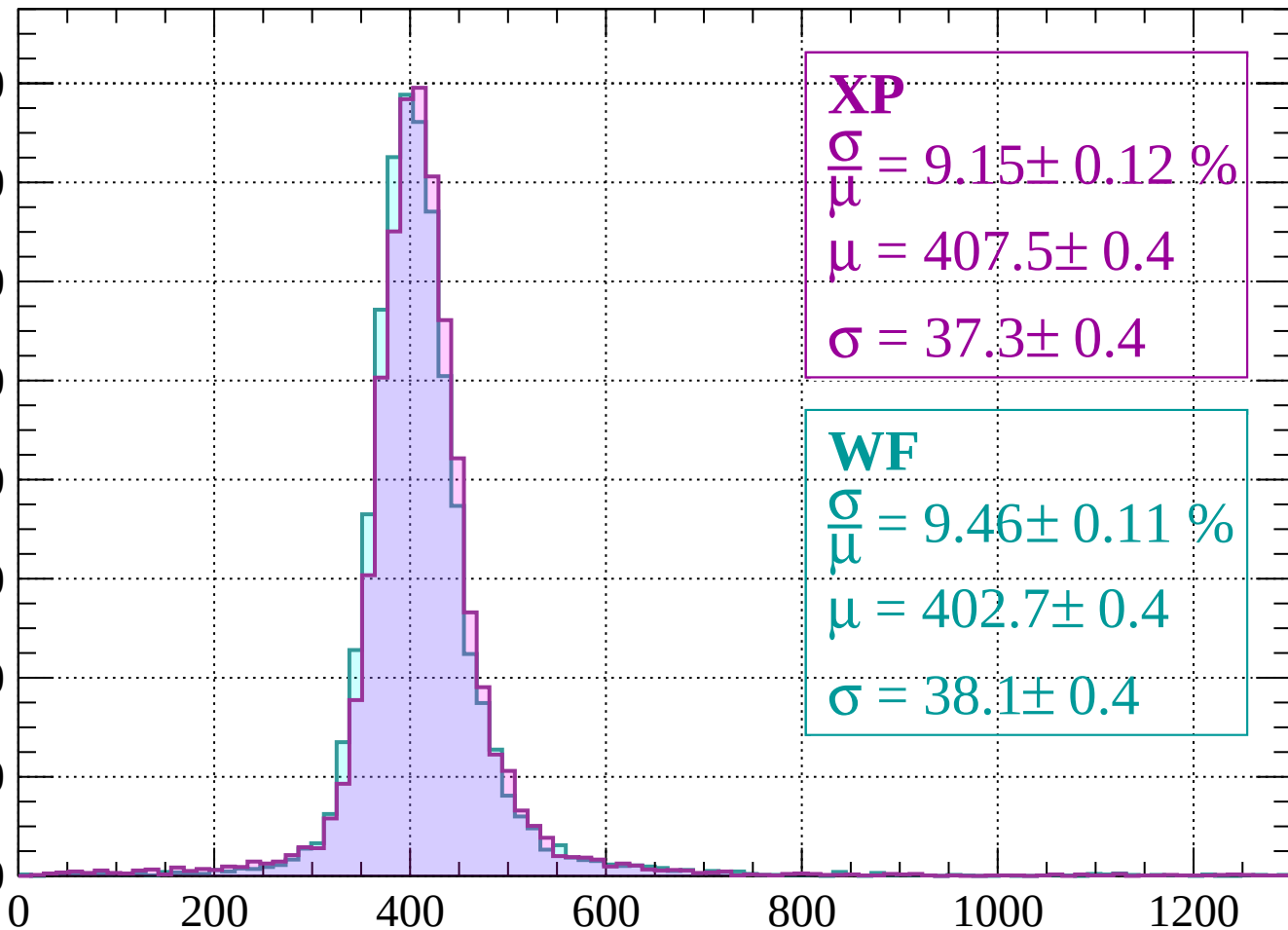
Energy loss | $-400 < p < -360$



Energy loss | $-360 < p < -320$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.12 \%$$

$$\mu = 407.5 \pm 0.4$$

$$\sigma = 37.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 9.46 \pm 0.11 \%$$

$$\mu = 402.7 \pm 0.4$$

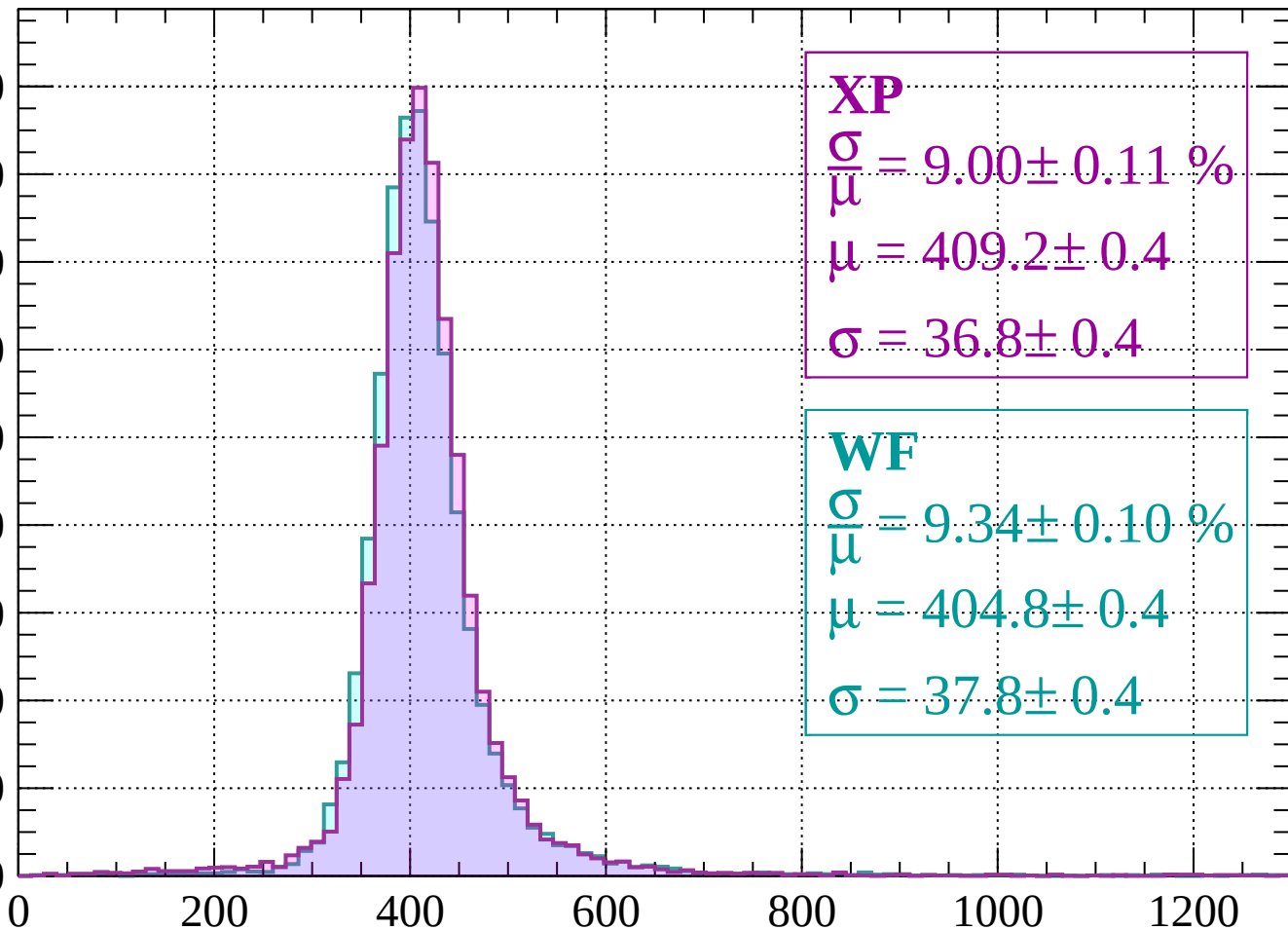
$$\sigma = 38.1 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count

1800
1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 9.00 \pm 0.11 \%$$

$$\mu = 409.2 \pm 0.4$$

$$\sigma = 36.8 \pm 0.4$$

WF

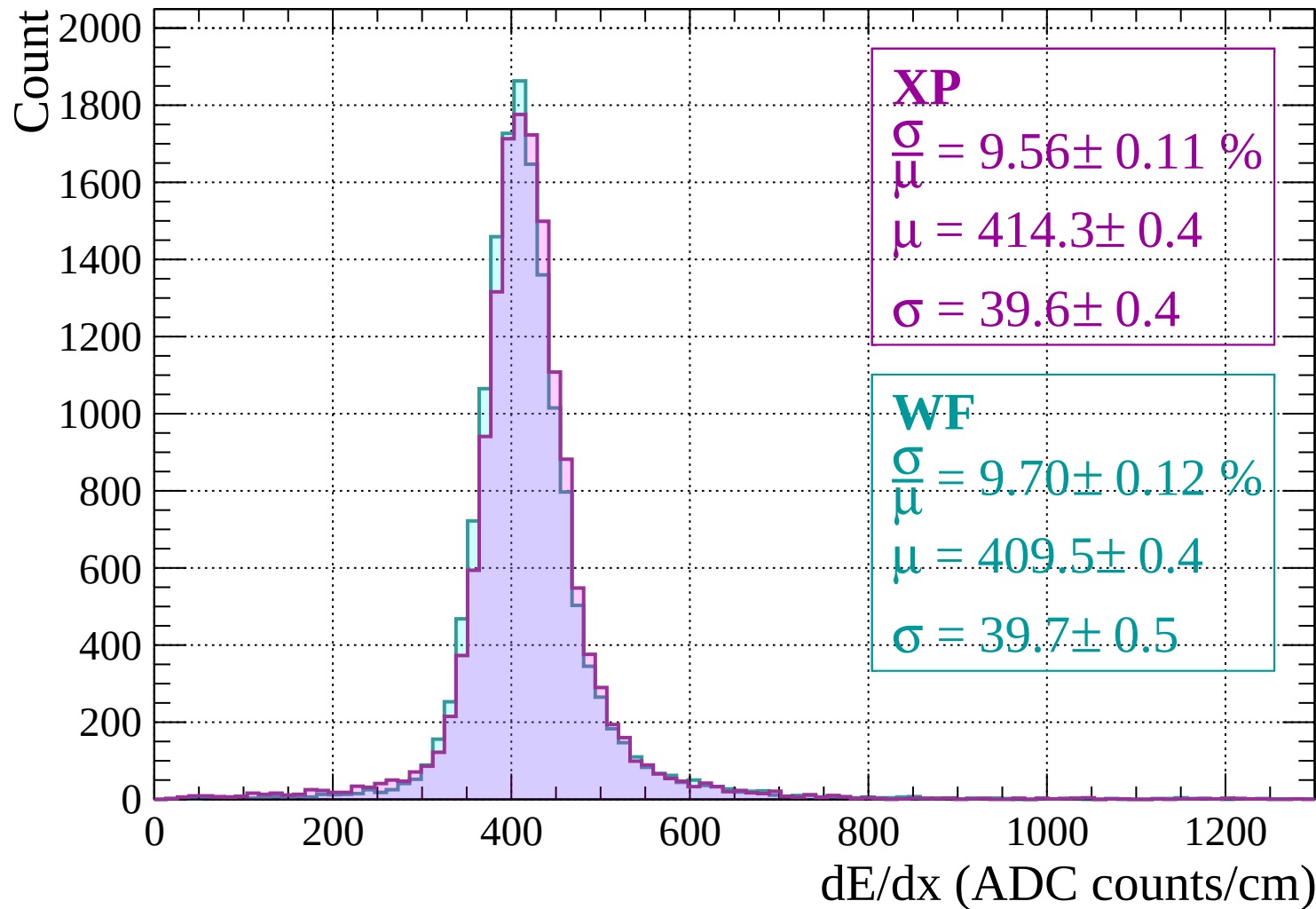
$$\frac{\sigma}{\mu} = 9.34 \pm 0.10 \%$$

$$\mu = 404.8 \pm 0.4$$

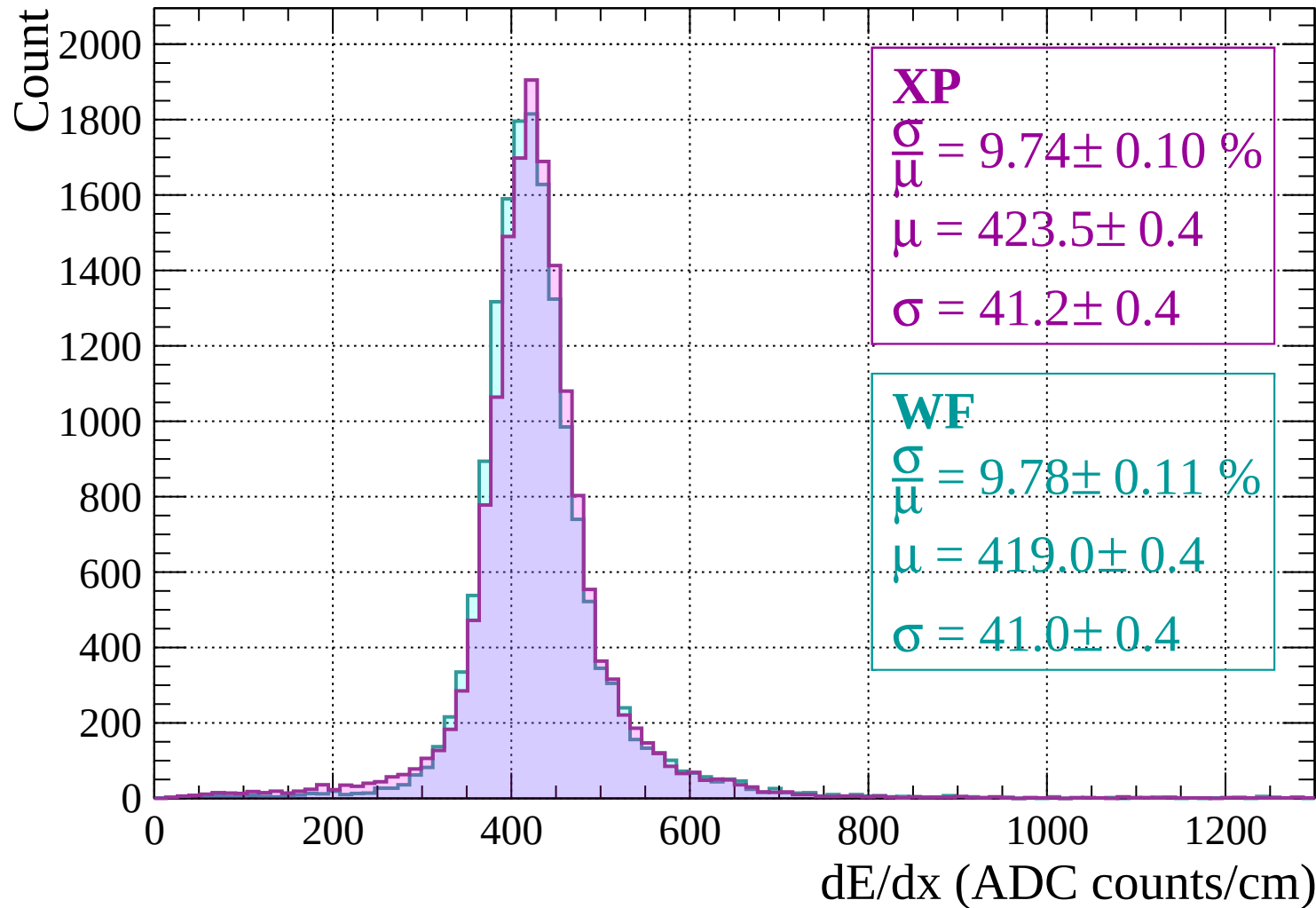
$$\sigma = 37.8 \pm 0.4$$

dE/dx (ADC counts/cm)

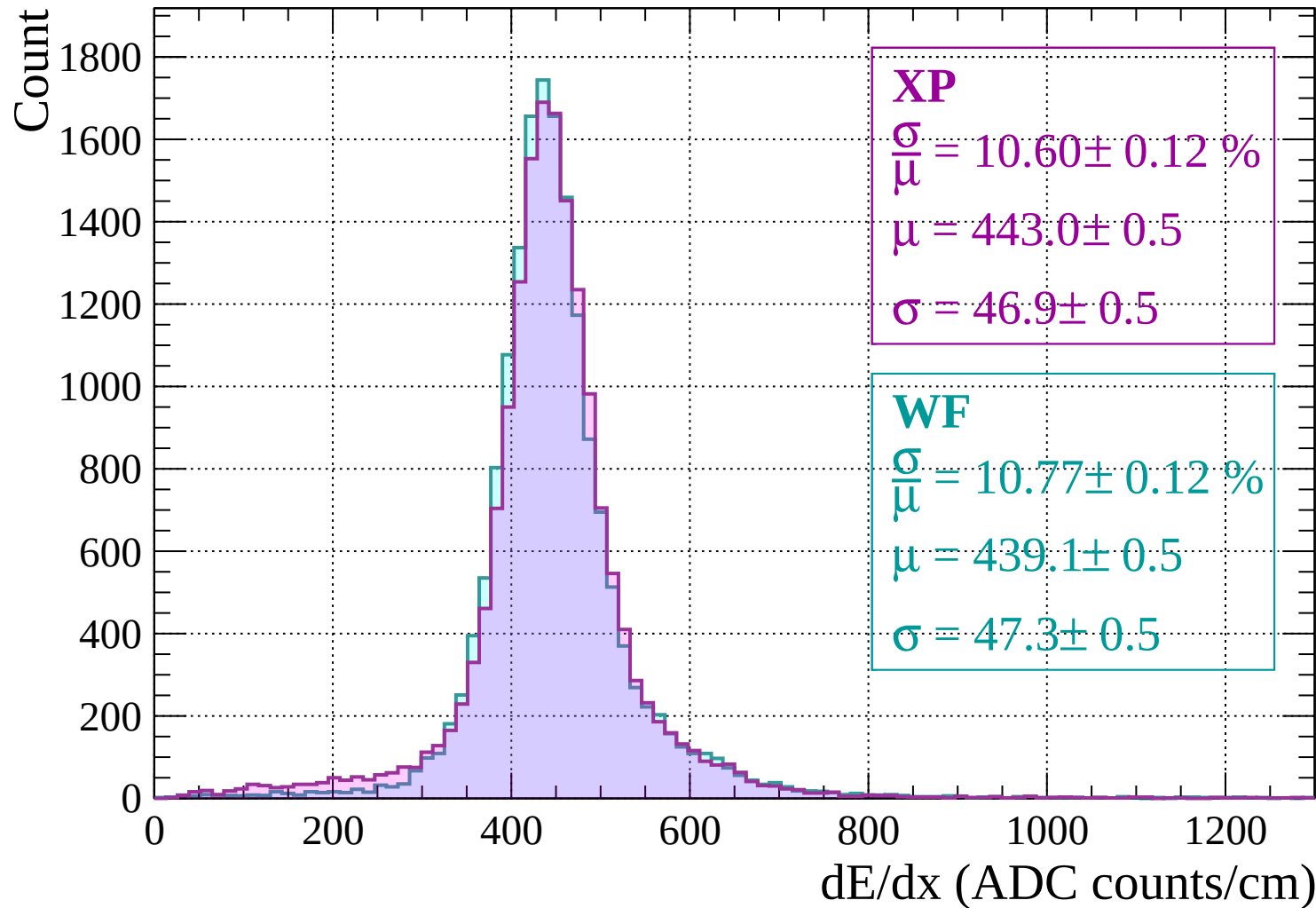
Energy loss | $-280 < p < -240$



Energy loss | $-240 < p < -200$



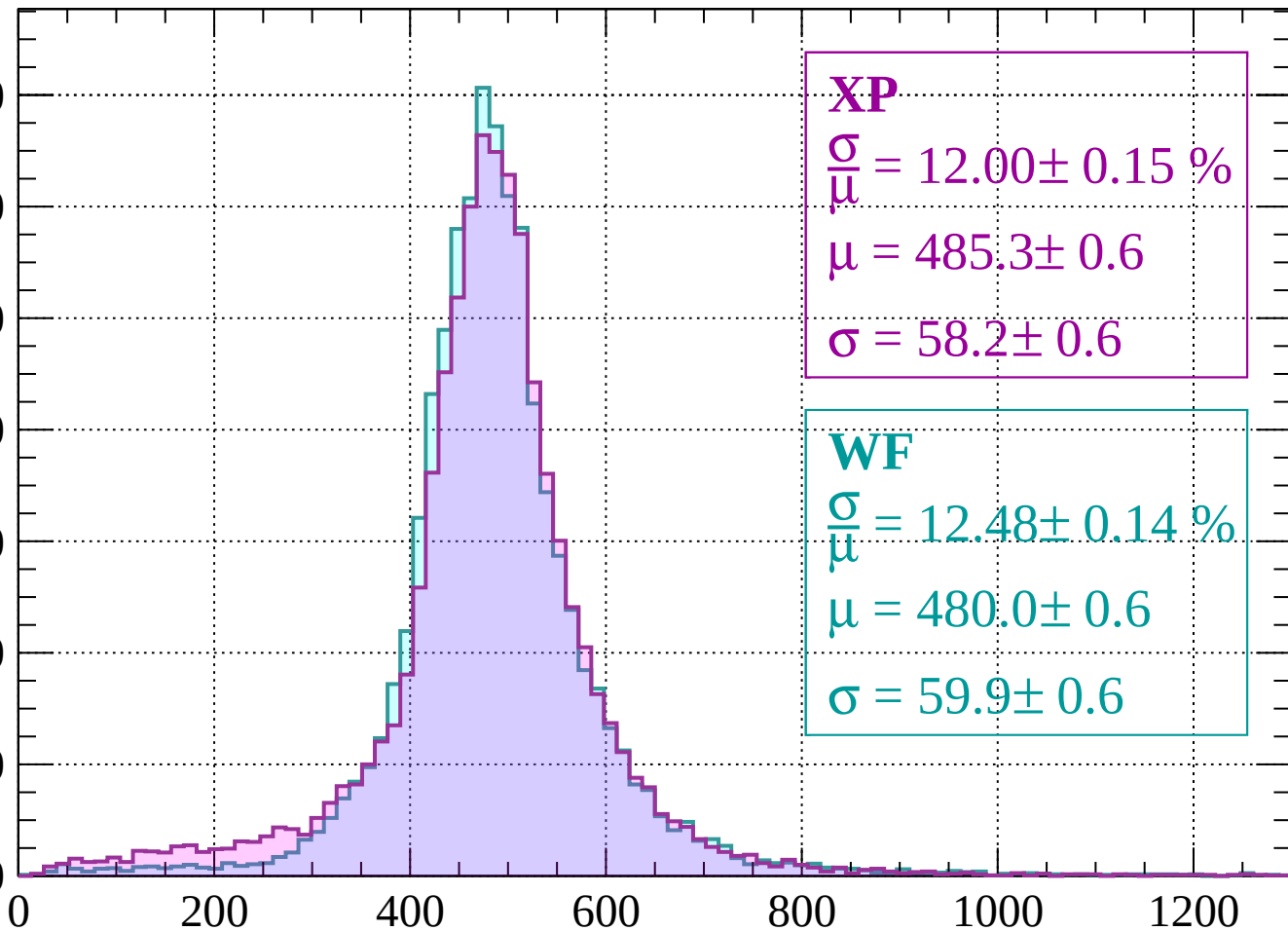
Energy loss | $-200 < p < -160$



Energy loss | -160 < p < -120

Count

1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$$

$$\mu = 485.3 \pm 0.6$$

$$\sigma = 58.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 12.48 \pm 0.14 \%$$

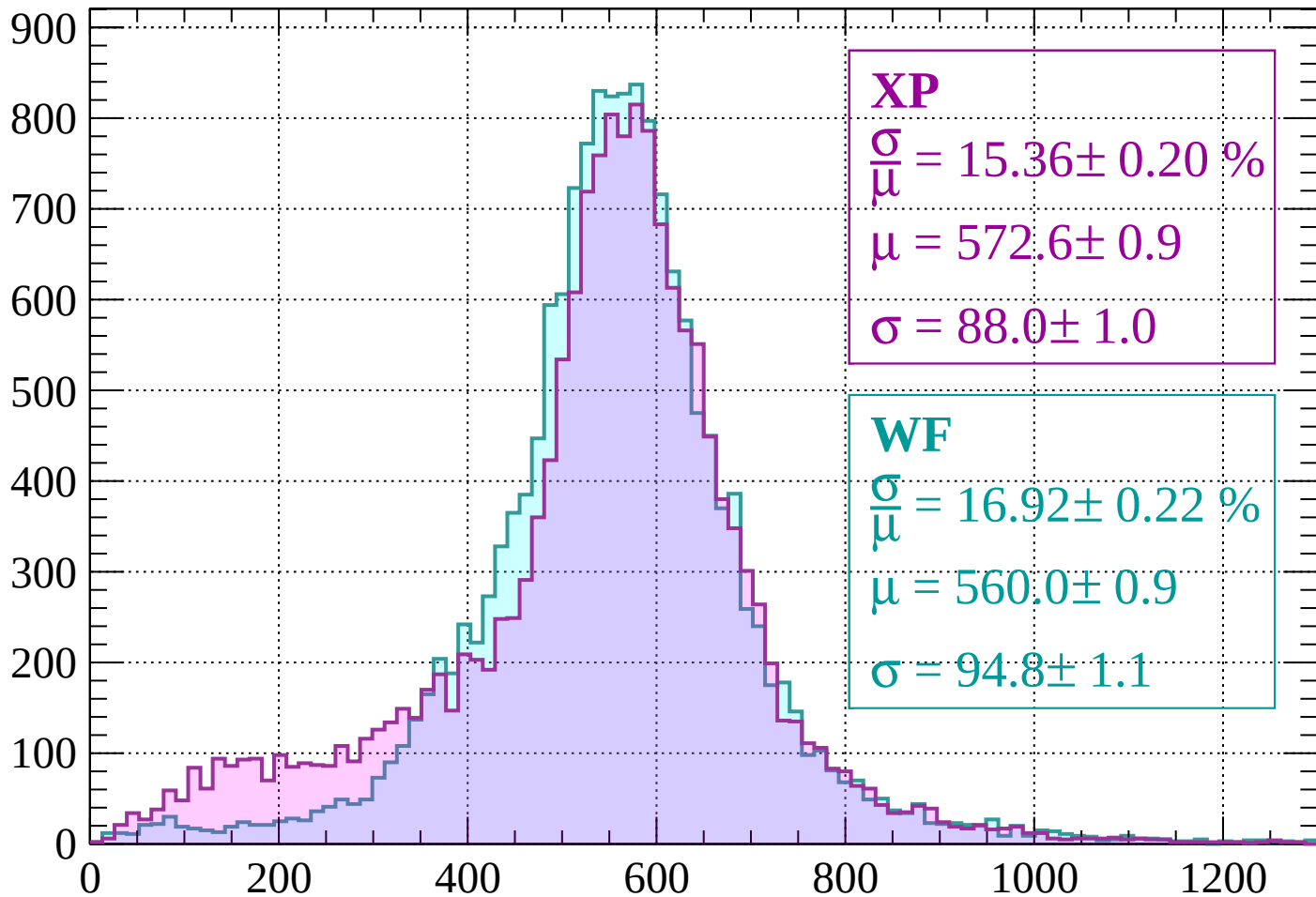
$$\mu = 480.0 \pm 0.6$$

$$\sigma = 59.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count



XP

$$\frac{\sigma}{\mu} = 15.36 \pm 0.20 \%$$

$$\mu = 572.6 \pm 0.9$$

$$\sigma = 88.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 16.92 \pm 0.22 \%$$

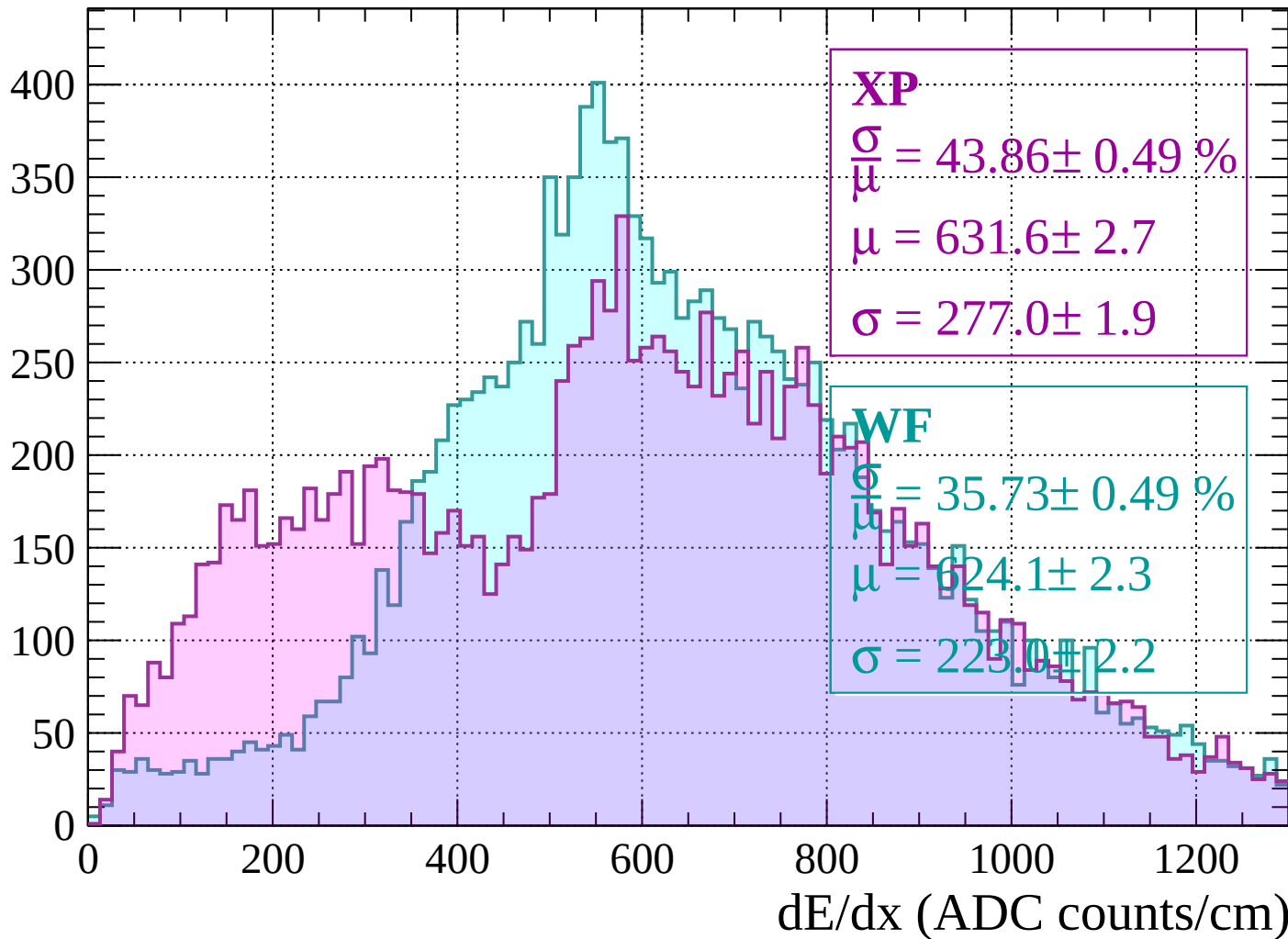
$$\mu = 560.0 \pm 0.9$$

$$\sigma = 94.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-80 < p < -40$

Count



Energy loss | $-40 < p < 0$

Count

1400
1200
1000
800
600
400
200
0

XP

$$\frac{\sigma}{\mu} = 45.11 \pm 0.43 \%$$

$$\mu = 407.8 \pm 1.6$$

$$\sigma = 184.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 15.45 \pm 0.17 \%$$

$$\mu = 518.3 \pm 0.7$$

$$\sigma = 80.1 \pm 0.8$$

0

200

400

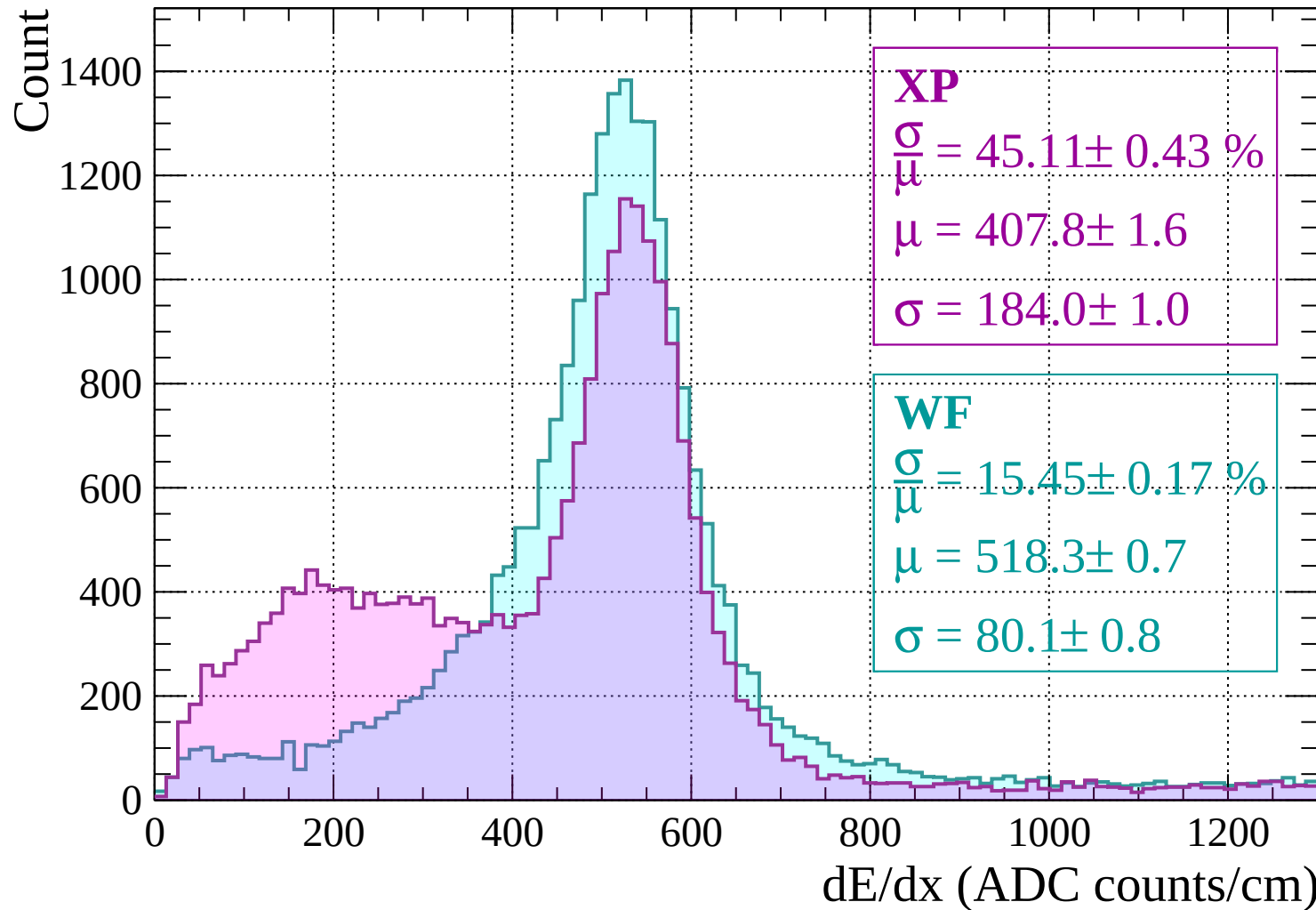
600

800

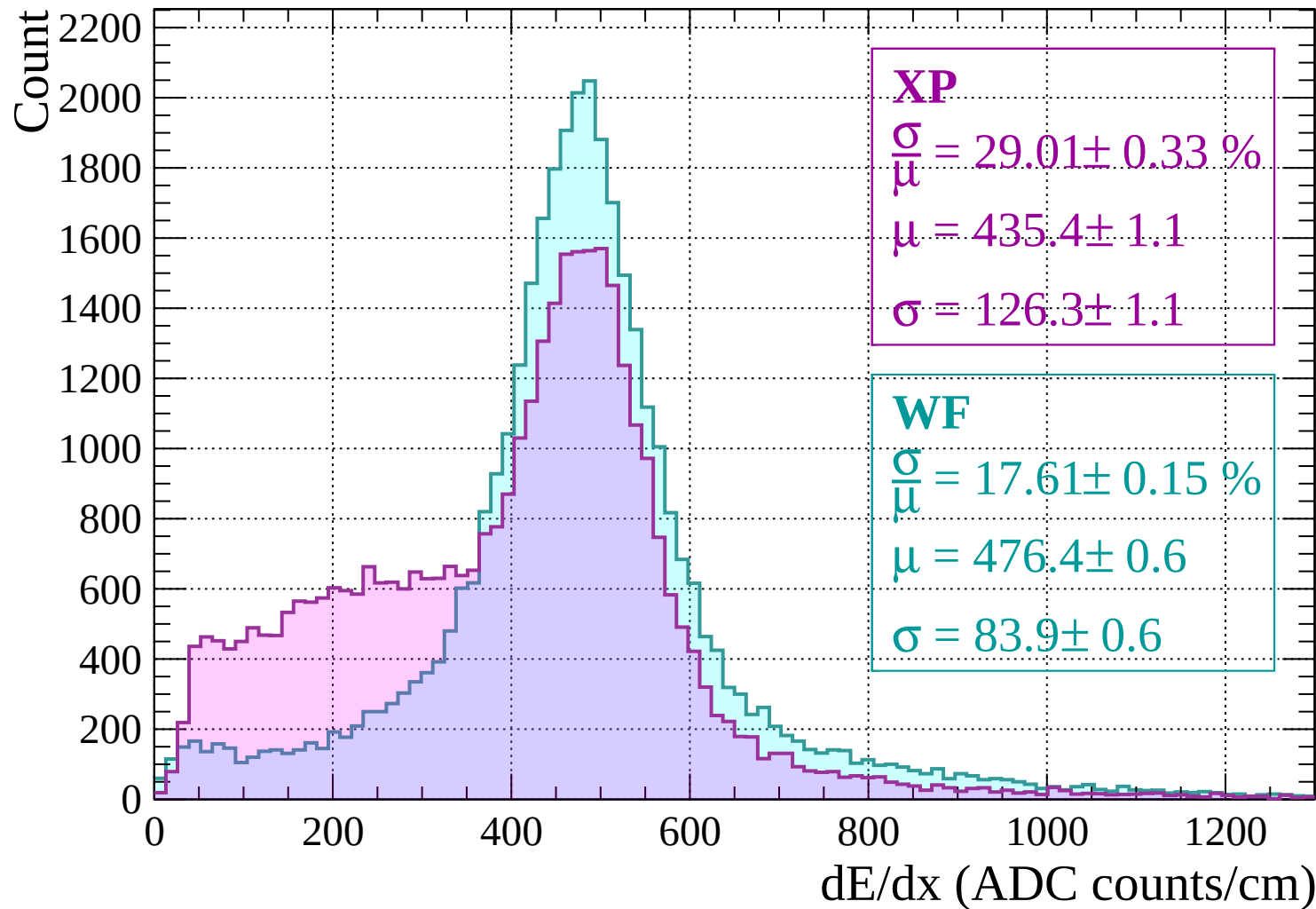
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 40$



Energy loss | $40 < p < 80$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 60.66 \pm 1.10 \%$$

$$\mu = 363.9 \pm 2.9$$

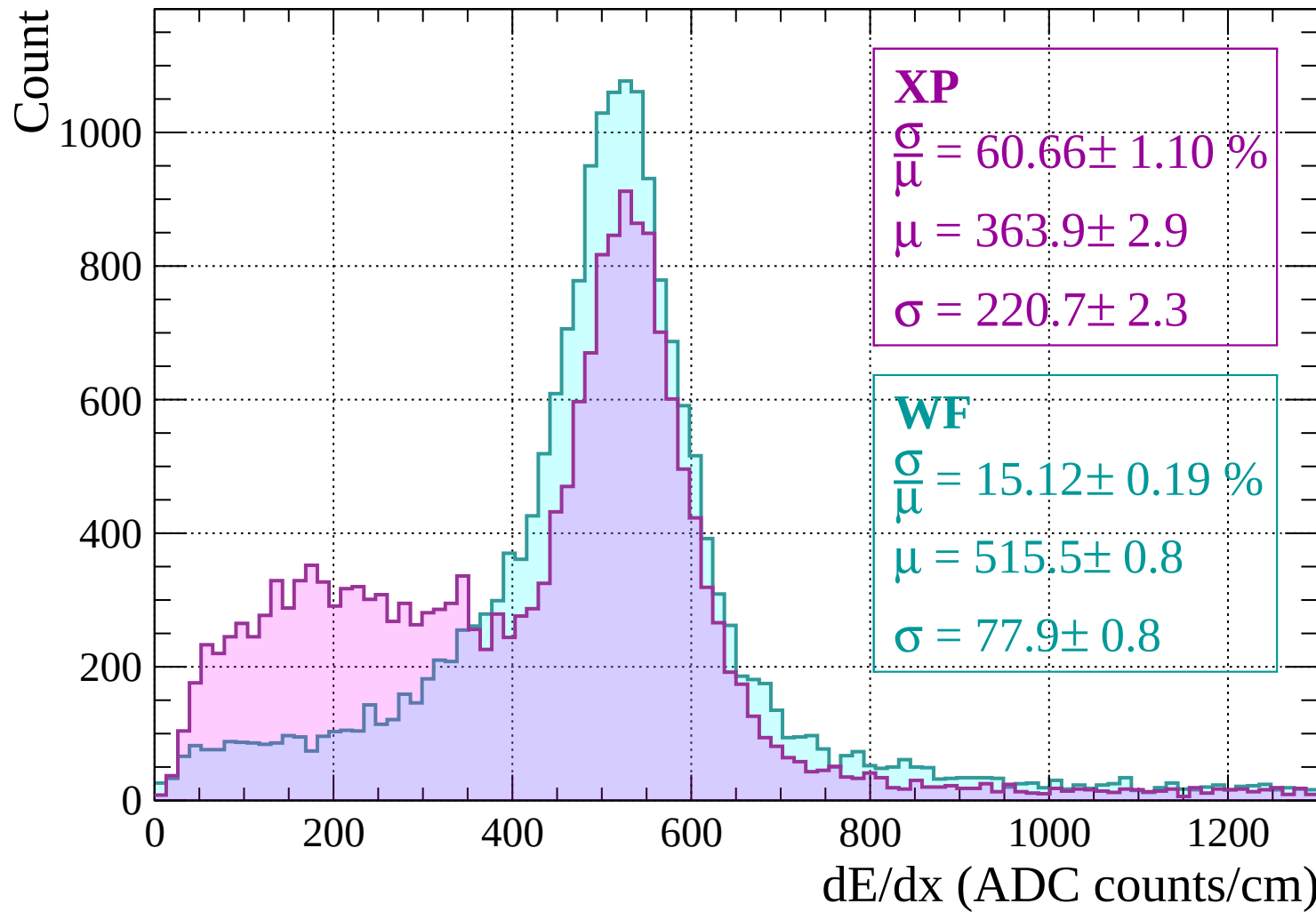
$$\sigma = 220.7 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 15.12 \pm 0.19 \%$$

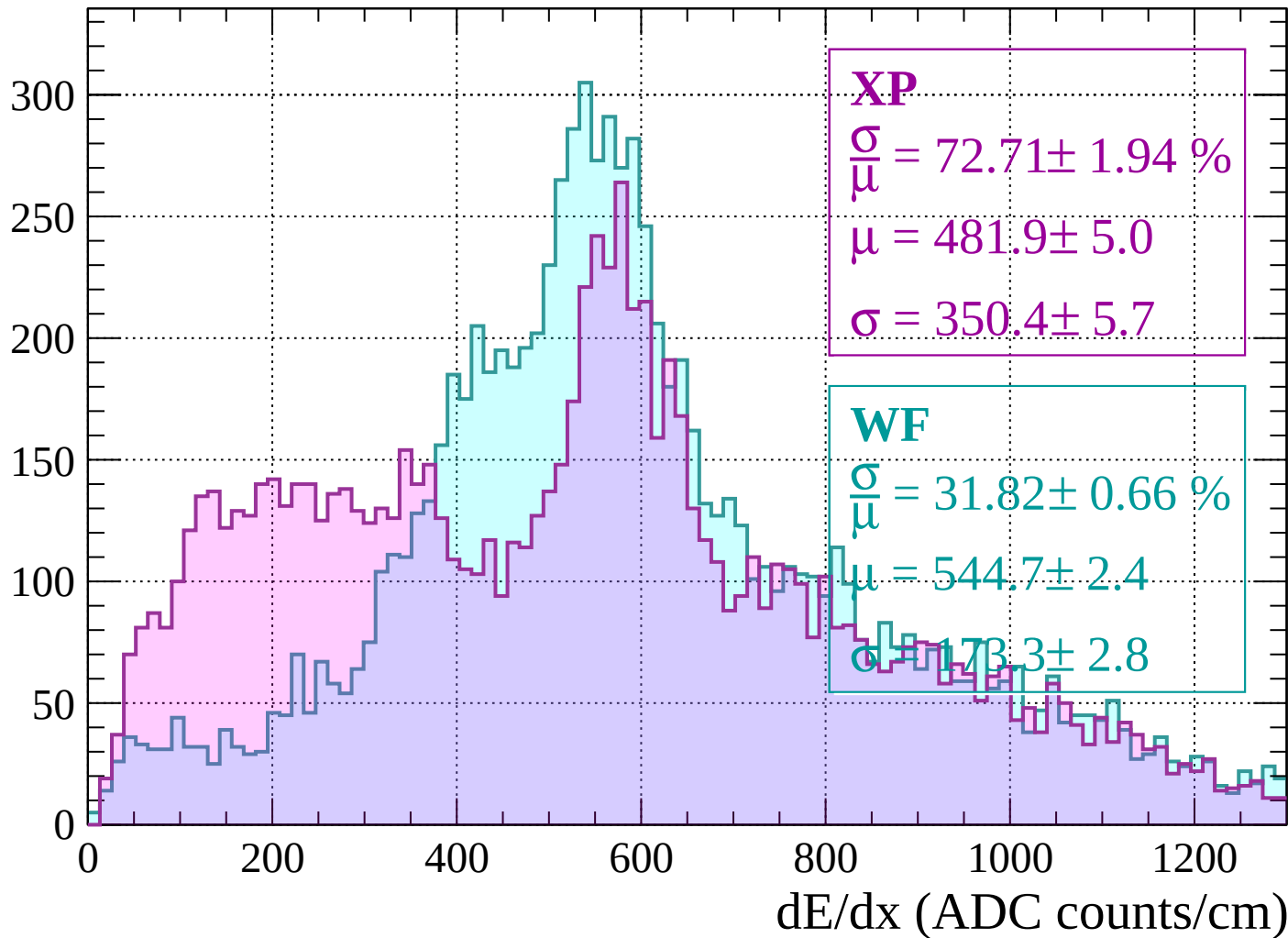
$$\mu = 515.5 \pm 0.8$$

$$\sigma = 77.9 \pm 0.8$$



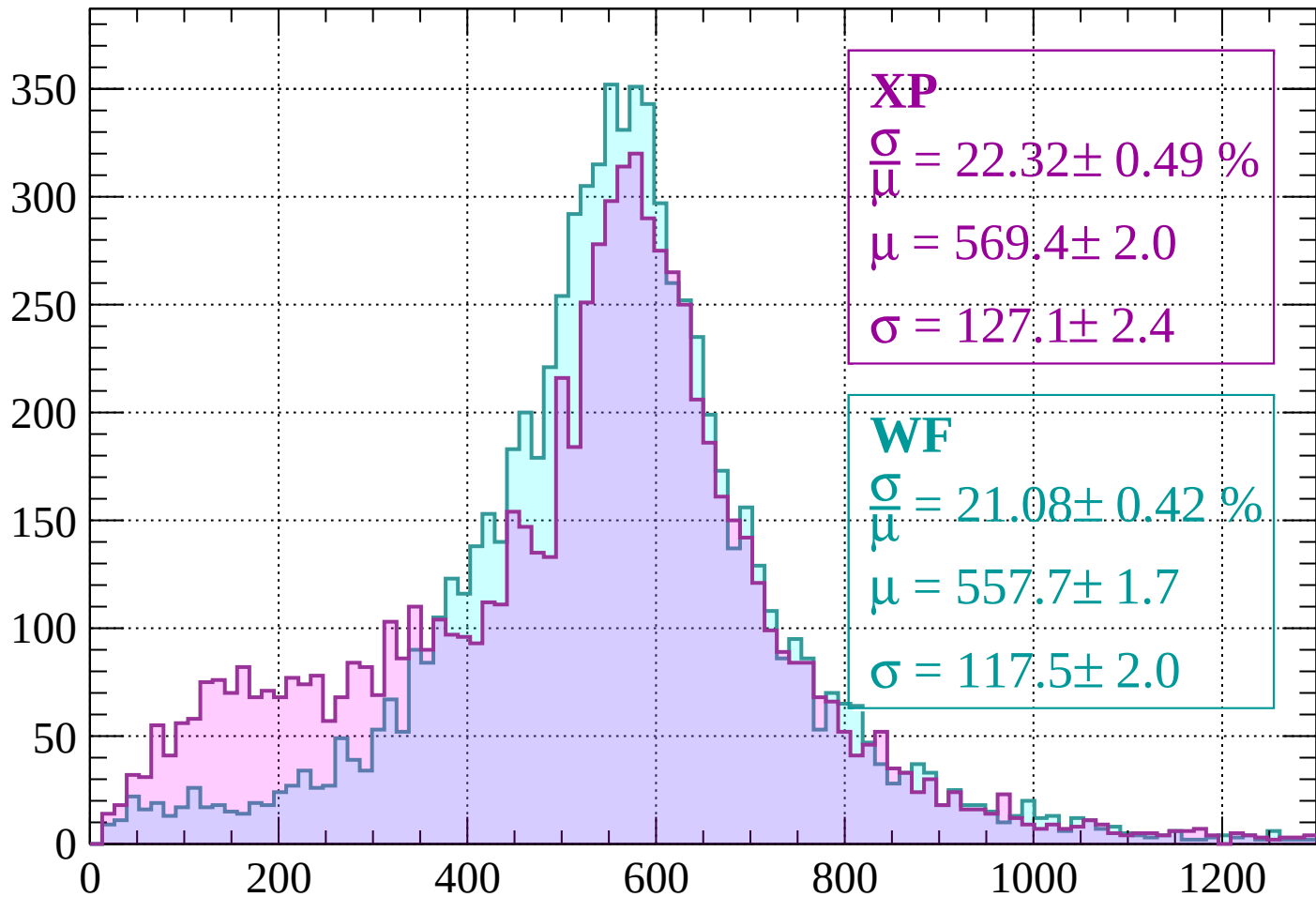
Energy loss | $80 < p < 120$

Count



Energy loss | $120 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 22.32 \pm 0.49 \%$$

$$\mu = 569.4 \pm 2.0$$

$$\sigma = 127.1 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 21.08 \pm 0.42 \%$$

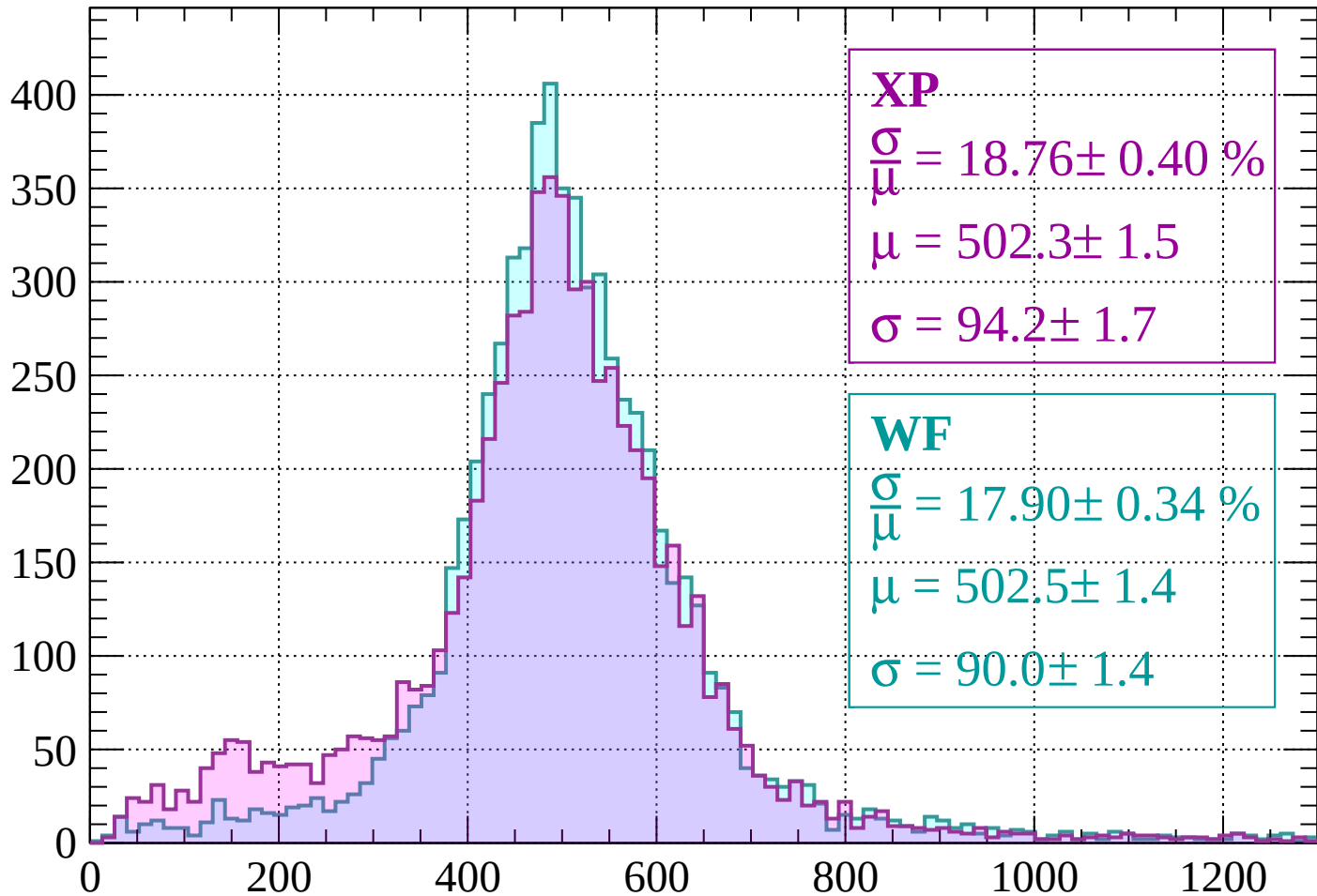
$$\mu = 557.7 \pm 1.7$$

$$\sigma = 117.5 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 18.76 \pm 0.40 \%$$

$$\mu = 502.3 \pm 1.5$$

$$\sigma = 94.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 17.90 \pm 0.34 \%$$

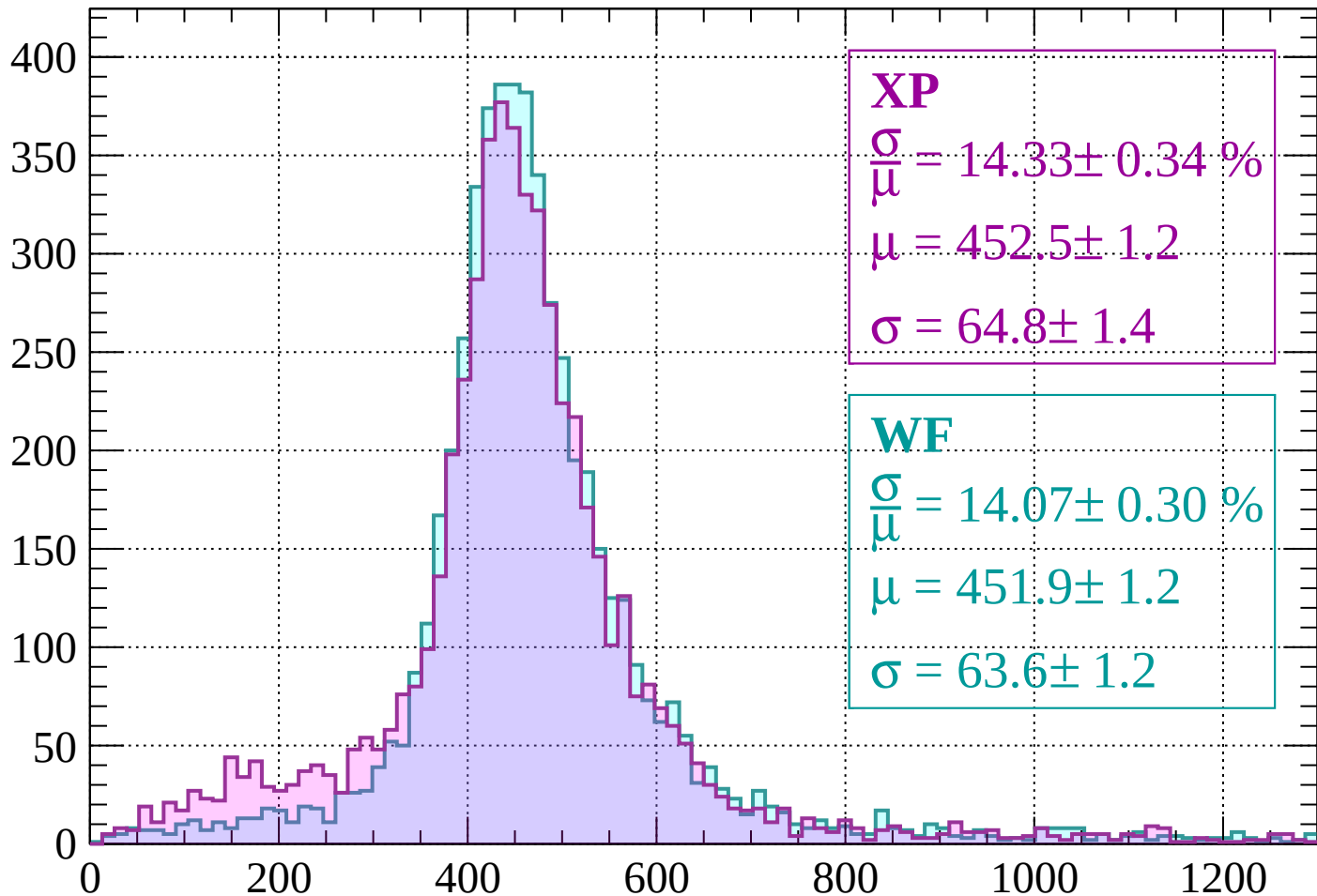
$$\mu = 502.5 \pm 1.4$$

$$\sigma = 90.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 14.33 \pm 0.34 \%$$

$$\mu = 452.5 \pm 1.2$$

$$\sigma = 64.8 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 14.07 \pm 0.30 \%$$

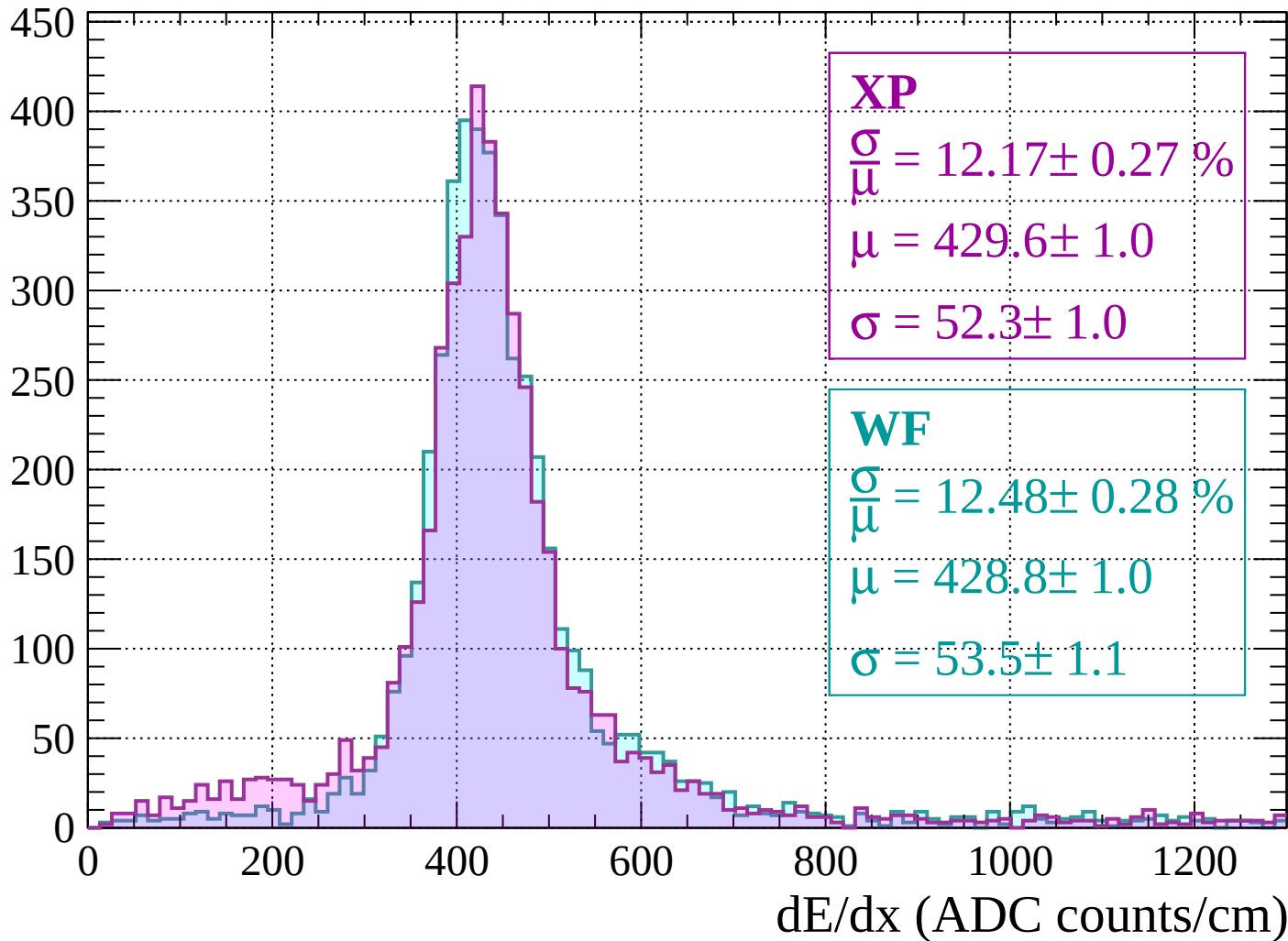
$$\mu = 451.9 \pm 1.2$$

$$\sigma = 63.6 \pm 1.2$$

dE/dx (ADC counts/cm)

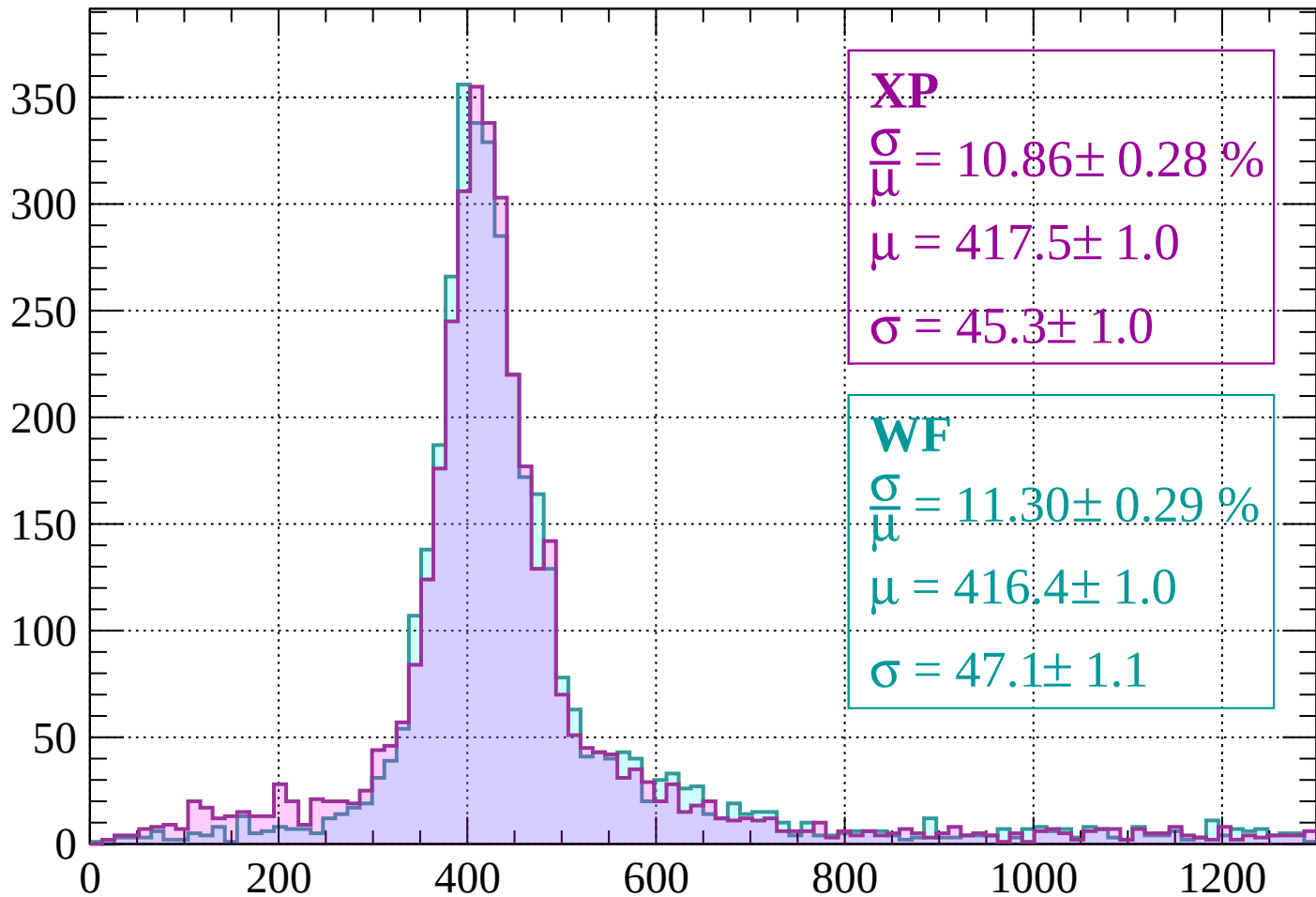
Energy loss | $240 < p < 280$

Count



Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 10.86 \pm 0.28 \%$$

$$\mu = 417.5 \pm 1.0$$

$$\sigma = 45.3 \pm 1.0$$

WF

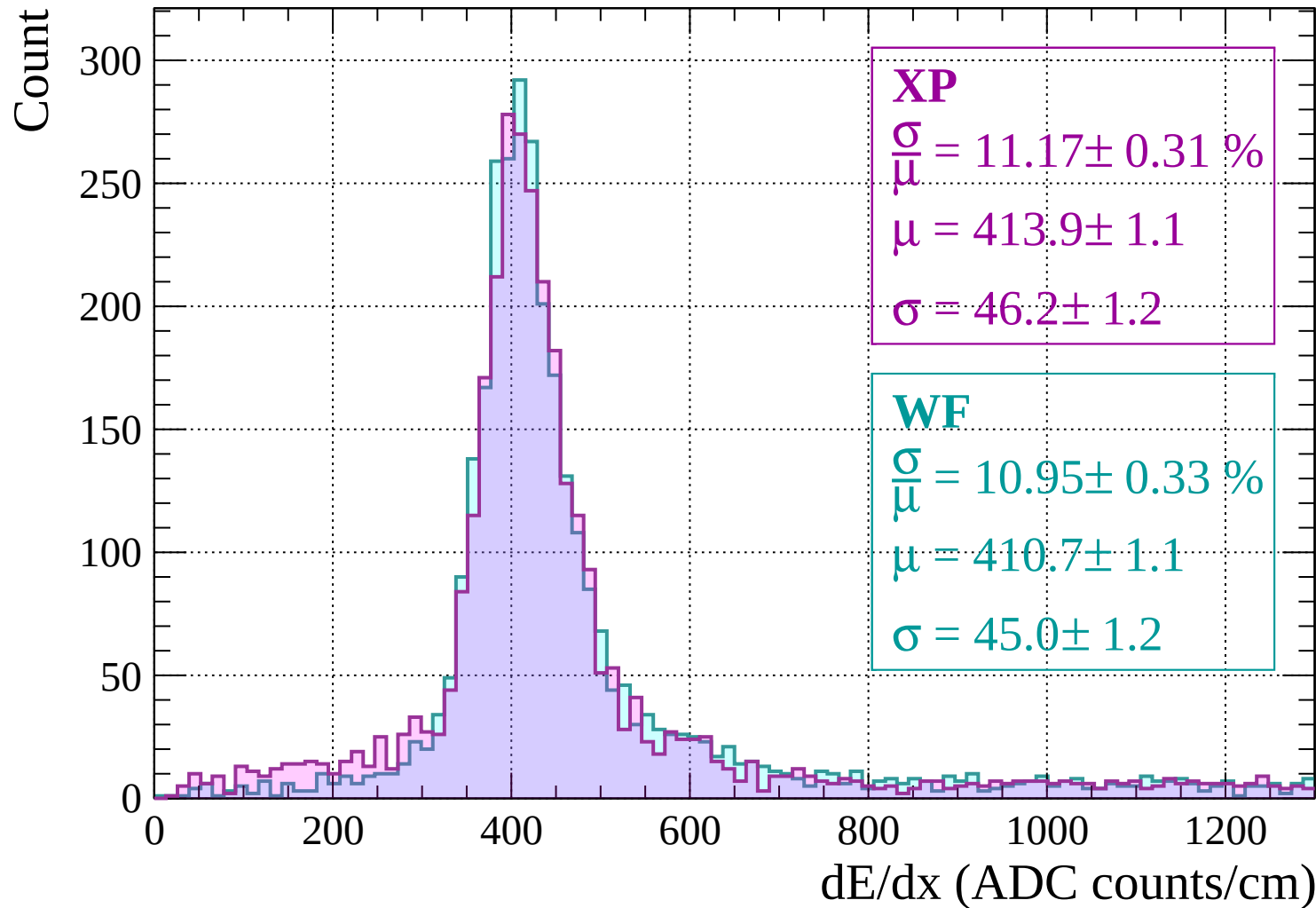
$$\frac{\sigma}{\mu} = 11.30 \pm 0.29 \%$$

$$\mu = 416.4 \pm 1.0$$

$$\sigma = 47.1 \pm 1.1$$

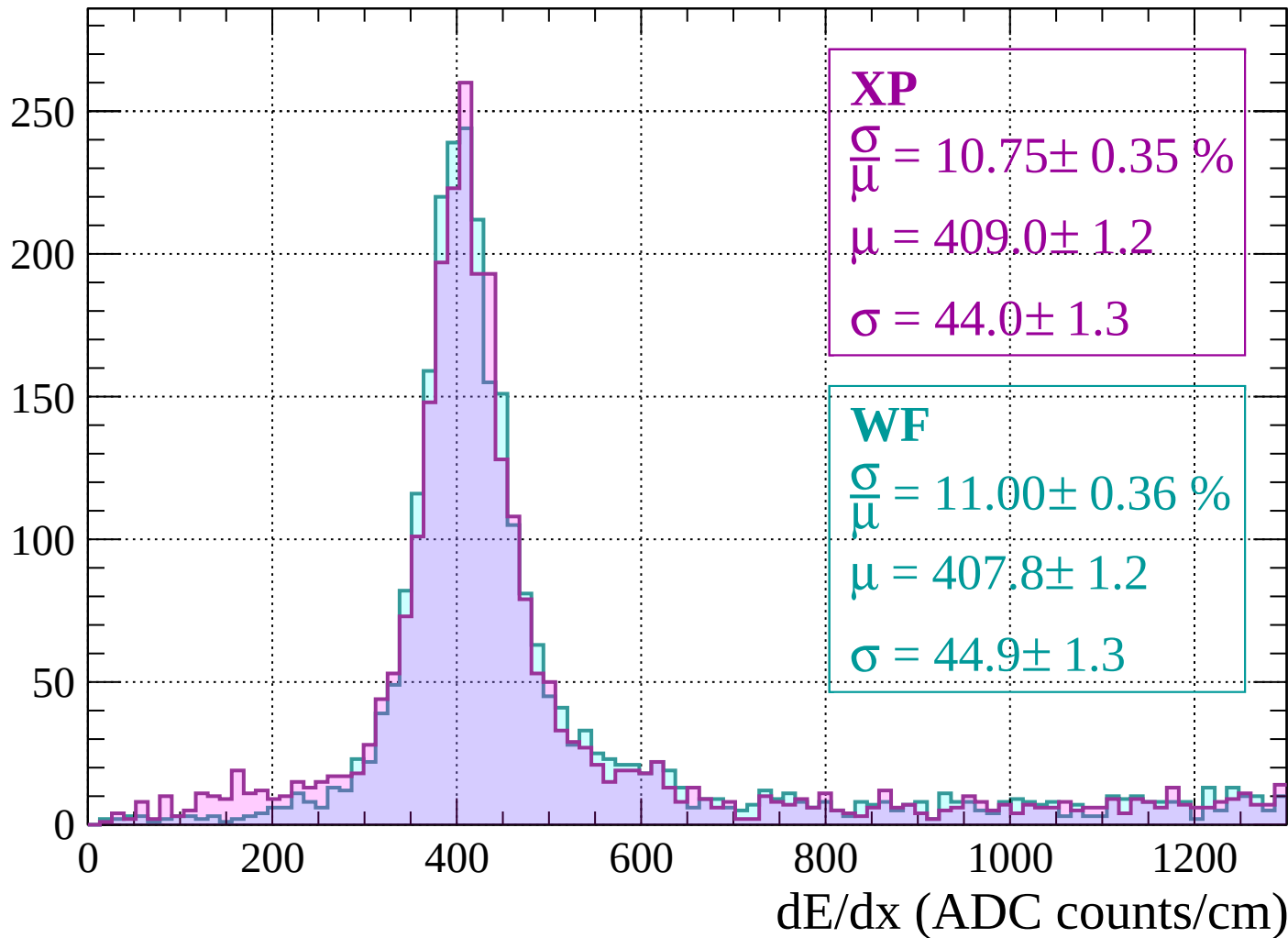
dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$



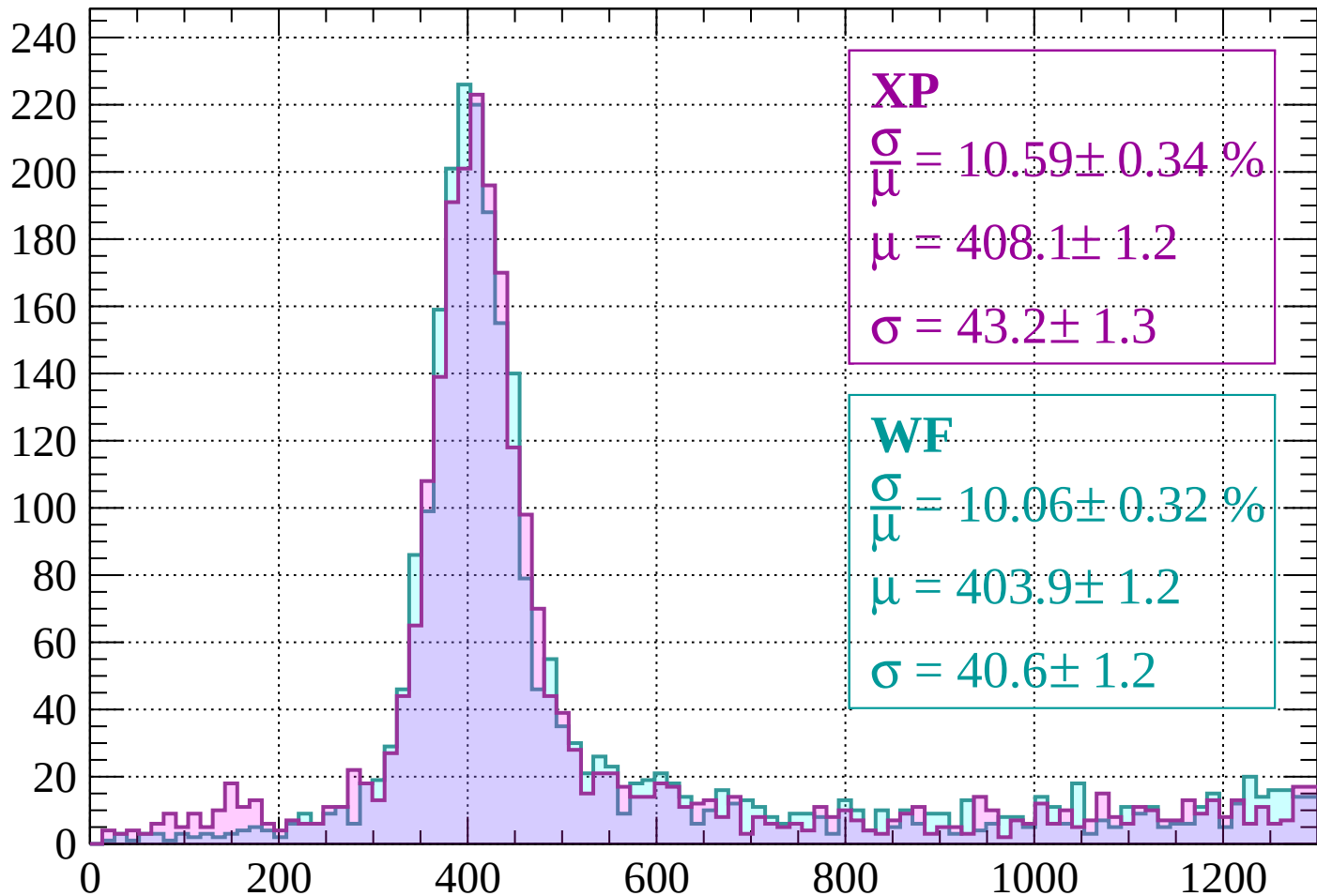
Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 10.59 \pm 0.34 \%$$

$$\mu = 408.1 \pm 1.2$$

$$\sigma = 43.2 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.06 \pm 0.32 \%$$

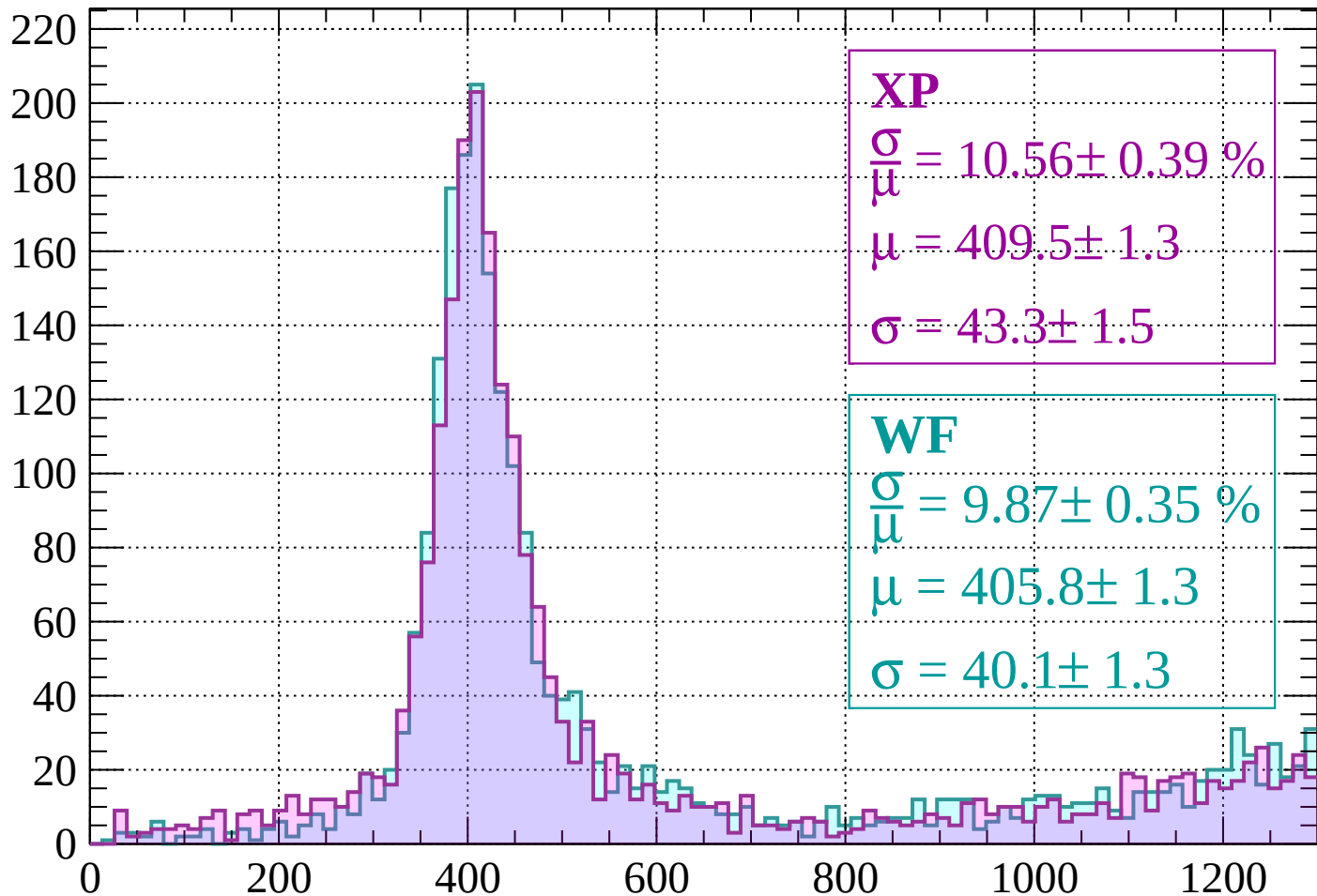
$$\mu = 403.9 \pm 1.2$$

$$\sigma = 40.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 10.56 \pm 0.39 \%$$

$$\mu = 409.5 \pm 1.3$$

$$\sigma = 43.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.87 \pm 0.35 \%$$

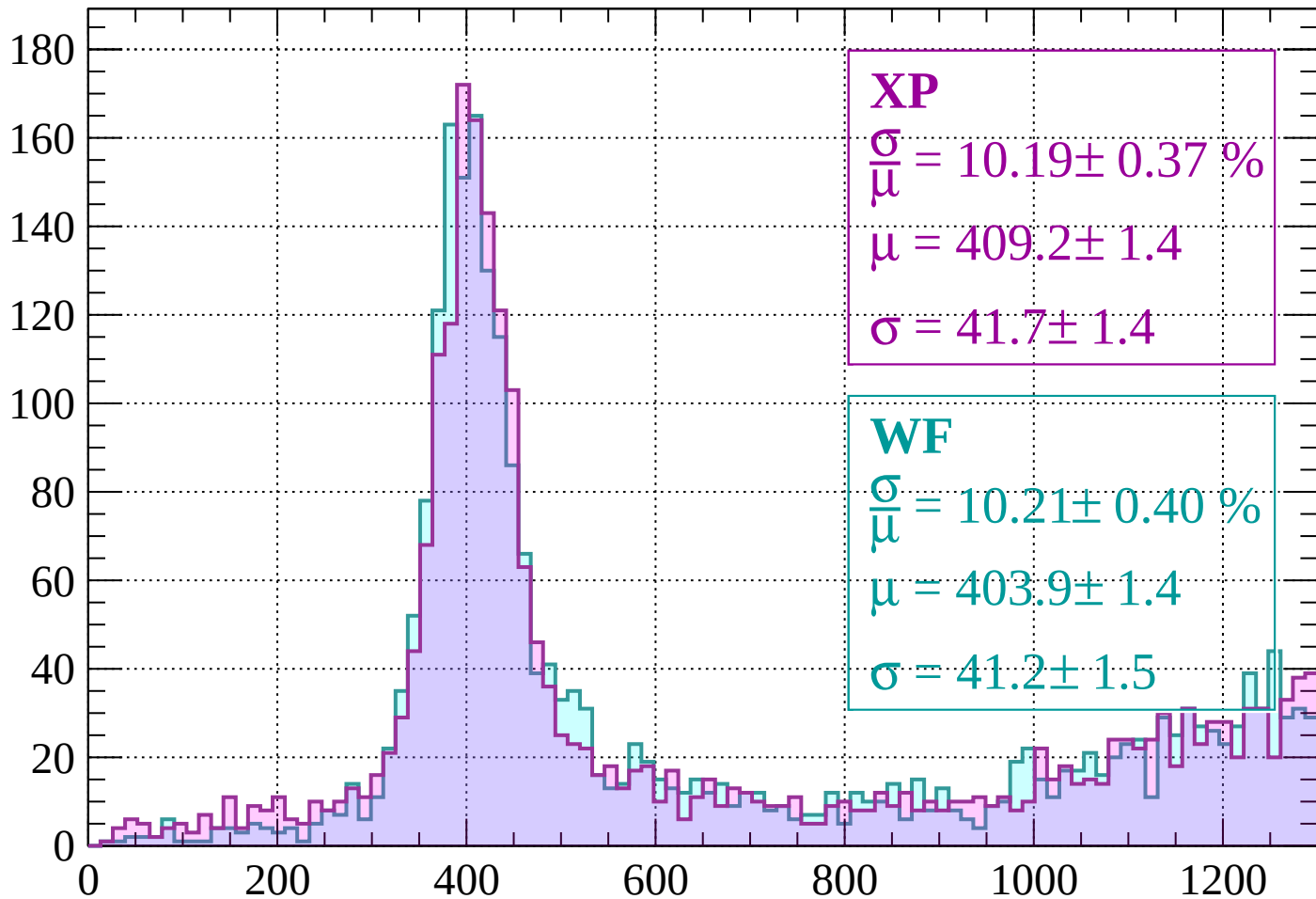
$$\mu = 405.8 \pm 1.3$$

$$\sigma = 40.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 10.19 \pm 0.37 \%$$

$$\mu = 409.2 \pm 1.4$$

$$\sigma = 41.7 \pm 1.4$$

WF

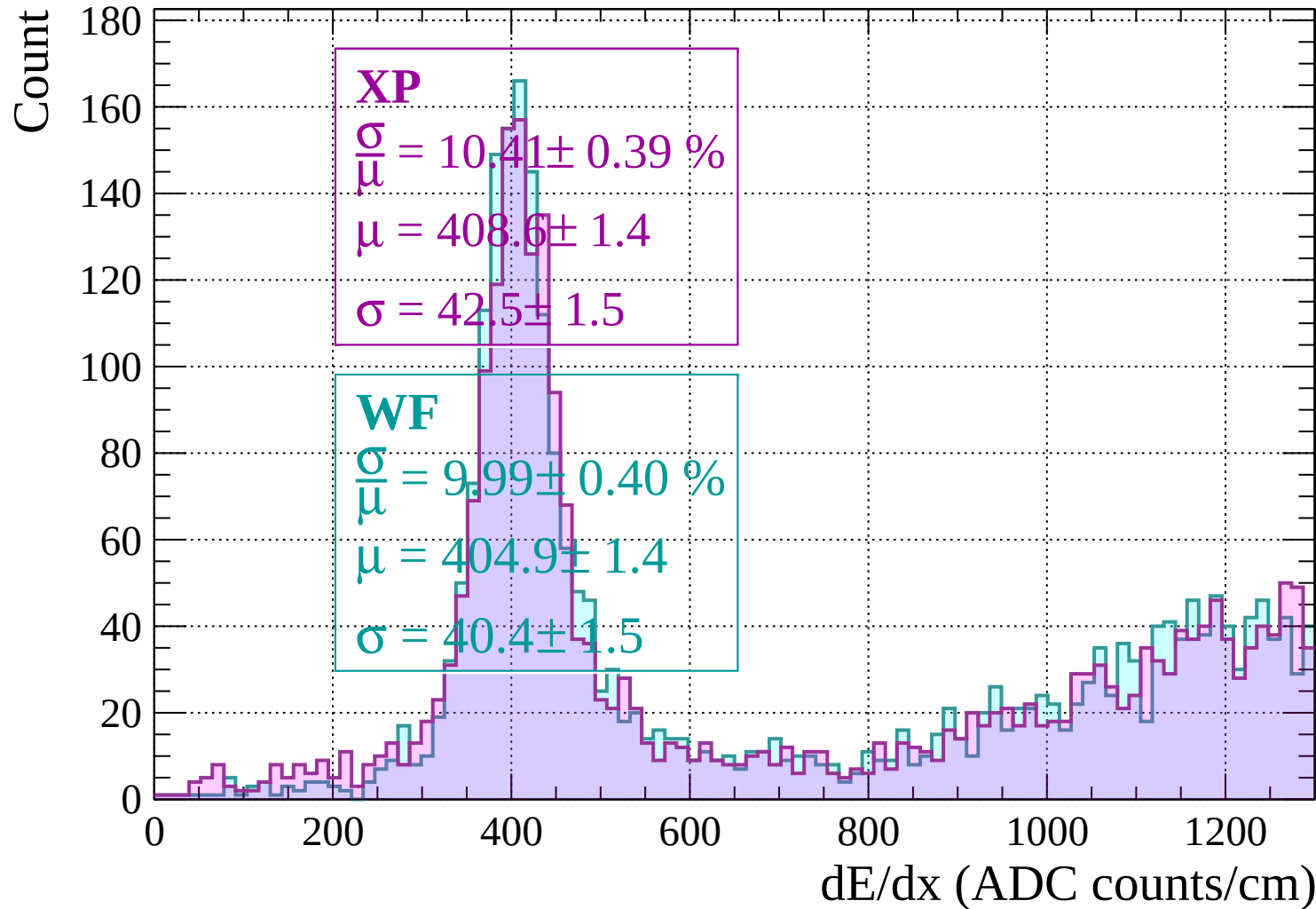
$$\frac{\sigma}{\mu} = 10.21 \pm 0.40 \%$$

$$\mu = 403.9 \pm 1.4$$

$$\sigma = 41.2 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $520 < p < 560$



Energy loss | $560 < p < 600$

Count

160
140
120
100
80
60
40
20
0

XP

$$\frac{\sigma}{\mu} = 10.72 \pm 0.46 \%$$

$$\mu = 411.5 \pm 1.6$$

$$\sigma = 44.1 \pm 1.7$$

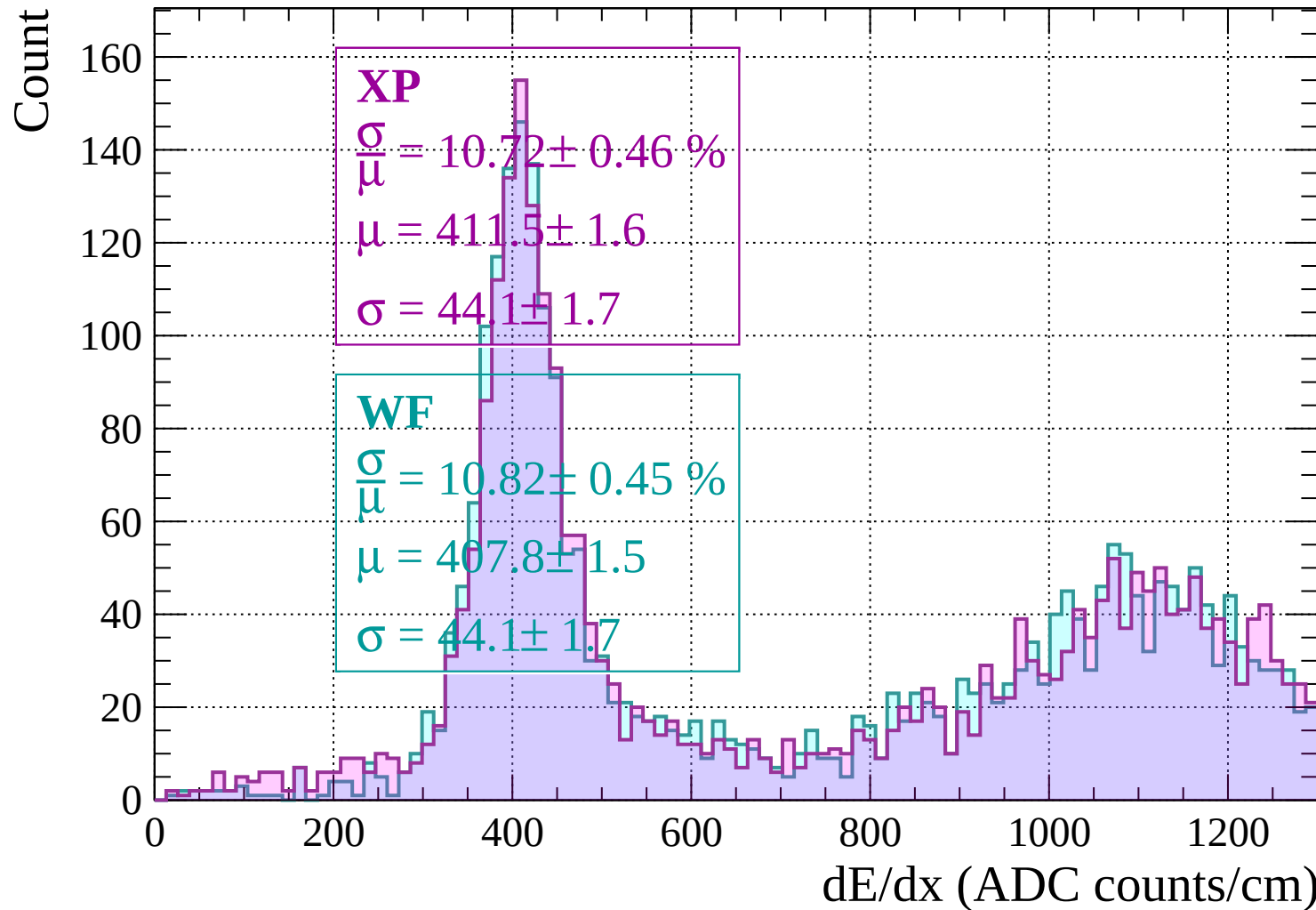
WF

$$\frac{\sigma}{\mu} = 10.82 \pm 0.45 \%$$

$$\mu = 407.8 \pm 1.5$$

$$\sigma = 44.1 \pm 1.7$$

dE/dx (ADC counts/cm)



Energy loss | $600 < p < 640$

Count

140

120

100

80

60

40

20

0

XP

$$\frac{\sigma}{\mu} = 16.31 \pm 0.72 \%$$

$$\mu = 1027.9 \pm 5.5$$

$$\sigma = 167.6 \pm 6.5$$

WF

$$\frac{\sigma}{\mu} = 43.61 \pm 3.19 \%$$

$$\mu = 1012.4 \pm 24.1$$

$$\sigma = 441.5 \pm 21.8$$

0

200

400

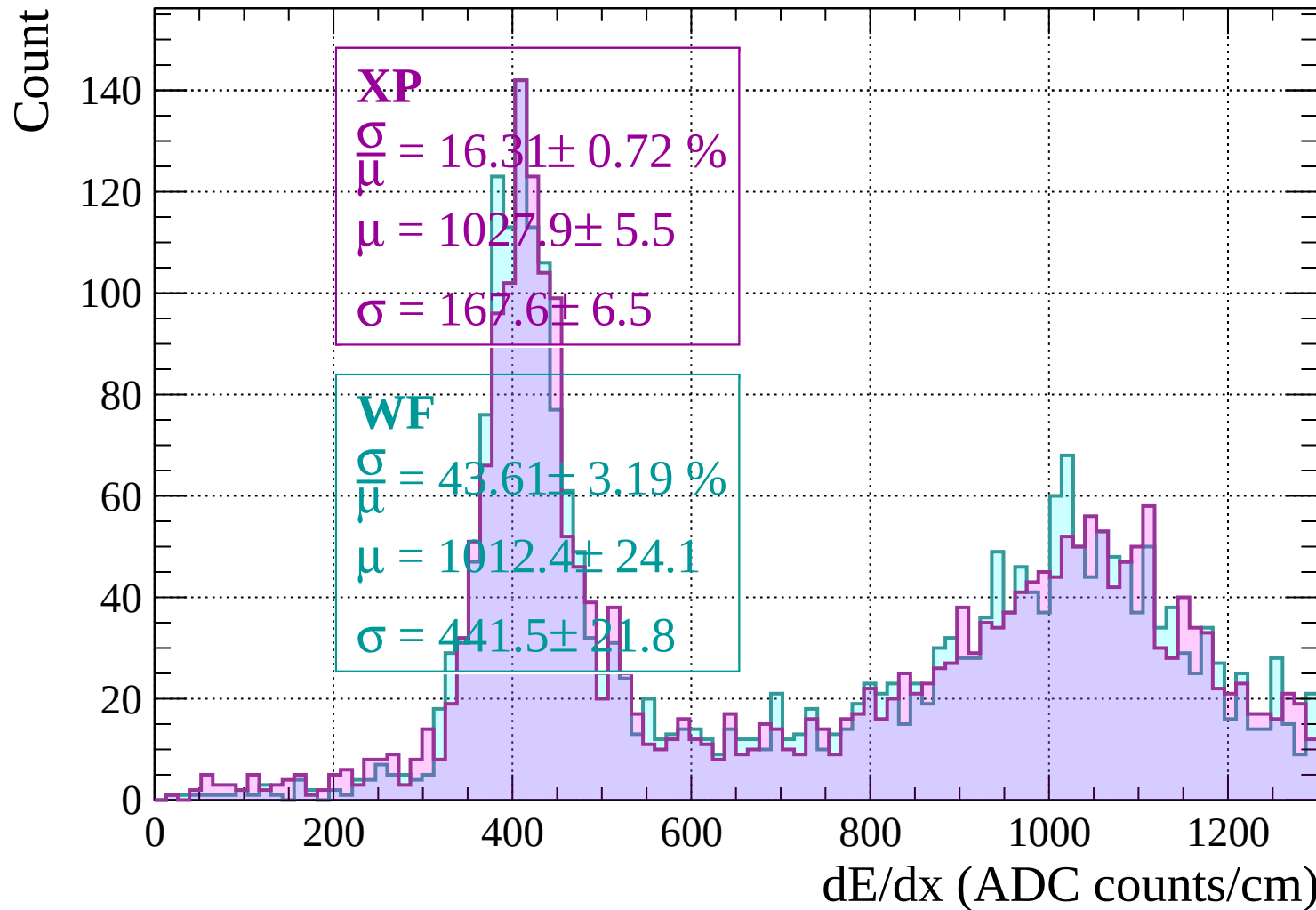
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $640 < p < 680$

Count

120

100

80

60

40

20

0

XP

$$\frac{\sigma}{\mu} = 28.10 \pm 4.43 \%$$

$$\mu = 3579.2 \pm 340.8$$

$$\sigma = 1005.9 \pm 62.6$$

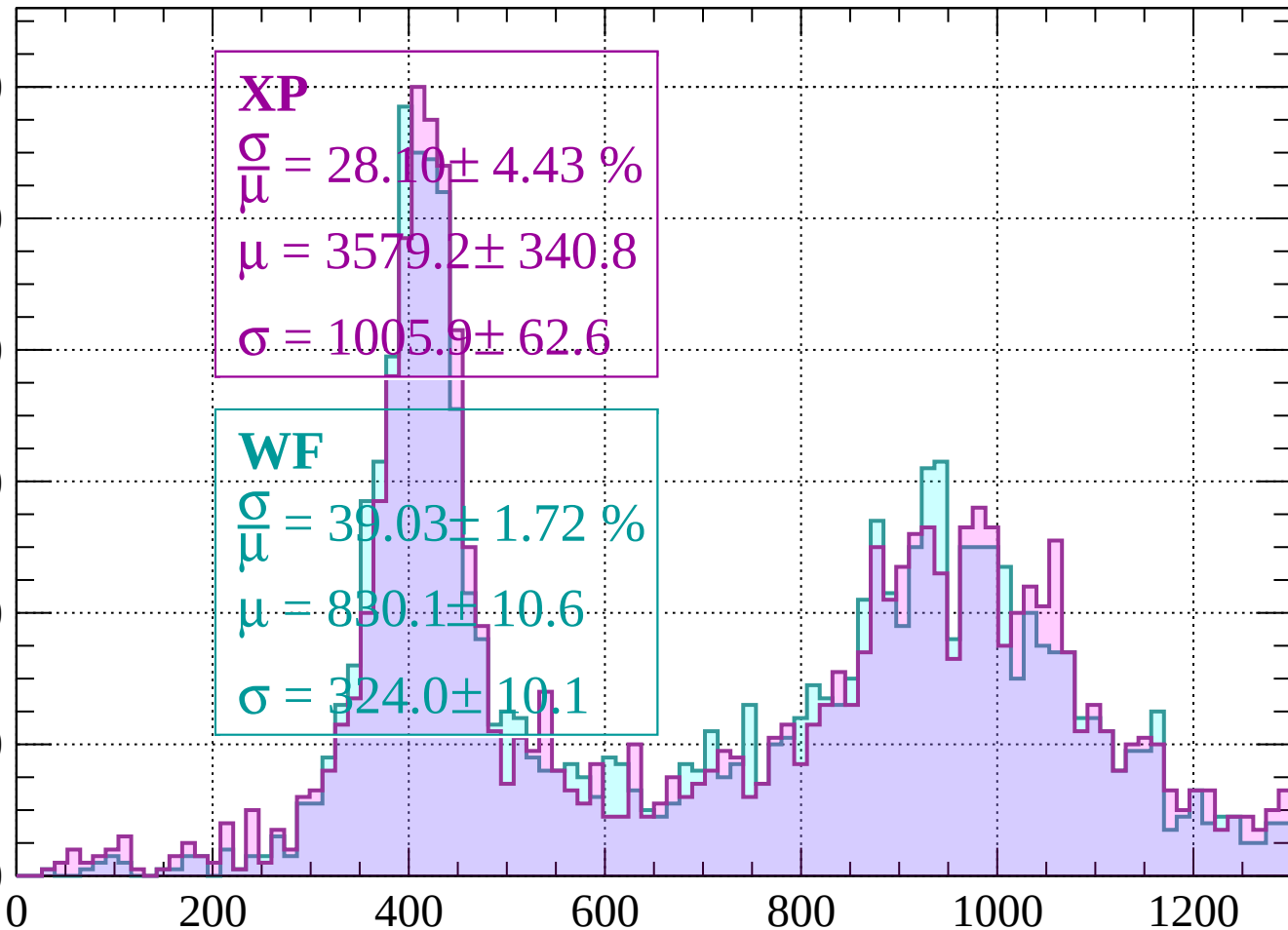
WF

$$\frac{\sigma}{\mu} = 39.03 \pm 1.72 \%$$

$$\mu = 830.1 \pm 10.6$$

$$\sigma = 324.0 \pm 10.1$$

dE/dx (ADC counts/cm)



Energy loss | $680 < p < 720$

Count

120
100
80
60
40
20
0

XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.49 \%$$

$$\mu = 421.8 \pm 1.7$$

$$\sigma = 39.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 23.61 \pm 5.89 \%$$

$$\mu = 4414.1 \pm 684.5$$

$$\sigma = 1042.1 \pm 98.4$$

dE/dx (ADC counts/cm)

0

200

400

600

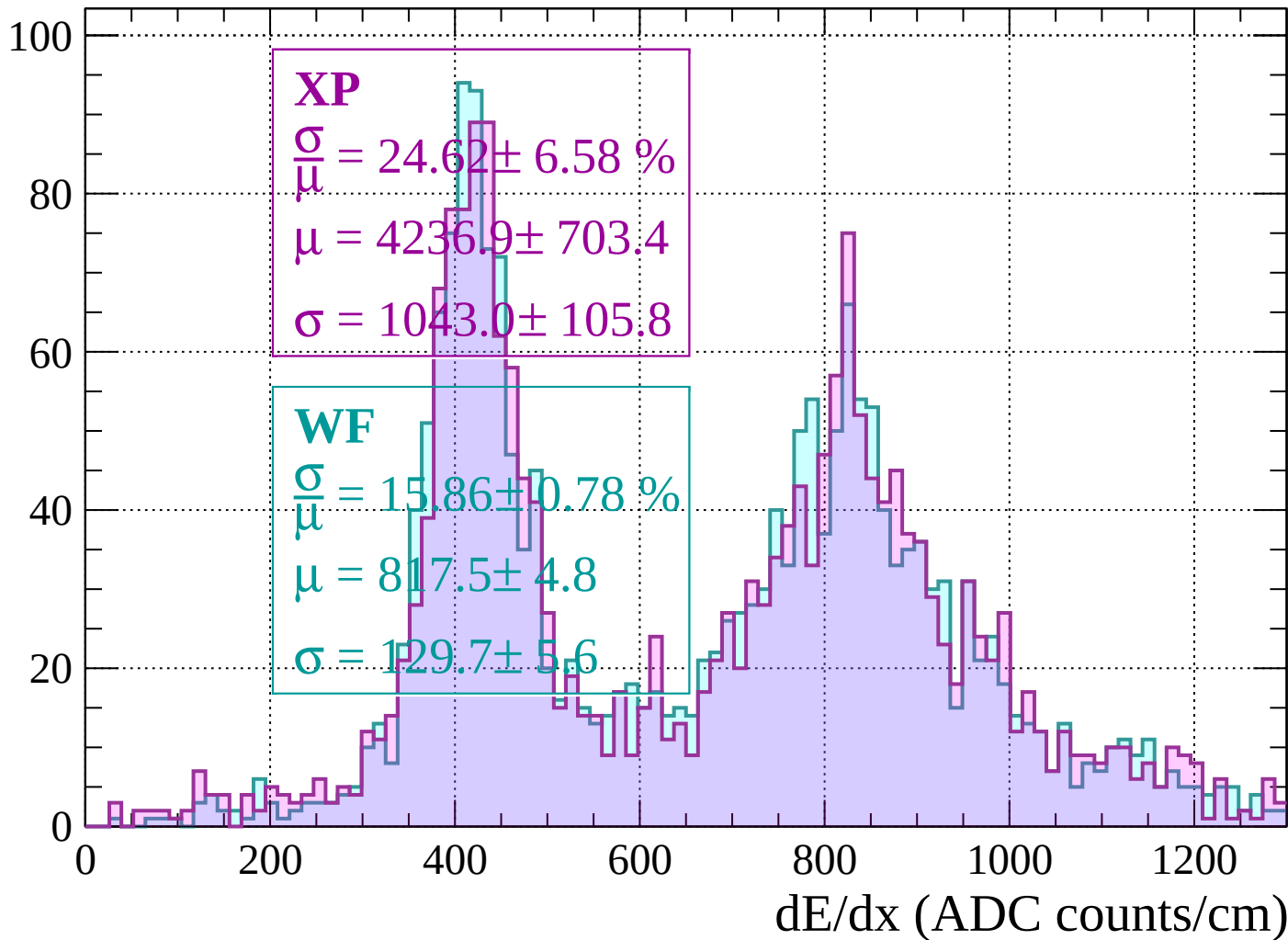
800

1000

1200

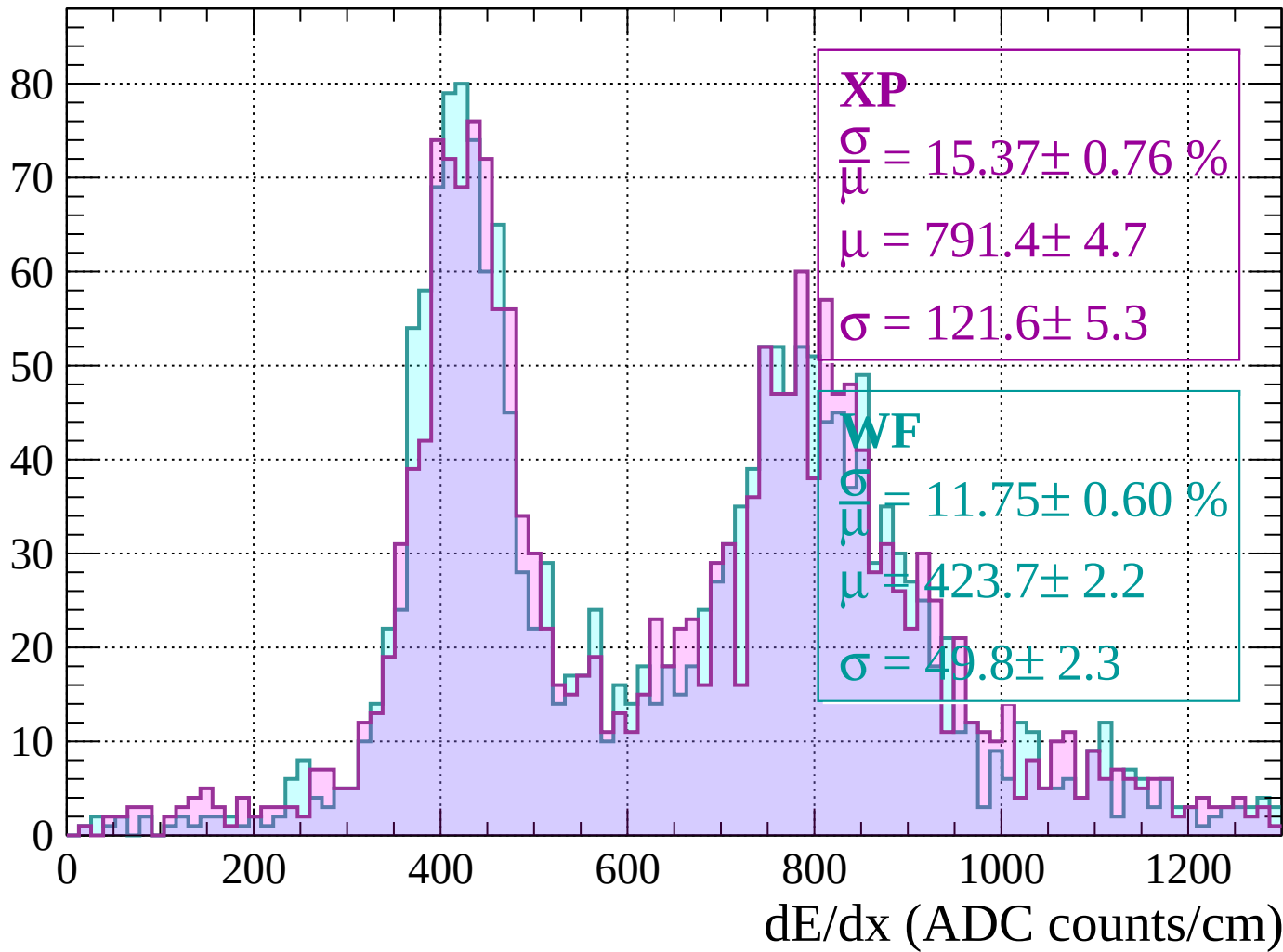
Energy loss | $720 < p < 760$

Count



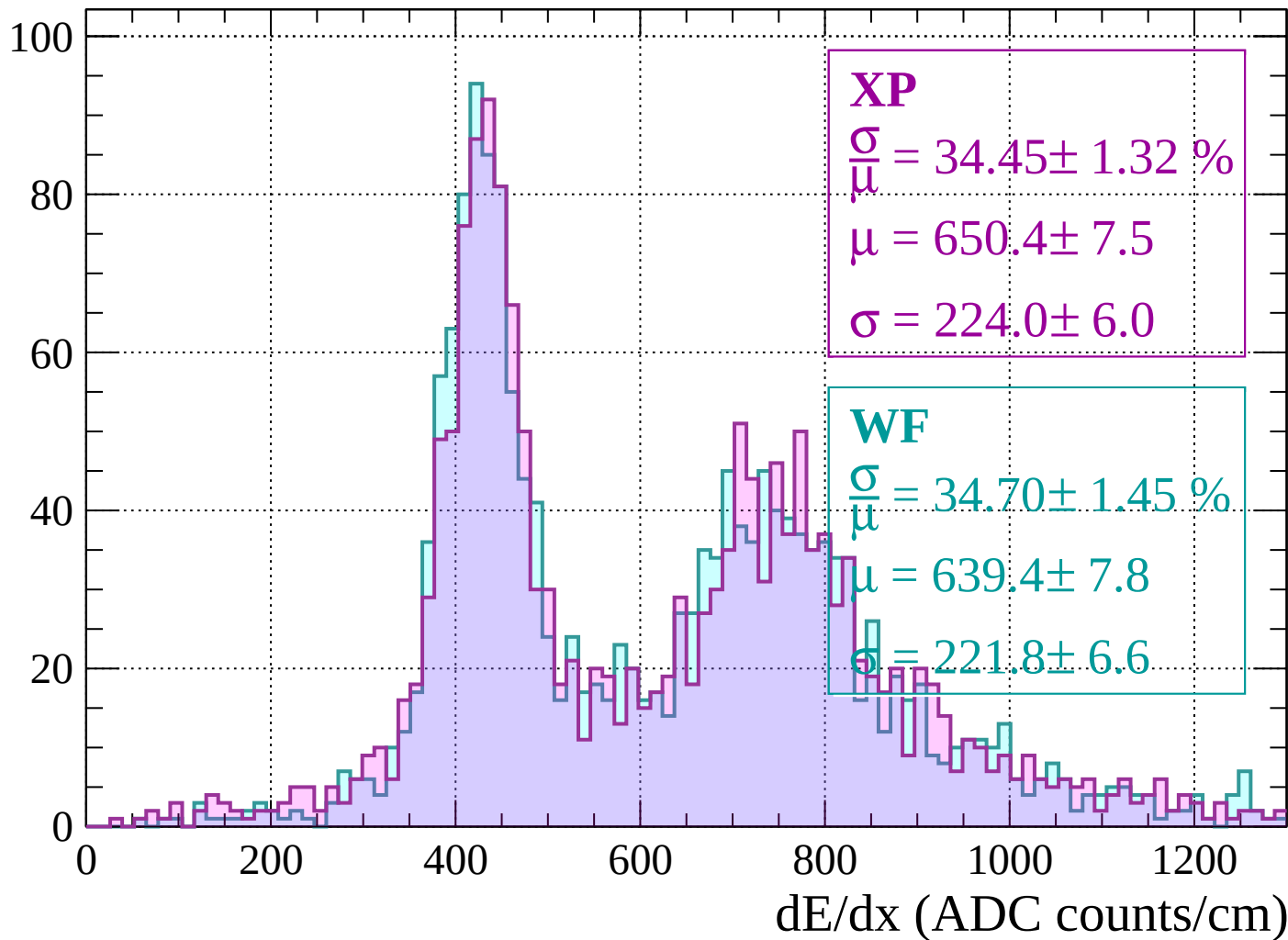
Energy loss | $760 < p < 800$

Count



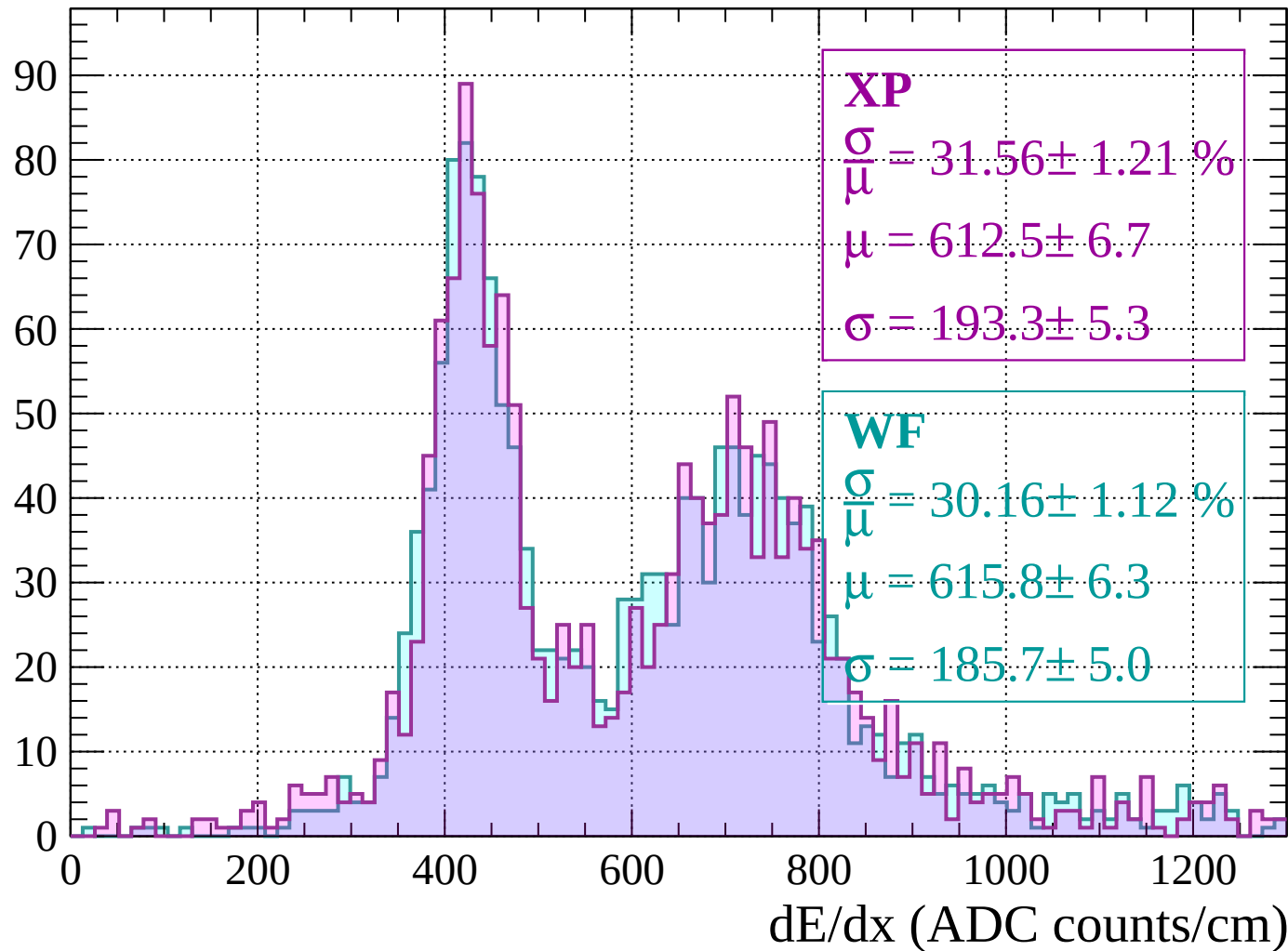
Energy loss | $800 < p < 840$

Count



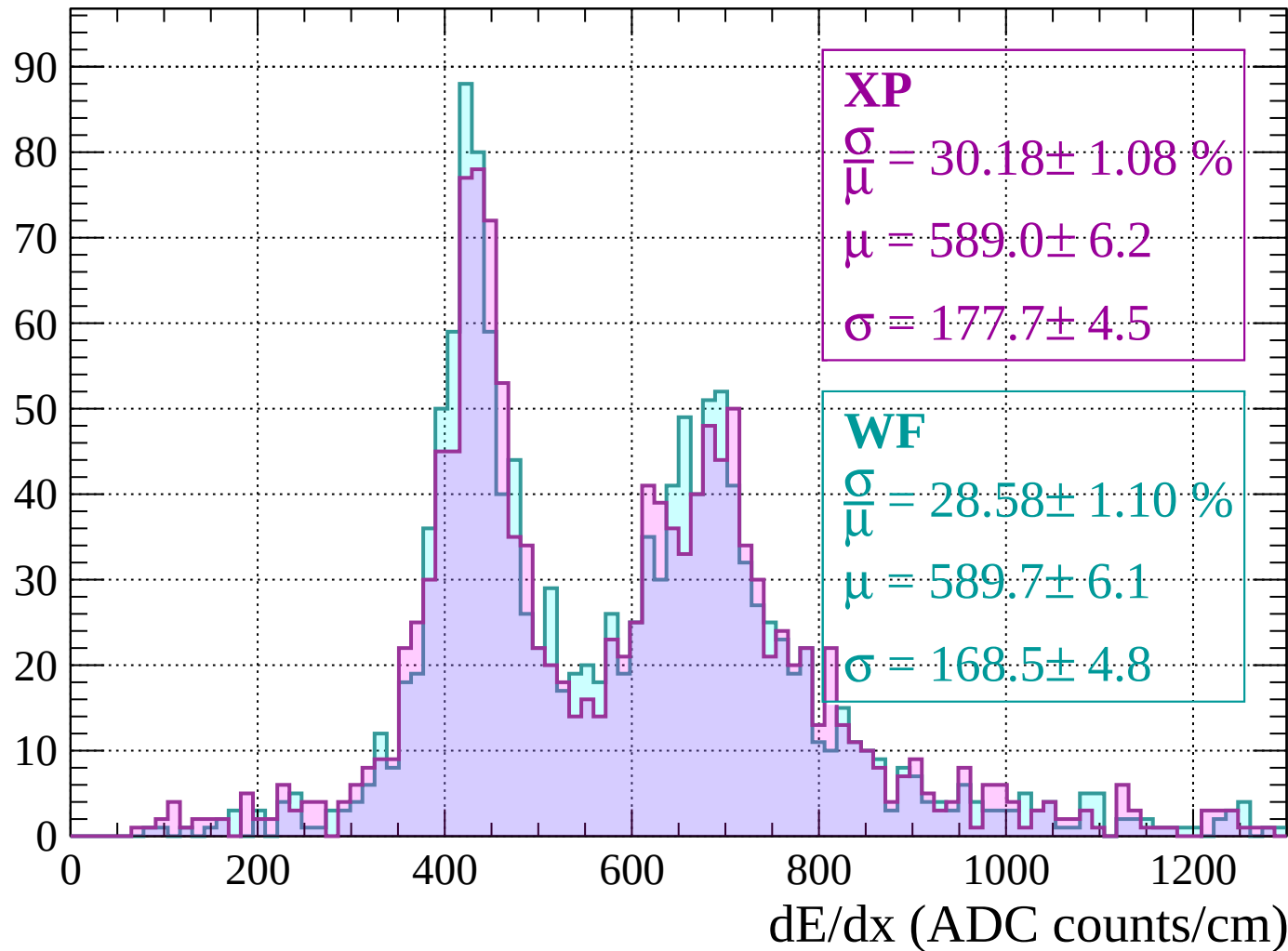
Energy loss | $840 < p < 880$

Count



Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count

XP

$$\frac{\sigma}{\mu} = 26.46 \pm 0.97 \%$$

$$\mu = 562.0 \pm 5.4$$

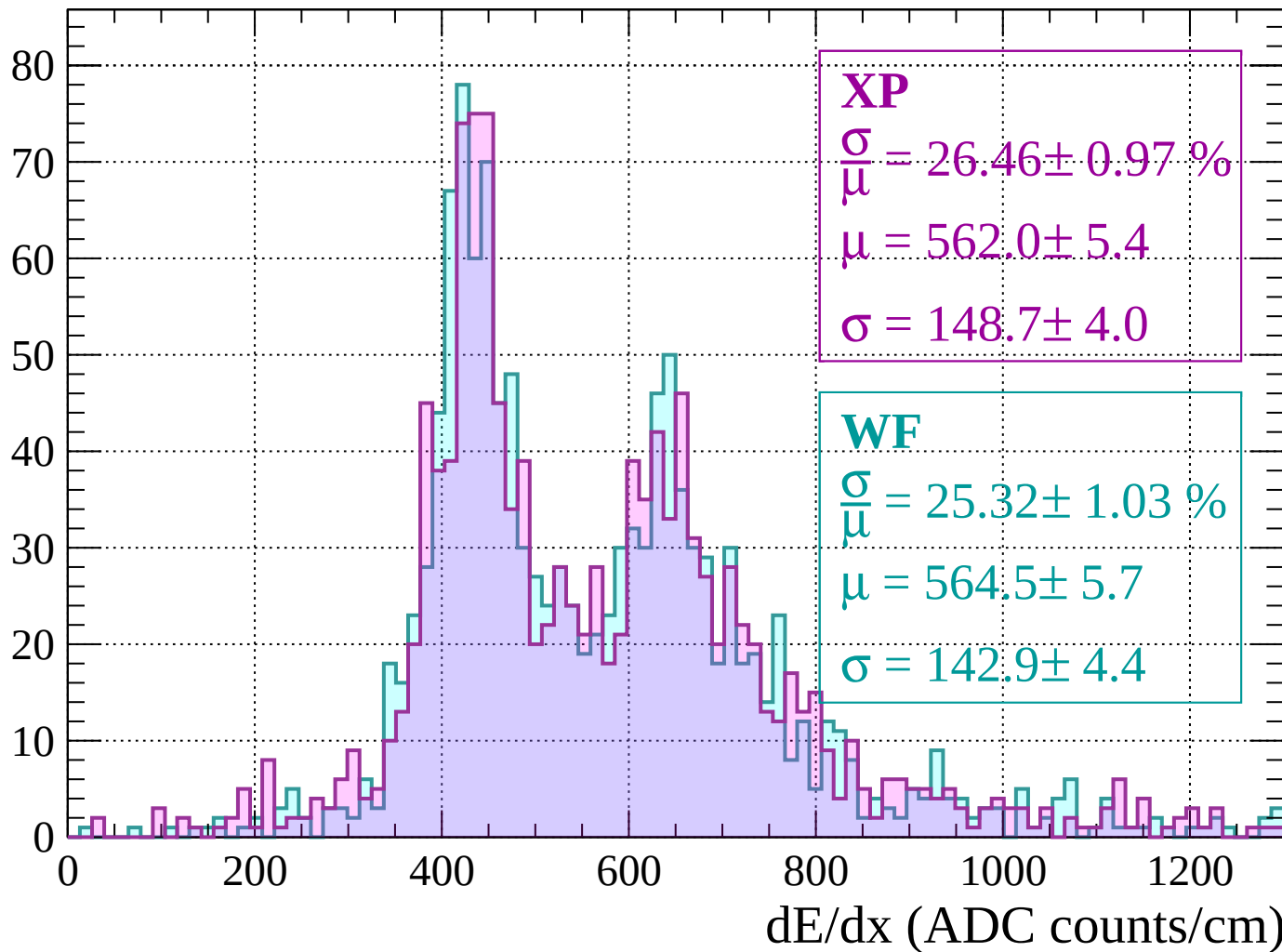
$$\sigma = 148.7 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 25.32 \pm 1.03 \%$$

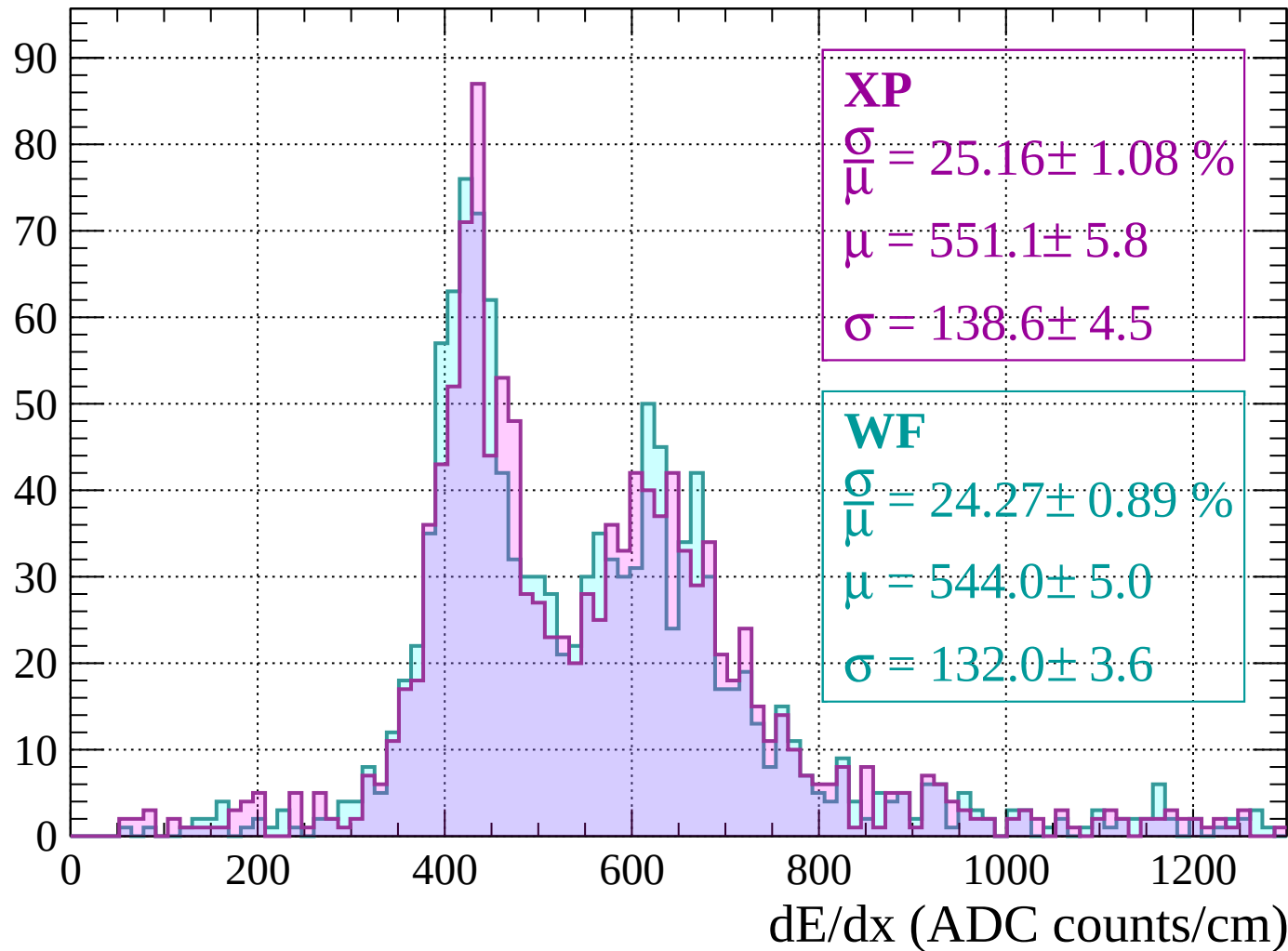
$$\mu = 564.5 \pm 5.7$$

$$\sigma = 142.9 \pm 4.4$$



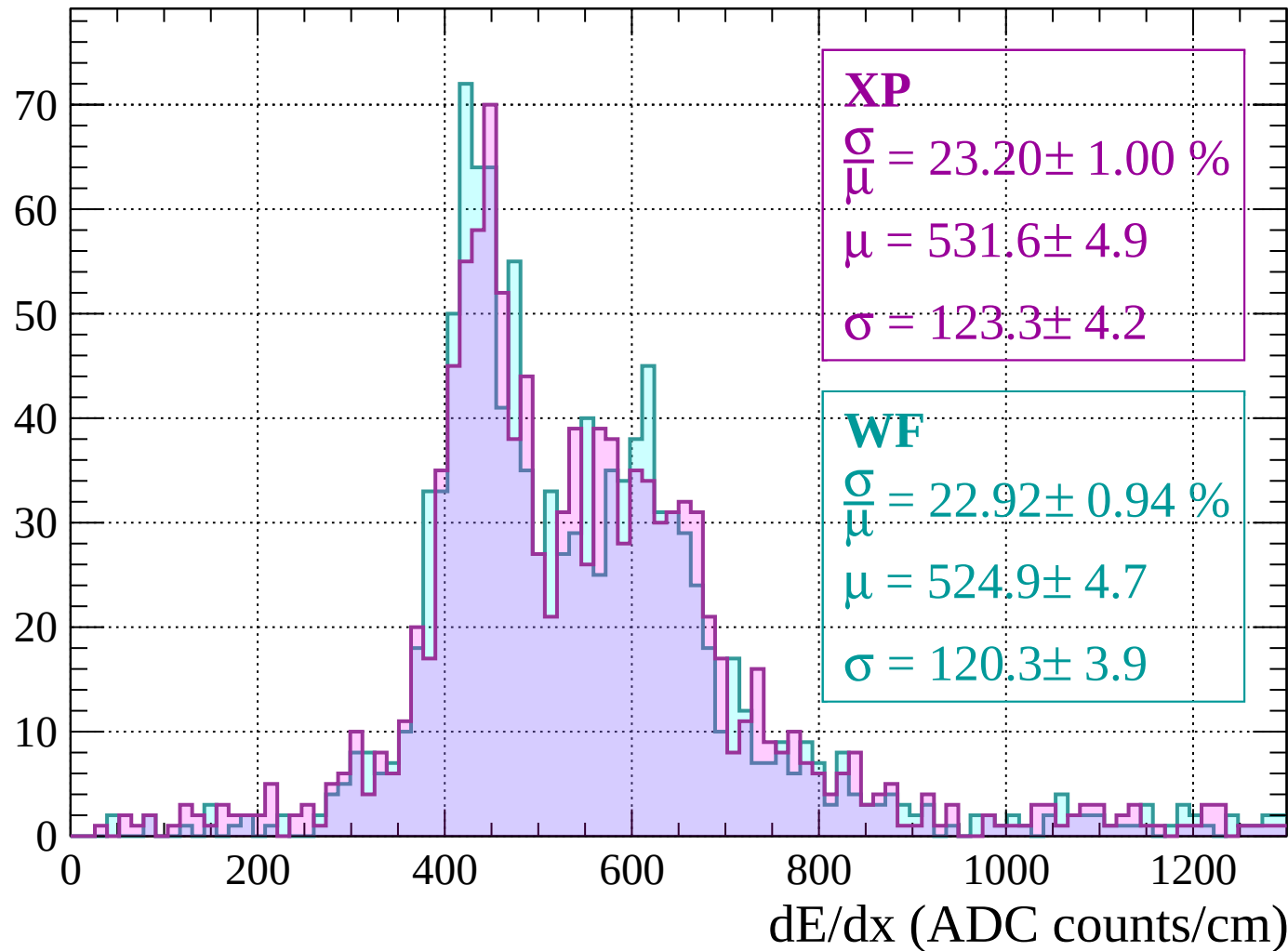
Energy loss | $960 < p < 1000$

Count



Energy loss | $1000 < p < 1040$

Count



Energy loss | $1040 < p < 1080$

Count

XP

$$\frac{\sigma}{\mu} = 20.78 \pm 0.92 \%$$

$$\mu = 509.9 \pm 4.4$$

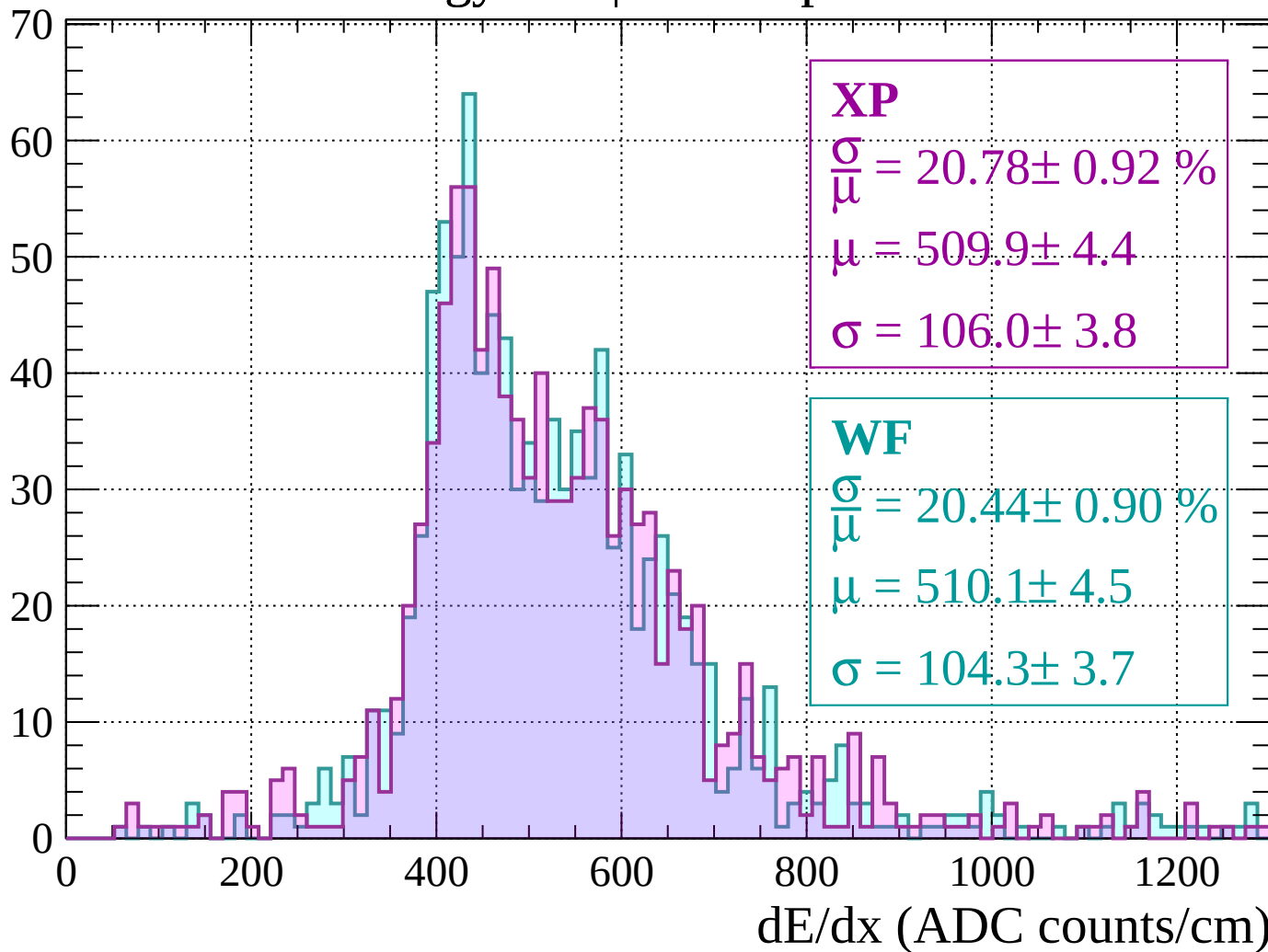
$$\sigma = 106.0 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 20.44 \pm 0.90 \%$$

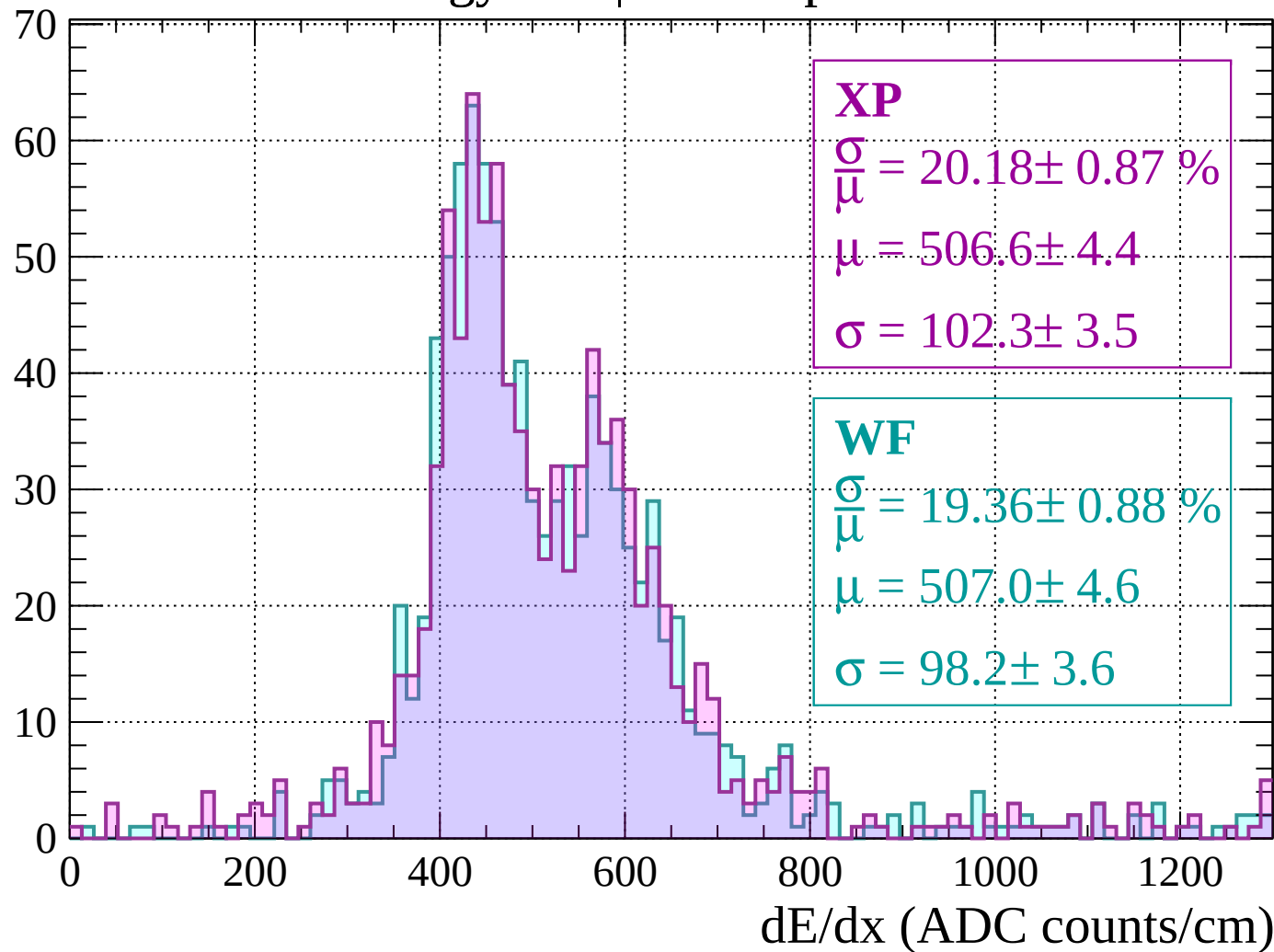
$$\mu = 510.1 \pm 4.5$$

$$\sigma = 104.3 \pm 3.7$$



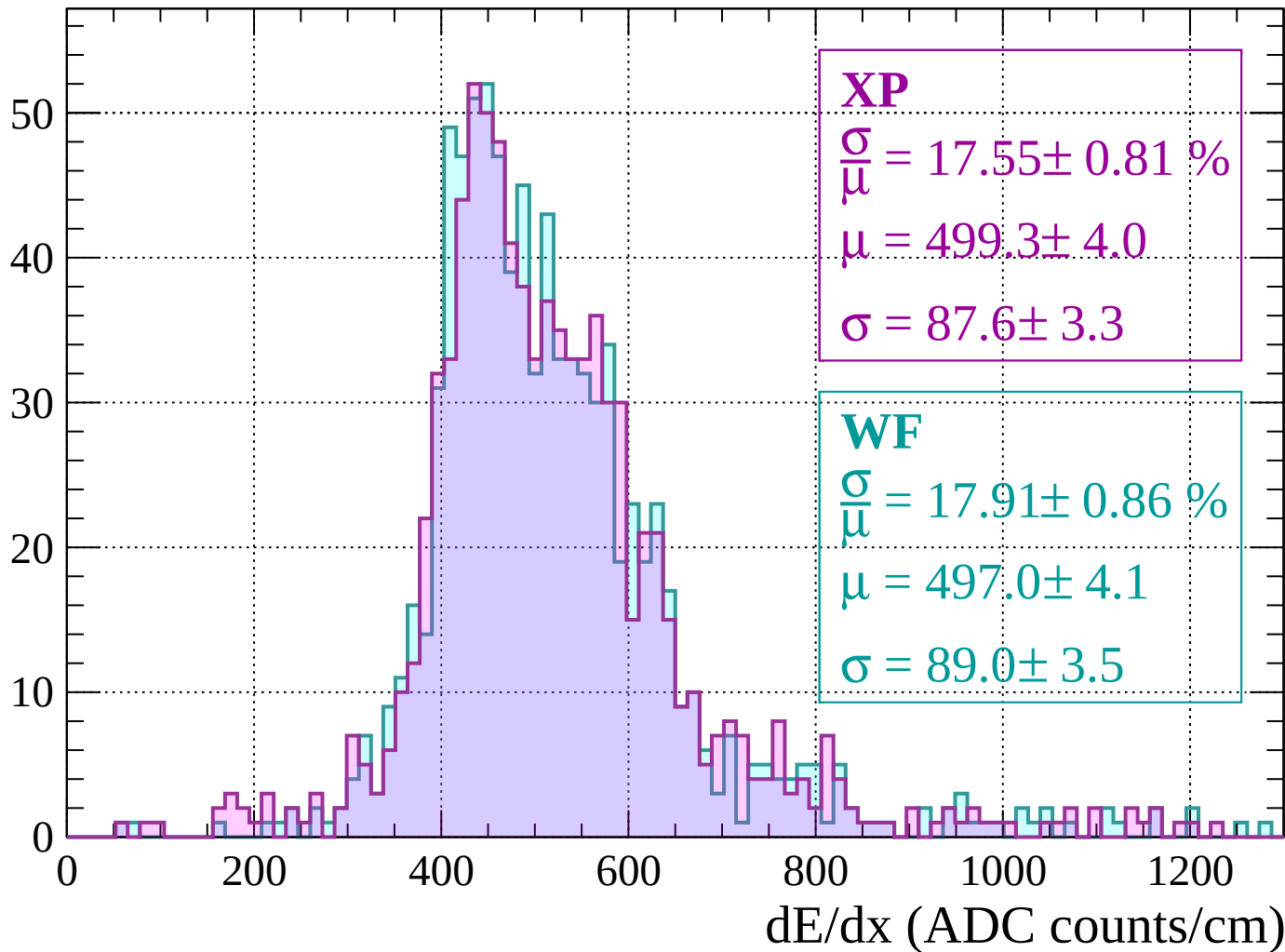
Energy loss | $1080 < p < 1120$

Count



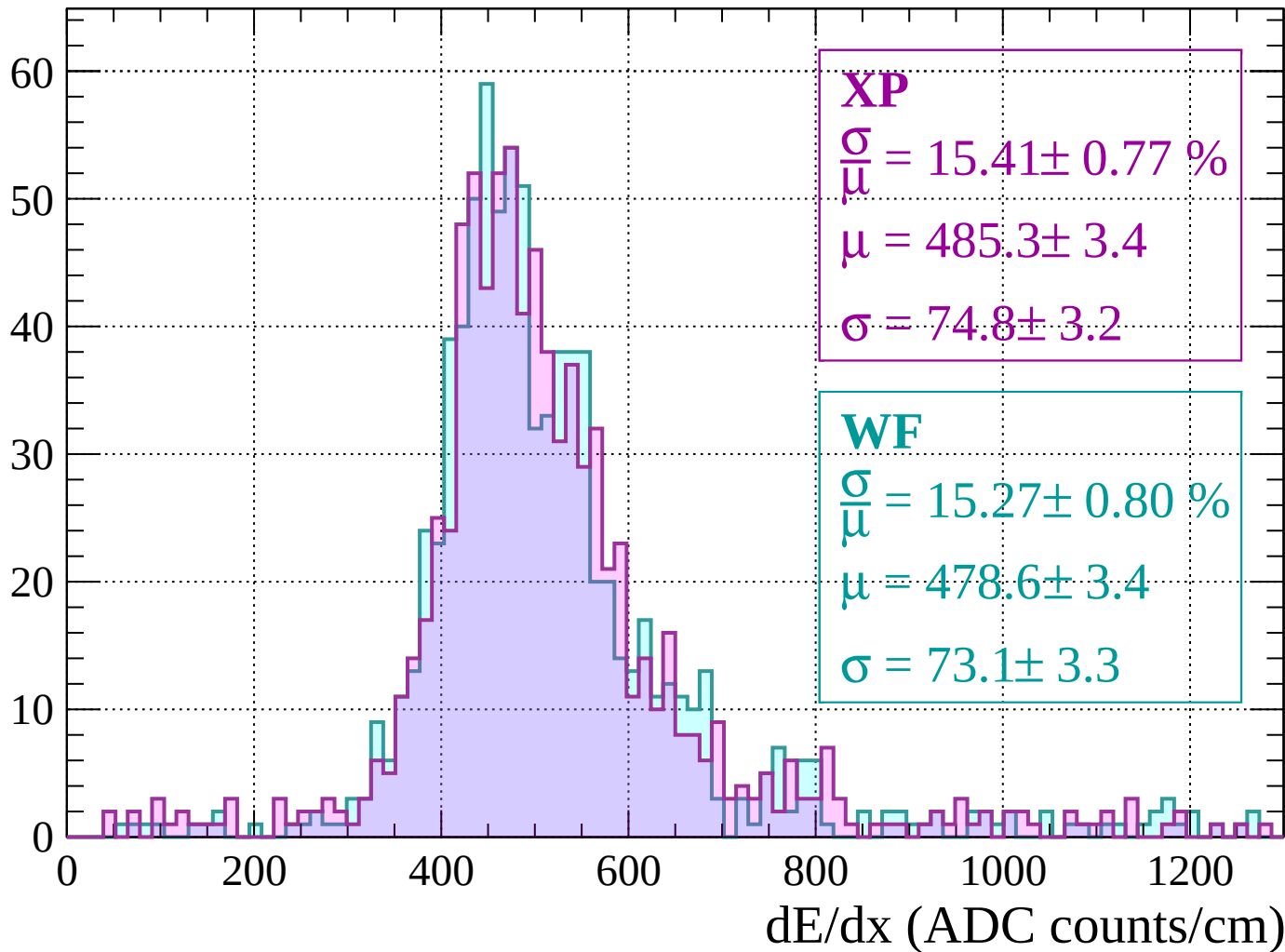
Energy loss | $1120 < p < 1160$

Count



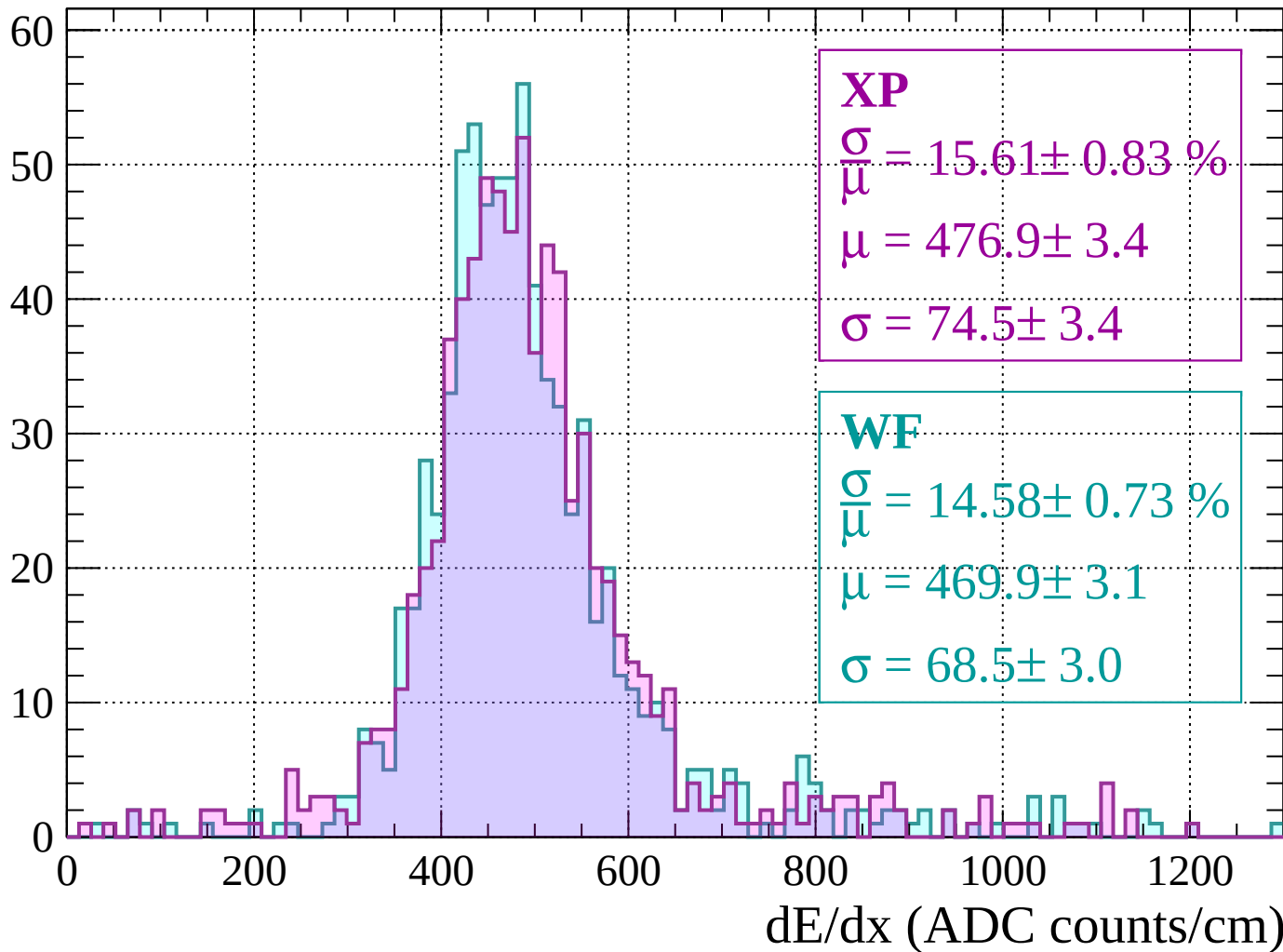
Energy loss | $1160 < p < 1200$

Count



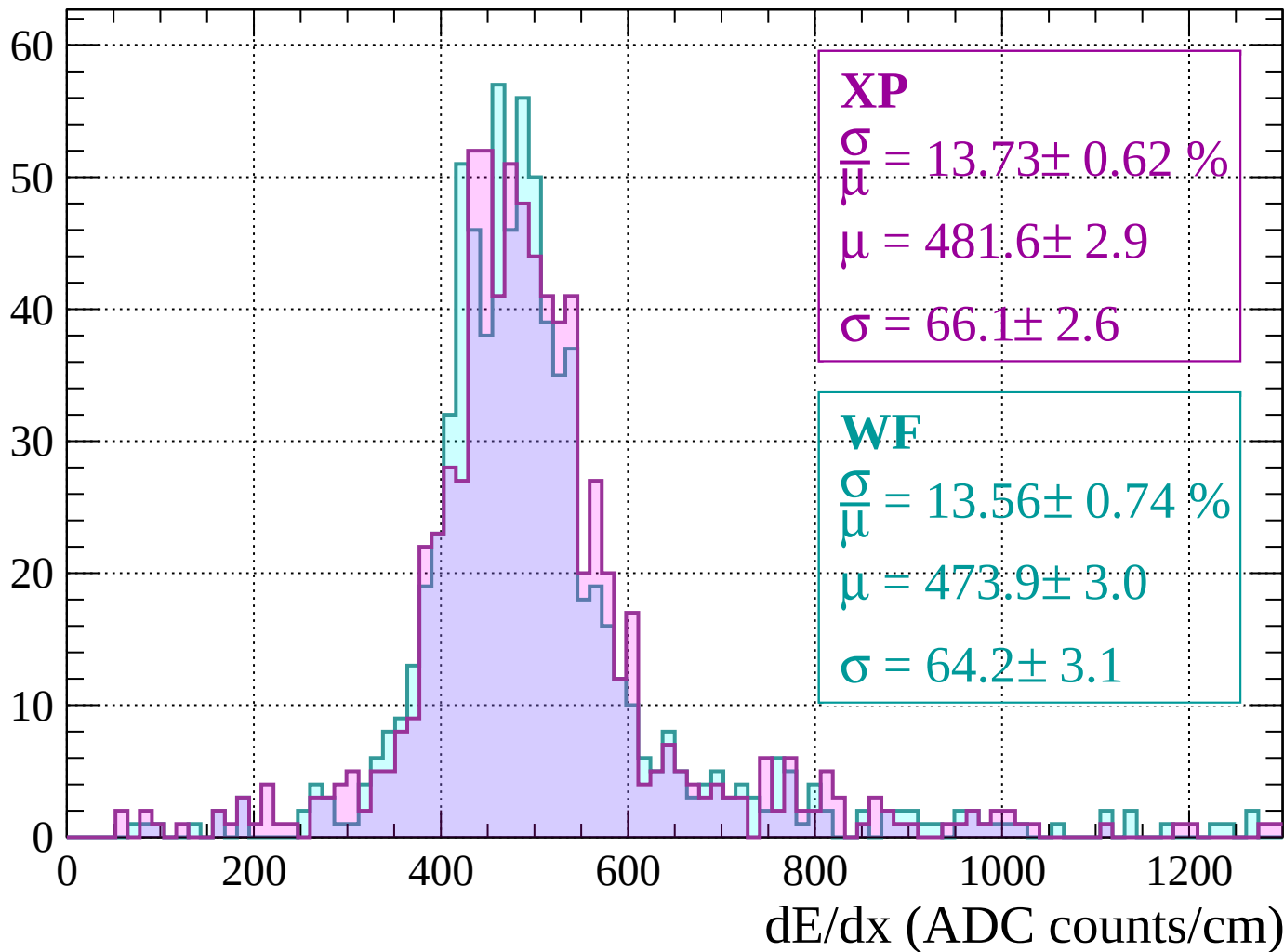
Energy loss | $1200 < p < 1240$

Count



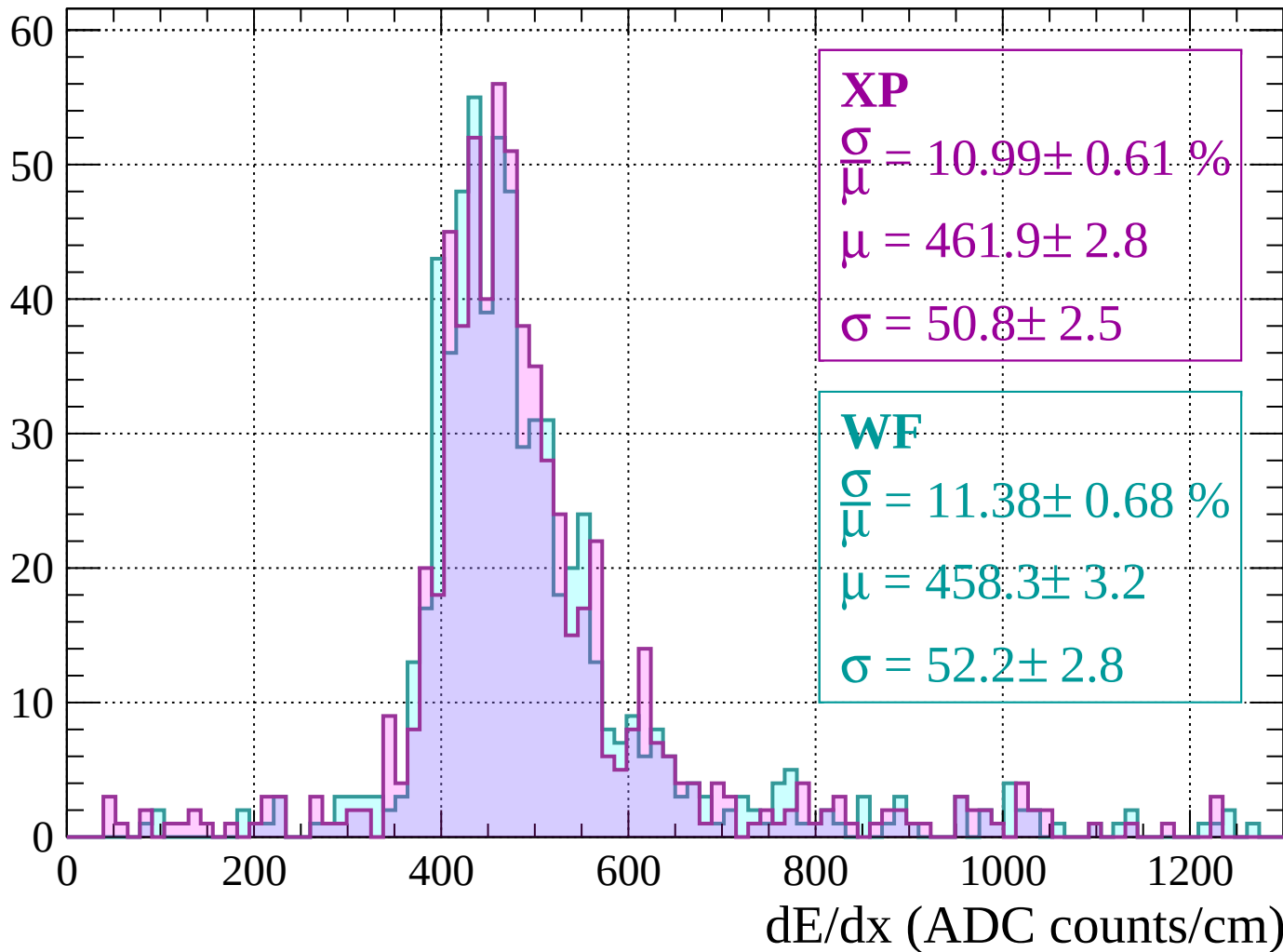
Energy loss | $1240 < p < 1280$

Count



Energy loss | $1280 < p < 1320$

Count



Energy loss | $1320 < p < 1360$

Count

XP

$$\frac{\sigma}{\mu} = 12.65 \pm 0.74 \%$$

$$\mu = 471.1 \pm 3.2$$

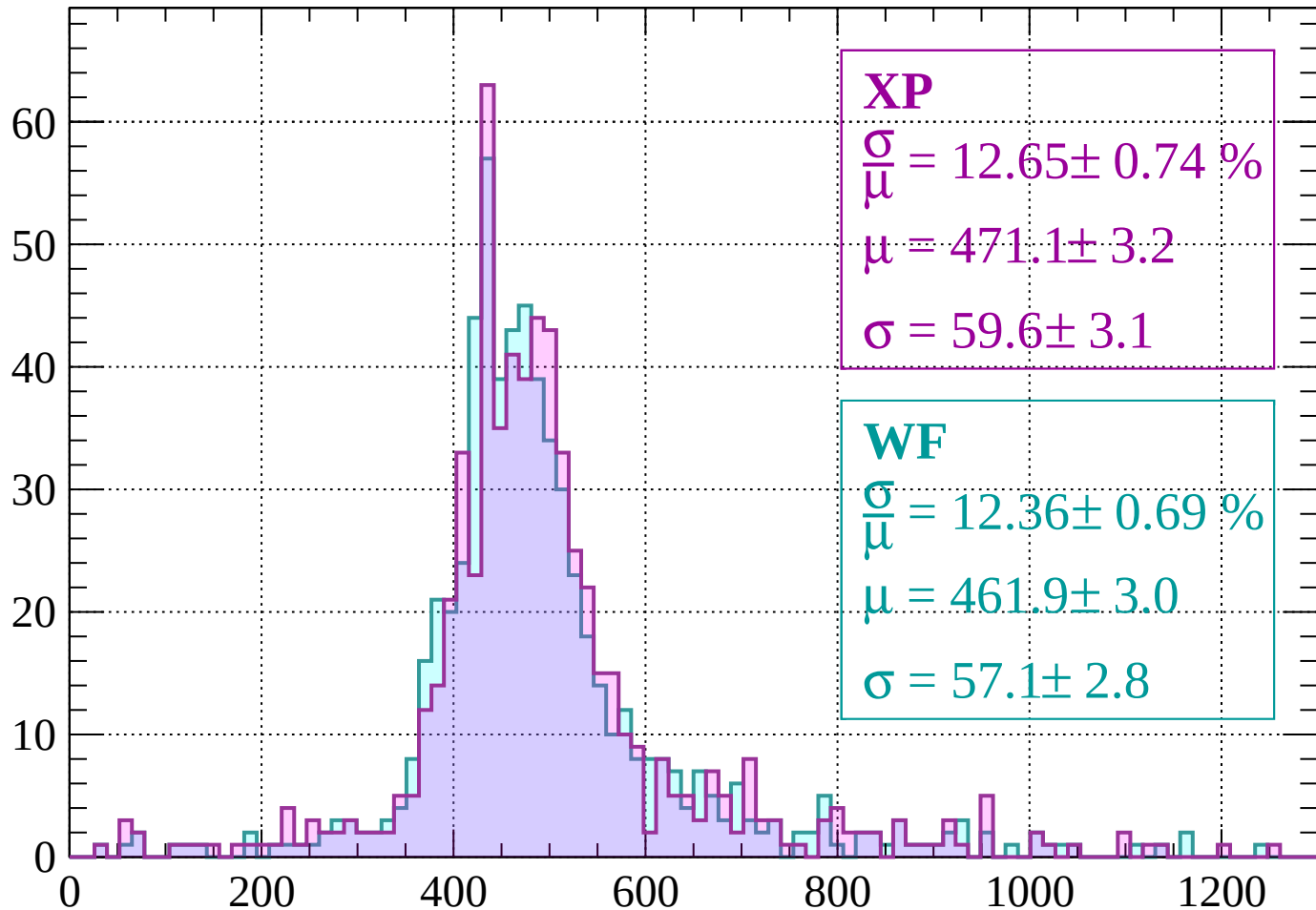
$$\sigma = 59.6 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 12.36 \pm 0.69 \%$$

$$\mu = 461.9 \pm 3.0$$

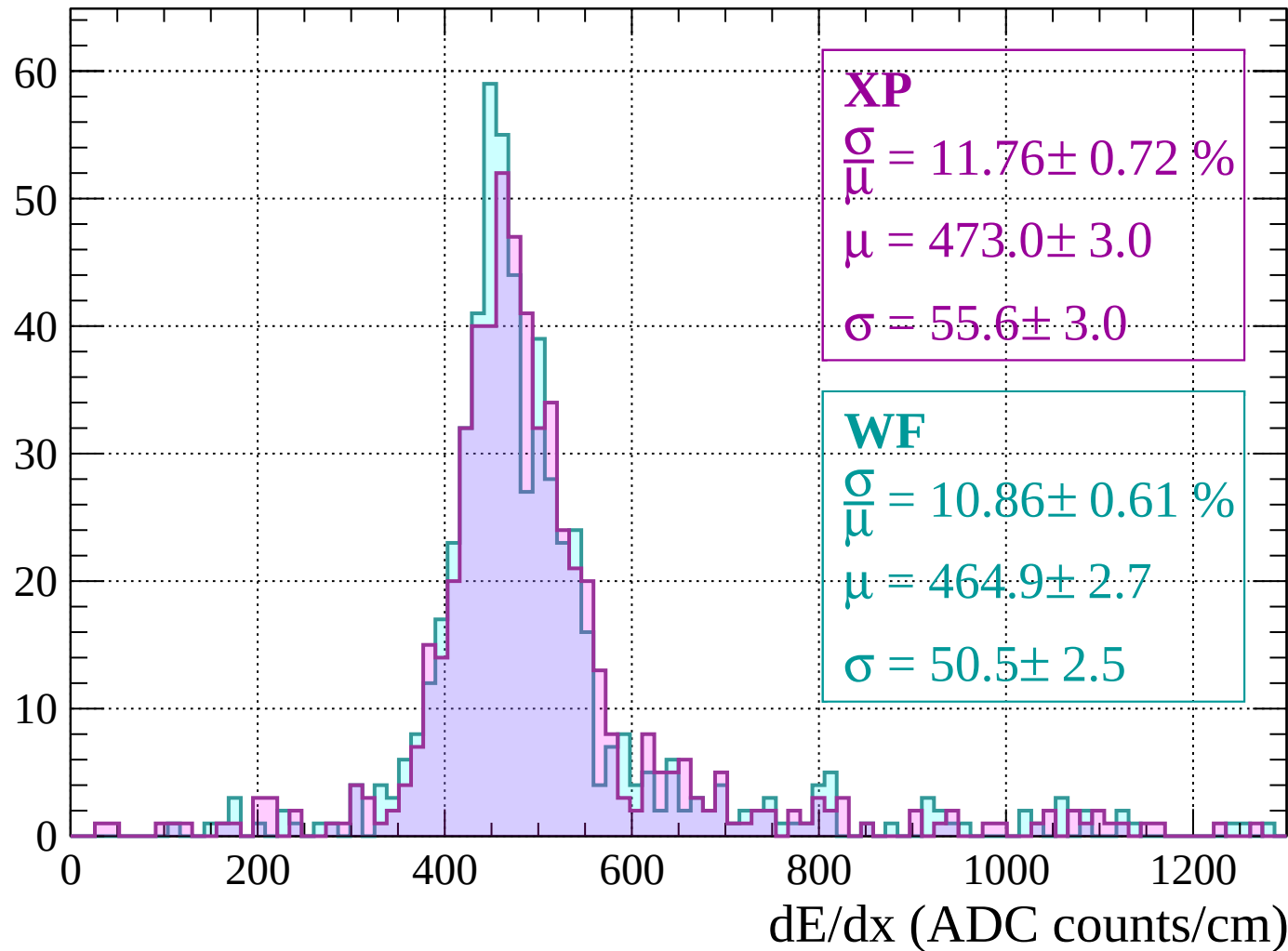
$$\sigma = 57.1 \pm 2.8$$



dE/dx (ADC counts/cm)

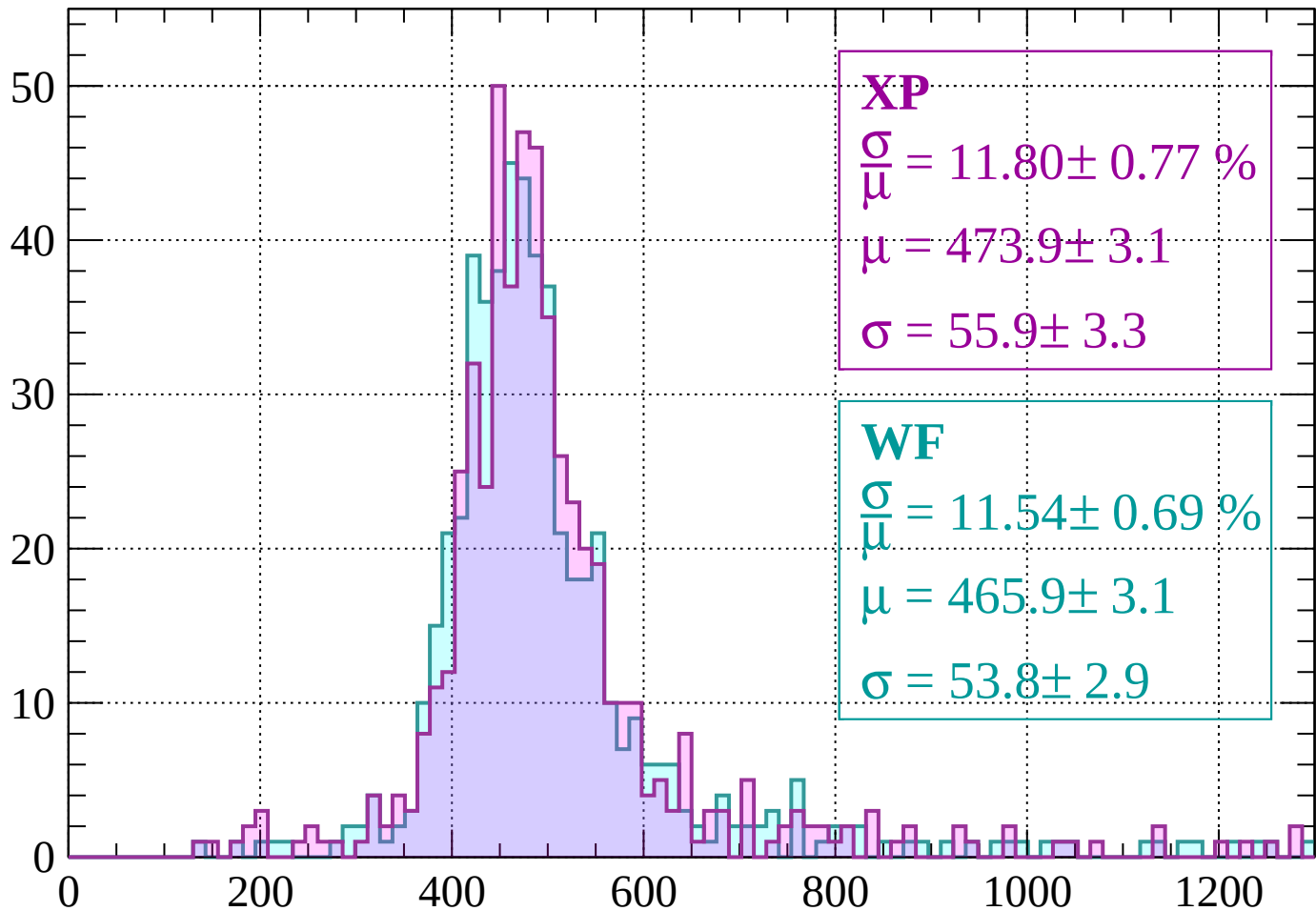
Energy loss | $1360 < p < 1400$

Count



Energy loss | $1400 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 11.80 \pm 0.77 \%$$

$$\mu = 473.9 \pm 3.1$$

$$\sigma = 55.9 \pm 3.3$$

WF

$$\frac{\sigma}{\mu} = 11.54 \pm 0.69 \%$$

$$\mu = 465.9 \pm 3.1$$

$$\sigma = 53.8 \pm 2.9$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1480$

Count

XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.72 \%$$

$$\mu = 463.4 \pm 2.8$$

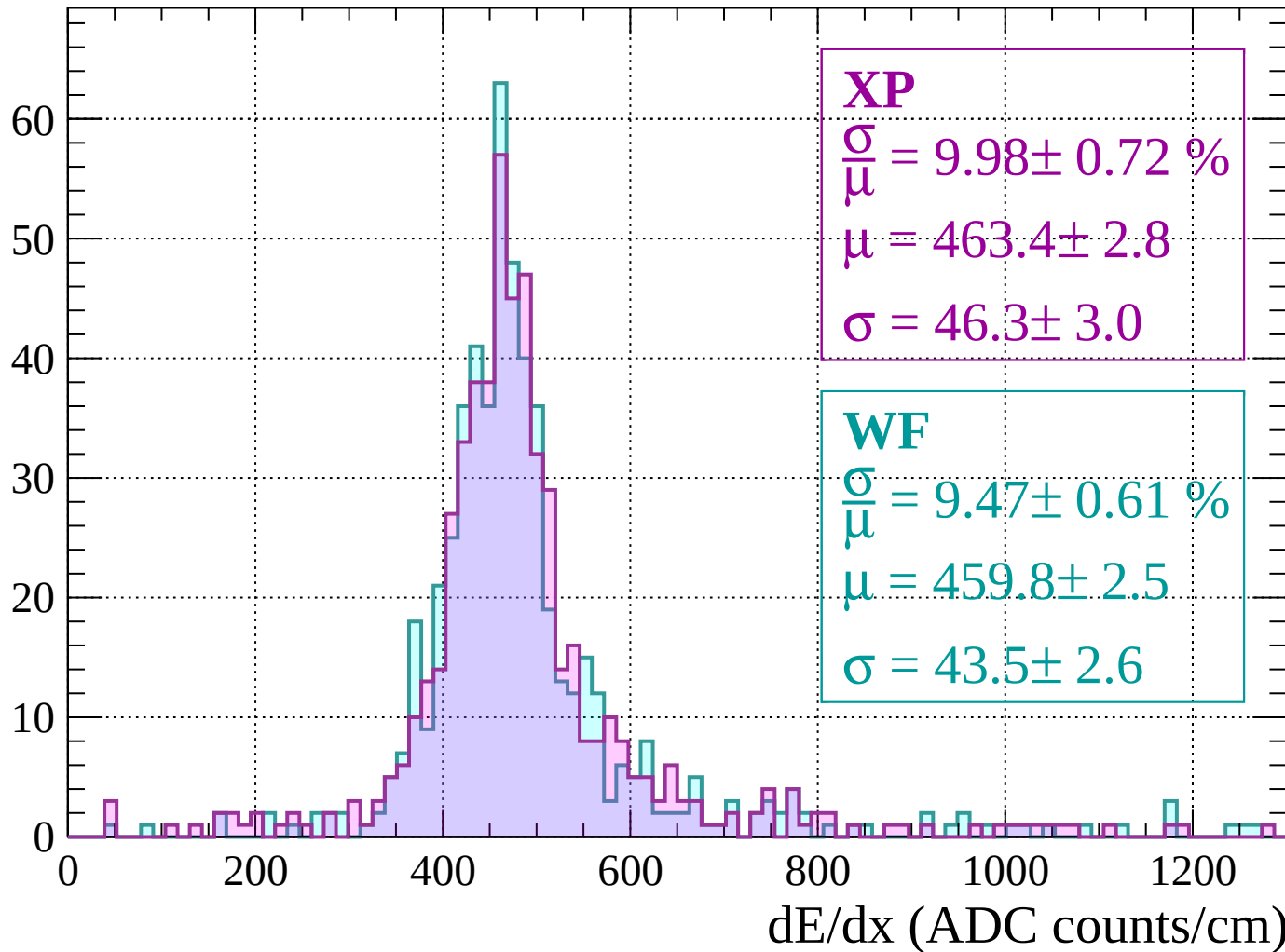
$$\sigma = 46.3 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 9.47 \pm 0.61 \%$$

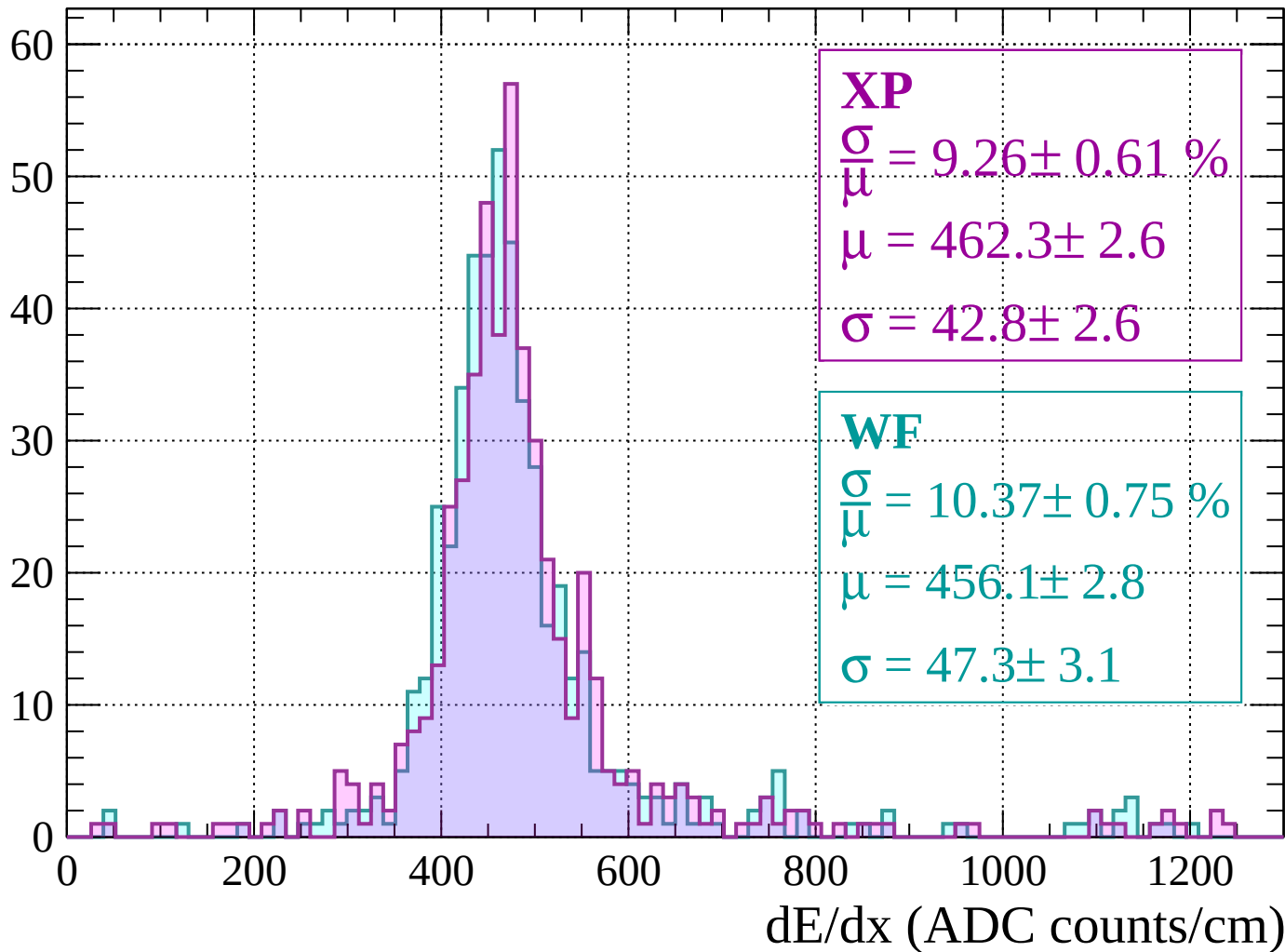
$$\mu = 459.8 \pm 2.5$$

$$\sigma = 43.5 \pm 2.6$$



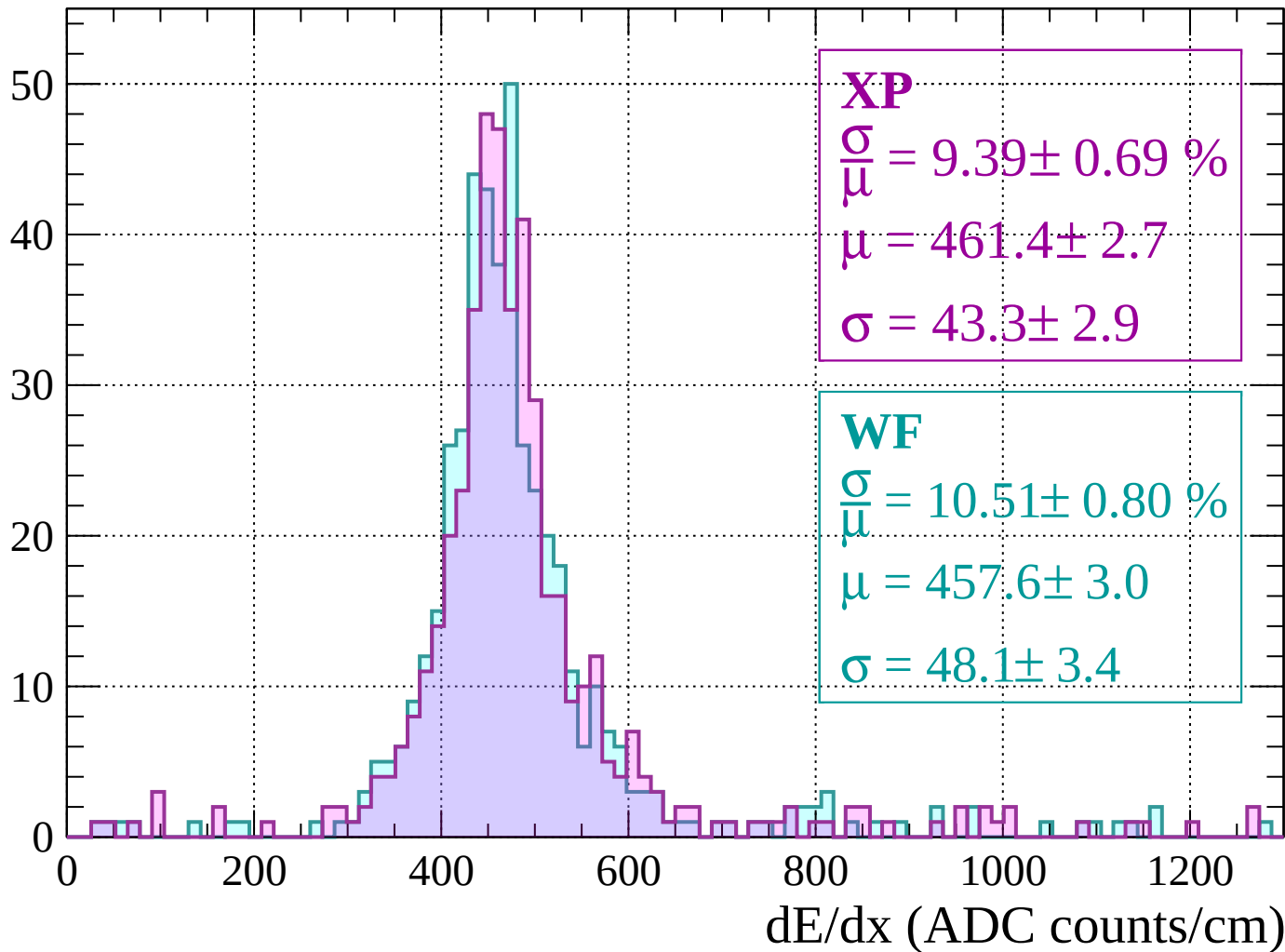
Energy loss | $1480 < p < 1520$

Count



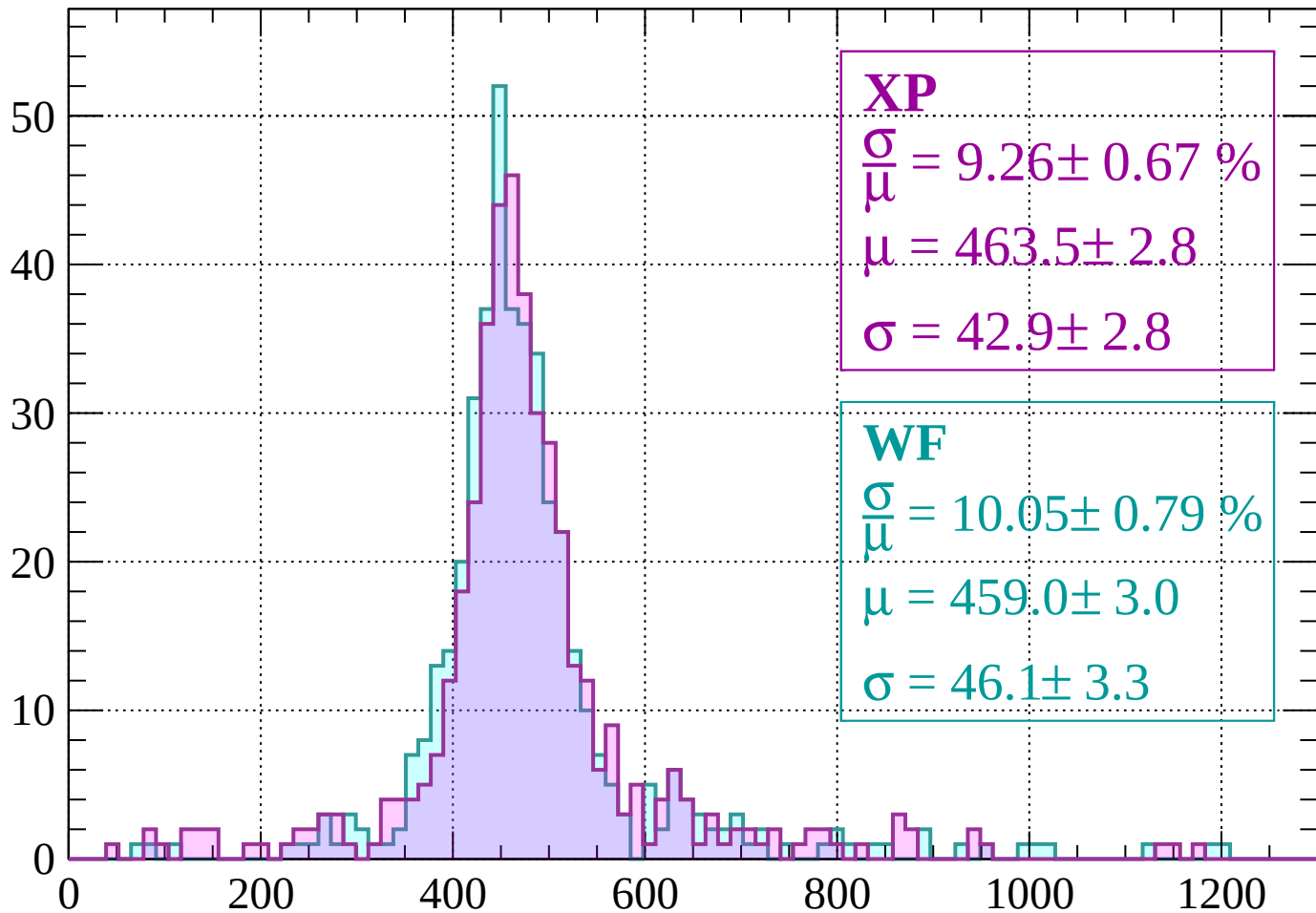
Energy loss | $1520 < p < 1560$

Count



Energy loss | $1560 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.67 \%$$

$$\mu = 463.5 \pm 2.8$$

$$\sigma = 42.9 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 10.05 \pm 0.79 \%$$

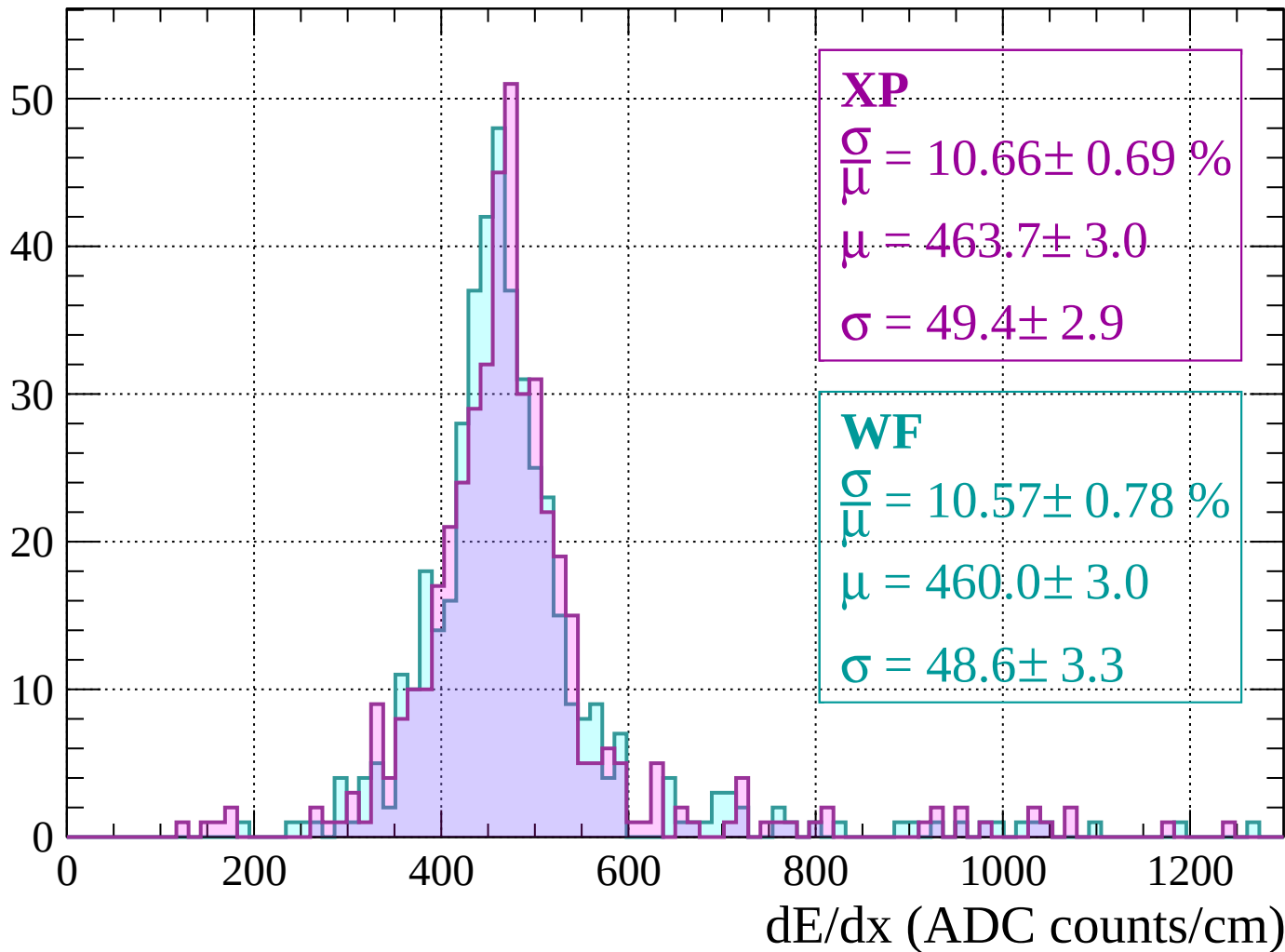
$$\mu = 459.0 \pm 3.0$$

$$\sigma = 46.1 \pm 3.3$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1640$

Count



Energy loss | $1640 < p < 1680$

Count

XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.65 \%$$

$$\mu = 462.9 \pm 2.9$$

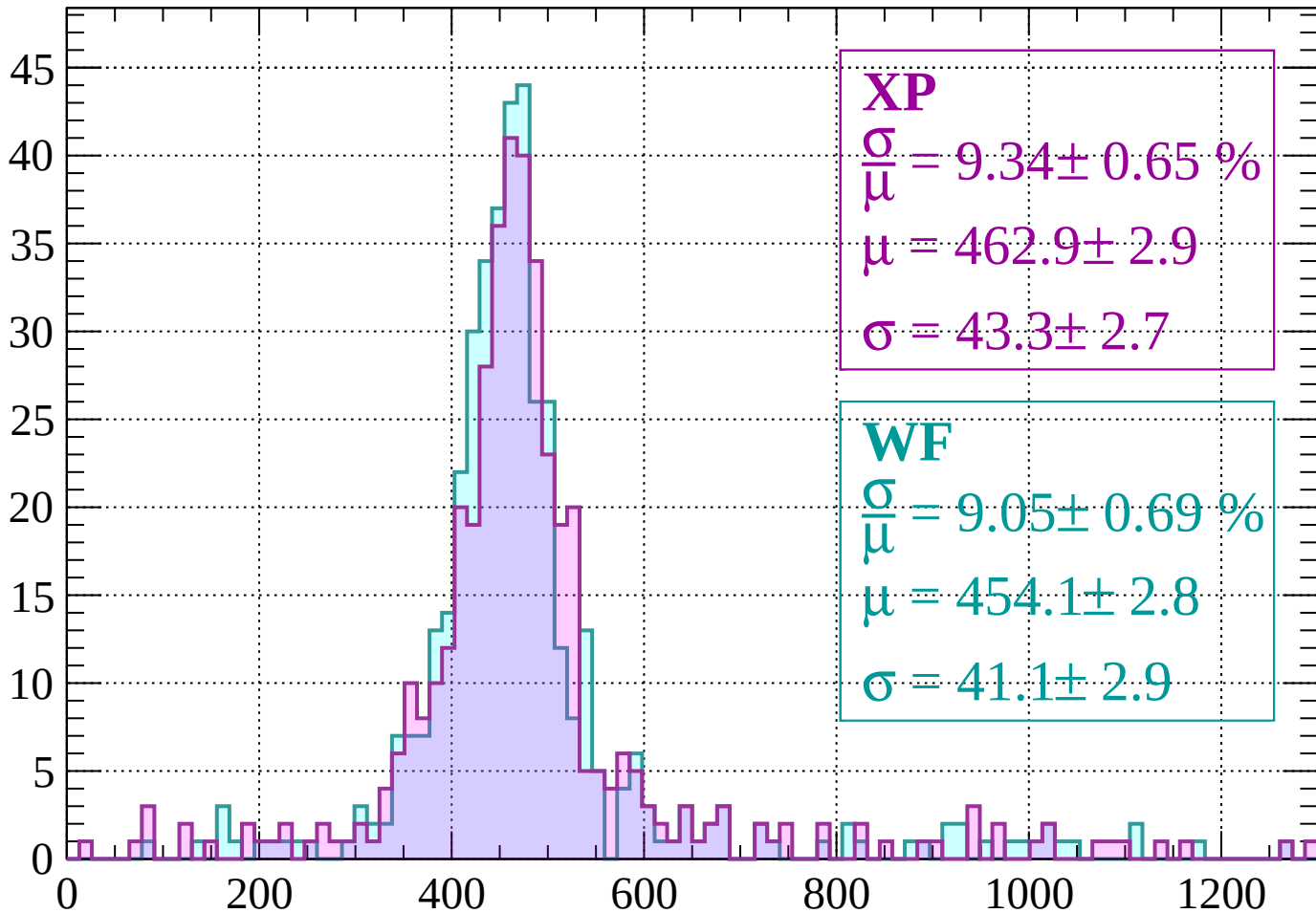
$$\sigma = 43.3 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.69 \%$$

$$\mu = 454.1 \pm 2.8$$

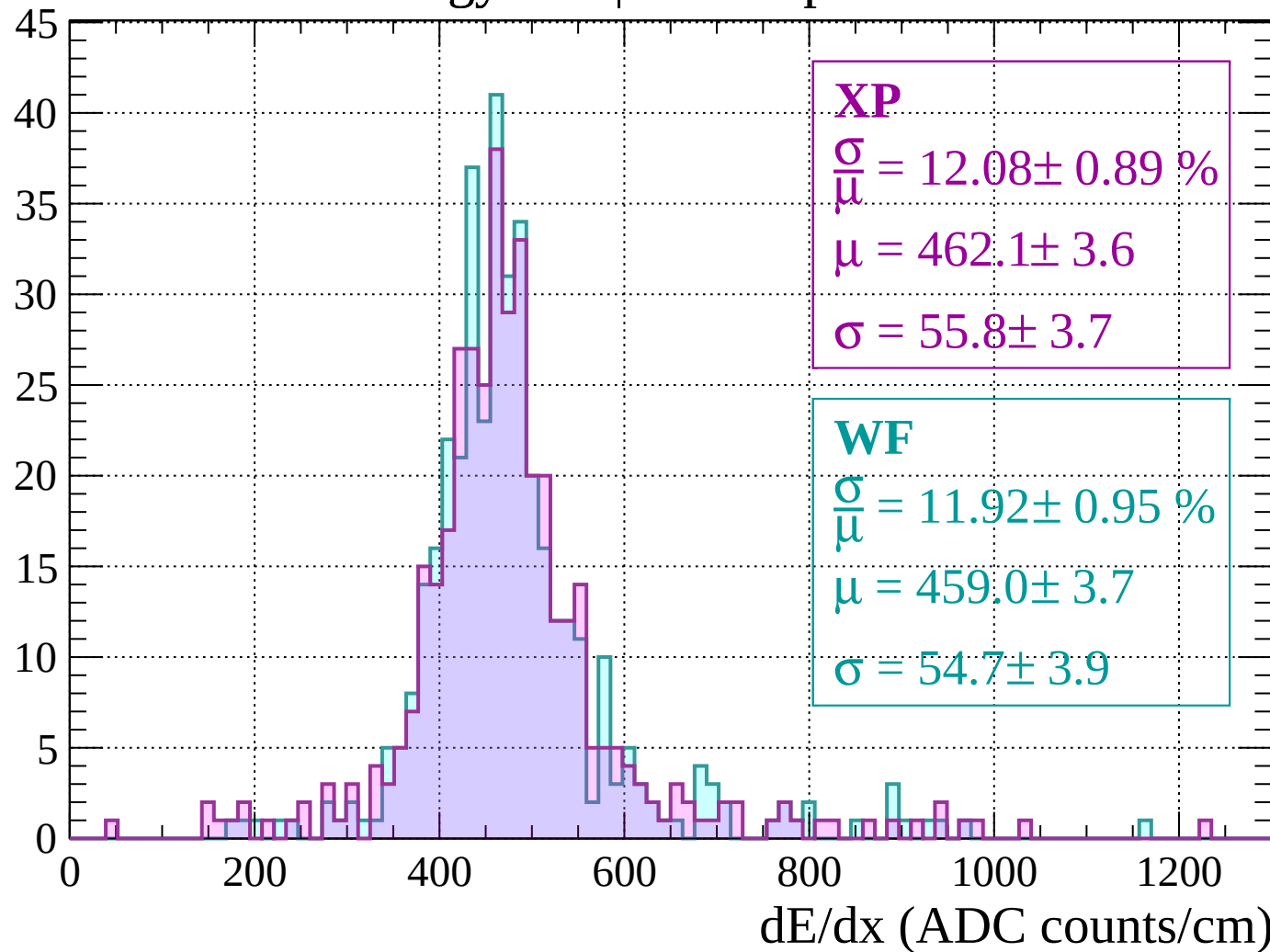
$$\sigma = 41.1 \pm 2.9$$



dE/dx (ADC counts/cm)

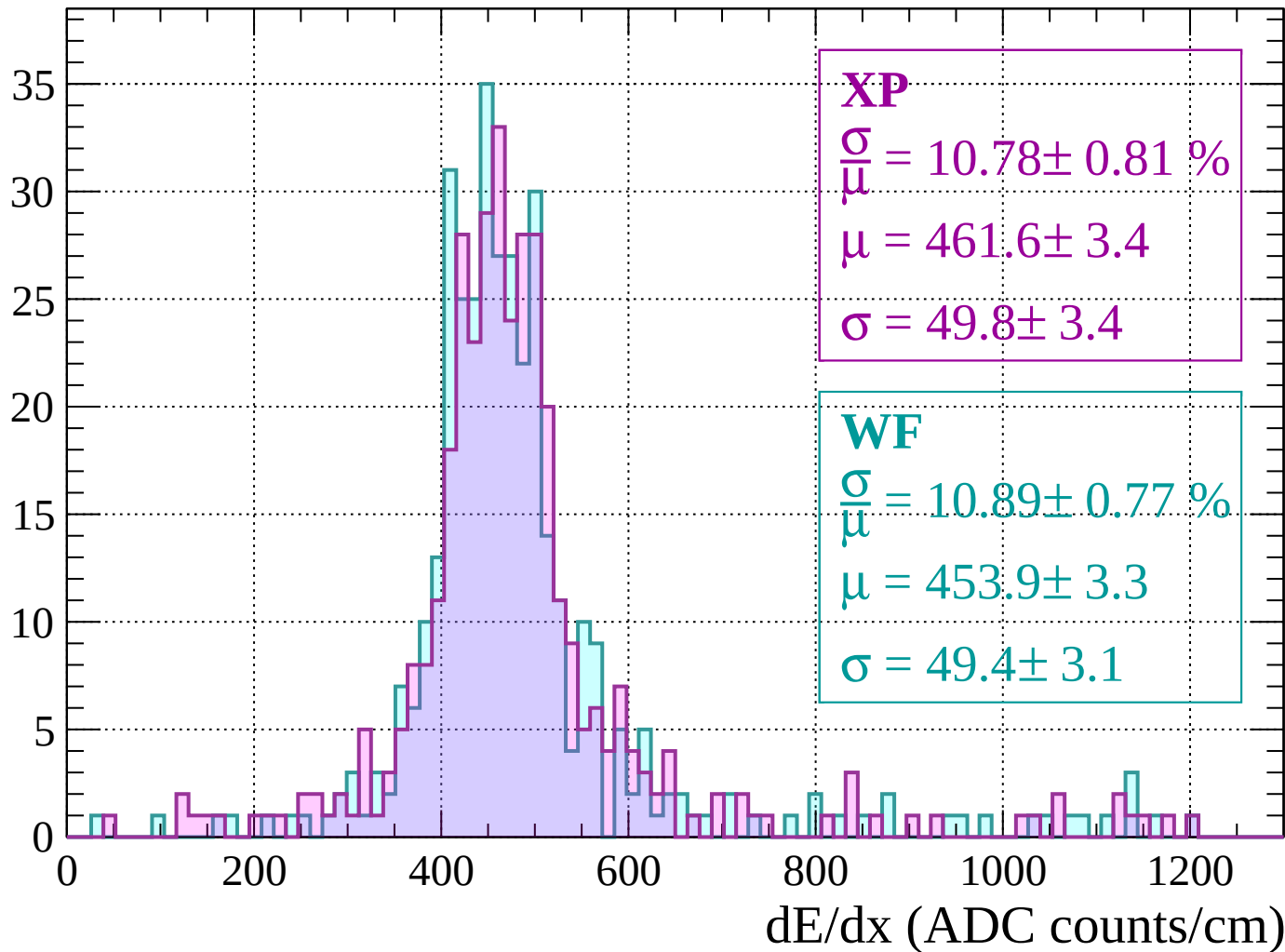
Energy loss | $1680 < p < 1720$

Count



Energy loss | $1720 < p < 1760$

Count



Energy loss | $1760 < p < 1800$

Count

XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.65 \%$$

$$\mu = 462.6 \pm 3.2$$

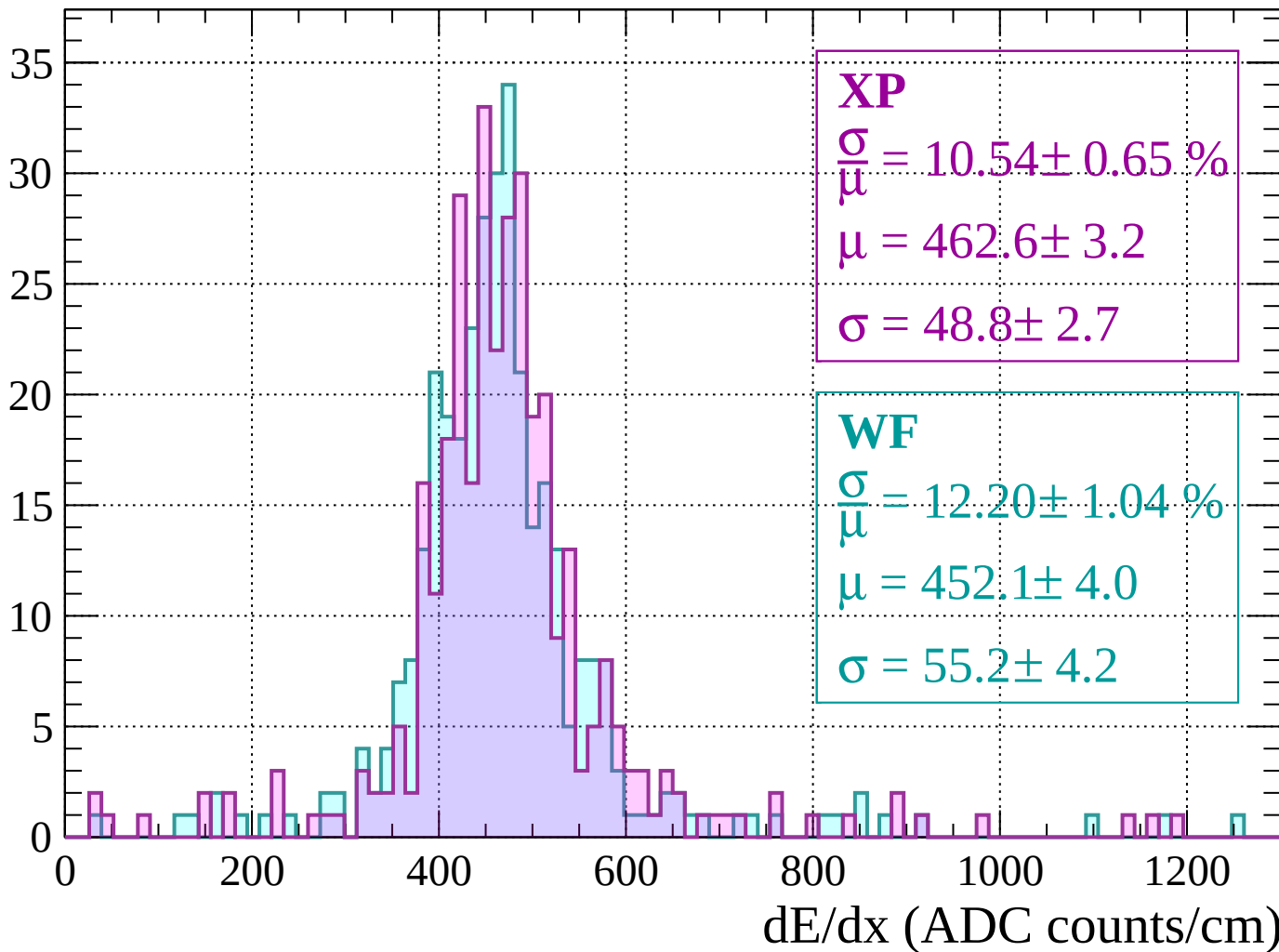
$$\sigma = 48.8 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 12.20 \pm 1.04 \%$$

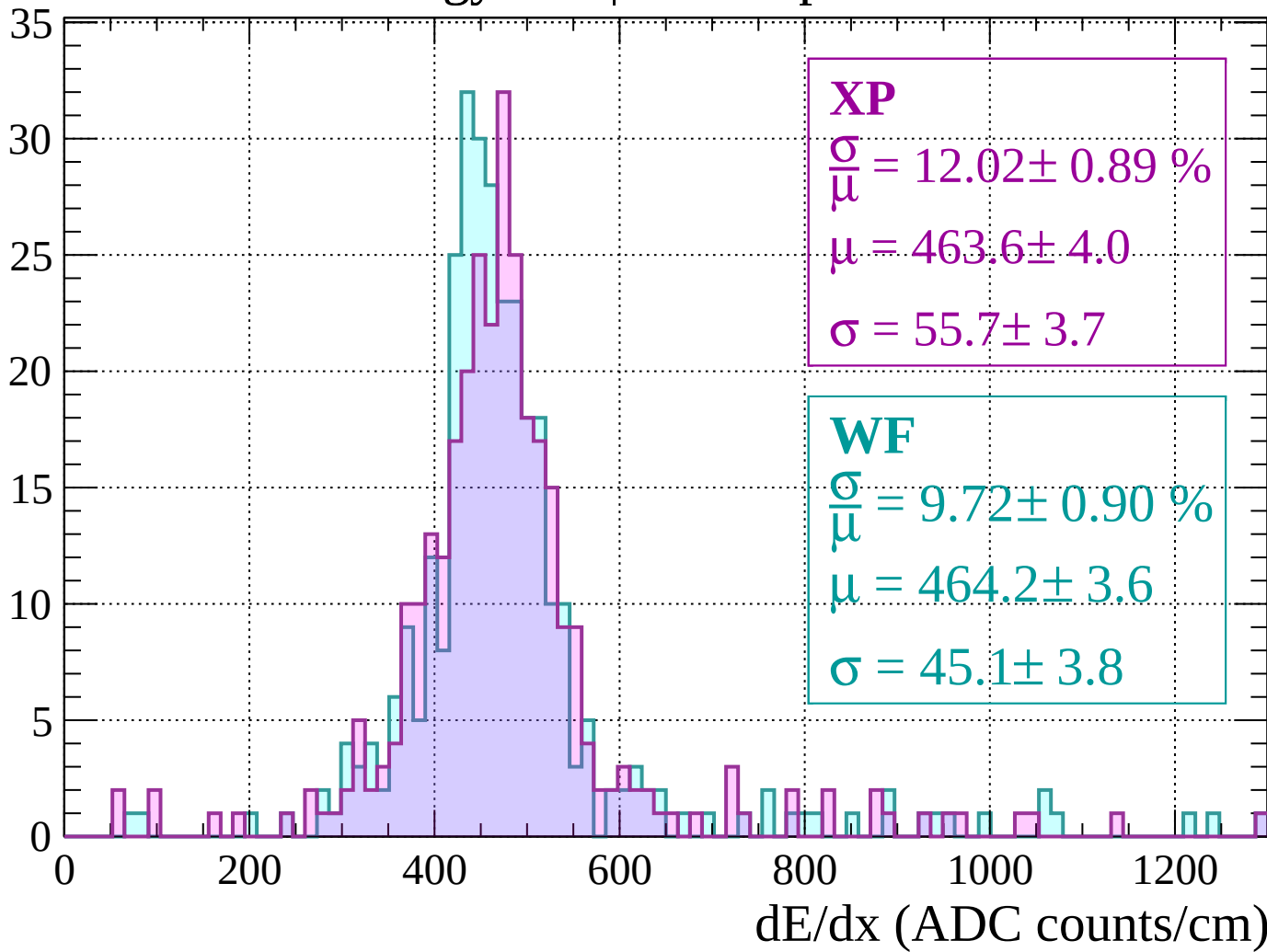
$$\mu = 452.1 \pm 4.0$$

$$\sigma = 55.2 \pm 4.2$$



Energy loss | $1800 < p < 1840$

Count



Energy loss | $1840 < p < 1880$

Count

XP

$$\frac{\sigma}{\mu} = 11.45 \pm 0.83 \%$$

$$\mu = 461.7 \pm 3.8$$

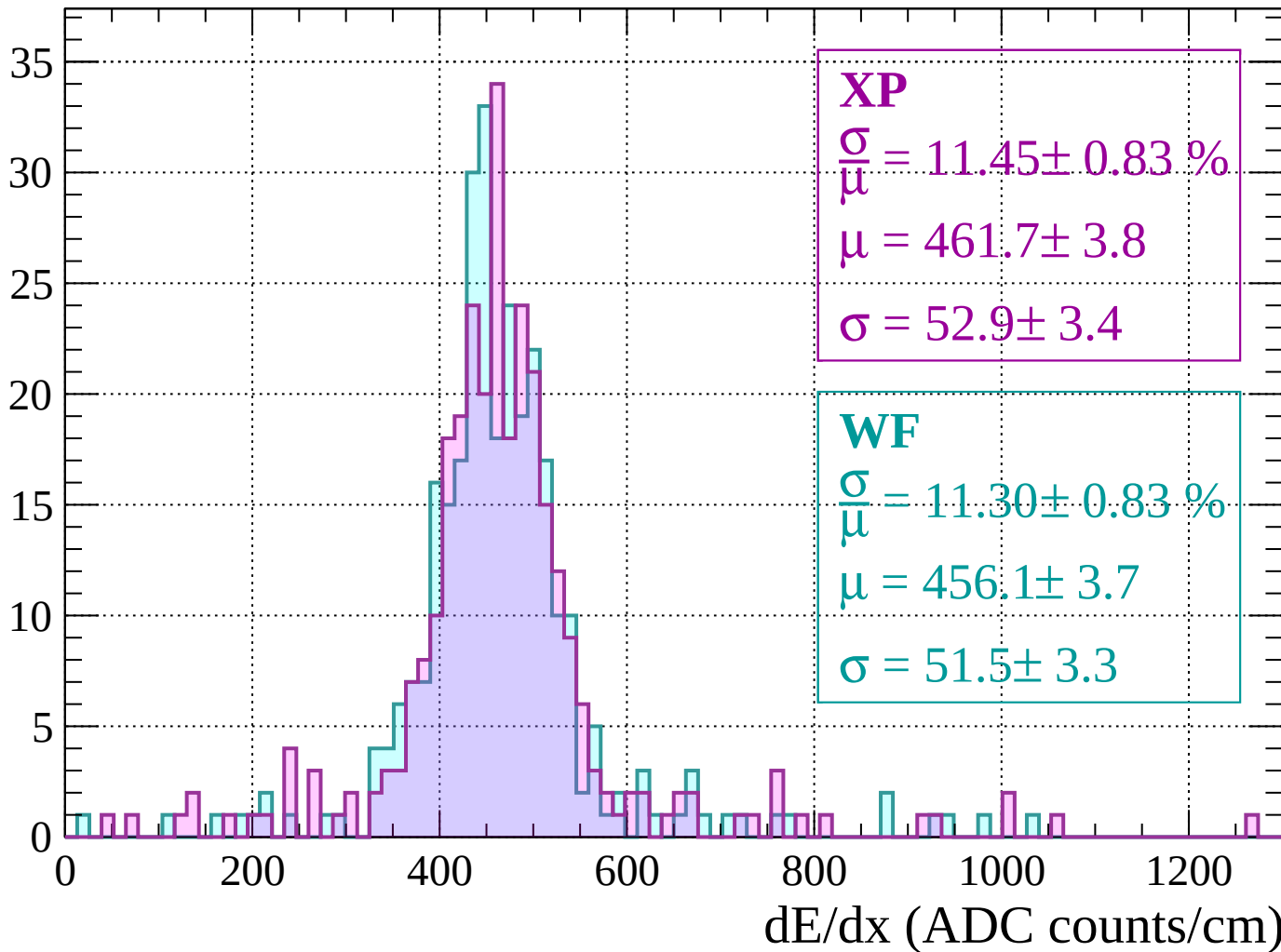
$$\sigma = 52.9 \pm 3.4$$

WF

$$\frac{\sigma}{\mu} = 11.30 \pm 0.83 \%$$

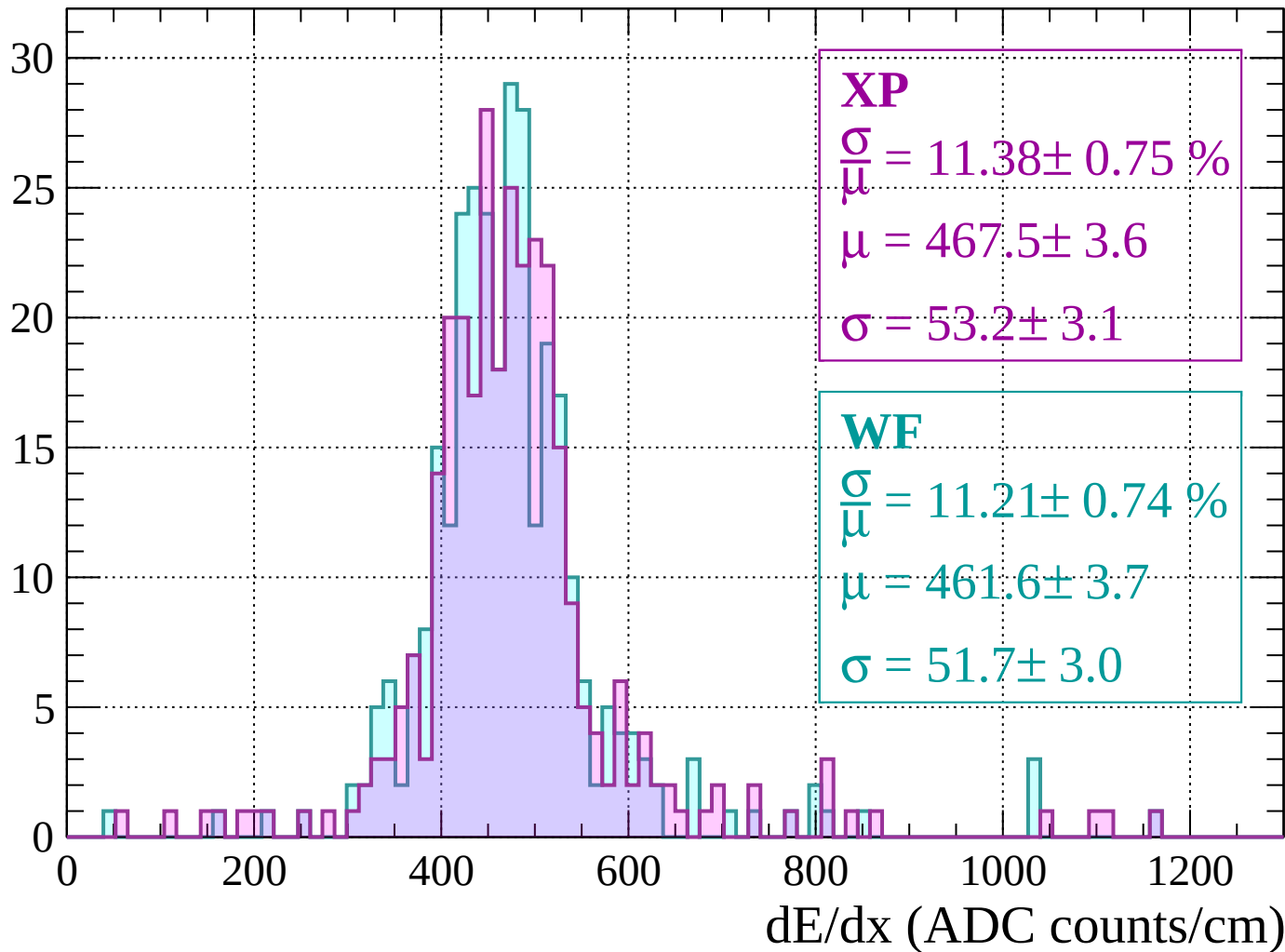
$$\mu = 456.1 \pm 3.7$$

$$\sigma = 51.5 \pm 3.3$$



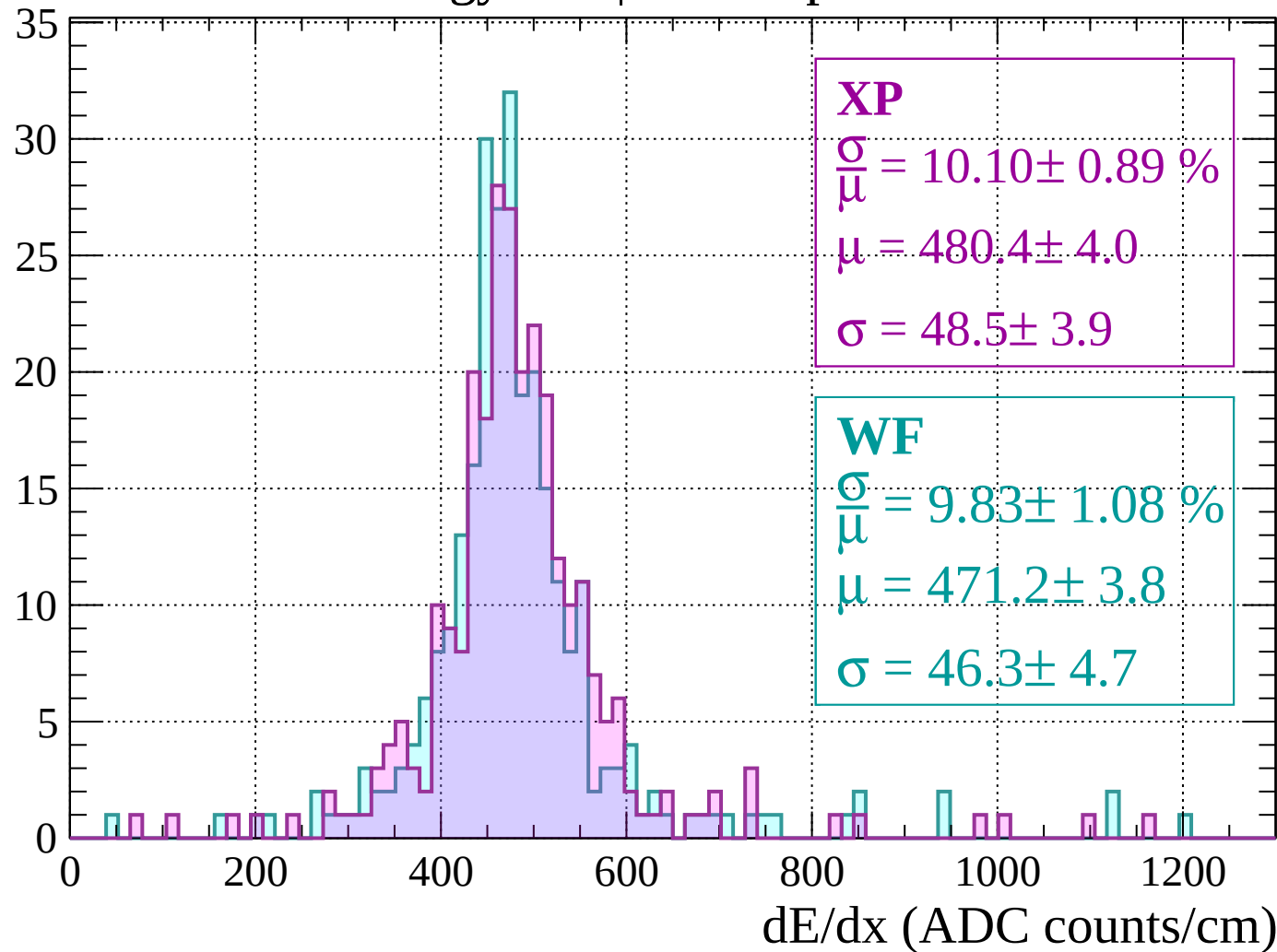
Energy loss | $1880 < p < 1920$

Count



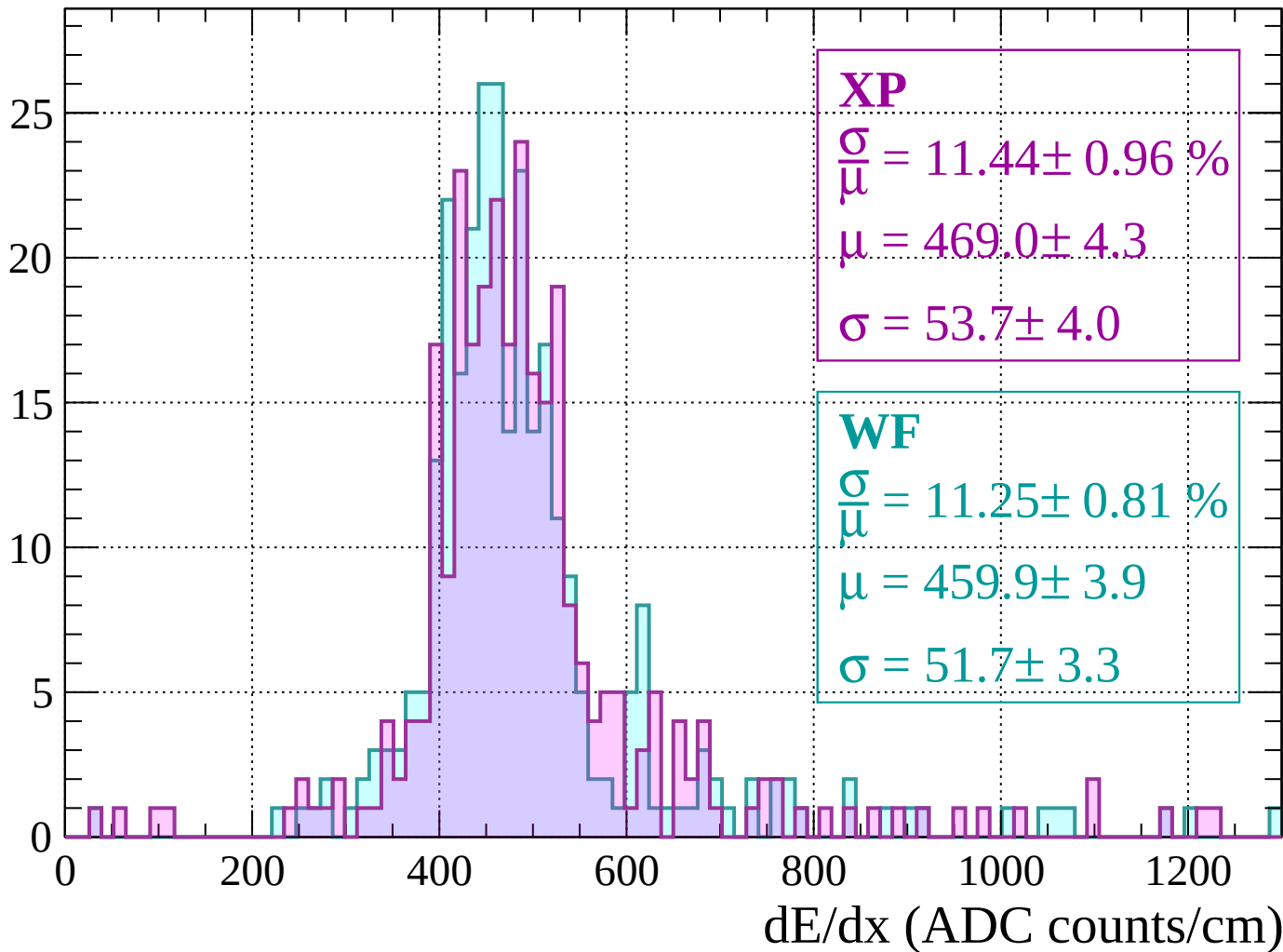
Energy loss | $1920 < p < 1960$

Count



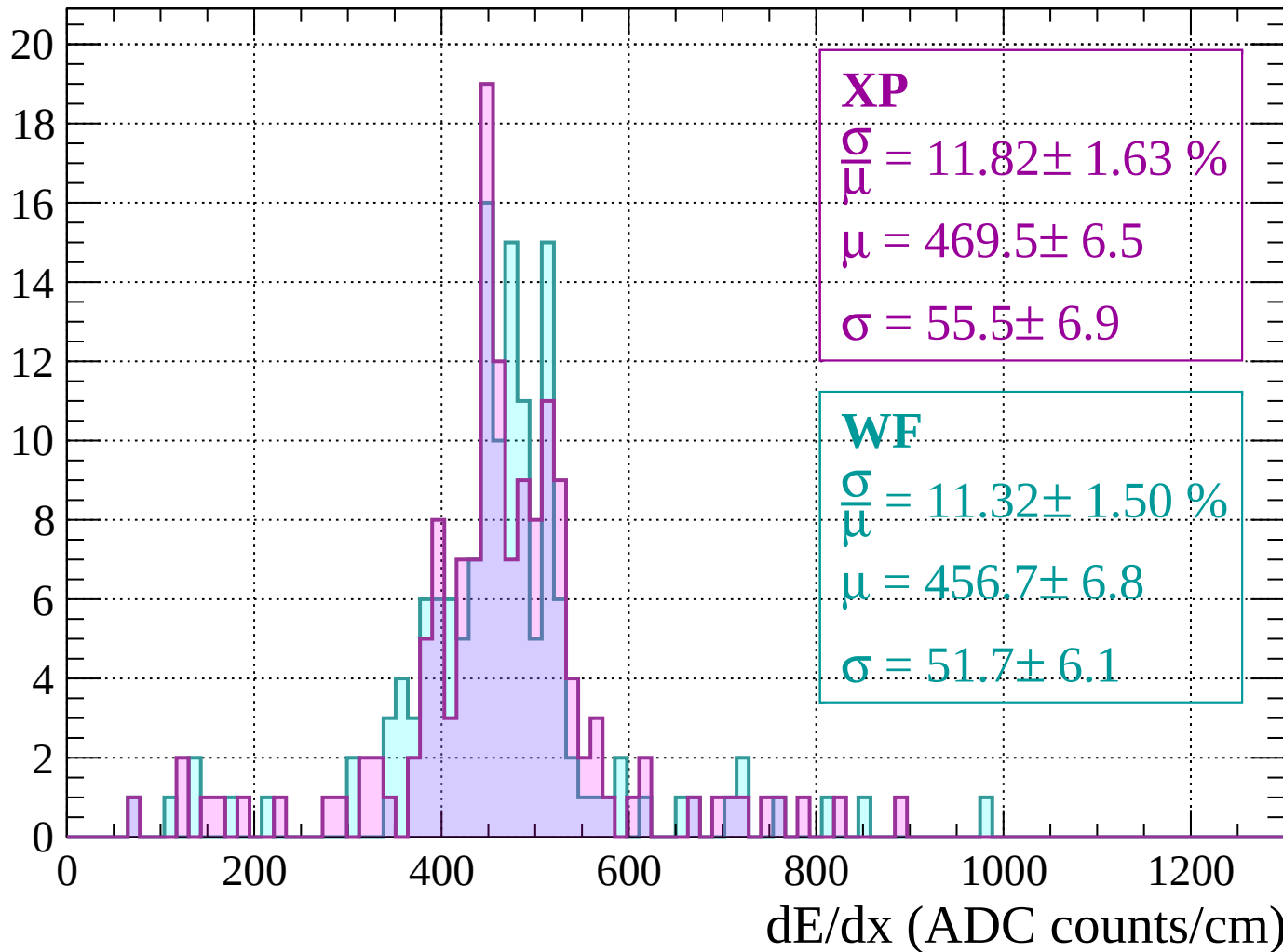
Energy loss | 1960 < p < 2000

Count



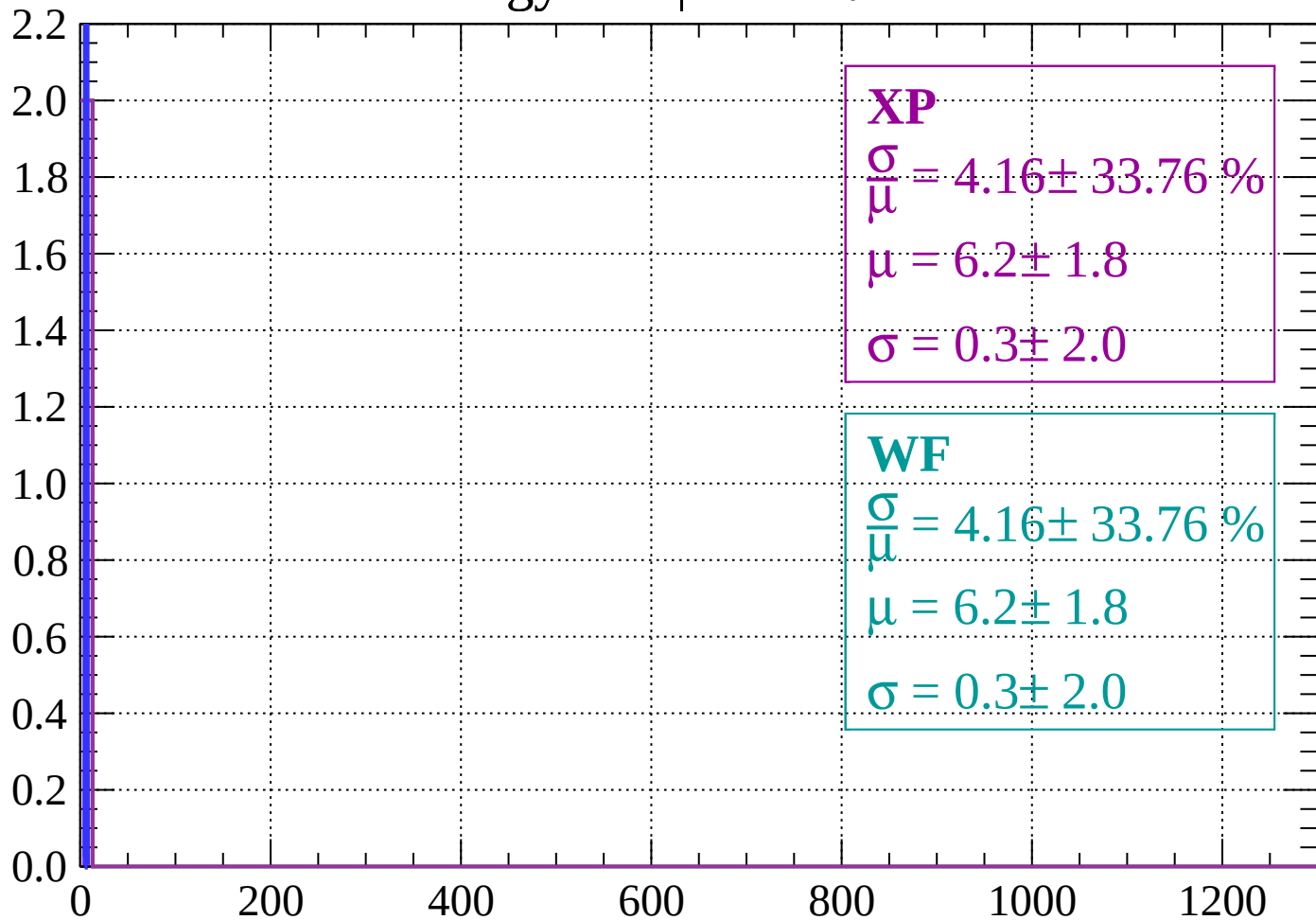
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

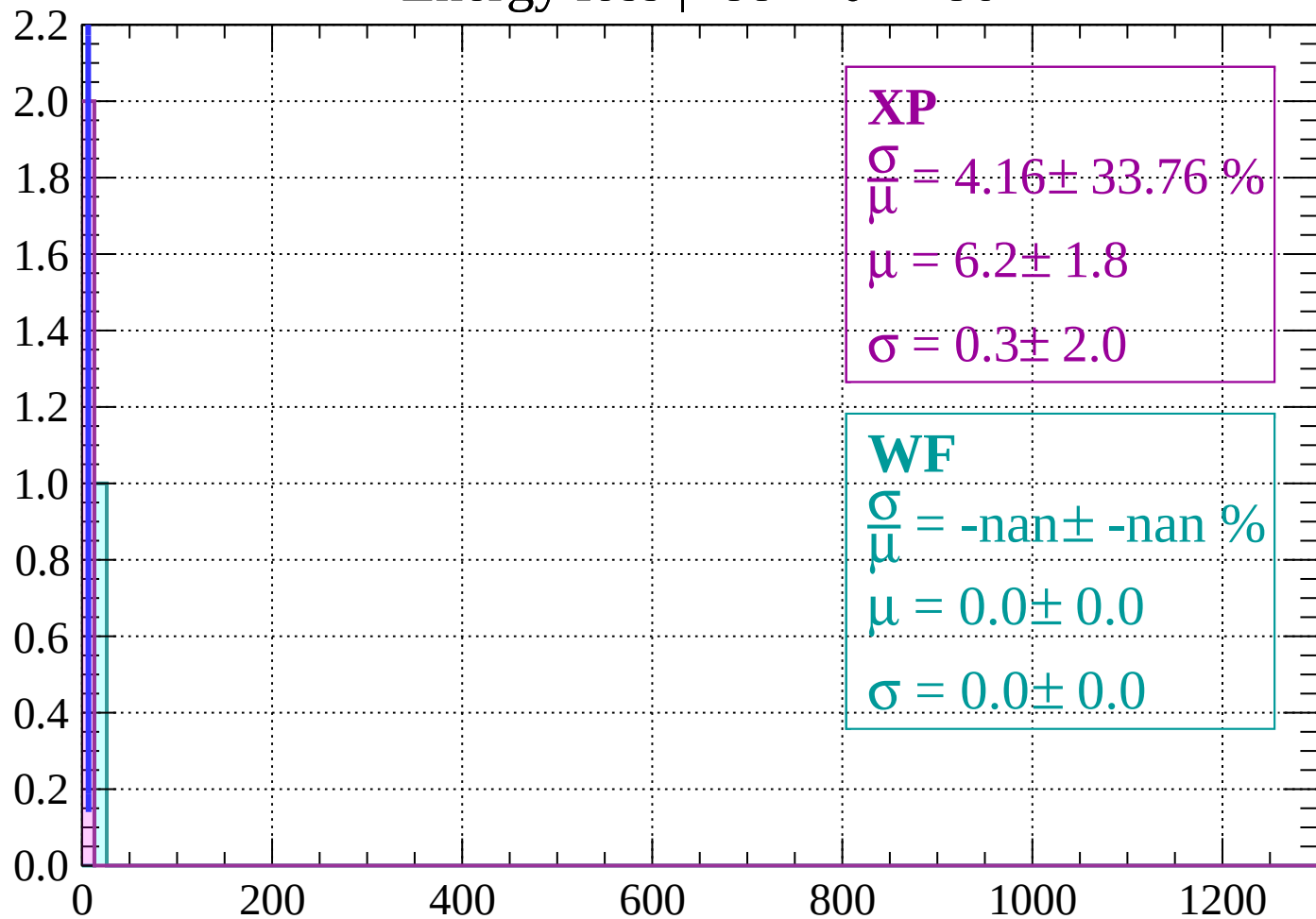
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

Count



XP

$$\frac{\sigma}{\mu} = 4.16 \pm 33.76 \%$$

$$\mu = 6.2 \pm 1.8$$

$$\sigma = 0.3 \pm 2.0$$

WF

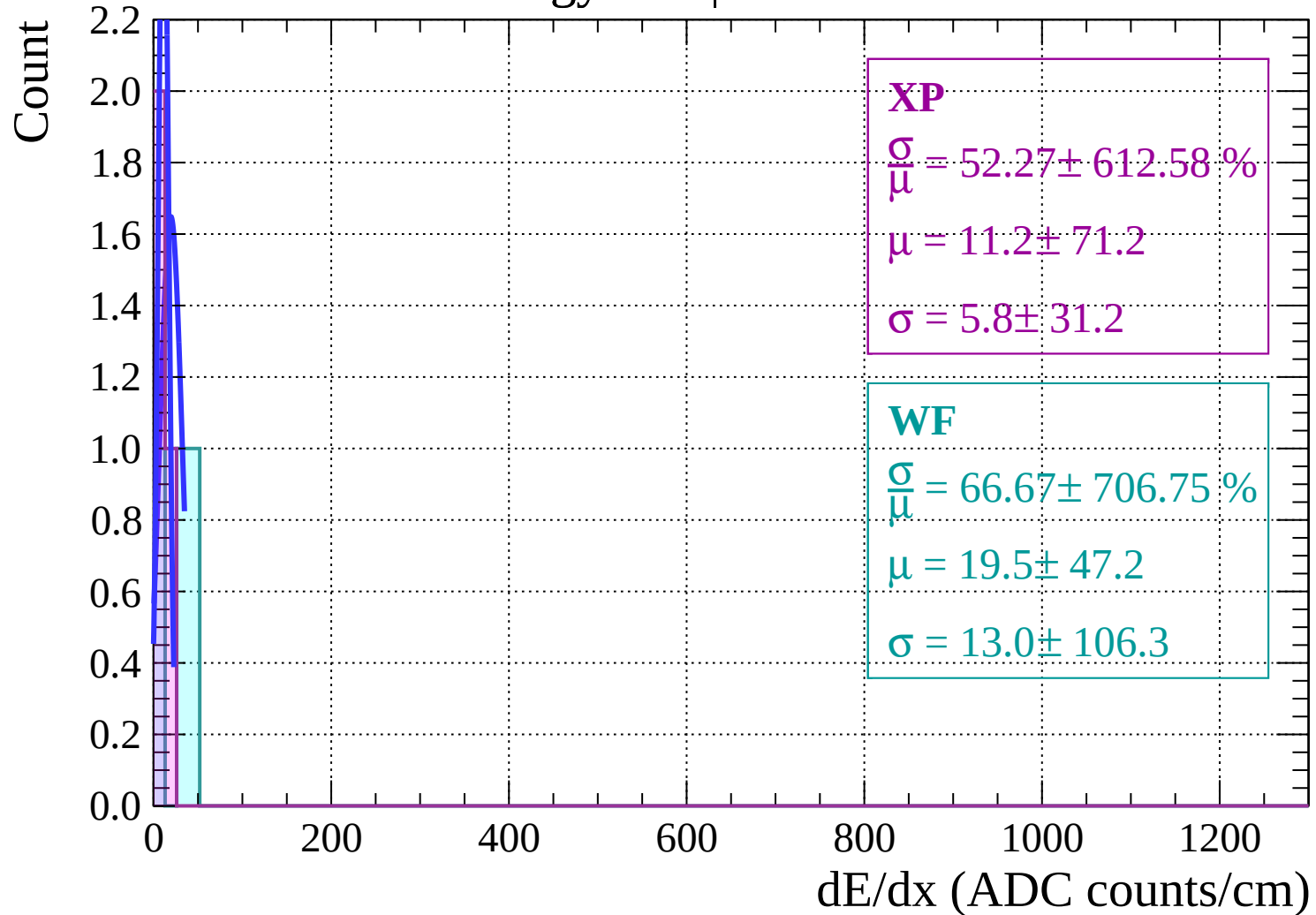
$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

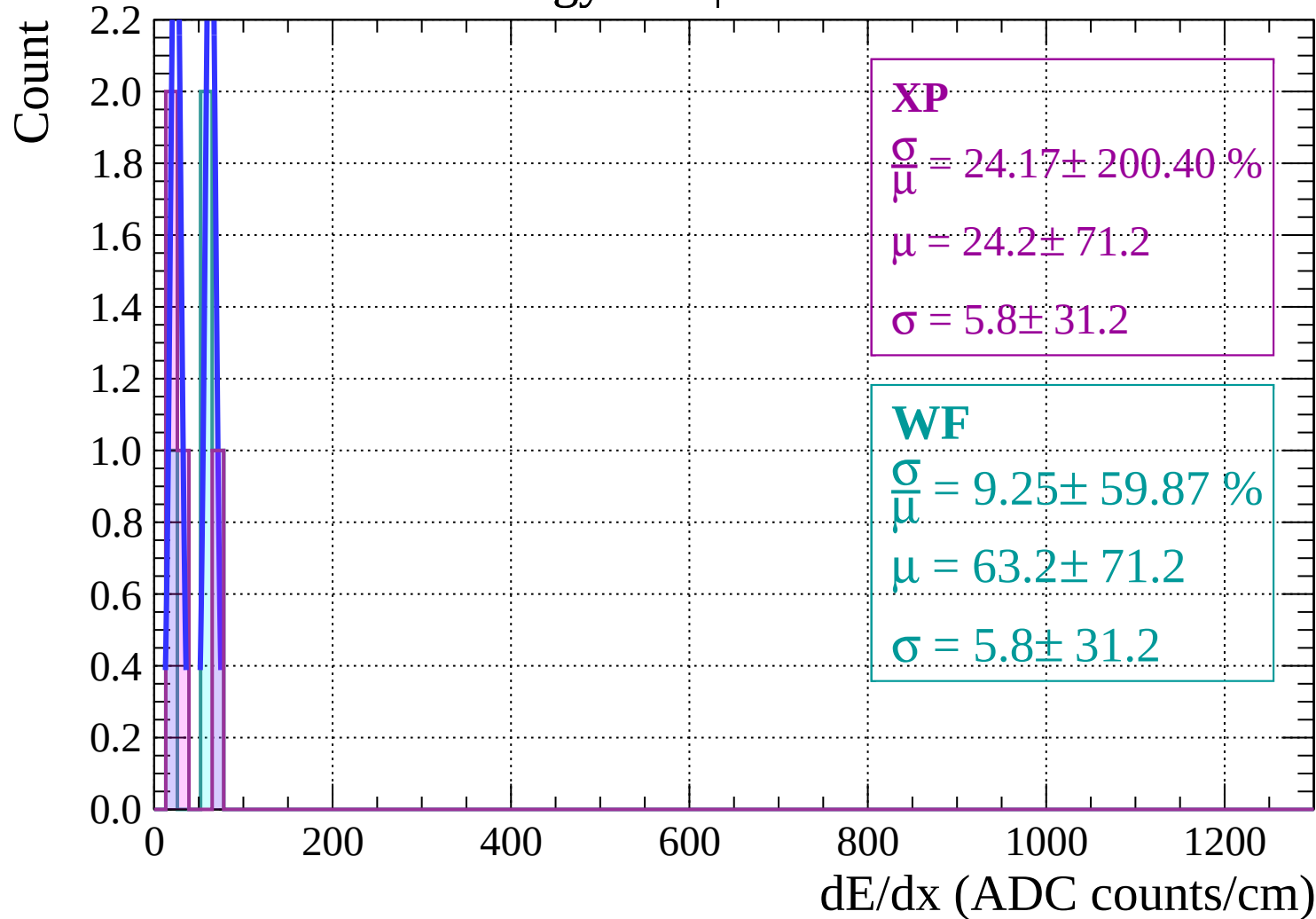
$$\sigma = 0.0 \pm 0.0$$

dE/dx (ADC counts/cm)

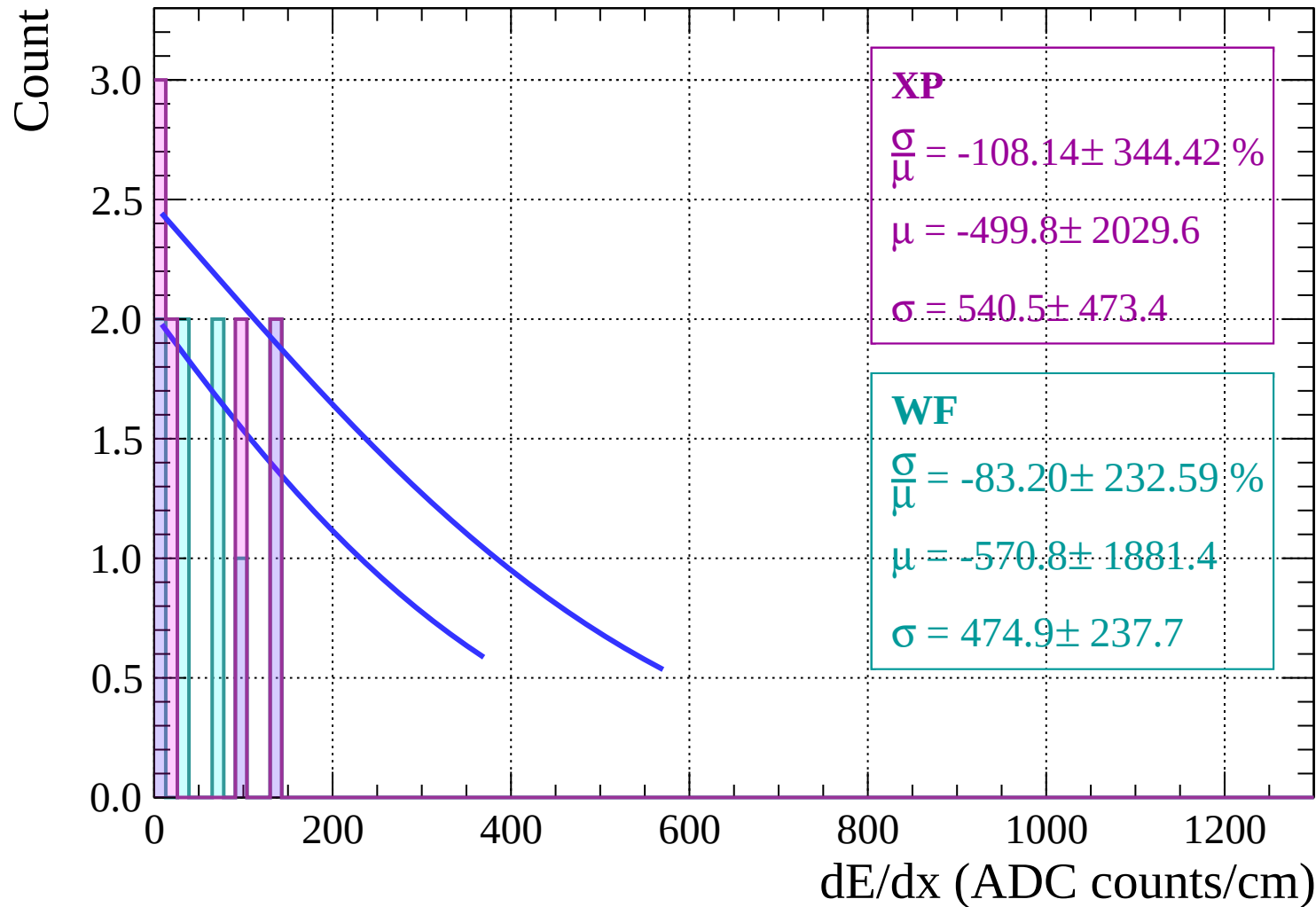
Energy loss | $-86 < \theta < -84$



Energy loss | $-84 < \theta < -82$

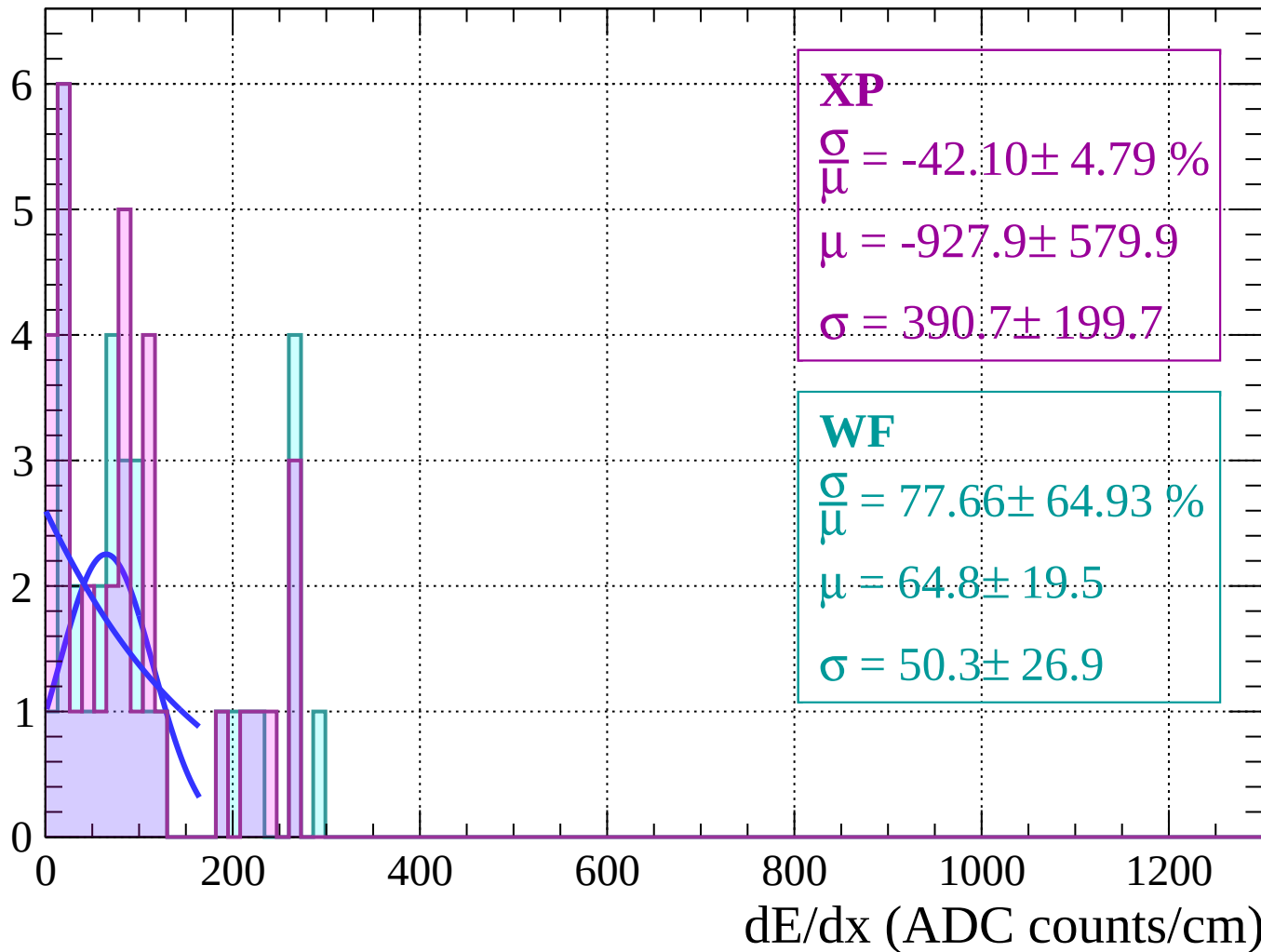


Energy loss | $-82 < \theta < -80$



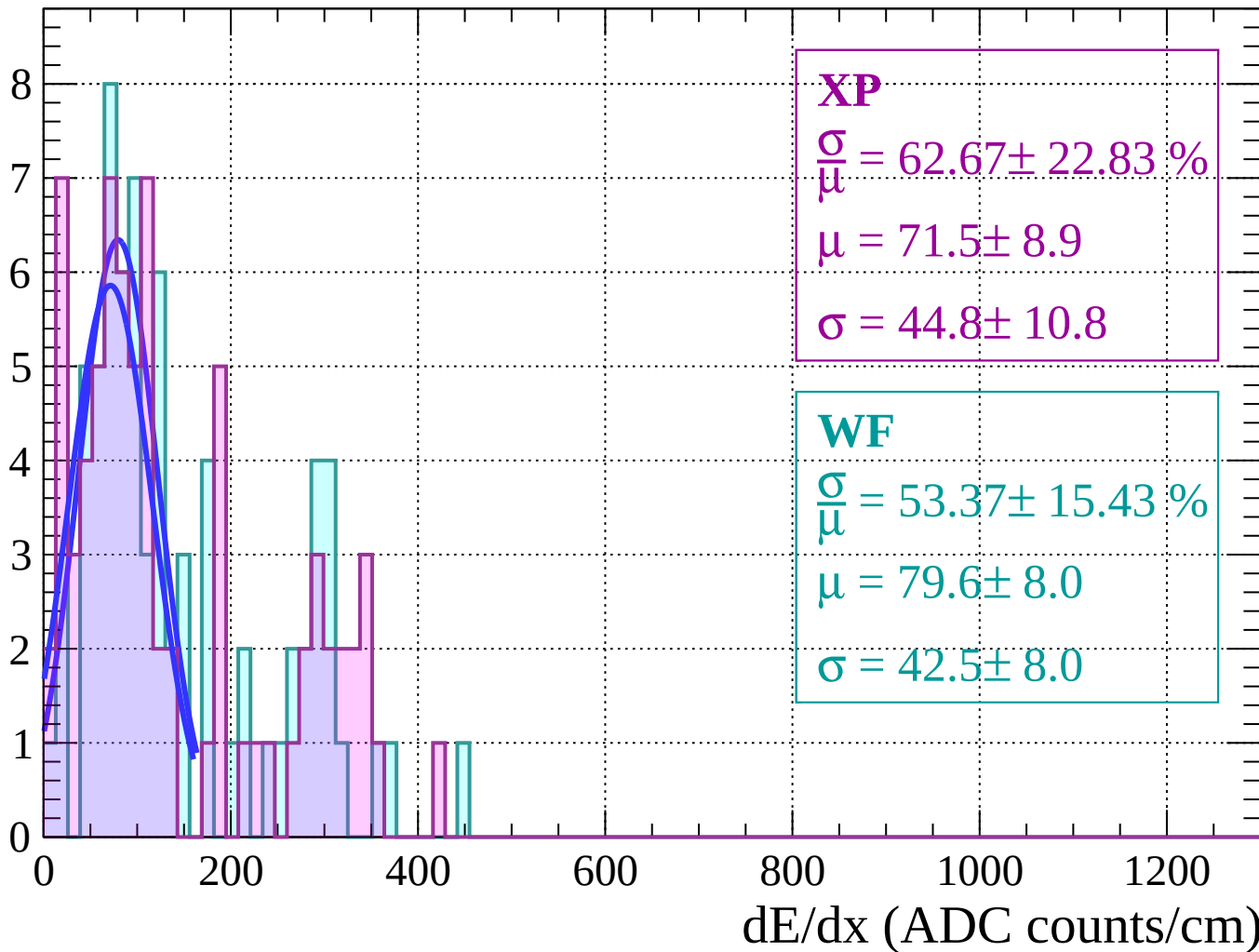
Energy loss | $-80 < \theta < -78$

Count



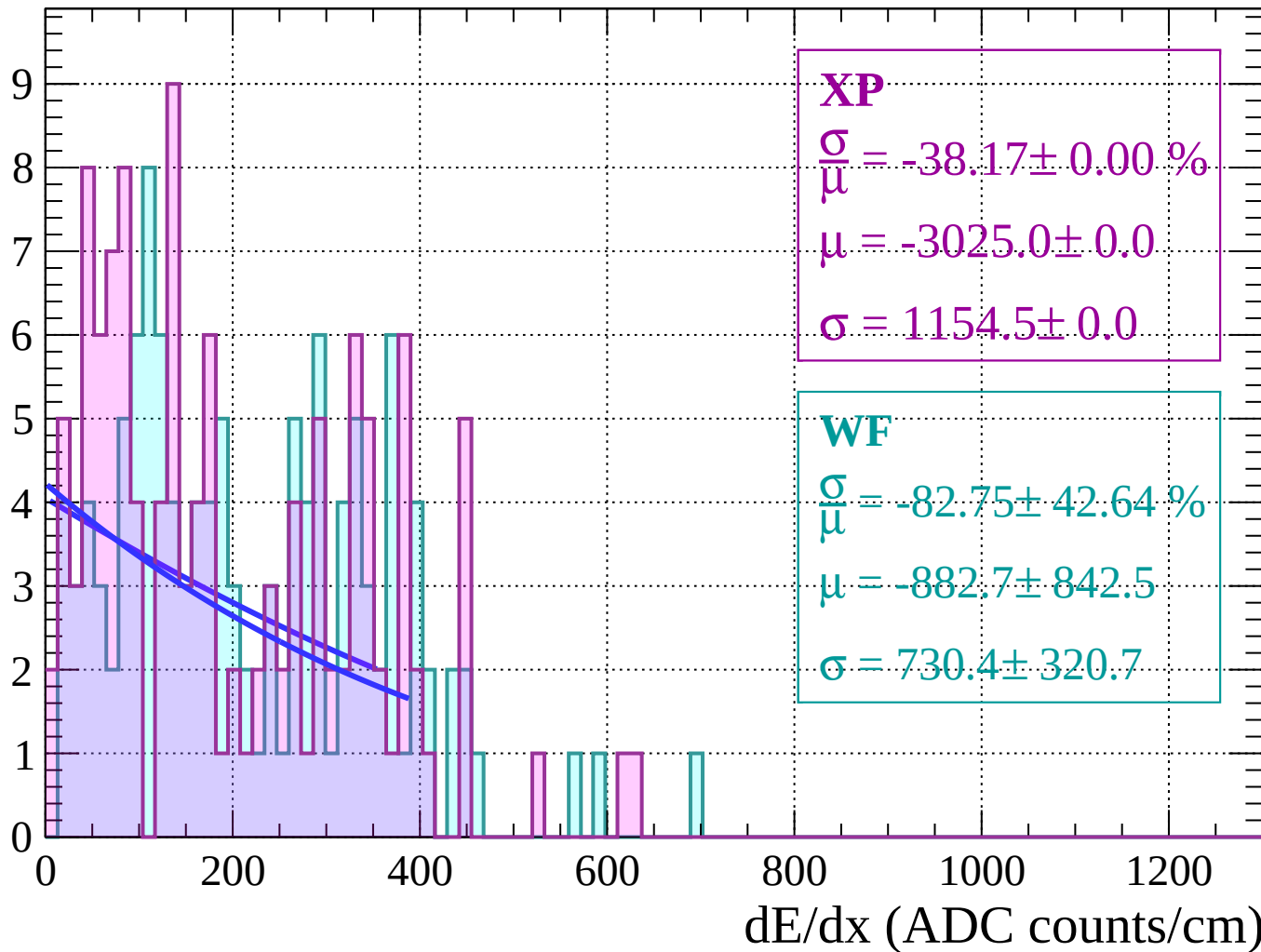
Energy loss | $-78 < \theta < -76$

Count



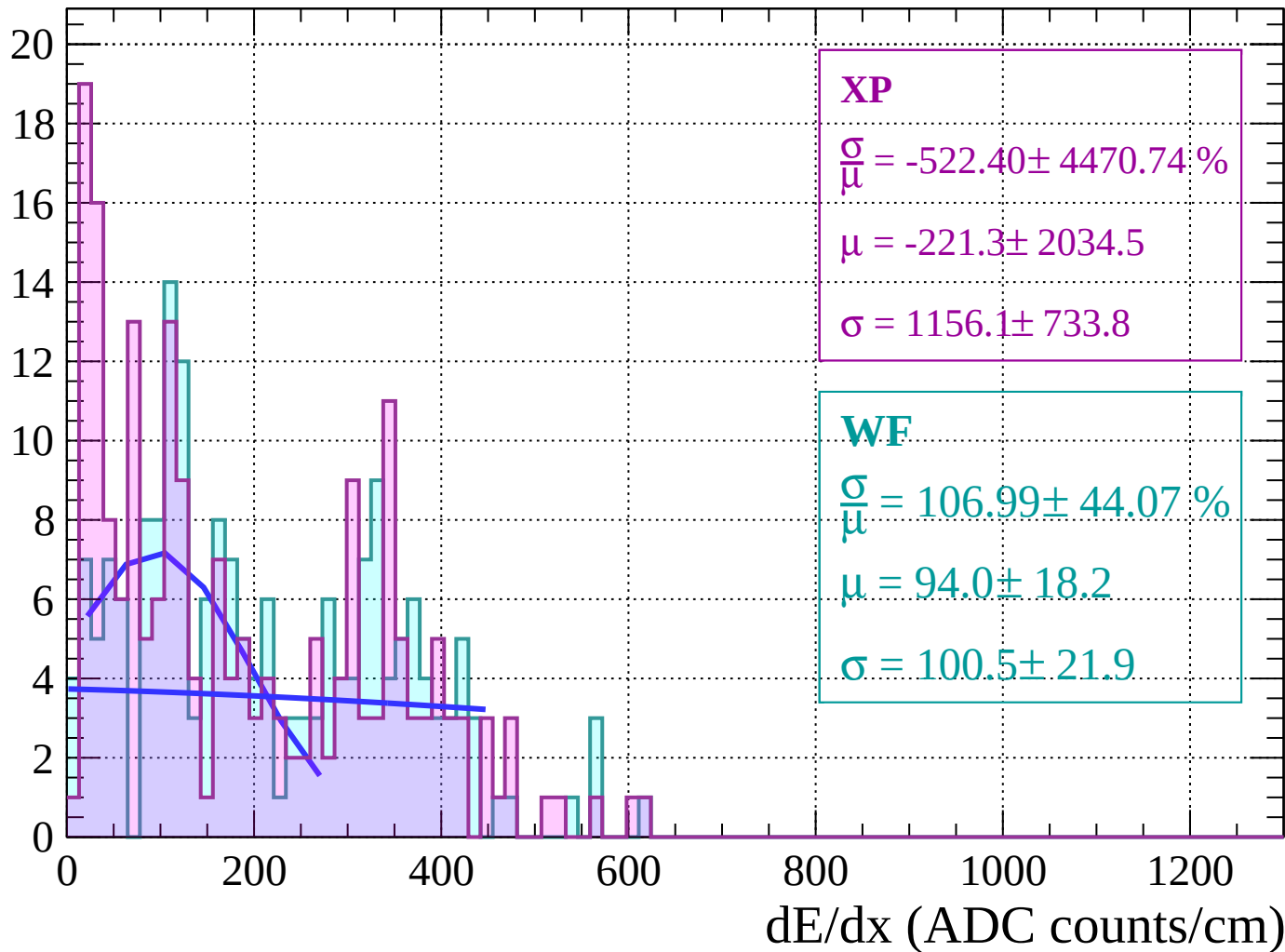
Energy loss $|-76 < \theta < -74$

Count



Energy loss | $-74 < \theta < -72$

Count



Energy loss | $-72 < \theta < -70$

Count

XP

$$\frac{\sigma}{\mu} = -30.36 \pm 6.97 \%$$

$$\mu = -3319.8 \pm 1650.6$$

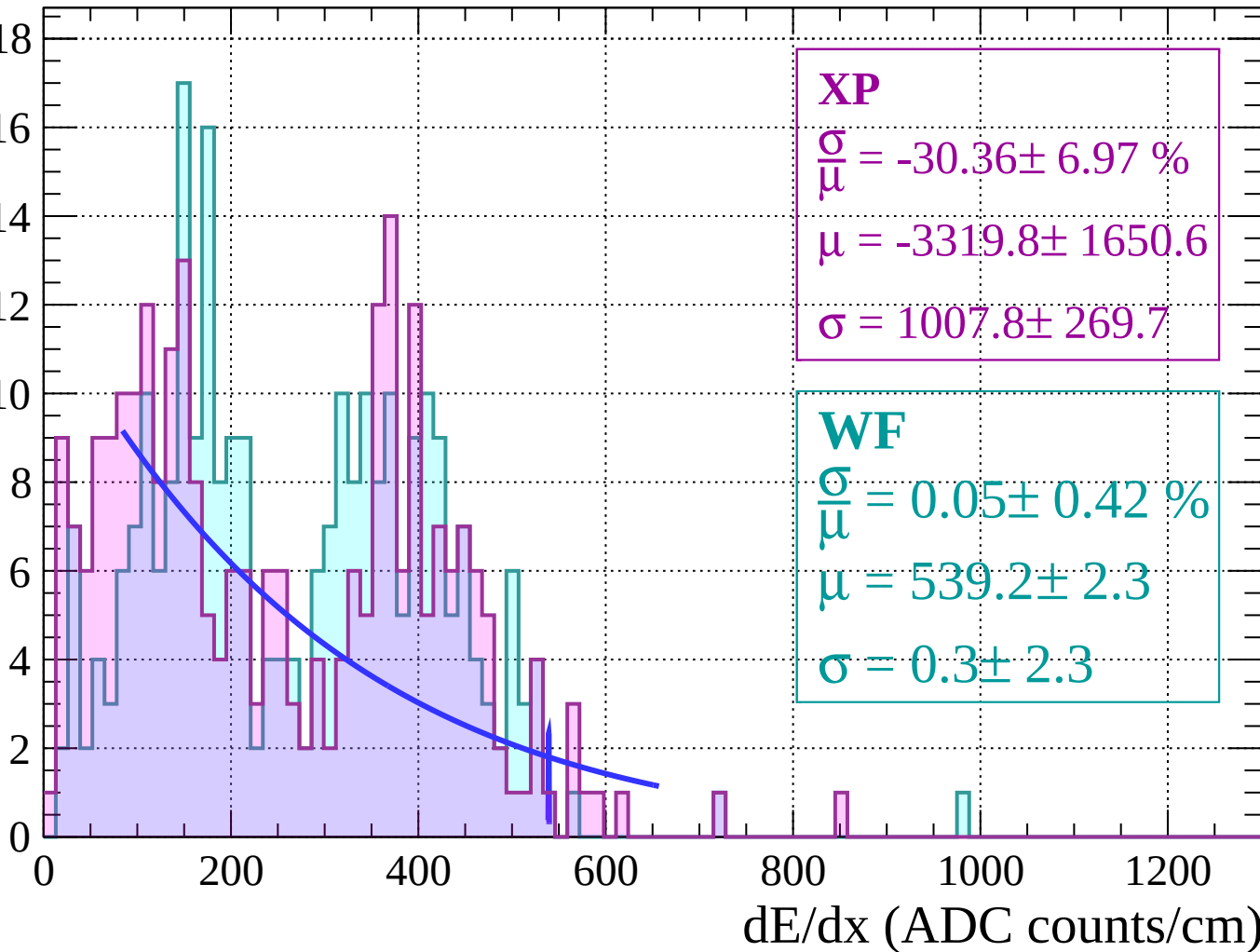
$$\sigma = 1007.8 \pm 269.7$$

WF

$$\frac{\sigma}{\mu} = 0.05 \pm 0.42 \%$$

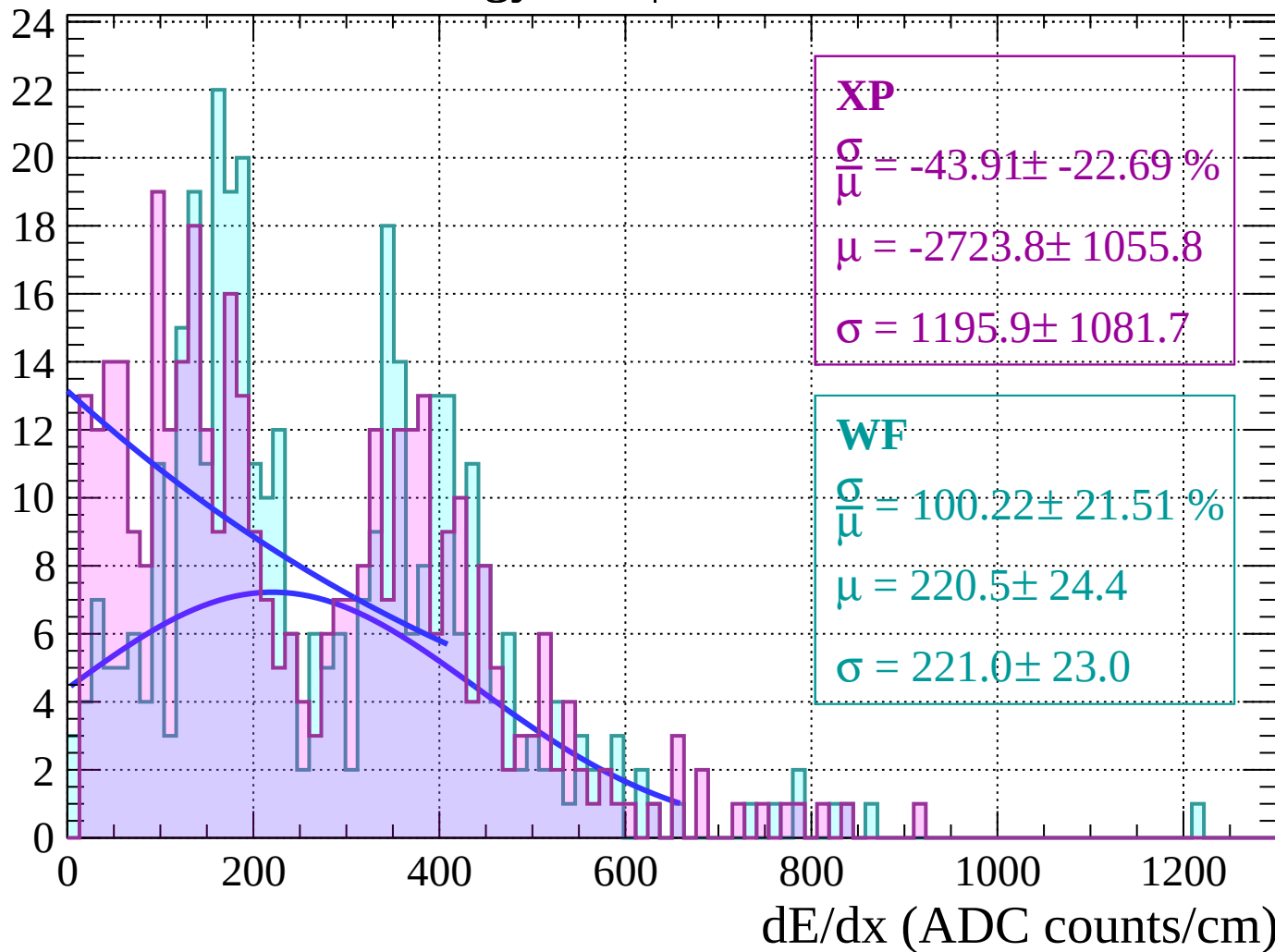
$$\mu = 539.2 \pm 2.3$$

$$\sigma = 0.3 \pm 2.3$$



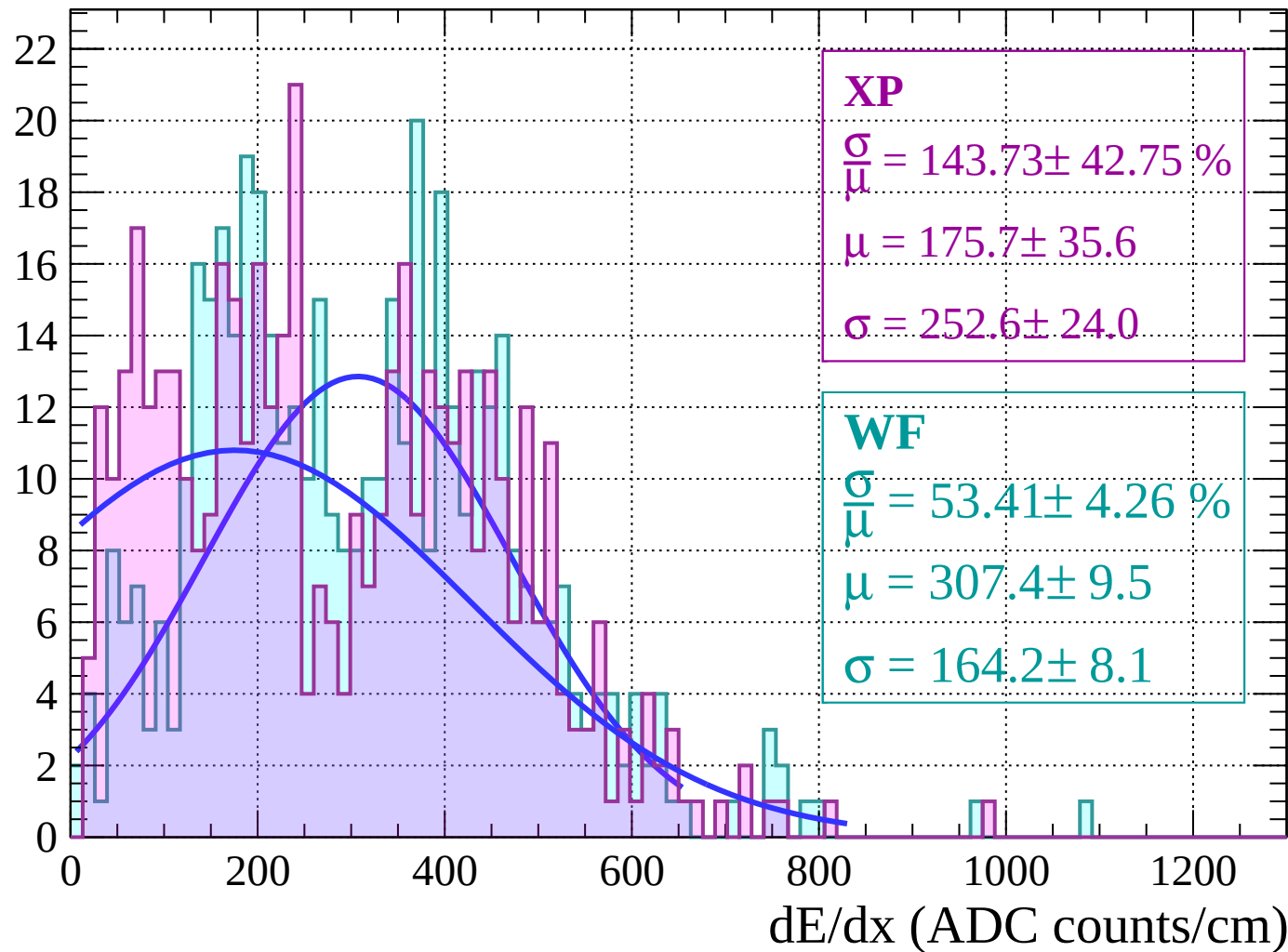
Energy loss | $-70 < \theta < -68$

Count



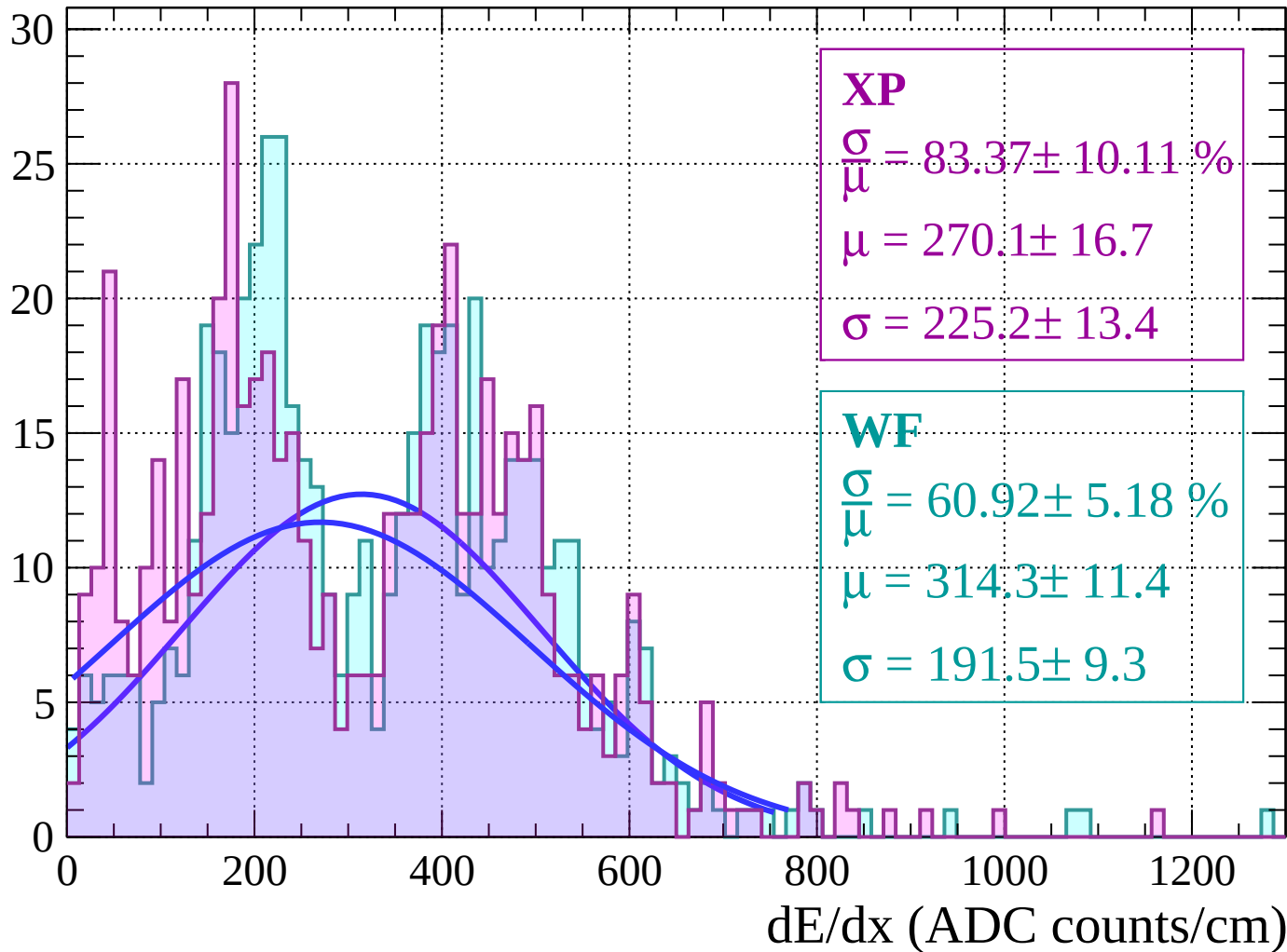
Energy loss | $-68 < \theta < -66$

Count



Energy loss | $-66 < \theta < -64$

Count



Energy loss $|-64 < \theta < -62$

Count

XP

$$\frac{\sigma}{\mu} = 82.45 \pm 10.01 \%$$

$$\mu = 269.3 \pm 15.2$$

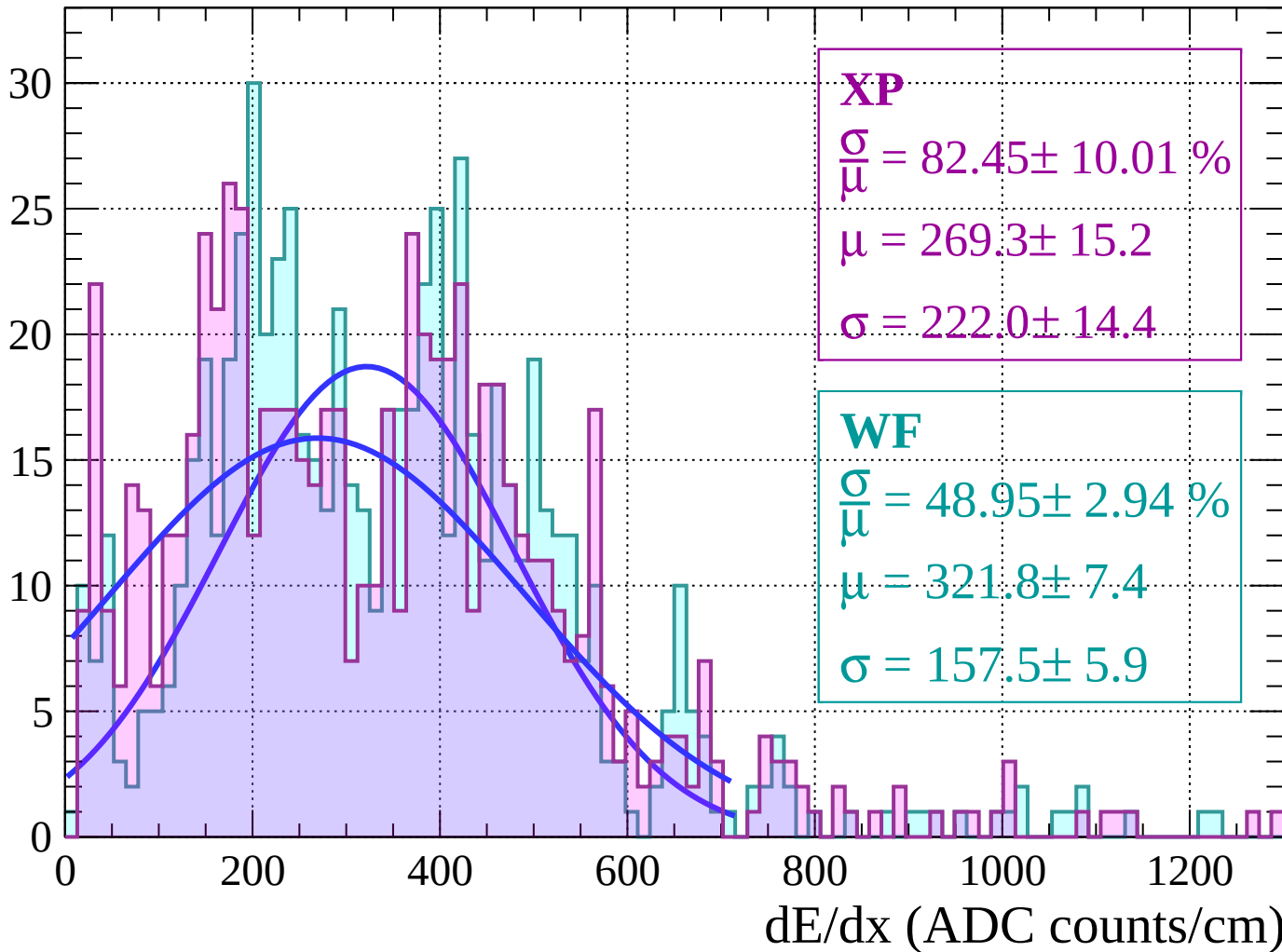
$$\sigma = 222.0 \pm 14.4$$

WF

$$\frac{\sigma}{\mu} = 48.95 \pm 2.94 \%$$

$$\mu = 321.8 \pm 7.4$$

$$\sigma = 157.5 \pm 5.9$$



Energy loss | $-62 < \theta < -60$

Count

XP

$$\frac{\sigma}{\mu} = 58.65 \pm 4.21 \%$$

$$\mu = 330.6 \pm 9.6$$

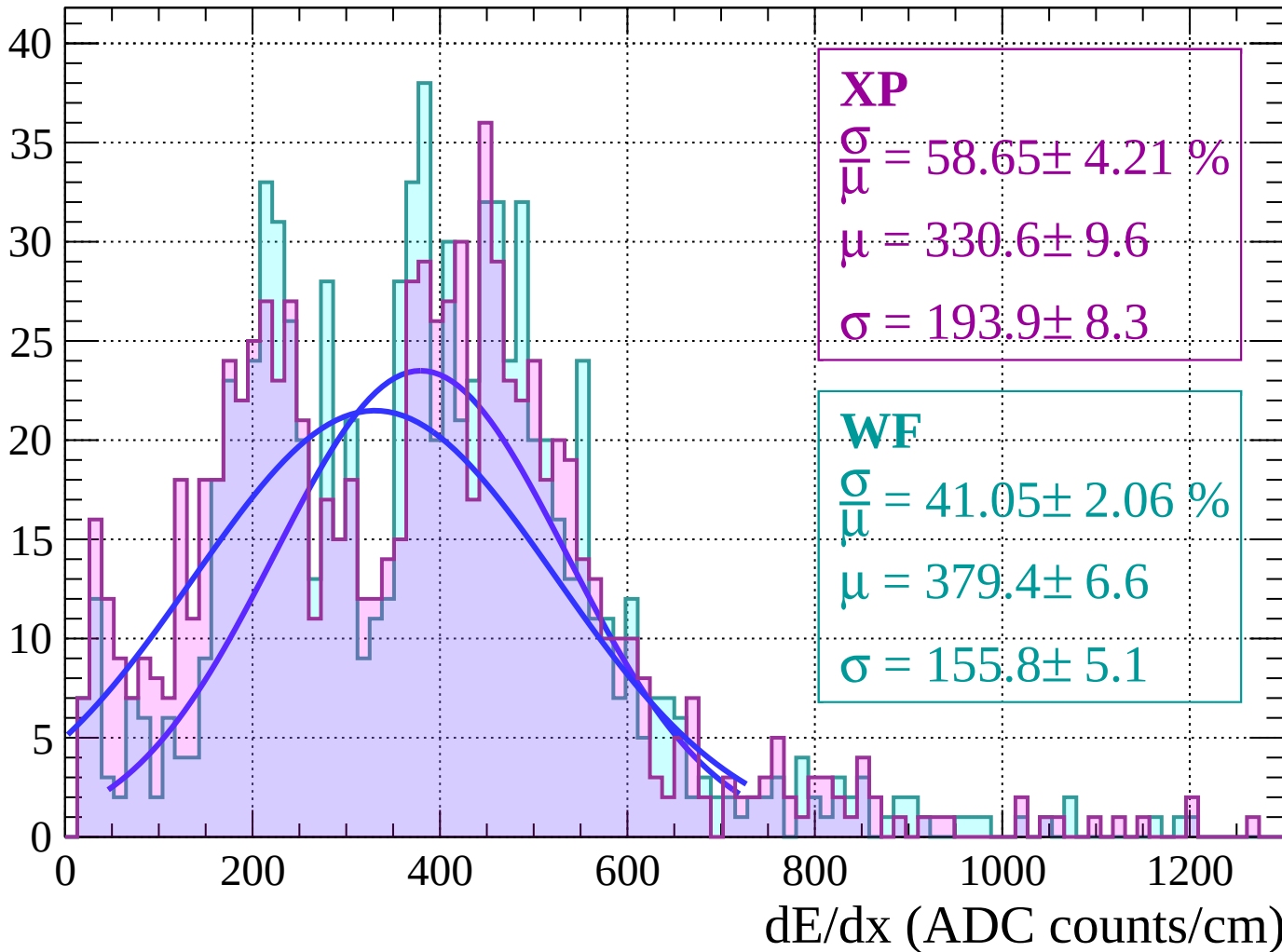
$$\sigma = 193.9 \pm 8.3$$

WF

$$\frac{\sigma}{\mu} = 41.05 \pm 2.06 \%$$

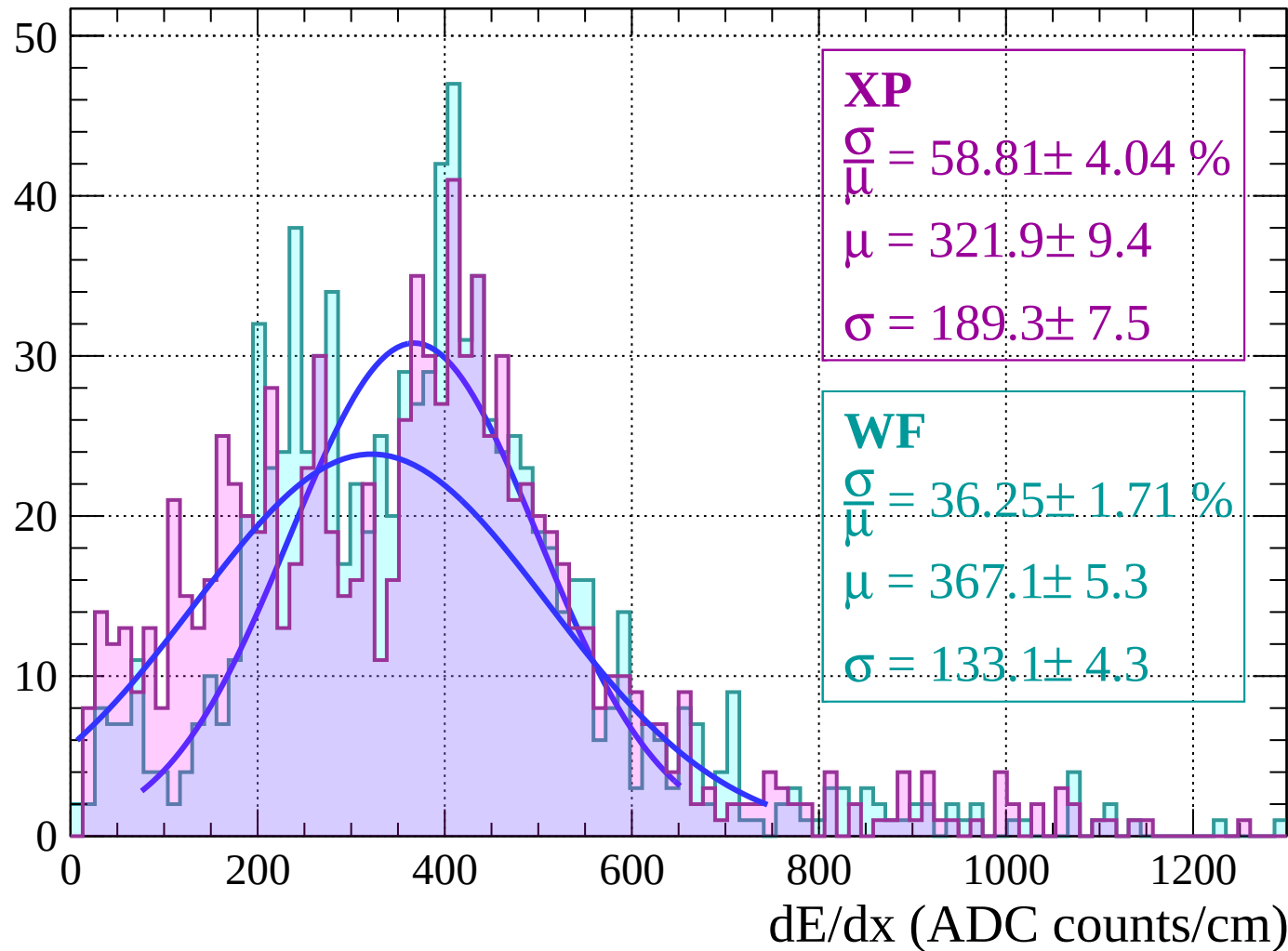
$$\mu = 379.4 \pm 6.6$$

$$\sigma = 155.8 \pm 5.1$$



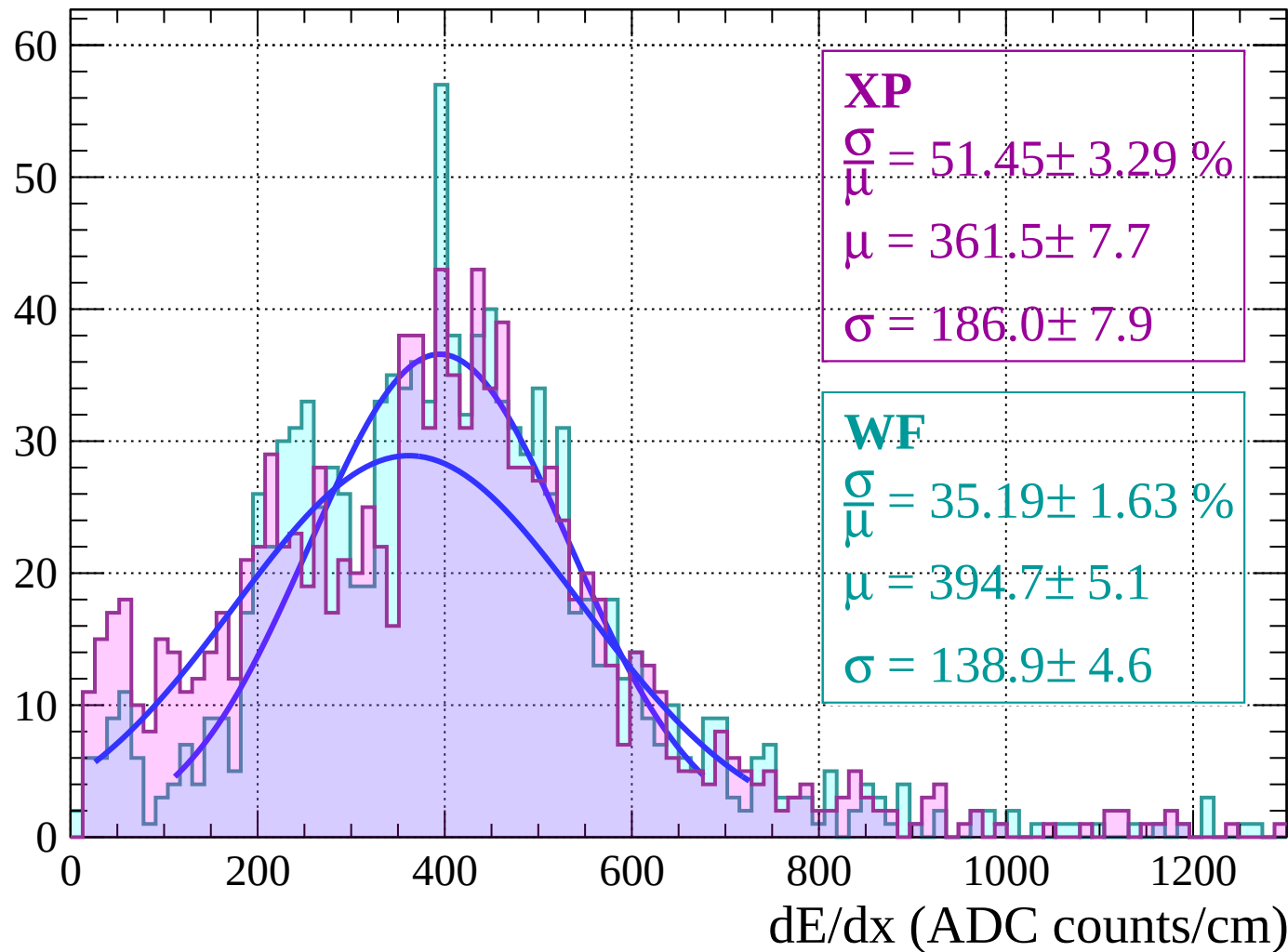
Energy loss | $-60 < \theta < -58$

Count



Energy loss | $-58 < \theta < -56$

Count



Energy loss | $-56 < \theta < -54$

Count

XP

$$\frac{\sigma}{\mu} = 42.30 \pm 2.22 \%$$

$$\mu = 379.1 \pm 6.0$$

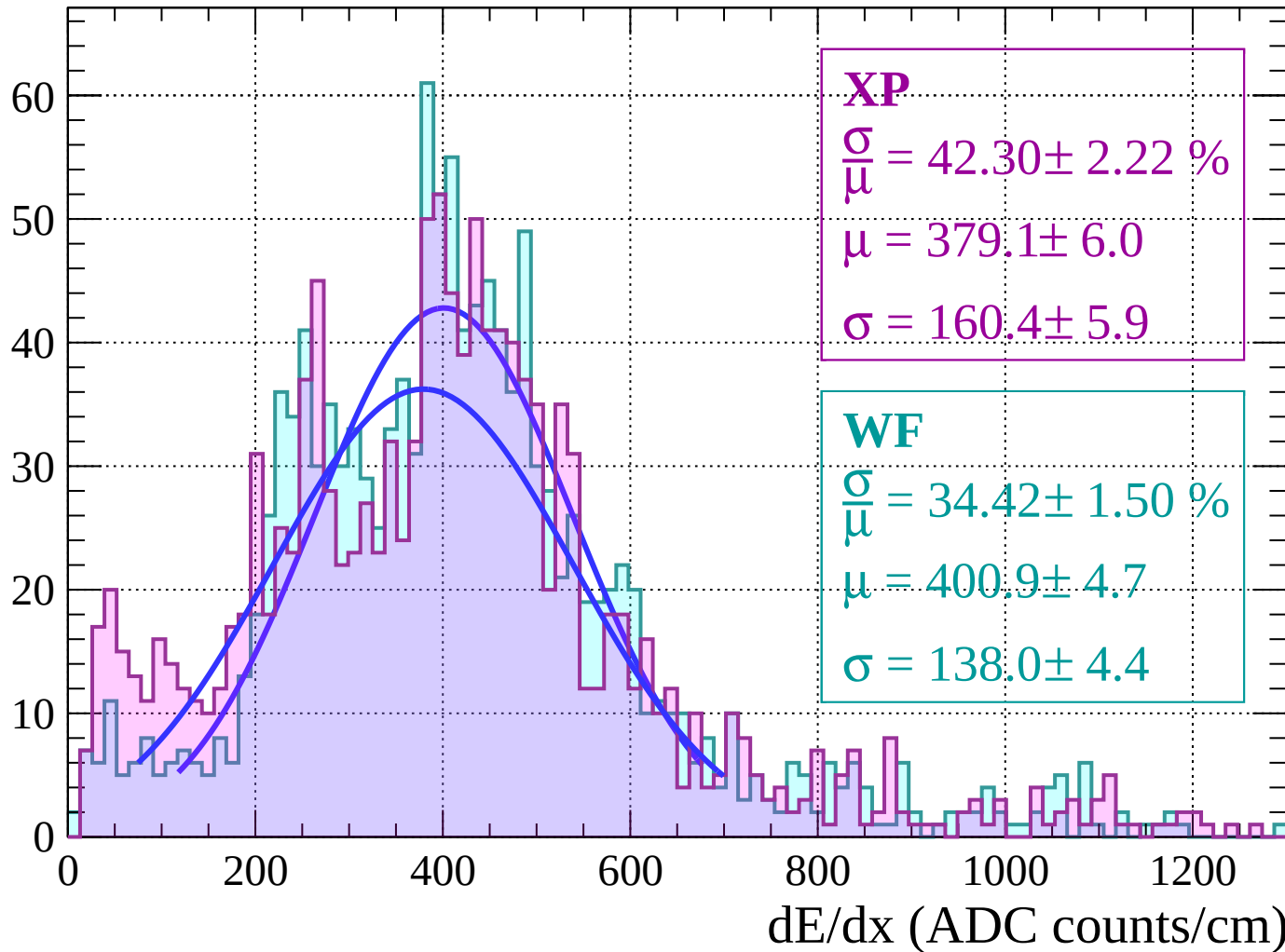
$$\sigma = 160.4 \pm 5.9$$

WF

$$\frac{\sigma}{\mu} = 34.42 \pm 1.50 \%$$

$$\mu = 400.9 \pm 4.7$$

$$\sigma = 138.0 \pm 4.4$$



Energy loss $|-54 < \theta < -52$

Count

XP

$$\frac{\sigma}{\mu} = 41.15 \pm 2.21 \%$$

$$\mu = 391.9 \pm 5.6$$

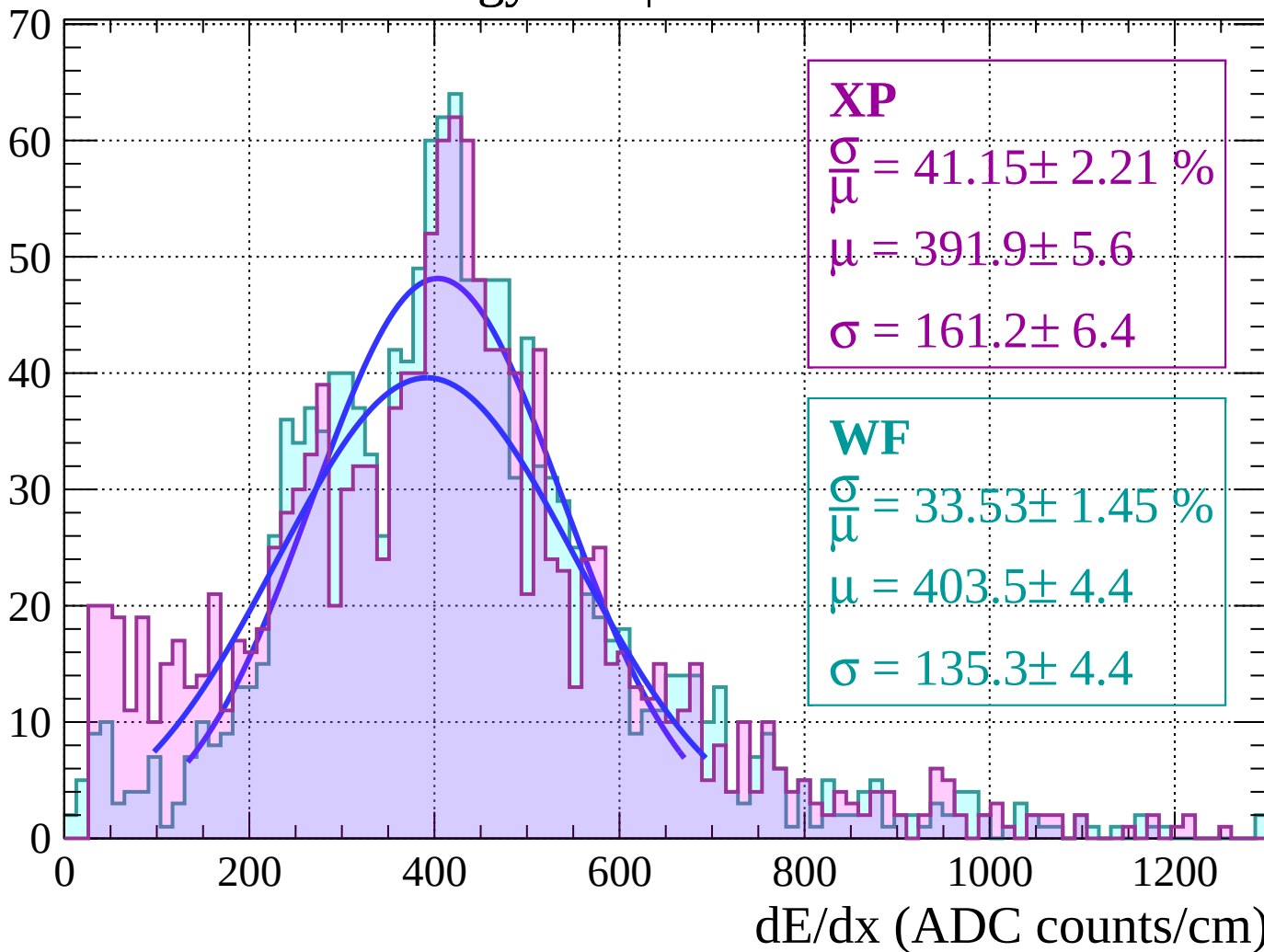
$$\sigma = 161.2 \pm 6.4$$

WF

$$\frac{\sigma}{\mu} = 33.53 \pm 1.45 \%$$

$$\mu = 403.5 \pm 4.4$$

$$\sigma = 135.3 \pm 4.4$$



Energy loss $|-52 < \theta < -50$

Count

XP

$$\frac{\sigma}{\mu} = 32.04 \pm 1.52 \%$$

$$\mu = 400.4 \pm 4.3$$

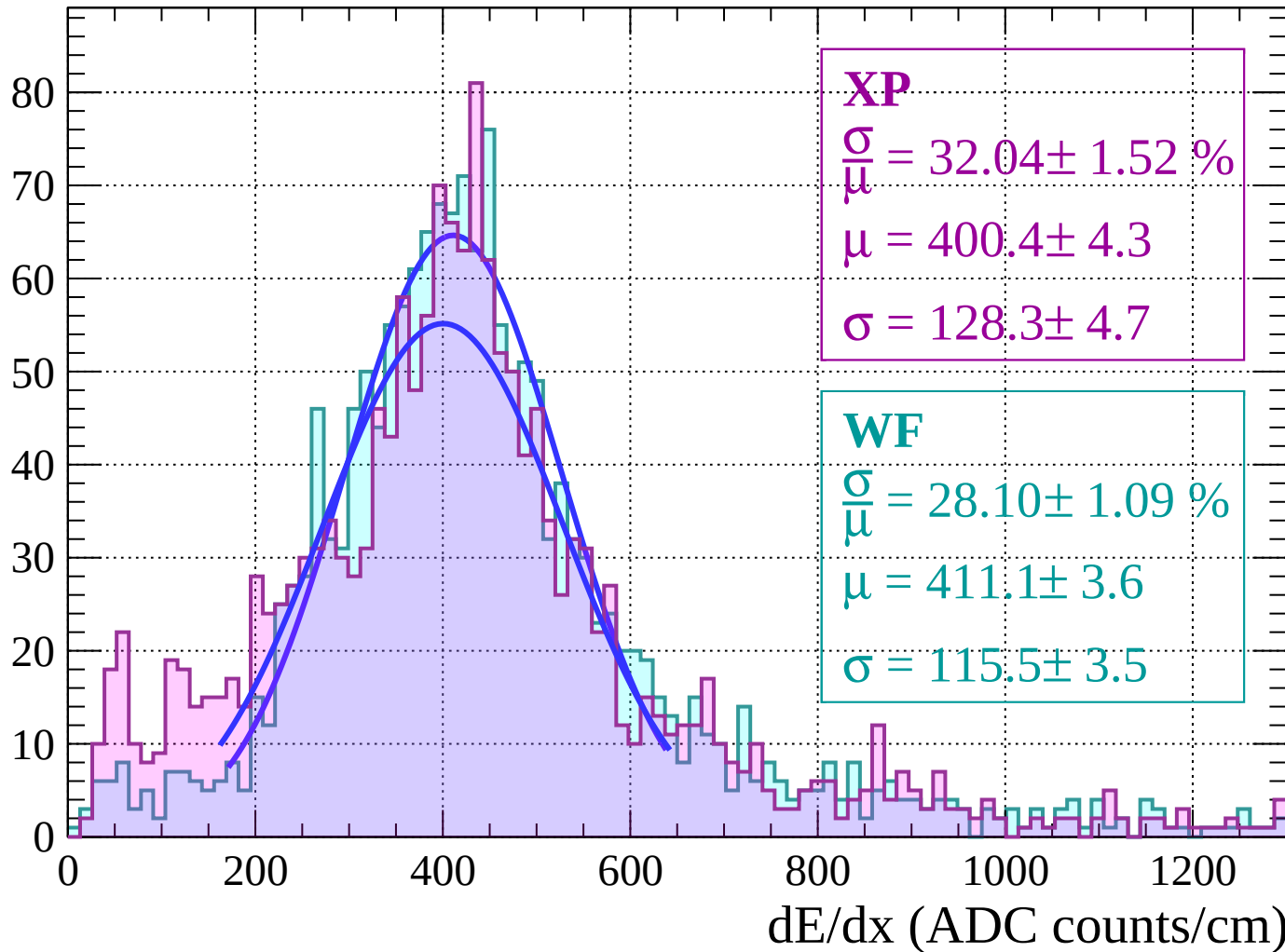
$$\sigma = 128.3 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 28.10 \pm 1.09 \%$$

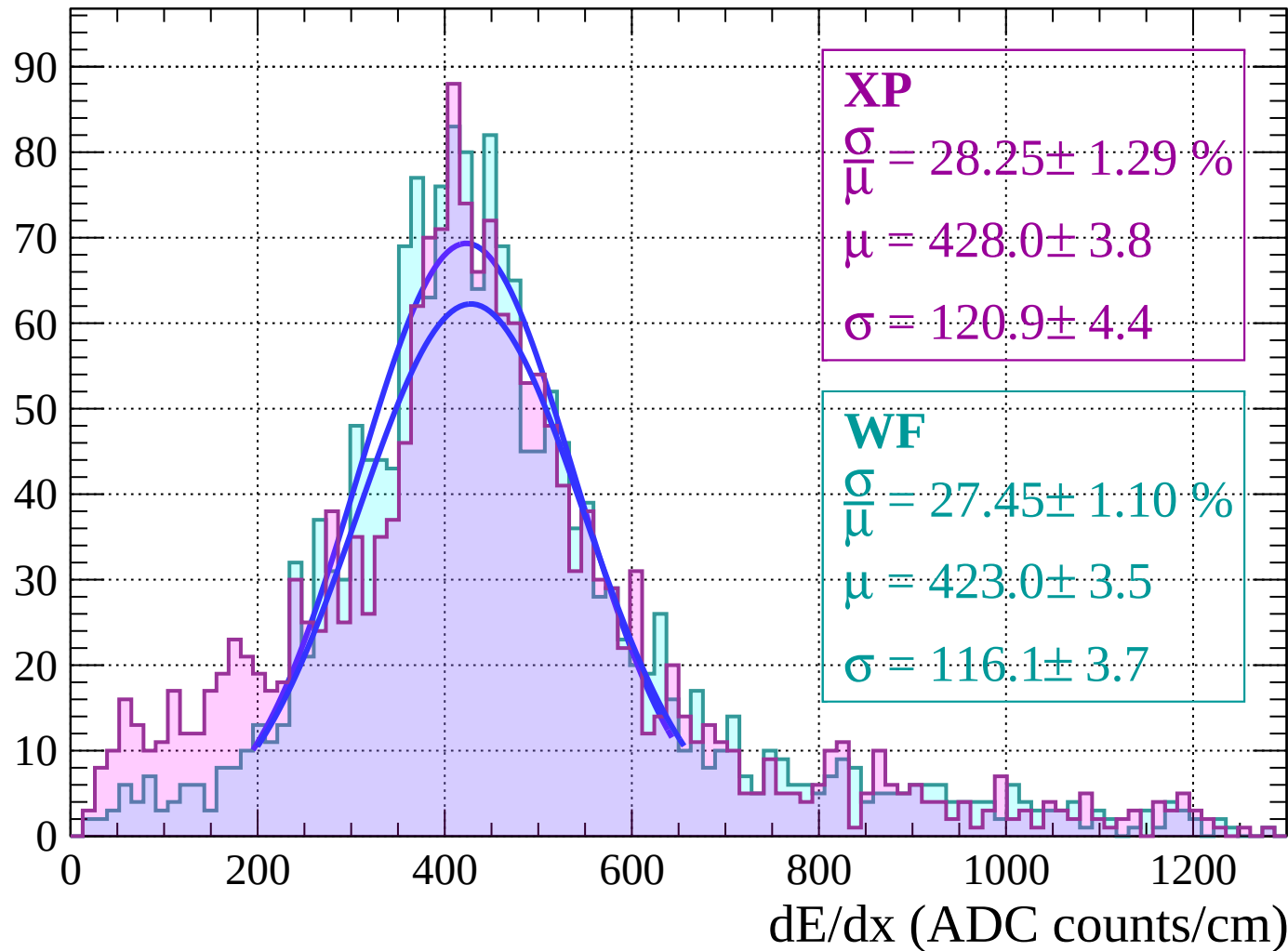
$$\mu = 411.1 \pm 3.6$$

$$\sigma = 115.5 \pm 3.5$$



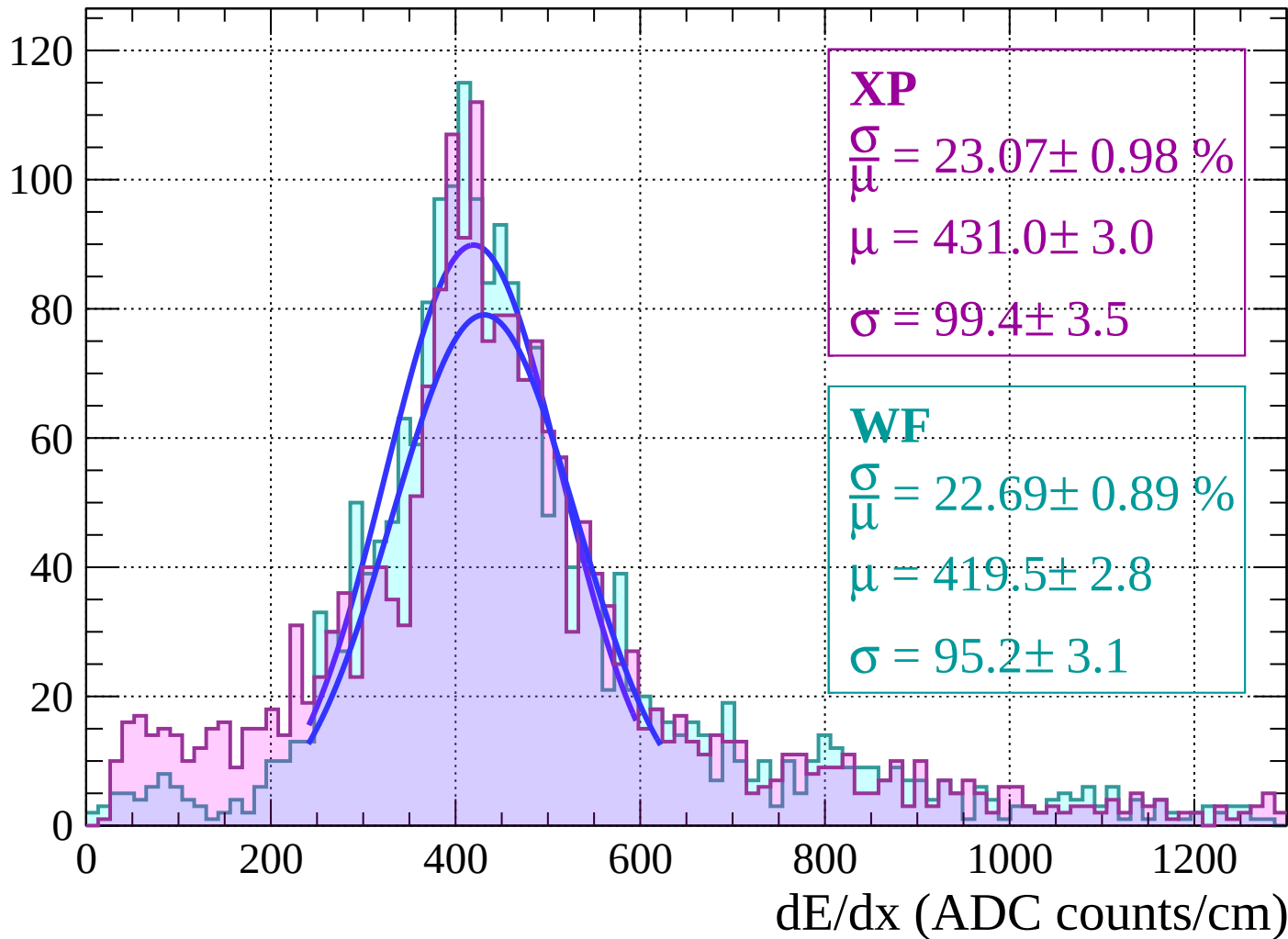
Energy loss | $-50 < \theta < -48$

Count



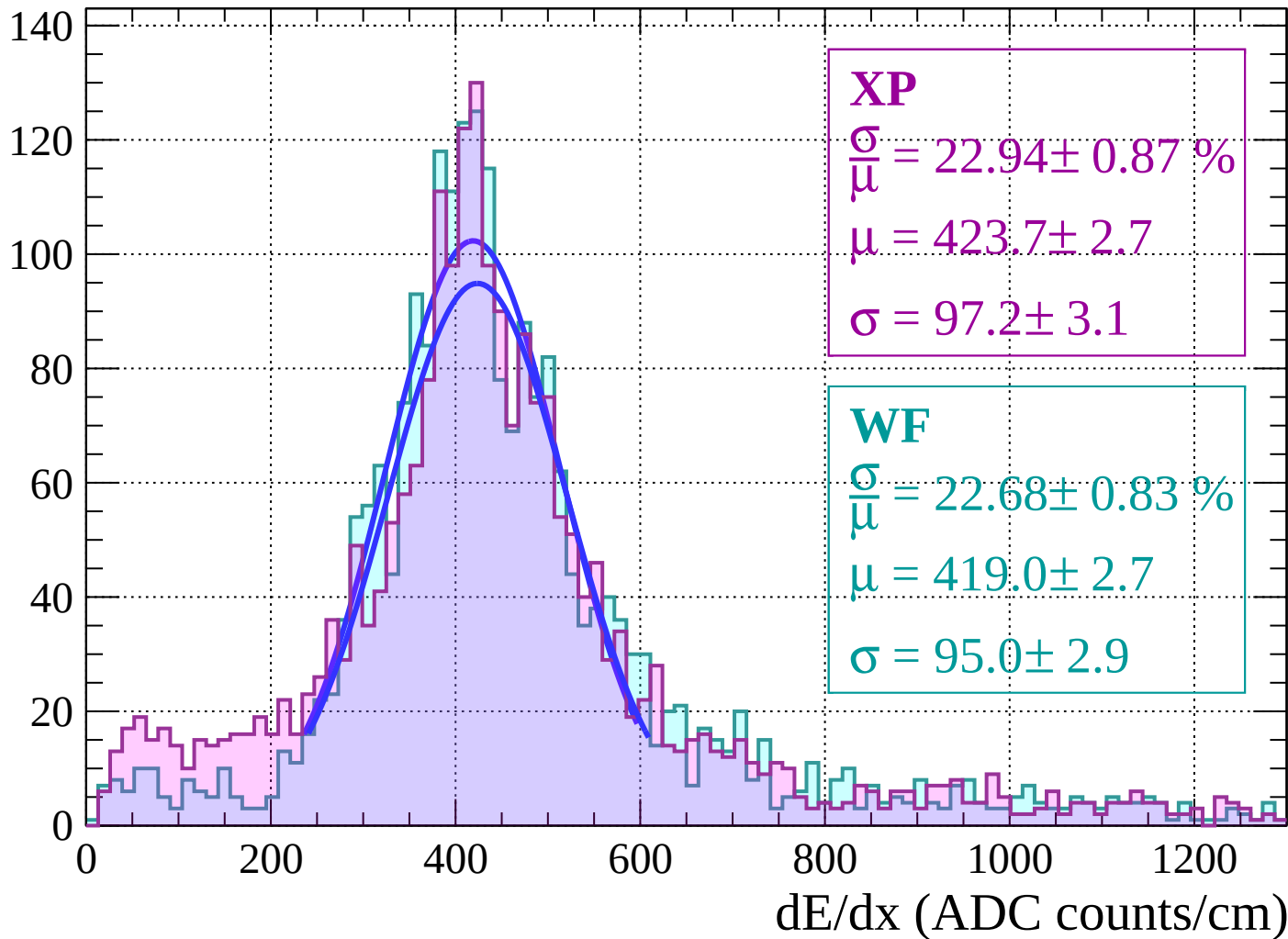
Energy loss | $-48 < \theta < -46$

Count

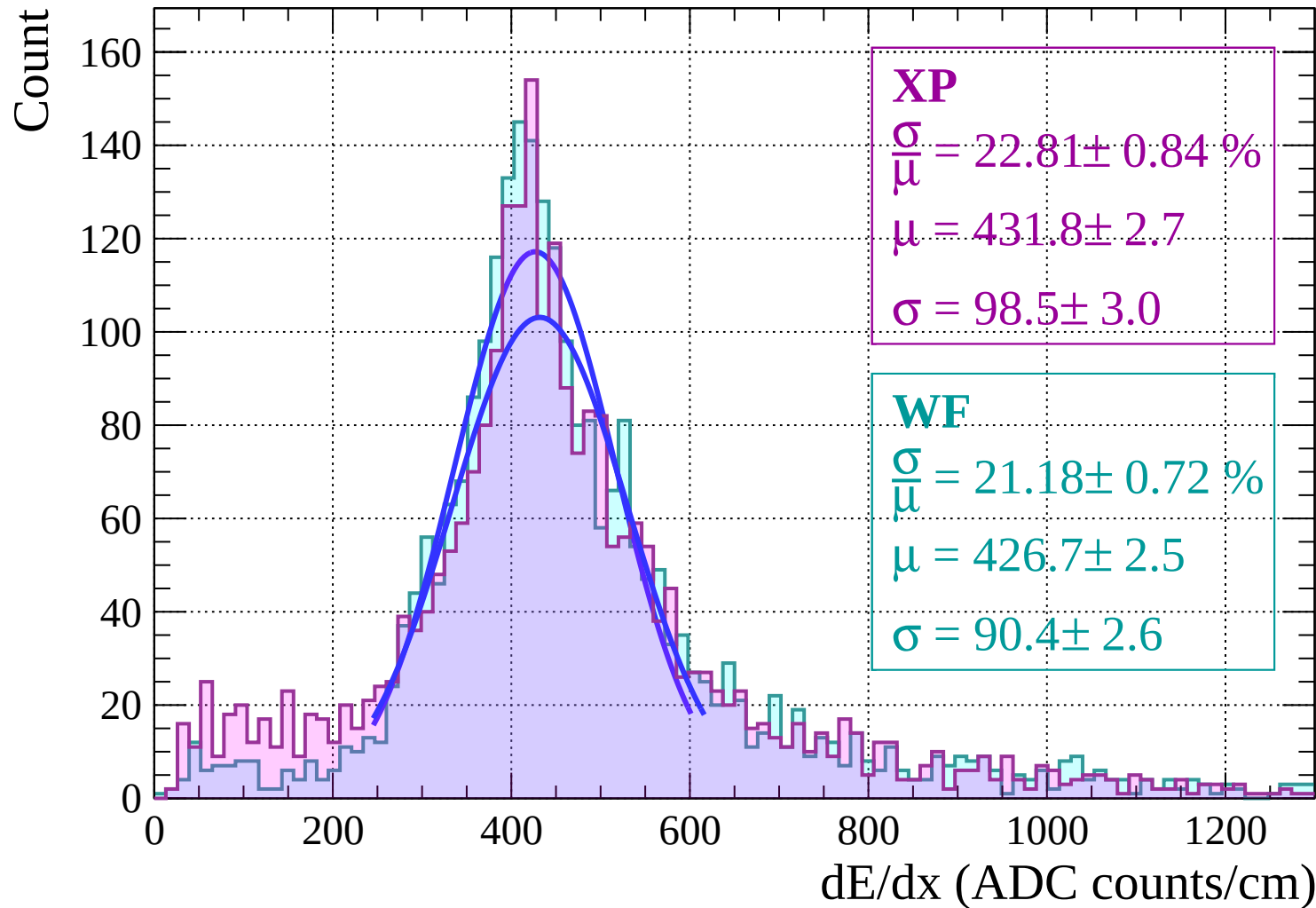


Energy loss | $-46 < \theta < -44$

Count

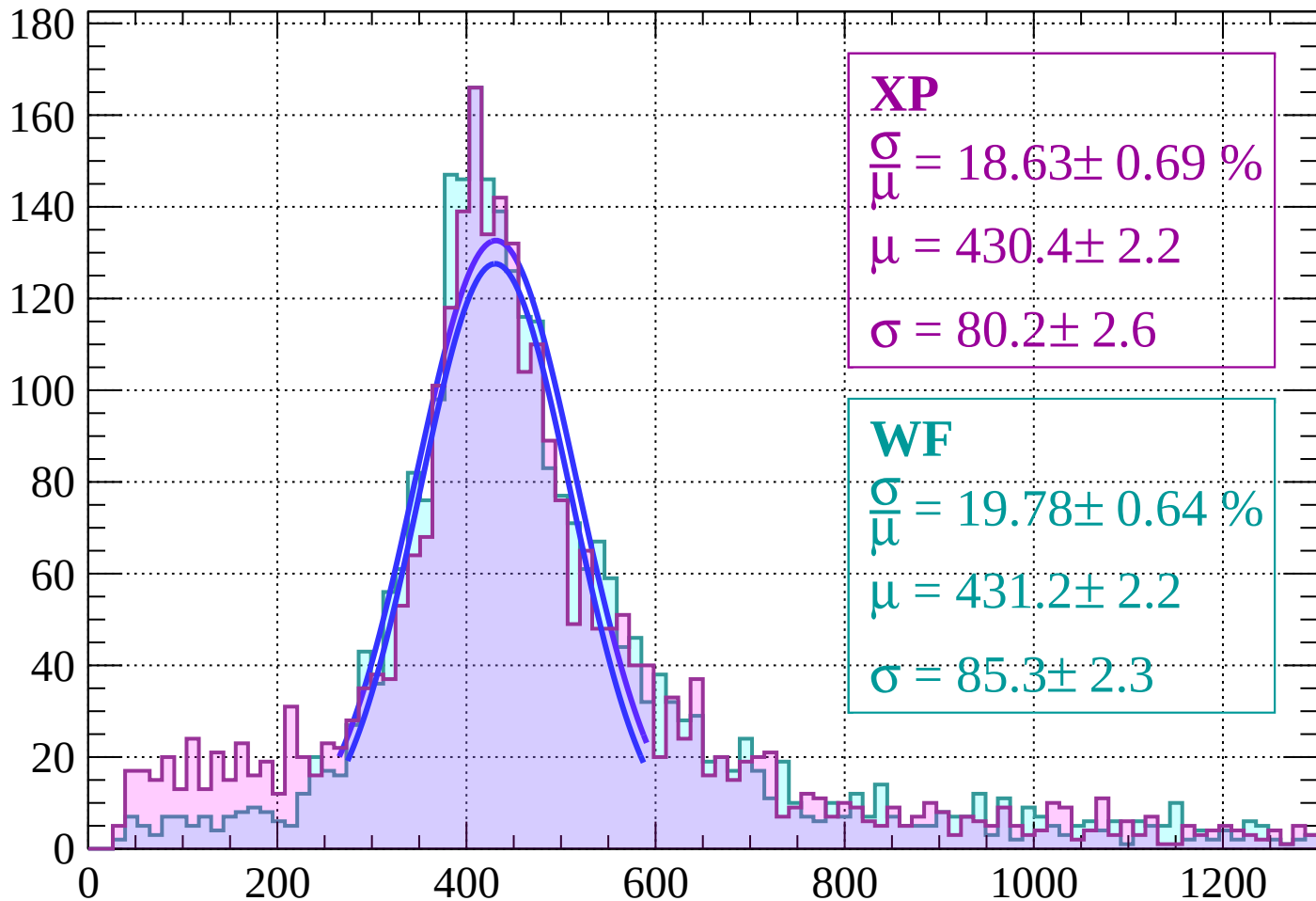


Energy loss | $-44 < \theta < -42$



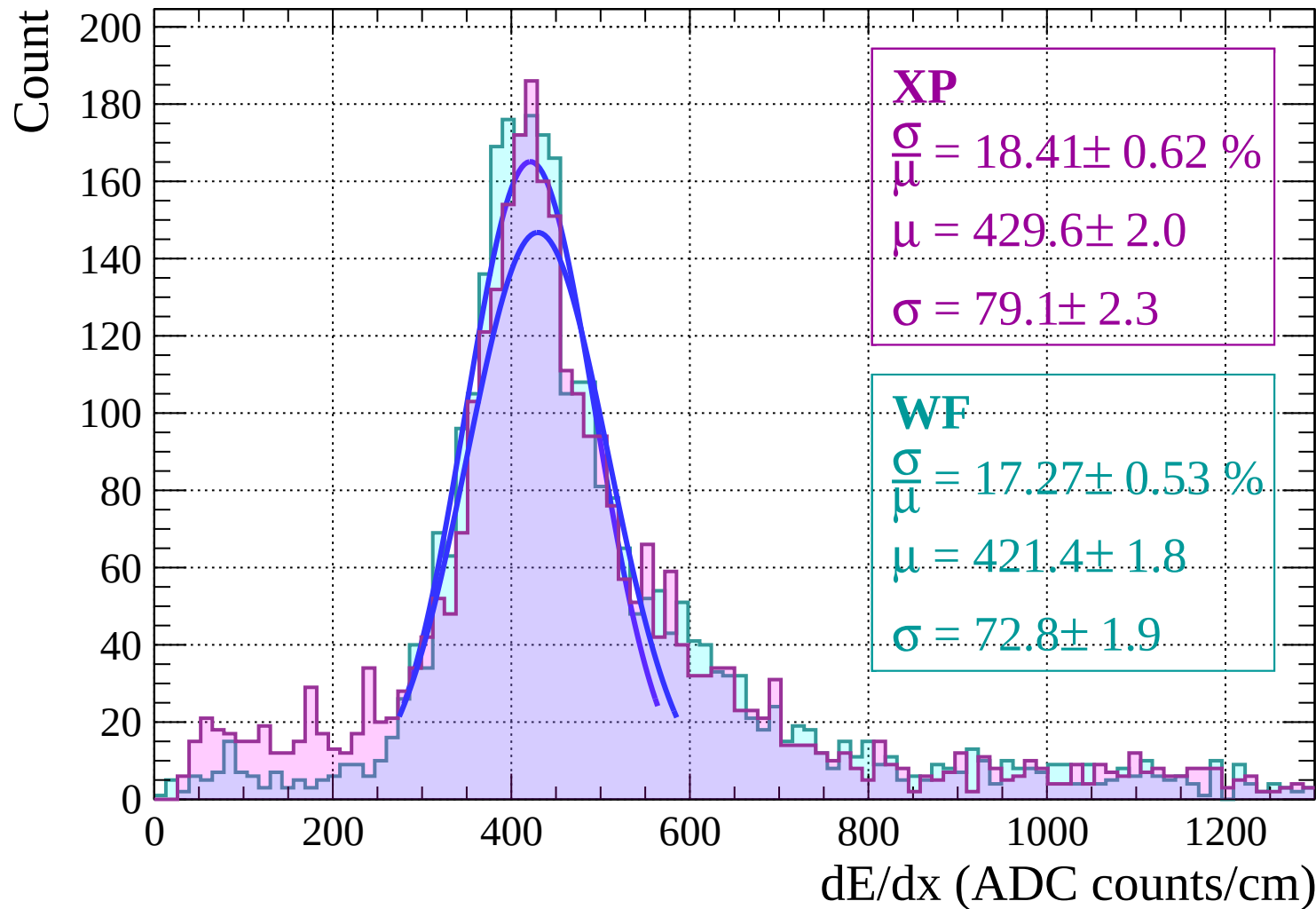
Energy loss | $-42 < \theta < -40$

Count

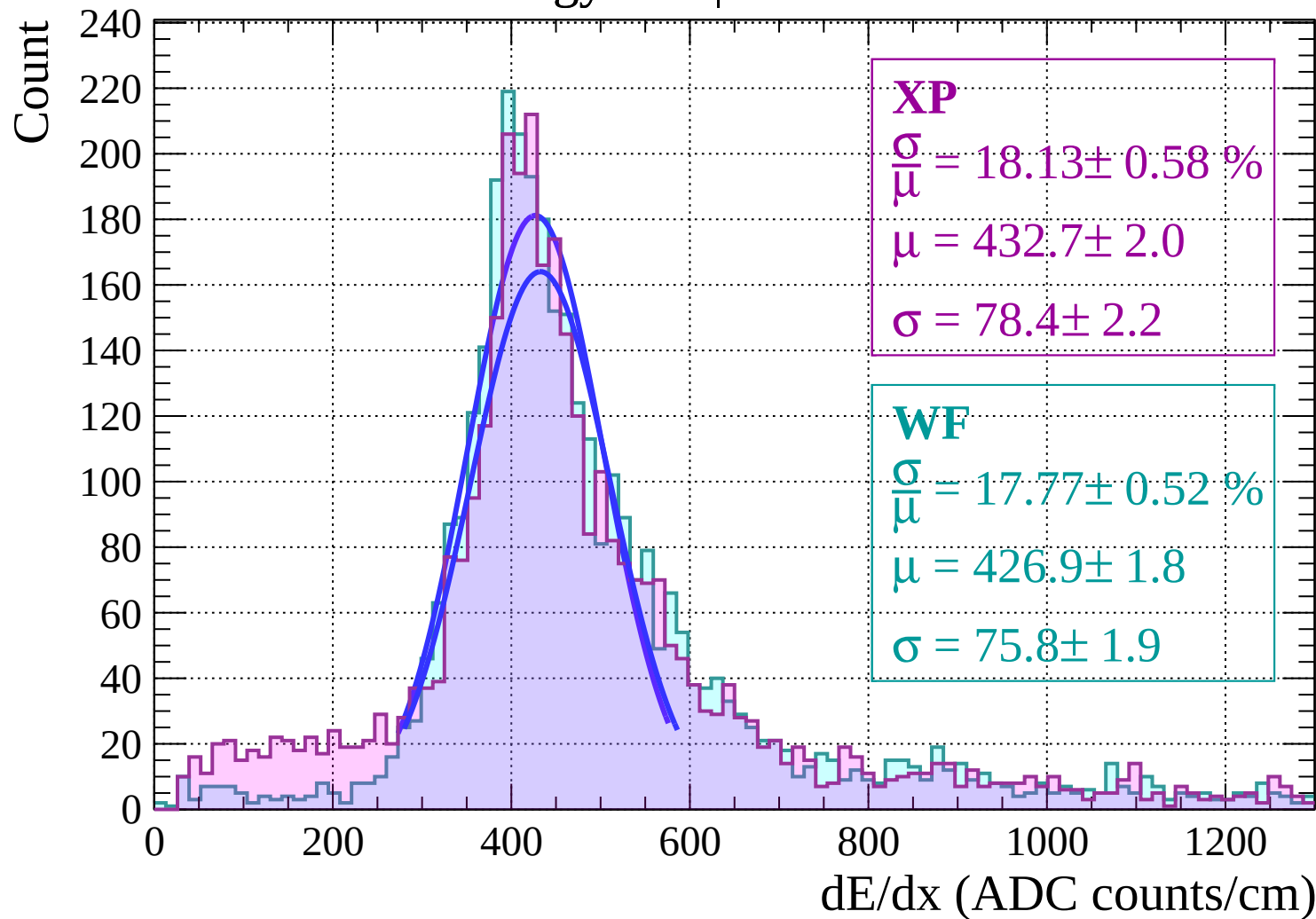


dE/dx (ADC counts/cm)

Energy loss | $-40 < \theta < -38$

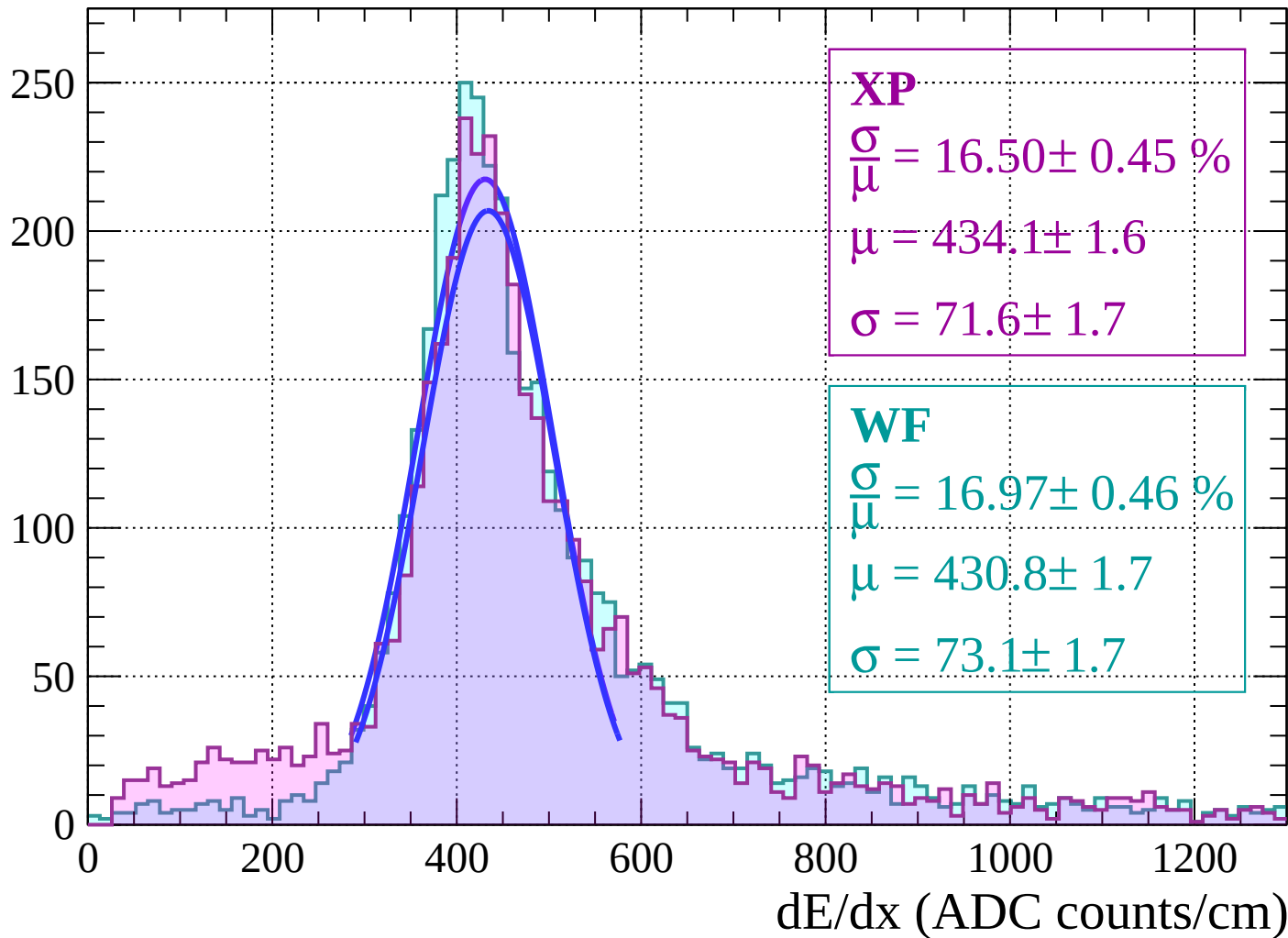


Energy loss $|-38 < \theta < -36$

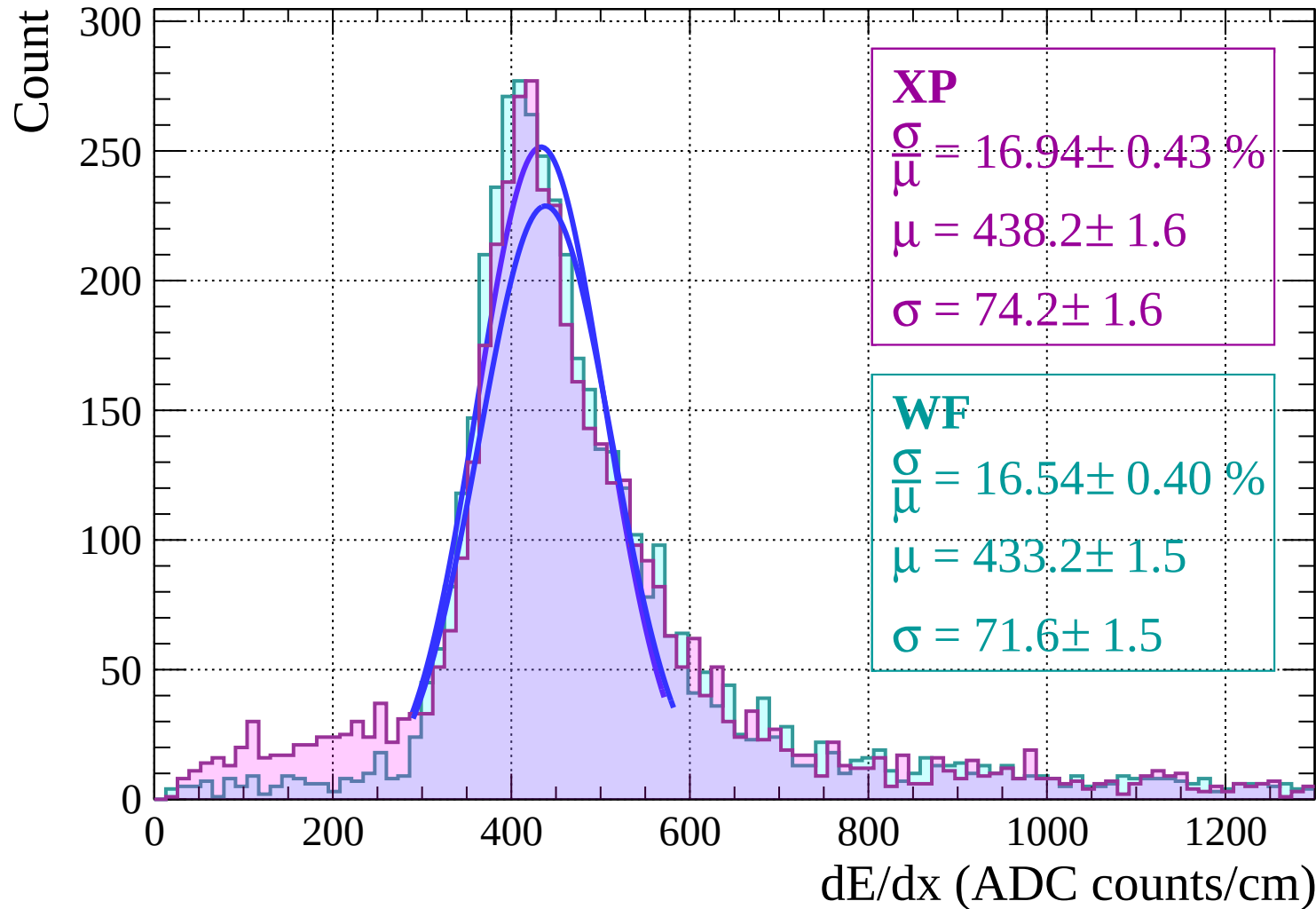


Energy loss | $-36 < \theta < -34$

Count

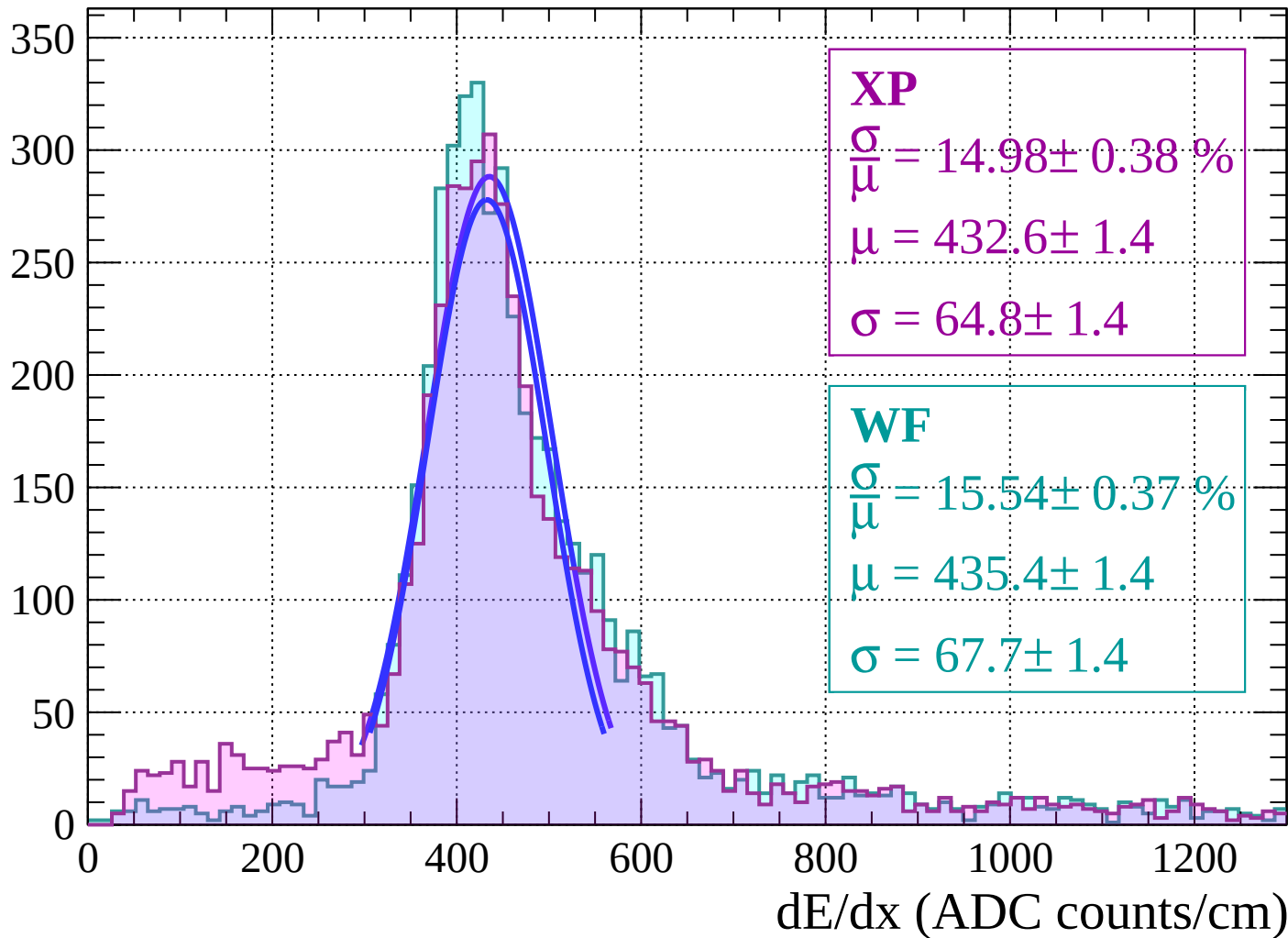


Energy loss $|-34 < \theta < -32$



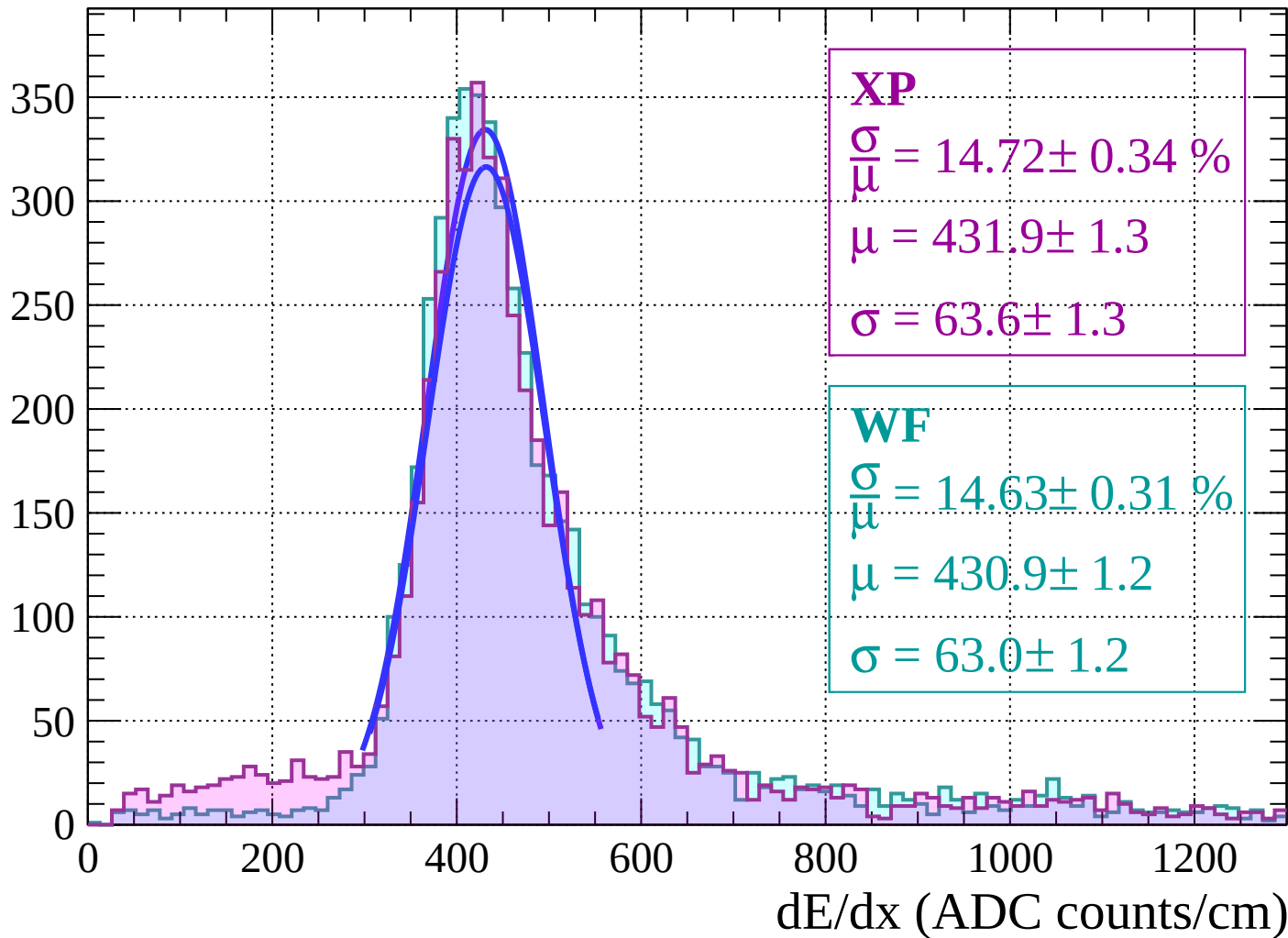
Energy loss | $-32 < \theta < -30$

Count



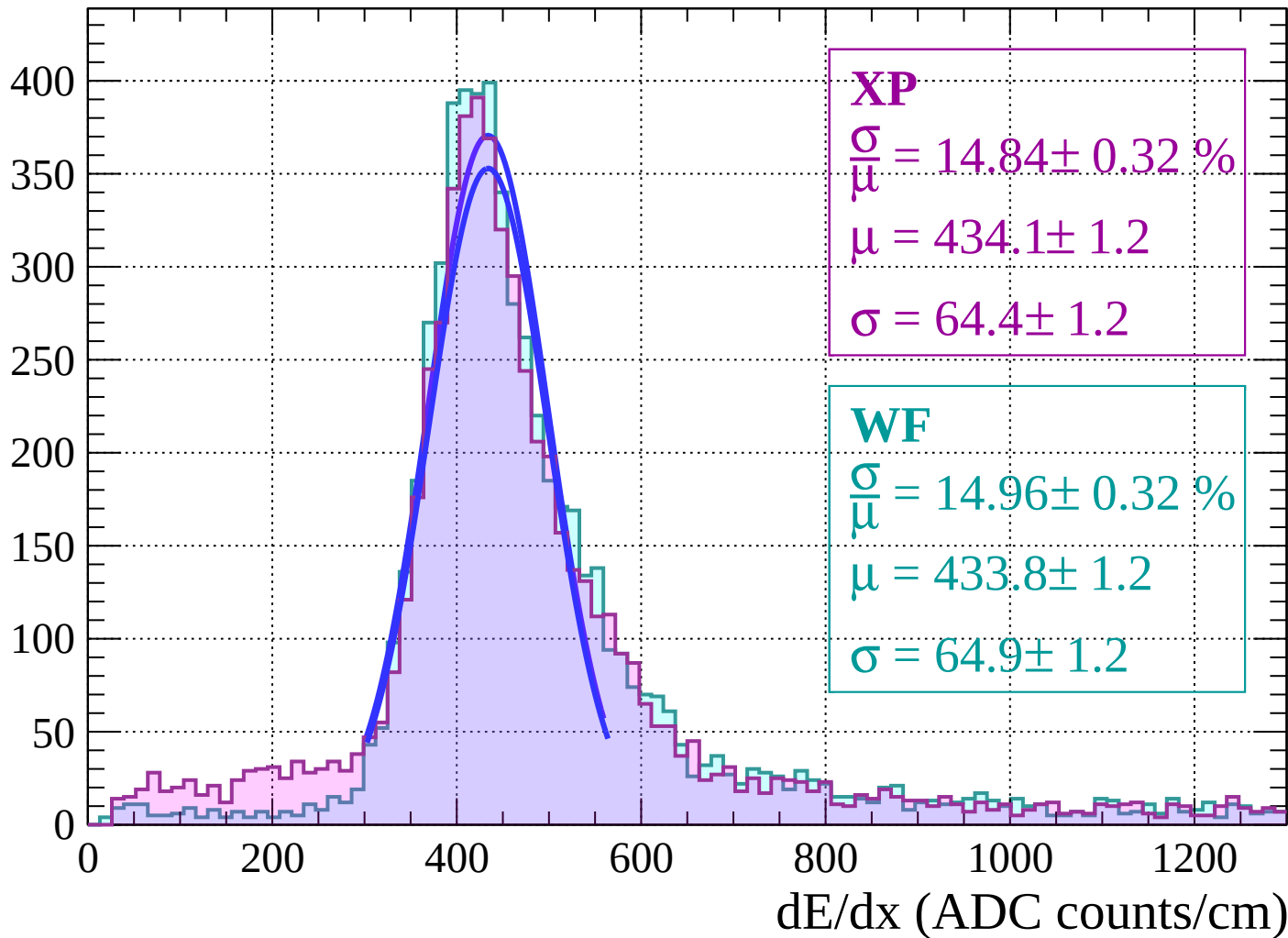
Energy loss | $-30 < \theta < -28$

Count

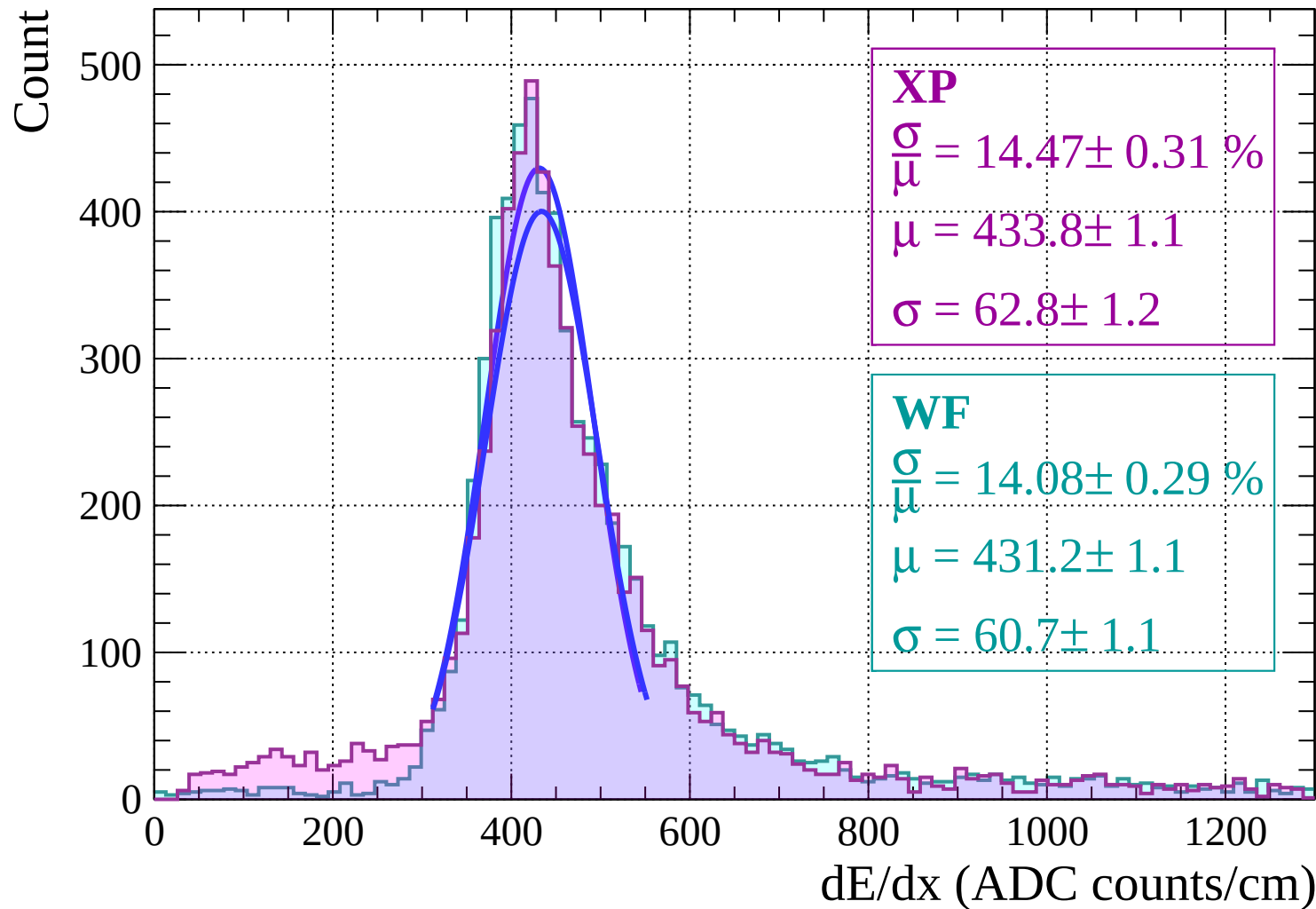


Energy loss | $-28 < \theta < -26$

Count

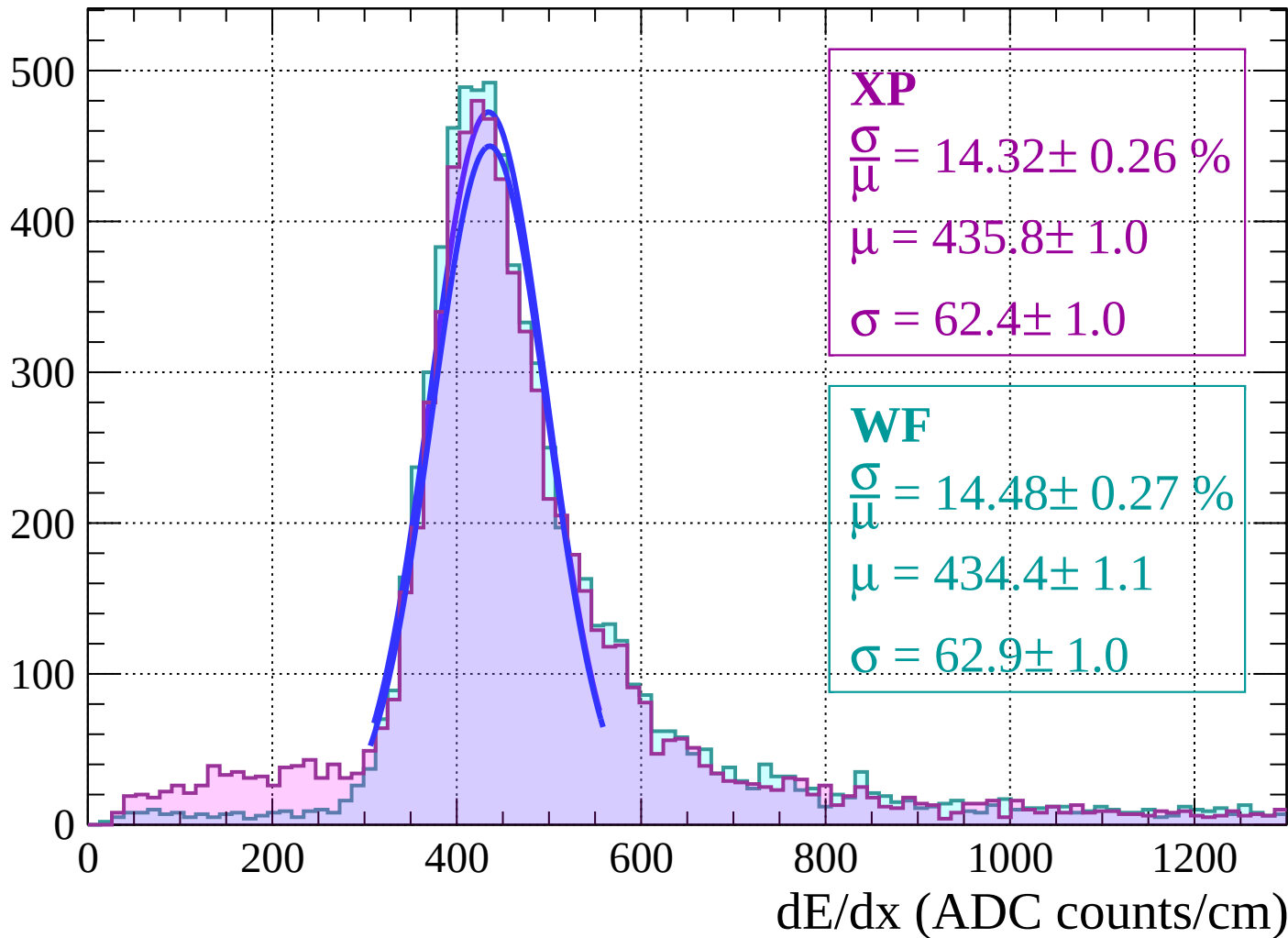


Energy loss | $-26 < \theta < -24$



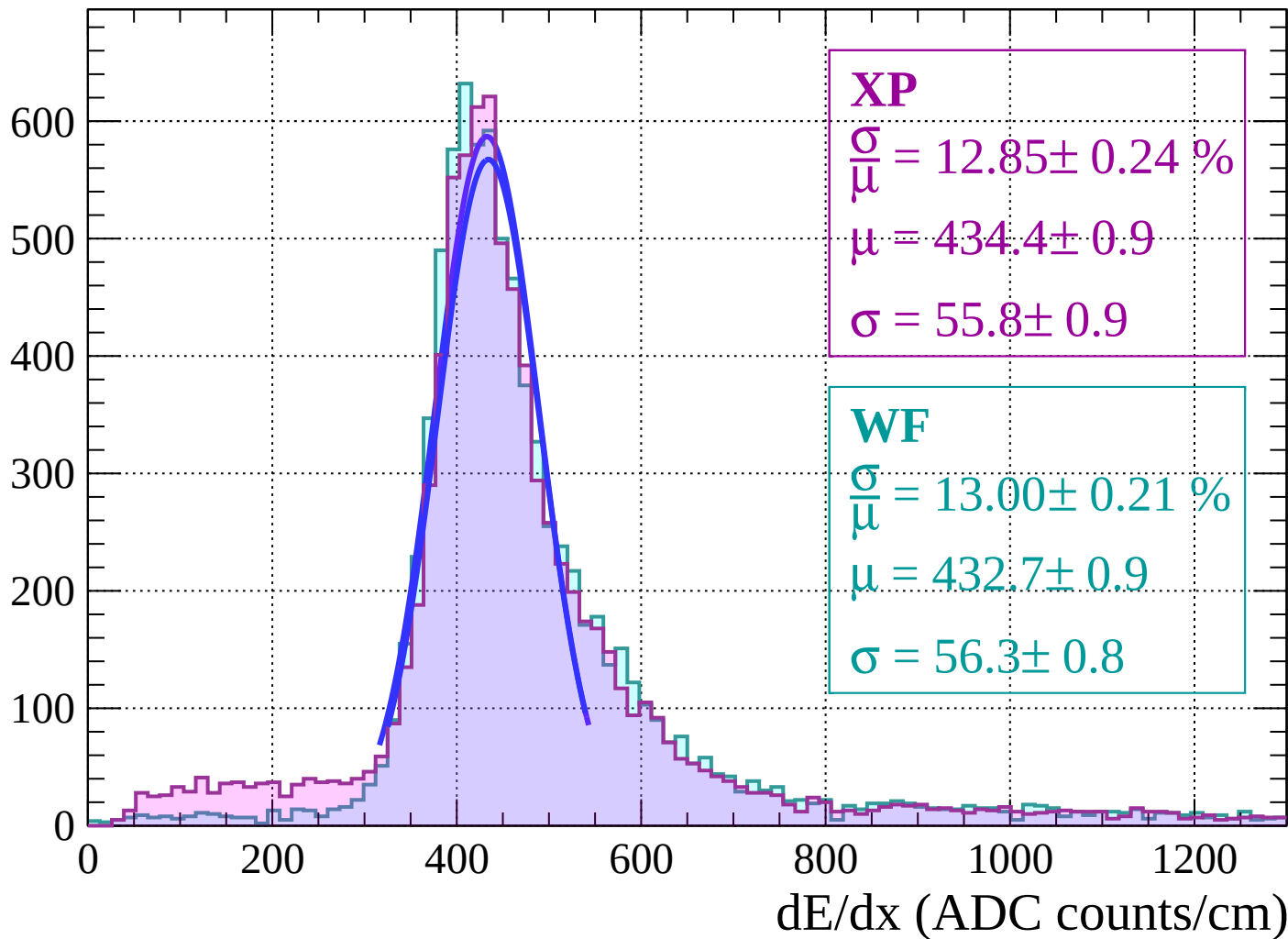
Energy loss $|-24 < \theta < -22$

Count



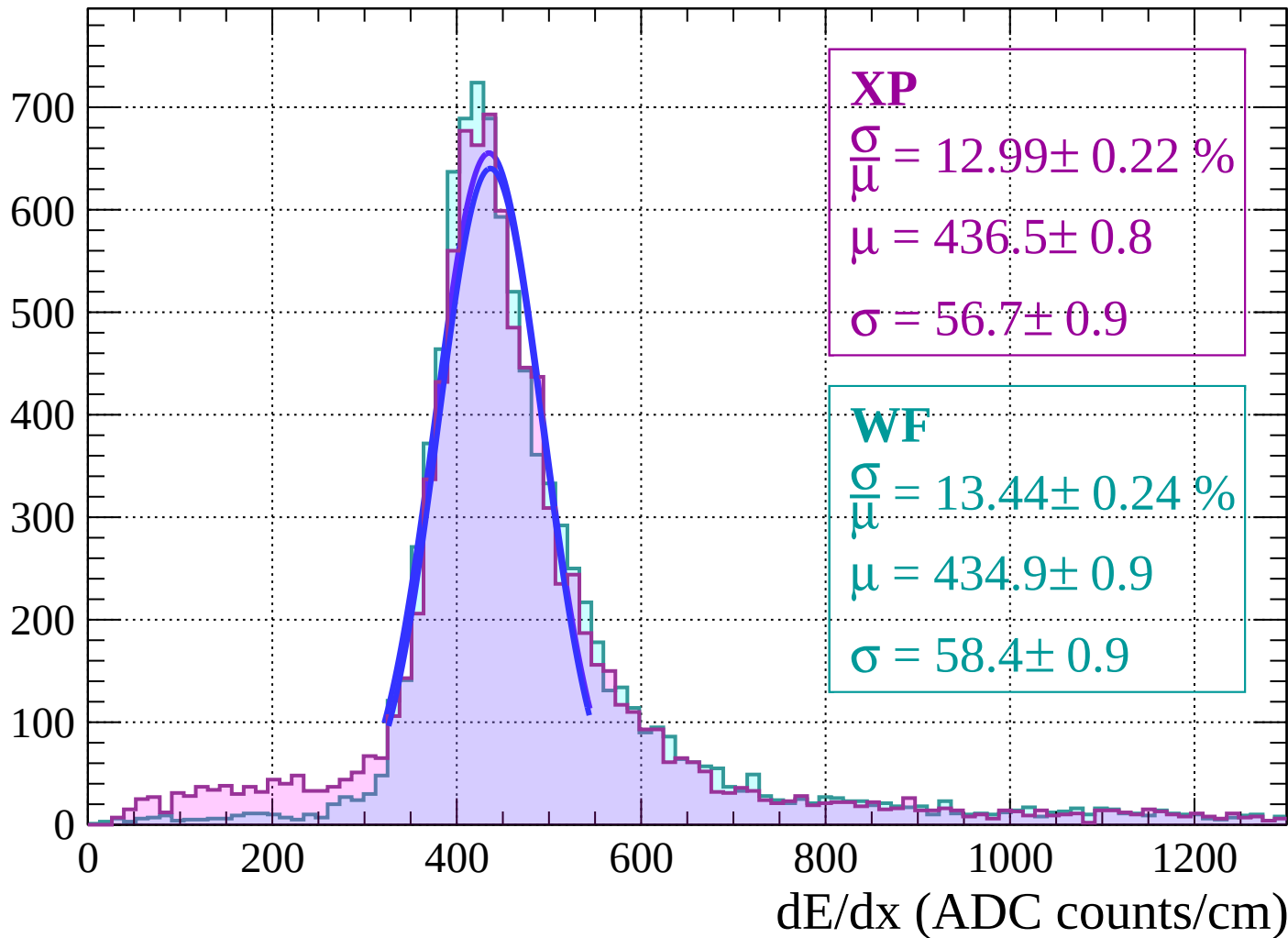
Energy loss | $-22 < \theta < -20$

Count



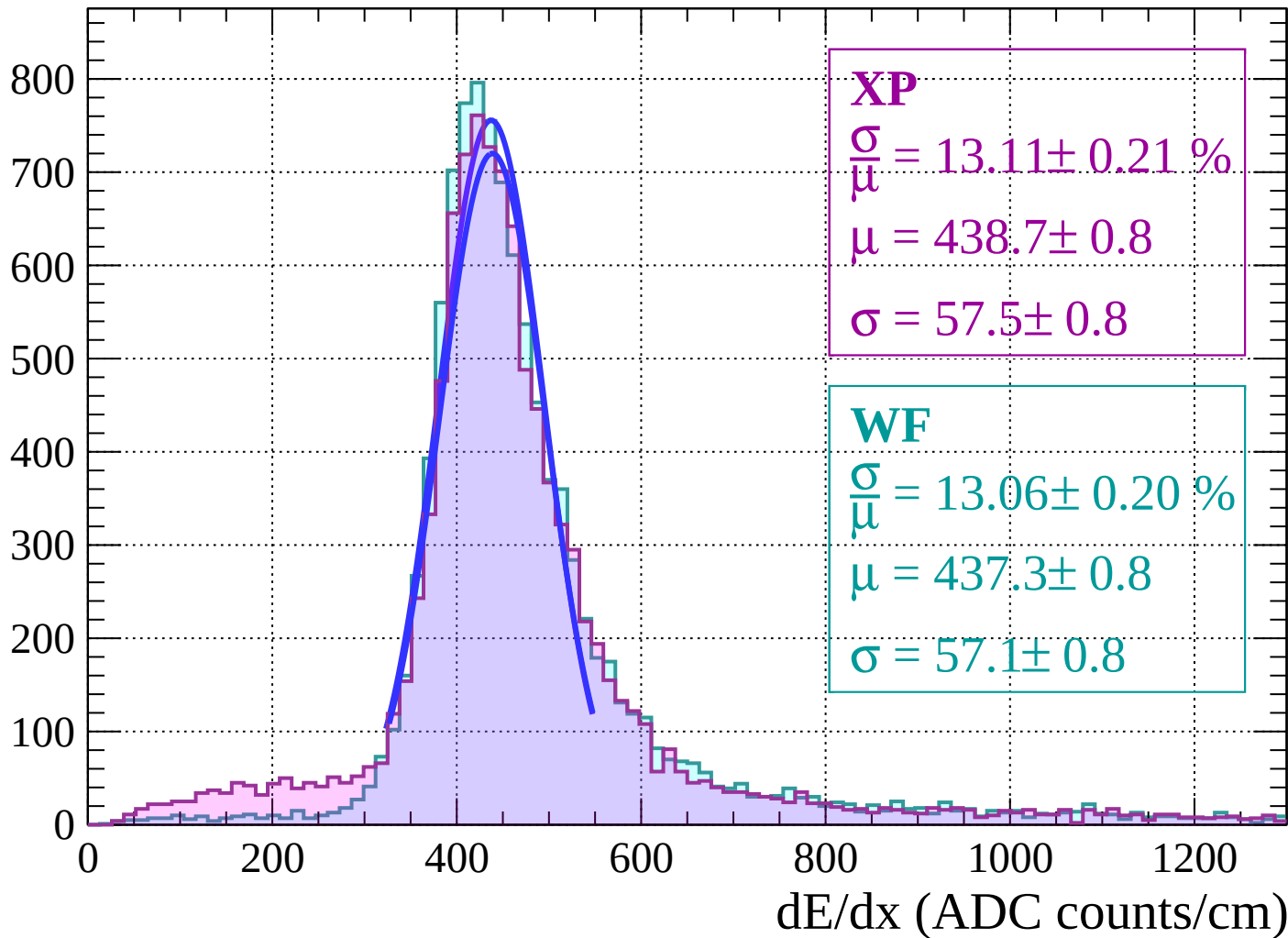
Energy loss | $-20 < \theta < -18$

Count

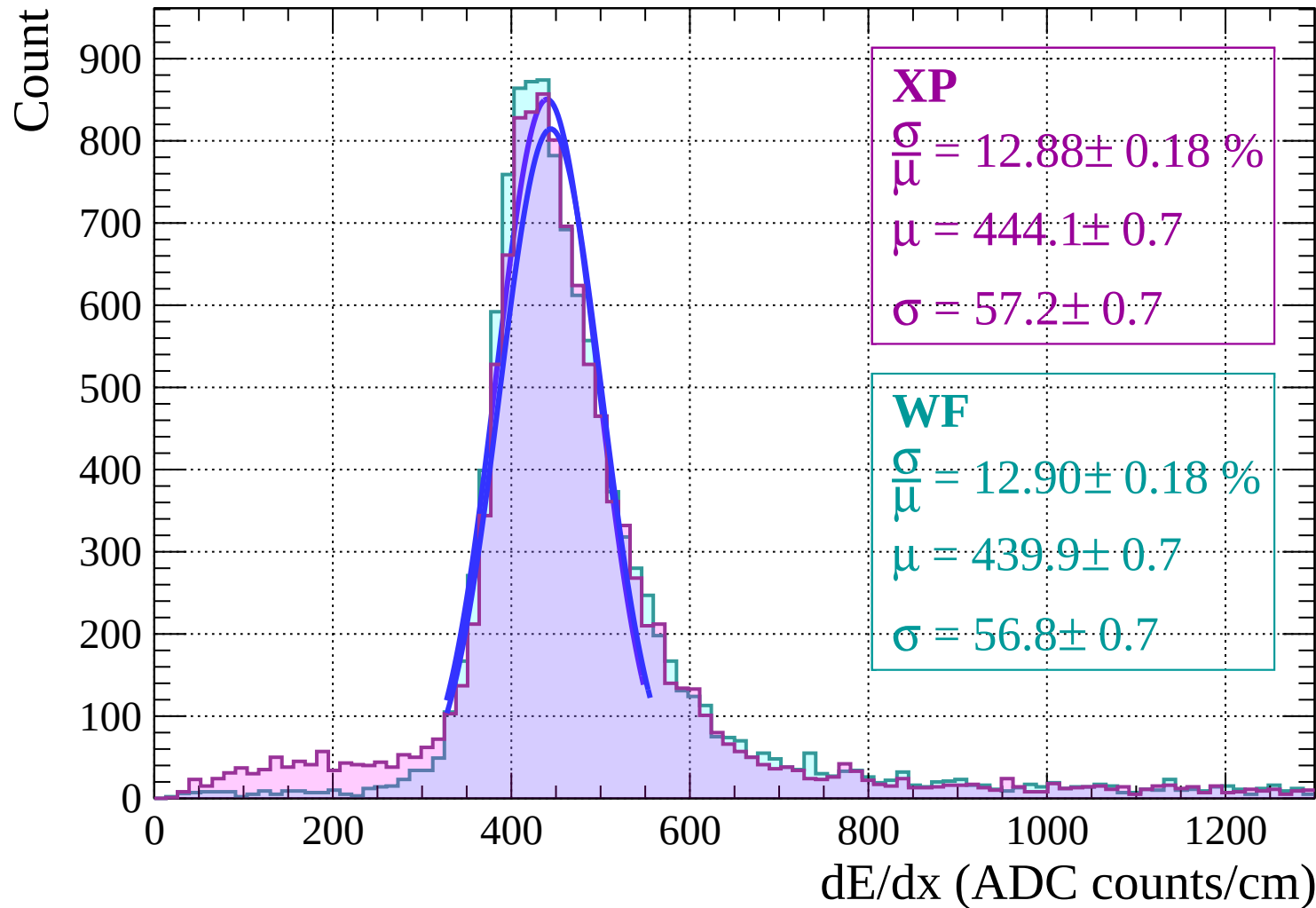


Energy loss $|-18 < \theta < -16$

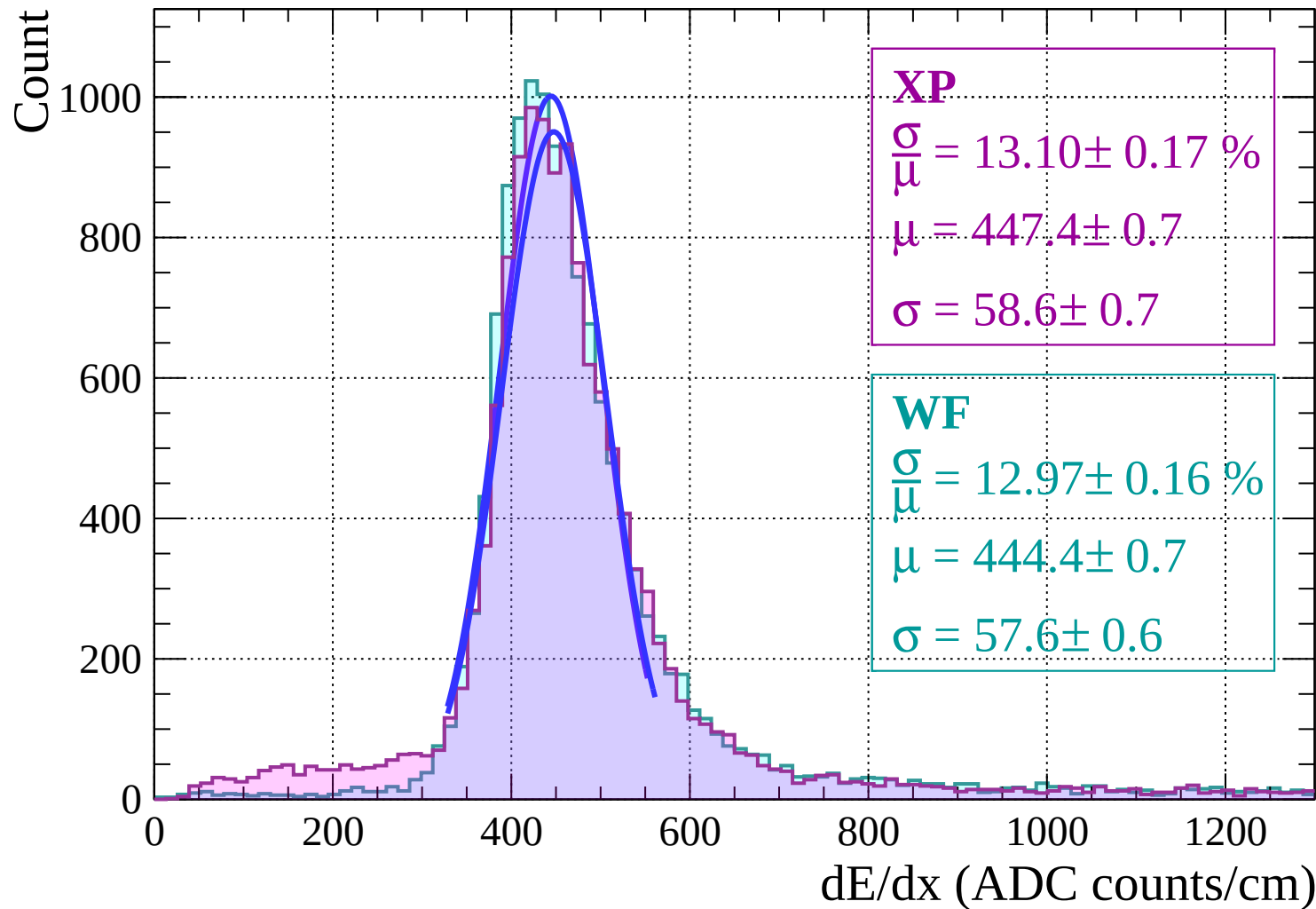
Count



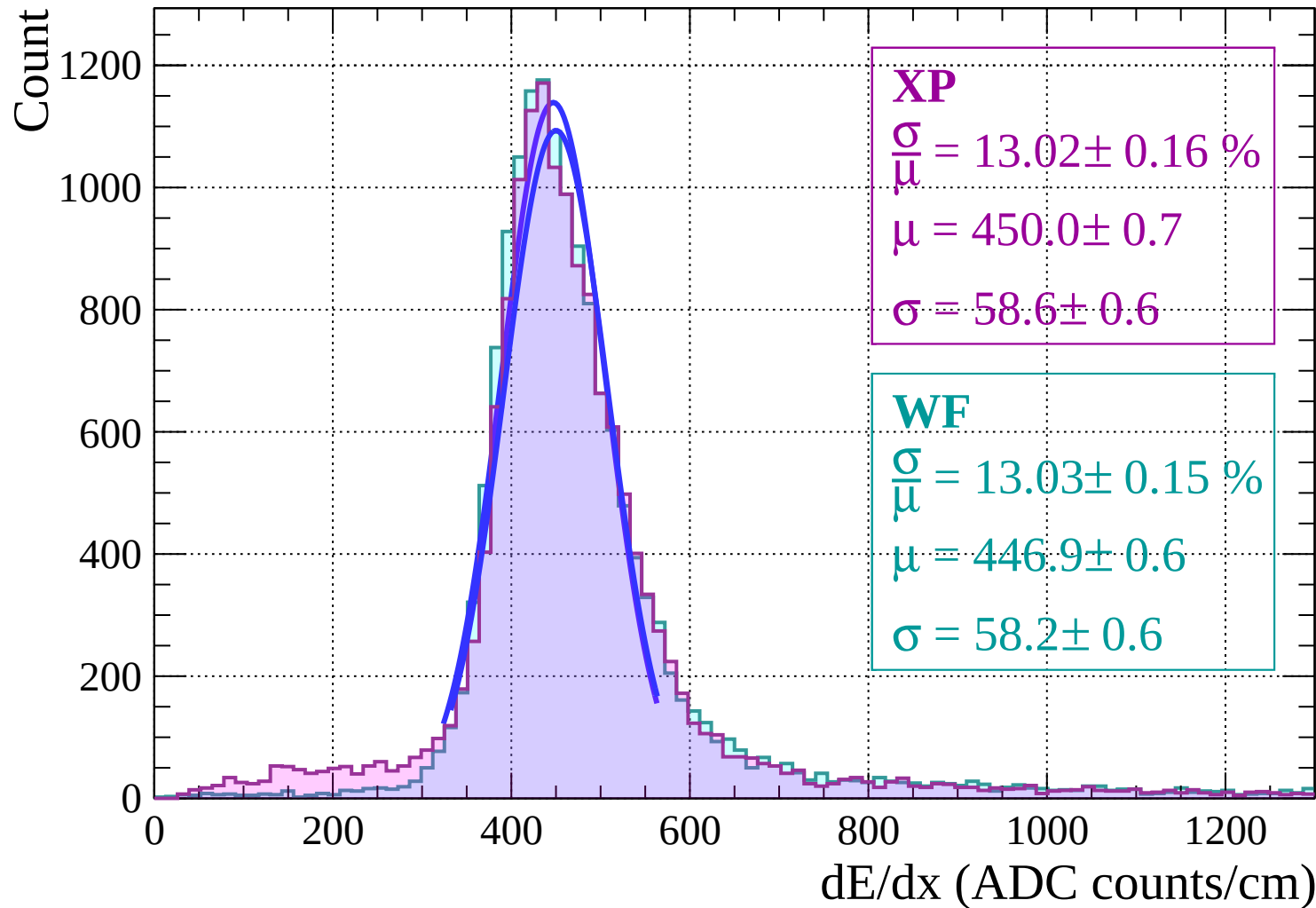
Energy loss | $-16 < \theta < -14$



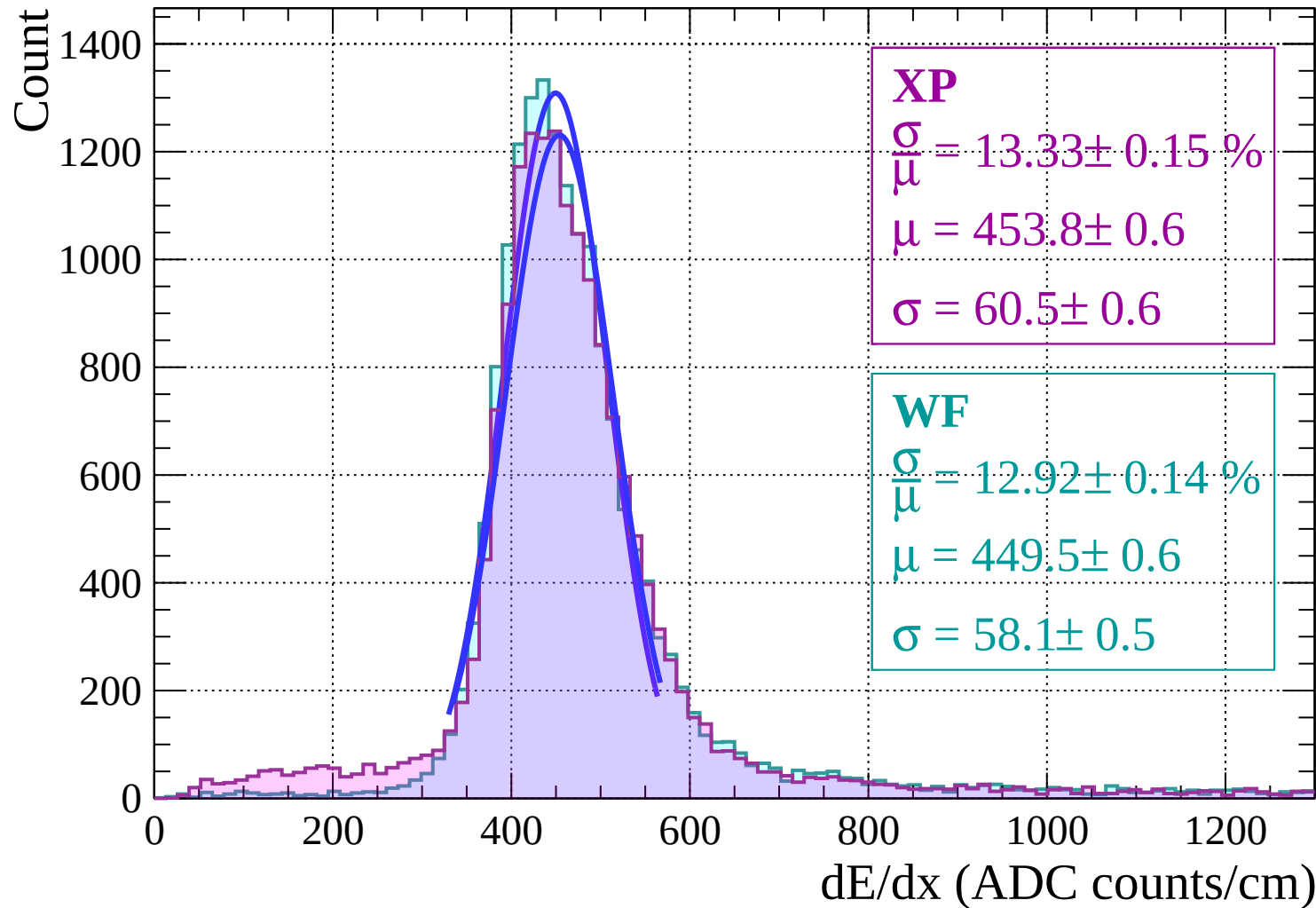
Energy loss | $-14 < \theta < -12$



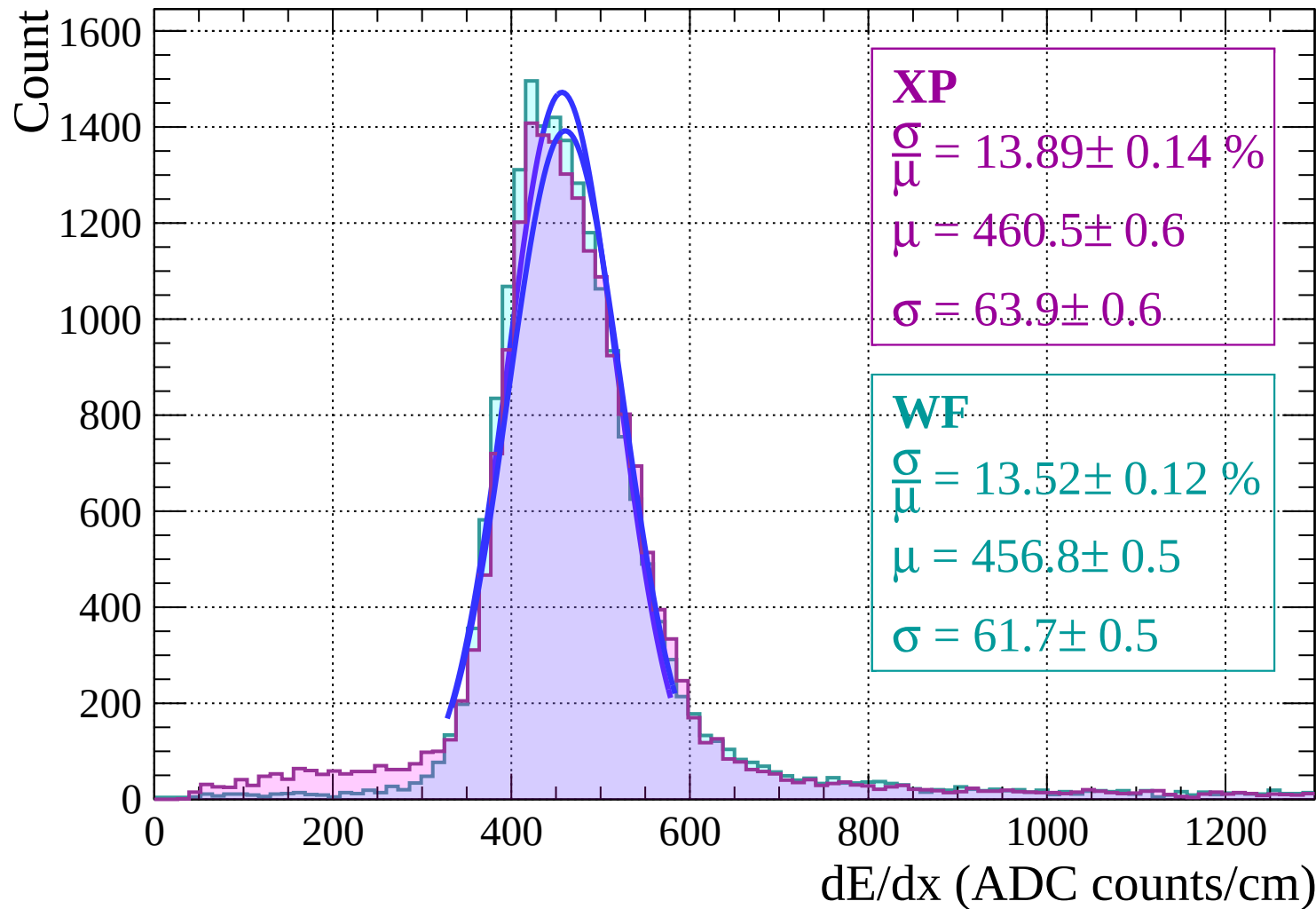
Energy loss | $-12 < \theta < -10$



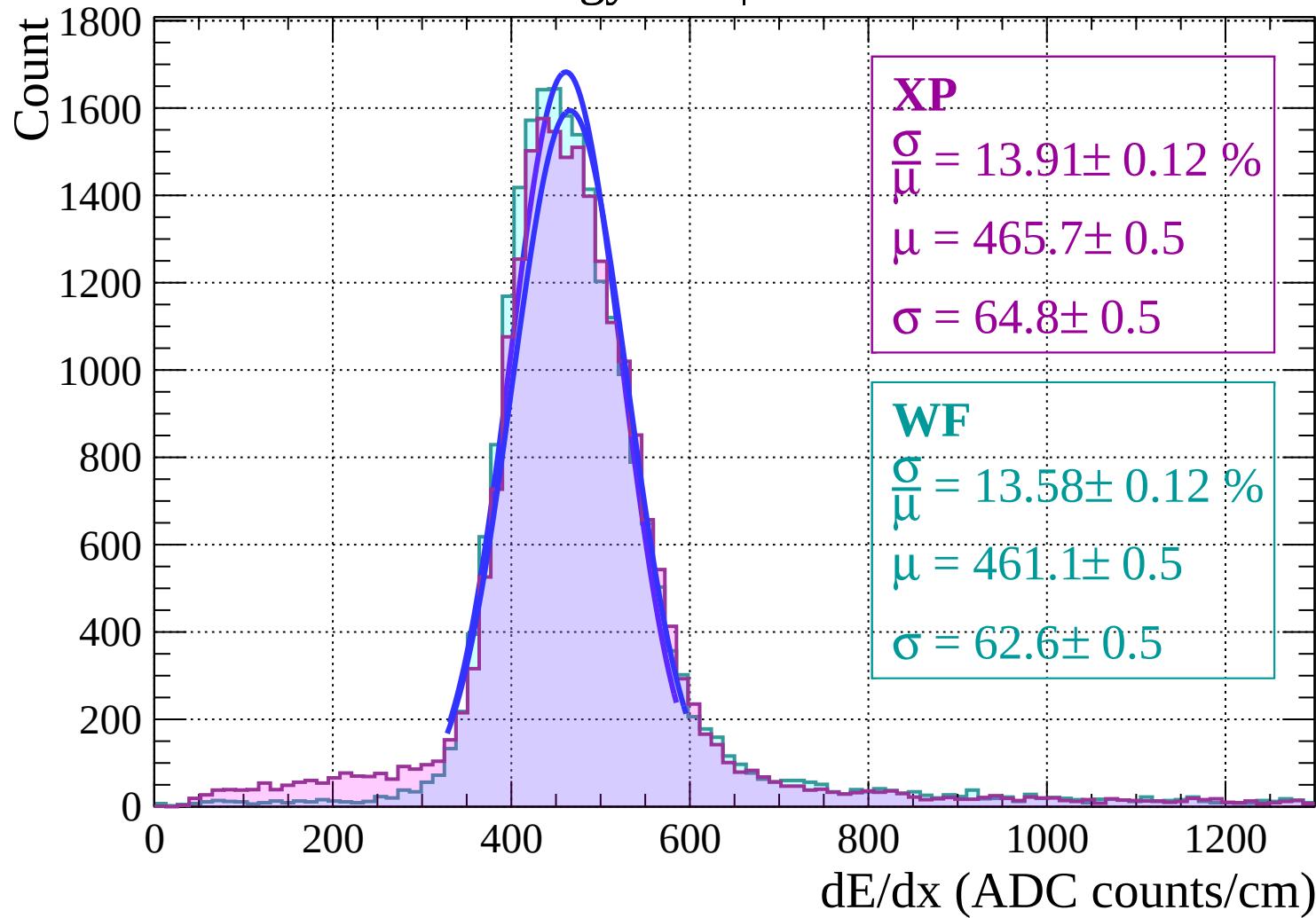
Energy loss | $-10 < \theta < -8$



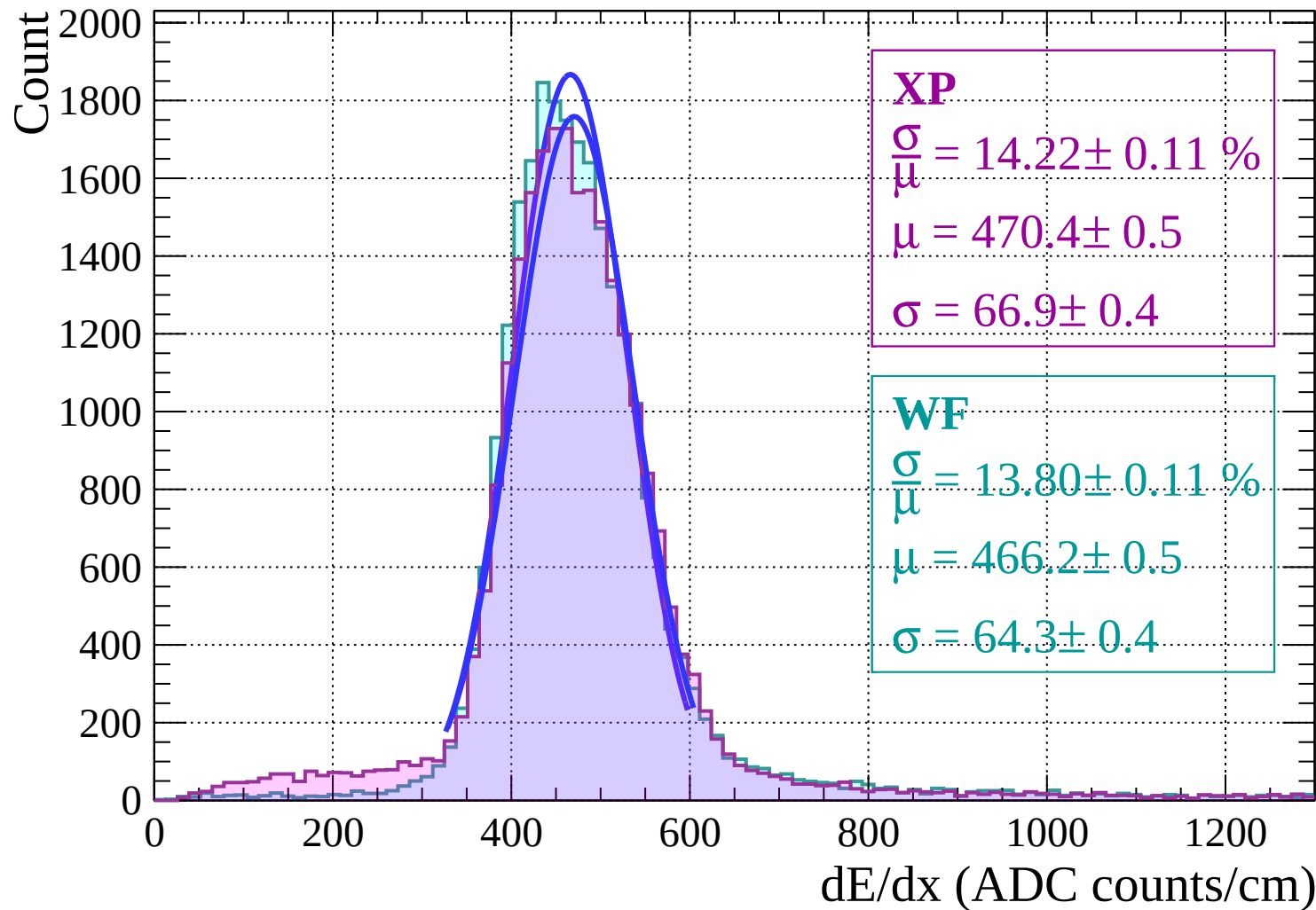
Energy loss | $-8 < \theta < -6$



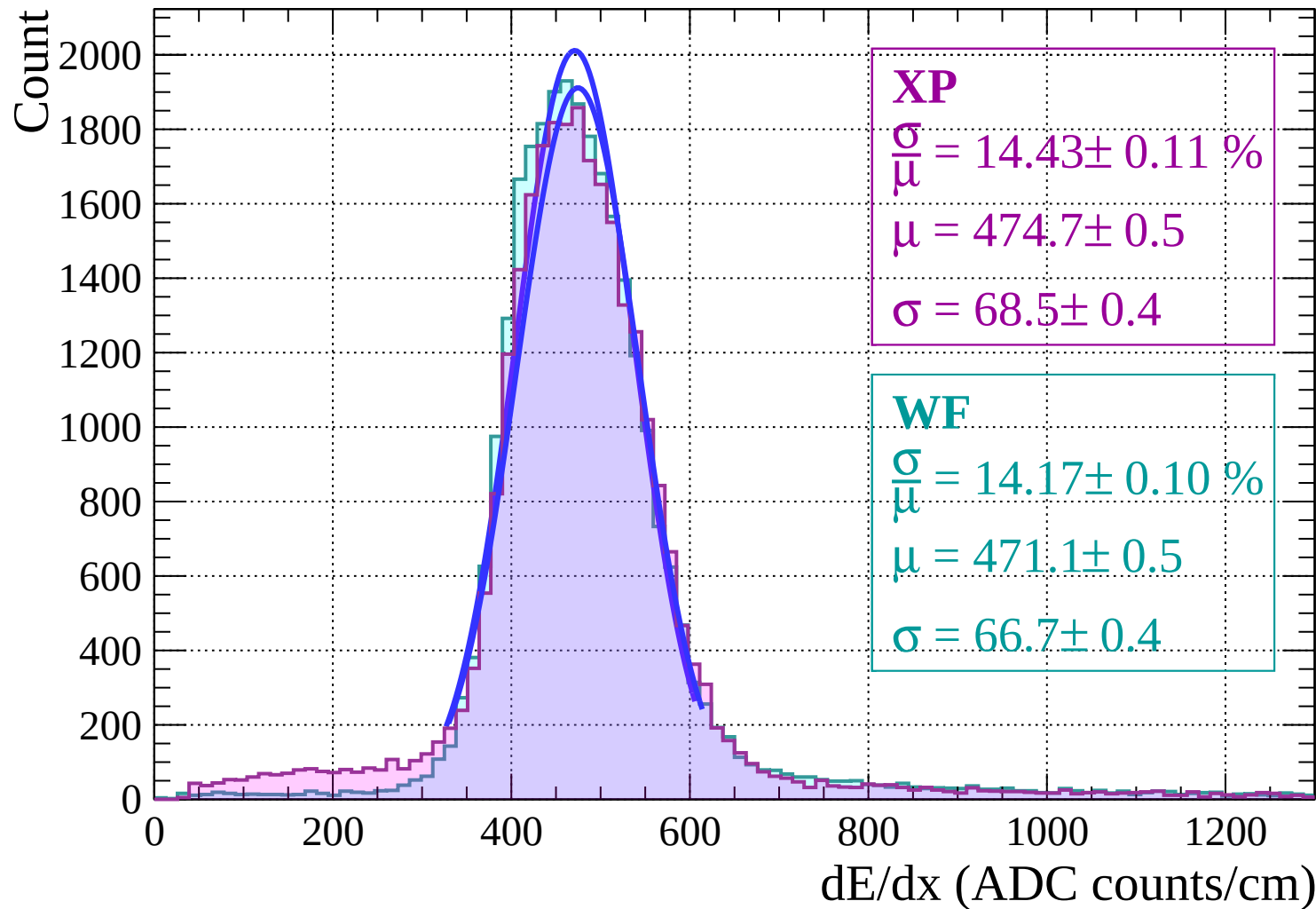
Energy loss $|-6 < \theta < -4|$



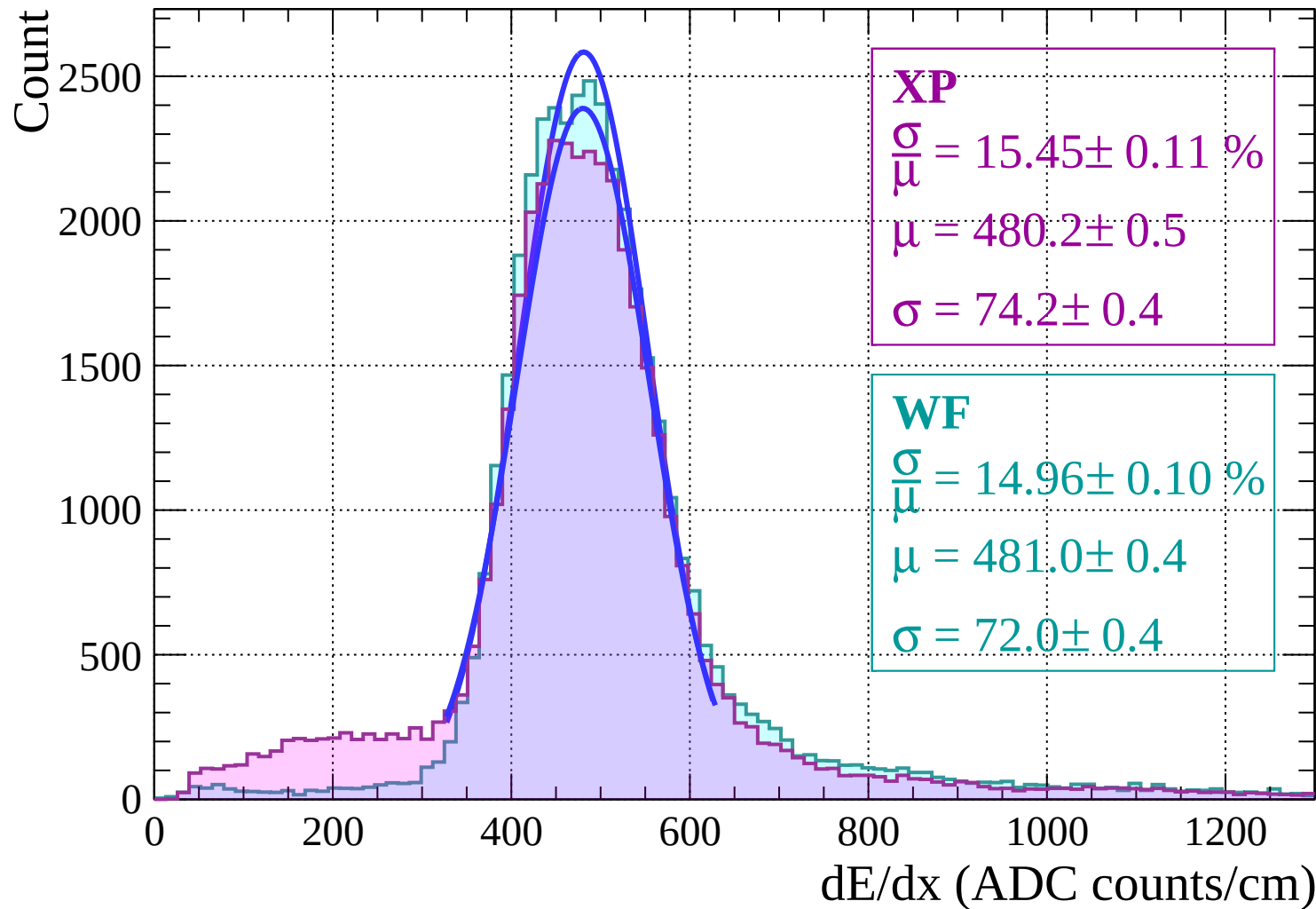
Energy loss | $-4 < \theta < -2$



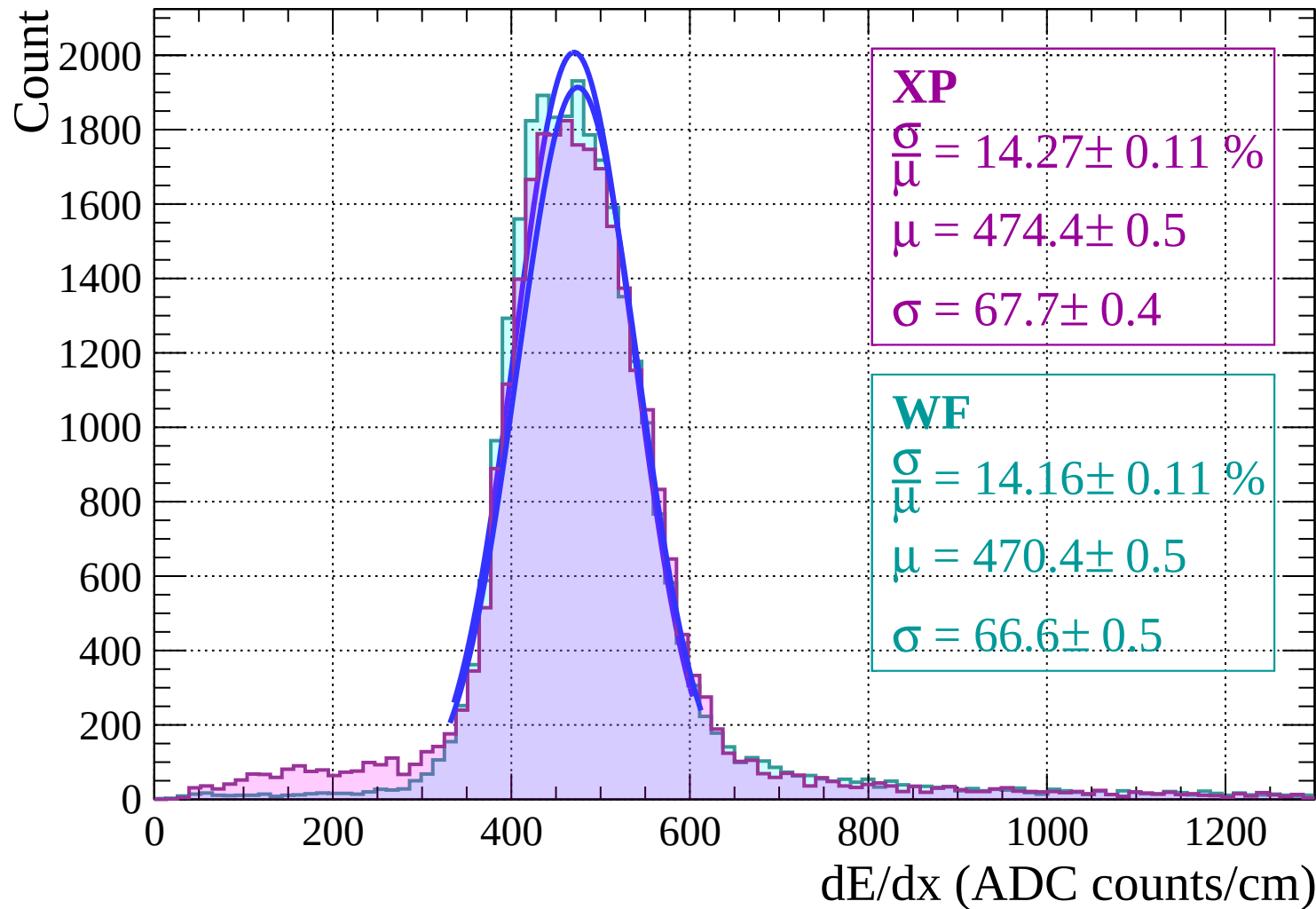
Energy loss | $-2 < \theta < 0$



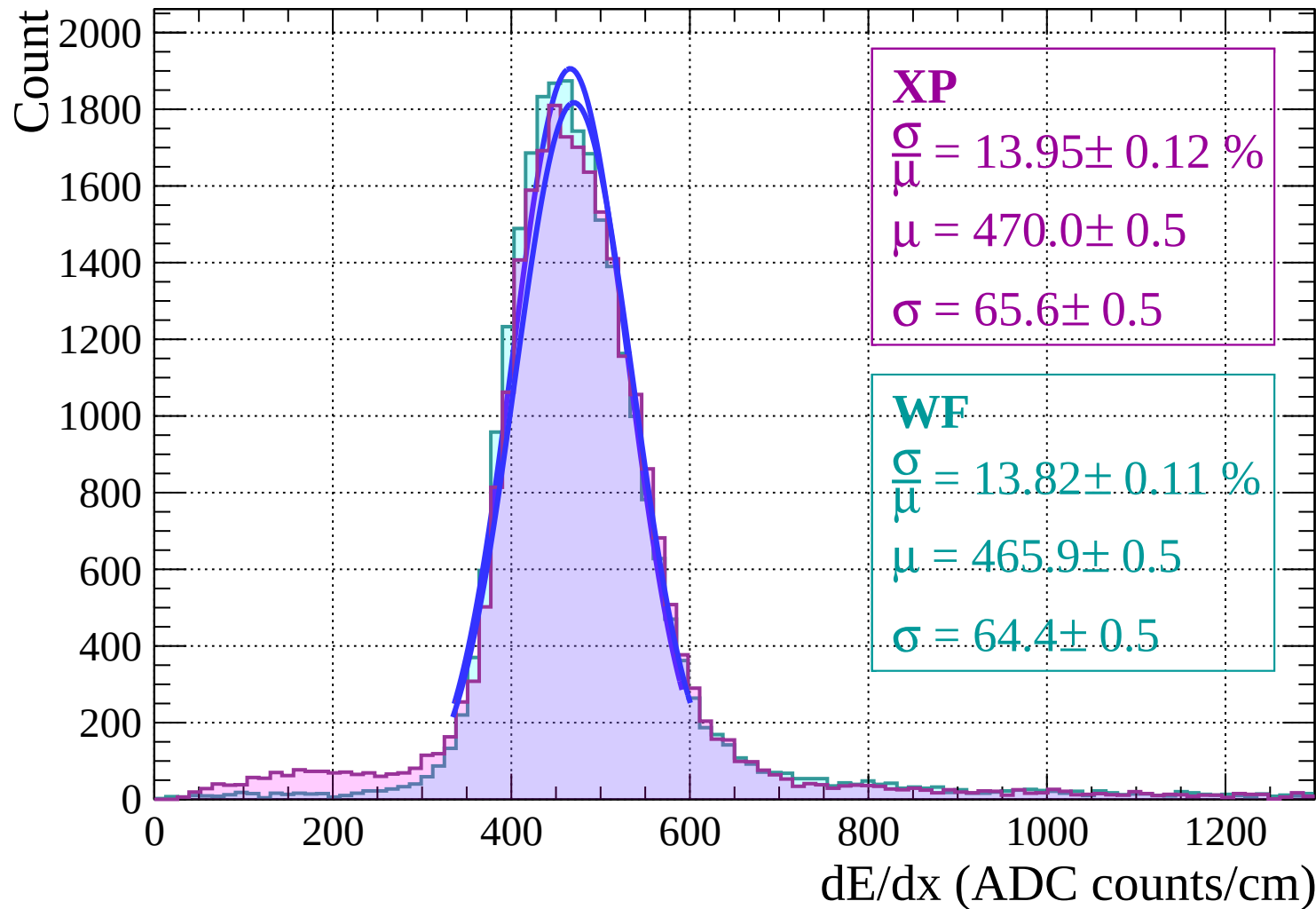
Energy loss | $0 < \theta < 2$



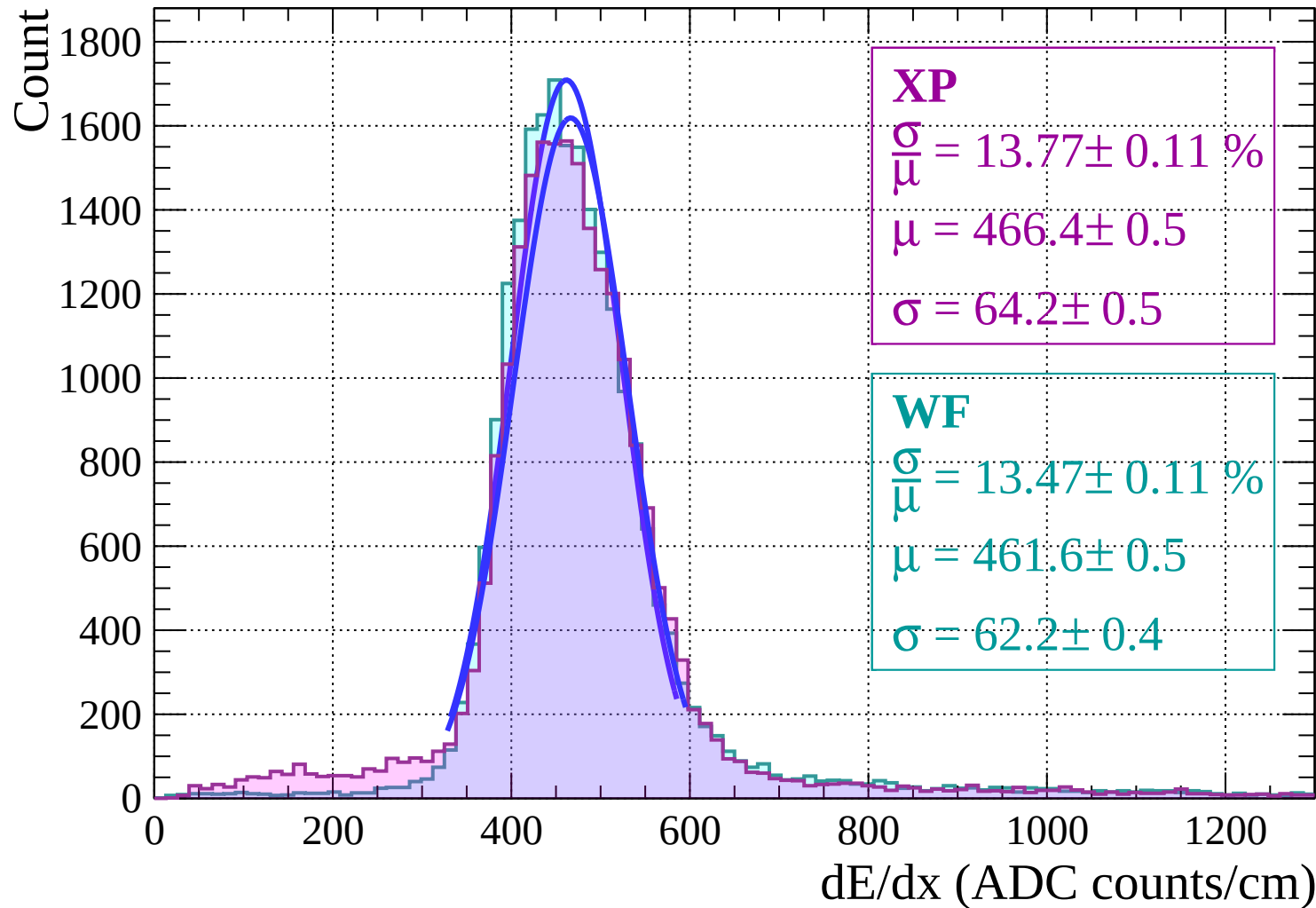
Energy loss | $2 < \theta < 4$



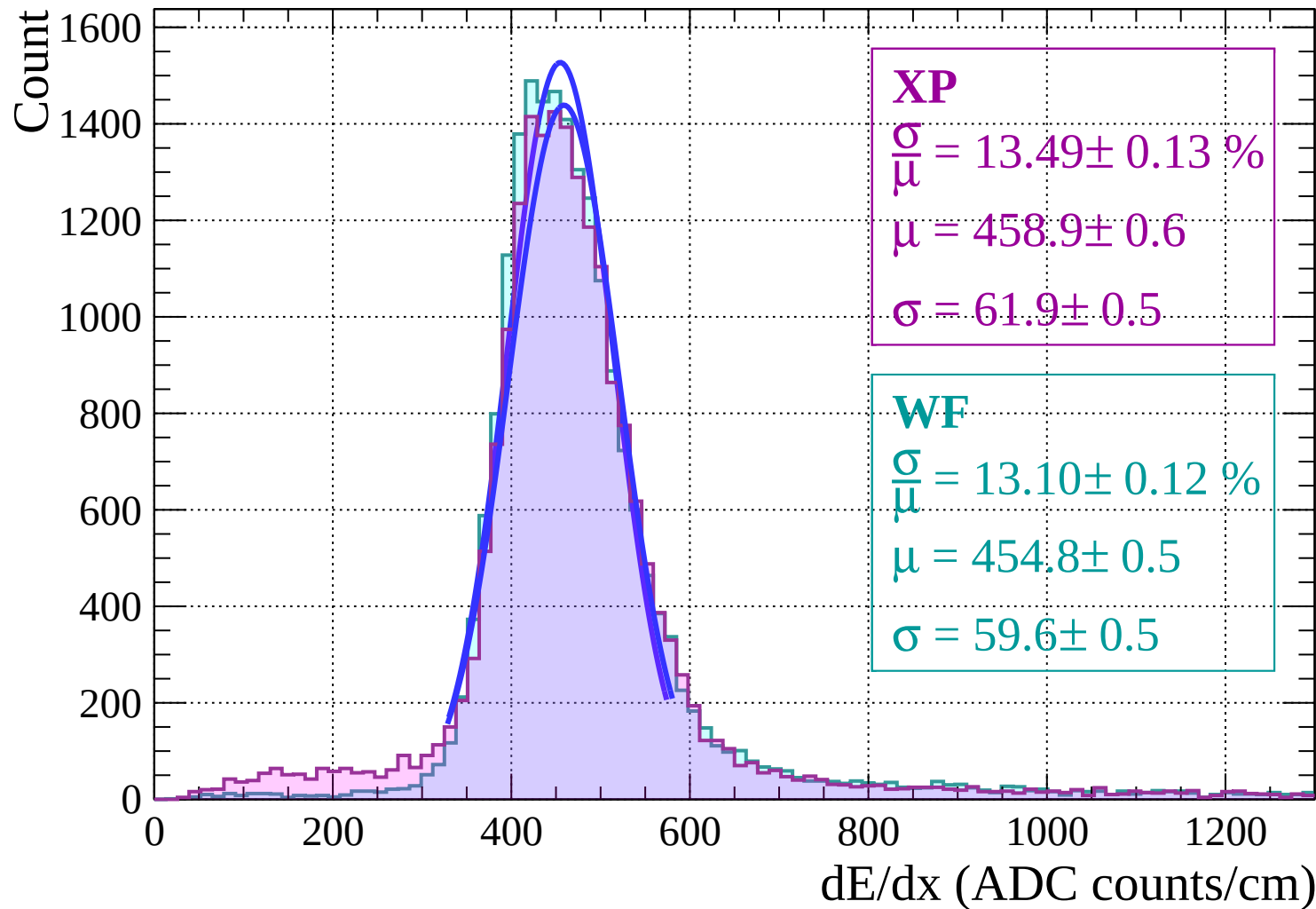
Energy loss | $4 < \theta < 6$



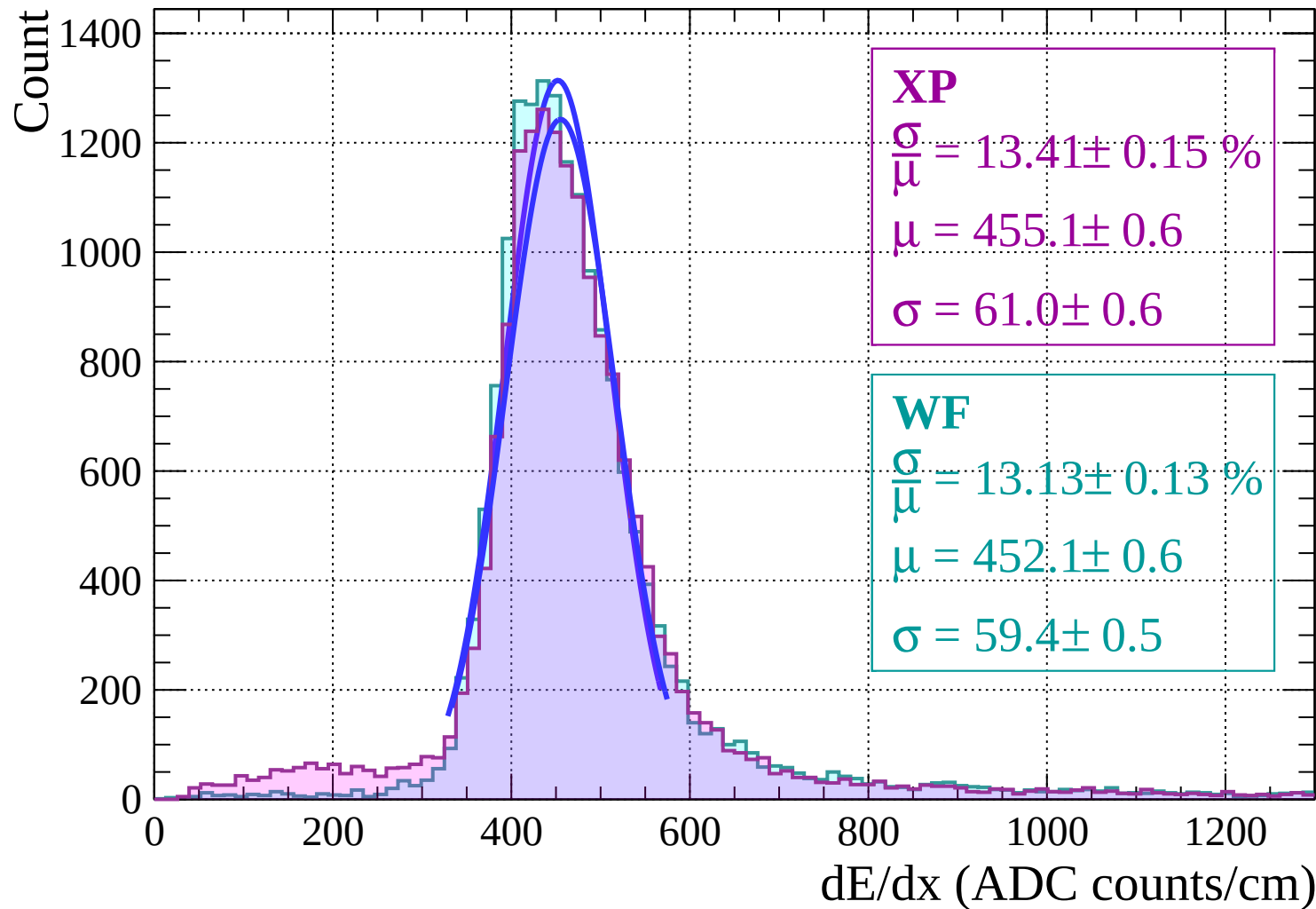
Energy loss | $6 < \theta < 8$



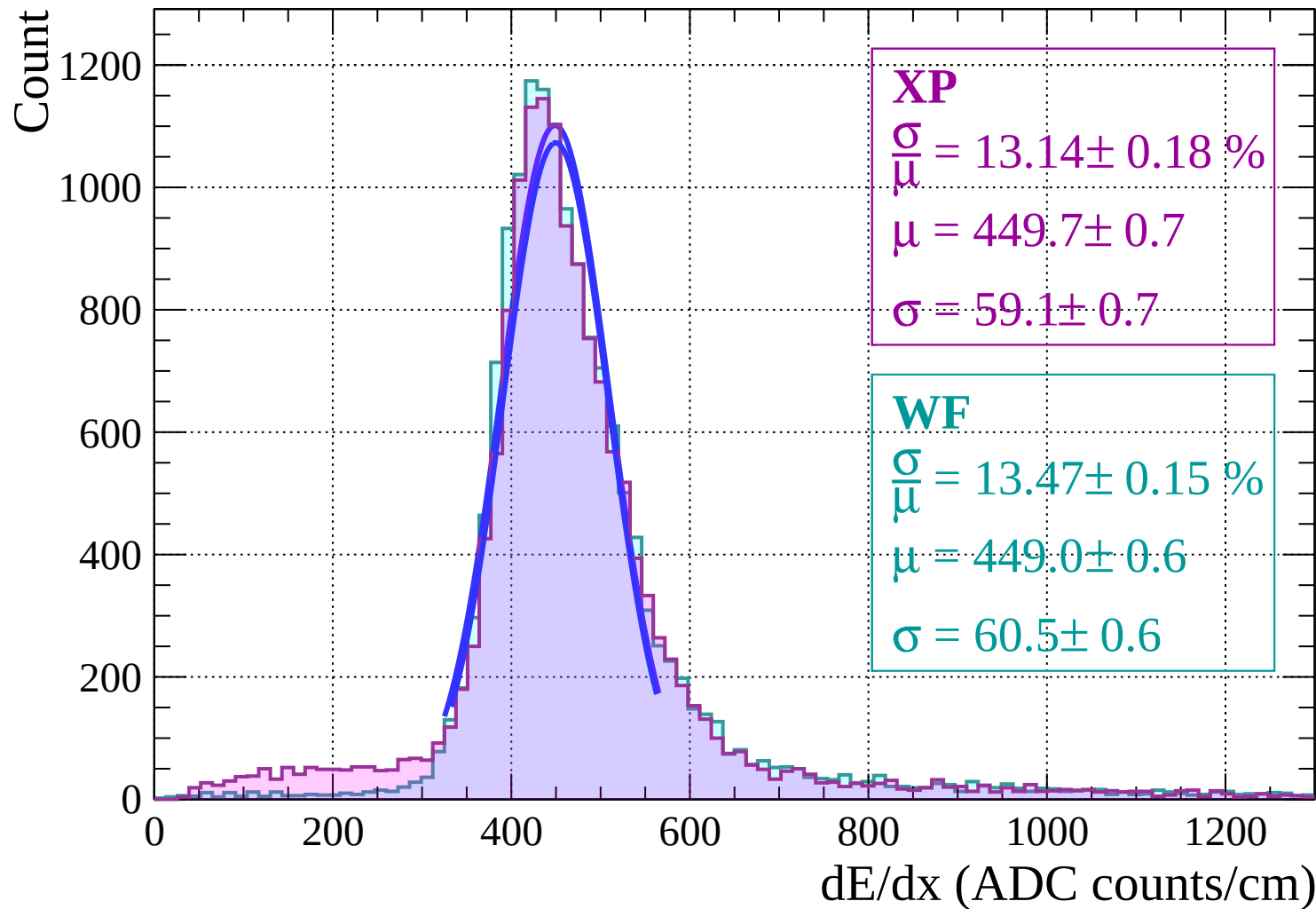
Energy loss | $8 < \theta < 10$



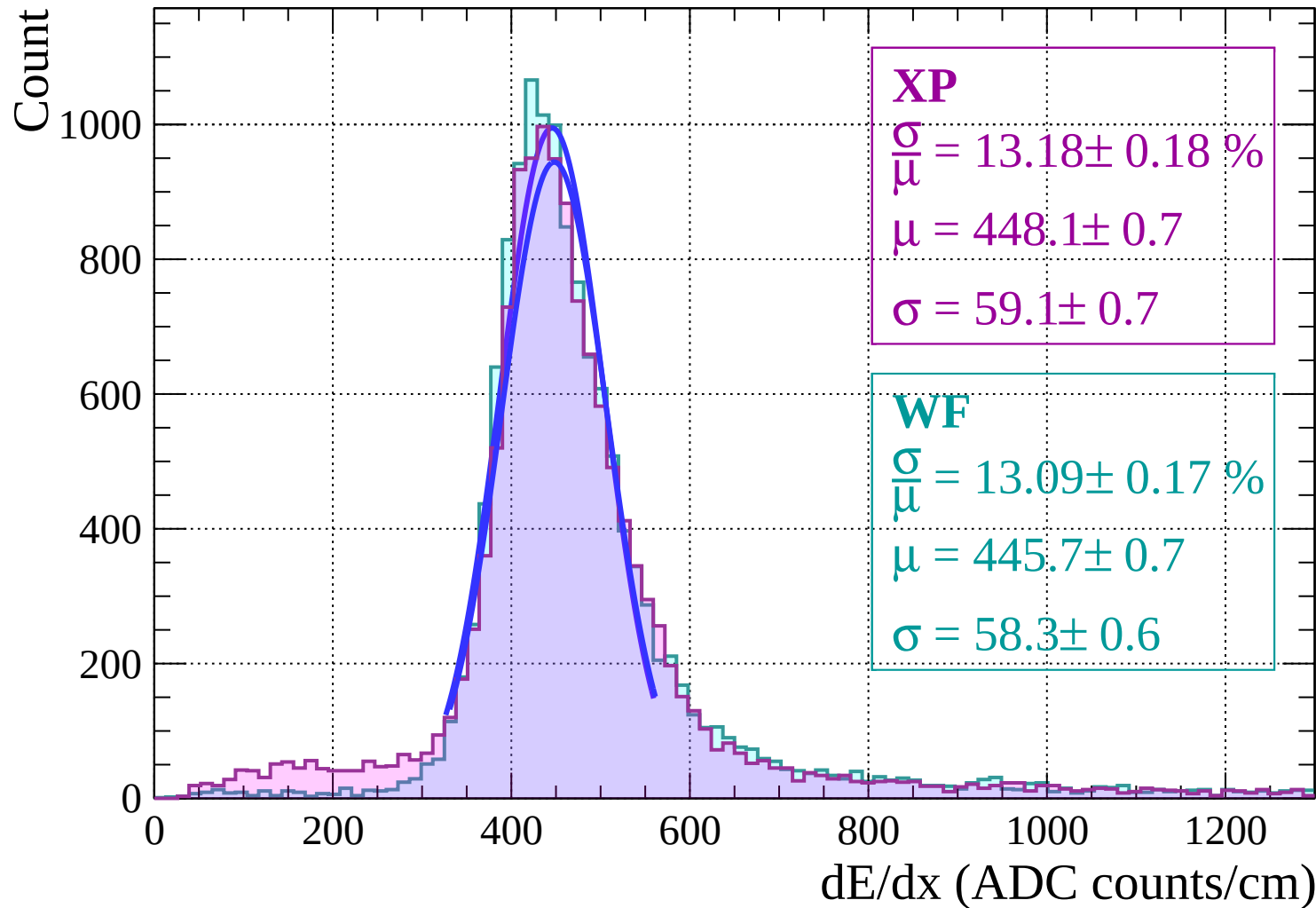
Energy loss | $10 < \theta < 12$



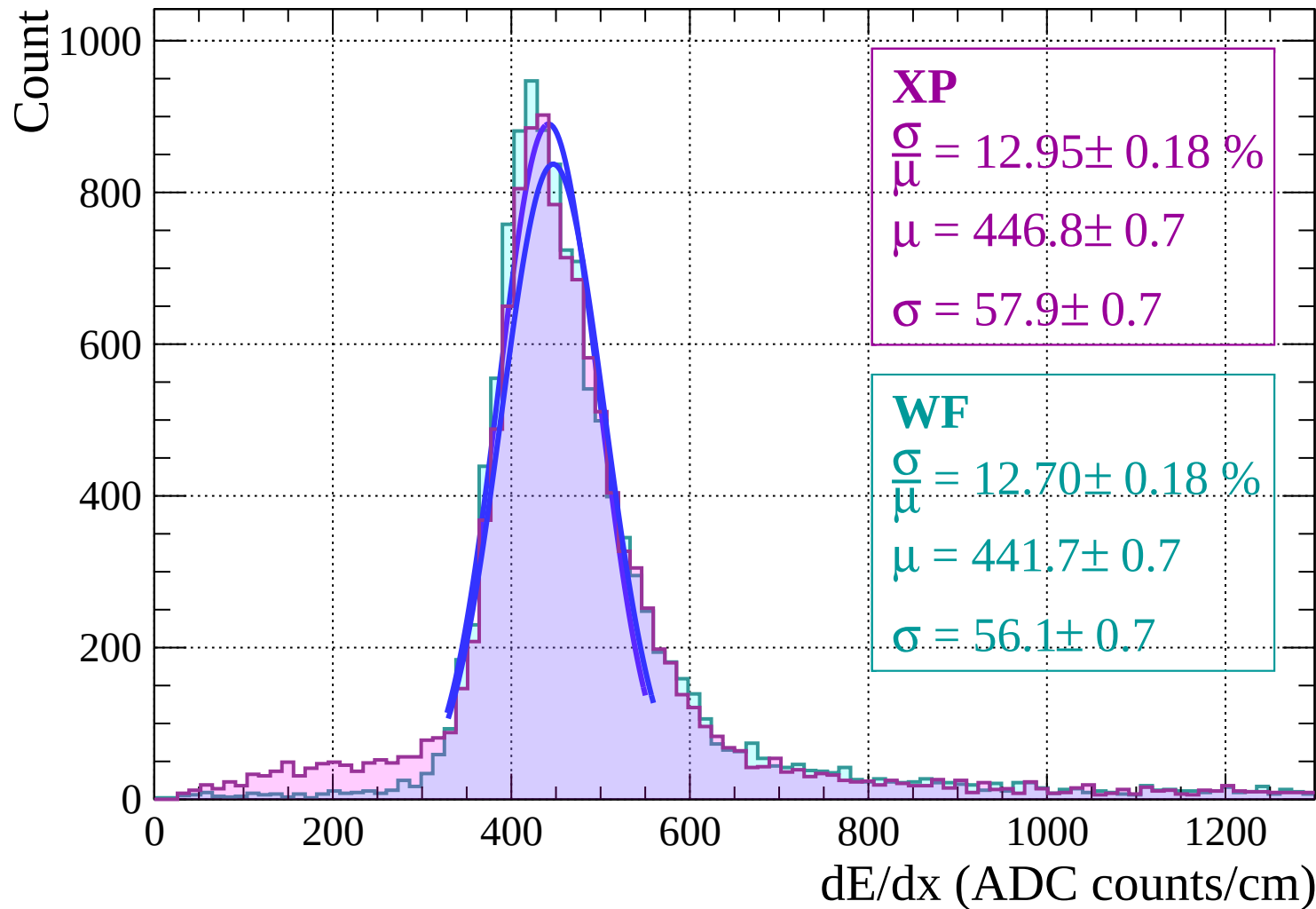
Energy loss | $12 < \theta < 14$



Energy loss | $14 < \theta < 16$

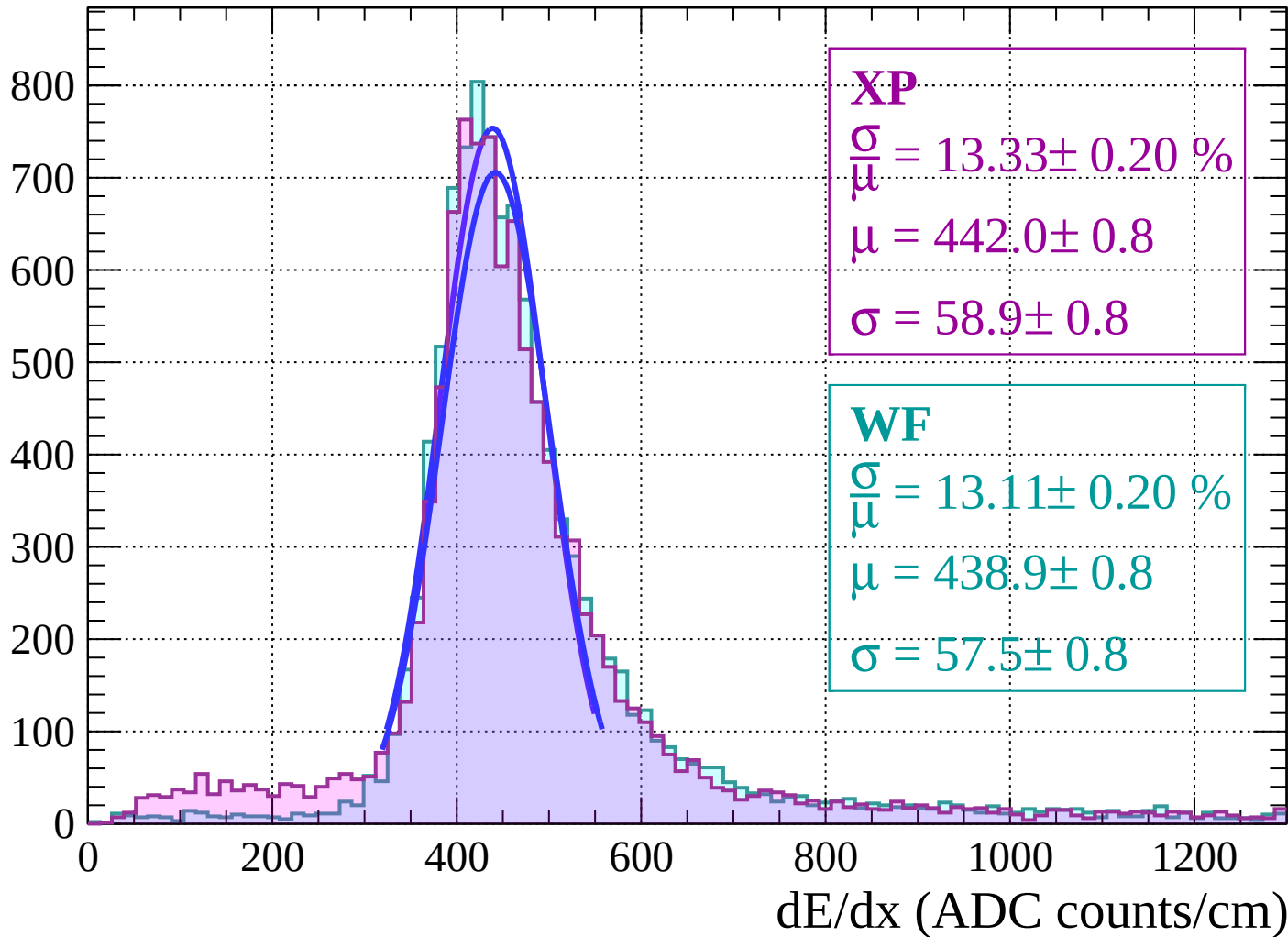


Energy loss | $16 < \theta < 18$



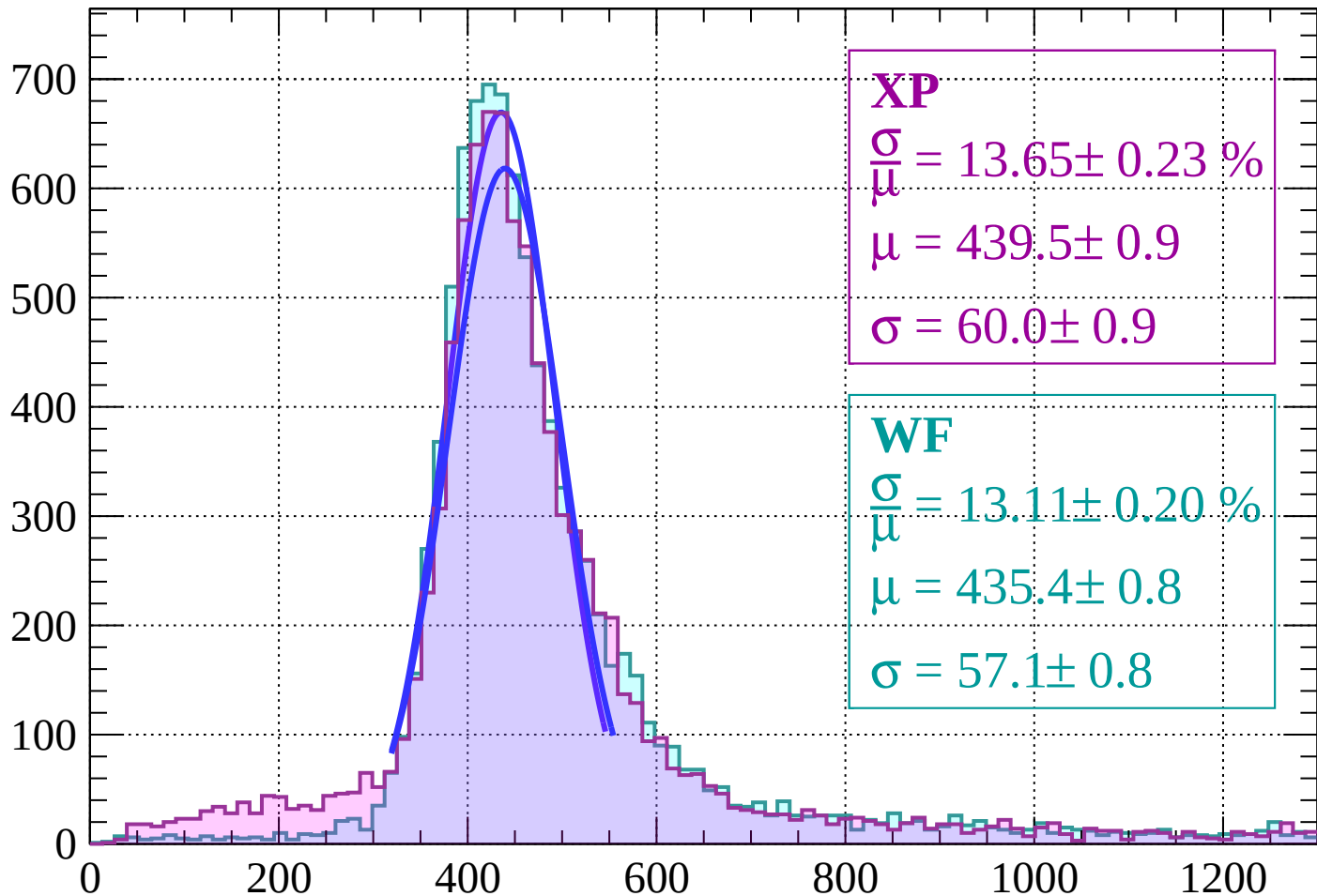
Energy loss | $18 < \theta < 20$

Count



Energy loss | $20 < \theta < 22$

Count



XP

$$\frac{\sigma}{\mu} = 13.65 \pm 0.23 \%$$

$$\mu = 439.5 \pm 0.9$$

$$\sigma = 60.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 13.11 \pm 0.20 \%$$

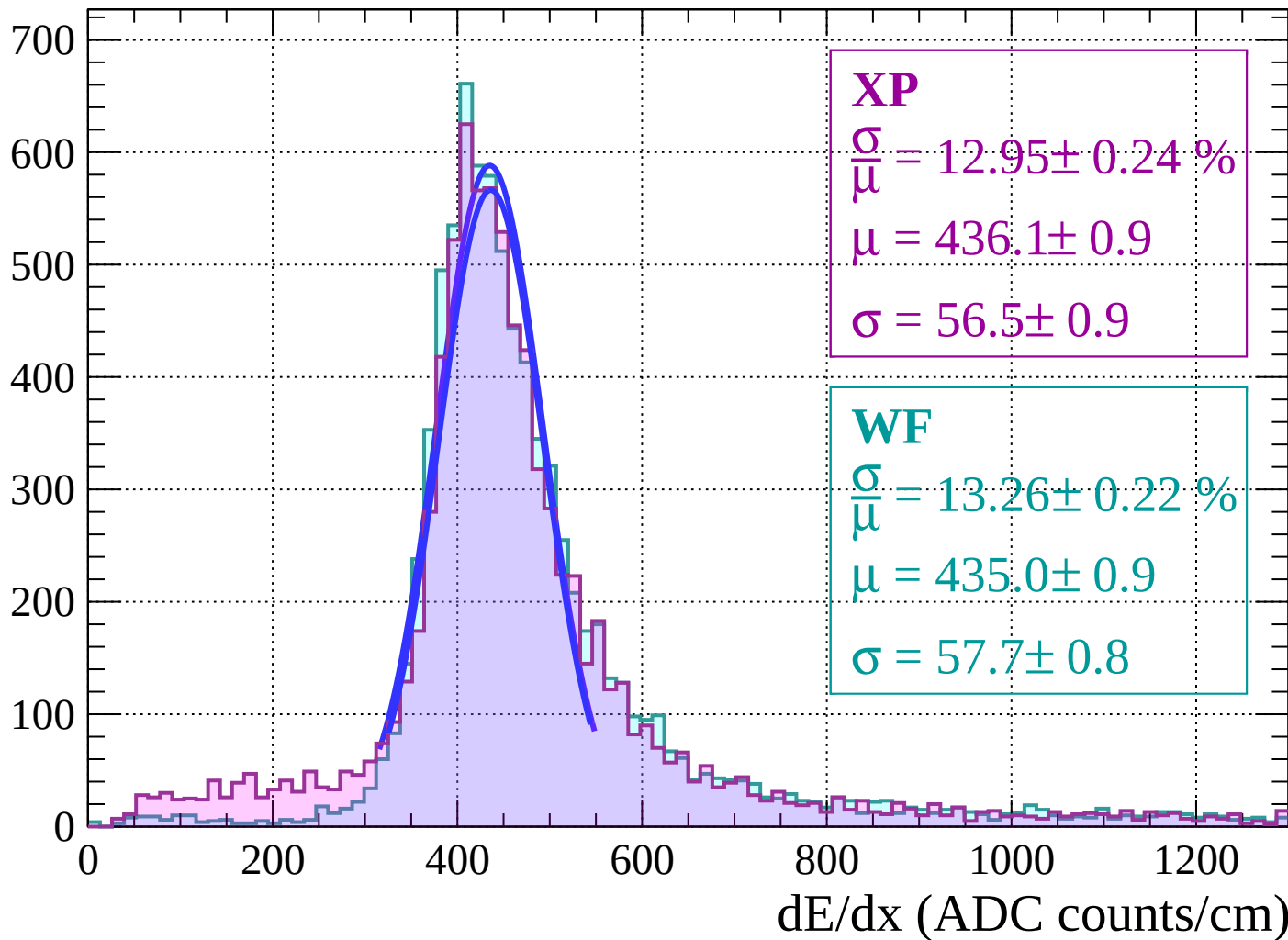
$$\mu = 435.4 \pm 0.8$$

$$\sigma = 57.1 \pm 0.8$$

dE/dx (ADC counts/cm)

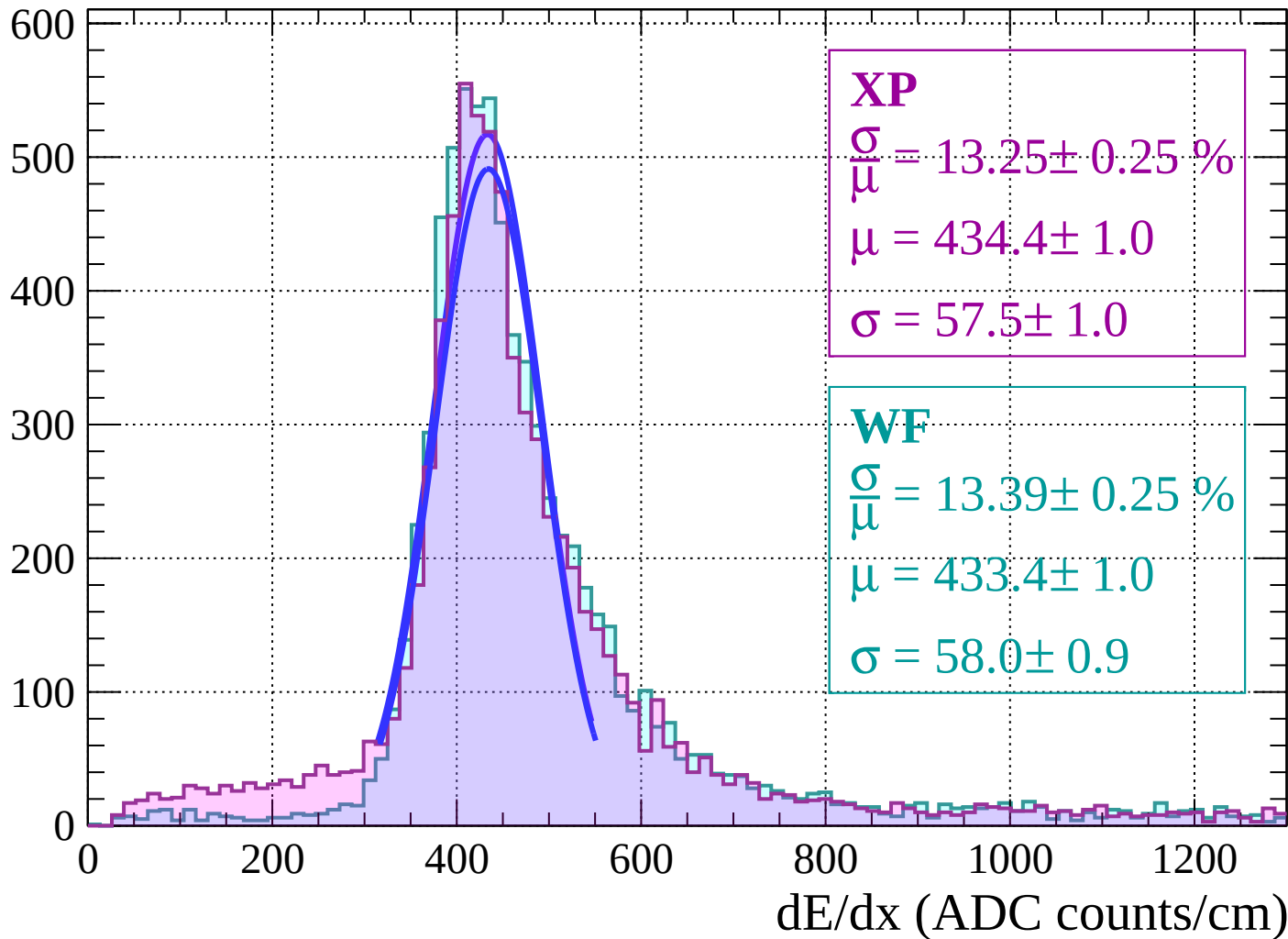
Energy loss | $22 < \theta < 24$

Count



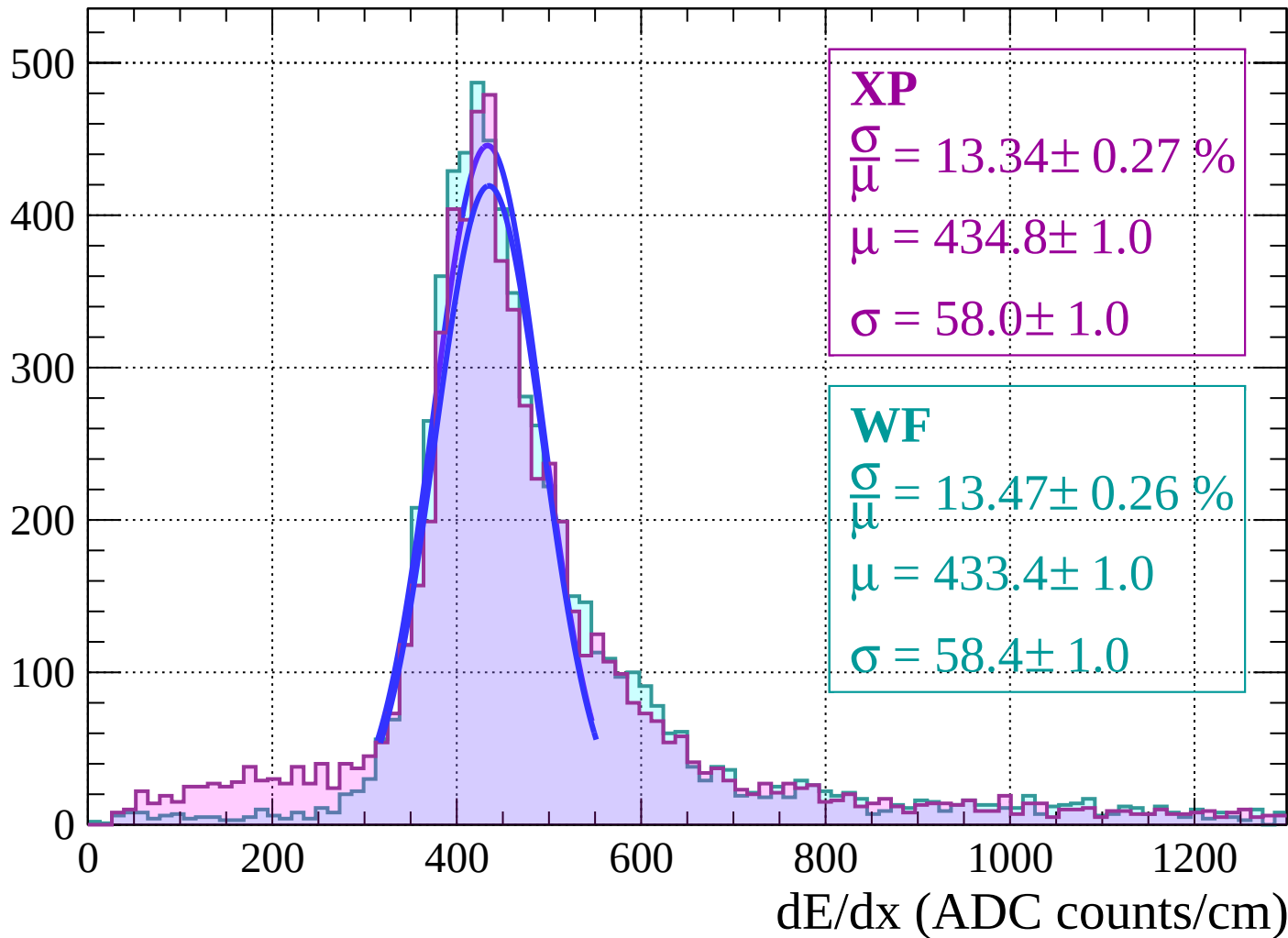
Energy loss | $24 < \theta < 26$

Count

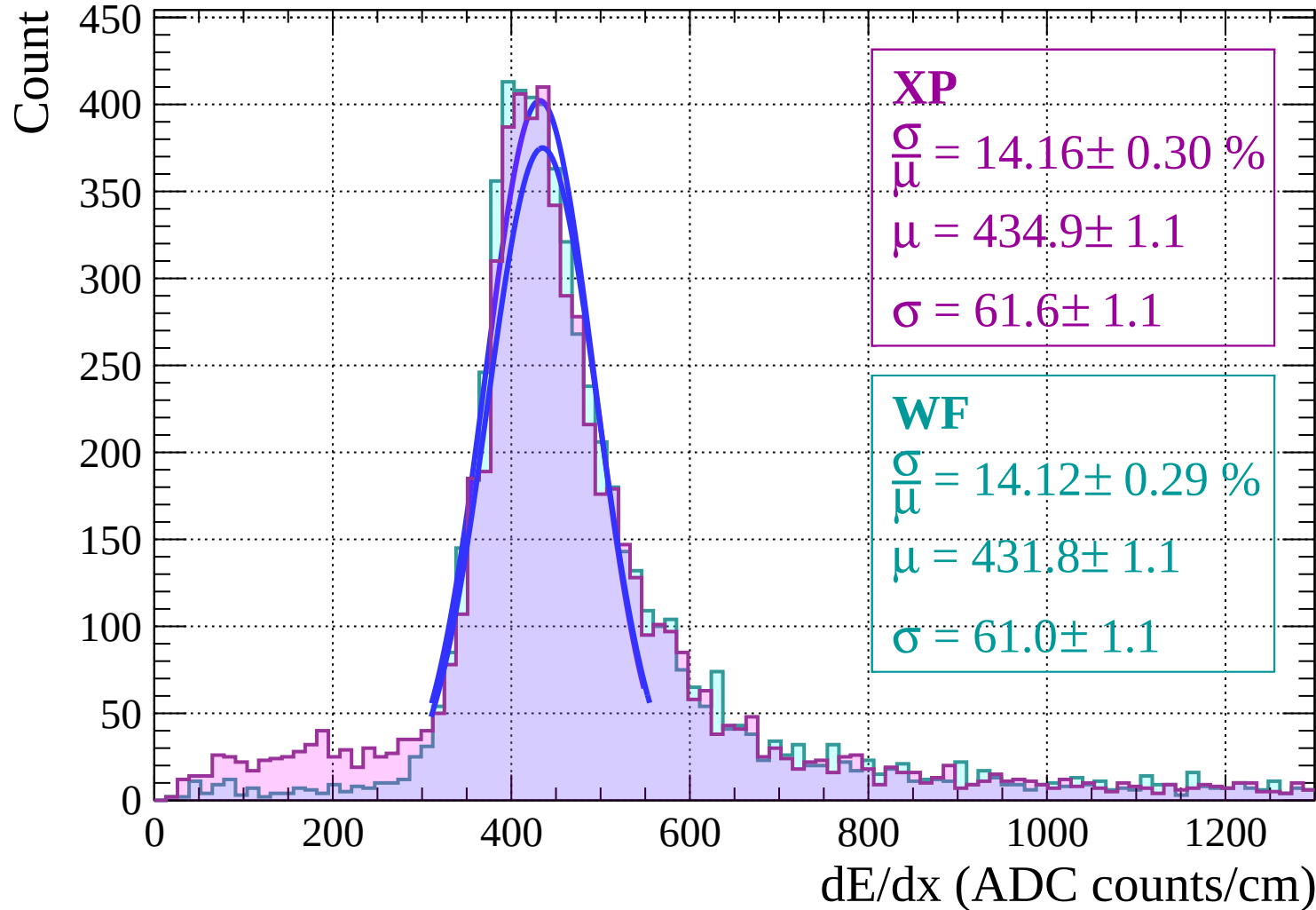


Energy loss | $26 < \theta < 28$

Count

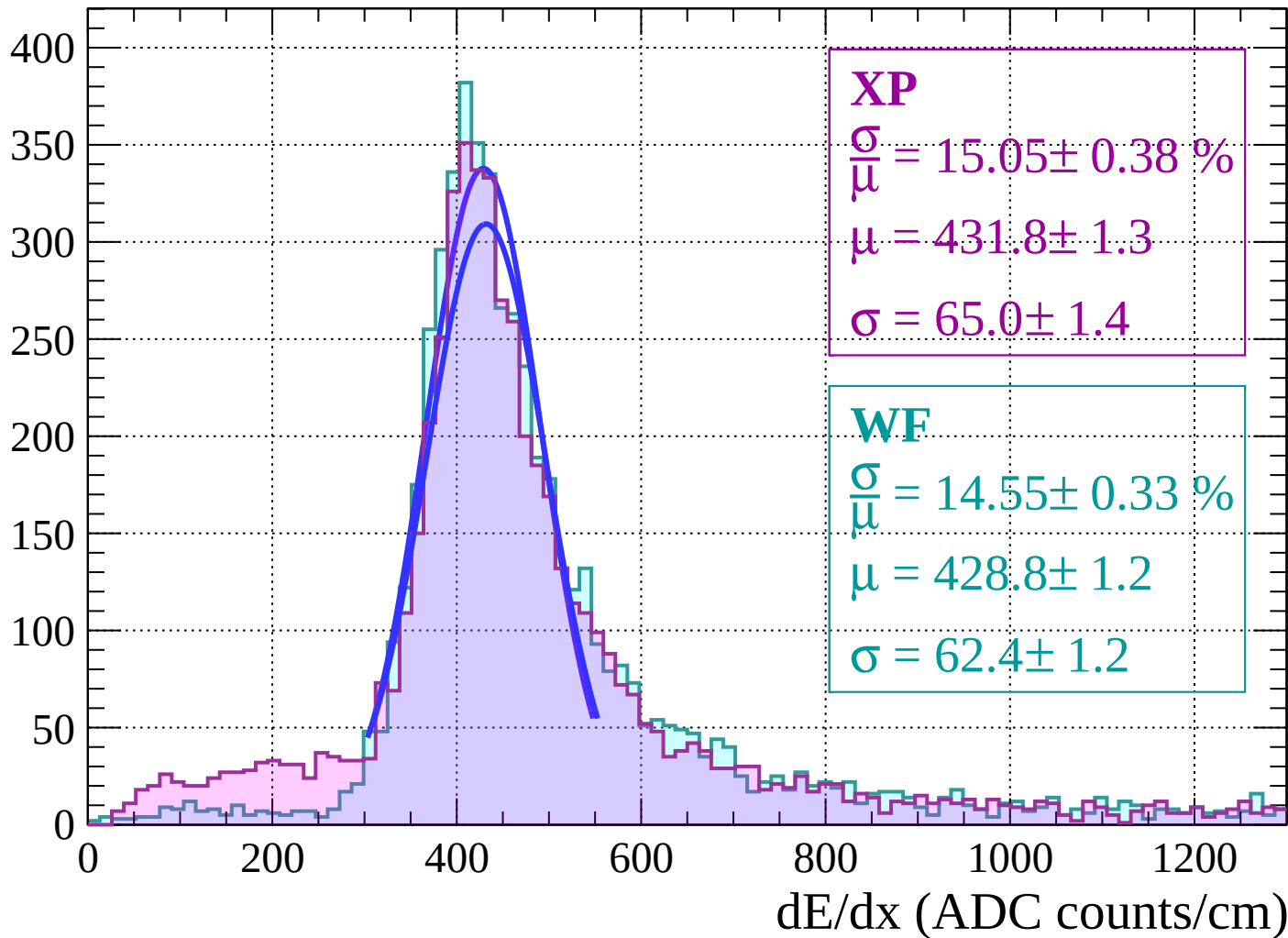


Energy loss | $28 < \theta < 30$



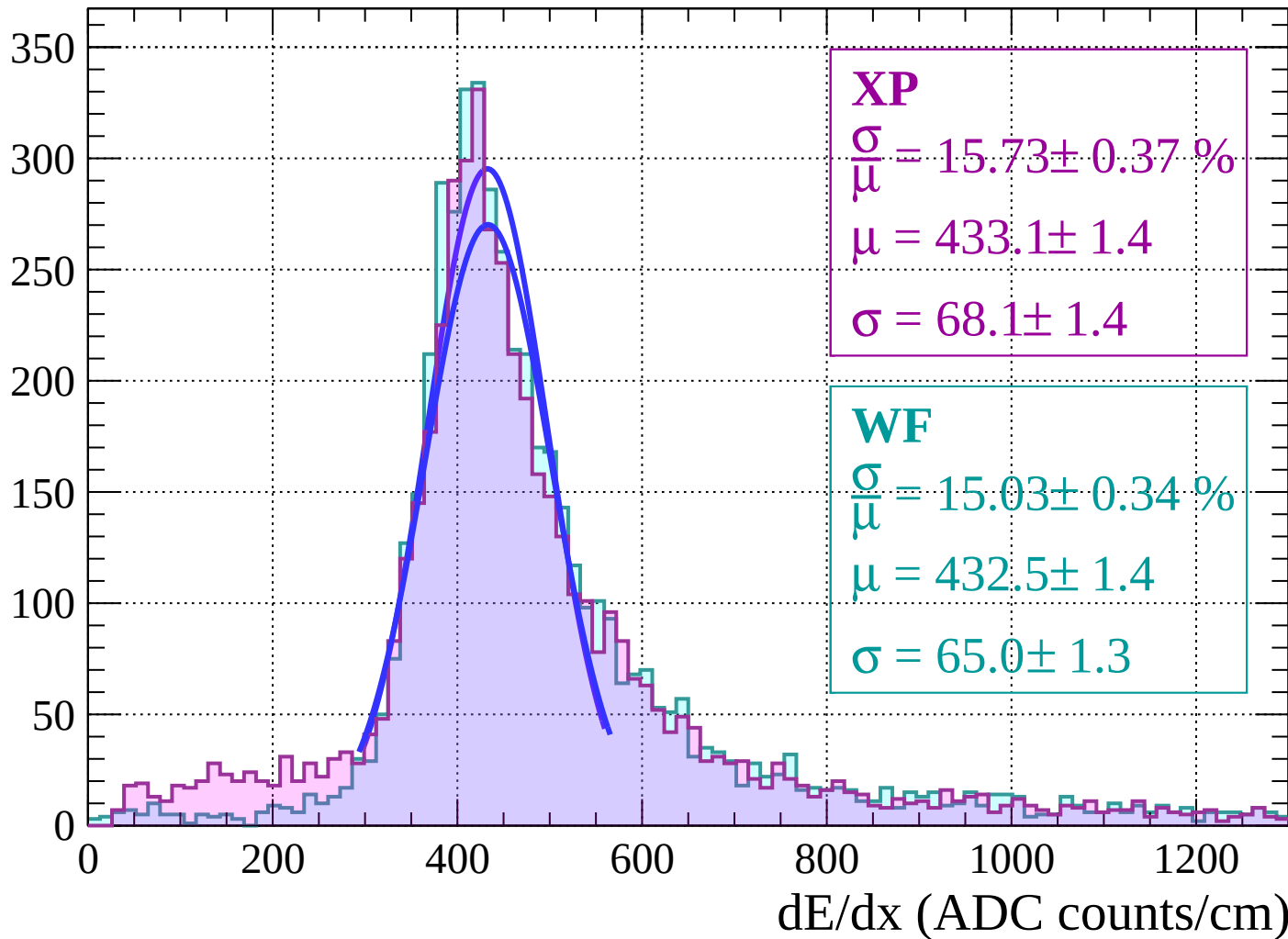
Energy loss | $30 < \theta < 32$

Count



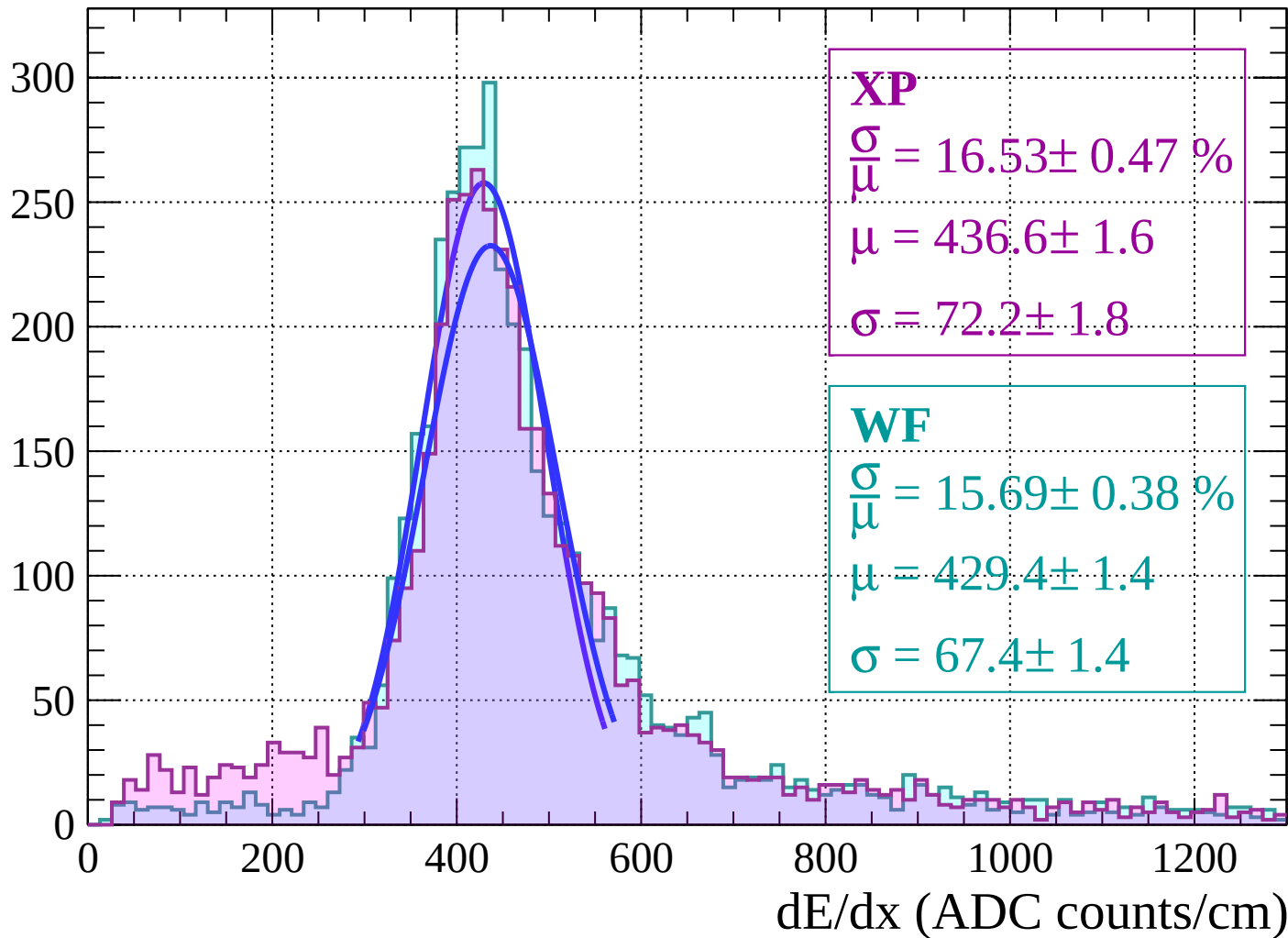
Energy loss | $32 < \theta < 34$

Count



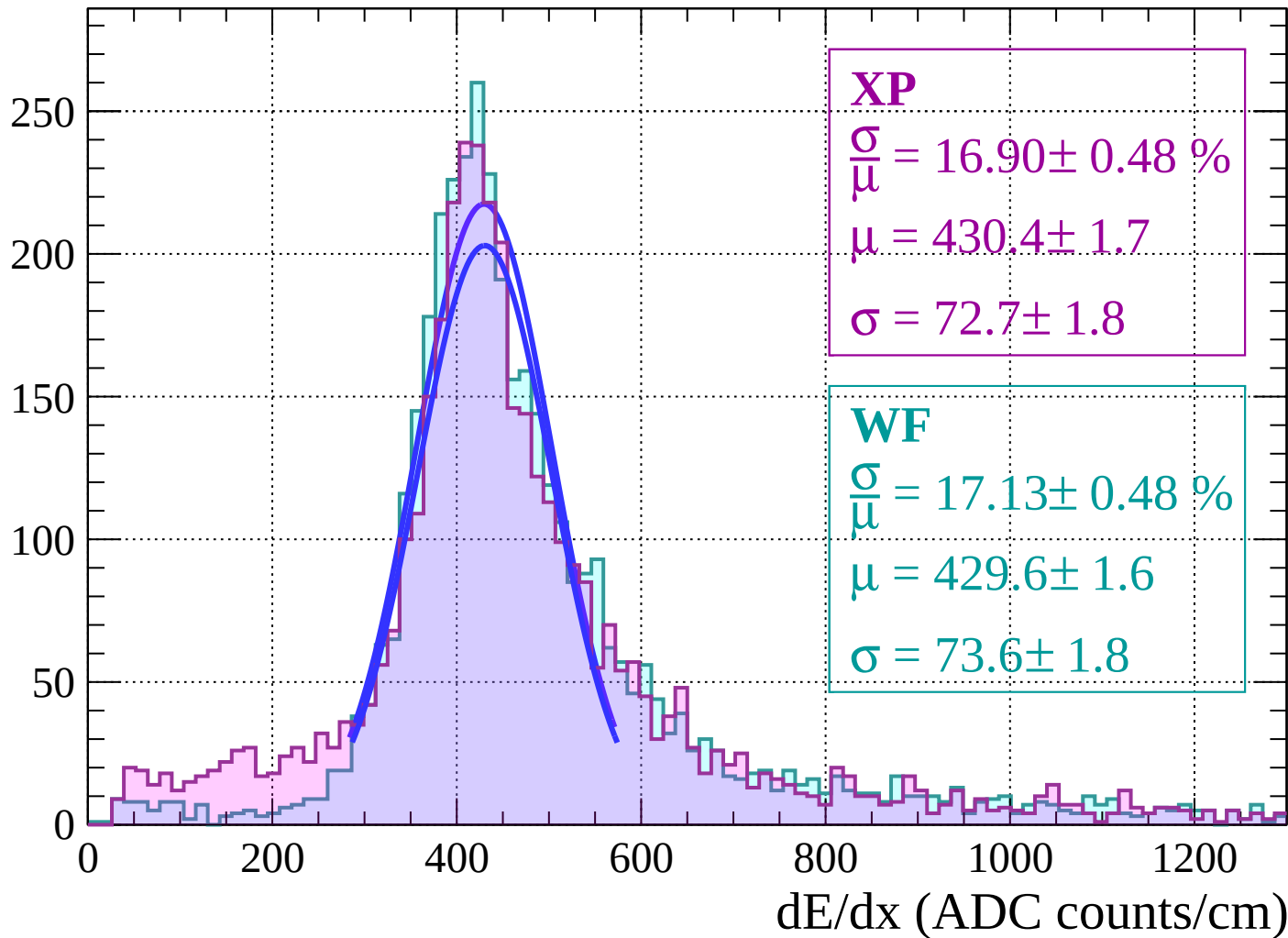
Energy loss | $34 < \theta < 36$

Count



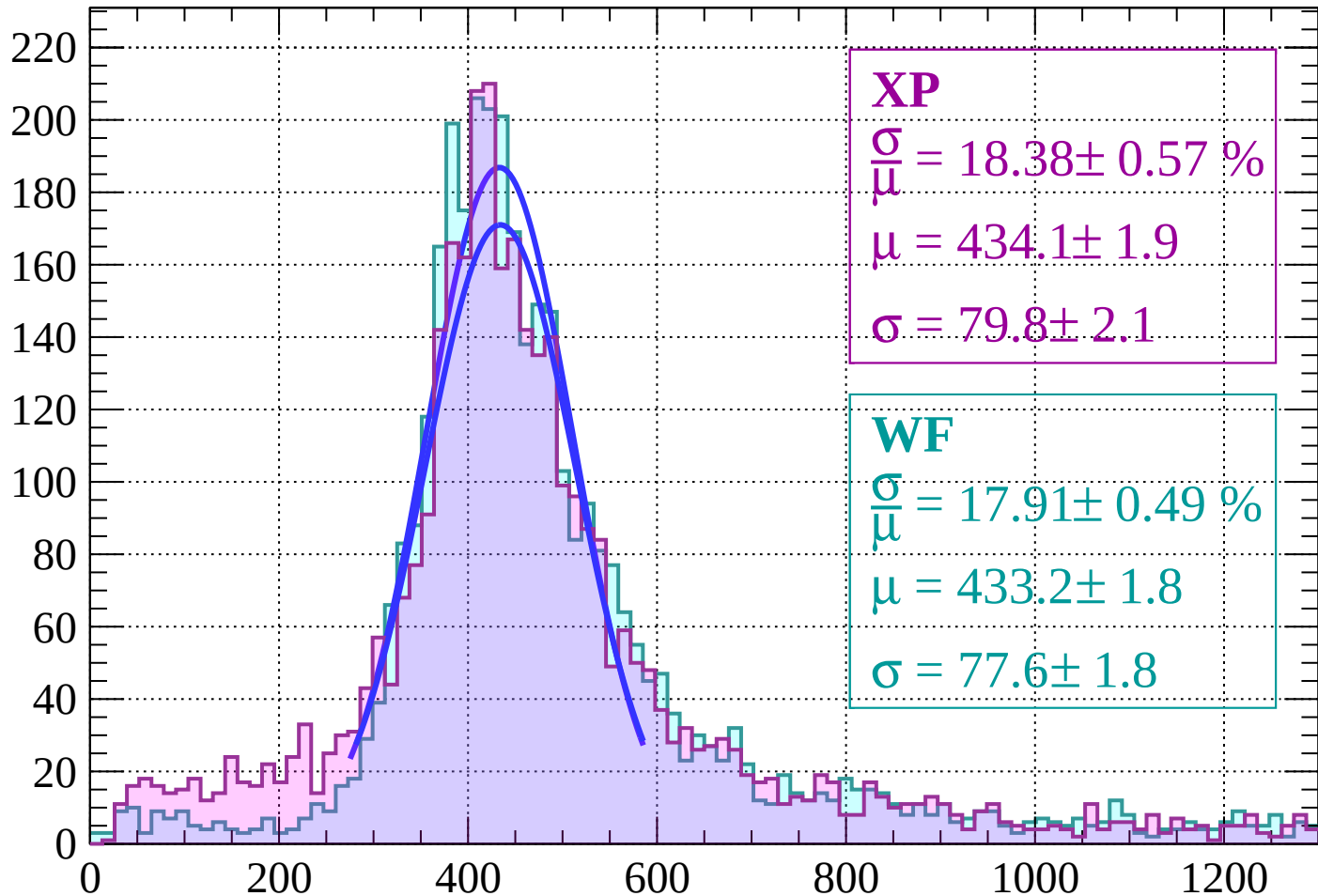
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



XP

$$\frac{\sigma}{\mu} = 18.38 \pm 0.57 \%$$

$$\mu = 434.1 \pm 1.9$$

$$\sigma = 79.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 17.91 \pm 0.49 \%$$

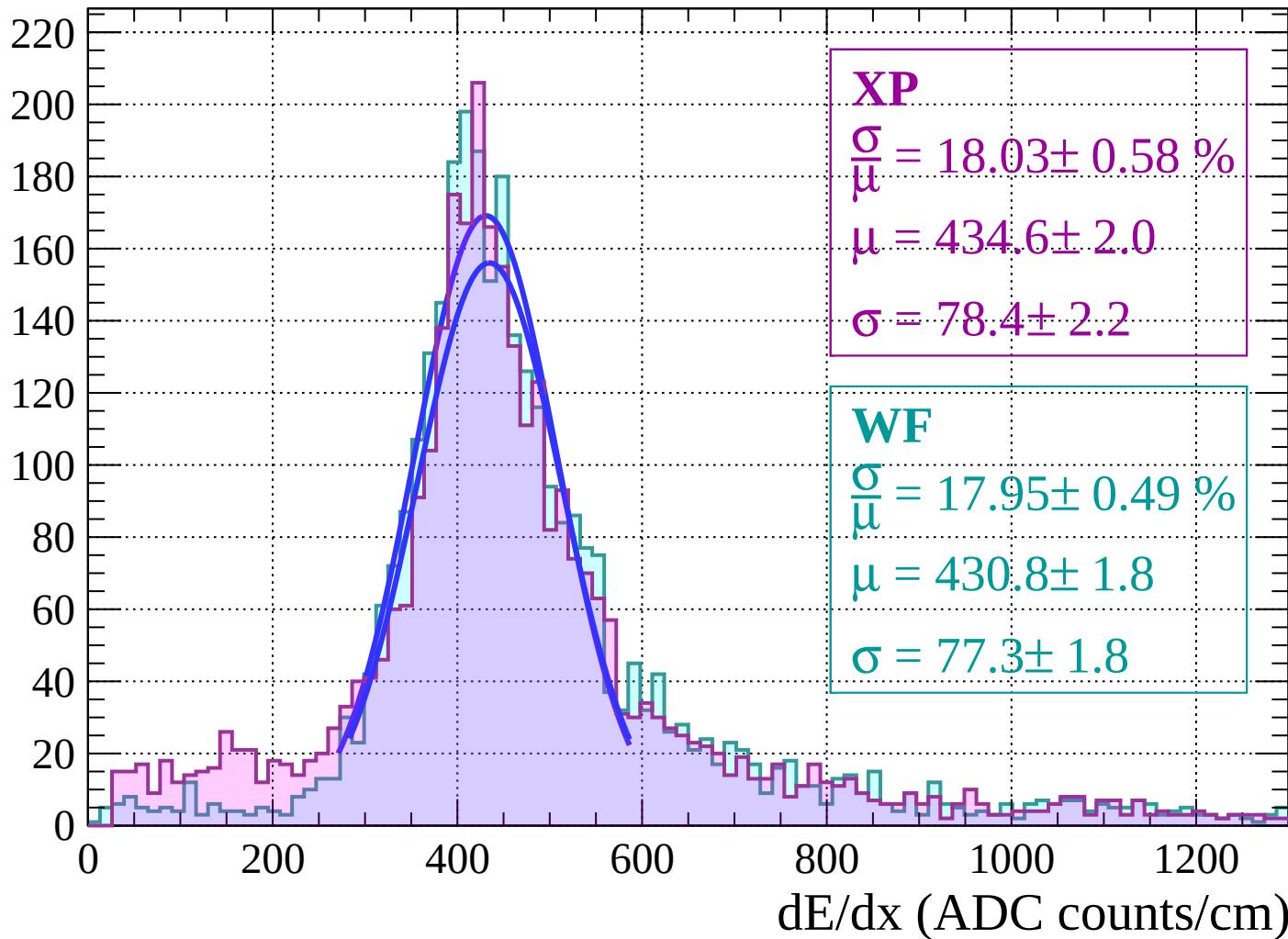
$$\mu = 433.2 \pm 1.8$$

$$\sigma = 77.6 \pm 1.8$$

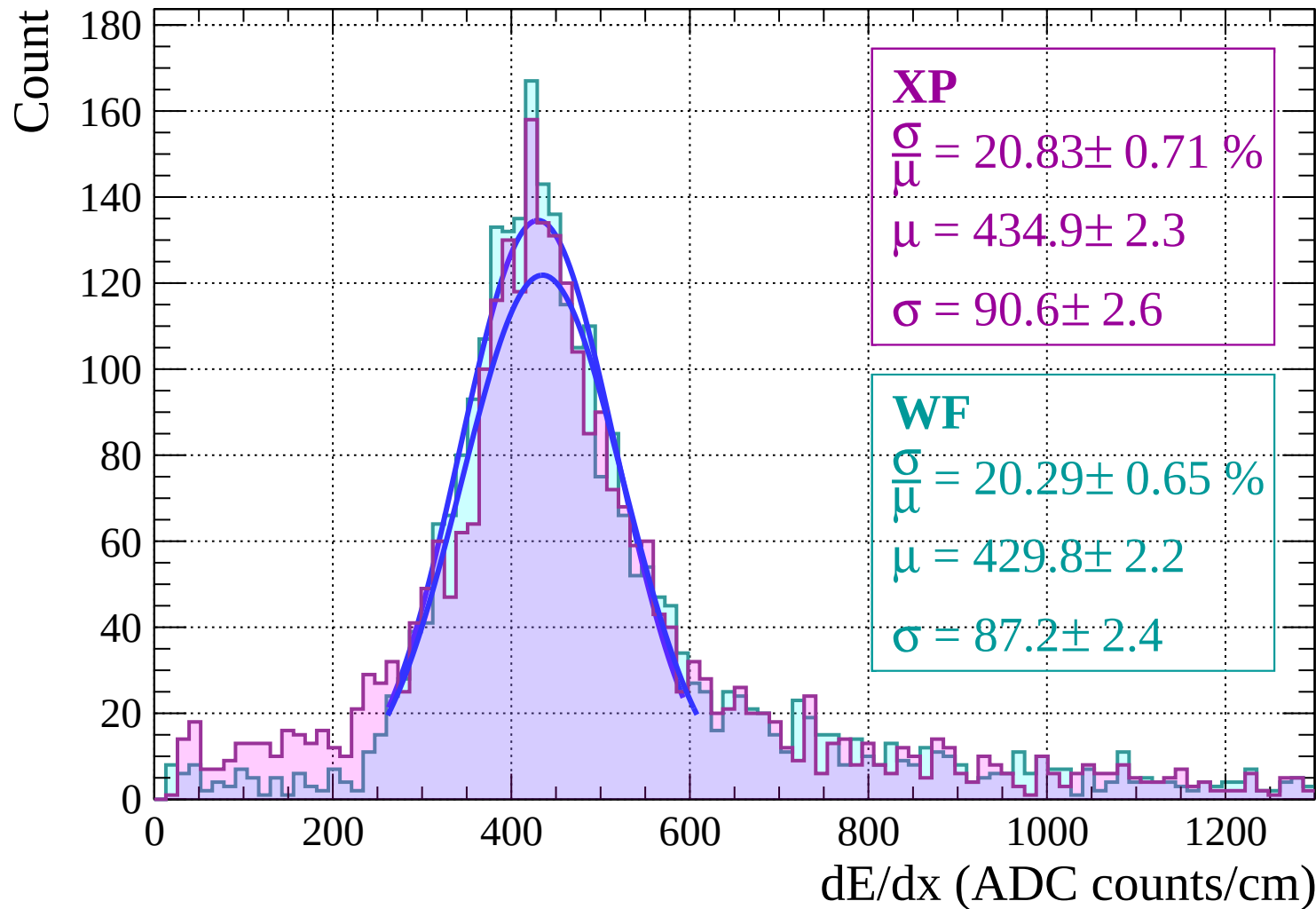
dE/dx (ADC counts/cm)

Energy loss | $40 < \theta < 42$

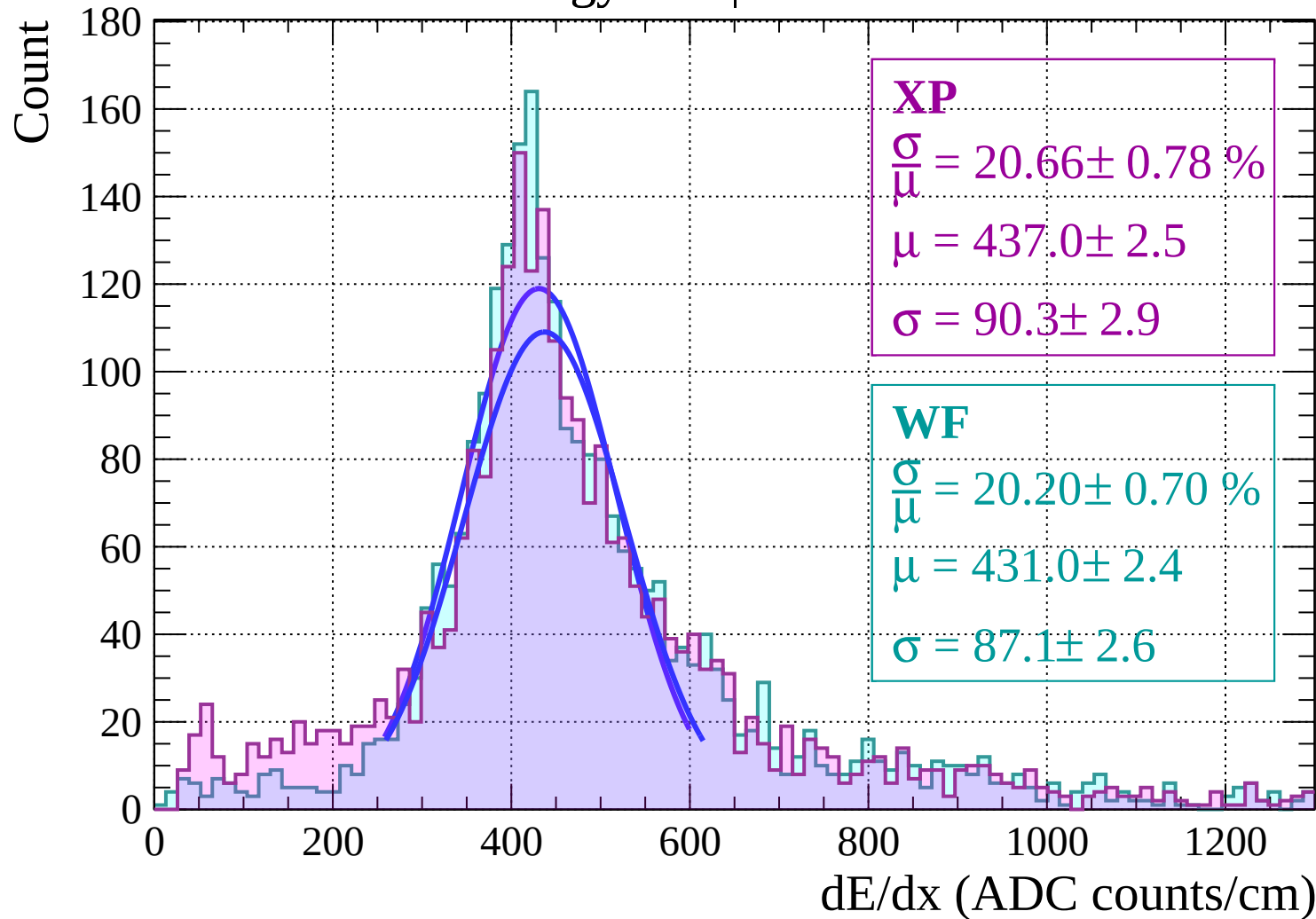
Count



Energy loss | $42 < \theta < 44$

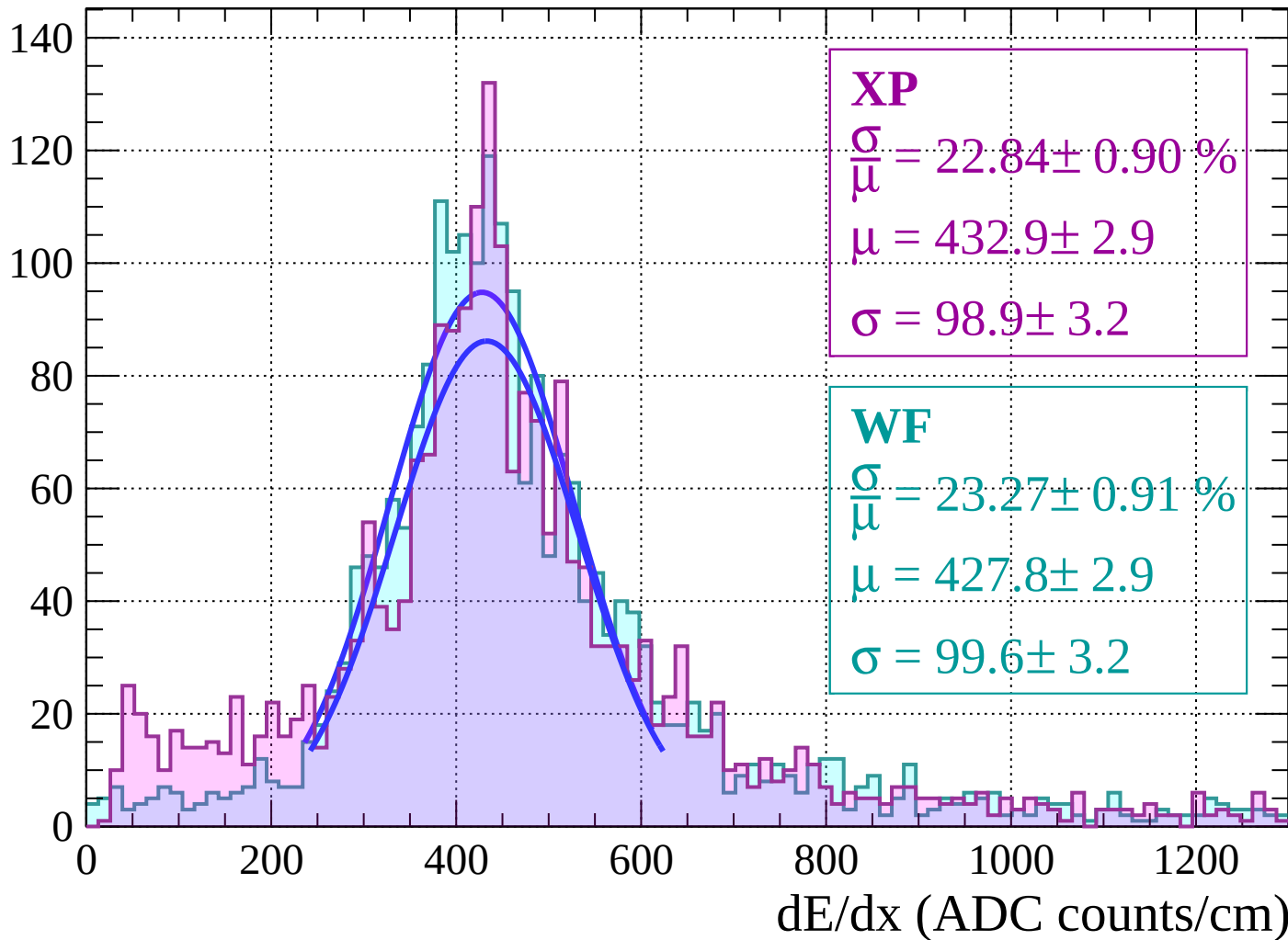


Energy loss | $44 < \theta < 46$



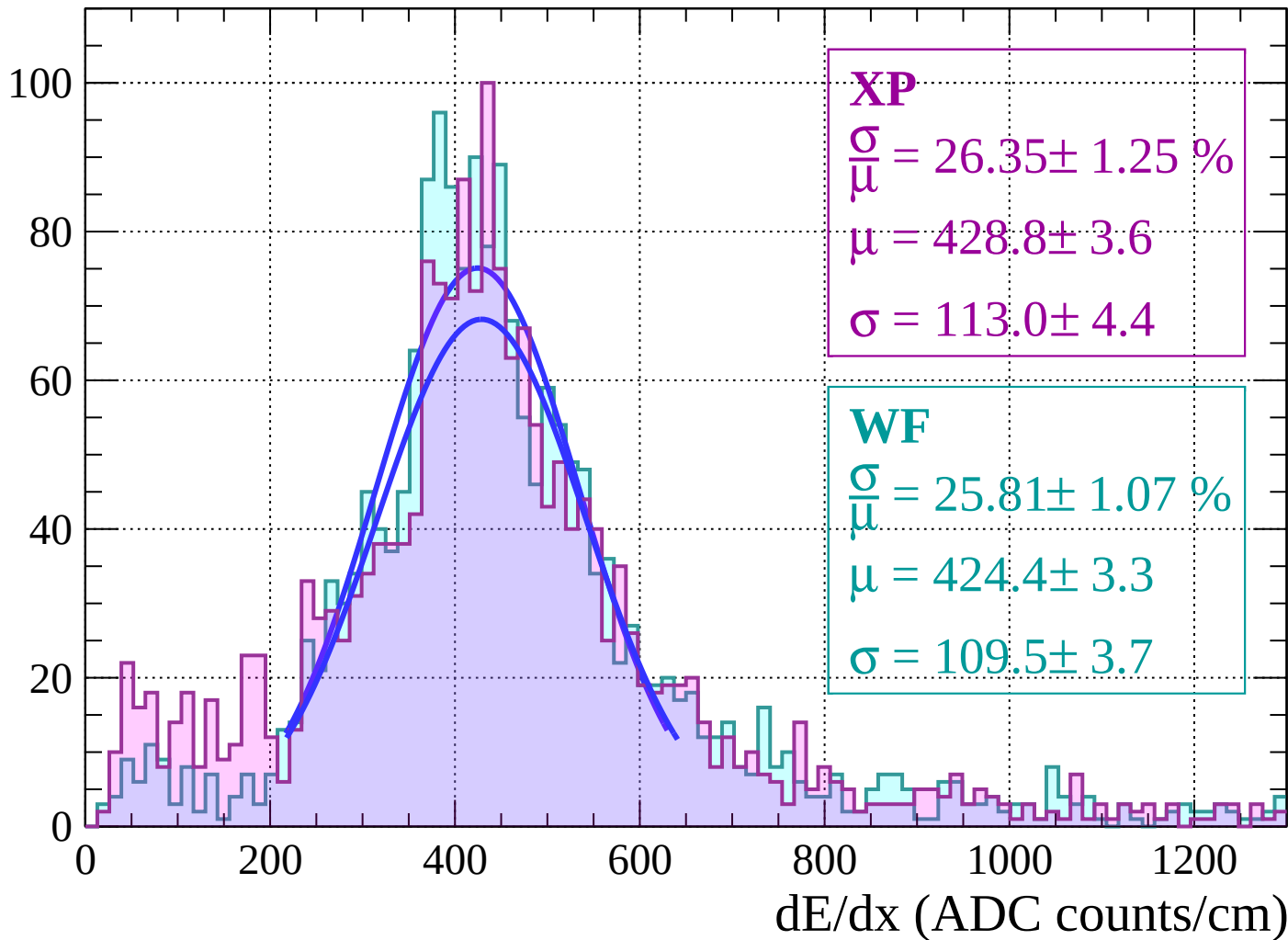
Energy loss | $46 < \theta < 48$

Count



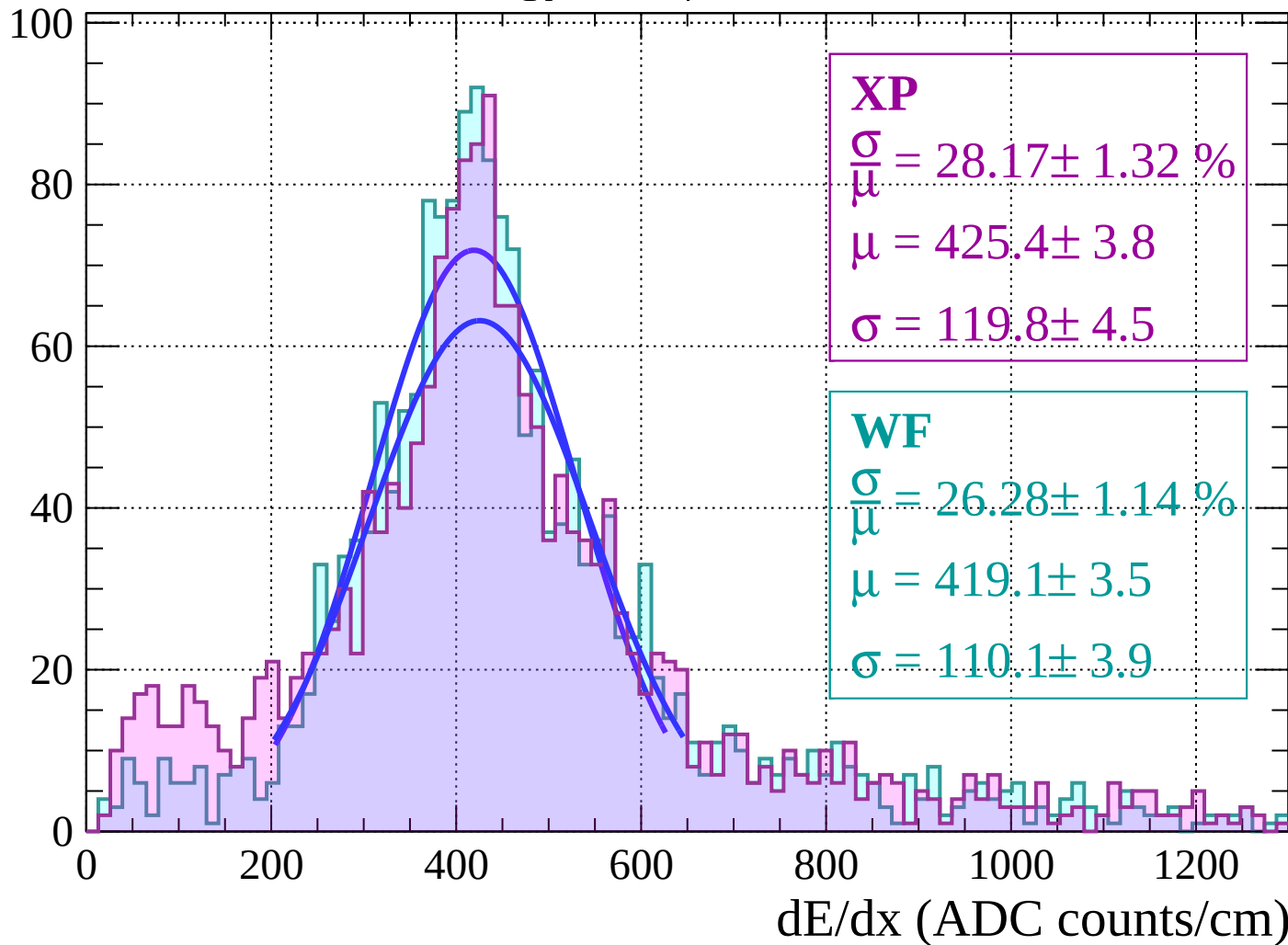
Energy loss | $48 < \theta < 50$

Count



Energy loss | $50 < \theta < 52$

Count



Energy loss | $52 < \theta < 54$

Count

XP

$$\frac{\sigma}{\mu} = 29.49 \pm 1.44 \%$$

$$\mu = 415.6 \pm 4.1$$

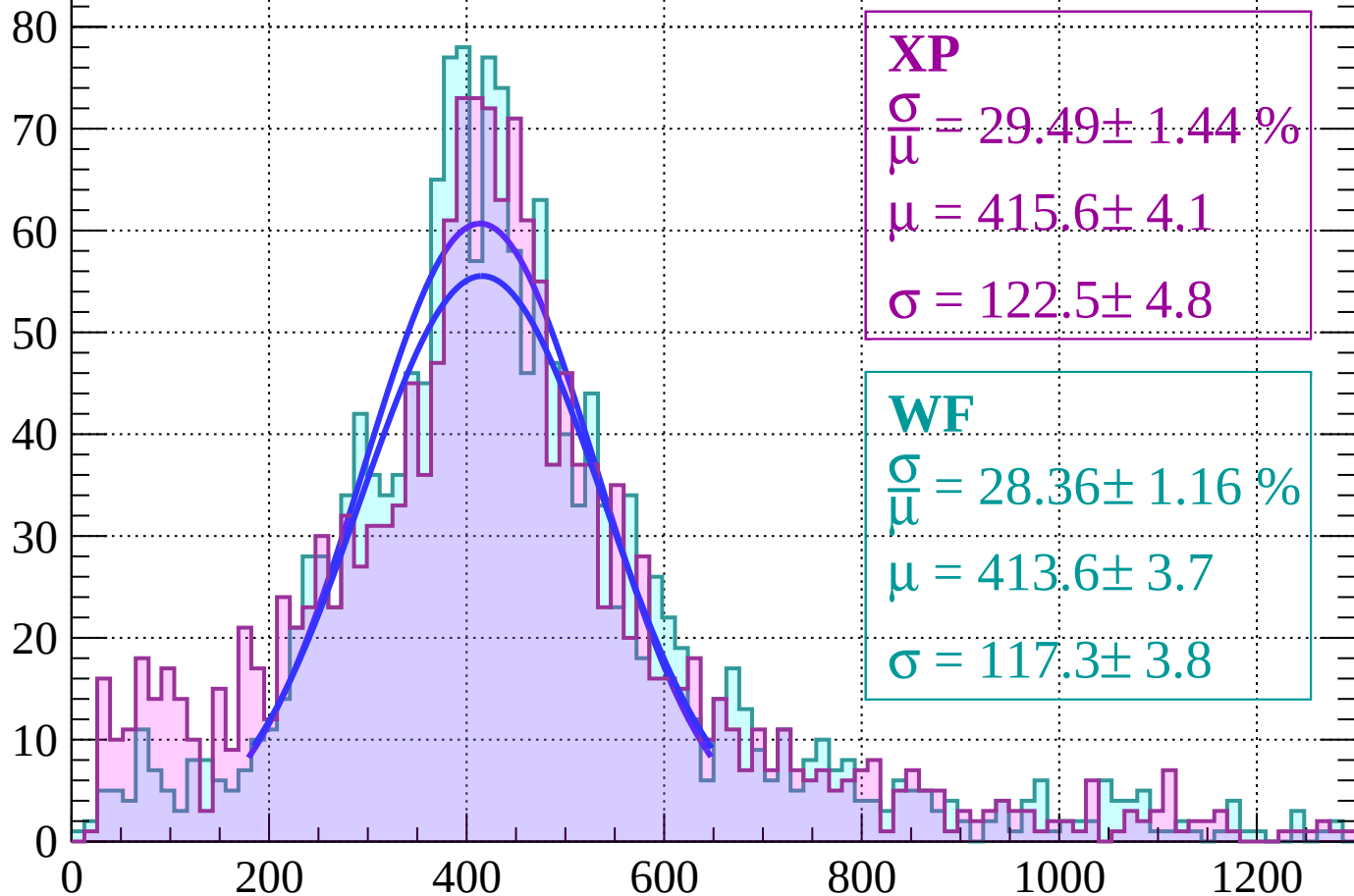
$$\sigma = 122.5 \pm 4.8$$

WF

$$\frac{\sigma}{\mu} = 28.36 \pm 1.16 \%$$

$$\mu = 413.6 \pm 3.7$$

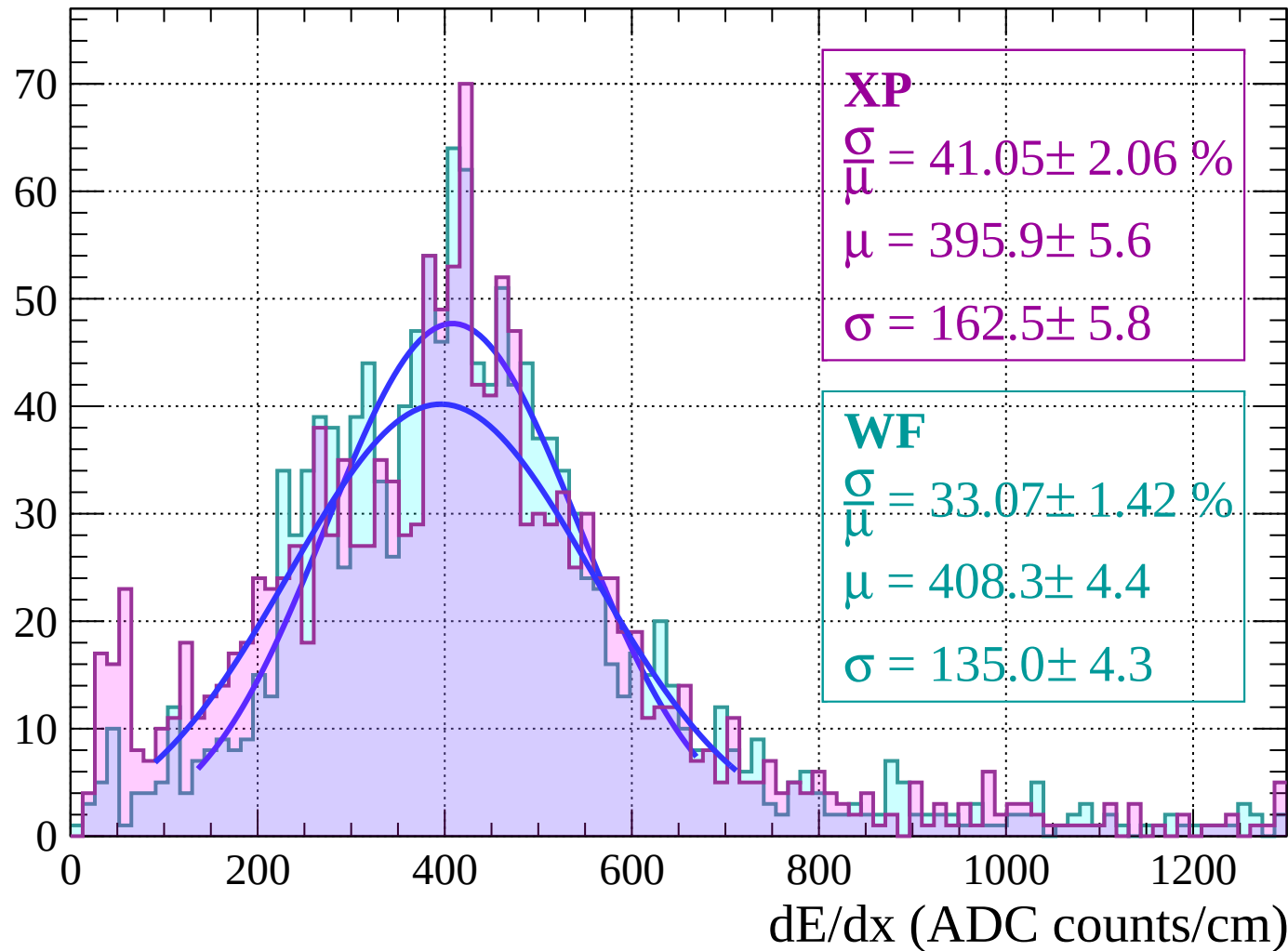
$$\sigma = 117.3 \pm 3.8$$



dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

Count



Energy loss | $56 < \theta < 58$

Count

XP

$$\frac{\sigma}{\mu} = 43.70 \pm 2.36 \%$$

$$\mu = 398.5 \pm 6.5$$

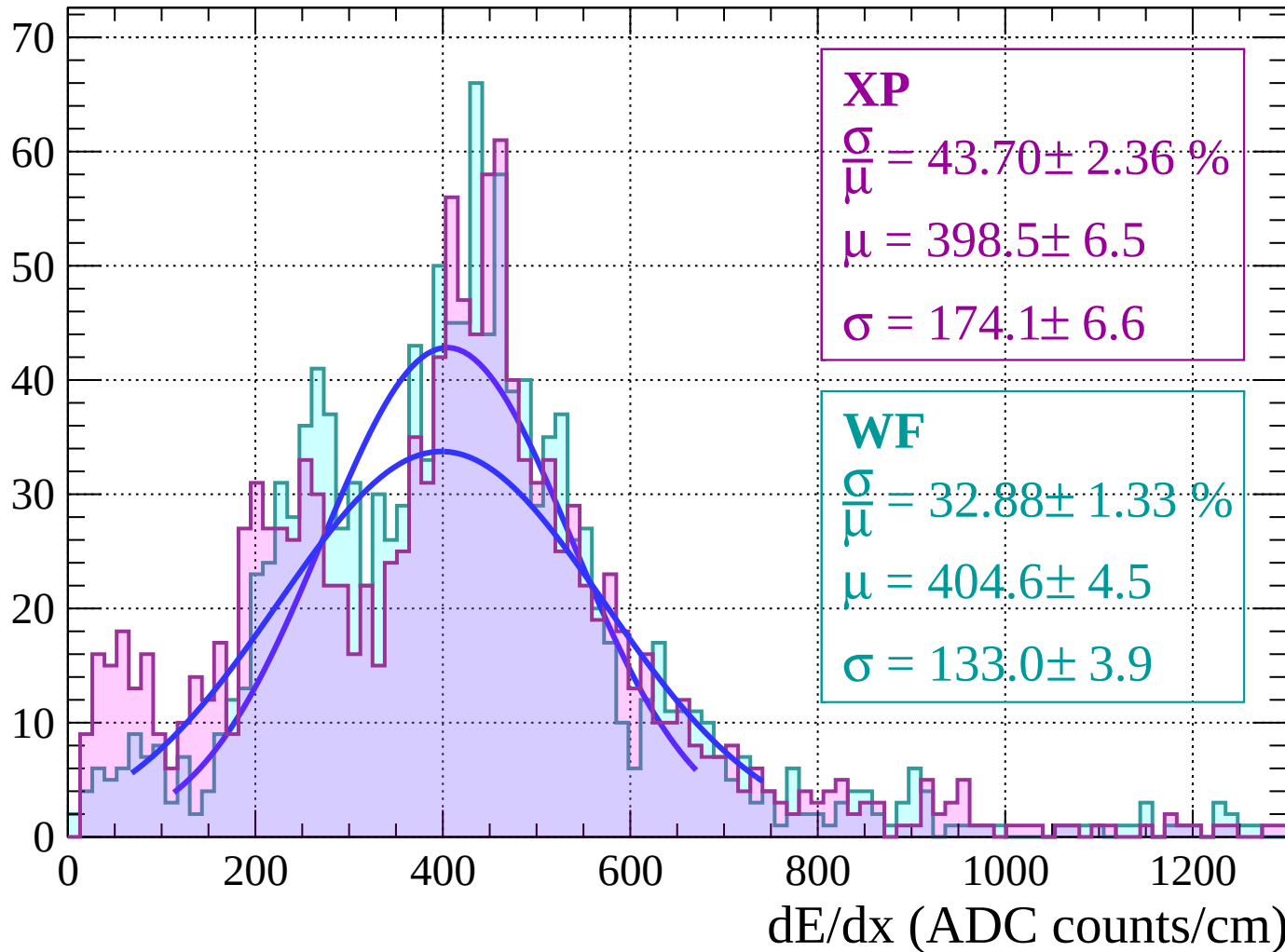
$$\sigma = 174.1 \pm 6.6$$

WF

$$\frac{\sigma}{\mu} = 32.88 \pm 1.33 \%$$

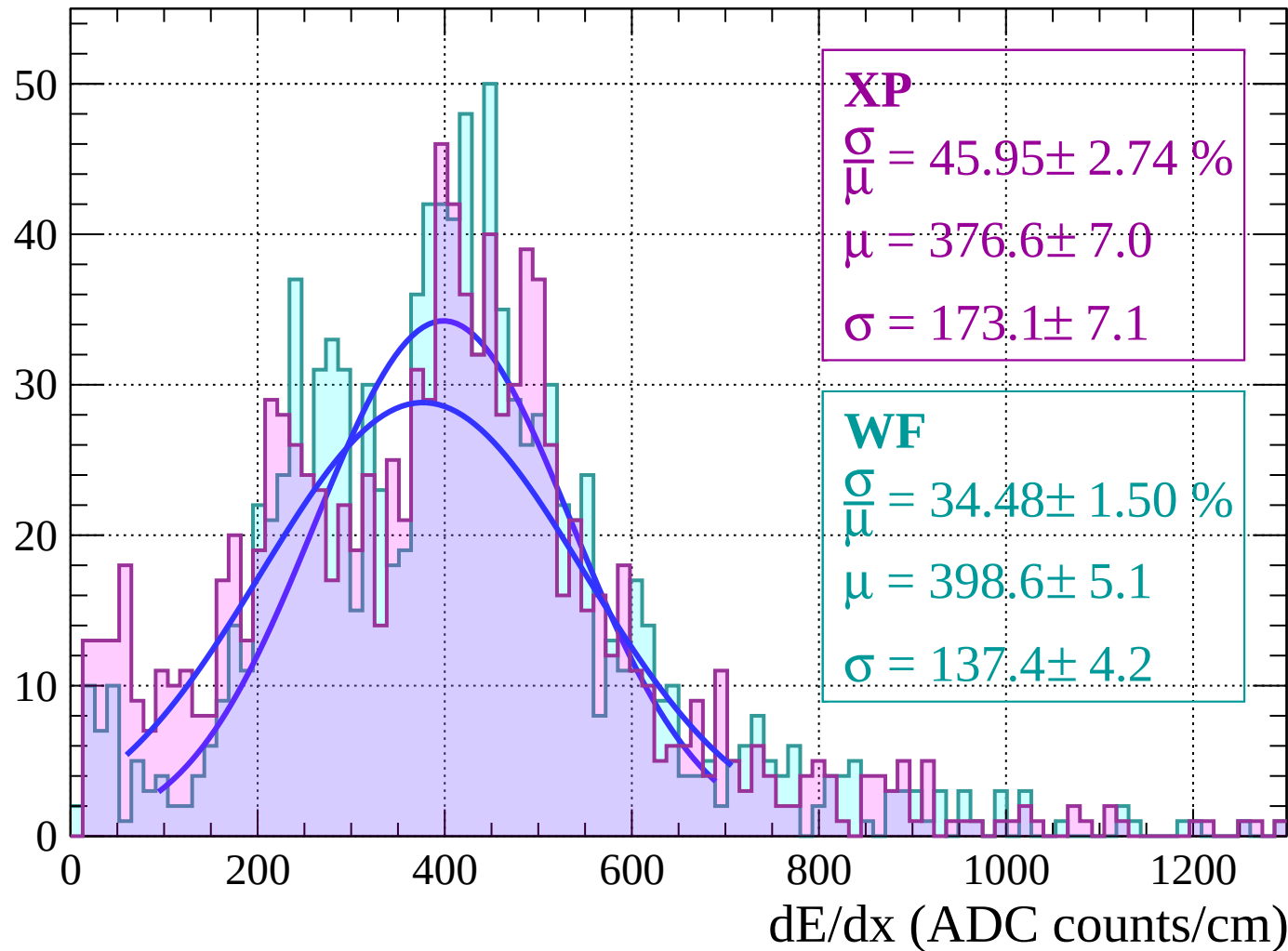
$$\mu = 404.6 \pm 4.5$$

$$\sigma = 133.0 \pm 3.9$$



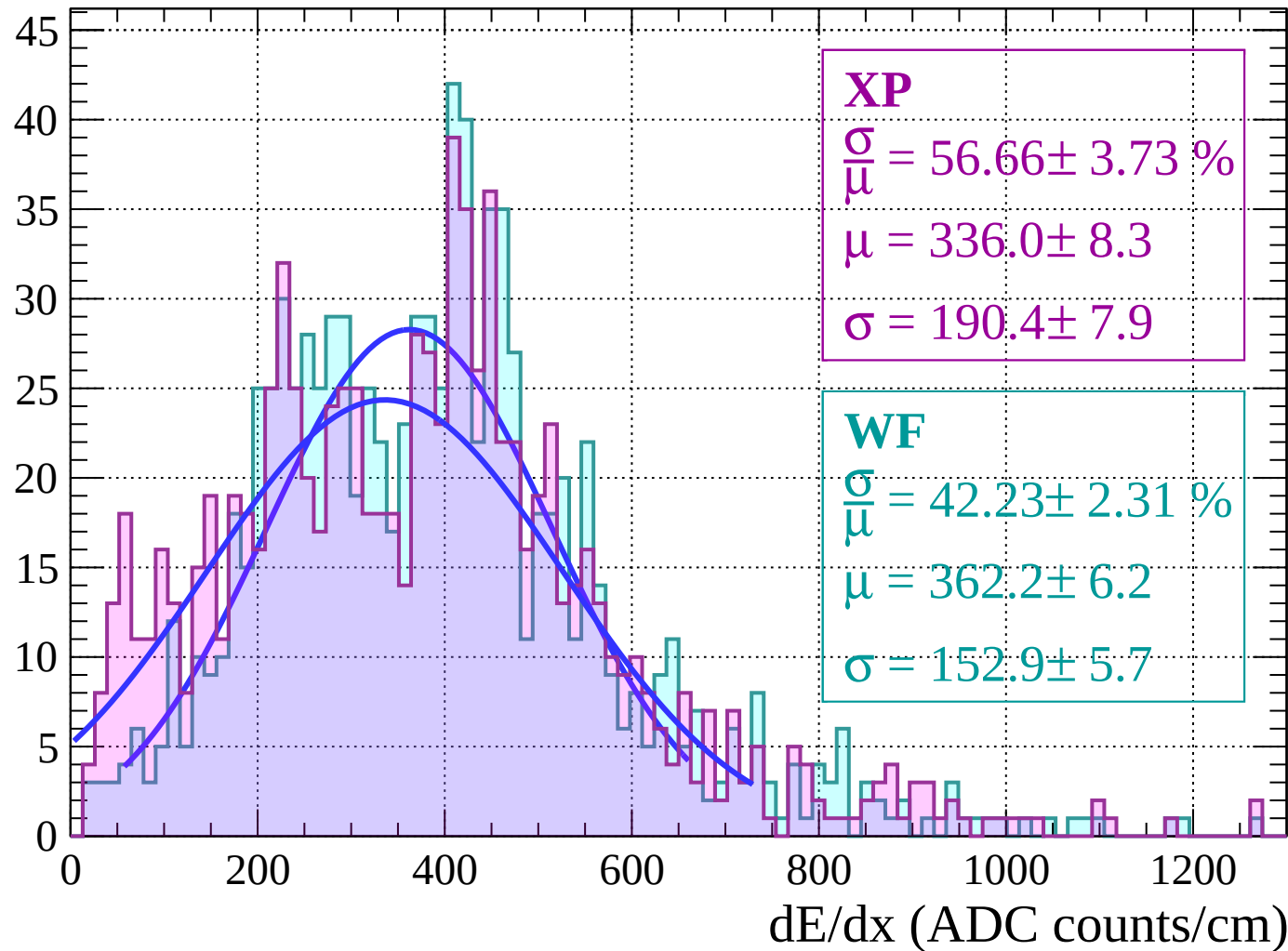
Energy loss | $58 < \theta < 60$

Count



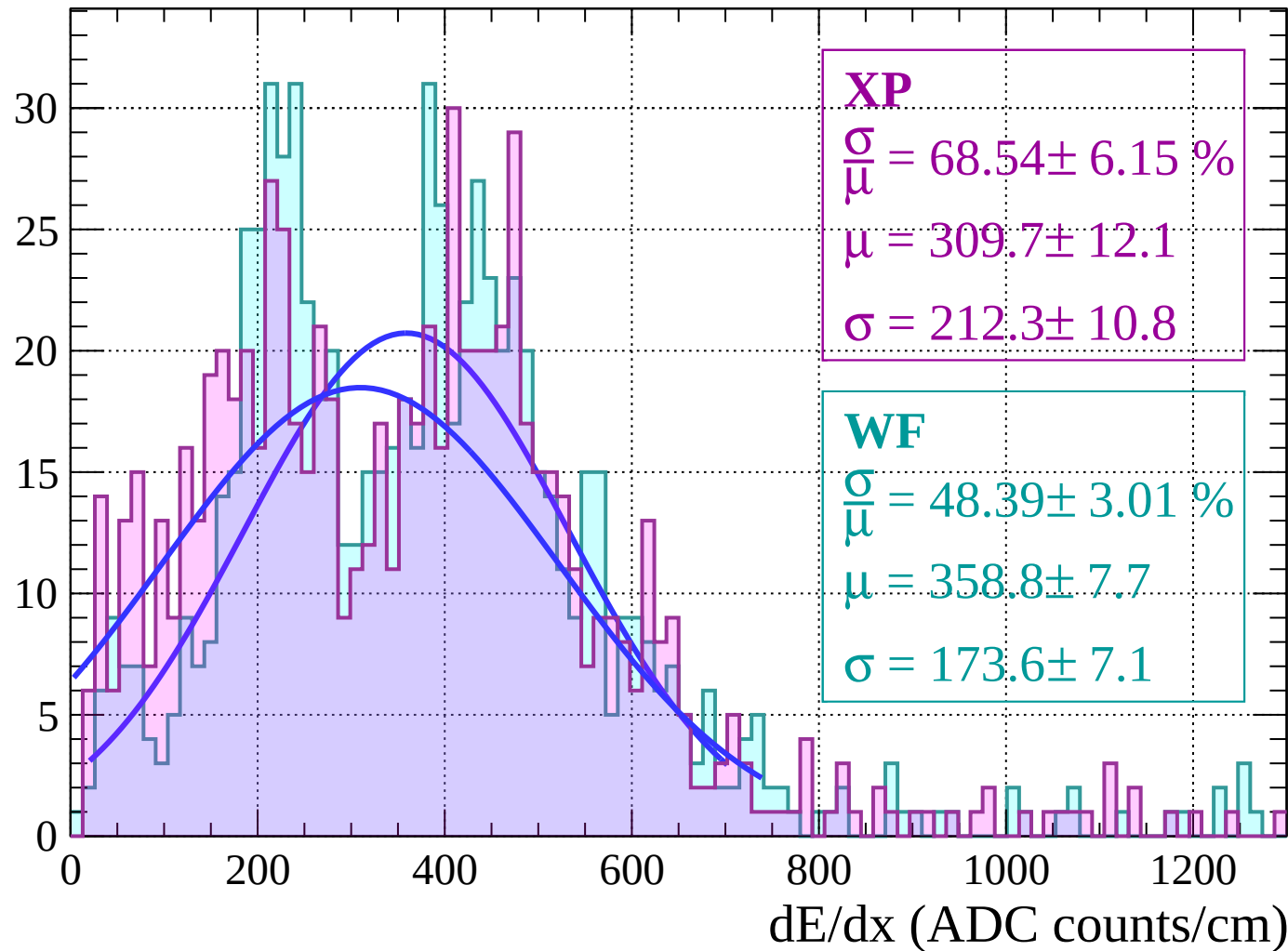
Energy loss | $60 < \theta < 62$

Count



Energy loss | $62 < \theta < 64$

Count



Energy loss | $64 < \theta < 66$

Count

XP

$$\frac{\sigma}{\mu} = 85.40 \pm 10.77 \%$$

$$\mu = 274.1 \pm 17.3$$

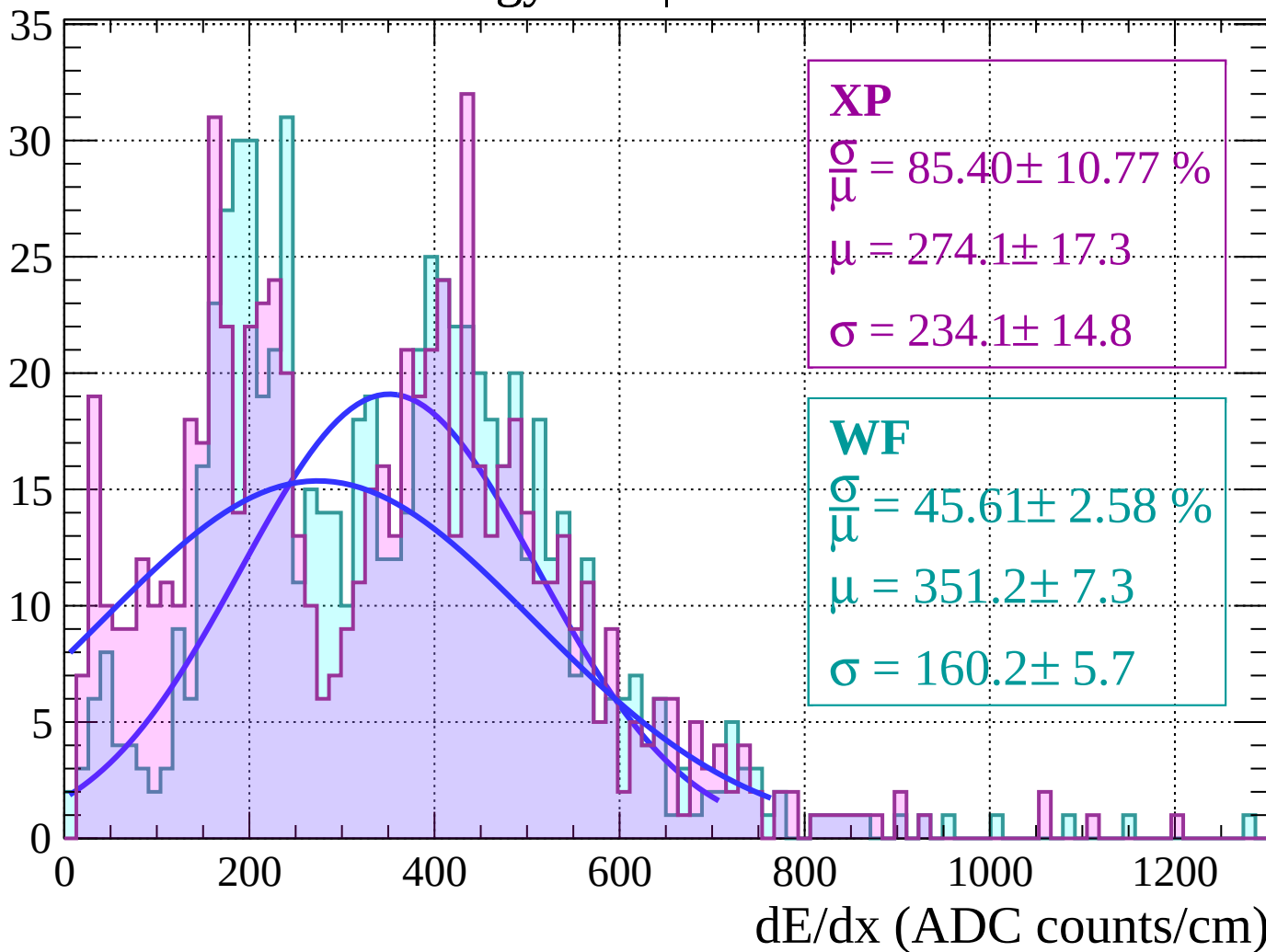
$$\sigma = 234.1 \pm 14.8$$

WF

$$\frac{\sigma}{\mu} = 45.61 \pm 2.58 \%$$

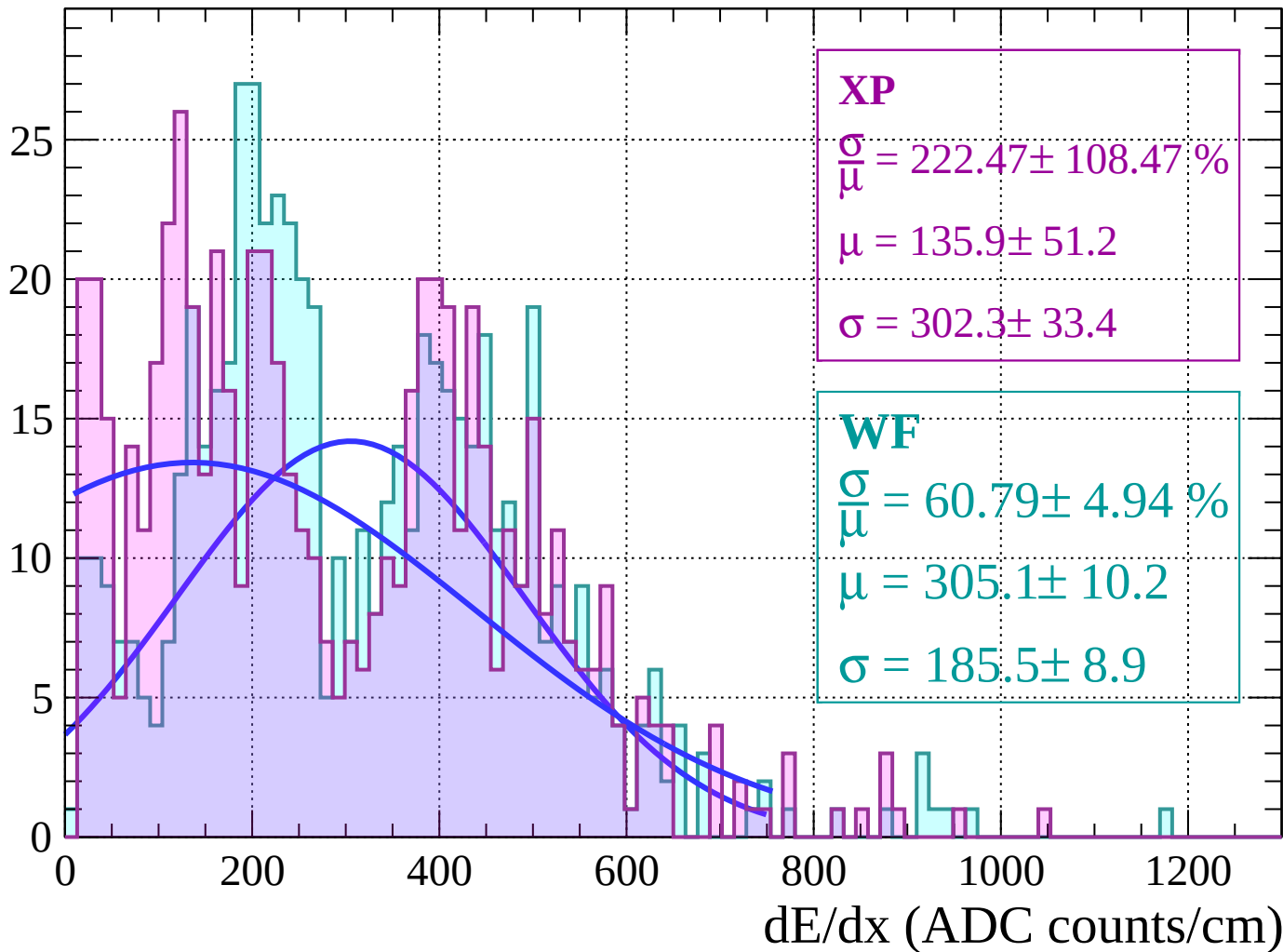
$$\mu = 351.2 \pm 7.3$$

$$\sigma = 160.2 \pm 5.7$$



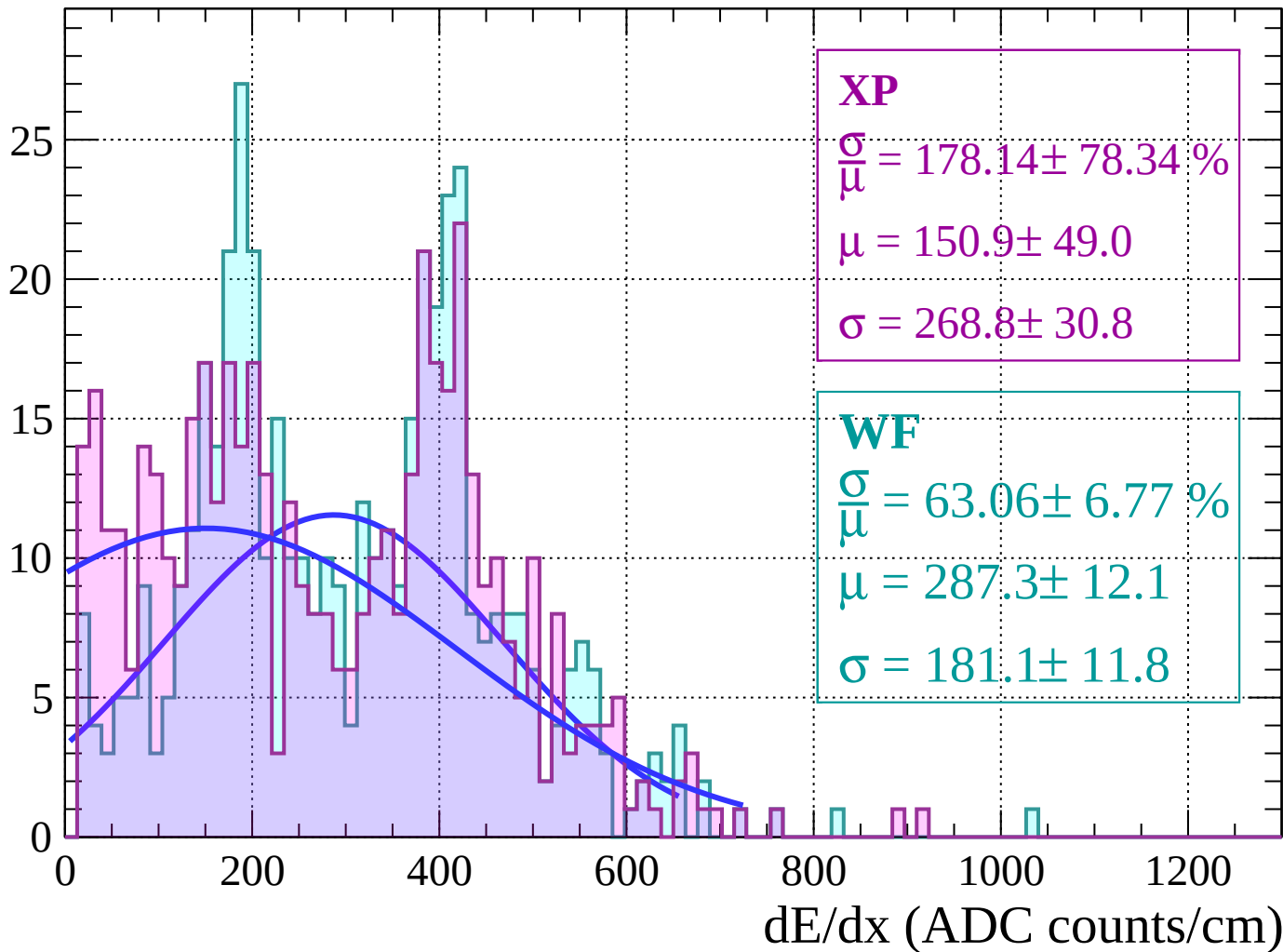
Energy loss | $66 < \theta < 68$

Count



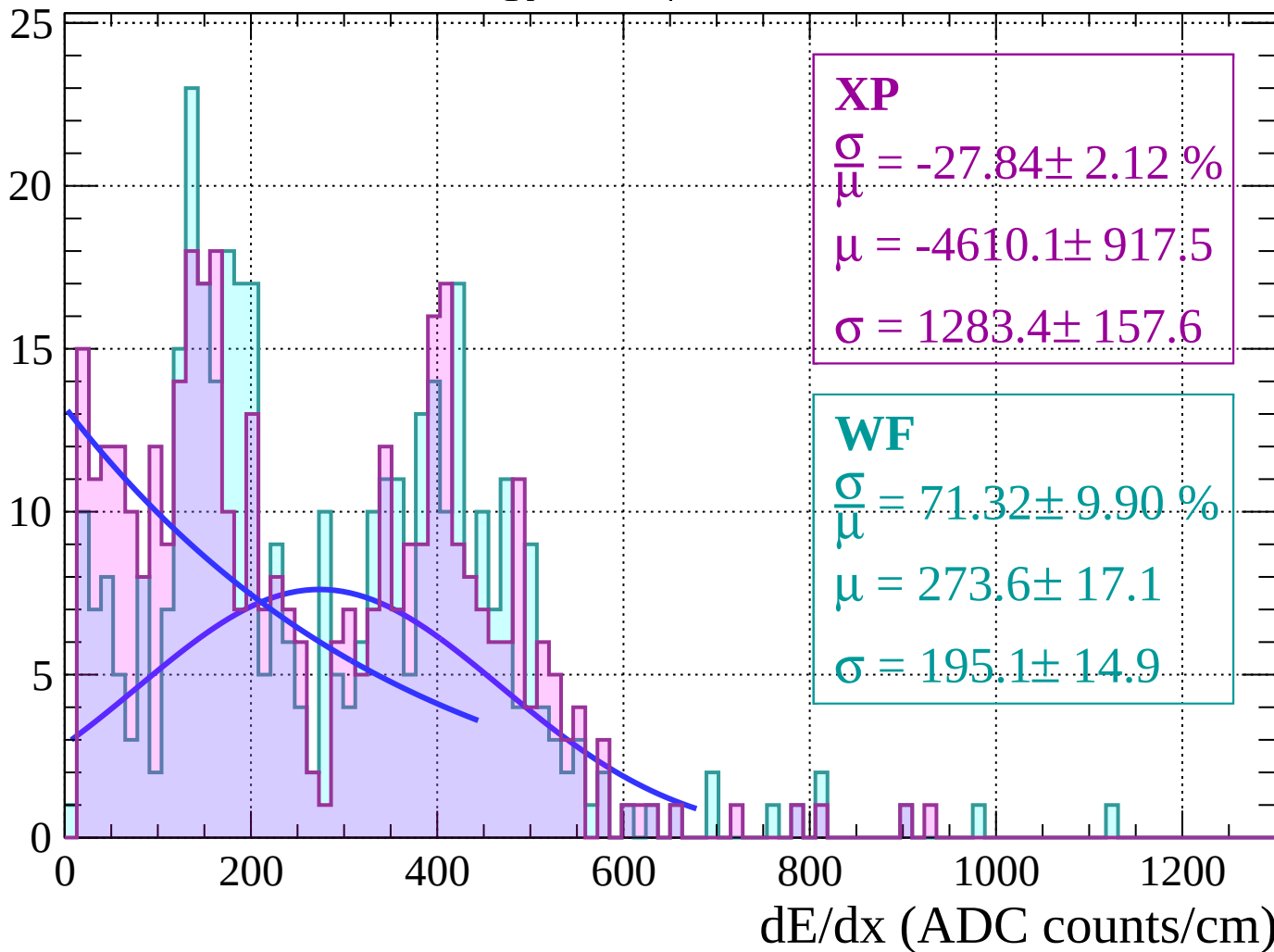
Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

Count



Energy loss | $72 < \theta < 74$

Count

XP

$$\frac{\sigma}{\mu} = -29.33 \pm 2.74 \%$$

$$\mu = -3345.8 \pm 840.3$$

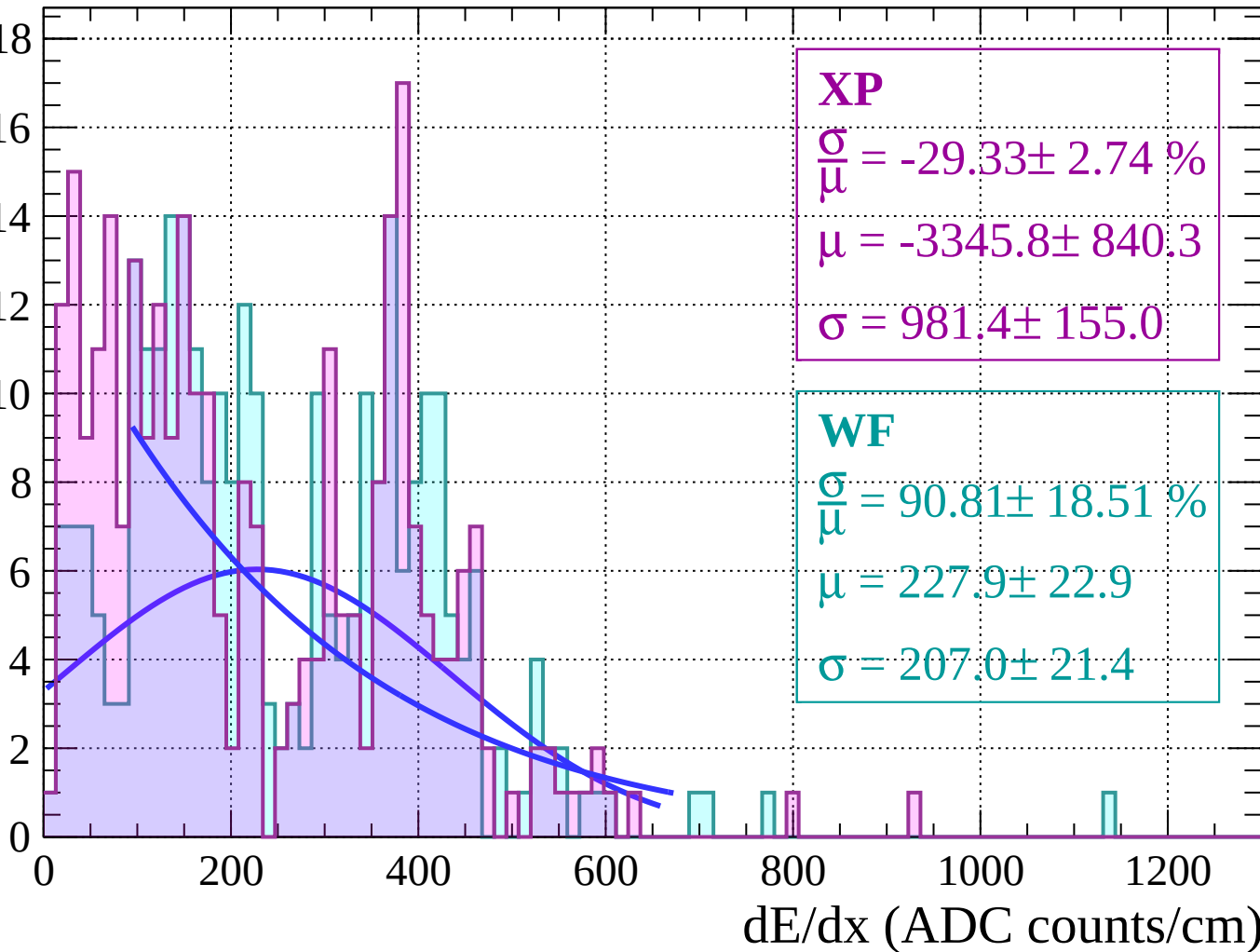
$$\sigma = 981.4 \pm 155.0$$

WF

$$\frac{\sigma}{\mu} = 90.81 \pm 18.51 \%$$

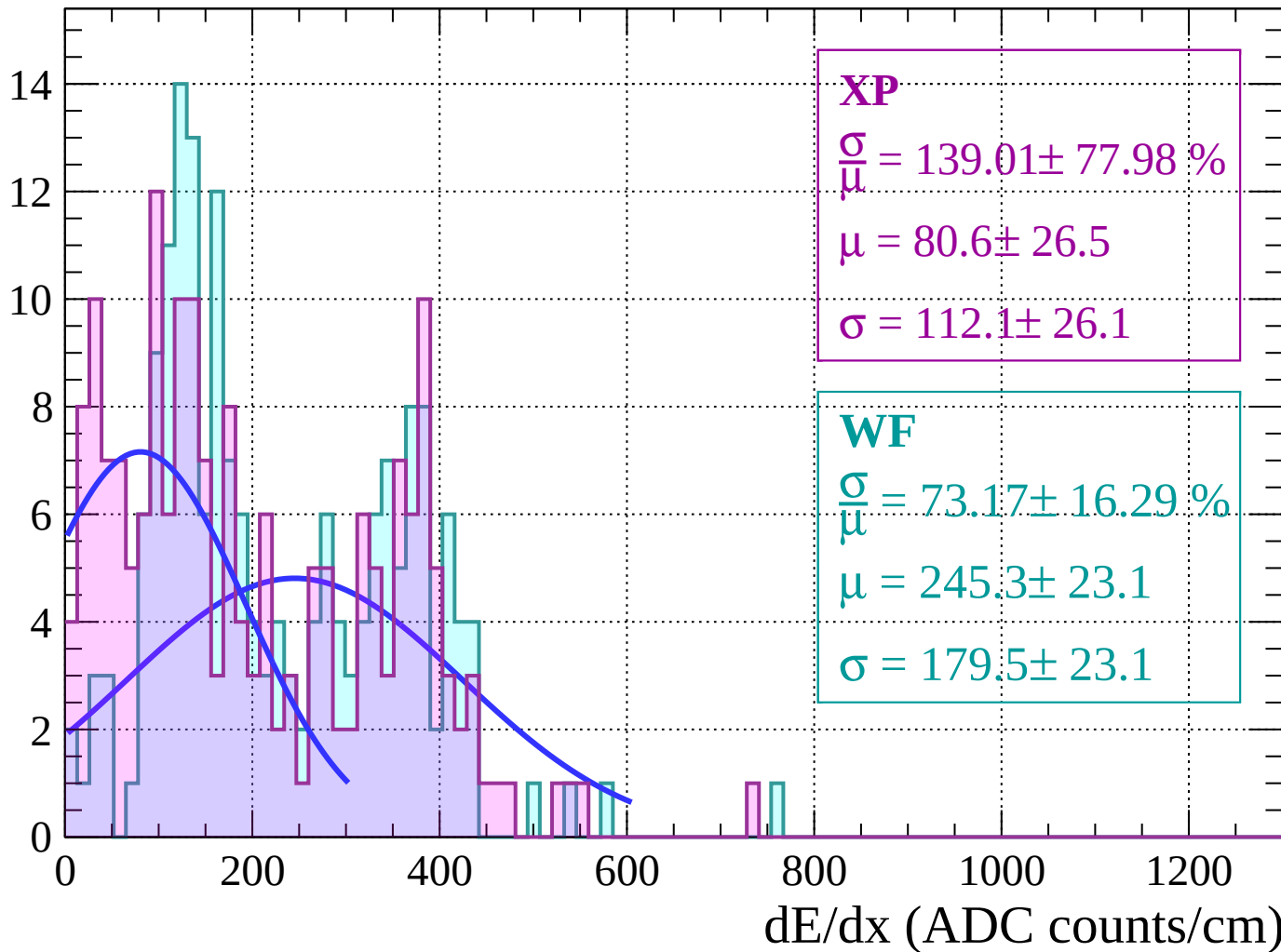
$$\mu = 227.9 \pm 22.9$$

$$\sigma = 207.0 \pm 21.4$$



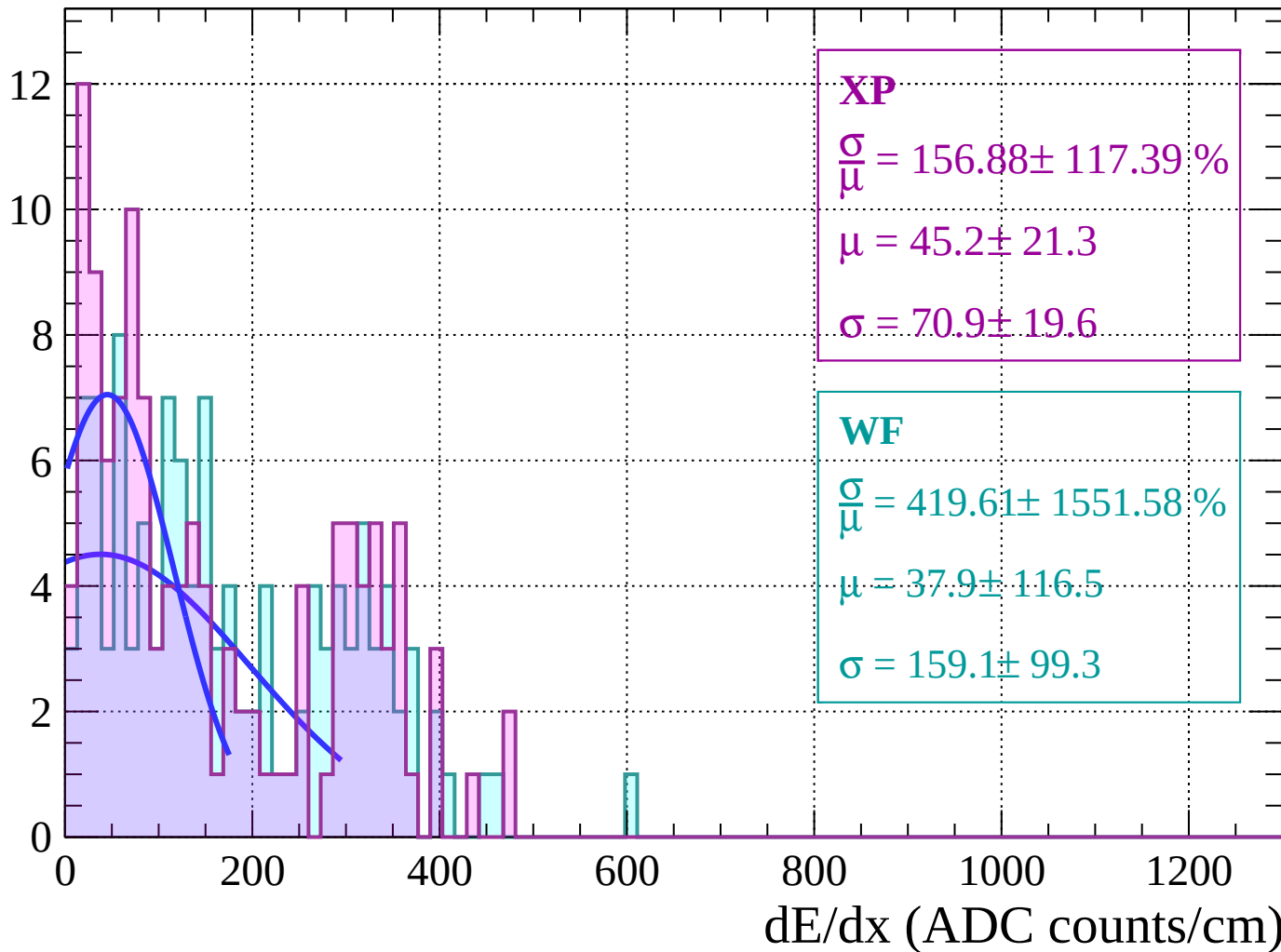
Energy loss | $74 < \theta < 76$

Count



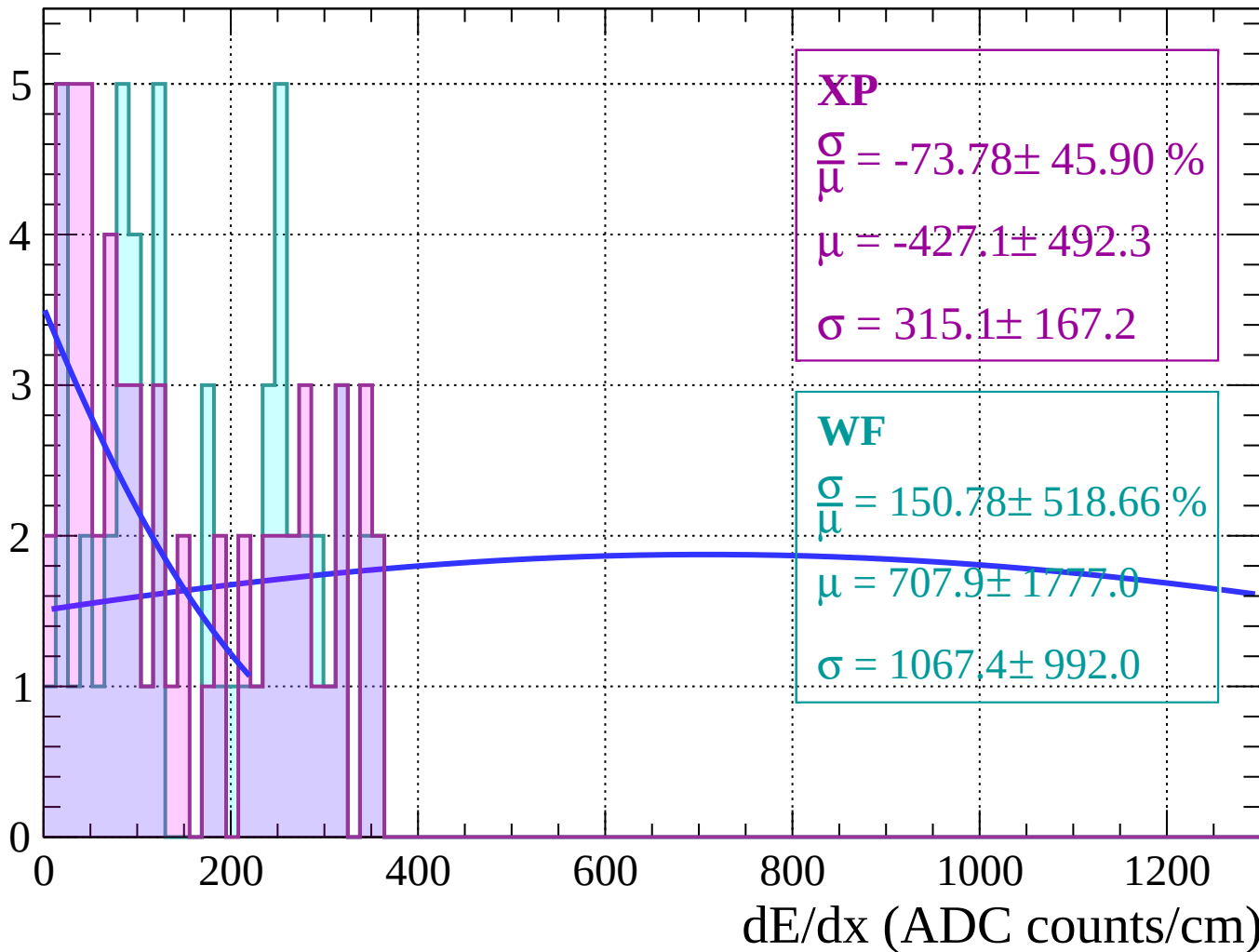
Energy loss | $76 < \theta < 78$

Count



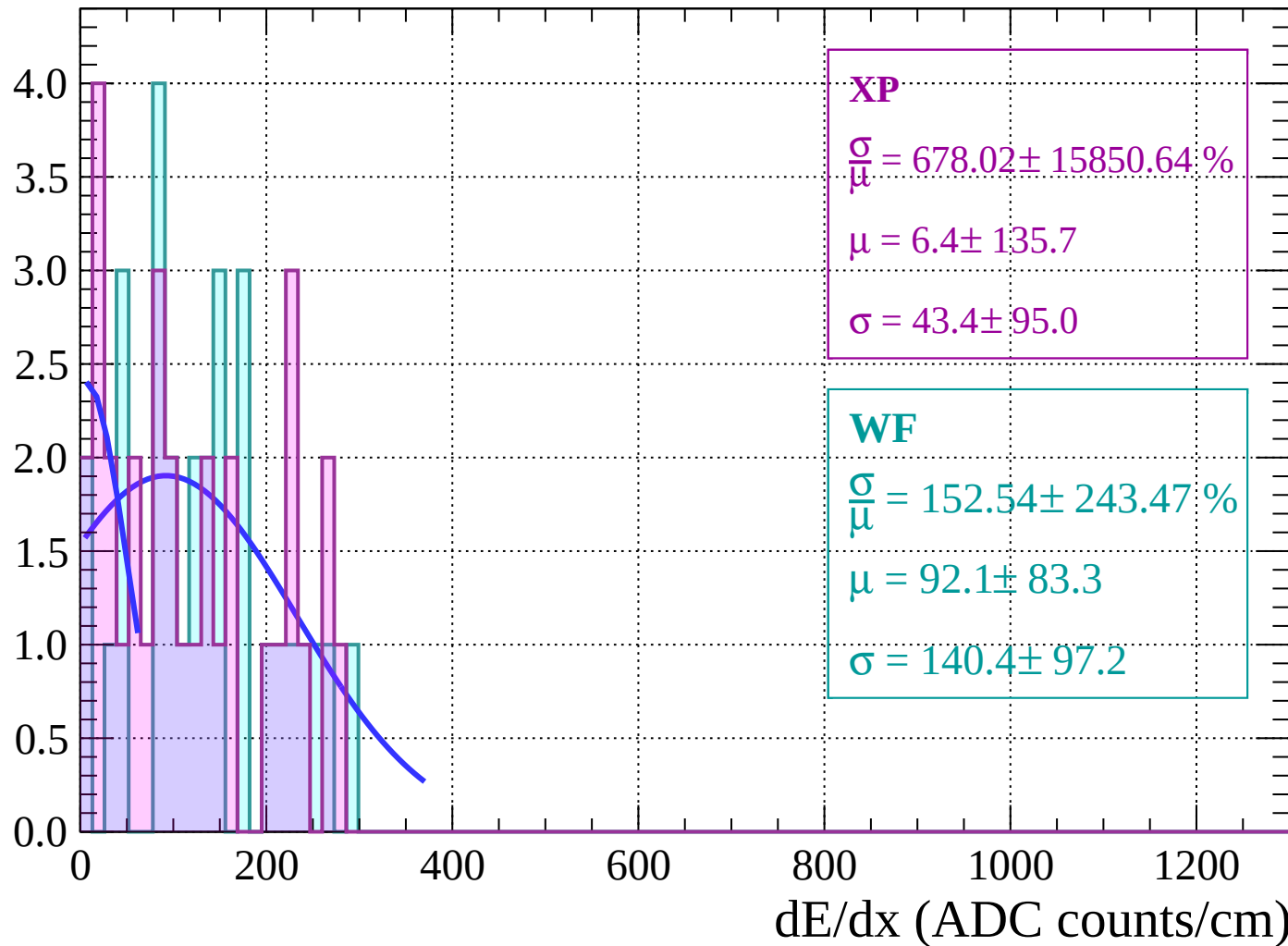
Energy loss | $78 < \theta < 80$

Count



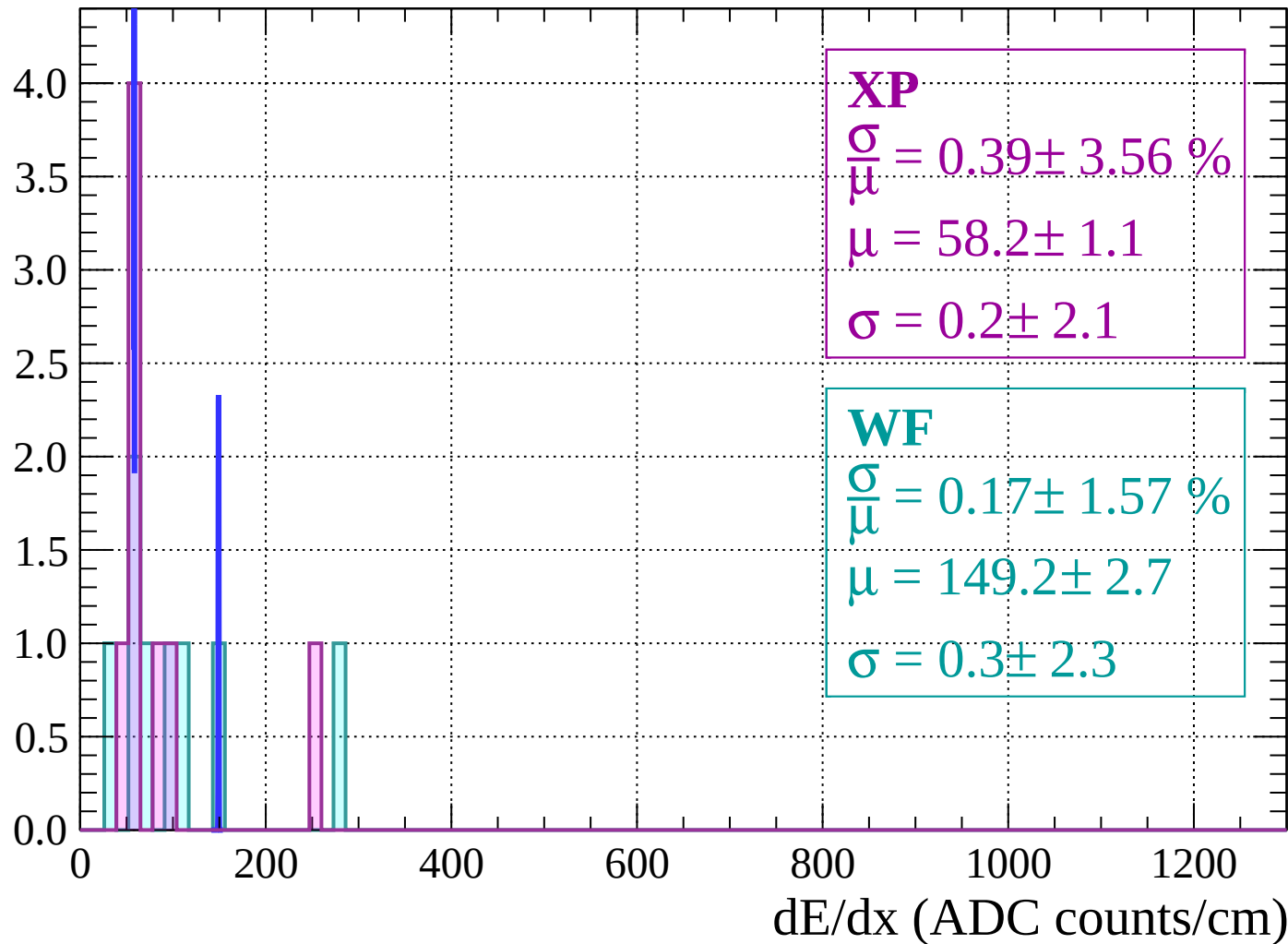
Energy loss | $80 < \theta < 82$

Count



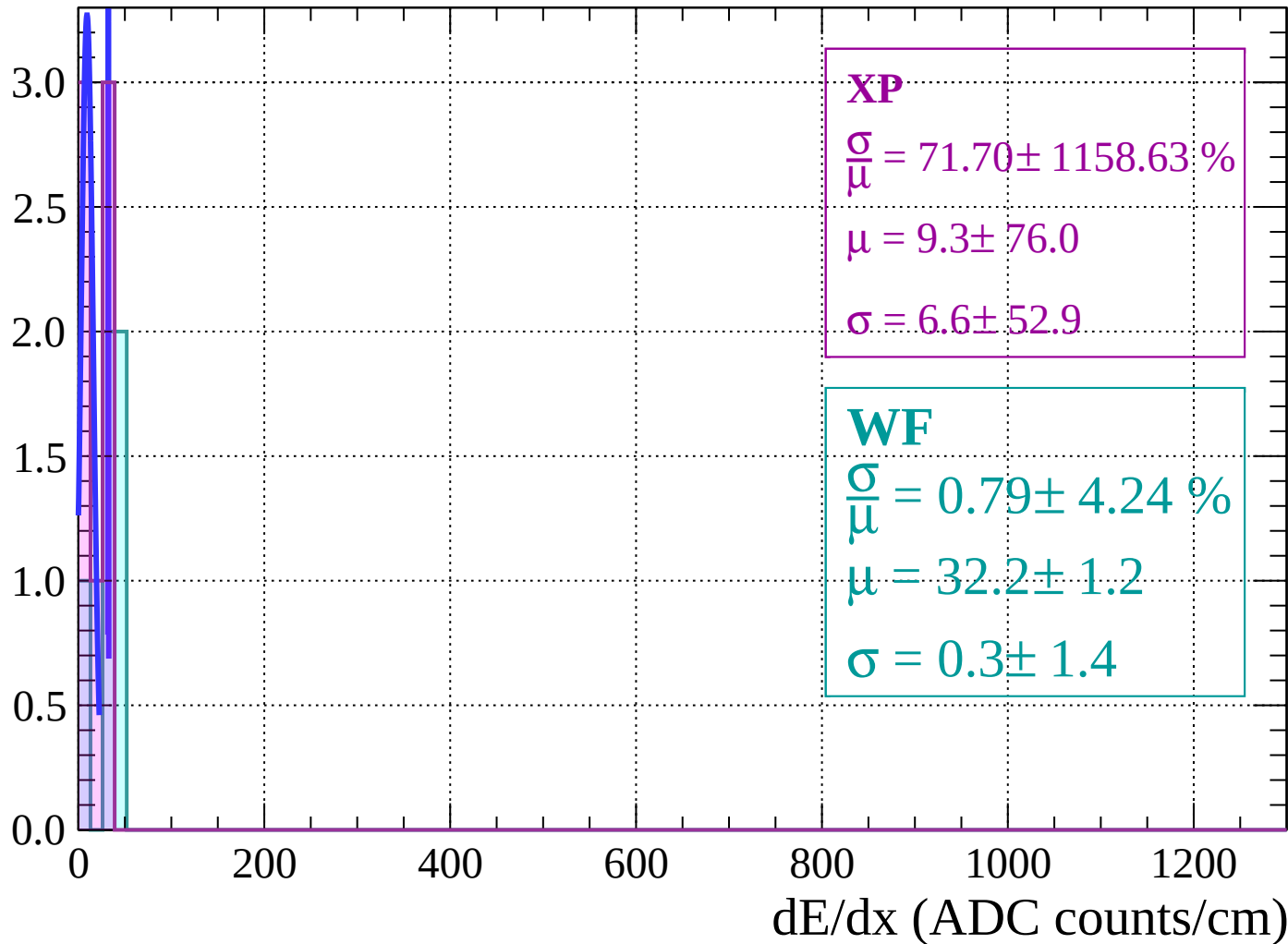
Energy loss | $82 < \theta < 84$

Count



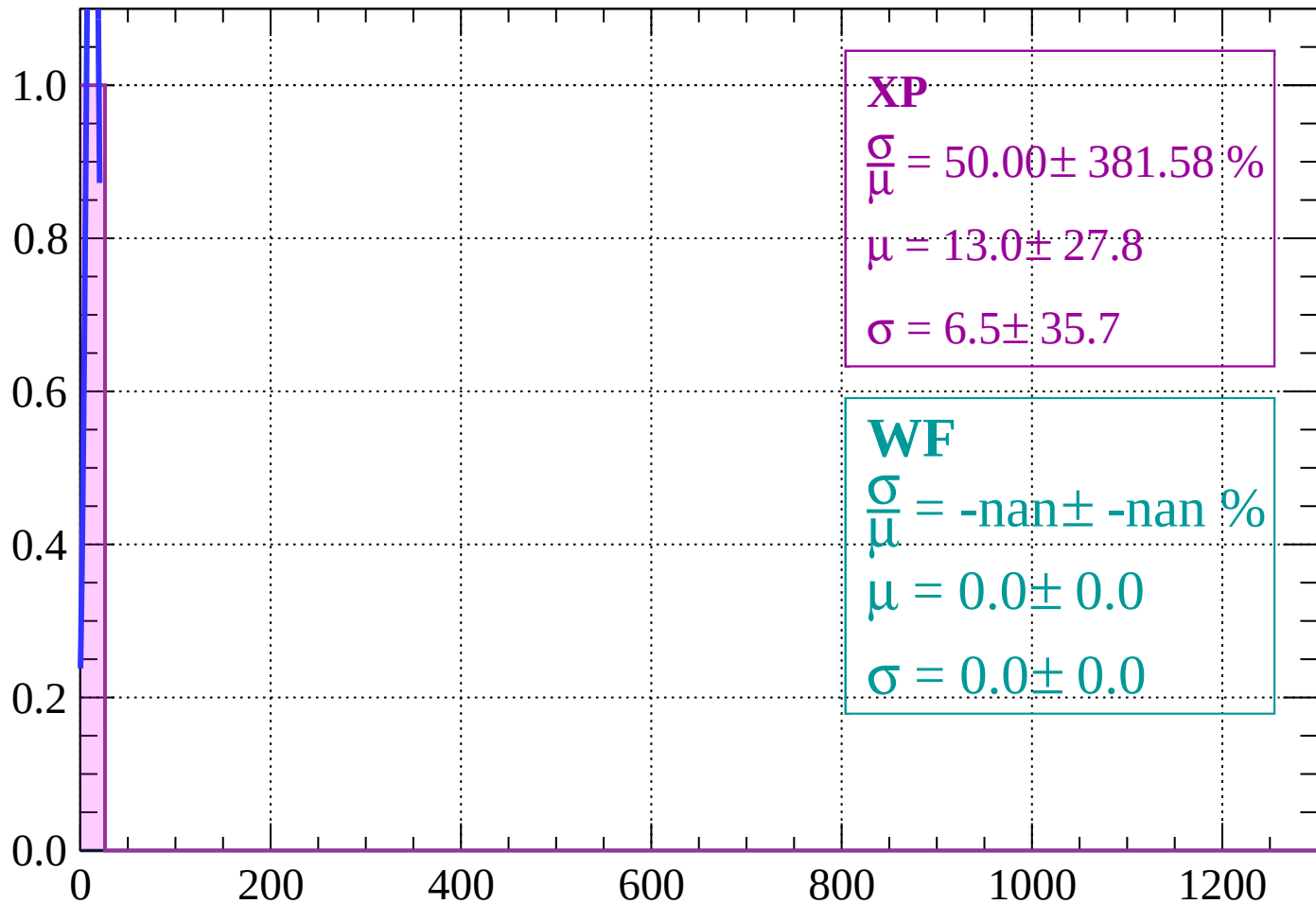
Energy loss | $84 < \theta < 86$

Count



Energy loss | $86 < \theta < 88$

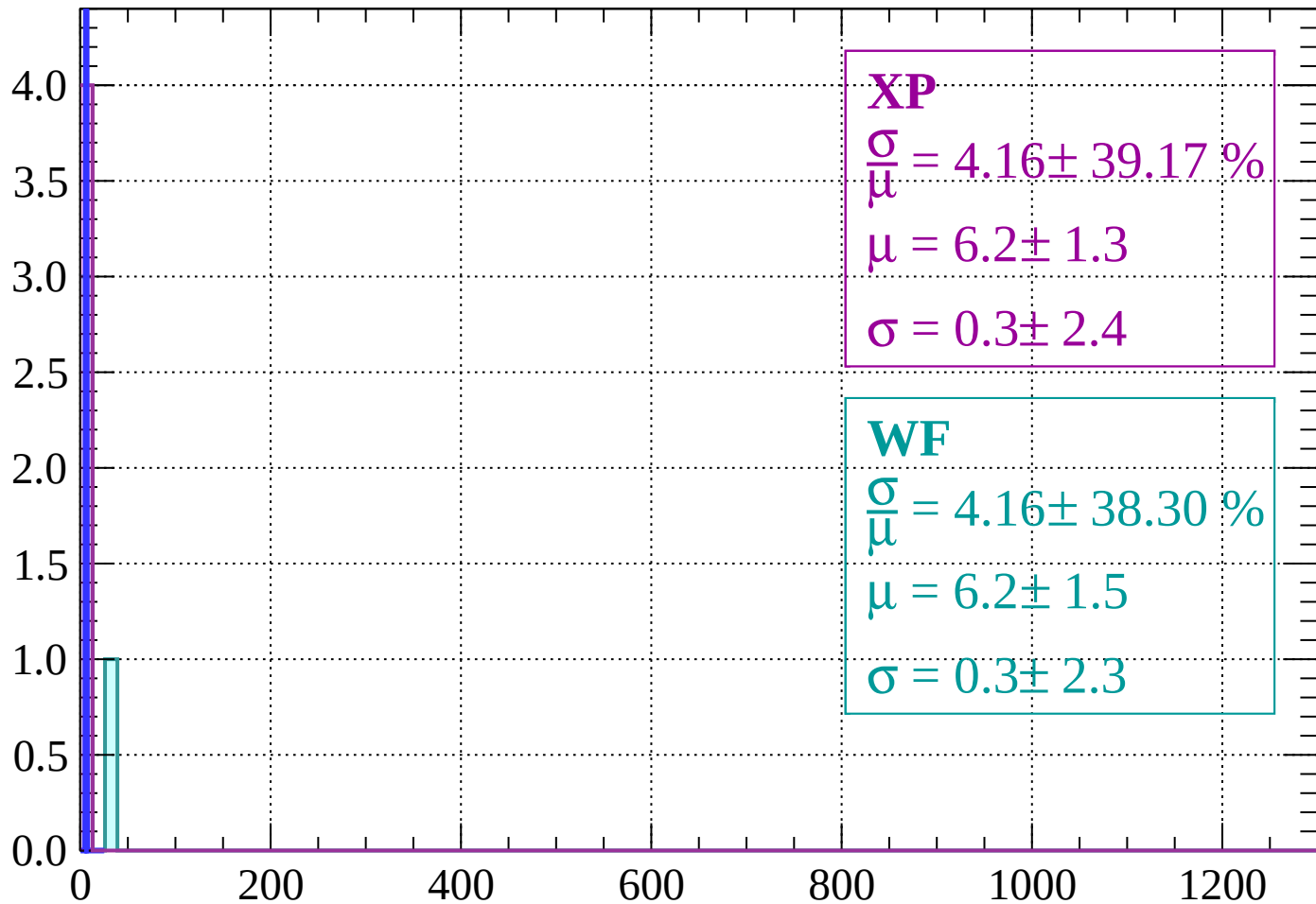
Count



dE/dx (ADC counts/cm)

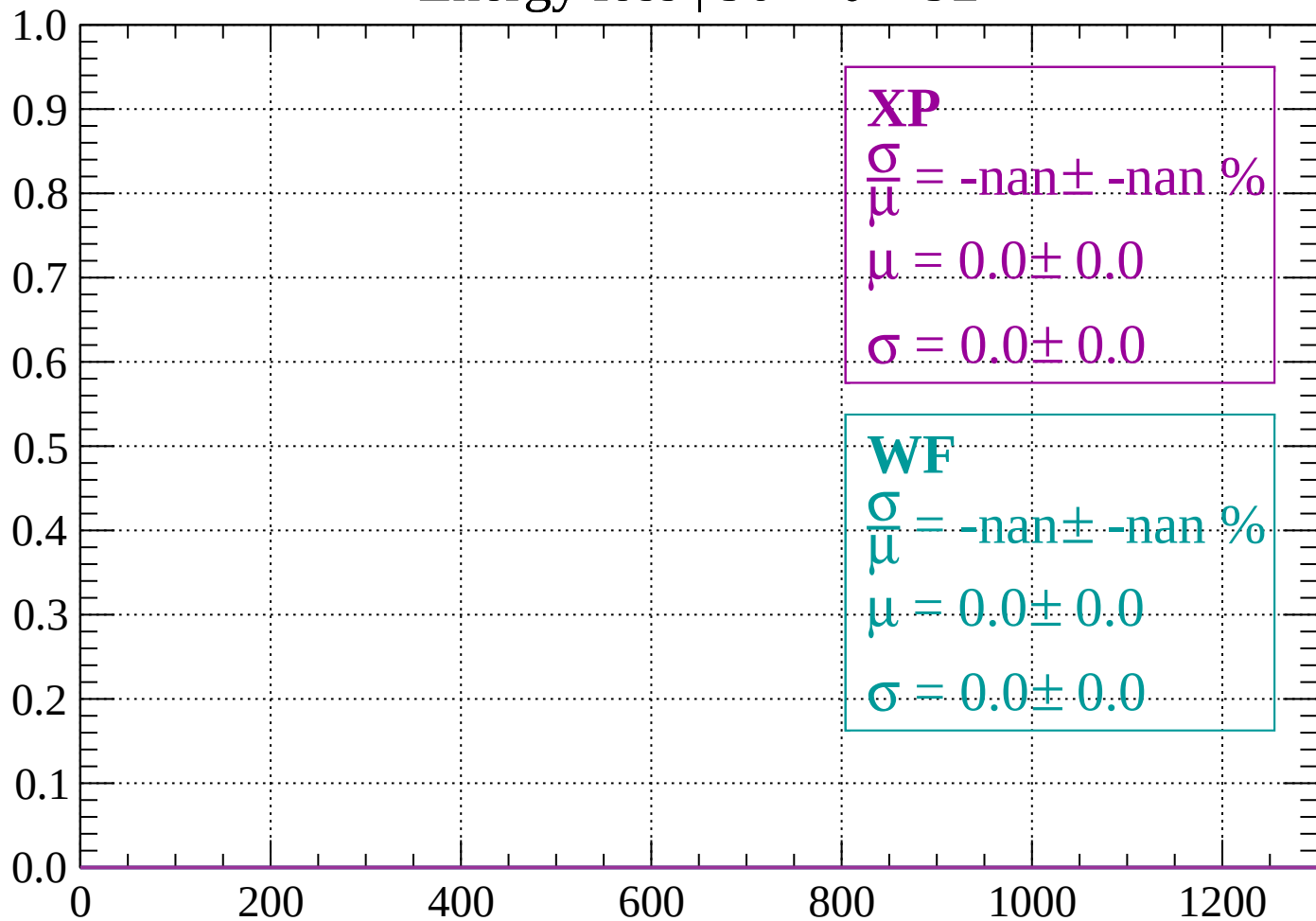
Energy loss | $88 < \theta < 90$

Count

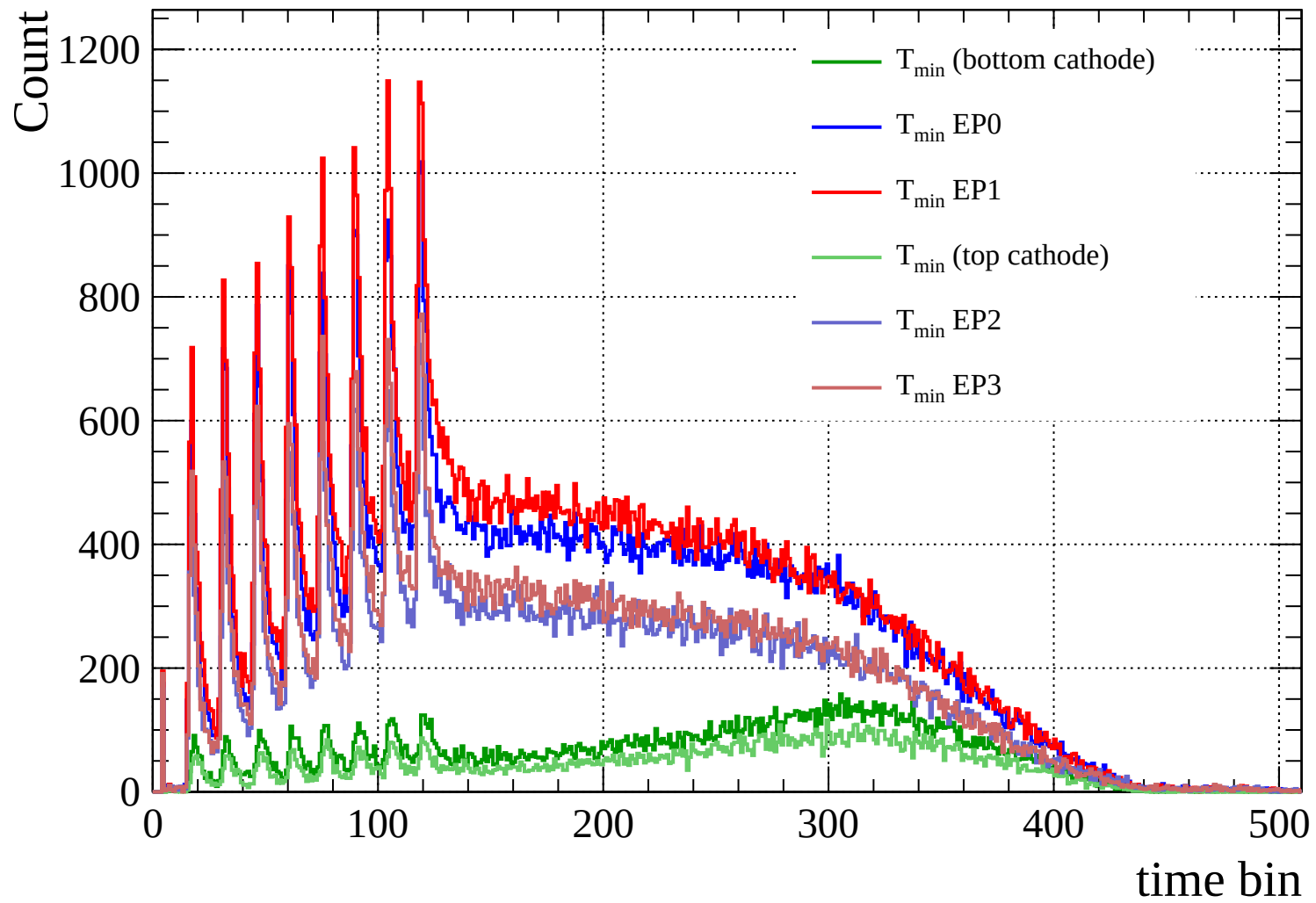


Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

